

# JESS Thermodynamic Database v8.9

## Aluminium

24-Jan-23

| Reaction No. 1315         |    |       |               |                  |   |       |        |
|---------------------------|----|-------|---------------|------------------|---|-------|--------|
| CO3-2 + Al+3 = Al+3_CO3-2 |    |       |               |                  |   |       |        |
| No                        | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                         | 25 | 0     | Inf. Dilution | lgK:7.14 (3SF)   | 2 | EAE   |        |
| 2                         | 37 | 0.15  | Blood Plasma  | lgK:5.7 (0.05SD) | 4 | ENS   | 94     |

| Reaction No. 1316                |    |       |               |                  |   |       |        |
|----------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 2<CO3-2> + Al+3 = Al+3_CO3-2 (2) |    |       |               |                  |   |       |        |
| No                               | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                | 25 | 0     | Inf. Dilution | lgK:10.3 (3SF)   | 2 | EAE   |        |
| 2                                | 37 | 0.15  | Blood Plasma  | lgK:8.4 (0.05SD) | 4 | ENS   | 94     |

| Reaction No. 1317                   |    |       |               |                   |   |       |        |
|-------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| H+1 + CO3-2 + Al+3 = Al+3_H+1_CO3-2 |    |       |               |                   |   |       |        |
| No                                  | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                   | 25 | 0     | Inf. Dilution | lgK:15.54 (4SF)   | 3 | EFE   | 30072  |
| 2                                   | 37 | 0.15  | Blood Plasma  | lgK:10.0 (0.05SD) | 4 | ENS   | 94     |

| Reaction No. 1434         |    |       |               |                  |   |       |        |
|---------------------------|----|-------|---------------|------------------|---|-------|--------|
| SO4-2 + Al+3 = Al+3_SO4-2 |    |       |               |                  |   |       |        |
| No                        | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                         | 5  | 0     | Inf. Dilution | lgK:1.60 (3SF)   | 0 | MEF   | 8374   |
| 2                         | 15 | 0     | Inf. Dilution | lgK:1.75 (3SF)   | 0 | MEF   | 8374   |
| 3                         | 25 | 0     | Inf. Dilution | lgK:3.2 (2SF)    | 3 | MEF   | 140    |
| 4                         | 25 | 0     | Inf. Dilution | lgK:3.89 (3SF)   | 4 | CRV   | 552    |
| 5                         | 25 | 0     | Inf. Dilution | lgK:4. (0.5SD)   | 3 | CMN   | 2171   |
| 6                         | 25 | 0     | Inf. Dilution | lgK:3.5 (0.5SD)  | 4 | ENS   | 2171   |
| 7                         | 25 | 0     | Inf. Dilution | lgK:3.70 (0.1SD) | 4 | ENS   | 5573   |
| 8                         | 25 | 0     | Inf. Dilution | lgK:3.02 (3SF)   | 3 | ENS   | 6496   |
| 9                         | 25 | 0     | Inf. Dilution | lgK:3.73 (3SF)   | 4 | MCN   | 7616   |
| 10                        | 25 | 0     | Inf. Dilution | lgK:1.90 (3SF)   | 0 | MEF   | 8374   |
| 11                        | 25 | 0     | Inf. Dilution | lgK:3.3 (2SF)    | 5 | MKN   | 8400   |

|    |    |        |               |                  |   |     |       |
|----|----|--------|---------------|------------------|---|-----|-------|
| 12 | 25 | 0      | Inf. Dilution | lgK:3.01(0.08SD) | 3 | MCL | 8402  |
| 13 | 25 | 0      | Inf. Dilution | lgK:3.20(3SF)    | 3 | MSL | 8620  |
| 14 | 25 | 0      | Inf. Dilution | lgK:3.53(0.45SD) | 5 | EOD | 22958 |
| 15 | 25 | 0      | Inf. Dilution | lgK:3.02(3SF)    | 4 | CRV | 32224 |
| 16 | 25 | 0      | CHEMVAL1      | lgK:3.89(0.01SD) | 0 | ENS | 5156  |
| 17 | 25 | 0      | CHEMVAL2      | lgK:3.89(0.01SD) | 0 | ENS | 5156  |
| 18 | 25 | 0      | CHEMVAL3      | lgK:3.01(0.01SD) | 0 | ENS | 5156  |
| 19 | 25 | 0      | CHEMVAL4      | lgK:3.01(0.01SD) | 3 | ENS | 5156  |
| 20 | 25 | 0      | Many Media    | lgK:3.5(2SF)     | 4 | EOD | 6953  |
| 21 | 25 | 0      | UCT_Sea       | lgK:3.90(0.01SD) | 4 | EDH | 5390  |
| 22 | 25 | 0      | Unknown       | lgK:3.02(3SF)    | 3 | ENS | 7373  |
| 23 | 25 | 0.0005 | K+1 salt      | lgK:3.606(4SF)   | 3 | MCN | 7616  |
| 24 | 25 | 0.0009 | K+1 salt      | lgK:3.588(4SF)   | 2 | MCN | 7616  |
| 25 | 25 | 0.0013 | K+1 salt      | lgK:3.523(4SF)   | 3 | MCN | 7616  |
| 26 | 25 | 0.0019 | K+1 salt      | lgK:3.535(4SF)   | 2 | MCN | 7616  |
| 27 | 25 | 0.003  | K+1 salt      | lgK:3.484(4SF)   | 2 | MCN | 7616  |
| 28 | 25 | 0.005  | K+1 salt      | lgK:3.444(4SF)   | 2 | MCN | 7616  |
| 29 | 25 | 0.007  | K+1 salt      | lgK:3.458(4SF)   | 0 | MCN | 7616  |
| 30 | 25 | 0.009  | K+1 salt      | lgK:3.498(4SF)   | 0 | MCN | 7616  |
| 31 | 25 | 0.011  | K+1 salt      | lgK:3.455(4SF)   | 0 | MCN | 7616  |
| 32 | 25 | 0.018  | K+1 salt      | lgK:3.618(4SF)   | 0 | MCN | 7616  |
| 33 | 25 | 0.1    | UCT_Sea       | lgK:3.45(0.01SD) | 0 | EDH | 5390  |
| 34 | 25 | 0.1    | Unknown       | lgK:1.46(3SF)    | 0 | CRV | 552   |
| 35 | 25 | 0.1    | Unknown       | lgK:1.28(3SF)    | 0 | MKN | 931   |
| 36 | 25 | 0.2    | UCT_Sea       | lgK:2.46(0.01SD) | 0 | EDH | 5390  |
| 37 | 25 | 0.3    | UCT_Sea       | lgK:2.90(0.01SD) | 0 | EDH | 5390  |
| 38 | 25 | 0.4    | UCT_Sea       | lgK:2.75(0.01SD) | 0 | EDH | 5390  |
| 39 | 25 | 0.5    | UCT_Sea       | lgK:2.63(0.01SD) | 0 | EDH | 5390  |
| 40 | 25 | 0.6    | NaClO4        | lgK:1.30(3SF)    | 3 | MEF | 621   |
| 41 | 25 | 0.6    | UCT_Sea       | lgK:2.53(0.01SD) | 0 | EDH | 5390  |
| 42 | 25 | 0.7    | UCT_Sea       | lgK:2.45(0.01SD) | 0 | EDH | 5390  |
| 43 | 25 | 1      | KCl/m         | lgK:3.35(3SF)    | 0 | MEF | 8387  |
| 44 | 25 | 1      | NaClO4        | lgK:1.5(0.05SD)  | 0 | MSP | 500   |
| 45 | 25 | 1      | NaClO4        | lgK:0.453(3SF)   | 0 | MCL | 8684  |
| 46 | 25 | 3      | NaClO4        | lgK:1.2(0.05SD)  | 3 | MSP | 500   |
| 47 | 25 | 5      | NaClO4        | lgK:1.2(0.05SD)  | 3 | MSP | 500   |

|                                             |       |       |               |                   |   |     |      |
|---------------------------------------------|-------|-------|---------------|-------------------|---|-----|------|
| 48                                          | 29:30 | 0     | Inf. Dilution | lgK:2.04 (3SF)    | 0 | MSP | 140  |
| Note (48): Low wgt for internal consistency |       |       |               |                   |   |     |      |
| 49                                          | 35    | 0     | Inf. Dilution | lgK:2.08 (3SF)    | 0 | MEF | 8374 |
| 50                                          | 35    | 1     | NaClO4        | lgK:0.617 (3SF)   | 0 | MCL | 8684 |
| 51                                          | 37    | 0.15  | Blood Plasma  | lgK:2.0 (0.05SD)  | 3 | ENS | 94   |
| 52                                          | 49.8  | 0.1   | NaCl/m        | lgK:2.30 (0.02SD) | 3 | MHY | 7702 |
| 53                                          | 49.9  | 0.099 | NaCl/m        | lgK:2.51 (0.03SD) | 2 | MHY | 7702 |
| 54                                          | 49.9  | 0.307 | NaCl/m        | lgK:1.88 (0.01SD) | 3 | MHY | 7702 |
| 55                                          | 50    | 0     | NaCl/m        | lgK:3.7 (0.4MD)   | 3 | MHY | 7702 |
| 56                                          | 50    | 0.1   | NaCl/m        | lgK:2.3 (0.2MD)   | 3 | MHY | 7702 |
| 57                                          | 50    | 0.2   | NaCl/m        | lgK:2.0 (0.2MD)   | 6 | MHY | 7702 |
| 58                                          | 50    | 0.306 | NaCl/m        | lgK:1.67 (0.02SD) | 3 | MHY | 7702 |
| 59                                          | 50    | 0.307 | NaCl/m        | lgK:1.57 (0.04SD) | 2 | MHY | 7702 |
| 60                                          | 50    | 0.5   | NaCl/m        | lgK:1.6 (0.3MD)   | 3 | MHY | 7702 |
| 61                                          | 50    | 1     | KCl/m         | lgK:3.59 (3SF)    | 0 | MEF | 8387 |
| 62                                          | 50    | 1     | NaCl/m        | lgK:1.2 (0.4MD)   | 3 | MHY | 7702 |
| 63                                          | 50    | 1     | NaClO4        | lgK:0.842 (3SF)   | 0 | MCL | 8684 |
| 64                                          | 50    | 1.016 | NaCl/m        | lgK:1.28 (0.04SD) | 3 | MHY | 7702 |
| 65                                          | 50    | 1.026 | NaCl/m        | lgK:1.36 (0.05SD) | 0 | MHY | 7702 |
| 66                                          | 70    | 1     | NaClO4        | lgK:1.11 (3SF)    | 0 | MCL | 8684 |
| 67                                          | 74.8  | 0.099 | NaCl/m        | lgK:2.35 (0.02SD) | 0 | MHY | 7702 |
| 68                                          | 74.8  | 0.309 | NaCl/m        | lgK:1.40 (0.07SD) | 0 | MHY | 7702 |
| 69                                          | 74.8  | 1.016 | NaCl/m        | lgK:1.32 (0.04SD) | 2 | MHY | 7702 |
| 70                                          | 74.9  | 0.312 | NaCl/m        | lgK:1.94 (0.01SD) | 3 | MHY | 7702 |
| 71                                          | 75    | 0     | NaCl/m        | lgK:3.9 (0.4MD)   | 0 | MHY | 7702 |
| 72                                          | 75    | 0.1   | NaCl/m        | lgK:2.4 (0.2MD)   | 0 | MHY | 7702 |
| 73                                          | 75    | 0.2   | NaCl/m        | lgK:2.0 (0.2MD)   | 0 | MHY | 7702 |
| 74                                          | 75    | 0.5   | NaCl/m        | lgK:1.6 (0.3MD)   | 2 | MHY | 7702 |
| 75                                          | 75    | 1     | KCl/m         | lgK:4.08 (3SF)    | 0 | MEF | 8387 |
| 76                                          | 75    | 1     | NaCl/m        | lgK:1.2 (0.4MD)   | 2 | MHY | 7702 |
| 77                                          | 99.5  | 0.098 | NaCl/m        | lgK:2.58 (0.04SD) | 0 | MHY | 7702 |
| 78                                          | 99.7  | 0.098 | NaCl/m        | lgK:2.18 (0.11SD) | 0 | MHY | 7702 |
| 79                                          | 99.7  | 0.306 | NaCl/m        | lgK:1.66 (0.07SD) | 0 | MHY | 7702 |
| 80                                          | 99.7  | 1.021 | NaCl/m        | lgK:1.28 (0.06SD) | 0 | MHY | 7702 |
| 81                                          | 99.7  | 1.024 | NaCl/m        | lgK:1.73 (0.04SD) | 5 | MHY | 7702 |
| 82                                          | 99.9  | 0.311 | NaCl/m        | lgK:2.22 (0.03SD) | 0 | MHY | 7702 |

|                                              |       |       |                 |                       |   |     |      |
|----------------------------------------------|-------|-------|-----------------|-----------------------|---|-----|------|
| 83                                           | 100   | 0     | NaCl/m          | lgK:4.5 (2SF)         | 0 | MHY | 7702 |
| 84                                           | 100   | 0.1   | NaCl/m          | lgK:2.9 (0.2MD)       | 0 | MHY | 7702 |
| 85                                           | 100   | 0.2   | NaCl/m          | lgK:2.6 (0.2MD)       | 3 | MHY | 7702 |
| 86                                           | 100   | 0.5   | NaCl/m          | lgK:2.1 (0.3MD)       | 6 | MHY | 7702 |
| 87                                           | 100   | 1     | KCl/m           | lgK:4.66 (3SF)        | 0 | MEF | 8387 |
| 88                                           | 100   | 1     | NaCl/m          | lgK:1.6 (0.4MD)       | 3 | MHY | 7702 |
| 89                                           | 124.3 | 0.098 | NaCl/m          | lgK:3.98 (0.03SD)     | 0 | MHY | 7702 |
| 90                                           | 124.5 | 0.098 | NaCl/m          | lgK:4.18 (0.06SD)     | 0 | MHY | 7702 |
| 91                                           | 124.5 | 0.304 | NaCl/m          | lgK:3.56 (0.06SD)     | 0 | MHY | 7702 |
| 92                                           | 124.7 | 1.007 | NaCl/m          | lgK:2.03 (0.03SD)     | 5 | MHY | 7702 |
| 93                                           | 125   | 0     | NaCl/m          | lgK:5.6 (2SF)         | 0 | MHY | 7702 |
| 94                                           | 125   | 0.1   | NaCl/m          | lgK:3.8 (0.4MD)       | 0 | MHY | 7702 |
| 95                                           | 125   | 0.2   | NaCl/m          | lgK:3.4 (0.3MD)       | 2 | MHY | 7702 |
| 96                                           | 125   | 0.301 | NaCl/m          | lgK:3.18 (0.03SD)     | 2 | MHY | 7702 |
| 97                                           | 125   | 0.5   | NaCl/m          | lgK:2.9 (0.4MD)       | 0 | MHY | 7702 |
| 98                                           | 125   | 1     | KCl/m           | lgK:5.34 (3SF)        | 0 | MEF | 8387 |
| 99                                           | 125   | 1     | NaCl/m          | lgK:2.3 (0.4MD)       | 3 | MHY | 7702 |
| 100                                          | 25    | 0     | Inf. Dilution   | dH:2.29 (0.02SD)      | 4 | ENS | 2171 |
| 101                                          | 25    | 0     | Inf. Dilution   | dH:2.15 (3SF)         | 4 | ENS | 6496 |
| 102                                          | 25    | 0     | Inf. Dilution   | dH:2.29 (0.08SD)      | 4 | MCL | 8402 |
| 103                                          | 25    | 0     | Unknown         | dH:2.15 (3SF)         | 3 | ENS | 7373 |
| 104                                          | 25    | 1     | KCl/m           | dH:6.61 (3SF) kJ      | 0 | MTD | 8387 |
| 105                                          | 25    | 1     | NaClO4          | dH:6.93 (3SF)         | 0 | MCL | 8684 |
| 106                                          | 25    | 3     | Unknown         | dH:6.9 (2SF)          | 0 | CRV | 7271 |
| Note (106): Low wgt for internal consistency |       |       |                 |                       |   |     |      |
| 107                                          | 35    | 1     | NaClO4          | dH:6.85 (3SF)         | 0 | MCL | 8684 |
| 108                                          | 50    | 0     | Inf. Dilution/m | dH:-10. (30.MD) kJ    | 0 | MTD | 7702 |
| 109                                          | 50    | 1     | KCl/m           | dH:30.3 (3SF) kJ      | 0 | MTD | 8387 |
| 110                                          | 50    | 1     | NaClO4          | dH:6.78 (3SF)         | 0 | MCL | 8684 |
| 111                                          | 70    | 1     | NaClO4          | dH:6.78 (3SF)         | 0 | MCL | 8684 |
| 112                                          | 75    | 1     | KCl/m           | dH:50.9 (3SF) kJ      | 0 | MTD | 8387 |
| 113                                          | 100   | 1     | KCl/m           | dH:68.6 (3SF) kJ      | 0 | MTD | 8387 |
| 114                                          | 125   | 1     | KCl/m           | dH:83.3 (3SF) kJ      | 0 | MTD | 8387 |
| 115                                          | 50    | 0     | Inf. Dilution/m | Cp:1900. (800.MD) J/K | 5 | MTD | 7702 |

Reaction No. 1494

| Citric-3 + Al+3 = Al+3_Citric-3 |       |       |               |                      |   |       |        |
|---------------------------------|-------|-------|---------------|----------------------|---|-------|--------|
| No                              | t     | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
| 1                               | 10    | 0.249 | NaCl          | lgK:7.19 (3SF)       | 4 | MGL   | 30072  |
| 2                               | 20    | 0.1   | Unknown       | lgK:8.54 (3SF)       | 3 | ENS   | 1539   |
| 3                               | 23    | 1     | KNO3          | lgK:7.38 (3SF)       | 0 | MSP   | 3783   |
| 4                               | 25    | 0     | Inf. Dilution | lgK:10.72 (4SF)      | 0 | MSF   | 1717   |
| 5                               | 25    | 0     | Inf. Dilution | lgK:10.16 (0.03SD)   | 3 | EDH   | 30072  |
| 6                               | 25    | 0     | Inf. Dilution | lgK:9.91 (3SF)       | 4 | CRV   | 31301  |
| 7                               | 25    | 0.1   | Inert         | lgK:8.35 (0.01SD)    | 2 | MGL   | 1151   |
| 8                               | 25    | 0.1   | KCl           | lgK:8. (0.2SD)       | 3 | MGL   | 1151   |
| 9                               | 25    | 0.1   | KNO3          | lgK:7.98 (3SF)       | 3 | MGL   | 1244   |
| 10                              | 25    | 0.1   | NaClO4        | lgK:7.14 (3SF)       | 0 | ENS   | 2171   |
| 11                              | 25    | 0.1   | NaClO4        | lgK:8.65 (3SF) w     | 0 | MPH   | 3180   |
| 12                              | 25    | 0.1   | Unknown       | lgK:7.98 (3SF)       | 2 | CMN   | 1084   |
| 13                              | 25    | 0.249 | NaCl          | lgK:7.57 (0.02SD)    | 5 | MGL   | 30072  |
| 14                              | 25    | 0.956 | NaCl          | lgK:6.554 (0.002SD)  | 2 | MGL   | 30072  |
| 15                              | 25:37 | 0.15  | KNO3          | lgK:8.0 (2SF)        | 5 | MPT   | 3607   |
| 16                              | 37    | 0.15  | Blood Plasma  | lgK:8.3 (0.05SD)     | 2 | ENS   | 94     |
| 17                              | 37    | 0.15  | NaCl          | lgK:8.25 (0.004SD)   | 3 | MGL   | 925    |
| 18                              | 37    | 0.15  | NaCl          | lgK:7.87 (0.02SD)    | 3 | MGL   | 3574   |
| 19                              | 37    | 0.15  | NaCl          | lgK:8.34 (3SF)       | 2 | MGL   | 10029  |
| 20                              | 45    | 0.249 | NaCl          | lgK:7.82 (0.03SD)    | 3 | MGL   | 30072  |
| 21                              | 25    | 0.25  | Unknown       | dG:-43.2 (0.17SD) kJ | 3 | EFE   | 30072  |
| 22                              | 25    | 0.25  | Unknown       | dH:29. (9.SD) kJ     | 3 | EFE   | 30072  |

| Reaction No. 1495                     |    |       |               |                    |   |       |        |
|---------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| 2<Citric-3> + Al+3 = Al+3_Citric-3(2) |    |       |               |                    |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                     | 10 | 0.249 | NaCl          | lgK:11.40 (0.04SD) | 4 | MGL   | 30072  |
| 2                                     | 25 | 0     | Inf. Dilution | lgK:14.77 (0.04SD) | 3 | EDH   | 30072  |
| 3                                     | 25 | 0     | Inf. Dilution | lgK:14.12 (4SF)    | 3 | CRV   | 31301  |
| 4                                     | 25 | 0.1   | Inert         | lgK:13.36 (0.02SD) | 0 | MGL   | 1151   |
| 5                                     | 25 | 0.1   | KCl           | lgK:13. (0.3SD)    | 3 | MGL   | 1151   |
| 6                                     | 25 | 0.249 | NaCl          | lgK:12.12 (0.04SD) | 3 | MGL   | 30072  |
| 7                                     | 25 | 0.6   | NaCl          | lgK:11.7 (3SF)     | 2 | ENS   | 1116   |
| 8                                     | 25 | 0.956 | NaCl          | lgK:10.85 (0.02SD) | 0 | MGL   | 30072  |

|    |       |       |              |                      |   |     |       |
|----|-------|-------|--------------|----------------------|---|-----|-------|
| 9  | 25:37 | 0.15  | NaCl         | lgK:13.0 (3SF)       | 2 | MPT | 3607  |
| 10 | 37    | 0.15  | Blood Plasma | lgK:13.1 (0.05SD)    | 2 | ENS | 94    |
| 11 | 37    | 0.15  | NaCl         | lgK:13.1 (0.04SD)    | 4 | MGL | 925   |
| 12 | 37    | 0.15  | NaCl         | lgK:13.56 (4SF)      | 0 | MGL | 10029 |
| 13 | 45    | 0.249 | NaCl         | lgK:12.74 (0.05SD)   | 2 | MGL | 30072 |
| 14 | 25    | 0.25  | Unknown      | dG:-69.2 (0.23SD) kJ | 3 | EFE | 30072 |
| 15 | 25    | 0.25  | Unknown      | dH:64. (9.SD) kJ     | 3 | EFE | 30072 |

| Reaction No. 1496                         |    |       |               |                      |   |       |        |
|-------------------------------------------|----|-------|---------------|----------------------|---|-------|--------|
| Al+3 + Citric-3 + H+1 = Al+3_H+1_Citric-3 |    |       |               |                      |   |       |        |
| No                                        | t  | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
| 1                                         | 10 | 0.249 | NaCl          | lgK:9.55 (0.05SD)    | 3 | MGL   | 30072  |
| 2                                         | 25 | 0     | Inf. Dilution | lgK:12.59 (0.04SD)   | 3 | EDH   | 30072  |
| 3                                         | 25 | 0.1   | Inert         | lgK:11.01 (0.01SD)   | 0 | MGL   | 1151   |
| 4                                         | 25 | 0.1   | KCl           | lgK:10.81 (0.02SD)   | 2 | MGL   | 1151   |
| 5                                         | 25 | 0.249 | NaCl          | lgK:9.91 (0.05SD)    | 4 | MGL   | 30072  |
| 6                                         | 25 | 0.956 | NaCl          | lgK:9.32 (0.01SD)    | 3 | MGL   | 30072  |
| 7                                         | 37 | 0.15  | Blood Plasma  | lgK:10.5 (0.05SD)    | 2 | ENS   | 94     |
| 8                                         | 37 | 0.15  | NaCl          | lgK:10.50 (0.008SD)  | 3 | MGL   | 925    |
| 9                                         | 37 | 0.15  | NaCl          | lgK:10.1 (0.05SD)    | 3 | MGL   | 3574   |
| 10                                        | 45 | 0.249 | NaCl          | lgK:9.93 (0.017SD)   | 3 | MGL   | 30072  |
| 11                                        | 25 | 0.25  | Unknown       | dG:-56.6 (0.23SD) kJ | 3 | EFE   | 30072  |
| 12                                        | 25 | 0.25  | Unknown       | dH:16. (12.SD) kJ    | 3 | EFE   | 30072  |

| Reaction No. 1497                                |    |       |               |                      |   |       |        |
|--------------------------------------------------|----|-------|---------------|----------------------|---|-------|--------|
| Al+3 + Citric-3 + H2O = Al+3_OH-1_Citric-3 + H+1 |    |       |               |                      |   |       |        |
| No                                               | t  | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
| 1                                                | 10 | 0.249 | NaCl          | lgK:2.66 (0.04SD)    | 4 | MGL   | 30072  |
| 2                                                | 25 | 0     | Inf. Dilution | lgK:6.52 (3SF)       | 2 | EAE   |        |
| 3                                                | 25 | 0     | Inf. Dilution | lgK:6.41 (0.02SD)    | 3 | EDH   | 30072  |
| 4                                                | 25 | 0.249 | NaCl          | lgK:3.81 (0.04SD)    | 4 | MGL   | 30072  |
| 5                                                | 25 | 0.956 | NaCl          | lgK:3.167 (0.003SD)  | 3 | MGL   | 30072  |
| 6                                                | 37 | 0.15  | Blood Plasma  | lgK:3.5 (0.05SD)     | 0 | ENS   | 94     |
| 7                                                | 37 | 0.15  | NaCl          | lgK:4.6 (0.04SD)     | 3 | MGL   | 3574   |
| 8                                                | 45 | 0.249 | NaCl          | lgK:4.44 (0.05SD)    | 3 | MGL   | 30072  |
| 9                                                | 25 | 0.25  | Unknown       | dG:-21.7 (0.11SD) kJ | 3 | EFE   | 30072  |

|    |    |      |         |                  |   |     |       |
|----|----|------|---------|------------------|---|-----|-------|
| 10 | 25 | 0.25 | Unknown | dH:82. (17.SD)kJ | 3 | EFE | 30072 |
|----|----|------|---------|------------------|---|-----|-------|

| Reaction No. 1498                                                                                             |    |       |               |                |   |       |        |
|---------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $\text{Al}+3 + 2\text{<Citric-3>} + \text{H}_2\text{O} = \text{Al}+3_{\text{OH-1\_Citric-3(2)}} + \text{H}+1$ |    |       |               |                |   |       |        |
| No                                                                                                            | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                             | 25 | 0     | Inf. Dilution | lgK:8.2(3SF)   | 2 | EAE   |        |
| 2                                                                                                             | 37 | 0.15  | NaCl          | lgK:7. (0.3SD) | 4 | MGL   | 925    |
| 3                                                                                                             | 37 | 0.15  | NaCl          | lgK:8.8(0.2SD) | 0 | MGL   | 3574   |

| Reaction No. 1499                                                                              |    |       |               |                  |   |       |        |
|------------------------------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| $2\text{<Citric-3>} + \text{Al}+3 - 2\text{<H+1>} = \text{Al}+3_{\text{H+1(-2)\_Citric-3(2)}}$ |    |       |               |                  |   |       |        |
| No                                                                                             | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                              | 25 | 0     | Inf. Dilution | lgK:-0.2(3SF)    | 2 | EAE   |        |
| 2                                                                                              | 37 | 0.15  | NaCl          | lgK:-0.2(0.05SD) | 6 | MGL   | 925    |

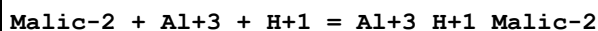
| Reaction No. 1500                                                                                      |    |       |               |                   |   |       |        |
|--------------------------------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $2\text{<Citric-3>} + 2\text{<Al+3>} - 2\text{<H+1>} = \text{Al}+3_{\text{(2)\_H+1(-2)\_Citric-3(2)}}$ |    |       |               |                   |   |       |        |
| No                                                                                                     | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                      | 25 | 0     | Inf. Dilution | lgK:16.3(3SF)     | 2 | EAE   |        |
| 2                                                                                                      | 37 | 0.15  | NaCl          | lgK:12.69(0.01SD) | 6 | MGL   | 925    |

| Reaction No. 1501                                                                                      |    |       |               |                  |   |       |        |
|--------------------------------------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| $3\text{<Citric-3>} + 3\text{<Al+3>} = 4\text{<H+1>} + \text{Al}+3_{\text{(3)\_H+1(-4)\_Citric-3(3)}}$ |    |       |               |                  |   |       |        |
| No                                                                                                     | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                      | 25 | 0     | Inf. Dilution | lgK:20.5(3SF)    | 1 | ERG   |        |
| 2                                                                                                      | 37 | 0.15  | NaCl          | lgK:15.1(0.07SD) | 0 | MGL   | 925    |

| Reaction No. 1588                                             |    |       |               |                  |   |       |        |
|---------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| $\text{Malic-2} + \text{Al}+3 = \text{Al}+3_{\text{Malic-2}}$ |    |       |               |                  |   |       |        |
| No                                                            | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                             | 20 | 0.2   | KNO3          | lgK:5.34(3SF)    | 0 | ENS   | 1246   |
| 2                                                             | 25 | 0     | Inf. Dilution | lgK:5.9(0.1SD)   | 2 | EDH   | 29190  |
| 3                                                             | 25 | 0     | UCT_Sea       | lgK:4.87(0.01SD) | 2 | EDH   | 5390   |
| 4                                                             | 25 | 0.1   | UCT_Sea       | lgK:3.84(0.01SD) | 2 | EDH   | 5390   |
| 5                                                             | 25 | 0.15  | NaCl          | lgK:4.25(0.03SD) | 3 | MGL   | 29190  |
| 6                                                             | 25 | 0.2   | KCl           | lgK:4. (0.2SD)   | 6 | MGL   | 4371   |
| 7                                                             | 25 | 0.2   | UCT_Sea       | lgK:3.66(0.01SD) | 2 | EDH   | 5390   |

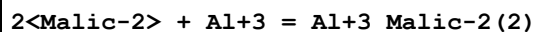
|    |       |      |         |                  |   |     |       |
|----|-------|------|---------|------------------|---|-----|-------|
| 8  | 25    | 0.3  | UCT_Sea | lgK:3.55(0.01SD) | 4 | EDH | 5390  |
| 9  | 25    | 0.4  | UCT_Sea | lgK:3.49(0.01SD) | 4 | EDH | 5390  |
| 10 | 25    | 0.5  | NaCl    | lgK:3.85(0.02SD) | 2 | MGL | 29190 |
| 11 | 25    | 0.5  | UCT_Sea | lgK:3.44(0.01SD) | 4 | EDH | 5390  |
| 12 | 25    | 0.6  | UCT_Sea | lgK:3.40(0.01SD) | 4 | EDH | 5390  |
| 13 | 25    | 0.7  | UCT_Sea | lgK:3.37(0.01SD) | 4 | EDH | 5390  |
| 14 | 25    | 1    | NaCl    | lgK:3.11(0.02SD) | 4 | MGL | 29190 |
| 15 | 27:30 | 1    | NaClO4  | lgK:3.32(3SF)    | 6 | MSP | 999   |
| 16 | 29    | 1    | Unknown | lgK:3.32(3SF)    | 6 | CMN | 1084  |
| 17 | 31    | 0    | Unknown | lgK:6.00(0.24SD) | 0 | EOD | 6004  |
| 18 | 37    | 0.15 | NaCl    | lgK:4.60(3SF)    | 3 | MGL | 2737  |
| 19 | 37    | 0.15 | NaCl    | lgK:4.37(3SF)    | 5 | MGL | 10029 |

## Reaction No. 1589



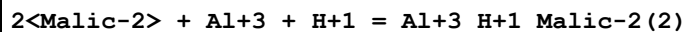
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:7.3(0.09SD) | 6 | MGL   | 4371   |
| 2  | 37 | 0.15  | NaCl   | lgK:6.87(3SF)   | 6 | MGL   | 2737   |

## Reaction No. 1590



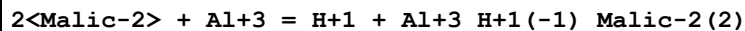
| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 20 | 0.2   | KNO3          | lgK:9.32(3SF) | 0 | ENS   | 1246   |
| 2  | 25 | 0     | Inf. Dilution | lgK:9.81(3SF) | 2 | EAE   |        |
| 3  | 37 | 0.15  | NaCl          | lgK:7.62(3SF) | 5 | MGL   | 2737   |
| 4  | 37 | 0.15  | NaCl          | lgK:8.17(3SF) | 0 | MGL   | 10029  |

## Reaction No. 1591



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:13.5(3SF)  | 2 | EAE   |        |
| 2  | 37 | 0.15  | NaCl          | lgK:11.31(4SF) | 6 | MGL   | 2737   |

## Reaction No. 1592



| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:5.53(0.09SD) | 2 | EDH   | 29190  |



|   |    |      |      |                  |   |     |       |
|---|----|------|------|------------------|---|-----|-------|
| 2 | 25 | 0.15 | NaCl | lgK:4.03(0.04SD) | 3 | MGL | 29190 |
| 3 | 25 | 0.2  | KCl  | lgK:4.74(0.02SD) | 0 | MGL | 4371  |
| 4 | 25 | 0.5  | NaCl | lgK:3.63(0.02SD) | 3 | MGL | 29190 |
| 5 | 25 | 1    | NaCl | lgK:3.22(0.01SD) | 3 | MGL | 29190 |
| 6 | 37 | 0.15 | NaCl | lgK:4.31(3SF)    | 3 | MGL | 2737  |

| Reaction No. 1593                                          |    |       |               |               |   |       |        |
|------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| 2<Malic-2> + 2<Al+3> - 2<H+1> = Al+3(2)_H+1(-2)_Malic-2(2) |    |       |               |               |   |       |        |
| No                                                         | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                          | 25 | 0     | Inf. Dilution | lgK:8.46(3SF) | 2 | EAE   |        |
| 2                                                          | 37 | 0.15  | NaCl          | lgK:5.59(3SF) | 6 | MGL   | 2737   |

| Reaction No. 1594                                          |    |       |               |               |   |       |        |
|------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| 2<Malic-2> + 2<Al+3> - 3<H+1> = Al+3(2)_H+1(-3)_Malic-2(2) |    |       |               |               |   |       |        |
| No                                                         | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                          | 25 | 0     | Inf. Dilution | lgK:5.14(3SF) | 2 | EAE   |        |
| 2                                                          | 37 | 0.15  | NaCl          | lgK:2.50(3SF) | 6 | MGL   | 2737   |

| Reaction No. 1636               |    |       |               |                  |   |       |        |
|---------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Oxalic-2 + Al+3 = Al+3_Oxalic-2 |    |       |               |                  |   |       |        |
| No                              | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                               | 20 | 0     | Inf. Dilution | lgK:7.06(3SF)    | 0 | MGL   | 3727   |
| 2                               | 20 | 0.1   | Unknown       | lgK:7.30(3SF)    | 0 | ENS   | 1539   |
| 3                               | 20 |       |               | lgK:7.3(2SF)     | 0 | MRM   | 3727   |
| 4                               | 20 |       |               | lgK:6.1(0.2SD)   | 0 | MPT   | 3727   |
| 5                               | 25 | 0     | Inf. Dilution | lgK:7.18(3SF)    | 3 | MMR   | 3727   |
| 6                               | 25 | 0     | Inf. Dilution | lgK:7.72(3SF)    | 4 | CRV   | 31301  |
| 7                               | 25 | 0.2   | KCl           | lgK:6.2(0.2SD)   | 4 | MGL   | 4371   |
| 8                               | 25 | 0.6   | NaCl          | lgK:5.97(3SF)    | 3 | ENS   | 6953   |
| 9                               | 25 | 1     | (Na)ClO4      | lgK:6.06(3SF)    | 0 | MEF   | 906    |
| 10                              | 25 | 1     | NaClO4        | lgK:4.85(0.03SD) | 0 | MGL   | 906    |
| 11                              | 25 | 1     | NaClO4        | lgK:4.90(0.02SD) | 0 | MAM   | 906    |
| 12                              | 25 | 1     | NaClO4        | lgK:6.06(3SF)    | 3 | EOD   | 26686  |
| 13                              | 31 | 0     | Unknown       | lgK:6.53(0.13SD) | 2 | EOD   | 6004   |
| 14                              | 37 | 0.15  | Blood Plasma  | lgK:5.2(0.05SD)  | 3 | ENS   | 94     |
| 15                              | 37 | 0.15  | NaCl          | lgK:5.0(0.09SD)  | 2 | MGL   | 2372   |
| 16                              | 37 | 0.15  | NaCl          | lgK:4.19(3SF)    | 0 | MGL   | 2737   |

| Reaction No. 1637                     |    |       |               |                  |   |       |        |
|---------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 2<Oxalic-2> + Al+3 = Al+3_Oxalic-2(2) |    |       |               |                  |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                     | 18 | 0     | Inf. Dilution | lgK:13.(2SF)     | 0 | MGL   | 140    |
| 2                                     | 20 | 0     | Inf. Dilution | lgK:13.49(4SF)   | 3 | MGL   | 3727   |
| 3                                     | 20 | 0.1   | Unknown       | lgK:12.70(4SF)   | 3 | ENS   | 1539   |
| 4                                     | 25 | 0     | Inf. Dilution | lgK:13.49(4SF)   | 5 | MMR   | 3727   |
| 5                                     | 25 | 0     | Inf. Dilution | lgK:13.20(4SF)   | 4 | CRV   | 31301  |
| 6                                     | 25 | 0.2   | KCl           | lgK:11.4(0.04SD) | 4 | MGL   | 4371   |
| 7                                     | 25 | 1     | NaClO4        | lgK:11.09(4SF)   | 4 | MEF   | 906    |
| 8                                     | 25 | 1     | NaClO4        | lgK:11.09(4SF)   | 3 | EOD   | 26686  |
| 9                                     | 37 | 0.15  | Blood Plasma  | lgK:10.5(0.05SD) | 3 | ENS   | 94     |
| 10                                    | 37 | 0.15  | NaCl          | lgK:9.3(0.06SD)  | 0 | MGL   | 2372   |
| 11                                    | 37 | 0.15  | NaCl          | lgK:9.55(3SF)    | 0 | MGL   | 2737   |

| Reaction No. 1638                     |       |       |               |                  |   |       |        |
|---------------------------------------|-------|-------|---------------|------------------|---|-------|--------|
| 3<Oxalic-2> + Al+3 = Al+3_Oxalic-2(3) |       |       |               |                  |   |       |        |
| No                                    | t     | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                     | 18    | 0     | Inf. Dilution | lgK:16.3(3SF)    | 3 | MGL   | 140    |
| 2                                     | 20    | 0     | Inf. Dilution | lgK:17.53(4SF)   | 0 | MGL   | 3727   |
| 3                                     | 20    | 0.1   | KClO4         | lgK:15.6(0.05SD) | 3 | MDS   | 931    |
| 4                                     | 20    | 0.1   | Unknown       | lgK:15.60(4SF)   | 3 | ENS   | 1539   |
| 5                                     | 25    | 0     | Inf. Dilution | lgK:16.74(4SF)   | 4 | CRV   | 31301  |
| 6                                     | 25    | 0.2   | KCl           | lgK:15.4(0.08SD) | 4 | MGL   | 4371   |
| 7                                     | 25    | 0.6   | NaCl          | lgK:10.93(4SF)   | 0 | ENS   | 6953   |
| 8                                     | 25    | 0.6   | NaCl          | lgK:14.88(4SF)   | 2 | ENS   | 6953   |
| 9                                     | 25    | 1     | NaClO4        | lgK:15.12(4SF)   | 4 | MEF   | 906    |
| 10                                    | 25    | 1     | NaClO4        | lgK:15.12(4SF)   | 3 | EOD   | 26686  |
| 11                                    | 37    | 0.15  | Blood Plasma  | lgK:14.0(0.05SD) | 0 | ENS   | 94     |
| 12                                    | 37    | 0.15  | NaCl          | lgK:12.4(0.2SD)  | 0 | MGL   | 2372   |
| 13                                    | 37    | 0.15  | NaCl          | lgK:13.06(4SF)   | 0 | MGL   | 2737   |
| 14                                    | 20:70 | 0.1   | Unknown       | dH:4.(1SF)       | 4 | CRV   | 2556   |

| Reaction No. 1639                     |   |       |        |             |   |       |        |
|---------------------------------------|---|-------|--------|-------------|---|-------|--------|
| Oxalic-2 + 2<Al+3> = Al+3(2)_Oxalic-2 |   |       |        |             |   |       |        |
| No                                    | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

|   |    |      |               |                |   |     |      |
|---|----|------|---------------|----------------|---|-----|------|
| 1 | 25 | 0    | Inf. Dilution | lgK:8.19 (3SF) | 2 | EAE |      |
| 2 | 37 | 0.15 | NaCl          | lgK:7.47 (3SF) | 4 | MGL | 2737 |

| Reaction No. 1702                           |       |       |                 |                    |   |       |        |
|---------------------------------------------|-------|-------|-----------------|--------------------|---|-------|--------|
| Salicylic-2 + Al+3 = Al+3_Salicylic-2       |       |       |                 |                    |   |       |        |
| No                                          | t     | I-Str | Medium          | Value (Dev)        | W | Tech. | Ref. # |
| 1                                           | 20    | 0.1   | KNO3            | lgK:12.9 (3SF)     | 3 | MGL   | 3494   |
| 2                                           | 20    | 0.1   | Unknown         | lgK:12.9 (3SF)     | 4 | ENS   | 773    |
| 3                                           | 25    | 0     | Inf. Dilution   | lgK:14.2 (3SF)     | 4 | ENS   | 2152   |
| 4                                           | 25    | 0     | UCT_Sea         | lgK:14.33 (0.01SD) | 4 | EDH   | 5390   |
| 5                                           | 25    | 0.1   | Inf. Dilution/m | lgK:0.18 (2SF)     | 0 | EOD   | 6366   |
| Note (5): NPD problem (indicated in source) |       |       |                 |                    |   |       |        |
| 6                                           | 25    | 0.1   | NaClO4          | lgK:14.5 (3SF)     | 0 | ENS   | 2171   |
| Note (6): Low wgt for internal consistency  |       |       |                 |                    |   |       |        |
| 7                                           | 25    | 0.1   | UCT_Sea         | lgK:13.33 (0.01SD) | 4 | EDH   | 5390   |
| 8                                           | 25    | 0.12  | NaCl            | lgK:13.7 (3SF)     | 0 | MGL   | 5987   |
| 9                                           | 25    | 0.12  | NaCl            | lgK:13.1 (3SF)     | 6 | MGL   | 5987   |
| 10                                          | 25    | 0.2   | KCl             | lgK:13.22 (0.01SD) | 5 | MGL   | 2470   |
| 11                                          | 25    | 0.2   | UCT_Sea         | lgK:13.11 (0.01SD) | 4 | EDH   | 5390   |
| 12                                          | 25    | 0.3   | UCT_Sea         | lgK:13.01 (0.01SD) | 4 | EDH   | 5390   |
| 13                                          | 25    | 0.4   | UCT_Sea         | lgK:12.93 (0.01SD) | 4 | EDH   | 5390   |
| 14                                          | 25    | 0.5   | UCT_Sea         | lgK:12.88 (0.01SD) | 4 | EDH   | 5390   |
| 15                                          | 25    | 0.6   | NaCl            | lgK:13.0 (3SF)     | 4 | ENS   | 1245   |
| 16                                          | 25    | 0.6   | UCT_Sea         | lgK:12.83 (0.01SD) | 4 | EDH   | 5390   |
| 17                                          | 25    | 0.7   | UCT_Sea         | lgK:12.80 (0.01SD) | 4 | EDH   | 5390   |
| 18                                          | 26:28 | 0     | Inf. Dilution   | lgK:14.11 (4SF)    | 4 | MSP   | 140    |
| 19                                          | 31    | 0     | Unknown         | lgK:4.74 (0.14SD)  | 0 | EOD   | 6004   |
| 20                                          | 37    | 0.15  | Blood Plasma    | lgK:12.5 (0.05SD)  | 3 | ENS   | 94     |

| Reaction No. 1703                            |    |       |               |                    |   |       |        |
|----------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| 2<Salicylic-2> + Al+3 = Al+3_Salicylic-2 (2) |    |       |               |                    |   |       |        |
| No                                           | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                            | 20 | 0.1   | KNO3          | lgK:23.2 (3SF)     | 3 | ENS   | 2171   |
| 2                                            | 25 | 0     | Inf. Dilution | lgK:24.0 (3SF)     | 4 | ENS   | 2152   |
| 3                                            | 25 | 0     | Inf. Dilution | lgK:25.1 (3SF)     | 4 | ENS   | 6405   |
| 4                                            | 25 | 0     | UCT_Sea       | lgK:25.49 (0.01SD) | 3 | EDH   | 5390   |

|                                                             |    |      |              |                    |   |     |       |
|-------------------------------------------------------------|----|------|--------------|--------------------|---|-----|-------|
| 5                                                           | 25 | 0.1  | UCT_Sea      | lgK:24.14 (0.01SD) | 3 | EDH | 5390  |
| 6                                                           | 25 | 0.12 | NaCl         | lgK:26.70 (4SF)    | 0 | MGL | 11516 |
| Note (6): Rajan K et al., J. Inorg. Biochem., 1981, 14, 339 |    |      |              |                    |   |     |       |
| 7                                                           | 25 | 0.2  | KCl          | lgK:23.7 (0.04SD)  | 0 | MGL | 2470  |
| 8                                                           | 25 | 0.2  | UCT_Sea      | lgK:23.81 (0.01SD) | 3 | EDH | 5390  |
| 9                                                           | 25 | 0.3  | UCT_Sea      | lgK:23.66 (0.01SD) | 3 | EDH | 5390  |
| 10                                                          | 25 | 0.4  | UCT_Sea      | lgK:23.56 (0.01SD) | 3 | EDH | 5390  |
| 11                                                          | 25 | 0.5  | UCT_Sea      | lgK:23.48 (0.01SD) | 3 | EDH | 5390  |
| 12                                                          | 25 | 0.6  | NaCl         | lgK:23.6 (3SF)     | 5 | ENS | 2171  |
| 13                                                          | 25 | 0.6  | UCT_Sea      | lgK:23.41 (0.01SD) | 3 | EDH | 5390  |
| 14                                                          | 25 | 0.7  | UCT_Sea      | lgK:23.37 (0.01SD) | 3 | EDH | 5390  |
| 15                                                          | 37 | 0.15 | Blood Plasma | lgK:22.0 (0.05SD)  | 4 | ENS | 94    |

| Reaction No. 1704                           |    |       |               |                   |   |       |        |
|---------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| 3<Salicylic-2> + Al+3 = Al+3_Salicylic-2(3) |    |       |               |                   |   |       |        |
| No                                          | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                           | 20 | 0.1   | KNO3          | lgK:29.8 (3SF)    | 3 | ENS   | 2171   |
| 2                                           | 25 | 0     | Inf. Dilution | lgK:31.8 (3SF)    | 1 | ERG   |        |
| 3                                           | 25 | 0     | Inf. Dilution | lgK:32.8 (3SF)    | 0 | ENS   | 2152   |
| 4                                           | 25 | 0     | Inf. Dilution | lgK:31.1 (3SF)    | 0 | ENS   | 6405   |
| 5                                           | 37 | 0.15  | Blood Plasma  | lgK:28.2 (0.05SD) | 2 | ENS   | 94     |

| Reaction No. 1721                   |    |       |         |                   |   |       |        |
|-------------------------------------|----|-------|---------|-------------------|---|-------|--------|
| Succinic-2 + Al+3 = Al+3_Succinic-2 |    |       |         |                   |   |       |        |
| No                                  | t  | I-Str | Medium  | Value (Dev)       | W | Tech. | Ref. # |
| 1                                   | 25 | 0     | UCT_Sea | lgK:4.70 (0.01SD) | 3 | EDH   | 5390   |
| 2                                   | 25 | 0.1   | UCT_Sea | lgK:3.64 (0.01SD) | 4 | EDH   | 5390   |
| 3                                   | 25 | 0.2   | KCl     | lgK:3.63 (3SF)    | 5 | MGL   | 10434  |
| 4                                   | 25 | 0.2   | UCT_Sea | lgK:3.44 (0.01SD) | 4 | EDH   | 5390   |
| 5                                   | 25 | 0.3   | UCT_Sea | lgK:3.33 (0.01SD) | 4 | EDH   | 5390   |
| 6                                   | 25 | 0.4   | UCT_Sea | lgK:3.25 (0.01SD) | 4 | EDH   | 5390   |
| 7                                   | 25 | 0.5   | NaClO4  | lgK:3.2 (2SF)     | 4 | ENS   | 6953   |
| 8                                   | 25 | 0.5   | UCT_Sea | lgK:3.20 (0.01SD) | 4 | EDH   | 5390   |
| 9                                   | 25 | 0.6   | UCT_Sea | lgK:3.16 (0.01SD) | 4 | EDH   | 5390   |
| 10                                  | 25 | 0.7   | UCT_Sea | lgK:3.13 (0.01SD) | 4 | EDH   | 5390   |
| 11                                  | 25 | 1     | NaClO4  | lgK:3.84 (3SF)    | 0 | MGL   | 1243   |

|                                              |    |      |           |                  |   |     |       |
|----------------------------------------------|----|------|-----------|------------------|---|-----|-------|
| 12                                           | 25 | 1    | Unknown/m | lgK:3.83(3SF)    | 0 | EOD | 6366  |
| Note (12): NDP problem (indicated in source) |    |      |           |                  |   |     |       |
| 13                                           | 31 | 0    | Unknown   | lgK:4.62(0.08SD) | 4 | EOD | 6004  |
| 14                                           | 35 | 1    | NaClO4    | lgK:3.78(3SF)    | 0 | MGL | 1243  |
| 15                                           | 37 | 0.15 | NaCl      | lgK:3.92(3SF)    | 5 | MGL | 2737  |
| 16                                           | 37 | 0.15 | NaCl      | lgK:3.9(0.04SD)  | 6 | MGL | 5979  |
| 17                                           | 37 | 0.15 | NaCl      | lgK:3.753(4SF)   | 5 | MGL | 7681  |
| 18                                           | 37 | 0.15 | NaCl      | lgK:4.17(3SF)    | 5 | MGL | 10029 |
| 19                                           | 25 | 1    | Unknown/m | dH:-2.56(3SF)    | 0 | EOD | 6366  |
| Note (19): NDP problem (indicated in source) |    |      |           |                  |   |     |       |

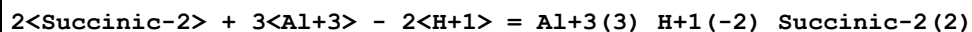
| Reaction No. 1722                                 |    |       |              |                   |   |       |        |
|---------------------------------------------------|----|-------|--------------|-------------------|---|-------|--------|
| Succinic-2 + Al+3 - H+1 = Al+3_H+1(-1)_Succinic-2 |    |       |              |                   |   |       |        |
| No                                                | t  | I-Str | Medium       | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                 | 25 | 0     | UCT_Sea      | lgK:1.05(0.01SD)  | 4 | EDH   | 5390   |
| 2                                                 | 25 | 0.1   | UCT_Sea      | lgK:-0.05(0.01SD) | 4 | EDH   | 5390   |
| 3                                                 | 25 | 0.2   | UCT_Sea      | lgK:-0.26(0.01SD) | 4 | EDH   | 5390   |
| 4                                                 | 25 | 0.3   | UCT_Sea      | lgK:-0.39(0.01SD) | 4 | EDH   | 5390   |
| 5                                                 | 25 | 0.4   | UCT_Sea      | lgK:-0.49(0.01SD) | 4 | EDH   | 5390   |
| 6                                                 | 25 | 0.5   | UCT_Sea      | lgK:-0.55(0.01SD) | 4 | EDH   | 5390   |
| 7                                                 | 25 | 0.6   | UCT_Sea      | lgK:-0.60(0.01SD) | 4 | EDH   | 5390   |
| 8                                                 | 25 | 0.7   | UCT_Sea      | lgK:-0.65(0.01SD) | 4 | EDH   | 5390   |
| 9                                                 | 37 | 0.15  | Blood Plasma | lgK:-1.0(0.05SD)  | 6 | ENS   | 94     |
| 10                                                | 37 | 0.15  | NaCl         | lgK:-0.2(0.03SD)  | 5 | MGL   | 2372   |

| Reaction No. 1723                                    |    |       |               |                  |   |       |        |
|------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Succinic-2 + Al+3 - 2<H+1> = Al+3_H+1(-2)_Succinic-2 |    |       |               |                  |   |       |        |
| No                                                   | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                    | 25 | 0     | Inf. Dilution | lgK:-3.99(3SF)   | 2 | EAE   |        |
| 2                                                    | 37 | 0.15  | Blood Plasma  | lgK:-5.19(3SF)   | 6 | ENS   | 1083   |
| 3                                                    | 37 | 0.15  | NaCl          | lgK:-5.2(0.06SD) | 5 | MGL   | 2372   |

| Reaction No. 1724                                          |    |       |               |                |   |       |        |
|------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Succinic-2 + 2<Al+3> - 3<H+1> = Al+3(2)_H+1(-3)_Succinic-2 |    |       |               |                |   |       |        |
| No                                                         | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                          | 25 | 0     | Inf. Dilution | lgK:-3.16(3SF) | 2 | EAE   |        |
| 2                                                          | 37 | 0.15  | NaCl          | lgK:-5.23(3SF) | 5 | MGL   | 2737   |

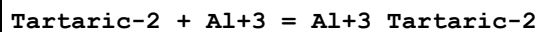
|   |    |      |      |                  |   |     |      |
|---|----|------|------|------------------|---|-----|------|
| 3 | 37 | 0.15 | NaCl | lgK:-5.4 (0.2SD) | 5 | MGL | 5979 |
|---|----|------|------|------------------|---|-----|------|

## Reaction No. 1725



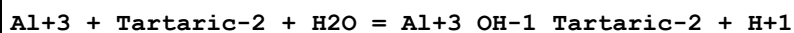
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:8.51 (3SF)   | 2 | EAE   |        |
| 2  | 37 | 0.15  | NaCl          | lgK:5.34 (3SF)   | 3 | MGL   | 2737   |
| 3  | 37 | 0.15  | NaCl          | lgK:5.8 (0.06SD) | 4 | MGL   | 5979   |

## Reaction No. 1757



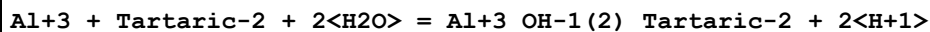
| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 20 | 0     | Inf. Dilution | lgK:6.35 (3SF)    | 3 | MOR   | 931    |
| 2  | 25 | 0.1   | NaClO4        | lgK:5.32 (3SF)    | 0 | MGL   | 906    |
| 3  | 25 | 0.1   | NaClO4        | lgK:5.62 (3SF)    | 5 | MGL   | 7544   |
| 4  | 25 | 0.6   | Unknown       | lgK:5.66 (3SF)    | 5 | MGL   | 5971   |
| 5  | 31 | 0     | Unknown       | lgK:6.21 (0.07SD) | 3 | EOD   | 6004   |
| 6  | 37 | 0.15  | Blood Plasma  | lgK:5.0 (0.05SD)  | 3 | ENS   | 94     |
| 7  | 37 | 0.15  | NaCl          | lgK:3.85 (3SF)    | 0 | MGL   | 5981   |

## Reaction No. 1758



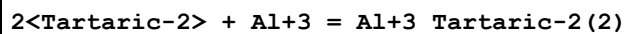
| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3    | lgK:1.18 (3SF) | 6 | MGL   | 1244   |
| 2  | 25 | 0.1   | Unknown | lgK:1.18 (3SF) | 6 | CMN   | 1084   |

## Reaction No. 1759



| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:-3.97 (3SF) | 6 | MGL   | 1217   |

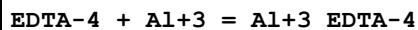
## Reaction No. 1760



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 20 | 0.1   | KClO4  | lgK:9.6 (0.04SD)  | 0 | MDS   | 931    |
| 2  | 25 | 0.1   | NaClO4 | lgK:11.9 (0.06SD) | 0 | MGL   | 906    |
| 3  | 25 | 0.1   | NaClO4 | lgK:9.95 (3SF)    | 0 | MGL   | 7544   |

|   |    |     |         |                |   |     |      |
|---|----|-----|---------|----------------|---|-----|------|
| 4 | 25 | 0.1 | Unknown | lgK:7.65 (3SF) | 3 | CMN | 1084 |
|---|----|-----|---------|----------------|---|-----|------|

## Reaction No. 1927



| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 15 | 0.1   | KNO3          | lgK:16.11 (4SF)   | 3 | MGL   | 140    |
| 2  | 20 | 0.1   | KNO3          | lgK:16.13 (4SF)   | 0 | MVM   | 197    |
| 3  | 20 | 0.1   | KNO3          | lgK:16.7 (0.1SD)  | 3 | EIU   | 903    |
| 4  | 20 | 0.1   | KNO3          | lgK:16.17 (4SF)   | 0 | MPL   | 903    |
| 5  | 20 | 0.1   | KNO3          | lgK:16.7 (0.1SD)  | 3 | MEF   | 2095   |
| 6  | 20 | 0.1   | Unknown       | lgK:16.3 (3SF)    | 4 | ENS   | 773    |
| 7  | 25 | 0     | Inf. Dilution | lgK:19.20 (4SF)   | 6 | ENS   | 1384   |
| 8  | 25 | 0     | Inf. Dilution | lgK:18.9 (3SF)    | 4 | ENS   | 6405   |
| 9  | 25 | 0.1   | KNO3          | lgK:16.5 (3SF)    | 4 | MEF   | 931    |
| 10 | 25 | 0.1   | KNO3          | lgK:16.8 (0.03SD) | 3 | MEF   | 2095   |
| 11 | 25 | 0.1   | Unknown       | lgK:16.5 (3SF)    | 4 | ENS   | 2171   |
| 12 | 25 | 0.12  | NaCl          | lgK:15.3 (3SF)    | 0 | MCL   | 11516  |

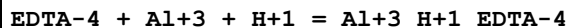
Note (12): Rajan K et al., J. Inorg. Biochem., 1981, 14, 339

|    |    |      |        |                   |   |     |       |
|----|----|------|--------|-------------------|---|-----|-------|
| 13 | 25 | 0.2  | NaClO4 | lgK:16.01 (4SF)   | 3 | MSP | 931   |
| 14 | 25 | 1.08 | K2SO4  | lgK:16.13 (4SF)   | 5 | MPL | 903   |
| 15 | 25 | 1.08 | K2SO4  | lgK:15.9 (3SF)    | 3 | MPL | 906   |
| 16 | 30 | 0.1  | KNO3   | lgK:16.8 (0.06SD) | 5 | MHG | 2095  |
| 17 | 35 | 0.1  | KNO3   | lgK:16.95 (4SF)   | 5 | MGL | 11516 |

Note (17): Khan M et al., Indian J. Chem., 1980, 19A, 44

|    |    |      |              |                   |   |     |      |
|----|----|------|--------------|-------------------|---|-----|------|
| 18 | 37 | 0.15 | Blood Plasma | lgK:16.6 (0.05SD) | 4 | ENS | 94   |
| 19 | 40 | 0.1  | KNO3         | lgK:17.3 (0.06SD) | 3 | MHG | 2095 |
| 20 | 25 | 0.1  | KNO3         | dG:-22.9 (0.04SD) | 3 | MEF | 2095 |
| 21 | 20 | 0.1  | KNO3         | dH:12.58 (4SF)    | 4 | MCL | 931  |
| 22 | 25 | 0.1  | KNO3         | dH:12.1 (3SF)     | 5 | MCL | 2095 |
| 23 | 25 | 0.1  | KNO3         | dS:117.4 (4SF)    | 5 | EFE | 2095 |

## Reaction No. 1928



| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:21.6 (3SF)    | 4 | ENS   | 6405   |
| 2  | 37 | 0.15  | Blood Plasma  | lgK:19.0 (0.05SD) | 4 | ENS   | 94     |

| Reaction No. 1929                                                                                                       |    |       |               |                  |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| $\text{Al}^{+3} + \text{EDTA}^{-4} + \text{H}_2\text{O} = \text{Al}^{+3}\text{OH}^{-1}\text{EDTA}^{-4} + \text{H}^{+1}$ |    |       |               |                  |   |       |        |
| No                                                                                                                      | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                                       | 25 | 0     | Inf. Dilution | lgK:13.1(3SF)    | 2 | EAE   |        |
| 2                                                                                                                       | 37 | 0.15  | Blood Plasma  | lgK:10.7(0.05SD) | 3 | ENS   | 94     |

| Reaction No. 1930                                                                                                            |    |       |               |                  |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| $\text{Al}^{+3} + \text{EDTA}^{-4} + 2\text{H}_2\text{O} = \text{Al}^{+3}\text{OH}^{-1}(2)\text{EDTA}^{-4} + 2\text{H}^{+1}$ |    |       |               |                  |   |       |        |
| No                                                                                                                           | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                                            | 25 | 0     | Inf. Dilution | lgK:1.38(3SF)    | 2 | EAE   |        |
| 2                                                                                                                            | 37 | 0.15  | Blood Plasma  | lgK:-0.3(0.05SD) | 4 | ENS   | 94     |

| Reaction No. 1998                                                        |    |       |               |                   |   |       |        |
|--------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $\text{Al}^{+3} + \text{Desfer}^{-2} = \text{Al}^{+3}\text{Desfer}^{-2}$ |    |       |               |                   |   |       |        |
| No                                                                       | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                        | 25 | 0     | Inf. Dilution | lgK:24.0(3SF)     | 1 | ELC   |        |
| 2                                                                        | 25 | 0.1   | KCl           | lgK:24.14(0.01SD) | 0 | MGL   | 1382   |

| Reaction No. 1999                                                                                     |    |       |               |                |   |       |        |
|-------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $\text{Desfer}^{-2} + \text{Al}^{+3} + \text{H}^{+1} = \text{Al}^{+3}\text{H}^{+1}\text{Desfer}^{-2}$ |    |       |               |                |   |       |        |
| No                                                                                                    | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                     | 25 | 0     | Inf. Dilution | lgK:25.11(4SF) | 1 | ELC   |        |

| Reaction No. 2000                                                                                         |    |       |               |                   |   |       |        |
|-----------------------------------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $\text{Desfer}^{-2} + \text{Al}^{+3} - \text{H}^{+1} = \text{Al}^{+3}\text{H}^{+1}(-1)\text{Desfer}^{-2}$ |    |       |               |                   |   |       |        |
| No                                                                                                        | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                         | 25 | 0     | Inf. Dilution | lgK:14.7(3SF)     | 1 | ELC   |        |
| 2                                                                                                         | 25 | 0.1   | KCl           | lgK:24.50(0.01SD) | 0 | MGL   | 1382   |
| Note (2): Wrong NDP?                                                                                      |    |       |               |                   |   |       |        |
| 3                                                                                                         | 37 | 0.15  | Blood Plasma  | lgK:6.0(0.05SD)   | 0 | ENS   | 94     |

| Reaction No. 2025                                                    |    |       |         |                 |   |       |        |
|----------------------------------------------------------------------|----|-------|---------|-----------------|---|-------|--------|
| $\text{DTPA}^{-5} + \text{Al}^{+3} = \text{Al}^{+3}\text{DTPA}^{-5}$ |    |       |         |                 |   |       |        |
| No                                                                   | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                    | 20 | 0.1   | KNO3    | lgK:18.4(0.1SD) | 6 | MEF   | 2095   |
| 2                                                                    | 20 | 0.1   | KNO3    | lgK:18.4(0.1SD) | 5 | CRV   | 13242  |
| 3                                                                    | 20 | 0.1   | Unknown | lgK:18.6(3SF)   | 4 | ENS   | 773    |



|                                                         |    |      |              |                   |   |     |       |
|---------------------------------------------------------|----|------|--------------|-------------------|---|-----|-------|
| 4                                                       | 25 | 0.1  | KNO3         | lgK:18.51(0.02SD) | 6 | MEF | 2095  |
| 5                                                       | 25 | 0.1  | KNO3         | lgK:18.7(3SF)     | 4 | ENS | 6953  |
| 6                                                       | 25 | 0.1  | KNO3         | lgK:18.5(0.1SD)   | 5 | CRV | 13242 |
| 7                                                       | 30 | 0.1  | KNO3         | lgK:18.6(0.06SD)  | 6 | MEF | 2095  |
| 8                                                       | 35 | 0.1  | KNO3         | lgK:20.66(4SF)    | 0 | MGL | 11516 |
| Note (8): Khan M et al., Indian J. Chem., 1980, 19A, 44 |    |      |              |                   |   |     |       |
| 9                                                       | 37 | 0.15 | Blood Plasma | lgK:17.2(0.05SD)  | 0 | ENS | 94    |
| 10                                                      | 40 | 0.1  | KNO3         | lgK:18.8(0.08SD)  | 6 | MEF | 2095  |
| 11                                                      | 25 | 0.1  | KNO3         | dG:-25.3(0.03SD)  | 5 | MEF | 2095  |
| 12                                                      | 25 | 0.1  | KNO3         | dH:8.(1SF)        | 6 | MEF | 2095  |
| 13                                                      | 25 | 0.1  | KNO3         | dS:113.0(4SF)     | 5 | EFE | 2095  |

| Reaction No. 2026                     |    |       |               |                  |   |       |        |
|---------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| DTPA-5 + Al+3 + H+1 = Al+3_H+1_DTPA-5 |    |       |               |                  |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                     | 25 | 0     | Inf. Dilution | lgK:26.1(3SF)    | 2 | EAE   |        |
| 2                                     | 37 | 0.15  | Blood Plasma  | lgK:22.0(0.05SD) | 4 | ENS   | 94     |

| Reaction No. 2027                            |    |       |               |                  |   |       |        |
|----------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Al+3 + DTPA-5 + H2O = Al+3_OH-1_DTPA-5 + H+1 |    |       |               |                  |   |       |        |
| No                                           | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                            | 25 | 0     | Inf. Dilution | lgK:13.4(3SF)    | 2 | EAE   |        |
| 2                                            | 37 | 0.15  | Blood Plasma  | lgK:12.0(0.05SD) | 4 | ENS   | 94     |

| Reaction No. 2106           |    |       |              |                  |   |       |        |
|-----------------------------|----|-------|--------------|------------------|---|-------|--------|
| EGTA-4 + Al+3 = Al+3_EGTA-4 |    |       |              |                  |   |       |        |
| No                          | t  | I-Str | Medium       | Value (Dev)      | W | Tech. | Ref. # |
| 1                           | 25 | 0.2   | NaClO4       | lgK:13.90(4SF)   | 5 | MSP   | 6953   |
| 2                           | 37 | 0.15  | Blood Plasma | lgK:13.5(0.05SD) | 4 | ENS   | 94     |

| Reaction No. 2107                     |    |       |               |                  |   |       |        |
|---------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| EGTA-4 + Al+3 + H+1 = Al+3_H+1_EGTA-4 |    |       |               |                  |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                     | 25 | 0     | Inf. Dilution | lgK:21.1(3SF)    | 1 | ELC   |        |
| 2                                     | 37 | 0.15  | Blood Plasma  | lgK:17.8(0.05SD) | 4 | ENS   | 94     |

| Reaction No. 2108 |  |  |  |  |  |  |  |
|-------------------|--|--|--|--|--|--|--|
|-------------------|--|--|--|--|--|--|--|

| Al+3 + EGTA-4 + H2O = Al+3_OH-1_EGTA-4 + H+1 |    |       |               |                 |   |       |        |
|----------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| No                                           | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                            | 25 | 0     | Inf. Dilution | lgK:11.2(3SF)   | 1 | ELC   |        |
| 2                                            | 37 | 0.15  | Blood Plasma  | lgK:5.5(0.05SD) | 0 | ENS   | 94     |

| Reaction No. 2364             |    |       |               |                  |   |       |        |
|-------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 23DHB-3 + Al+3 = Al+3_23DHB-3 |    |       |               |                  |   |       |        |
| No                            | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                             | 25 | 0     | Inf. Dilution | lgK:23.8(3SF)    | 1 | ELC   |        |
| 2                             | 25 | 0.2   | KCl           | lgK:10.32(4SF)   | 0 | MGL   | 2470   |
| 3                             | 37 | 0.15  | Blood Plasma  | lgK:17.0(0.05SD) | 0 | ENS   | 94     |

| Reaction No. 2365                   |    |       |               |                  |   |       |        |
|-------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 2<23DHB-3> + Al+3 = Al+3_23DHB-3(2) |    |       |               |                  |   |       |        |
| No                                  | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                   | 25 | 0     | Inf. Dilution | lgK:31.7(3SF)    | 1 | ELC   |        |
| 2                                   | 25 | 0.2   | KCl           | lgK:18.26(4SF)   | 0 | MGL   | 2470   |
| 3                                   | 37 | 0.15  | Blood Plasma  | lgK:30.0(0.05SD) | 3 | ENS   | 94     |

| Reaction No. 2366                   |    |       |               |                  |   |       |        |
|-------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 3<23DHB-3> + Al+3 = Al+3_23DHB-3(3) |    |       |               |                  |   |       |        |
| No                                  | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                   | 25 | 0     | Inf. Dilution | lgK:38.5(3SF)    | 1 | ESC   |        |
| 2                                   | 37 | 0.15  | Blood Plasma  | lgK:38.0(0.05SD) | 4 | ENS   | 94     |

| Reaction No. 2367                       |    |       |               |                  |   |       |        |
|-----------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 23DHB-3 + Al+3 + H+1 = Al+3_H+1_23DHB-3 |    |       |               |                  |   |       |        |
| No                                      | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                       | 25 | 0     | Inf. Dilution | lgK:26.8(3SF)    | 1 | ELC   |        |
| 2                                       | 37 | 0.15  | Blood Plasma  | lgK:22.6(0.05SD) | 4 | ENS   | 94     |

| Reaction No. 2368                                   |    |       |               |                  |   |       |        |
|-----------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 2<23DHB-3> + Al+3 + 2<H+1> = Al+3_H+1(2)_23DHB-3(2) |    |       |               |                  |   |       |        |
| No                                                  | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                   | 25 | 0     | Inf. Dilution | lgK:50.2(3SF)    | 1 | ELC   |        |
| 2                                                   | 37 | 0.15  | Blood Plasma  | lgK:42.3(0.05SD) | 0 | ENS   | 94     |

| Reaction No. 2369                             |    |       |               |                  |   |       |        |
|-----------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 23DHB-3 + Al+3 + 2<H+1> = Al+3_H+1(2)_23DHB-3 |    |       |               |                  |   |       |        |
| No                                            | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                             | 25 | 0     | Inf. Dilution | lgK:27.2(3SF)    | 1 | ESC   |        |
| 2                                             | 37 | 0.15  | Blood Plasma  | lgK:24.8(0.05SD) | 4 | ENS   | 94     |

| Reaction No. 2370                                   |    |       |               |                  |   |       |        |
|-----------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 2<23DHB-3> + Al+3 + 3<H+1> = Al+3_H+1(3)_23DHB-3(2) |    |       |               |                  |   |       |        |
| No                                                  | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                   | 25 | 0     | Inf. Dilution | lgK:46.5(3SF)    | 1 | ESC   |        |
| 2                                                   | 37 | 0.15  | Blood Plasma  | lgK:46.1(0.05SD) | 4 | ENS   | 94     |

| Reaction No. 2371                                   |    |       |               |                  |   |       |        |
|-----------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 3<23DHB-3> + Al+3 + 4<H+1> = Al+3_H+1(4)_23DHB-3(3) |    |       |               |                  |   |       |        |
| No                                                  | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                   | 25 | 0     | Inf. Dilution | lgK:66.9(3SF)    | 1 | ESC   |        |
| 2                                                   | 37 | 0.15  | Blood Plasma  | lgK:66.3(0.05SD) | 4 | ENS   | 94     |

| Reaction No. 2389                                 |    |       |        |                 |   |       |        |
|---------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Al+3_Malonic-2(2) + Malonic-2 = Al+3_Malonic-2(3) |    |       |        |                 |   |       |        |
| No                                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                 | 25 | 0.1   | KCl    | lgK:2.6(0.04SD) | 6 | MGL   | 2356   |
| 2                                                 | 37 | 0.15  | NaCl   | lgK:2.189(4SF)  | 4 | MGL   | 2372   |

| Reaction No. 2390         |       |       |         |                   |   |       |        |
|---------------------------|-------|-------|---------|-------------------|---|-------|--------|
| Cat-2 + Al+3 = Al+3_Cat-2 |       |       |         |                   |   |       |        |
| No                        | t     | I-Str | Medium  | Value (Dev)       | W | Tech. | Ref. # |
| 1                         | 20    | 0.1   | KNO3    | lgK:16.9(3SF)     | 4 | MGL   | 3494   |
| 2                         | 20    | 0.1   | Unknown | lgK:16.3(3SF)     | 4 | ENS   | 773    |
| 3                         | 20    | 0.2   | KCl     | lgK:16.27(4SF)    | 6 | MGL   | 999    |
| 4                         | 22:25 | 0.2   | KNO3    | lgK:15.31(0.10SD) | 0 | MGL   | 2349   |
| 5                         | 23    |       |         | lgK:17.1(3SF)     | 4 | EES   | 5741   |
| 6                         | 25    | 0     | UCT_Sea | lgK:18.03(0.01SD) | 4 | EDH   | 5390   |
| 7                         | 25    | 0.1   | KCl     | lgK:16.9(0.03SD)  | 2 | MGL   | 1148   |
| 8                         | 25    | 0.1   | KNO3    | lgK:16.1(3SF)     | 0 | ENS   | 1244   |
| 9                         | 25    | 0.1   | UCT_Sea | lgK:16.60(0.01SD) | 4 | EDH   | 5390   |
| 10                        | 25    | 0.2   | KCl     | lgK:16.20(0.02SD) | 6 | MGL   | 2470   |

|    |    |      |              |                   |   |     |      |
|----|----|------|--------------|-------------------|---|-----|------|
| 11 | 25 | 0.2  | UCT_Sea      | lgK:16.33(0.01SD) | 4 | EDH | 5390 |
| 12 | 25 | 0.3  | UCT_Sea      | lgK:16.17(0.01SD) | 4 | EDH | 5390 |
| 13 | 25 | 0.4  | UCT_Sea      | lgK:16.05(0.01SD) | 4 | EDH | 5390 |
| 14 | 25 | 0.5  | UCT_Sea      | lgK:15.98(0.01SD) | 4 | EDH | 5390 |
| 15 | 25 | 0.6  | NaCl         | lgK:15.31(4SF)    | 4 | MGL | 6957 |
| 16 | 25 | 0.6  | UCT_Sea      | lgK:15.91(0.01SD) | 3 | EDH | 5390 |
| 17 | 25 | 0.7  | UCT_Sea      | lgK:15.87(0.01SD) | 3 | EDH | 5390 |
| 18 | 37 | 0.15 | Blood Plasma | lgK:16.3(0.05SD)  | 4 | ENS | 94   |

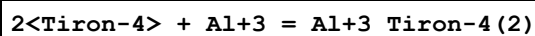
| Reaction No. 2391               |    |       |              |                   |   |       |        |
|---------------------------------|----|-------|--------------|-------------------|---|-------|--------|
| 2<Cat-2> + Al+3 = Al+3_Cat-2(2) |    |       |              |                   |   |       |        |
| No                              | t  | I-Str | Medium       | Value (Dev)       | W | Tech. | Ref. # |
| 1                               | 20 | 0.1   | Unknown      | lgK:29.3(3SF)     | 4 | ENS   | 773    |
| 2                               | 25 | 0     | UCT_Sea      | lgK:32.29(0.01SD) | 2 | EDH   | 5390   |
| 3                               | 25 | 0.1   | KNO3         | lgK:29.1(3SF)     | 0 | ENS   | 1244   |
| 4                               | 25 | 0.1   | UCT_Sea      | lgK:30.10(0.01SD) | 3 | EDH   | 5390   |
| 5                               | 25 | 0.2   | KCl          | lgK:29.3(0.06SD)  | 6 | MGL   | 2470   |
| 6                               | 25 | 0.2   | UCT_Sea      | lgK:29.69(0.01SD) | 3 | EDH   | 5390   |
| 7                               | 25 | 0.3   | UCT_Sea      | lgK:29.46(0.01SD) | 4 | EDH   | 5390   |
| 8                               | 25 | 0.4   | UCT_Sea      | lgK:29.28(0.01SD) | 4 | EDH   | 5390   |
| 9                               | 25 | 0.5   | UCT_Sea      | lgK:29.20(0.01SD) | 4 | EDH   | 5390   |
| 10                              | 25 | 0.6   | NaCl         | lgK:27.67(4SF)    | 0 | MGL   | 6957   |
| 11                              | 25 | 0.6   | UCT_Sea      | lgK:29.11(0.01SD) | 4 | EDH   | 5390   |
| 12                              | 25 | 0.7   | UCT_Sea      | lgK:29.07(0.01SD) | 4 | EDH   | 5390   |
| 13                              | 37 | 0.15  | Blood Plasma | lgK:29.3(0.05SD)  | 2 | ENS   | 94     |

| Reaction No. 2392               |    |       |              |                  |   |       |        |
|---------------------------------|----|-------|--------------|------------------|---|-------|--------|
| 3<Cat-2> + Al+3 = Al+3_Cat-2(3) |    |       |              |                  |   |       |        |
| No                              | t  | I-Str | Medium       | Value (Dev)      | W | Tech. | Ref. # |
| 1                               | 20 | 0.1   | Unknown      | lgK:37.6(3SF)    | 4 | ENS   | 773    |
| 2                               | 25 | 0.1   | KNO3         | lgK:37.8(3SF)    | 4 | ENS   | 1244   |
| 3                               | 25 | 0.2   | KCl          | lgK:38.0(0.07SD) | 6 | MGL   | 2470   |
| 4                               | 25 | 0.6   | NaCl         | lgK:14.45(4SF)   | 0 | ENS   | 6953   |
| 5                               | 37 | 0.15  | Blood Plasma | lgK:37.6(0.05SD) | 4 | ENS   | 94     |

| Reaction No. 2406             |  |  |  |  |  |  |  |
|-------------------------------|--|--|--|--|--|--|--|
| Tiron-4 + Al+3 = Al+3_Tiron-4 |  |  |  |  |  |  |  |

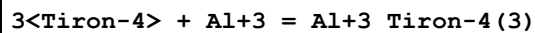
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3          | lgK:16.7(3SF)    | 4 | MGL   | 3494   |
| 2  | 20 | 1     | Unknown       | lgK:14.9(3SF)    | 3 | ENS   | 773    |
| 3  | 25 | 0     | Inf. Dilution | lgK:19.02(4SF)   | 6 | MGL   | 999    |
| 4  | 25 | 0     | Inf. Dilution | lgK:19.06(4SF)   | 5 | ENS   | 2171   |
| 5  | 25 | 0.1   | NaCl          | lgK:16.6(3SF)    | 5 | ENS   | 6953   |
| 6  | 25 | 0.2   | KCl           | lgK:16.3(0.03SD) | 4 | MGL   | 1391   |
| 7  | 25 | 0.2   | NaNO3         | lgK:16.65(4SF)   | 5 | MGL   | 999    |
| 8  | 37 | 0.15  | Blood Plasma  | lgK:16.9(0.05SD) | 4 | ENS   | 94     |

## Reaction No. 2407



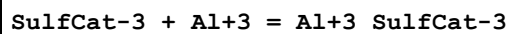
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 20 | 0     | Inf. Dilution | lgK:31.10(4SF)   | 5 | ENS   | 773    |
| 2  | 20 | 0.1   | Unknown       | lgK:30.0(3SF)    | 4 | ENS   | 773    |
| 3  | 20 | 1     | Unknown       | lgK:28.4(3SF)    | 4 | ENS   | 773    |
| 4  | 25 | 0     | Inf. Dilution | lgK:31.10(4SF)   | 5 | ENS   | 2171   |
| 5  | 25 | 0.1   | NaCl          | lgK:30.0(3SF)    | 5 | ENS   | 6953   |
| 6  | 37 | 0.15  | Blood Plasma  | lgK:31.2(0.05SD) | 4 | ENS   | 94     |

## Reaction No. 2408



| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 20 | 0     | Unknown       | lgK:33.5(3SF)    | 0 | ENS   | 773    |
| 2  | 20 | 0.1   | Unknown       | lgK:40.0(3SF)    | 0 | ENS   | 773    |
| 3  | 25 | 0     | Inf. Dilution | lgK:37.5(3SF)    | 1 | ELC   |        |
| 4  | 25 | 0     | Inf. Dilution | lgK:33.5(3SF)    | 0 | ENS   | 2171   |
| 5  | 37 | 0.15  | Blood Plasma  | lgK:39.0(0.05SD) | 0 | ENS   | 94     |

## Reaction No. 2422



| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3          | lgK:16.6(3SF)    | 4 | MGL   | 3494   |
| 2  | 25 | 0     | Inf. Dilution | lgK:16.8(3SF)    | 1 | ESC   |        |
| 3  | 37 | 0.15  | Blood Plasma  | lgK:16.3(0.05SD) | 4 | ENS   | 94     |

## Reaction No. 2423

| 2<SulfCat-3> + Al+3 = Al+3_SulfCat-3(2) |    |       |               |                  |   |       |        |
|-----------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| No                                      | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                       | 20 | 0.1   | KNO3          | lgK:29.9(3SF)    | 5 | ENS   | 2171   |
| 2                                       | 25 | 0     | Inf. Dilution | lgK:30.2(3SF)    | 1 | ESC   |        |
| 3                                       | 37 | 0.15  | Blood Plasma  | lgK:28.5(0.05SD) | 4 | ENS   | 94     |

| Reaction No. 2424                       |    |       |               |                  |   |       |        |
|-----------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 3<SulfCat-3> + Al+3 = Al+3_SulfCat-3(3) |    |       |               |                  |   |       |        |
| No                                      | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                       | 20 | 0.1   | KNO3          | lgK:39.2(3SF)    | 5 | ENS   | 2171   |
| 2                                       | 25 | 0     | Inf. Dilution | lgK:39.4(3SF)    | 1 | ESC   |        |
| 3                                       | 37 | 0.15  | Blood Plasma  | lgK:38.2(0.05SD) | 4 | ENS   | 94     |

| Reaction No. 2719     |    |       |               |                  |   |       |        |
|-----------------------|----|-------|---------------|------------------|---|-------|--------|
| F-1 + Al+3 = Al+3_F-1 |    |       |               |                  |   |       |        |
| No                    | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                     | 23 | 0.1   | HNO3          | lgK:6.32(3SF)    | 3 | MCO   | 140    |
| 2                     | 23 |       |               | lgK:6.4(2SF)     | 0 | MSP   | 140    |
| 3                     | 25 | 0     | Inf. Dilution | lgK:6.69(3SF)    | 3 | MQH   | 298    |
| 4                     | 25 | 0     | Inf. Dilution | lgK:7.0(0.05SD)  | 5 | CMN   | 2171   |
| 5                     | 25 | 0     | Inf. Dilution | lgK:7.01(3SF)    | 4 | EMU   | 5398   |
| 6                     | 25 | 0     | Inf. Dilution | lgK:6.98(3SF)    | 4 | MIS   | 5398   |
| 7                     | 25 | 0     | Inf. Dilution | lgK:7.0(2SF)     | 4 | ENS   | 6496   |
| 8                     | 25 | 0     | Inf. Dilution | lgK:6.13(3SF)    | 0 | CRV   | 22551  |
| 9                     | 25 | 0     | Inf. Dilution | lgK:7.03(0.02SD) | 3 | EFE   | 30072  |
| 10                    | 25 | 0     | Inf. Dilution | lgK:5.740(4SF)   | 0 | CRV   | 32224  |
| 11                    | 25 | 0     | CHEMVAL1      | lgK:7.01(0.01SD) | 0 | ENS   | 5156   |
| 12                    | 25 | 0     | CHEMVAL2      | lgK:6.92(0.01SD) | 0 | ENS   | 5156   |
| 13                    | 25 | 0     | CHEMVAL3      | lgK:6.98(0.01SD) | 0 | ENS   | 5156   |
| 14                    | 25 | 0     | CHEMVAL4      | lgK:6.98(0.01SD) | 5 | ENS   | 5156   |
| 15                    | 25 | 0     | Many Media    | lgK:7.0(0.1SD)   | 7 | EOD   | 6953   |
| 16                    | 25 | 0     | SOLMNEQ       | lgK:6.97(3SF)    | 4 | CMN   | 5416   |
| 17                    | 25 | 0     | SOLVEQ        | lgK:7.00(3SF)    | 4 | CMN   | 5416   |
| 18                    | 25 | 0     | UCT_Sea       | lgK:7.00(0.01SD) | 4 | EDH   | 5390   |
| 19                    | 25 | 0     | WATCH         | lgK:6.97(3SF)    | 4 | CMN   | 5416   |
| 20                    | 25 | 0.01  | NH4NO3        | lgK:6.65(3SF)    | 4 | MIS   | 5398   |

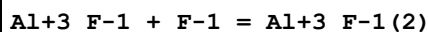
|    |     |      |               |                   |   |     |       |
|----|-----|------|---------------|-------------------|---|-----|-------|
| 21 | 25  | 0.05 | KNO3          | lgK:6.51 (3SF)    | 3 | MQH | 298   |
| 22 | 25  | 0.1  | KNO3          | lgK:6.45 (3SF)    | 3 | MQH | 298   |
| 23 | 25  | 0.1  | NH4NO3        | lgK:6.40 (3SF)    | 4 | MIS | 5398  |
| 24 | 25  | 0.1  | UCT_Sea       | lgK:6.43 (0.01SD) | 4 | EDH | 5390  |
| 25 | 25  | 0.1  | Unknown       | lgK:6.42 (3SF)    | 6 | CRV | 552   |
| 26 | 25  | 0.15 | NaCl          | lgK:6.33 (0.01SD) | 3 | MGL | 30072 |
| 27 | 25  | 0.2  | KNO3          | lgK:6.32 (3SF)    | 3 | MQH | 298   |
| 28 | 25  | 0.2  | UCT_Sea       | lgK:6.25 (0.01SD) | 4 | EDH | 5390  |
| 29 | 25  | 0.3  | NH4NO3        | lgK:6.29 (3SF)    | 3 | MIS | 5398  |
| 30 | 25  | 0.3  | UCT_Sea       | lgK:6.16 (0.01SD) | 4 | EDH | 5390  |
| 31 | 25  | 0.4  | UCT_Sea       | lgK:6.12 (0.01SD) | 4 | EDH | 5390  |
| 32 | 25  | 0.5  | KNO3          | lgK:6.14 (3SF)    | 3 | MQH | 298   |
| 33 | 25  | 0.5  | NH4NO3        | lgK:6.08 (3SF)    | 4 | MIS | 437   |
| 34 | 25  | 0.5  | UCT_Sea       | lgK:6.11 (0.01SD) | 4 | EDH | 5390  |
| 35 | 25  | 0.5  | Unknown       | lgK:6.13 (3SF)    | 6 | CRV | 552   |
| 36 | 25  | 0.5  | Unknown       | lgK:6.32 (3SF)    | 5 | CRV | 6761  |
| 37 | 25  | 0.53 | KNO3          | lgK:6.13 (3SF)    | 4 | MQH | 140   |
| 38 | 25  | 0.53 | KNO3          | lgK:6.16 (3SF)    | 4 | MQH | 140   |
| 39 | 25  | 0.53 | KNO3          | lgK:6.1 (0.05SD)  | 5 | CMN | 437   |
| 40 | 25  | 0.6  | UCT_Sea       | lgK:6.10 (0.01SD) | 4 | EDH | 5390  |
| 41 | 25  | 0.7  | UCT_Sea       | lgK:6.10 (0.01SD) | 4 | EDH | 5390  |
| 42 | 25  | 1    | HClO4         | lgK:3.33 (3SF)    | 0 | MAM | 437   |
| 43 | 25  | 1    | NaCl          | lgK:6.12 (0.01SD) | 3 | MGL | 30072 |
| 44 | 25  | 1    | Unknown       | lgK:6.09 (3SF)    | 6 | CRV | 552   |
| 45 | 25  |      |               | lgK:6.08 (3SF)    | 0 | MSC | 931   |
| 46 | 37  | 0    | Inf. Dilution | lgK:6.68 (3SF)    | 3 | MQH | 298   |
| 47 | 37  | 0.05 | KNO3          | lgK:6.71 (3SF)    | 0 | MQH | 298   |
| 48 | 37  | 0.1  | KNO3          | lgK:6.49 (3SF)    | 3 | MQH | 298   |
| 49 | 37  | 0.2  | KNO3          | lgK:6.39 (3SF)    | 3 | MQH | 298   |
| 50 | 37  | 0.5  | KNO3          | lgK:6.29 (3SF)    | 3 | MQH | 298   |
| 51 | 50  | 0    | SOLMNEQ       | lgK:7.07 (3SF)    | 4 | CMN | 5416  |
| 52 | 50  | 0    | SOLVEQ        | lgK:7.11 (3SF)    | 4 | CMN | 5416  |
| 53 | 50  | 0    | WATCH         | lgK:7.07 (3SF)    | 4 | CMN | 5416  |
| 54 | 75  | 0    | SOLMNEQ       | lgK:7.22 (3SF)    | 4 | CMN | 5416  |
| 55 | 75  | 0    | WATCH         | lgK:7.23 (3SF)    | 4 | CMN | 5416  |
| 56 | 100 | 0    | SOLMNEQ       | lgK:7.43 (3SF)    | 4 | CMN | 5416  |

|    |     |      |               |                       |   |     |       |
|----|-----|------|---------------|-----------------------|---|-----|-------|
| 57 | 100 | 0    | SOLVEQ        | lgK:7.54 (3SF)        | 4 | CMN | 5416  |
| 58 | 100 | 0    | WATCH         | lgK:7.44 (3SF)        | 4 | CMN | 5416  |
| 59 | 125 | 0    | SOLMNEQ       | lgK:7.71 (3SF)        | 4 | CMN | 5416  |
| 60 | 125 | 0    | WATCH         | lgK:7.71 (3SF)        | 4 | CMN | 5416  |
| 61 | 150 | 0    | SOLMNEQ       | lgK:8.04 (3SF)        | 4 | CMN | 5416  |
| 62 | 150 | 0    | SOLVEQ        | lgK:8.21 (3SF)        | 4 | CMN | 5416  |
| 63 | 150 | 0    | WATCH         | lgK:8.03 (3SF)        | 4 | CMN | 5416  |
| 64 | 175 | 0    | WATCH         | lgK:8.38 (3SF)        | 4 | CMN | 5416  |
| 65 | 200 | 0    | SOLMNEQ       | lgK:8.92 (3SF)        | 4 | CMN | 5416  |
| 66 | 200 | 0    | SOLVEQ        | lgK:9.17 (3SF)        | 3 | CMN | 5416  |
| 67 | 200 | 0    | WATCH         | lgK:8.78 (3SF)        | 4 | CMN | 5416  |
| 68 | 225 | 0    | WATCH         | lgK:9.22 (3SF)        | 3 | CMN | 5416  |
| 69 | 250 | 0    | SOLMNEQ       | lgK:10.25 (4SF)       | 0 | CMN | 5416  |
| 70 | 250 | 0    | SOLVEQ        | lgK:10.59 (4SF)       | 0 | CMN | 5416  |
| 71 | 250 | 0    | WATCH         | lgK:9.69 (3SF)        | 3 | CMN | 5416  |
| 72 | 275 | 0    | WATCH         | lgK:10.20 (4SF)       | 3 | CMN | 5416  |
| 73 | 300 | 0    | SOLMNEQ       | lgK:12.52 (4SF)       | 0 | CMN | 5416  |
| 74 | 300 | 0    | SOLVEQ        | lgK:13.00 (4SF)       | 0 | CMN | 5416  |
| 75 | 300 | 0    | WATCH         | lgK:10.74 (4SF)       | 0 | CMN | 5416  |
| 76 | 325 | 0    | WATCH         | lgK:11.32 (4SF)       | 0 | CMN | 5416  |
| 77 | 350 | 0    | SOLMNEQ       | lgK:17.53 (4SF)       | 0 | CMN | 5416  |
| 78 | 350 | 0    | WATCH         | lgK:11.93 (4SF)       | 0 | CMN | 5416  |
| 79 | 25  | 0    | Inf. Dilution | dG:-35.02 (4SF) kJ    | 0 | CRV | 6434  |
| 80 | 25  | 0    | Inf. Dilution | dG:-40.11 (0.11SD) kJ | 3 | EFE | 30072 |
| 81 | 25  | 0    | Inf. Dilution | dH:2.1 (0.05SD)       | 4 | CMN | 300   |
| 82 | 25  | 0    | Inf. Dilution | dH:4.44 (3SF) kJ      | 4 | CRV | 2171  |
| 83 | 25  | 0    | Inf. Dilution | dH:4.60 (3SF) kJ      | 6 | CRV | 6434  |
| 84 | 25  | 0    | Inf. Dilution | dH:1.1 (2SF)          | 4 | ENS | 6496  |
| 85 | 25  | 0    | Inf. Dilution | dH:6.3 (0.2SD) kJ     | 3 | EFE | 30072 |
| 86 | 25  | 0.07 | HClO4         | dH:1.0 (5.SD)         | 3 | MCL | 437   |
| 87 | 25  | 0.1  | Unknown       | dH:1.1 (2SF)          | 6 | CRV | 552   |
| 88 | 25  | 0.5  | HClO4         | dH:0.6 (5.SD)         | 3 | MCL | 437   |
| 89 | 25  | 0.5  | Unknown       | dH:0.6 (1SF)          | 6 | CRV | 552   |
| 90 | 25  | 1    | NaClO4        | dH:0.7 (5.SD)         | 5 | MCL | 437   |
| 91 | 25  |      |               | dH:1.15 (3SF)         | 0 | MCL | 140   |
| 92 | 25  |      |               | dH:1.17 (3SF)         | 0 | MCL | 140   |



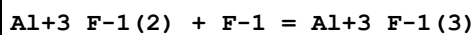
|    |    |     |         |              |   |     |     |
|----|----|-----|---------|--------------|---|-----|-----|
| 93 | 35 | 0.5 | Unknown | dH:0.9 (1SF) | 6 | CRV | 552 |
|----|----|-----|---------|--------------|---|-----|-----|

## Reaction No. 2720



| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 23 |       |               | lgK:5.31 (3SF)   | 0 | MSC   | 140    |
| 2  | 25 | 0     | Inf. Dilution | lgK:5.35 (3SF)   | 5 | MQH   | 298    |
| 3  | 25 | 0     | Inf. Dilution | lgK:5.62 (3SF)   | 5 | MIS   | 5398   |
| 4  | 25 | 0.01  | NH4NO3        | lgK:5.44 (3SF)   | 5 | MIS   | 5398   |
| 5  | 25 | 0.05  | KNO3          | lgK:5.29 (3SF)   | 5 | MQH   | 298    |
| 6  | 25 | 0.1   | KNO3          | lgK:5.21 (3SF)   | 5 | MQH   | 298    |
| 7  | 25 | 0.1   | NH4NO3        | lgK:5.19 (3SF)   | 5 | MIS   | 5398   |
| 8  | 25 | 0.2   | KNO3          | lgK:5.16 (3SF)   | 5 | MQH   | 298    |
| 9  | 25 | 0.3   | NH4NO3        | lgK:4.97 (3SF)   | 5 | MIS   | 5398   |
| 10 | 25 | 0.5   | KNO3          | lgK:5.09 (3SF)   | 5 | MQH   | 298    |
| 11 | 25 | 0.5   | NH4NO3        | lgK:4.93 (3SF)   | 5 | MIS   | 5398   |
| 12 | 25 | 0.53  | KNO3          | lgK:5.02 (3SF)   | 4 | MQH   | 140    |
| 13 | 25 | 0.53  | KNO3          | lgK:5.05 (3SF)   | 4 | MQH   | 140    |
| 14 | 25 | 0.53  | KNO3          | lgK:5.0 (0.05SD) | 5 | CMN   | 437    |
| 15 | 25 |       |               | lgK:5.02 (3SF)   | 0 | MSC   | 931    |
| 16 | 37 | 0     | Inf. Dilution | lgK:5.34 (3SF)   | 5 | MQH   | 298    |
| 17 | 37 | 0.05  | KNO3          | lgK:5.26 (3SF)   | 5 | MQH   | 298    |
| 18 | 37 | 0.1   | KNO3          | lgK:5.24 (3SF)   | 5 | MQH   | 298    |
| 19 | 37 | 0.2   | KNO3          | lgK:5.17 (3SF)   | 5 | MQH   | 298    |
| 20 | 37 | 0.5   | KNO3          | lgK:5.09 (3SF)   | 5 | MQH   | 298    |
| 21 | 25 | 0.07  | HClO4         | dH:0.91 (5.SD)   | 5 | MCL   | 437    |
| 22 | 25 | 0.5   | HClO4         | dH:1.17 (5.SD)   | 5 | MCL   | 437    |
| 23 | 25 |       |               | dH:0.78 (2SF)    | 0 | MCL   | 140    |
| 24 | 25 |       |               | dH:1.97 (3SF)    | 0 | MCL   | 140    |

## Reaction No. 2721



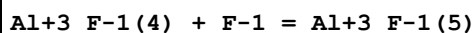
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 23 |       |               | lgK:3.82 (3SF) | 0 | MSC   | 140    |
| 2  | 25 | 0     | Inf. Dilution | lgK:3.68 (3SF) | 5 | MQH   | 298    |
| 3  | 25 | 0     | Inf. Dilution | lgK:4.05 (3SF) | 5 | MIS   | 5398   |

|    |    |      |               |                  |   |     |      |
|----|----|------|---------------|------------------|---|-----|------|
| 4  | 25 | 0.01 | NH4NO3        | lgK:3.92 (3SF)   | 5 | MIS | 5398 |
| 5  | 25 | 0.05 | KNO3          | lgK:3.76 (3SF)   | 5 | MQH | 298  |
| 6  | 25 | 0.1  | KNO3          | lgK:3.79 (3SF)   | 5 | MQH | 298  |
| 7  | 25 | 0.1  | NH4NO3        | lgK:3.91 (3SF)   | 5 | MIS | 5398 |
| 8  | 25 | 0.2  | KNO3          | lgK:3.85 (3SF)   | 5 | MQH | 298  |
| 9  | 25 | 0.3  | NH4NO3        | lgK:3.73 (3SF)   | 5 | MIS | 5398 |
| 10 | 25 | 0.5  | KNO3          | lgK:3.93 (3SF)   | 5 | MQH | 298  |
| 11 | 25 | 0.5  | NH4NO3        | lgK:3.69 (3SF)   | 5 | MIS | 5398 |
| 12 | 25 | 0.53 | KNO3          | lgK:3.85 (3SF)   | 4 | MQH | 140  |
| 13 | 25 | 0.53 | KNO3          | lgK:3.91 (3SF)   | 4 | MQH | 140  |
| 14 | 25 | 0.53 | KNO3          | lgK:3.9 (0.05SD) | 5 | CMN | 437  |
| 15 | 25 | 0.53 | KNO3          | lgK:3.85 (3SF)   | 4 | MQH | 437  |
| 16 | 25 | 0.53 | NH4NO3        | lgK:3.57 (3SF)   | 4 | MQH | 140  |
| 17 | 37 | 0    | Inf. Dilution | lgK:3.94 (3SF)   | 5 | MQH | 298  |
| 18 | 37 | 0.05 | KNO3          | lgK:3.92 (3SF)   | 5 | MQH | 298  |
| 19 | 37 | 0.1  | KNO3          | lgK:3.86 (3SF)   | 5 | MQH | 298  |
| 20 | 37 | 0.2  | KNO3          | lgK:3.86 (3SF)   | 5 | MQH | 298  |
| 21 | 37 | 0.5  | KNO3          | lgK:3.84 (3SF)   | 5 | MQH | 298  |
| 22 | 25 | 0.07 | HC1O4         | dH:0.17 (5.SD)   | 5 | MCL | 437  |
| 23 | 25 | 0.5  | HC1O4         | dH:1.27 (5.SD)   | 5 | MCL | 437  |
| 24 | 25 |      |               | dH:0.19 (2SF)    | 0 | MCL | 140  |
| 25 | 25 |      |               | dH:2.18 (3SF)    | 0 | MCL | 140  |

| Reaction No. 2722                 |    |       |               |                |   |       |        |
|-----------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Al+3_F-1 (3) + F-1 = Al+3_F-1 (4) |    |       |               |                |   |       |        |
| No                                | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                 | 23 |       |               | lgK:2.14 (3SF) | 0 | MSC   | 140    |
| 2                                 | 25 | 0     | Inf. Dilution | lgK:2.75 (3SF) | 0 | MQH   | 298    |
| 3                                 | 25 | 0     | Inf. Dilution | lgK:2.38 (3SF) | 3 | MIS   | 437    |
| 4                                 | 25 | 0.01  | NH4NO3        | lgK:2.38 (3SF) | 5 | MIS   | 5398   |
| 5                                 | 25 | 0.05  | KNO3          | lgK:3.05 (3SF) | 0 | MQH   | 298    |
| 6                                 | 25 | 0.1   | KNO3          | lgK:3.18 (3SF) | 0 | MQH   | 298    |
| 7                                 | 25 | 0.1   | NH4NO3        | lgK:2.42 (3SF) | 5 | MIS   | 5398   |
| 8                                 | 25 | 0.2   | KNO3          | lgK:3.30 (3SF) | 0 | MQH   | 298    |
| 9                                 | 25 | 0.3   | NH4NO3        | lgK:2.50 (3SF) | 5 | MIS   | 5398   |
| 10                                | 25 | 0.5   | KNO3          | lgK:3.68 (3SF) | 0 | MQH   | 298    |

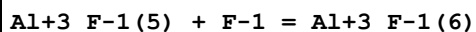
|    |    |      |               |                  |   |     |     |
|----|----|------|---------------|------------------|---|-----|-----|
| 11 | 25 | 0.5  | KNO3          | lgK:3.68 (3SF)   | 0 | MIS | 437 |
| 12 | 25 | 0.5  | NH4NO3        | lgK:2.50 (3SF)   | 3 | MIS | 437 |
| 13 | 25 | 0.53 | KNO3          | lgK:2.71 (3SF)   | 4 | MQH | 140 |
| 14 | 25 | 0.53 | KNO3          | lgK:2.7 (0.07SD) | 5 | CMN | 437 |
| 15 | 25 | 0.53 | KNO3          | lgK:2.74 (3SF)   | 3 | MQH | 437 |
| 16 | 25 | 0.53 | NH4NO3        | lgK:2.64 (3SF)   | 4 | MQH | 140 |
| 17 | 37 | 0    | Inf. Dilution | lgK:3.29 (3SF)   | 0 | MQH | 298 |
| 18 | 37 | 0.05 | KNO3          | lgK:3.29 (3SF)   | 0 | MQH | 298 |
| 19 | 37 | 0.1  | KNO3          | lgK:3.38 (3SF)   | 0 | MQH | 298 |
| 20 | 37 | 0.2  | KNO3          | lgK:3.38 (3SF)   | 0 | MQH | 298 |
| 21 | 37 | 0.5  | KNO3          | lgK:3.43 (3SF)   | 0 | MQH | 298 |
| 22 | 25 | 0.07 | HClO4         | dH:0.05 (5.SD)   | 5 | MCL | 437 |
| 23 | 25 | 0.07 | Unknown       | dH:0.04 (1SF)    | 3 | MCL | 140 |
| 24 | 25 |      |               | dH:0.28 (2SF)    | 0 | MCL | 140 |
| 25 | 25 |      |               | dH:2.14 (3SF)    | 0 | MCL | 140 |

## Reaction No. 2723



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 23 |       |        | lgK:0.43 (2SF)   | 0 | MSC   | 140    |
| 2  | 25 | 0.53  | KNO3   | lgK:1.46 (3SF)   | 0 | MQH   | 140    |
| 3  | 25 | 0.53  | KNO3   | lgK:1.6 (0.07SD) | 5 | CMN   | 437    |
| 4  | 25 | 0.53  | KNO3   | lgK:1.63 (3SF)   | 3 | MQH   | 437    |
| 5  | 25 | 0.53  | NH4NO3 | lgK:1.46 (3SF)   | 4 | MQH   | 140    |
| 6  | 25 | 0.07  | HClO4  | dH:-0.36 (5.SD)  | 5 | MCL   | 437    |
| 7  | 25 |       |        | dH:-0.75 (2SF)   | 0 | MCL   | 140    |
| 8  | 25 |       |        | dH:2.27 (3SF)    | 0 | MCL   | 140    |

## Reaction No. 2724

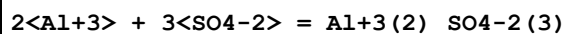


| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.53  | KNO3   | lgK:0.5 (0.07SD) | 5 | CMN   | 437    |
| 2  | 25 | 0.53  | KNO3   | lgK:0.47 (2SF)   | 3 | MQH   | 437    |
| 3  | 25 | 0.53  | NH4NO3 | lgK:0.04 (1SF)   | 0 | MQH   | 140    |
| 4  | 25 |       |        | dH:-1.55 (3SF)   | 0 | MCL   | 140    |
| 5  | 25 |       |        | dH:-1.24 (3SF)   | 0 | MCL   | 140    |

| Reaction No. 2868               |       |       |                 |                   |   |       |        |
|---------------------------------|-------|-------|-----------------|-------------------|---|-------|--------|
| 2<SO4-2> + Al+3 = Al+3_SO4-2(2) |       |       |                 |                   |   |       |        |
| No                              | t     | I-Str | Medium          | Value (Dev)       | W | Tech. | Ref. # |
| 1                               | 15    | 0     | Inf. Dilution   | lgK:2.25 (3SF)    | 0 | MEF   | 8374   |
| 2                               | 25    | 0     | Inf. Dilution   | lgK:5. (0.5SD)    | 3 | CMN   | 2171   |
| 3                               | 25    | 0     | Inf. Dilution   | lgK:5.0 (0.5SD)   | 4 | ENS   | 2171   |
| 4                               | 25    | 0     | Inf. Dilution   | lgK:4.92 (3SF)    | 4 | ENS   | 6496   |
| 5                               | 25    | 0     | Inf. Dilution   | lgK:4.90 (3SF)    | 5 | MCL   | 8402   |
| 6                               | 25    | 0     | Inf. Dilution   | lgK:4.92 (3SF)    | 4 | CRV   | 32224  |
| 7                               | 25    | 0     | CHEMVAL1        | lgK:4.92 (0.01SD) | 0 | ENS   | 5156   |
| 8                               | 25    | 0     | CHEMVAL2        | lgK:4.92 (0.01SD) | 0 | ENS   | 5156   |
| 9                               | 25    | 0     | CHEMVAL3        | lgK:4.90 (0.01SD) | 0 | ENS   | 5156   |
| 10                              | 25    | 0     | CHEMVAL4        | lgK:4.90 (0.01SD) | 5 | ENS   | 5156   |
| 11                              | 25    | 0     | Many Media      | lgK:5. (0.5SD)    | 6 | EOD   | 6953   |
| 12                              | 25    | 0     | Unknown         | lgK:4.92 (3SF)    | 3 | ENS   | 7373   |
| 13                              | 25    | 0.6   | NaClO4          | lgK:2.30 (3SF)    | 0 | MEF   | 621    |
| 14                              | 49.9  | 0.099 | NaCl/m          | lgK:3.86 (0.22SD) | 2 | MHY   | 7702   |
| 15                              | 50    | 0     | Inf. Dilution/m | lgK:5.6 (0.7MD)   | 4 | MHY   | 7702   |
| 16                              | 50    | 1.016 | NaCl/m          | lgK:2.84 (0.07SD) | 0 | MHY   | 7702   |
| 17                              | 50    | 1.026 | NaCl/m          | lgK:2.36 (0.31SD) | 0 | MHY   | 7702   |
| 18                              | 74.8  | 0.309 | NaCl/m          | lgK:3.43 (0.10SD) | 6 | MHY   | 7702   |
| 19                              | 74.8  | 1.016 | NaCl/m          | lgK:2.16 (0.43SD) | 3 | MHY   | 7702   |
| 20                              | 99.5  | 0.098 | NaCl/m          | lgK:5.31 (0.03SD) | 2 | MHY   | 7702   |
| 21                              | 99.7  | 0.098 | NaCl/m          | lgK:5.23 (0.04SD) | 2 | MHY   | 7702   |
| 22                              | 99.7  | 0.306 | NaCl/m          | lgK:4.68 (0.02SD) | 4 | MHY   | 7702   |
| 23                              | 99.7  | 1.021 | NaCl/m          | lgK:3.30 (0.04SD) | 2 | MHY   | 7702   |
| 24                              | 99.7  | 1.024 | NaCl/m          | lgK:2.52 (0.58SD) | 2 | MHY   | 7702   |
| 25                              | 99.9  | 0.311 | NaCl/m          | lgK:4.36 (0.05SD) | 2 | MHY   | 7702   |
| 26                              | 124.3 | 0.098 | NaCl/m          | lgK:6.90 (0.03SD) | 4 | MHY   | 7702   |
| 27                              | 124.5 | 0.098 | NaCl/m          | lgK:7.00 (0.08SD) | 6 | MHY   | 7702   |
| 28                              | 124.5 | 0.304 | NaCl/m          | lgK:6.29 (0.05SD) | 2 | MHY   | 7702   |
| 29                              | 124.7 | 1.007 | NaCl/m          | lgK:4.29 (0.03SD) | 6 | MHY   | 7702   |
| 30                              | 125   | 0.301 | NaCl/m          | lgK:6.09 (0.02SD) | 3 | MHY   | 7702   |
| 31                              | 25    | 0     | Inf. Dilution   | dH:3.1 (0.2SD)    | 4 | ENS   | 2171   |
| 32                              | 25    | 0     | Inf. Dilution   | dH:2.84 (3SF)     | 4 | ENS   | 6496   |
| 33                              | 25    | 0     | Unknown         | dH:2.84 (3SF)     | 3 | ENS   | 7373   |

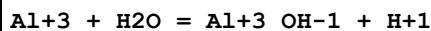
|    |    |   |                 |                     |   |     |      |
|----|----|---|-----------------|---------------------|---|-----|------|
| 34 | 50 | 0 | Inf. Dilution/m | dH:10.(50.MD)kJ     | 0 | MTD | 7702 |
| 35 | 50 | 0 | Inf. Dilution/m | Cp:2800.(800.SD)J/K | 5 | MTD | 7702 |

## Reaction No. 2869



| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-1.9(0.05SD) | 4 | ENS   | 557    |

## Reaction No. 2870



| No | t    | I-Str | Medium                 | Value (Dev)        | W | Tech. | Ref. # |
|----|------|-------|------------------------|--------------------|---|-------|--------|
| 1  | 25   | 0     | P=100<br>Inf. Dilution | lgK:-5.04(3SF)     | 3 | ENS   | 18070  |
| 2  | 200  | 0     | P=300<br>Inf. Dilution | lgK:-1.446(4SF)    | 4 | EOD   | 29559  |
| 3  | 0    | 0     | NaCl/m                 | lgK:-5.84(0.04MD)  | 2 | MEF   | 4278   |
| 4  | 0    | 0     | NaCl/m                 | lgK:-5.84(3SF)     | 2 | MSL   | 7025   |
| 5  | 0    | 0.1   | NaCl/m                 | lgK:-6.16(0.04MD)  | 3 | MEF   | 4278   |
| 6  | 0    | 0.1   | NaCl/m                 | lgK:-6.16(3SF)     | 3 | MSL   | 7025   |
| 7  | 0    | 0.3   | NaCl/m                 | lgK:-6.22(0.06MD)  | 5 | MEF   | 4278   |
| 8  | 0    | 0.5   | NaCl/m                 | lgK:-6.23(3SF)     | 7 | MSL   | 7025   |
| 9  | 0    | 1     | NaCl/m                 | lgK:-6.28(0.07MD)  | 5 | MEF   | 4278   |
| 10 | 0    | 1     | NaCl/m                 | lgK:-6.28(3SF)     | 5 | MSL   | 7025   |
| 11 | 0    | 5     | NaCl/m                 | lgK:-6.42(0.12MD)  | 5 | MEF   | 4278   |
| 12 | 0    | 5     | NaCl/m                 | lgK:-6.42(3SF)     | 7 | MSL   | 7025   |
| 13 | 10   | 0     | Inf. Dilution          | lgK:-5.5(0.05SD)   | 4 | ENS   | 4998   |
| 14 | 10   | 1     | NaNO3                  | lgK:-5.56(0.05SD)  | 1 | MGL   | 30097  |
| 15 | 15   | 0     | Inf. Dilution          | lgK:-5.28(0.02SD)w | 0 | MPH   | 5637   |
| 16 | 20   | 0     | Inf. Dilution          | lgK:-5.15(0.02SD)w | 0 | MPH   | 5637   |
| 17 | 21.5 | 0     | Inf. Dilution          | lgK:-5.13(0.02SD)w | 0 | MPH   | 5637   |
| 18 | 25   | 0     | Inf. Dilution          | lgK:-5.00(0.04SD)  | 4 | ENS   | 2171   |
| 19 | 25   | 0     | Inf. Dilution          | lgK:-4.99(3SF)     | 6 | CRV   | 5623   |
| 20 | 25   | 0     | Inf. Dilution          | lgK:-4.98(0.02SD)w | 3 | MPH   | 5637   |
| 21 | 25   | 0     | Inf. Dilution          | lgK:-4.97(3SF)     | 5 | CRV   | 6478   |
| 22 | 25   | 0     | Inf. Dilution          | lgK:-5.00(3SF)     | 4 | ENS   | 6496   |
| 23 | 25   | 0     | Inf. Dilution          | lgK:-4.99(3SF)     | 3 | ENS   | 7373   |
| 24 | 25   | 0     | Inf. Dilution          | lgK:-5.02(3SF)     | 4 | MSL   | 8620   |

|                                             |    |       |               |                     |   |     |       |
|---------------------------------------------|----|-------|---------------|---------------------|---|-----|-------|
| 25                                          | 25 | 0     | Inf. Dilution | lgK:-5.00 (3SF)     | 3 | ENS | 18070 |
| 26                                          | 25 | 0     | Inf. Dilution | lgK:-4.85 (3SF)     | 4 | CRV | 22551 |
| 27                                          | 25 | 0     | Inf. Dilution | lgK:-4.964 (4SF)    | 4 | EOD | 29559 |
| 28                                          | 25 | 0     | Inf. Dilution | lgK:-4.84 (0.14SD)  | 4 | EDH | 30072 |
| 29                                          | 25 | 0     | Inf. Dilution | lgK:-4.969 (4SF)    | 4 | CRV | 32224 |
| 30                                          | 25 | 0     | CHEMVAL1      | lgK:-4.99 (0.01SD)  | 0 | ENS | 5156  |
| 31                                          | 25 | 0     | CHEMVAL2      | lgK:-4.99 (0.01SD)  | 0 | ENS | 5156  |
| 32                                          | 25 | 0     | CHEMVAL3      | lgK:-4.96 (0.01SD)  | 0 | ENS | 5156  |
| 33                                          | 25 | 0     | CHEMVAL4      | lgK:-4.96 (0.01SD)  | 3 | ENS | 5156  |
| 34                                          | 25 | 0     | Inert         | lgK:-4.97 (3SF)     | 0 | EOD | 5613  |
| 35                                          | 25 | 0     | Many Media    | lgK:-5.00 (0.04MD)  | 5 | CRV | 2171  |
| 36                                          | 25 | 0     | Many Media    | lgK:-5.0 (0.04SD)   | 5 | EOD | 6953  |
| 37                                          | 25 | 0     | NaCl/m        | lgK:-4.95 (0.03MD)  | 3 | MEF | 4278  |
| 38                                          | 25 | 0     | NaCl/m        | lgK:-4.95 (3SF)     | 3 | MSL | 7025  |
| 39                                          | 25 | 0     | UCT_Sea       | lgK:-4.99 (0.01SD)  | 3 | EDH | 5390  |
| 40                                          | 25 | 0     | UCT_Sea       | lgK:-4.99 (0.01SD)  | 3 | EDT | 5390  |
| 41                                          | 25 | 0.01  | NaClO4        | lgK:-5.19 (3SF)     | 2 | MMR | 7045  |
| 42                                          | 25 | 0.06  | Cl-1 salt     | lgK:-5.26 (0.02SD)  | 5 | MLC | 5588  |
| 43                                          | 25 | 0.065 | Cl-1 salt     | lgK:-4.7 (0.05SD) w | 0 | MPH | 5637  |
| Note (43): Low wgt for internal consistency |    |       |               |                     |   |     |       |
| 44                                          | 25 | 0.1   | Inert         | lgK:-5.41 (3SF)     | 0 | EOD | 5613  |
| 45                                          | 25 | 0.1   | KNO3          | lgK:-5.00 (3SF) w   | 0 | MPH | 5625  |
| Note (45): Low wgt for internal consistency |    |       |               |                     |   |     |       |
| 46                                          | 25 | 0.1   | NaCl          | lgK:-5.25 (3SF)     | 5 | MGL | 29939 |
| 47                                          | 25 | 0.1   | NaCl/m        | lgK:-5.31 (0.03MD)  | 6 | MEF | 4278  |
| 48                                          | 25 | 0.1   | NaCl/m        | lgK:-5.31 (3SF)     | 5 | MSL | 7025  |
| 49                                          | 25 | 0.1   | NaClO4        | lgK:-5.3 (0.05SD)   | 4 | ENS | 4998  |
| 50                                          | 25 | 0.1   | NaClO4        | lgK:-4.90 (3SF)     | 0 | MLC | 5620  |
| Note (50): Low wgt for internal consistency |    |       |               |                     |   |     |       |
| 51                                          | 25 | 0.1   | NaNO3         | lgK:-5.33 (0.009SD) | 4 | MPT | 4878  |
| 52                                          | 25 | 0.1   | UCT_Sea       | lgK:-5.31 (0.01SD)  | 3 | EDH | 5390  |
| 53                                          | 25 | 0.1   | UCT_Sea       | lgK:-5.31 (0.01SD)  | 3 | EDT | 5390  |
| 54                                          | 25 | 0.2   | Inert         | lgK:-5.50 (3SF)     | 3 | EOD | 5613  |
| 55                                          | 25 | 0.2   | UCT_Sea       | lgK:-5.39 (0.01SD)  | 3 | EDH | 5390  |
| 56                                          | 25 | 0.2   | UCT_Sea       | lgK:-5.39 (0.01SD)  | 3 | EDT | 5390  |
| 57                                          | 25 | 0.25  | NaCl          | lgK:-5.36 (3SF)     | 5 | MGL | 29939 |

|    |    |     |               |                    |   |     |       |
|----|----|-----|---------------|--------------------|---|-----|-------|
| 58 | 25 | 0.3 | Inert         | lgK:-5.54 (3SF)    | 3 | EOD | 5613  |
| 59 | 25 | 0.3 | NaCl/m        | lgK:-5.39 (0.04MD) | 6 | MEF | 4278  |
| 60 | 25 | 0.3 | NaClO4        | lgK:-4.60 (3SF)    | 0 | MLC | 5620  |
| 61 | 25 | 0.3 | UCT_Sea       | lgK:-5.47 (0.01SD) | 3 | EDH | 5390  |
| 62 | 25 | 0.3 | UCT_Sea       | lgK:-5.47 (0.01SD) | 3 | EDT | 5390  |
| 63 | 25 | 0.4 | UCT_Sea       | lgK:-5.54 (0.01SD) | 3 | EDH | 5390  |
| 64 | 25 | 0.4 | UCT_Sea       | lgK:-5.54 (0.01SD) | 3 | EDT | 5390  |
| 65 | 25 | 0.5 | Inert         | lgK:-5.56 (3SF)    | 5 | EOD | 5613  |
| 66 | 25 | 0.5 | NaCl          | lgK:-5.43 (3SF)    | 5 | MGL | 29939 |
| 67 | 25 | 0.5 | NaCl/m        | lgK:-5.43 (3SF)    | 7 | MSL | 7025  |
| 68 | 25 | 0.5 | NaClO4        | lgK:-4.30 (3SF)    | 0 | MLC | 5620  |
| 69 | 25 | 0.5 | UCT_Sea       | lgK:-5.53 (0.01SD) | 3 | EDH | 5390  |
| 70 | 25 | 0.5 | UCT_Sea       | lgK:-5.53 (0.01SD) | 3 | EDT | 5390  |
| 71 | 25 | 0.6 | NaCl          | lgK:-5.52 (0.04SD) | 7 | MGL | 5447  |
| 72 | 25 | 0.6 | UCT_Sea       | lgK:-5.52 (0.01SD) | 3 | EDH | 5390  |
| 73 | 25 | 0.6 | UCT_Sea       | lgK:-5.52 (0.01SD) | 3 | EDT | 5390  |
| 74 | 25 | 0.7 | NaCl          | lgK:-5.45 (3SF)    | 5 | MGL | 29939 |
| 75 | 25 | 0.7 | Seawater      | lgK:-5.51 (3SF)    | 0 | ENS | 6496  |
| 76 | 25 | 0.7 | Seawater      | lgK:-5.51 (3SF)    | 0 | EDT | 9100  |
| 77 | 25 | 0.7 | UCT_Sea       | lgK:-5.52 (0.01SD) | 3 | EDH | 5390  |
| 78 | 25 | 0.7 | UCT_Sea       | lgK:-5.52 (0.01SD) | 3 | EDT | 5390  |
| 79 | 25 | 1   | Inert         | lgK:-5.47 (3SF)    | 3 | EOD | 5613  |
| 80 | 25 | 1   | NaCl/m        | lgK:-5.51 (0.05MD) | 6 | MEF | 4278  |
| 81 | 25 | 1   | NaCl/m        | lgK:-5.51 (3SF)    | 5 | MSL | 7025  |
| 82 | 25 | 1   | NaClO4        | lgK:-5.5 (0.05SD)  | 4 | ENS | 4998  |
| 83 | 25 | 1   | NaClO4        | lgK:-3.60 (3SF)    | 0 | MLC | 5620  |
| 84 | 25 | 1   | NaNO3         | lgK:-5.01 (0.03SD) | 1 | MGL | 30097 |
| 85 | 25 | 2   | Inert         | lgK:-5.13 (3SF)    | 0 | EOD | 5613  |
| 86 | 25 | 3   | Inert         | lgK:-4.71 (3SF)    | 0 | EOD | 5613  |
| 87 | 25 | 3   | Na (Cl)       | lgK:-5.52 (0.05SD) | 3 | MGL | 5494  |
| 88 | 25 | 5   | Inert         | lgK:-3.78 (3SF)    | 0 | EOD | 5613  |
| 89 | 25 | 5   | NaCl/m        | lgK:-5.64 (0.09MD) | 6 | MEF | 4278  |
| 90 | 25 | 5   | NaCl/m        | lgK:-5.64 (3SF)    | 7 | MSL | 7025  |
| 91 | 40 | 0   | Inf. Dilution | lgK:-4.6 (0.05SD)  | 4 | ENS | 4998  |
| 92 | 50 | 0   | NaCl/m        | lgK:-4.20 (0.03MD) | 5 | MEF | 4278  |
| 93 | 50 | 0   | NaCl/m        | lgK:-4.20 (3SF)    | 3 | MSL | 7025  |

|     |     |     |               |                   |   |     |       |
|-----|-----|-----|---------------|-------------------|---|-----|-------|
| 94  | 50  | 0.1 | NaCl/m        | lgK:-4.60(0.02MD) | 6 | MEF | 4278  |
| 95  | 50  | 0.1 | NaCl/m        | lgK:-4.60(3SF)    | 7 | MSL | 7025  |
| 96  | 50  | 0.3 | NaCl/m        | lgK:-4.71(0.03MD) | 6 | MEF | 4278  |
| 97  | 50  | 0.5 | NaCl/m        | lgK:-4.77(3SF)    | 7 | MSL | 7025  |
| 98  | 50  | 1   | NaCl/m        | lgK:-4.88(0.04MD) | 6 | MEF | 4278  |
| 99  | 50  | 1   | NaCl/m        | lgK:-4.88(3SF)    | 7 | MSL | 7025  |
| 100 | 50  | 1   | NaNO3         | lgK:-4.37(0.04SD) | 1 | MGL | 30097 |
| 101 | 50  | 5   | NaCl/m        | lgK:-4.99(0.07MD) | 6 | MEF | 4278  |
| 102 | 50  | 5   | NaCl/m        | lgK:-4.99(3SF)    | 7 | MSL | 7025  |
| 103 | 70  | 1   | NaNO3         | lgK:-4.08(0.05SD) | 1 | MGL | 30097 |
| 104 | 75  | 0   | NaCl/m        | lgK:-3.56(0.03MD) | 5 | MEF | 4278  |
| 105 | 75  | 0   | NaCl/m        | lgK:-3.56(3SF)    | 6 | MSL | 7025  |
| 106 | 75  | 0.1 | NaCl/m        | lgK:-4.00(0.02MD) | 6 | MEF | 4278  |
| 107 | 75  | 0.1 | NaCl/m        | lgK:-4.00(3SF)    | 7 | MSL | 7025  |
| 108 | 75  | 0.3 | NaCl/m        | lgK:-4.14(0.03MD) | 6 | MEF | 4278  |
| 109 | 75  | 0.5 | NaCl/m        | lgK:-4.22(3SF)    | 7 | MSL | 7025  |
| 110 | 75  | 1   | NaCl/m        | lgK:-4.36(0.04MD) | 6 | MEF | 4278  |
| 111 | 75  | 1   | NaCl/m        | lgK:-4.35(3SF)    | 7 | MSL | 7025  |
| 112 | 75  | 5   | NaCl/m        | lgK:-4.45(0.07MD) | 6 | MEF | 4278  |
| 113 | 75  | 5   | NaCl/m        | lgK:-4.45(3SF)    | 7 | MSL | 7025  |
| 114 | 90  | 0   | Inf. Dilution | lgK:-3.248(4SF)   | 4 | EOD | 29559 |
| 115 | 100 | 0   | NaCl/m        | lgK:-3.00(0.04MD) | 4 | MEF | 4278  |
| 116 | 100 | 0   | NaCl/m        | lgK:-3.00(3SF)    | 6 | MSL | 7025  |
| 117 | 100 | 0.1 | NaCl/m        | lgK:-3.49(0.03MD) | 6 | MEF | 4278  |
| 118 | 100 | 0.1 | NaCl/m        | lgK:-3.49(3SF)    | 7 | MSL | 7025  |
| 119 | 100 | 0.3 | NaCl/m        | lgK:-3.67(0.03MD) | 6 | MEF | 4278  |
| 120 | 100 | 0.5 | NaCl/m        | lgK:-3.76(3SF)    | 7 | MSL | 7025  |
| 121 | 100 | 1   | NaCl/m        | lgK:-3.93(0.05MD) | 4 | MEF | 4278  |
| 122 | 100 | 1   | NaCl/m        | lgK:-3.93(3SF)    | 5 | MSL | 7025  |
| 123 | 100 | 5   | NaCl/m        | lgK:-4.01(0.1MD)  | 6 | MEF | 4278  |
| 124 | 100 | 5   | NaCl/m        | lgK:-4.01(3SF)    | 7 | MSL | 7025  |
| 125 | 125 | 0   | NaCl/m        | lgK:-2.51(0.04MD) | 4 | MEF | 4278  |
| 126 | 125 | 0   | NaCl/m        | lgK:-2.51(3SF)    | 6 | MSL | 7025  |
| 127 | 125 | 0.1 | NaCl/m        | lgK:-3.05(0.04MD) | 6 | MEF | 4278  |
| 128 | 125 | 0.1 | NaCl/m        | lgK:-3.05(3SF)    | 7 | MSL | 7025  |
| 129 | 125 | 0.3 | NaCl/m        | lgK:-3.27(0.04MD) | 5 | MEF | 4278  |



|     |        |     |               |                    |   |     |       |
|-----|--------|-----|---------------|--------------------|---|-----|-------|
| 130 | 125    | 0.5 | NaCl/m        | lgK:-3.38 (3SF)    | 7 | MSL | 7025  |
| 131 | 125    | 1   | NaCl/m        | lgK:-3.58 (0.05MD) | 3 | MEF | 4278  |
| 132 | 125    | 5   | NaCl/m        | lgK:-3.65 (0.13MD) | 6 | MEF | 4278  |
| 133 | 125    | 5   | NaCl/m        | lgK:-3.65 (3SF)    | 7 | MSL | 7025  |
| 134 | 150    | 0   | Inf. Dilution | lgK:-2.129 (4SF)   | 4 | EOD | 29559 |
| 135 | 150    | 0   | NaCl/m        | lgK:-2.08 (0.05MD) | 4 | MEF | 4278  |
| 136 | 150    | 0   | NaCl/m        | lgK:-2.1 (2SF)     | 6 | MSL | 7025  |
| 137 | 150    |     |               | lgK:-2.1 (2SF)     | 0 | MSL | 4281  |
| 138 | 175    | 0   | NaCl/m        | lgK:-1.70 (0.05MD) | 4 | MEF | 4278  |
| 139 | 175    | 0   | NaCl/m        | lgK:-1.7 (2SF)     | 6 | MSL | 7025  |
| 140 | 200    | 0   | NaCl/m        | lgK:-1.36 (0.06MD) | 4 | MEF | 4278  |
| 141 | 200    | 0   | NaCl/m        | lgK:-1.4 (2SF)     | 6 | MSL | 7025  |
| 142 | 200    |     |               | lgK:-1.1 (2SF)     | 0 | MSL | 4281  |
| 143 | 210    | 0   | Inf. Dilution | lgK:-1.265 (4SF)   | 4 | EOD | 29559 |
| 144 | 225    | 0   | NaCl/m        | lgK:-1.1 (2SF)     | 6 | MSL | 7025  |
| 145 | 250    | 0   | NaCl/m        | lgK:-0.8 (1SF)     | 6 | MSL | 7025  |
| 146 | 250    |     |               | lgK:0.4 (1SF)      | 0 | MSL | 4281  |
| 147 | 0      | 1   | NaCl/m        | dH:50.8 (1.5MD) kJ | 5 | MTD | 4278  |
| 148 | 0      | 5   | NaCl/m        | dH:53.0 (2.1MD) kJ | 5 | MTD | 4278  |
| 149 | 0:125  | 0.1 | NaCl/m        | dH:54.4 (1.0MD) kJ | 5 | MTD | 4278  |
| 150 | 0:200  | 0   | NaCl/m        | dH:54.4 (1.0MD) kJ | 5 | MTD | 4278  |
| 151 | 25     | 0   | Inf. Dilution | dH:11.5 (0.1SD)    | 5 | ENS | 5484  |
| 152 | 25     | 0   | Inf. Dilution | dH:13.0 (3SF)      | 3 | CRV | 6478  |
| 153 | 25     | 0   | Inf. Dilution | dH:11.9 (0.7SD)    | 5 | CRV | 6478  |
| 154 | 25     | 0   | Inf. Dilution | dH:46.9 (1.8SD) kJ | 3 | EDH | 30072 |
| 155 | 25     | 0   | Unknown       | dH:11.90 (4SF)     | 3 | ENS | 7373  |
| 156 | 25     | 0.1 | KNO3          | dH:11.8 (3SF)      | 4 | MTD | 5625  |
| 157 | 25     | 0.3 | NaCl/m        | dH:53.0 (1.1MD) kJ | 6 | MTD | 4278  |
| 158 | 25     | 0.7 | Seawater      | dH:11.9 (3SF)      | 0 | ENS | 6496  |
| 159 | 25     | 0.7 | Seawater      | dH:11.9 (3SF)      | 0 | EDT | 9100  |
| 160 | 25     | 5   | NaCl/m        | dH:54.0 (2.6MD) kJ | 6 | MTD | 4278  |
| 161 | 25:50  | 1   | NaCl/m        | dH:50.9 (1.4MD) kJ | 4 | MTD | 4278  |
| 162 | 25:125 | 0.3 | NaCl/m        | dH:53.1 (1.1MD) kJ | 4 | MTD | 4278  |
| 163 | 50     | 5   | NaCl/m        | dH:55.2 (3.3MD) kJ | 6 | MTD | 4278  |
| 164 | 75     | 1   | NaCl/m        | dH:51.0 (1.4MD) kJ | 3 | MTD | 4278  |
| 165 | 75     | 5   | NaCl/m        | dH:56.4 (4.0MD) kJ | 6 | MTD | 4278  |

|     |         |     |        |                    |   |     |      |
|-----|---------|-----|--------|--------------------|---|-----|------|
| 166 | 100     | 5   | NaCl/m | dH:13.79 (4.8MD)   | 4 | MTD | 4278 |
| 167 | 100:125 | 1   | NaCl/m | dH:51.1 (1.4MD) kJ | 0 | MTD | 4278 |
| 168 | 125     | 5   | NaCl/m | dH:59.2 (5.6MD) kJ | 0 | MTD | 4278 |
| 169 | 0:25    | 5   | NaCl/m | Cp:40. (30.MD) J/K | 5 | MTD | 4278 |
| 170 | 0:50    | 1   | NaCl/m | Cp:2. (10.MD) J/K  | 5 | MTD | 4278 |
| 171 | 0:125   | 0.1 | NaCl/m | Cp:0.0 (1SF) J/K   | 5 | MTD | 4278 |
| 172 | 0:125   | 0.3 | NaCl/m | Cp:0.0 (1SF) J/K   | 5 | MTD | 4278 |
| 173 | 0:200   | 0   | NaCl/m | Cp:0.0 (1SF) J/K   | 5 | MTD | 4278 |
| 174 | 50:100  | 5   | NaCl/m | Cp:50. (30.MD) J/K | 4 | MTD | 4278 |
| 175 | 75      | 1   | NaCl/m | Cp:2. (20.MD) J/K  | 5 | MTD | 4278 |
| 176 | 100:125 | 1   | NaCl/m | Cp:3. (20.MD) J/K  | 4 | MTD | 4278 |
| 177 | 125     | 5   | NaCl/m | Cp:60. (40.MD) J/K | 3 | MTD | 4278 |

| Reaction No. 2871                     |     |       |                        |                     |   |       |        |
|---------------------------------------|-----|-------|------------------------|---------------------|---|-------|--------|
| Al+3 + 2<H2O> = Al+3_OH-1(2) + 2<H+1> |     |       |                        |                     |   |       |        |
| No                                    | t   | I-Str | Medium                 | Value (Dev)         | W | Tech. | Ref. # |
| 1                                     | 200 | 0     | P=300<br>Inf. Dilution | lgK:-3.63 (3SF)     | 4 | EOD   | 29559  |
| 2                                     | 25  | 0     | Inf. Dilution          | lgK:-10.5 (3SF)     | 1 | ELC   |        |
| 3                                     | 25  | 0     | Inf. Dilution          | lgK:-10.1 (0.05SD)  | 0 | CMN   | 2171   |
| 4                                     | 25  | 0     | Inf. Dilution          | lgK:-10.1 (3SF)     | 0 | ENS   | 2171   |
| 5                                     | 25  | 0     | Inf. Dilution          | lgK:-9.90 (3SF)     | 0 | CRV   | 5623   |
| 6                                     | 25  | 0     | Inf. Dilution          | lgK:-10.1 (3SF)     | 0 | ENS   | 6496   |
| 7                                     | 25  | 0     | Inf. Dilution          | lgK:-10.921 (5SF)   | 0 | EOD   | 29559  |
| 8                                     | 25  | 0     | Inf. Dilution          | lgK:-10.49 (0.16SD) | 4 | EDH   | 30072  |
| 9                                     | 25  | 0     | Inf. Dilution          | lgK:-10.158 (5SF)   | 0 | CRV   | 32224  |
| 10                                    | 25  | 0     | CHEMVAL1               | lgK:-9.30 (0.01SD)  | 0 | ENS   | 5156   |
| 11                                    | 25  | 0     | CHEMVAL2               | lgK:-9.30 (0.01SD)  | 0 | ENS   | 5156   |
| 12                                    | 25  | 0     | CHEMVAL3               | lgK:-9.25 (0.01SD)  | 0 | ENS   | 5156   |
| 13                                    | 25  | 0     | CHEMVAL4               | lgK:-9.25 (0.01SD)  | 0 | ENS   | 5156   |
| 14                                    | 25  | 0     | Inert                  | lgK:-9.30 (3SF)     | 0 | EOD   | 5613   |
| 15                                    | 25  | 0     | Many Media             | lgK:-10.1 (3SF)     | 0 | CRV   | 2171   |
| 16                                    | 25  | 0     | Many Media             | lgK:-10.1 (3SF)     | 0 | EOD   | 6953   |
| 17                                    | 25  | 0     | UCT_Sea                | lgK:-10.10 (0.01SD) | 0 | EDH   | 5390   |
| 18                                    | 25  | 0     | Unknown                | lgK:-10.1 (3SF)     | 0 | ENS   | 7373   |
| 19                                    | 25  | 0.01  | NaClO4                 | lgK:-10.4 (3SF)     | 0 | MMR   | 7045   |
| 20                                    | 25  | 0.1   | Inert                  | lgK:-9.98 (3SF)     | 0 | EOD   | 5613   |

|    |     |      |               |                     |   |     |       |
|----|-----|------|---------------|---------------------|---|-----|-------|
| 21 | 25  | 0.1  | NaCl          | lgK:-11.12 (4SF)    | 5 | MGL | 29939 |
| 22 | 25  | 0.1  | NaClO4        | lgK:-9.9 (0.1SD)    | 0 | ENS | 4998  |
| 23 | 25  | 0.1  | NaNO3         | lgK:-10.91 (0.04SD) | 0 | MPT | 4878  |
| 24 | 25  | 0.1  | UCT_Sea       | lgK:-10.62 (0.01SD) | 0 | EDH | 5390  |
| 25 | 25  | 0.2  | Inert         | lgK:-10.14 (4SF)    | 0 | EOD | 5613  |
| 26 | 25  | 0.2  | UCT_Sea       | lgK:-10.76 (0.01SD) | 0 | EDH | 5390  |
| 27 | 25  | 0.25 | NaCl          | lgK:-11.30 (4SF)    | 5 | MGL | 29939 |
| 28 | 25  | 0.3  | Inert         | lgK:-10.22 (4SF)    | 0 | EOD | 5613  |
| 29 | 25  | 0.3  | UCT_Sea       | lgK:-10.88 (0.01SD) | 0 | EDH | 5390  |
| 30 | 25  | 0.4  | UCT_Sea       | lgK:-10.98 (0.01SD) | 0 | EDH | 5390  |
| 31 | 25  | 0.5  | Inert         | lgK:-10.29 (4SF)    | 0 | EOD | 5613  |
| 32 | 25  | 0.5  | NaCl          | lgK:-11.41 (4SF)    | 5 | MGL | 29939 |
| 33 | 25  | 0.5  | UCT_Sea       | lgK:-10.99 (0.01SD) | 0 | EDH | 5390  |
| 34 | 25  | 0.6  | NaCl          | lgK:-11.3 (0.05SD)  | 0 | MPS | 5075  |
| 35 | 25  | 0.6  | UCT_Sea       | lgK:-11.00 (0.01SD) | 0 | EDH | 5390  |
| 36 | 25  | 0.7  | NaCl          | lgK:-11.46 (4SF)    | 5 | MGL | 29939 |
| 37 | 25  | 0.7  | Seawater      | lgK:-10.13 (4SF)    | 0 | ENS | 6496  |
| 38 | 25  | 0.7  | UCT_Sea       | lgK:-11.02 (0.01SD) | 0 | EDH | 5390  |
| 39 | 25  | 1    | Inert         | lgK:-10.28 (4SF)    | 0 | EOD | 5613  |
| 40 | 25  | 2    | Inert         | lgK:-10.00 (4SF)    | 0 | EOD | 5613  |
| 41 | 25  | 3    | Inert         | lgK:-9.59 (3SF)     | 0 | EOD | 5613  |
| 42 | 25  | 5    | Inert         | lgK:-8.67 (3SF)     | 0 | EOD | 5613  |
| 43 | 37  | 0.15 | Blood Plasma  | lgK:-9.5 (0.05SD)   | 0 | ENS | 94    |
| 44 | 90  | 0    | Inf. Dilution | lgK:-7.360 (4SF)    | 4 | EOD | 29559 |
| 45 | 150 | 0    | Inf. Dilution | lgK:-5.045 (4SF)    | 4 | EOD | 29559 |
| 46 | 150 |      |               | lgK:-4.8 (2SF)      | 0 | MSL | 4281  |
| 47 | 200 |      |               | lgK:-3.4 (2SF)      | 0 | MSL | 4281  |
| 48 | 210 | 0    | Inf. Dilution | lgK:-3.264 (4SF)    | 4 | EOD | 29559 |
| 49 | 250 |      |               | lgK:-1.7 (2SF)      | 0 | MSL | 4281  |
| 50 | 25  | 0    | Inf. Dilution | dH:26.9 (0.1SD)     | 0 | ENS | 5484  |
| 51 | 25  | 0    | Inf. Dilution | dH:26.90 (4SF)      | 0 | ENS | 6496  |
| 52 | 25  | 0    | Inf. Dilution | dH:96.7 (6.8SD) kJ  | 3 | EDH | 30072 |
| 53 | 25  | 0    | Unknown       | dH:0.0 (1SF)        | 0 | ENS | 7373  |
| 54 | 25  | 0.7  | Seawater      | dH:19.8 (3SF)       | 0 | ENS | 6496  |
| 55 | 25  | 0.7  | Seawater      | dH:19.8 (3SF)       | 0 | EDT | 9100  |

| Reaction No. 2872                     |     |       |                        |                    |   |       |        |
|---------------------------------------|-----|-------|------------------------|--------------------|---|-------|--------|
| Al+3 + 3<H2O> = Al+3_OH-1(3) + 3<H+1> |     |       |                        |                    |   |       |        |
| No                                    | t   | I-Str | Medium                 | Value (Dev)        | W | Tech. | Ref. # |
| 1                                     | 200 | 0     | P=300<br>Inf. Dilution | lgK:-7.301(4SF)    | 4 | EOD   | 29559  |
| 2                                     | 25  | 0     | Inf. Dilution          | lgK:-17.2(3SF)     | 1 | ELC   |        |
| 3                                     | 25  | 0     | Inf. Dilution          | lgK:-16.3(3SF)     | 0 | ENS   | 2171   |
| 4                                     | 25  | 0     | Inf. Dilution          | lgK:-16.9(0.1SD)   | 0 | ENS   | 5484   |
| 5                                     | 25  | 0     | Inf. Dilution          | lgK:-15.60(4SF)    | 0 | CRV   | 5623   |
| 6                                     | 25  | 0     | Inf. Dilution          | lgK:-15.2(3SF)     | 0 | EOD   | 5629   |
| 7                                     | 25  | 0     | Inf. Dilution          | lgK:-16.9(3SF)     | 0 | ENS   | 6496   |
| 8                                     | 25  | 0     | Inf. Dilution          | lgK:-17.044(5SF)   | 2 | EOD   | 29559  |
| 9                                     | 25  | 0     | Inf. Dilution          | lgK:-17.1(0.2SD)   | 4 | EDH   | 30072  |
| 10                                    | 25  | 0     | CHEMVAL1               | lgK:-15.00(0.01SD) | 0 | ENS   | 5156   |
| 11                                    | 25  | 0     | CHEMVAL2               | lgK:-15.00(0.01SD) | 0 | ENS   | 5156   |
| 12                                    | 25  | 0     | CHEMVAL3               | lgK:-15.90(0.01SD) | 0 | ENS   | 5156   |
| 13                                    | 25  | 0     | CHEMVAL4               | lgK:-15.90(0.01SD) | 0 | ENS   | 5156   |
| 14                                    | 25  | 0     | Inert                  | lgK:-15.00(4SF)    | 0 | EOD   | 5613   |
| 15                                    | 25  | 0     | Many Media             | lgK:-16.8(3SF)     | 0 | CRV   | 2171   |
| 16                                    | 25  | 0     | Many Media             | lgK:-16.8(3SF)     | 0 | EOD   | 6953   |
| 17                                    | 25  | 0     | UCT_Sea                | lgK:-16.30(0.01SD) | 0 | EDH   | 5390   |
| 18                                    | 25  | 0     | Unknown                | lgK:-16.0(3SF)     | 0 | ENS   | 7373   |
| 19                                    | 25  | 0.01  | NaClO4                 | lgK:-17.1(3SF)     | 0 | MMR   | 7045   |
| 20                                    | 25  | 0.1   | Inert                  | lgK:-15.69(4SF)    | 0 | EOD   | 5613   |
| 21                                    | 25  | 0.1   | NaCl                   | lgK:-17.77(4SF)    | 5 | MGL   | 29939  |
| 22                                    | 25  | 0.1   | UCT_Sea                | lgK:-16.87(0.01SD) | 0 | EDH   | 5390   |
| 23                                    | 25  | 0.2   | Inert                  | lgK:-15.86(4SF)    | 0 | EOD   | 5613   |
| 24                                    | 25  | 0.2   | UCT_Sea                | lgK:-17.04(0.01SD) | 0 | EDH   | 5390   |
| 25                                    | 25  | 0.25  | NaCl                   | lgK:-17.94(4SF)    | 5 | MGL   | 29939  |
| 26                                    | 25  | 0.3   | Inert                  | lgK:-15.95(4SF)    | 0 | EOD   | 5613   |
| 27                                    | 25  | 0.3   | UCT_Sea                | lgK:-17.18(0.01SD) | 0 | EDH   | 5390   |
| 28                                    | 25  | 0.4   | UCT_Sea                | lgK:-17.29(0.01SD) | 0 | EDH   | 5390   |
| 29                                    | 25  | 0.5   | Inert                  | lgK:-16.04(4SF)    | 0 | EOD   | 5613   |
| 30                                    | 25  | 0.5   | NaCl                   | lgK:-18.04(4SF)    | 5 | MGL   | 29939  |
| Note (30): sign changed               |     |       |                        |                    |   |       |        |
| 31                                    | 25  | 0.5   | UCT_Sea                | lgK:-17.32(0.01SD) | 0 | EDH   | 5390   |

|    |     |     |               |                    |   |     |       |
|----|-----|-----|---------------|--------------------|---|-----|-------|
| 32 | 25  | 0.6 | NaCl          | lgK:-17.3(0.05SD)  | 0 | MPS | 5075  |
| 33 | 25  | 0.6 | UCT_Sea       | lgK:-17.35(0.01SD) | 0 | EDH | 5390  |
| 34 | 25  | 0.7 | NaCl          | lgK:-18.08(4SF)    | 5 | MGL | 29939 |
| 35 | 25  | 0.7 | Seawater      | lgK:-15.83(4SF)    | 0 | ENS | 6496  |
| 36 | 25  | 0.7 | Seawater      | lgK:-15.83(4SF)    | 0 | EDT | 9100  |
| 37 | 25  | 0.7 | UCT_Sea       | lgK:-17.37(0.01SD) | 0 | EDH | 5390  |
| 38 | 25  | 1   | Inert         | lgK:-16.08(4SF)    | 0 | EOD | 5613  |
| 39 | 25  | 2   | Inert         | lgK:-15.90(4SF)    | 0 | EOD | 5613  |
| 40 | 25  | 3   | Inert         | lgK:-15.59(4SF)    | 0 | EOD | 5613  |
| 41 | 25  | 5   | Inert         | lgK:-14.87(4SF)    | 0 | EOD | 5613  |
| 42 | 90  | 0   | Inf. Dilution | lgK:-12.252(5SF)   | 2 | EOD | 29559 |
| 43 | 150 | 0   | Inf. Dilution | lgK:-9.168(4SF)    | 2 | EOD | 29559 |
| 44 | 150 |     |               | lgK:-7.8(2SF)      | 0 | MSL | 4281  |
| 45 | 200 |     |               | lgK:-6.1(2SF)      | 0 | MSL | 4281  |
| 46 | 210 | 0   | Inf. Dilution | lgK:-6.823(4SF)    | 2 | EOD | 29559 |
| 47 | 250 |     |               | lgK:-4.3(2SF)      | 0 | MSL | 4281  |
| 48 | 25  | 0   | Inf. Dilution | dH:39.9(0.1SD)     | 0 | ENS | 5484  |
| 49 | 25  | 0   | Inf. Dilution | dH:39.89(4SF)      | 0 | ENS | 6496  |
| 50 | 25  | 0   | Inf. Dilution | dH:172.0(7.0SD) kJ | 3 | EDH | 30072 |
| 51 | 25  | 0   | Unknown       | dH:0.0(1SF)        | 0 | ENS | 7373  |
| 52 | 25  | 0.7 | Seawater      | dH:25.9(3SF)       | 0 | ENS | 6496  |
| 53 | 25  | 0.7 | Seawater      | dH:25.9(3SF)       | 0 | EDT | 9100  |

| Reaction No. 2873                     |     |       |                        |                    |   |       |        |
|---------------------------------------|-----|-------|------------------------|--------------------|---|-------|--------|
| Al+3 + 4<H2O> = Al+3_OH-1(4) + 4<H+1> |     |       |                        |                    |   |       |        |
| No                                    | t   | I-Str | Medium                 | Value (Dev)        | W | Tech. | Ref. # |
| 1                                     | 200 | 0     | P=300<br>Inf. Dilution | lgK:-11.572(5SF)   | 4 | EOD   | 29559  |
| 2                                     | 25  | 0     | Inf. Dilution          | lgK:-22.8(3SF)     | 1 | ELC   |        |
| 3                                     | 25  | 0     | Inf. Dilution          | lgK:-22.7(0.6SD)   | 0 | ENS   | 2171   |
| 4                                     | 25  | 0     | Inf. Dilution          | lgK:-22.7(0.1SD)   | 0 | ENS   | 5484   |
| 5                                     | 25  | 0     | Inf. Dilution          | lgK:-22.851(5SF)   | 2 | EOD   | 29559  |
| 6                                     | 25  | 0     | Inf. Dilution          | lgK:-22.83(0.10SD) | 4 | EDH   | 30072  |
| 7                                     | 25  | 0     | Inf. Dilution          | lgK:-22.10(4SF)    | 0 | CRV   | 32224  |
| 8                                     | 25  | 0     | CHEMVAL1               | lgK:-23.00(0.01SD) | 0 | ENS   | 5156   |
| 9                                     | 25  | 0     | CHEMVAL2               | lgK:-23.00(0.01SD) | 0 | ENS   | 5156   |
| 10                                    | 25  | 0     | CHEMVAL3               | lgK:-22.90(0.01SD) | 0 | ENS   | 5156   |

|    |     |      |               |                     |   |     |       |
|----|-----|------|---------------|---------------------|---|-----|-------|
| 11 | 25  | 0    | CHEMVAL4      | lgK:-22.90 (0.01SD) | 0 | ENS | 5156  |
| 12 | 25  | 0    | Inert         | lgK:-23.12 (4SF)    | 0 | EOD | 5613  |
| 13 | 25  | 0    | Many Media    | lgK:-22.7 (3SF)     | 0 | CRV | 2171  |
| 14 | 25  | 0    | Many Media    | lgK:-23.0 (0.06SD)  | 0 | EOD | 6953  |
| 15 | 25  | 0    | UCT_Sea       | lgK:-23.00 (0.01SD) | 0 | EDH | 5390  |
| 16 | 25  | 0    | Unknown       | lgK:-23.0 (3SF)     | 0 | ENS | 7373  |
| 17 | 25  | 0.01 | NaClO4        | lgK:-23.0 (3SF)     | 0 | MMR | 7045  |
| 18 | 25  | 0.1  | Inert         | lgK:-23.58 (4SF)    | 0 | EOD | 5613  |
| 19 | 25  | 0.1  | KNO3          | lgK:-22.20 (4SF)    | 0 | MDS | 5625  |
| 20 | 25  | 0.1  | NaCl          | lgK:-23.25 (4SF)    | 5 | MGL | 29939 |
| 21 | 25  | 0.1  | UCT_Sea       | lgK:-23.46 (0.01SD) | 0 | EDH | 5390  |
| 22 | 25  | 0.2  | Inert         | lgK:-23.68 (4SF)    | 0 | EOD | 5613  |
| 23 | 25  | 0.2  | UCT_Sea       | lgK:-23.56 (0.01SD) | 0 | EDH | 5390  |
| 24 | 25  | 0.25 | NaCl          | lgK:-23.38 (4SF)    | 5 | MGL | 29939 |
| 25 | 25  | 0.3  | Inert         | lgK:-23.74 (4SF)    | 0 | EOD | 5613  |
| 26 | 25  | 0.3  | UCT_Sea       | lgK:-23.62 (0.01SD) | 0 | EDH | 5390  |
| 27 | 25  | 0.4  | UCT_Sea       | lgK:-23.65 (0.01SD) | 0 | EDH | 5390  |
| 28 | 25  | 0.5  | Inert         | lgK:-23.79 (4SF)    | 0 | EOD | 5613  |
| 29 | 25  | 0.5  | NaCl          | lgK:-23.48 (4SF)    | 5 | MGL | 29939 |
| 30 | 25  | 0.5  | UCT_Sea       | lgK:-23.66 (0.01SD) | 0 | EDH | 5390  |
| 31 | 25  | 0.6  | NaCl          | lgK:-23.5 (0.1SD)   | 0 | MPS | 5075  |
| 32 | 25  | 0.6  | UCT_Sea       | lgK:-23.67 (0.01SD) | 0 | EDH | 5390  |
| 33 | 25  | 0.7  | NaCl          | lgK:-23.53 (4SF)    | 5 | MGL | 29939 |
| 34 | 25  | 0.7  | Seawater      | lgK:-23.53 (4SF)    | 0 | ENS | 6496  |
| 35 | 25  | 0.7  | Seawater      | lgK:-23.53 (4SF)    | 0 | EDT | 9100  |
| 36 | 25  | 0.7  | UCT_Sea       | lgK:-23.68 (0.01SD) | 0 | EDH | 5390  |
| 37 | 25  | 1    | Inert         | lgK:-23.78 (4SF)    | 0 | EOD | 5613  |
| 38 | 25  | 2    | Inert         | lgK:-23.60 (4SF)    | 0 | EOD | 5613  |
| 39 | 25  | 3    | Inert         | lgK:-23.34 (4SF)    | 0 | EOD | 5613  |
| 40 | 25  | 5    | Inert         | lgK:-22.73 (4SF)    | 0 | EOD | 5613  |
| 41 | 37  | 0.15 | Blood Plasma  | lgK:-23.9 (0.05SD)  | 0 | ENS | 94    |
| 42 | 90  | 0    | Inf. Dilution | lgK:-17.299 (5SF)   | 3 | EOD | 29559 |
| 43 | 150 | 0    | Inf. Dilution | lgK:-13.747 (5SF)   | 3 | EOD | 29559 |
| 44 | 150 |      |               | lgK:-13.6 (3SF)     | 0 | MSL | 4281  |
| 45 | 200 |      |               | lgK:-11.5 (3SF)     | 0 | MSL | 4281  |
| 46 | 210 | 0    | Inf. Dilution | lgK:-11.086 (5SF)   | 4 | EOD | 29559 |

|    |     |     |               |                     |   |     |       |
|----|-----|-----|---------------|---------------------|---|-----|-------|
| 47 | 250 |     |               | lgK:-9.5 (2SF)      | 0 | MSL | 4281  |
| 48 | 25  | 0   | Inf. Dilution | dH:42.3 (0.1SD)     | 0 | ENS | 5484  |
| 49 | 25  | 0   | Inf. Dilution | dH:42.30 (4SF)      | 0 | ENS | 6496  |
| 50 | 25  | 0   | Inf. Dilution | dH:179.9 (1.2SD) kJ | 3 | EDH | 30072 |
| 51 | 25  | 0   | Unknown       | dH:44.06 (4SF)      | 0 | ENS | 7373  |
| 52 | 25  | 0.1 | KNO3          | dH:42.45 (4SF)      | 0 | MTD | 5625  |
| 53 | 25  | 0.7 | Seawater      | dH:31.5 (3SF)       | 0 | ENS | 6496  |
| 54 | 25  | 0.7 | Seawater      | dH:31.5 (3SF)       | 0 | EDT | 9100  |

| Reaction No. 2874                             |    |       |               |                    |   |       |        |
|-----------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| 2<Al+3> + 2<H2O> = Al+3 (2)_OH-1 (2) + 2<H+1> |    |       |               |                    |   |       |        |
| No                                            | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                             | 25 | 0     | Inf. Dilution | lgK:-7.7 (0.05SD)  | 3 | CMN   | 2171   |
| 2                                             | 25 | 0     | Inf. Dilution | lgK:-6.27 (0.02SD) | 0 | CRV   | 5627   |
| 3                                             | 25 | 0     | Inf. Dilution | lgK:-7.55 (0.02SD) | 2 | CRV   | 5627   |
| 4                                             | 25 | 0     | Inf. Dilution | lgK:-6.95 (3SF)    | 0 | CRV   | 5629   |
| 5                                             | 25 | 0     | Inf. Dilution | lgK:-7.8 (0.6SD)   | 4 | EDH   | 30072  |
| 6                                             | 25 | 0     | CHEMVAL1      | lgK:-7.70 (0.01SD) | 0 | ENS   | 5156   |
| 7                                             | 25 | 0     | CHEMVAL2      | lgK:-7.70 (0.01SD) | 0 | ENS   | 5156   |
| 8                                             | 25 | 0     | CHEMVAL3      | lgK:-7.70 (0.01SD) | 0 | ENS   | 5156   |
| 9                                             | 25 | 0     | CHEMVAL4      | lgK:-7.70 (0.01SD) | 4 | ENS   | 5156   |
| 10                                            | 25 | 0     | Inert         | lgK:-7.70 (3SF)    | 4 | EOD   | 5613   |
| 11                                            | 25 | 0     | UCT_Sea       | lgK:-7.70 (0.01SD) | 4 | EDH   | 5390   |
| 12                                            | 25 | 0     | UCT_Sea       | lgK:-7.70 (0.01SD) | 4 | EDT   | 5390   |
| 13                                            | 25 | 0.06  | Cl-1 salt     | lgK:-7.42 (3SF)    | 0 | MLC   | 5588   |
| 14                                            | 25 | 0.1   | Inert         | lgK:-7.70 (3SF)    | 4 | EOD   | 5613   |
| 15                                            | 25 | 0.1   | NaCl          | lgK:-7.75 (3SF)    | 5 | MGL   | 29939  |
| 16                                            | 25 | 0.1   | UCT_Sea       | lgK:-7.78 (0.01SD) | 4 | EDH   | 5390   |
| 17                                            | 25 | 0.1   | UCT_Sea       | lgK:-7.78 (0.01SD) | 4 | EDT   | 5390   |
| 18                                            | 25 | 0.12  | Ba (NO3) 2    | lgK:-8.06 (0.02SD) | 0 | CRV   | 5627   |
| 19                                            | 25 | 0.2   | UCT_Sea       | lgK:-7.82 (0.01SD) | 4 | EDH   | 5390   |
| 20                                            | 25 | 0.2   | UCT_Sea       | lgK:-7.82 (0.01SD) | 4 | EDT   | 5390   |
| 21                                            | 25 | 0.25  | NaCl          | lgK:-7.74 (3SF)    | 5 | MGL   | 29939  |
| 22                                            | 25 | 0.3   | UCT_Sea       | lgK:-7.85 (0.01SD) | 4 | EDH   | 5390   |
| 23                                            | 25 | 0.3   | UCT_Sea       | lgK:-7.85 (0.01SD) | 4 | EDT   | 5390   |
| 24                                            | 25 | 0.4   | UCT_Sea       | lgK:-7.88 (0.01SD) | 4 | EDH   | 5390   |

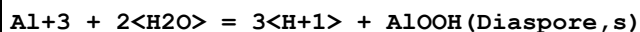
|    |    |     |               |                     |   |     |       |
|----|----|-----|---------------|---------------------|---|-----|-------|
| 25 | 25 | 0.4 | UCT_Sea       | lgK:-7.88 (0.01SD)  | 4 | EDT | 5390  |
| 26 | 25 | 0.5 | NaCl          | lgK:-7.72 (3SF)     | 5 | MGL | 29939 |
| 27 | 25 | 0.5 | UCT_Sea       | lgK:-7.90 (0.01SD)  | 4 | EDH | 5390  |
| 28 | 25 | 0.5 | UCT_Sea       | lgK:-7.90 (0.01SD)  | 4 | EDT | 5390  |
| 29 | 25 | 0.6 | Ba (NO3) 2    | lgK:-8.24 (0.02SD)  | 0 | CRV | 5627  |
| 30 | 25 | 0.6 | UCT_Sea       | lgK:-7.92 (0.01SD)  | 3 | EDH | 5390  |
| 31 | 25 | 0.6 | UCT_Sea       | lgK:-7.92 (0.01SD)  | 3 | EDT | 5390  |
| 32 | 25 | 0.7 | NaCl          | lgK:-7.71 (3SF)     | 5 | MGL | 29939 |
| 33 | 25 | 0.7 | UCT_Sea       | lgK:-7.94 (0.01SD)  | 2 | EDH | 5390  |
| 34 | 25 | 0.7 | UCT_Sea       | lgK:-7.94 (0.01SD)  | 2 | EDT | 5390  |
| 35 | 25 | 1   | Inert         | lgK:-7.70 (3SF)     | 3 | EOD | 5613  |
| 36 | 25 | 1   | NaClO4        | lgK:-8.0 (0.05SD)   | 2 | ENS | 4998  |
| 37 | 25 | 1   | NaClO4        | lgK:-7.1 (0.06SD)   | 0 | CRV | 5627  |
| 38 | 25 | 3   | Inert         | lgK:-7.7 (2SF)      | 3 | EOD | 5613  |
| 39 | 25 | 3   | NaCl          | lgK:-7.5 (0.2SD)    | 3 | MGL | 5061  |
| 40 | 25 | 3   | NaClO4        | lgK:-7.1 (0.1SD)    | 0 | MGL | 5485  |
| 41 | 25 | 3   | NaNO3         | lgK:-7.6 (0.2SD)    | 3 | MGL | 5060  |
| 42 | 25 | 0   | Inf. Dilution | dH:77.7 (20.4SD) kJ | 3 | EDH | 30072 |
| 43 | 25 | 1   | Unknown       | dH:19. (2.SD)       | 5 | ENS | 4998  |

| Reaction No. 2875                             |    |       |               |                     |   |       |        |
|-----------------------------------------------|----|-------|---------------|---------------------|---|-------|--------|
| 3<Al+3> + 4<H2O> = Al+3 (3)_OH-1 (4) + 4<H+1> |    |       |               |                     |   |       |        |
| No                                            | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                             | 10 | 1     | NaNO3         | lgK:-14.98 (0.14SD) | 2 | MGL   | 30097  |
| 2                                             | 25 | 0     | Inf. Dilution | lgK:-13.9 (0.05SD)  | 4 | ENS   | 4998   |
| 3                                             | 25 | 0     | Inf. Dilution | lgK:-13.9 (0.2SD)   | 4 | EDH   | 30072  |
| 4                                             | 25 | 0     | CHEMVAL1      | lgK:-13.90 (0.01SD) | 0 | ENS   | 5156   |
| 5                                             | 25 | 0     | CHEMVAL2      | lgK:-13.90 (0.01SD) | 0 | ENS   | 5156   |
| 6                                             | 25 | 0     | CHEMVAL3      | lgK:-13.84 (0.01SD) | 0 | ENS   | 5156   |
| 7                                             | 25 | 0     | CHEMVAL4      | lgK:-13.80 (0.01SD) | 3 | ENS   | 5156   |
| 8                                             | 25 | 0     | Inert         | lgK:-13.94 (4SF)    | 5 | EOD   | 5613   |
| 9                                             | 25 | 0     | UCT_Sea       | lgK:-13.94 (0.01SD) | 4 | EDH   | 5390   |
| 10                                            | 25 | 0     | UCT_Sea       | lgK:-13.94 (0.01SD) | 4 | EDT   | 5390   |
| 11                                            | 25 | 0.1   | Inert         | lgK:-13.69 (4SF)    | 5 | EOD   | 5613   |
| 12                                            | 25 | 0.1   | NaCl          | lgK:-13.7 (0.05SD)  | 4 | ENS   | 5494   |
| 13                                            | 25 | 0.1   | NaCl          | lgK:-13.69 (4SF)    | 5 | MGL   | 29939  |



|    |    |      |               |                     |   |     |       |
|----|----|------|---------------|---------------------|---|-----|-------|
| 14 | 25 | 0.1  | NaNO3         | lgK:-13.13(0.005SD) | 0 | MPT | 4878  |
| 15 | 25 | 0.1  | UCT_Sea       | lgK:-13.69(0.01SD)  | 4 | EDH | 5390  |
| 16 | 25 | 0.1  | UCT_Sea       | lgK:-13.69(0.01SD)  | 4 | EDT | 5390  |
| 17 | 25 | 0.2  | UCT_Sea       | lgK:-13.62(0.01SD)  | 4 | EDH | 5390  |
| 18 | 25 | 0.2  | UCT_Sea       | lgK:-13.62(0.01SD)  | 4 | EDT | 5390  |
| 19 | 25 | 0.25 | NaCl          | lgK:-13.63(4SF)     | 5 | MGL | 29939 |
| 20 | 25 | 0.3  | UCT_Sea       | lgK:-13.58(0.01SD)  | 4 | EDH | 5390  |
| 21 | 25 | 0.3  | UCT_Sea       | lgK:-13.58(0.01SD)  | 4 | EDT | 5390  |
| 22 | 25 | 0.4  | UCT_Sea       | lgK:-13.54(0.01SD)  | 4 | EDH | 5390  |
| 23 | 25 | 0.4  | UCT_Sea       | lgK:-13.54(0.01SD)  | 4 | EDT | 5390  |
| 24 | 25 | 0.5  | Inert         | lgK:-13.52(4SF)     | 4 | EOD | 5613  |
| 25 | 25 | 0.5  | NaCl          | lgK:-13.58(4SF)     | 5 | MGL | 29939 |
| 26 | 25 | 0.5  | UCT_Sea       | lgK:-13.52(0.01SD)  | 4 | EDH | 5390  |
| 27 | 25 | 0.5  | UCT_Sea       | lgK:-13.52(0.01SD)  | 4 | EDT | 5390  |
| 28 | 25 | 0.6  | NaCl          | lgK:-13.57(0.02SD)  | 6 | MGL | 5447  |
| 29 | 25 | 0.6  | UCT_Sea       | lgK:-13.49(0.01SD)  | 4 | EDH | 5390  |
| 30 | 25 | 0.6  | UCT_Sea       | lgK:-13.49(0.01SD)  | 4 | EDT | 5390  |
| 31 | 25 | 0.7  | NaCl          | lgK:-13.56(4SF)     | 5 | MGL | 29939 |
| 32 | 25 | 0.7  | UCT_Sea       | lgK:-13.47(0.01SD)  | 4 | EDH | 5390  |
| 33 | 25 | 0.7  | UCT_Sea       | lgK:-13.47(0.01SD)  | 4 | EDT | 5390  |
| 34 | 25 | 1    | Inert         | lgK:-13.43(4SF)     | 4 | EOD | 5613  |
| 35 | 25 | 1    | NaClO4        | lgK:-13.5(0.05SD)   | 4 | ENS | 4998  |
| 36 | 25 | 1    | NaNO3         | lgK:-13.71(0.03SD)  | 2 | MGL | 30097 |
| 37 | 25 | 3    | Inert         | lgK:-13.29(4SF)     | 4 | EOD | 5613  |
| 38 | 25 | 3    | Na (Cl)       | lgK:-13.96(0.01SD)  | 0 | MGL | 5494  |
| 39 | 25 | 3    | NaCl          | lgK:-13.4(0.05SD)   | 2 | MGL | 5061  |
| 40 | 25 | 3    | NaClO4        | lgK:-13.1(0.1SD)    | 2 | MGL | 5485  |
| 41 | 25 | 3    | NaNO3         | lgK:-13.2(0.08SD)   | 3 | MGL | 5060  |
| 42 | 50 | 1    | NaNO3         | lgK:-11.57(0.04SD)  | 2 | MGL | 30097 |
| 43 | 70 | 1    | NaNO3         | lgK:-10.20(0.08SD)  | 2 | MGL | 30097 |
| 44 | 25 | 0    | Inf. Dilution | dH:149.5(1.1SD) kJ  | 2 | EDH | 30072 |
| 45 | 25 | 1    | Unknown       | dH:40. (4. SD)      | 0 | ENS | 4998  |

## Reaction No. 2876



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

|                                                          |    |   |               |                   |   |     |       |
|----------------------------------------------------------|----|---|---------------|-------------------|---|-----|-------|
| 1                                                        | 25 | 0 | Inf. Dilution | lgK:-7.9(2SF)     | 1 | ELC |       |
| 2                                                        | 25 | 0 | Inf. Dilution | lgK:-8.922(4SF)   | 0 | MSL | 440   |
| 3                                                        | 25 | 0 | Inf. Dilution | lgK:-7.9(0.05SD)  | 0 | ENS | 557   |
| 4                                                        | 25 | 0 | Inf. Dilution | lgK:-6.75(0.24SD) | 0 | MSL | 5617  |
| 5                                                        | 25 | 0 | Inf. Dilution | lgK:-7.318(4SF)   | 0 | MSL | 5619  |
| 6                                                        | 25 | 0 | Inf. Dilution | lgK:-7.239(4SF)   | 0 | MSL | 5627  |
| 7                                                        | 25 | 0 | Inf. Dilution | lgK:-7.92(3SF)    | 0 | MGL | 6498  |
| 8                                                        | 25 | 0 | Inf. Dilution | lgK:-8.949(4SF)   | 0 | CRV | 8872  |
| 9                                                        | 25 | 0 | Inf. Dilution | lgK:-7.890(4SF)   | 0 | EOD | 8872  |
| Note (9): Data from Naumov, Ryzhenko & Khodakovsky, 1971 |    |   |               |                   |   |     |       |
| 10                                                       | 25 | 0 | Inf. Dilution | lgK:-7.528(4SF)   | 0 | EOD | 8872  |
| Note (10): Data from Khodakovsky, 1983                   |    |   |               |                   |   |     |       |
| 11                                                       | 25 | 0 | Inf. Dilution | lgK:-7.169(4SF)   | 0 | MSL | 8873  |
| 12                                                       | 25 | 0 | Inf. Dilution | lgK:-7.570(4SF)   | 0 | MSL | 8874  |
| 13                                                       | 25 | 0 | Inf. Dilution | lgK:-8.764(4SF)   | 0 | CRV | 32224 |
| 14                                                       | 25 | 0 | CHEMVAL1      | lgK:-8.66(0.01SD) | 0 | ENS | 5156  |
| 15                                                       | 25 | 0 | CHEMVAL2      | lgK:-8.66(0.01SD) | 0 | ENS | 5156  |
| 16                                                       | 25 | 0 | CHEMVAL3      | lgK:-8.66(0.01SD) | 0 | ENS | 5156  |
| 17                                                       | 25 | 0 | CHEMVAL4      | lgK:-7.93(0.01SD) | 0 | ENS | 5156  |
| 18                                                       | 25 | 0 | Inf. Dilution | dH:24.6(0.1SD)    | 2 | ENS | 766   |

## Reaction No. 2877

Al+3\_Na+1\_OH-1(2)\_Oxalic-2\_(s) + 2<Al+3\_OH-1(3)\_(am.,s)> + H+1 = Na+1 + H2O + Al+3(3)\_OH-1(7)\_Oxalic-2\_(s)

| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:6.54(3SF) | 4 | ENS   | 5982   |

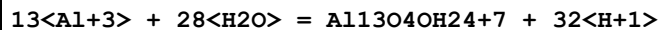
## Reaction No. 2878

Al+3 + 2<H2O> = 3<H+1> + AlOOH(Boehmite,s)

| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-8.1(2SF)     | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:-8.1(0.05SD)  | 0 | ENS   | 557    |
| 3  | 25 | 0     | Inf. Dilution | lgK:-7.49(0.09SD) | 0 | MSL   | 5617   |
| 4  | 25 | 0     | Inf. Dilution | lgK:-7.931(4SF)   | 0 | MSL   | 5619   |
| 5  | 25 | 0     | Inf. Dilution | lgK:-7.826(4SF)   | 0 | MSL   | 5627   |
| 6  | 25 | 0     | Inf. Dilution | lgK:-8.02(3SF)    | 0 | ENS   | 6333   |
| 7  | 25 | 0     | Inf. Dilution | lgK:-8.13(3SF)    | 0 | MGL   | 6498   |

|                                       |     |   |               |                   |   |     |       |
|---------------------------------------|-----|---|---------------|-------------------|---|-----|-------|
| 8                                     | 25  | 0 | Inf. Dilution | lgK:-8.205(4SF)   | 0 | CRV | 8872  |
| 9                                     | 25  | 0 | Inf. Dilution | lgK:-8.474(4SF)   | 0 | EOD | 8872  |
| Note (9): Data from Khodakovsky, 1983 |     |   |               |                   |   |     |       |
| 10                                    | 25  | 0 | Inf. Dilution | lgK:-9.180(4SF)   | 0 | MSL | 8874  |
| 11                                    | 25  | 0 | Inf. Dilution | lgK:-9.609(4SF)   | 0 | CRV | 32224 |
| 12                                    | 25  | 0 | CHEMVAL1      | lgK:-8.76(0.01SD) | 0 | ENS | 5156  |
| 13                                    | 25  | 0 | CHEMVAL2      | lgK:-8.76(0.01SD) | 0 | ENS | 5156  |
| 14                                    | 25  | 0 | CHEMVAL3      | lgK:-8.76(0.01SD) | 0 | ENS | 5156  |
| 15                                    | 25  | 0 | CHEMVAL4      | lgK:-7.30(0.01SD) | 0 | ENS | 5156  |
| 16                                    | 50  | 0 | Inf. Dilution | lgK:-7.64(3SF)    | 0 | ENS | 6333  |
| 17                                    | 60  | 0 | Inf. Dilution | lgK:-7.51(3SF)    | 0 | ENS | 6333  |
| 18                                    | 70  | 0 | Inf. Dilution | lgK:-7.38(3SF)    | 0 | ENS | 6333  |
| 19                                    | 75  | 0 | Inf. Dilution | lgK:-7.32(3SF)    | 0 | ENS | 6333  |
| 20                                    | 80  | 0 | Inf. Dilution | lgK:-7.26(3SF)    | 0 | ENS | 6333  |
| 21                                    | 90  | 0 | Inf. Dilution | lgK:-7.15(3SF)    | 0 | ENS | 6333  |
| 22                                    | 100 | 0 | Inf. Dilution | lgK:-7.09(3SF)    | 0 | ENS | 6333  |
| 23                                    | 125 | 0 | Inf. Dilution | lgK:-6.96(3SF)    | 0 | ENS | 6333  |
| 24                                    | 150 | 0 | Inf. Dilution | lgK:-6.84(3SF)    | 0 | ENS | 6333  |
| 25                                    | 150 |   |               | lgK:-1.53(3SF)    | 0 | MSL | 4281  |
| 26                                    | 170 | 0 | Inf. Dilution | lgK:-6.76(3SF)    | 0 | ENS | 6333  |
| 27                                    | 200 | 0 | Inf. Dilution | lgK:-6.65(3SF)    | 0 | ENS | 6333  |
| 28                                    | 200 |   |               | lgK:-0.05(1SF)    | 0 | MSL | 4281  |
| 29                                    | 250 | 0 | Inf. Dilution | lgK:-6.56(3SF)    | 0 | ENS | 6333  |
| 30                                    | 250 |   |               | lgK:1.67(3SF)     | 0 | MSL | 4281  |
| 31                                    | 300 | 0 | Inf. Dilution | lgK:-6.35(3SF)    | 0 | ENS | 6333  |
| 32                                    | 325 | 0 | Inf. Dilution | lgK:-6.22(3SF)    | 0 | ENS | 6333  |
| 33                                    | 350 | 0 | Inf. Dilution | lgK:-6.10(3SF)    | 0 | ENS | 6333  |
| 34                                    | 25  | 0 | Inf. Dilution | dH:24.6(0.1SD)    | 0 | ENS | 766   |

## Reaction No. 2879



| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-98.7(0.05SD) | 0 | ENS   | 4998   |
| 2  | 25 | 0     | Inert         | lgK:-98.73(4SF)   | 0 | EOD   | 5613   |
| 3  | 25 | 0.1   | Inert         | lgK:-102.8(4SF)   | 0 | EOD   | 5613   |
| 4  | 25 | 0.2   | Inert         | lgK:-103.7(4SF)   | 0 | EOD   | 5613   |

|                                            |    |     |               |                     |   |     |      |
|--------------------------------------------|----|-----|---------------|---------------------|---|-----|------|
| 5                                          | 25 | 0.3 | Inert         | lgK:-104.18 (5SF)   | 0 | EOD | 5613 |
| 6                                          | 25 | 0.5 | Inert         | lgK:-104.6 (4SF)    | 2 | EOD | 5613 |
| 7                                          | 25 | 0.6 | NaCl          | lgK:-109.2 (0.12SD) | 0 | MGL | 5447 |
| Note (7): Low wgt for internal consistency |    |     |               |                     |   |     |      |
| 8                                          | 25 | 1   | Inert         | lgK:-104.4 (4SF)    | 2 | EOD | 5613 |
| 9                                          | 25 | 1   | NaClO4        | lgK:-104.8 (0.05SD) | 0 | ENS | 4998 |
| 10                                         | 25 | 2   | Inert         | lgK:-102.4 (4SF)    | 2 | EOD | 5613 |
| 11                                         | 25 | 3   | Inert         | lgK:-99.75 (4SF)    | 2 | EOD | 5613 |
| 12                                         | 25 | 5   | Inert         | lgK:-93.7 (3SF)     | 2 | EOD | 5613 |
| 13                                         | 25 | 0   | Inf. Dilution | dH:279. (2.SD)      | 2 | ENS | 4998 |

| Reaction No. 2882           |    |       |               |                    |   |       |        |
|-----------------------------|----|-------|---------------|--------------------|---|-------|--------|
| 2<F-1> + Al+3 = Al+3_F-1(2) |    |       |               |                    |   |       |        |
| No                          | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                           | 25 | 0     | Inf. Dilution | lgK:12.6 (0.05SD)  | 4 | ENS   | 557    |
| 2                           | 25 | 0     | Inf. Dilution | lgK:12.7 (0.2SD)   | 5 | CMN   | 2171   |
| 3                           | 25 | 0     | Inf. Dilution | lgK:12.75 (4SF)    | 4 | EMU   | 5398   |
| 4                           | 25 | 0     | Inf. Dilution | lgK:12.7 (3SF)     | 4 | ENS   | 6496   |
| 5                           | 25 | 0     | Inf. Dilution | lgK:12.63 (0.02SD) | 3 | EFE   | 30072  |
| 6                           | 25 | 0     | Inf. Dilution | lgK:12.75 (4SF)    | 4 | CRV   | 32224  |
| 7                           | 25 | 0     | CHEMVAL1      | lgK:12.60 (0.01SD) | 0 | ENS   | 5156   |
| 8                           | 25 | 0     | CHEMVAL2      | lgK:12.54 (0.01SD) | 0 | ENS   | 5156   |
| 9                           | 25 | 0     | CHEMVAL3      | lgK:12.60 (0.01SD) | 0 | ENS   | 5156   |
| 10                          | 25 | 0     | CHEMVAL4      | lgK:12.60 (0.01SD) | 6 | ENS   | 5156   |
| 11                          | 25 | 0     | Many Media    | lgK:12.7 (0.2SD)   | 7 | EOD   | 6953   |
| 12                          | 25 | 0     | SOLMNEQ       | lgK:12.60 (4SF)    | 4 | CMN   | 5416   |
| 13                          | 25 | 0     | SOLVEQ        | lgK:12.60 (4SF)    | 4 | CMN   | 5416   |
| 14                          | 25 | 0     | UCT_Sea       | lgK:12.60 (0.01SD) | 4 | EDH   | 5390   |
| 15                          | 25 | 0     | WATCH         | lgK:12.57 (4SF)    | 4 | CMN   | 5416   |
| 16                          | 25 | 0.1   | UCT_Sea       | lgK:11.63 (0.01SD) | 4 | EDH   | 5390   |
| 17                          | 25 | 0.1   | Unknown       | lgK:11.63 (4SF)    | 6 | CRV   | 552    |
| 18                          | 25 | 0.15  | NaCl          | lgK:11.45 (0.02SD) | 3 | MGL   | 30072  |
| 19                          | 25 | 0.2   | UCT_Sea       | lgK:11.45 (0.01SD) | 4 | EDH   | 5390   |
| 20                          | 25 | 0.3   | UCT_Sea       | lgK:11.32 (0.01SD) | 4 | EDH   | 5390   |
| 21                          | 25 | 0.4   | UCT_Sea       | lgK:11.21 (0.01SD) | 4 | EDH   | 5390   |
| 22                          | 25 | 0.5   | UCT_Sea       | lgK:11.12 (0.01SD) | 4 | EDH   | 5390   |

|    |     |     |               |                       |   |     |       |
|----|-----|-----|---------------|-----------------------|---|-----|-------|
| 23 | 25  | 0.5 | Unknown       | lgK:11.13 (4SF)       | 6 | CRV | 552   |
| 24 | 25  | 0.6 | UCT_Sea       | lgK:11.04 (0.01SD)    | 4 | EDH | 5390  |
| 25 | 25  | 0.7 | UCT_Sea       | lgK:10.97 (0.01SD)    | 4 | EDH | 5390  |
| 26 | 25  | 1   | NaCl          | lgK:11.05 (0.03SD)    | 3 | MGL | 30072 |
| 27 | 25  | 1   | Unknown       | lgK:11.0 (3SF)        | 6 | CRV | 552   |
| 28 | 25  |     |               | lgK:9.06 (3SF)        | 0 | MSL | 140   |
| 29 | 50  | 0   | SOLMNEQ       | lgK:12.78 (4SF)       | 4 | CMN | 5416  |
| 30 | 50  | 0   | SOLVEQ        | lgK:12.81 (4SF)       | 4 | CMN | 5416  |
| 31 | 50  | 0   | WATCH         | lgK:12.76 (4SF)       | 4 | CMN | 5416  |
| 32 | 75  | 0   | SOLMNEQ       | lgK:13.06 (4SF)       | 4 | CMN | 5416  |
| 33 | 75  | 0   | WATCH         | lgK:13.05 (4SF)       | 4 | CMN | 5416  |
| 34 | 100 | 0   | SOLMNEQ       | lgK:13.45 (4SF)       | 4 | CMN | 5416  |
| 35 | 100 | 0   | SOLVEQ        | lgK:13.58 (4SF)       | 4 | CMN | 5416  |
| 36 | 100 | 0   | WATCH         | lgK:13.45 (4SF)       | 4 | CMN | 5416  |
| 37 | 125 | 0   | SOLMNEQ       | lgK:13.94 (4SF)       | 4 | CMN | 5416  |
| 38 | 125 | 0   | WATCH         | lgK:13.94 (4SF)       | 4 | CMN | 5416  |
| 39 | 150 | 0   | SOLMNEQ       | lgK:14.55 (4SF)       | 4 | CMN | 5416  |
| 40 | 150 | 0   | SOLVEQ        | lgK:14.77 (4SF)       | 4 | CMN | 5416  |
| 41 | 150 | 0   | WATCH         | lgK:14.51 (4SF)       | 4 | CMN | 5416  |
| 42 | 175 | 0   | WATCH         | lgK:15.16 (4SF)       | 4 | CMN | 5416  |
| 43 | 200 | 0   | SOLMNEQ       | lgK:16.16 (4SF)       | 3 | CMN | 5416  |
| 44 | 200 | 0   | SOLVEQ        | lgK:16.49 (4SF)       | 2 | CMN | 5416  |
| 45 | 200 | 0   | WATCH         | lgK:15.88 (4SF)       | 3 | CMN | 5416  |
| 46 | 225 | 0   | WATCH         | lgK:16.67 (4SF)       | 3 | CMN | 5416  |
| 47 | 250 | 0   | SOLMNEQ       | lgK:18.57 (4SF)       | 0 | CMN | 5416  |
| 48 | 250 | 0   | SOLVEQ        | lgK:19.04 (4SF)       | 0 | CMN | 5416  |
| 49 | 250 | 0   | WATCH         | lgK:17.53 (4SF)       | 3 | CMN | 5416  |
| 50 | 275 | 0   | WATCH         | lgK:18.46 (4SF)       | 3 | CMN | 5416  |
| 51 | 300 | 0   | SOLMNEQ       | lgK:22.70 (4SF)       | 0 | CMN | 5416  |
| 52 | 300 | 0   | SOLVEQ        | lgK:23.37 (4SF)       | 0 | CMN | 5416  |
| 53 | 300 | 0   | WATCH         | lgK:19.45 (4SF)       | 0 | CMN | 5416  |
| 54 | 325 | 0   | WATCH         | lgK:20.50 (4SF)       | 0 | CMN | 5416  |
| 55 | 350 | 0   | SOLMNEQ       | lgK:31.77 (4SF)       | 0 | CMN | 5416  |
| 56 | 350 | 0   | WATCH         | lgK:21.60 (4SF)       | 0 | CMN | 5416  |
| 57 | 25  | 0   | Inf. Dilution | dG:-72.07 (0.11SD) kJ | 3 | EFE | 30072 |
| 58 | 25  | 0   | Inf. Dilution | dH:8.28 (3SF) kJ      | 4 | CRV | 2171  |

|    |    |     |               |                    |   |     |       |
|----|----|-----|---------------|--------------------|---|-----|-------|
| 59 | 25 | 0   | Inf. Dilution | dH:1.98 (3SF)      | 4 | ENS | 6496  |
| 60 | 25 | 0   | Inf. Dilution | dH:10.9 (0.4SD) kJ | 3 | EFE | 30072 |
| 61 | 25 | 0.1 | Unknown       | dH:2.0 (2SF)       | 3 | CRV | 2556  |
| 62 | 25 | 0.5 | Unknown       | dH:1.2 (2SF)       | 0 | CRV | 2556  |
| 63 | 35 | 0.5 | Unknown       | dH:1.7 (2SF)       | 3 | CRV | 2556  |

| Reaction No. 2883            |    |       |               |                    |   |       |        |
|------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| 3<F-1> + Al+3 = Al+3_F-1 (3) |    |       |               |                    |   |       |        |
| No                           | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                            | 25 | 0     | Inf. Dilution | lgK:16.7 (0.05SD)  | 4 | ENS   | 557    |
| 2                            | 25 | 0     | Inf. Dilution | lgK:17. (0.3SD)    | 3 | CMN   | 2171   |
| 3                            | 25 | 0     | Inf. Dilution | lgK:17.02 (4SF)    | 3 | EMU   | 5398   |
| 4                            | 25 | 0     | Inf. Dilution | lgK:16.8 (3SF)     | 4 | ENS   | 6496   |
| 5                            | 25 | 0     | Inf. Dilution | lgK:16.75 (0.02SD) | 3 | EFE   | 30072  |
| 6                            | 25 | 0     | Inf. Dilution | lgK:17.02 (4SF)    | 4 | CRV   | 32224  |
| 7                            | 25 | 0     | CHEMVAL1      | lgK:16.70 (0.01SD) | 0 | ENS   | 5156   |
| 8                            | 25 | 0     | CHEMVAL2      | lgK:16.55 (0.01SD) | 0 | ENS   | 5156   |
| 9                            | 25 | 0     | CHEMVAL3      | lgK:16.65 (0.01SD) | 0 | ENS   | 5156   |
| 10                           | 25 | 0     | CHEMVAL4      | lgK:16.60 (0.01SD) | 4 | ENS   | 5156   |
| 11                           | 25 | 0     | Many Media    | lgK:17. (0.3SD)    | 3 | EOD   | 6953   |
| 12                           | 25 | 0     | SOLMNEQ       | lgK:16.65 (4SF)    | 4 | CMN   | 5416   |
| 13                           | 25 | 0     | SOLMNEQ       | lgK:16.86 (4SF)    | 4 | CMN   | 5416   |
| 14                           | 25 | 0     | SOLVEQ        | lgK:16.94 (4SF)    | 3 | CMN   | 5416   |
| 15                           | 25 | 0     | SOLVEQ        | lgK:16.70 (4SF)    | 4 | CMN   | 5416   |
| 16                           | 25 | 0     | UCT_Sea       | lgK:16.70 (0.01SD) | 4 | EDH   | 5390   |
| 17                           | 25 | 0     | WATCH         | lgK:16.62 (4SF)    | 4 | CMN   | 5416   |
| 18                           | 25 | 0     | WATCH         | lgK:16.84 (4SF)    | 4 | CMN   | 5416   |
| 19                           | 25 | 0.1   | UCT_Sea       | lgK:15.50 (0.01SD) | 4 | EDH   | 5390   |
| 20                           | 25 | 0.1   | Unknown       | lgK:15.5 (3SF)     | 6 | CRV   | 552    |
| 21                           | 25 | 0.15  | NaCl          | lgK:15.33 (0.01SD) | 3 | MGL   | 30072  |
| 22                           | 25 | 0.2   | UCT_Sea       | lgK:15.30 (0.01SD) | 4 | EDH   | 5390   |
| 23                           | 25 | 0.3   | UCT_Sea       | lgK:15.17 (0.01SD) | 4 | EDH   | 5390   |
| 24                           | 25 | 0.4   | UCT_Sea       | lgK:15.07 (0.01SD) | 4 | EDH   | 5390   |
| 25                           | 25 | 0.5   | UCT_Sea       | lgK:15.00 (0.01SD) | 4 | EDH   | 5390   |
| 26                           | 25 | 0.5   | Unknown       | lgK:15.0 (0.03SD)  | 6 | CRV   | 552    |
| 27                           | 25 | 0.6   | UCT_Sea       | lgK:14.94 (0.01SD) | 4 | EDH   | 5390   |

|    |     |     |               |                      |   |     |       |
|----|-----|-----|---------------|----------------------|---|-----|-------|
| 28 | 25  | 0.7 | UCT_Sea       | lgK:14.89(0.01SD)    | 4 | EDH | 5390  |
| 29 | 25  | 1   | NaCl          | lgK:14.80(0.02SD)    | 3 | MGL | 30072 |
| 30 | 75  | 0   | SOLMNEQ       | lgK:17.21(4SF)       | 3 | CMN | 5416  |
| 31 | 75  | 0   | WATCH         | lgK:17.20(4SF)       | 3 | CMN | 5416  |
| 32 | 100 | 0   | SOLMNEQ       | lgK:17.70(4SF)       | 2 | CMN | 5416  |
| 33 | 100 | 0   | SOLVEQ        | lgK:17.85(4SF)       | 3 | CMN | 5416  |
| 34 | 100 | 0   | WATCH         | lgK:17.70(4SF)       | 2 | CMN | 5416  |
| 35 | 125 | 0   | SOLMNEQ       | lgK:18.32(4SF)       | 2 | CMN | 5416  |
| 36 | 125 | 0   | WATCH         | lgK:18.32(4SF)       | 2 | CMN | 5416  |
| 37 | 150 | 0   | SOLMNEQ       | lgK:19.09(4SF)       | 3 | CMN | 5416  |
| 38 | 150 | 0   | SOLVEQ        | lgK:19.33(4SF)       | 4 | CMN | 5416  |
| 39 | 150 | 0   | WATCH         | lgK:19.05(4SF)       | 3 | CMN | 5416  |
| 40 | 175 | 0   | WATCH         | lgK:19.88(4SF)       | 3 | CMN | 5416  |
| 41 | 200 | 0   | SOLMNEQ       | lgK:21.16(4SF)       | 2 | CMN | 5416  |
| 42 | 200 | 0   | SOLVEQ        | lgK:21.49(4SF)       | 2 | CMN | 5416  |
| 43 | 200 | 0   | WATCH         | lgK:20.81(4SF)       | 4 | CMN | 5416  |
| 44 | 225 | 0   | WATCH         | lgK:21.83(4SF)       | 3 | CMN | 5416  |
| 45 | 250 | 0   | SOLMNEQ       | lgK:24.28(4SF)       | 0 | CMN | 5416  |
| 46 | 250 | 0   | SOLVEQ        | lgK:24.72(4SF)       | 0 | CMN | 5416  |
| 47 | 250 | 0   | WATCH         | lgK:22.94(4SF)       | 3 | CMN | 5416  |
| 48 | 275 | 0   | WATCH         | lgK:24.13(4SF)       | 3 | CMN | 5416  |
| 49 | 300 | 0   | SOLMNEQ       | lgK:29.61(4SF)       | 0 | CMN | 5416  |
| 50 | 300 | 0   | SOLVEQ        | lgK:30.24(4SF)       | 0 | CMN | 5416  |
| 51 | 300 | 0   | WATCH         | lgK:25.41(4SF)       | 0 | CMN | 5416  |
| 52 | 325 | 0   | WATCH         | lgK:26.77(4SF)       | 0 | CMN | 5416  |
| 53 | 350 | 0   | SOLMNEQ       | lgK:41.38(4SF)       | 0 | CMN | 5416  |
| 54 | 350 | 0   | WATCH         | lgK:28.20(4SF)       | 0 | CMN | 5416  |
| 55 | 25  | 0   | Inf. Dilution | dG:-95.58(0.11SD) kJ | 3 | EFE | 30072 |
| 56 | 25  | 0   | Inf. Dilution | dH:9.04(3SF) kJ      | 4 | CRV | 2171  |
| 57 | 25  | 0   | Inf. Dilution | dH:2.16(3SF)         | 4 | ENS | 6496  |
| 58 | 25  | 0   | Inf. Dilution | dH:13.2(0.6SD) kJ    | 3 | EFE | 30072 |
| 59 | 25  | 0.1 | Unknown       | dH:2.1(2SF)          | 2 | CRV | 2556  |
| 60 | 25  | 0.5 | Unknown       | dH:1.3(2SF)          | 0 | CRV | 2556  |
| 61 | 35  | 0.5 | Unknown       | dH:1.8(2SF)          | 0 | CRV | 2556  |

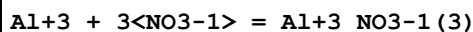
Reaction No. 2884

| 4<F-1> + Al+3 = Al+3_F-1(4) |     |       |               |                   |   |       |        |
|-----------------------------|-----|-------|---------------|-------------------|---|-------|--------|
| No                          | t   | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                           | 25  | 0     | Inf. Dilution | lgK:19.0(0.05SD)  | 4 | ENS   | 557    |
| 2                           | 25  | 0     | Inf. Dilution | lgK:19.(0.4SD)    | 3 | CMN   | 2171   |
| 3                           | 25  | 0     | Inf. Dilution | lgK:19.72(4SF)    | 3 | EMU   | 5398   |
| 4                           | 25  | 0     | Inf. Dilution | lgK:19.4(3SF)     | 4 | ENS   | 6496   |
| 5                           | 25  | 0     | Inf. Dilution | lgK:19.42(0.04SD) | 3 | EFE   | 30072  |
| 6                           | 25  | 0     | Inf. Dilution | lgK:19.72(4SF)    | 4 | CRV   | 32224  |
| 7                           | 25  | 0     | CHEMVAL1      | lgK:19.10(0.01SD) | 0 | ENS   | 5156   |
| 8                           | 25  | 0     | CHEMVAL2      | lgK:18.93(0.01SD) | 0 | ENS   | 5156   |
| 9                           | 25  | 0     | CHEMVAL3      | lgK:19.40(0.01SD) | 0 | ENS   | 5156   |
| 10                          | 25  | 0     | CHEMVAL4      | lgK:19.40(0.01SD) | 4 | ENS   | 5156   |
| 11                          | 25  | 0     | Many Media    | lgK:19.(0.4SD)    | 7 | EOD   | 6953   |
| 12                          | 25  | 0     | SOLMNEQ       | lgK:19.04(4SF)    | 4 | CMN   | 5416   |
| 13                          | 25  | 0     | SOLVEQ        | lgK:19.10(4SF)    | 4 | CMN   | 5416   |
| 14                          | 25  | 0     | UCT_Sea       | lgK:19.10(0.01SD) | 4 | EDH   | 5390   |
| 15                          | 25  | 0     | WATCH         | lgK:19.00(4SF)    | 4 | CMN   | 5416   |
| 16                          | 25  | 0.1   | UCT_Sea       | lgK:18.30(0.01SD) | 4 | EDH   | 5390   |
| 17                          | 25  | 0.1   | Unknown       | lgK:18.1(3SF)     | 6 | CRV   | 552    |
| 18                          | 25  | 0.2   | UCT_Sea       | lgK:18.06(0.01SD) | 4 | EDH   | 5390   |
| 19                          | 25  | 0.3   | UCT_Sea       | lgK:18.00(0.01SD) | 4 | EDH   | 5390   |
| 20                          | 25  | 0.4   | UCT_Sea       | lgK:17.99(0.01SD) | 4 | EDH   | 5390   |
| 21                          | 25  | 0.5   | UCT_Sea       | lgK:18.00(0.01SD) | 4 | EDH   | 5390   |
| 22                          | 25  | 0.5   | Unknown       | lgK:17.5(3SF)     | 2 | CRV   | 552    |
| 23                          | 25  | 0.6   | UCT_Sea       | lgK:18.02(0.01SD) | 4 | EDH   | 5390   |
| 24                          | 25  | 0.7   | UCT_Sea       | lgK:18.06(0.01SD) | 4 | EDH   | 5390   |
| 25                          | 50  | 0     | SOLMNEQ       | lgK:19.26(4SF)    | 4 | CMN   | 5416   |
| 26                          | 50  | 0     | SOLVEQ        | lgK:19.38(4SF)    | 4 | CMN   | 5416   |
| 27                          | 50  | 0     | WATCH         | lgK:19.23(4SF)    | 4 | CMN   | 5416   |
| 28                          | 75  | 0     | SOLMNEQ       | lgK:19.64(4SF)    | 4 | CMN   | 5416   |
| 29                          | 75  | 0     | WATCH         | lgK:19.64(4SF)    | 4 | CMN   | 5416   |
| 30                          | 100 | 0     | SOLMNEQ       | lgK:20.18(4SF)    | 4 | CMN   | 5416   |
| 31                          | 100 | 0     | SOLVEQ        | lgK:20.45(4SF)    | 4 | CMN   | 5416   |
| 32                          | 100 | 0     | WATCH         | lgK:20.19(4SF)    | 4 | CMN   | 5416   |
| 33                          | 125 | 0     | SOLMNEQ       | lgK:20.88(4SF)    | 4 | CMN   | 5416   |
| 34                          | 125 | 0     | WATCH         | lgK:20.88(4SF)    | 4 | CMN   | 5416   |



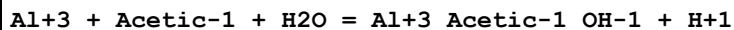
|    |     |     |               |                        |   |     |       |
|----|-----|-----|---------------|------------------------|---|-----|-------|
| 35 | 150 | 0   | SOLMNEQ       | lgK:21.75 (4SF)        | 4 | CMN | 5416  |
| 36 | 150 | 0   | SOLVEQ        | lgK:22.17 (4SF)        | 4 | CMN | 5416  |
| 37 | 150 | 0   | WATCH         | lgK:21.70 (4SF)        | 4 | CMN | 5416  |
| 38 | 175 | 0   | WATCH         | lgK:22.63 (4SF)        | 3 | CMN | 5416  |
| 39 | 200 | 0   | SOLMNEQ       | lgK:24.08 (4SF)        | 3 | CMN | 5416  |
| 40 | 200 | 0   | SOLVEQ        | lgK:24.67 (4SF)        | 2 | CMN | 5416  |
| 41 | 200 | 0   | WATCH         | lgK:23.68 (4SF)        | 3 | CMN | 5416  |
| 42 | 225 | 0   | WATCH         | lgK:24.83 (4SF)        | 3 | CMN | 5416  |
| 43 | 250 | 0   | SOLMNEQ       | lgK:27.59 (4SF)        | 0 | CMN | 5416  |
| 44 | 250 | 0   | SOLVEQ        | lgK:28.40 (4SF)        | 0 | CMN | 5416  |
| 45 | 250 | 0   | WATCH         | lgK:26.08 (4SF)        | 3 | CMN | 5416  |
| 46 | 275 | 0   | WATCH         | lgK:27.42 (4SF)        | 0 | CMN | 5416  |
| 47 | 300 | 0   | SOLMNEQ       | lgK:33.63 (4SF)        | 0 | CMN | 5416  |
| 48 | 300 | 0   | SOLVEQ        | lgK:34.76 (4SF)        | 0 | CMN | 5416  |
| 49 | 300 | 0   | WATCH         | lgK:28.86 (4SF)        | 0 | CMN | 5416  |
| 50 | 325 | 0   | WATCH         | lgK:30.40 (4SF)        | 0 | CMN | 5416  |
| 51 | 350 | 0   | SOLMNEQ       | lgK:46.95 (4SF)        | 0 | CMN | 5416  |
| 52 | 350 | 0   | WATCH         | lgK:32.02 (4SF)        | 0 | CMN | 5416  |
| 53 | 25  | 0   | Inf. Dilution | dG:-110.81 (0.23SD) kJ | 3 | EFE | 30072 |
| 54 | 25  | 0   | Inf. Dilution | dH:2.16 (0.02SD)       | 4 | ENS | 2171  |
| 55 | 25  | 0   | Inf. Dilution | dH:2.20 (3SF)          | 4 | ENS | 6496  |
| 56 | 25  | 0   | Inf. Dilution | dH:15.1 (0.8SD) kJ     | 3 | EFE | 30072 |
| 57 | 25  | 0.1 | Unknown       | dH:2.1 (2SF)           | 0 | CRV | 2556  |

## Reaction No. 2885



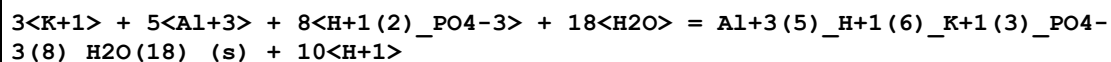
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:0.1 (0.05SD) | 4 | ENS   | 557    |

## Reaction No. 2886



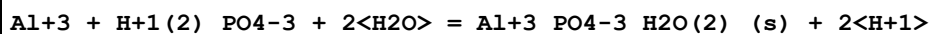
| No | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
|----|----|-------|--------|---------------------|---|-------|--------|
| 1  | 25 | 1     | NaNO3  | lgK:-1.6 (0.05SD) w | 3 | MPH   | 564    |

## Reaction No. 2923



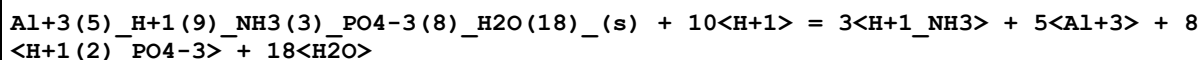
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:22.3(0.05SD) | 4 | ENS   | 557    |

## Reaction No. 2924



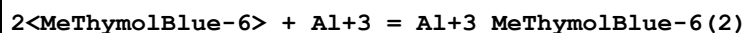
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:2.5(0.05SD) | 4 | ENS   | 557    |

## Reaction No. 2925



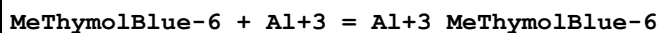
| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-19.8(3SF)    | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:-19.1(0.05SD) | 0 | ENS   | 557    |

## Reaction No. 3062



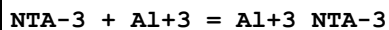
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 23 | 1     | KNO3          | lgK:11.468(5SF) | 4 | MSP   | 3783   |
| 2  | 25 | 0     | Inf. Dilution | lgK:11.5(3SF)   | 1 | ESC   |        |

## Reaction No. 3063



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 23 | 1     | KNO3          | lgK:6.817(4SF) | 0 | MSP   | 3783   |
| 2  | 25 | 0     | Inf. Dilution | lgK:26.2(3SF)  | 1 | ELC   |        |

## Reaction No. 3131



| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 20 | 0.001 | Unknown       | lgK:10.(2SF)    | 0 | MGL   | 1118   |
| 2  | 20 | 0.1   | KCl           | lgK:10.(2SF)    | 0 | MGL   | 140    |
| 3  | 20 | 0.1   | KClO4         | lgK:9.5(0.1SD)  | 0 | MDS   | 931    |
| 4  | 25 | 0     | Inf. Dilution | lgK:13.38(4SF)  | 0 | ENS   | 1384   |
| 5  | 25 | 0     | Inf. Dilution | lgK:13.4(3SF)   | 0 | ENS   | 6405   |
| 6  | 25 | 0     | Inf. Dilution | lgK:18.1(0.1SD) | 1 | EDH   | 30680  |
| 7  | 25 | 0.1   | NaClO4        | lgK:10.53(4SF)  | 0 | MGL   | 1411   |
| 8  | 25 | 0.1   | NaClO4        | lgK:12.72(4SF)w | 2 | MPH   | 3180   |

|    |    |      |                    |                    |   |     |       |
|----|----|------|--------------------|--------------------|---|-----|-------|
| 9  | 25 | 0.1  | Unknown            | lgK:11.4 (3SF)     | 0 | CRV | 2556  |
| 10 | 25 | 0.15 | NaCl               | lgK:12.52 (0.02SD) | 1 | MGL | 30680 |
| 11 | 25 | 0.2  | NaClO <sub>4</sub> | lgK:11.37 (4SF)    | 0 | MSP | 931   |
| 12 | 25 | 0.2  | NaClO <sub>4</sub> | lgK:11.37 (4SF)    | 0 | ENS | 6953  |
| 13 | 25 | 0.5  | Na+1 salt          | lgK:11.1 (3SF)     | 2 | CRV | 2556  |
| 14 | 25 | 0.5  | NaClO <sub>4</sub> | lgK:10.8 (3SF)     | 2 | ENS | 6953  |
| 15 | 25 | 0.6  | NaCl               | lgK:11.097 (5SF)   | 3 | MGL | 1087  |
| 16 | 35 | 0.1  | K+1 salt           | lgK:11.6 (3SF)     | 0 | CRV | 2556  |
| 17 | 35 | 0.1  | NaClO <sub>4</sub> | lgK:10.30 (4SF)    | 0 | MGL | 1411  |
| 18 | 35 | 0.1  | Unknown            | lgK:10.83 (4SF)    | 0 | EOD | 6223  |
| 19 | 45 | 0.1  | NaClO <sub>4</sub> | lgK:10.18 (4SF)    | 0 | MGL | 1411  |
| 20 | 25 | 0.15 | NaCl               | dG:-71.4 (3SF) kJ  | 1 | EDH | 30680 |
| 21 | 25 | 0.15 | NaCl               | dH:21. (1.SD) kJ   | 3 | EDH | 30680 |

| Reaction No. 3132                 |    |       |                    |                |   |       |        |
|-----------------------------------|----|-------|--------------------|----------------|---|-------|--------|
| Al+3_NTA-3 + H+1 = Al+3_H+1_NTA-3 |    |       |                    |                |   |       |        |
| No                                | t  | I-Str | Medium             | Value (Dev)    | W | Tech. | Ref. # |
| 1                                 | 25 | 0.2   | NaClO <sub>4</sub> | lgK:1.90 (3SF) | 6 | MGL   | 1118   |

| Reaction No. 3133                   |    |       |         |               |   |       |        |
|-------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Al+3_NTA-3 + OH-1 = Al+3_OH-1_NTA-3 |    |       |         |               |   |       |        |
| No                                  | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                   | 25 | 0.2   | Unknown | lgK:8.7 (2SF) | 4 | ENS   | 773    |

| Reaction No. 3134                           |    |       |         |               |   |       |        |
|---------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Al+3_OH-1_NTA-3 + OH-1 = Al+3_OH-1(2)_NTA-3 |    |       |         |               |   |       |        |
| No                                          | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 25 | 0.2   | Unknown | lgK:5.5 (2SF) | 4 | ENS   | 773    |

| Reaction No. 3183               |    |       |                  |                   |   |       |        |
|---------------------------------|----|-------|------------------|-------------------|---|-------|--------|
| HOEDTA-3 + Al+3 = Al+3_HOEDTA-3 |    |       |                  |                   |   |       |        |
| No                              | t  | I-Str | Medium           | Value (Dev)       | W | Tech. | Ref. # |
| 1                               | 20 | 0.1   | KNO <sub>3</sub> | lgK:12.4 (0.06SD) | 0 | MEF   | 2095   |
| 2                               | 20 | 0.1   | Unknown          | lgK:14.3 (3SF)    | 4 | ENS   | 773    |
| 3                               | 25 | 0.1   | KNO <sub>3</sub> | lgK:14.4 (3SF)    | 4 | MEF   | 931    |
| 4                               | 25 | 0.1   | KNO <sub>3</sub> | lgK:12.5 (0.05SD) | 0 | MEF   | 2095   |
| 5                               | 25 | 0.1   | Unknown          | lgK:14.4 (3SF)    | 5 | ENS   | 2171   |

|                                                         |       |     |                    |                  |   |     |       |
|---------------------------------------------------------|-------|-----|--------------------|------------------|---|-----|-------|
| 6                                                       | 25    | 0.2 | NaClO <sub>4</sub> | lgK:13.96(4SF)   | 5 | MSP | 931   |
| 7                                                       | 30    | 0.1 | KNO <sub>3</sub>   | lgK:13.(0.2SD)   | 0 | MEF | 2095  |
| 8                                                       | 35    | 0.1 | KNO <sub>3</sub>   | lgK:15.68(4SF)   | 0 | MGL | 11516 |
| Note (8): Khan M et al., Indian J. Chem., 1980, 19A, 44 |       |     |                    |                  |   |     |       |
| 9                                                       | 40    | 0.1 | KNO <sub>3</sub>   | lgK:12.9(0.1SD)  | 0 | MEF | 2095  |
| 10                                                      | 25    | 0.1 | KNO <sub>3</sub>   | dG:-17.1(0.07SD) | 0 | MEF | 2095  |
| 11                                                      | 10:40 | 0.1 | KNO <sub>3</sub>   | dH:9.4(2SF)      | 3 | MCL | 2095  |

| Reaction No. 3184                       |    |       |                    |               |   |       |        |
|-----------------------------------------|----|-------|--------------------|---------------|---|-------|--------|
| Al+3_HOEDTA-3 + H+1 = Al+3_H+1_HOEDTA-3 |    |       |                    |               |   |       |        |
| No                                      | t  | I-Str | Medium             | Value (Dev)   | W | Tech. | Ref. # |
| 1                                       | 20 | 0.1   | KNO <sub>3</sub>   | lgK:5.08(3SF) | 0 | MEF   | 2095   |
| 2                                       | 25 | 0.1   | KNO <sub>3</sub>   | lgK:2.4(2SF)  | 4 | MHG   | 999    |
| 3                                       | 25 | 0.1   | KNO <sub>3</sub>   | lgK:2.4(2SF)  | 4 | ENS   | 6953   |
| 4                                       | 25 | 0.2   | NaClO <sub>4</sub> | lgK:2.14(3SF) | 5 | MGL   | 999    |

| Reaction No. 3185                         |    |       |                  |               |   |       |        |
|-------------------------------------------|----|-------|------------------|---------------|---|-------|--------|
| Al+3_HOEDTA-3 + OH-1 = Al+3_OH-1_HOEDTA-3 |    |       |                  |               |   |       |        |
| No                                        | t  | I-Str | Medium           | Value (Dev)   | W | Tech. | Ref. # |
| 1                                         | 25 | 0.1   | KNO <sub>3</sub> | lgK:9.3(2SF)  | 4 | MEF   | 999    |
| 2                                         | 25 | 0.2   | Unknown          | lgK:8.88(3SF) | 6 | ENS   | 773    |

| Reaction No. 3186                                 |    |       |         |               |   |       |        |
|---------------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Al+3_OH-1_HOEDTA-3 + OH-1 = Al+3_OH-1(2)_HOEDTA-3 |    |       |         |               |   |       |        |
| No                                                | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                 | 25 | 0.2   | Unknown | lgK:4.58(3SF) | 6 | ENS   | 773    |

| Reaction No. 3405               |    |       |                  |               |   |       |        |
|---------------------------------|----|-------|------------------|---------------|---|-------|--------|
| Ferron-2 + Al+3 = Al+3_Ferron-2 |    |       |                  |               |   |       |        |
| No                              | t  | I-Str | Medium           | Value (Dev)   | W | Tech. | Ref. # |
| 1                               | 25 | 0.1   | KCl              | lgK:8.5(2SF)w | 0 | MPH   | 3955   |
| 2                               | 25 | 0.1   | KCl              | lgK:7.6(2SF)  | 4 | MGL   | 6953   |
| 3                               | 25 | 0.1   | KCl              | lgK:7.6(2SF)  | 5 | MPT   | 31140  |
| 4                               | 25 | 0.14  | KCl              | lgK:7.9(2SF)w | 2 | MPH   | 3955   |
| 5                               | 28 | 0.1   | KNO <sub>3</sub> | lgK:6.76(3SF) | 0 | MGL   | 6953   |

| Reaction No. 3406 |  |  |  |  |  |  |  |
|-------------------|--|--|--|--|--|--|--|
|-------------------|--|--|--|--|--|--|--|

| 2<Ferron-2> + Al+3 = Al+3_Ferron-2(2) |    |       |         |               |   |       |        |
|---------------------------------------|----|-------|---------|---------------|---|-------|--------|
| No                                    | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                     | 23 |       |         | lgK:12.5(3SF) | 0 | MSP   | 906    |
| 2                                     | 25 | 0.1   | Unknown | lgK:14.7(3SF) | 4 | ENS   | 2171   |
| 3                                     | 28 | 0.1   | Unknown | lgK:13.8(3SF) | 4 | ENS   | 2171   |

| Reaction No. 3407                     |    |       |         |               |   |       |        |
|---------------------------------------|----|-------|---------|---------------|---|-------|--------|
| 3<Ferron-2> + Al+3 = Al+3_Ferron-2(3) |    |       |         |               |   |       |        |
| No                                    | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | Unknown | lgK:20.3(3SF) | 4 | ENS   | 2171   |

| Reaction No. 3408                                    |    |       |        |              |   |       |        |
|------------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| Al+3_OH-1_Ferron-2(2) + H+1 = Al+3_Ferron-2(2) + H2O |    |       |        |              |   |       |        |
| No                                                   | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                    | 25 | 0.1   | KCl    | lgK:5.0(2SF) | 3 | MGL   | 999    |
| 2                                                    | 25 | 0.1   | KCl    | lgK:5.0(2SF) | 5 | MPT   | 31140  |

| Reaction No. 3691               |    |       |            |                 |   |       |        |
|---------------------------------|----|-------|------------|-----------------|---|-------|--------|
| Formic-1 + Al+3 = Al+3_Formic-1 |    |       |            |                 |   |       |        |
| No                              | t  | I-Str | Medium     | Value (Dev)     | W | Tech. | Ref. # |
| 1                               | 25 | 1     | NO3-1 salt | lgK:0.56(2SF)   | 3 | MPT   | 6953   |
| 2                               | 25 | 1     | NaClO4     | lgK:1.36(3SF)   | 6 | ENS   | 1241   |
| 3                               | 25 | 1     | NaNO3      | lgK:1.3(2SF)    | 5 | MGL   | 1241   |
| 4                               | 25 | 1     | Unknown    | lgK:1.8(0.03SD) | 0 | MIX   | 1539   |

| Reaction No. 3936                                    |    |       |         |               |   |       |        |
|------------------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Al+3 + 2<H+1_Ascorbic-2> = Al+3_H+1(2)_Ascorbic-2(2) |    |       |         |               |   |       |        |
| No                                                   | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                    | 25 | 0.1   | Unknown | lgK:3.55(3SF) | 6 | ENS   | 773    |

| Reaction No. 3937                           |    |       |         |               |   |       |        |
|---------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Al+3 + H+1_Ascorbic-2 = Al+3_H+1_Ascorbic-2 |    |       |         |               |   |       |        |
| No                                          | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | Unknown | lgK:1.89(3SF) | 3 | ENS   | 773    |

| Reaction No. 3987                             |  |  |  |  |  |  |  |
|-----------------------------------------------|--|--|--|--|--|--|--|
| Al+3 + OH-1 + Benzoic-1 = Al+3_Benzoic-1_OH-1 |  |  |  |  |  |  |  |

| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.5   | NaClO4 | lgK:12.1 (3SF) | 4 | MGL   | 5626   |

| Reaction No. 4030                 |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| SulfSal-3 + Al+3 = Al+3_SulfSal-3 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:12.3 (3SF)  | 2 | MGL   | 3494   |
| 2                                 | 25 | 0.1   | NaClO4 | lgK:13.20 (4SF) | 0 | MGL   | 2122   |
| 3                                 | 25 | 0.1   | NaClO4 | lgK:12.91 (4SF) | 2 | MSP   | 2122   |
| 4                                 | 25 | 0.5   | NaClO4 | lgK:11.8 (3SF)  | 4 | MGL   | 5987   |
| 5                                 | 30 | 0.2   | NaClO4 | lgK:12.20 (4SF) | 3 | MGL   | 2123   |

| Reaction No. 4031                       |    |       |               |                 |   |       |        |
|-----------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| 2<SulfSal-3> + Al+3 = Al+3_SulfSal-3(2) |    |       |               |                 |   |       |        |
| No                                      | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                       | 20 | 0.1   | KNO3          | lgK:20.0 (3SF)  | 0 | ENS   | 2171   |
| 2                                       | 20 | 0.1   | Unknown       | lgK:22.18 (4SF) | 0 | ENS   | 1539   |
| 3                                       | 25 | 0     | Inf. Dilution | lgK:24.3 (3SF)  | 1 | ESC   |        |

| Reaction No. 4032                       |    |       |               |                |   |       |        |
|-----------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| 3<SulfSal-3> + Al+3 = Al+3_SulfSal-3(3) |    |       |               |                |   |       |        |
| No                                      | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                       | 20 | 0.1   | KNO3          | lgK:25.8 (3SF) | 0 | ENS   | 2171   |
| 2                                       | 25 | 0     | Inf. Dilution | lgK:28.4 (3SF) | 1 | ELC   |        |

| Reaction No. 4045                   |    |       |               |                   |   |       |        |
|-------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| Phthalic-2 + Al+3 = Al+3_Phthalic-2 |    |       |               |                   |   |       |        |
| No                                  | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                   | 25 | 0     | Inf. Dilution | lgK:5.0 (2SF)     | 4 | ENS   | 2152   |
| 2                                   | 25 | 0     | UCT_Sea       | lgK:4.61 (0.01SD) | 4 | EDH   | 5390   |
| 3                                   | 25 | 0.1   | UCT_Sea       | lgK:3.58 (0.01SD) | 4 | EDH   | 5390   |
| 4                                   | 25 | 0.2   | UCT_Sea       | lgK:3.40 (0.01SD) | 4 | EDH   | 5390   |
| 5                                   | 25 | 0.3   | UCT_Sea       | lgK:3.30 (0.01SD) | 4 | EDH   | 5390   |
| 6                                   | 25 | 0.4   | UCT_Sea       | lgK:3.23 (0.01SD) | 4 | EDH   | 5390   |
| 7                                   | 25 | 0.5   | NaClO4        | lgK:3.18 (3SF)    | 5 | MGL   | 6953   |
| 8                                   | 25 | 0.5   | UCT_Sea       | lgK:3.18 (0.01SD) | 4 | EDH   | 5390   |
| 9                                   | 25 | 0.6   | NaCl          | lgK:2.94 (0.02MD) | 4 | MGL   | 1342   |

|    |    |     |         |                   |   |     |      |
|----|----|-----|---------|-------------------|---|-----|------|
| 10 | 25 | 0.6 | UCT_Sea | lgK:3.14 (0.01SD) | 4 | EDH | 5390 |
| 11 | 25 | 0.7 | UCT_Sea | lgK:3.12 (0.01SD) | 4 | EDH | 5390 |

| Reaction No. 4046                         |    |       |               |                   |   |       |        |
|-------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| 2<Phthalic-2> + Al+3 = Al+3_Phthalic-2(2) |    |       |               |                   |   |       |        |
| No                                        | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                         | 25 | 0     | Inf. Dilution | lgK:8.7 (2SF)     | 0 | ENS   | 2152   |
| 2                                         | 25 | 0     | Inf. Dilution | lgK:8.7 (2SF)     | 0 | ENS   | 6405   |
| 3                                         | 25 | 0     | UCT_Sea       | lgK:8.31 (0.01SD) | 3 | EDH   | 5390   |
| 4                                         | 25 | 0.1   | UCT_Sea       | lgK:6.89 (0.01SD) | 3 | EDH   | 5390   |
| 5                                         | 25 | 0.2   | UCT_Sea       | lgK:6.63 (0.01SD) | 3 | EDH   | 5390   |
| 6                                         | 25 | 0.3   | UCT_Sea       | lgK:6.48 (0.01SD) | 3 | EDH   | 5390   |
| 7                                         | 25 | 0.4   | UCT_Sea       | lgK:6.39 (0.01SD) | 3 | EDH   | 5390   |
| 8                                         | 25 | 0.5   | NaClO4        | lgK:6.32 (3SF)    | 3 | MGL   | 906    |
| 9                                         | 25 | 0.5   | UCT_Sea       | lgK:6.32 (0.01SD) | 3 | EDH   | 5390   |
| 10                                        | 25 | 0.6   | NaCl          | lgK:5.0 (0.1MD)   | 0 | MGL   | 1342   |
| 11                                        | 25 | 0.6   | UCT_Sea       | lgK:6.27 (0.01SD) | 3 | EDH   | 5390   |
| 12                                        | 25 | 0.7   | UCT_Sea       | lgK:6.23 (0.01SD) | 3 | EDH   | 5390   |

| Reaction No. 4234                   |    |       |         |                  |   |       |        |
|-------------------------------------|----|-------|---------|------------------|---|-------|--------|
| Al+3_EDTA-4 + H+1 = Al+3_H+1_EDTA-4 |    |       |         |                  |   |       |        |
| No                                  | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
| 1                                   | 15 | 0.1   | KNO3    | lgK:2.0 (2SF)    | 0 | MGL   | 140    |
| 2                                   | 15 |       |         | lgK:3. (1SF)     | 0 | MGL   | 903    |
| 3                                   | 20 | 0.1   | KNO3    | lgK:2.6 (0.2SD)  | 7 | EIU   | 903    |
| 4                                   | 20 | 0.1   | KNO3    | lgK:2.5 (2SF)    | 3 | MPL   | 903    |
| 5                                   | 20 | 0.1   | KNO3    | lgK:2.77 (3SF)   | 3 | MEF   | 2095   |
| 6                                   | 25 | 0.1   | KNO3    | lgK:3.4 (2SF)    | 0 | ENS   | 6953   |
| 7                                   | 25 | 0.2   | NaClO4  | lgK:2.63 (3SF)   | 3 | MGL   | 931    |
| 8                                   | 25 | 0.5   | KNO3    | lgK:2.4 (0.03SD) | 4 | MSP   | 1075   |
| 9                                   | 25 | 0.1   | Unknown | dH:-3.9 (2SF)    | 6 | CRV   | 552    |

| Reaction No. 4235                             |    |       |         |                |   |       |        |
|-----------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Al+3_OH-1_EDTA-4 + OH-1 = Al+3_OH-1(2)_EDTA-4 |    |       |         |                |   |       |        |
| No                                            | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                             | 20 | 0.1   | Unknown | lgK:3.98 (3SF) | 4 | ENS   | 773    |
| 2                                             | 25 | 0.2   | Unknown | lgK:3.47 (3SF) | 4 | ENS   | 773    |

| Reaction No. 4289                   |    |       |        |                |   |       |        |
|-------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Al+3_DTPA-5 + H+1 = Al+3_H+1_DTPA-5 |    |       |        |                |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                   | 20 | 0.1   | KNO3   | lgK:4.63(3SF)  | 3 | MEF   | 2095   |
| 2                                   | 20 | 0.1   | KNO3   | lgK:4.6(0.1SD) | 3 | CRV   | 13242  |
| 3                                   | 25 | 0.1   | KNO3   | lgK:4.3(2SF)   | 3 | MEF   | 6953   |

| Reaction No. 4290                     |    |       |        |              |   |       |        |
|---------------------------------------|----|-------|--------|--------------|---|-------|--------|
| Al+3_DTPA-5 + OH-1 = Al+3_OH-1_DTPA-5 |    |       |        |              |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | KNO3   | lgK:6.6(2SF) | 4 | MEF   | 999    |

| Reaction No. 4340                   |    |       |        |               |   |       |        |
|-------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Al+3_EGTA-4 + H+1 = Al+3_H+1_EGTA-4 |    |       |        |               |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                   | 25 | 0.2   | NaClO4 | lgK:3.97(3SF) | 6 | MGL   | 999    |

| Reaction No. 4341                     |    |       |         |              |   |       |        |
|---------------------------------------|----|-------|---------|--------------|---|-------|--------|
| Al+3_EGTA-4 + OH-1 = Al+3_OH-1_EGTA-4 |    |       |         |              |   |       |        |
| No                                    | t  | I-Str | Medium  | Value (Dev)  | W | Tech. | Ref. # |
| 1                                     | 25 | 0.2   | Unknown | lgK:8.6(2SF) | 4 | ENS   | 773    |

| Reaction No. 4342                             |    |       |         |               |   |       |        |
|-----------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Al+3_OH-1_EGTA-4 + OH-1 = Al+3_OH-1(2)_EGTA-4 |    |       |         |               |   |       |        |
| No                                            | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                             | 25 | 0.2   | Unknown | lgK:5.33(3SF) | 6 | ENS   | 773    |

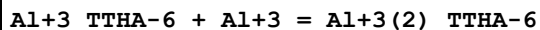
| Reaction No. 4440                                          |    |       |         |                |   |       |        |
|------------------------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| TTHA-6 + Al+3 = Al+3_TTHA-6                                |    |       |         |                |   |       |        |
| No                                                         | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                          | 25 | 0.1   | KNO3    | lgK:19.7(3SF)  | 4 | MGL   | 999    |
| 2                                                          | 25 | 0.1   | Unknown | lgK:21.0(3SF)  | 4 | ENS   | 2171   |
| 3                                                          | 35 | 0.1   | KNO3    | lgK:18.74(4SF) | 5 | MGL   | 11516  |
| Note (3): Khan M et al., Indian J. Chem., 1980, 19A, 44,50 |    |       |         |                |   |       |        |

| Reaction No. 4441                   |  |  |  |  |  |  |  |
|-------------------------------------|--|--|--|--|--|--|--|
| Al+3_TTHA-6 + H+1 = Al+3_H+1_TTHA-6 |  |  |  |  |  |  |  |



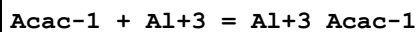
| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3    | lgK:5.85(3SF)  | 6 | MGL   | 999    |
| 2  | 25 | 0.1   | Me4NNO3 | lgK:5.9(0.1SD) | 5 | CRV   | 13242  |

## Reaction No. 4442



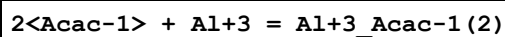
| No | t  | I-Str | Medium  | Value (Dev)  | W | Tech. | Ref. # |
|----|----|-------|---------|--------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:9.2(2SF) | 4 | ENS   | 773    |

## Reaction No. 4584



| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | UCT_Sea       | lgK:8.68(0.01SD) | 4 | EDH   | 5390   |
| 2  | 25 | 0.1   | UCT_Sea       | lgK:8.13(0.01SD) | 4 | EDH   | 5390   |
| 3  | 25 | 0.2   | UCT_Sea       | lgK:8.07(0.01SD) | 4 | EDH   | 5390   |
| 4  | 25 | 0.3   | UCT_Sea       | lgK:8.04(0.01SD) | 4 | EDH   | 5390   |
| 5  | 25 | 0.4   | UCT_Sea       | lgK:8.02(0.01SD) | 4 | EDH   | 5390   |
| 6  | 25 | 0.5   | UCT_Sea       | lgK:8.02(0.01SD) | 4 | EDH   | 5390   |
| 7  | 25 | 0.6   | UCT_Sea       | lgK:8.01(0.01SD) | 4 | EDH   | 5390   |
| 8  | 25 | 0.7   | UCT_Sea       | lgK:8.01(0.01SD) | 4 | EDH   | 5390   |
| 9  | 30 | 0     | Inf. Dilution | lgK:8.6(2SF)     | 4 | MGL   | 2171   |
| 10 | 30 | 0.1   | Unknown       | lgK:8.1(2SF)     | 4 | ENS   | 5971   |

## Reaction No. 4585



| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | UCT_Sea       | lgK:16.64(0.01SD) | 4 | EDH   | 5390   |
| 2  | 25 | 0.1   | UCT_Sea       | lgK:15.80(0.01SD) | 4 | EDH   | 5390   |
| 3  | 25 | 0.2   | UCT_Sea       | lgK:15.66(0.01SD) | 4 | EDH   | 5390   |
| 4  | 25 | 0.3   | UCT_Sea       | lgK:15.59(0.01SD) | 4 | EDH   | 5390   |
| 5  | 25 | 0.4   | UCT_Sea       | lgK:15.55(0.01SD) | 4 | EDH   | 5390   |
| 6  | 25 | 0.5   | UCT_Sea       | lgK:15.52(0.01SD) | 4 | EDH   | 5390   |
| 7  | 25 | 0.6   | UCT_Sea       | lgK:15.50(0.01SD) | 4 | EDH   | 5390   |
| 8  | 25 | 0.7   | UCT_Sea       | lgK:15.49(0.01SD) | 4 | EDH   | 5390   |
| 9  | 30 | 0     | Inf. Dilution | lgK:16.5(3SF)     | 5 | ENS   | 2171   |

## Reaction No. 4586

| 3<Acac-1> + Al+3 = Al+3_Acac-1(3) |    |       |               |               |   |       |        |
|-----------------------------------|----|-------|---------------|---------------|---|-------|--------|
| No                                | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                 | 25 | 0     | Inf. Dilution | lgK:22.3(3SF) | 5 | MGL   | 2360   |

| Reaction No. 4627             |       |       |         |              |   |       |        |
|-------------------------------|-------|-------|---------|--------------|---|-------|--------|
| Kojic-1 + Al+3 = Al+3_Kojic-1 |       |       |         |              |   |       |        |
| No                            | t     | I-Str | Medium  | Value (Dev)  | W | Tech. | Ref. # |
| 1                             | 20:23 | 0.1   | KCl     | lgK:7.7(2SF) | 4 | MHY   | 999    |
| 2                             | 25    | 0.1   | Unknown | lgK:7.7(2SF) | 4 | ENS   | 2171   |
| 3                             | 25    | 1     | NaCl    | lgK:7.6(2SF) | 5 | MGL   | 4159   |

| Reaction No. 4628                   |    |       |         |               |   |       |        |
|-------------------------------------|----|-------|---------|---------------|---|-------|--------|
| 2<Kojic-1> + Al+3 = Al+3_Kojic-1(2) |    |       |         |               |   |       |        |
| No                                  | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                   | 20 | 0.1   | Unknown | lgK:14.2(3SF) | 4 | ENS   | 773    |
| 2                                   | 25 | 0.1   | Unknown | lgK:14.2(3SF) | 4 | ENS   | 2171   |

| Reaction No. 4629                   |    |       |         |               |   |       |        |
|-------------------------------------|----|-------|---------|---------------|---|-------|--------|
| 3<Kojic-1> + Al+3 = Al+3_Kojic-1(3) |    |       |         |               |   |       |        |
| No                                  | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                   | 20 | 0.1   | Unknown | lgK:19.5(3SF) | 4 | ENS   | 773    |
| 2                                   | 25 | 0.1   | Unknown | lgK:19.5(3SF) | 4 | ENS   | 2171   |

| Reaction No. 4757     |    |       |               |                   |   |       |        |
|-----------------------|----|-------|---------------|-------------------|---|-------|--------|
| Al+3 + 3<e-1> = Al(s) |    |       |               |                   |   |       |        |
| No                    | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                     | 25 | 0     | Inf. Dilution | lgK:-85.78(4SF)   | 4 | CRV   | 32224  |
| 2                     | 25 | 0     | Inf. Dilution | dG:485.34(5SF) kJ | 0 | CRV   | 440    |
| 3                     | 25 | 0     | Inf. Dilution | dG:116.0(0.05SD)  | 0 | ENS   | 802    |
| 4                     | 25 | 0     | Inf. Dilution | dG:487.8(2.SD) kJ | 0 | EFE   | 2171   |
| 5                     | 25 | 0     | Inf. Dilution | dG:489.9(4SF) kJ  | 0 | EFE   | 5616   |
| 6                     | 25 | 0     | Inf. Dilution | dG:489.40(5SF) kJ | 0 | CRV   | 5619   |
| 7                     | 25 | 0     | Inf. Dilution | dG:485.34(5SF) kJ | 0 | CRV   | 5627   |
| 8                     | 25 | 0     | Inf. Dilution | dG:485.34(5SF) kJ | 0 | CRV   | 6421   |
| 9                     | 25 | 0     | Inf. Dilution | dG:115.38(5SF)    | 0 | CRV   | 6488   |
| 10                    | 25 | 0     | Inf. Dilution | dG:489.6(4SF) kJ  | 0 | CRV   | 6494   |
| 11                    | 25 | 0     | Inf. Dilution | dG:487.7(4SF) kJ  | 0 | EOD   | 6953   |

|                                                                                     |     |   |               |                      |   |     |       |
|-------------------------------------------------------------------------------------|-----|---|---------------|----------------------|---|-----|-------|
| 12                                                                                  | 25  | 0 | Inf. Dilution | dG:485.(3SF)kJ       | 0 | EFE | 7275  |
| 13                                                                                  | 25  | 0 | Inf. Dilution | dG:485.(3SF)kJ       | 0 | CRV | 7421  |
| 14                                                                                  | 25  | 0 | Inf. Dilution | dG:491.5(4SF)kJ      | 3 | CRV | 8746  |
| 15                                                                                  | 25  | 0 | Inf. Dilution | dG:487.7(1.5MD)kJ    | 0 | CRV | 8775  |
| 16                                                                                  | 25  | 0 | Inf. Dilution | dG:489.400(6SF)kJ    | 0 | CRV | 8872  |
| 17                                                                                  | 25  | 0 | Inf. Dilution | dG:492.00(5SF)kJ     | 0 | EOD | 8872  |
| Note (17): Data from Naumov, Ryzhenko & Khodakovsky, 1971 (low Wgt for consistency) |     |   |               |                      |   |     |       |
| 18                                                                                  | 25  | 0 | Inf. Dilution | dG:489.40(5SF)kJ     | 0 | EOD | 8872  |
| Note (18): Data from Khodakovsky, 1983                                              |     |   |               |                      |   |     |       |
| 19                                                                                  | 25  | 0 | Inf. Dilution | dG:481.16(5SF)kJ     | 0 | CRV | 8873  |
| 20                                                                                  | 25  | 0 | Inf. Dilution | dG:116.970(6SF)      | 0 | CRV | 8875  |
| 21                                                                                  | 25  | 0 | Inf. Dilution | dG:115.610(6SF)      | 0 | CRV | 9029  |
| 22                                                                                  | 25  | 0 | Inf. Dilution | dG:116.54(5SF)       | 0 | CRV | 10174 |
| 23                                                                                  | 25  | 0 | Inf. Dilution | dG:491.51(3.338SD)kJ | 3 | CRV | 14923 |
| 24                                                                                  | 25  | 0 | Inf. Dilution | dG:483.71(5SF)kJ     | 0 | EOD | 23101 |
| 25                                                                                  | 25  | 0 | Inf. Dilution | dG:491.510(1.9SD)kJ  | 4 | CRV | 27292 |
| 26                                                                                  | 25  | 0 | Inf. Dilution | dG:486.63(5SF)kJ     | 3 | EFE | 28693 |
| 27                                                                                  | 25  | 0 | Inf. Dilution | dG:116.90(5SF)       | 0 | EOD | 29489 |
| 28                                                                                  | 25  | 0 | Inf. Dilution | dG:490.12(0.81SD)kJ  | 7 | CRV | 29520 |
| 29                                                                                  | 25  | 0 | Inf. Dilution | dG:491.51(5SF)kJ     | 5 | EOD | 29520 |
| 30                                                                                  | 25  | 0 | Inf. Dilution | dG:485.0(4SF)kJ      | 2 | EFE | 32152 |
| 31                                                                                  | 50  | 0 | Inf. Dilution | dG:486.1(4SF)kJ      | 0 | EFE | 5616  |
| 32                                                                                  | 50  | 0 | Inf. Dilution | dG:114.49(5SF)       | 0 | CRV | 10174 |
| 33                                                                                  | 75  | 0 | Inf. Dilution | dG:112.39(5SF)       | 0 | CRV | 10174 |
| 34                                                                                  | 100 | 0 | Inf. Dilution | dG:477.4(4SF)kJ      | 0 | EFE | 5616  |
| 35                                                                                  | 100 | 0 | Inf. Dilution | dG:110.23(5SF)       | 0 | CRV | 10174 |
| 36                                                                                  | 125 | 0 | Inf. Dilution | dG:108.02(5SF)       | 0 | CRV | 10174 |
| 37                                                                                  | 150 | 0 | Inf. Dilution | dG:105.75(5SF)       | 0 | CRV | 10174 |
| 38                                                                                  | 175 | 0 | Inf. Dilution | dG:103.41(5SF)       | 0 | CRV | 10174 |
| 39                                                                                  | 200 | 0 | Inf. Dilution | dG:100.99(5SF)       | 0 | CRV | 10174 |
| 40                                                                                  | 225 | 0 | Inf. Dilution | dG:98.48(4SF)        | 0 | CRV | 10174 |
| 41                                                                                  | 250 | 0 | Inf. Dilution | dG:95.86(4SF)        | 0 | CRV | 10174 |
| 42                                                                                  | 300 | 0 | Inf. Dilution | dG:90.28(4SF)        | 0 | CRV | 10174 |
| 43                                                                                  | 25  | 0 | Inf. Dilution | E:-1.662(4SF)        | 0 | CRV | 6330  |
| 44                                                                                  | 25  | 0 | Inf. Dilution | E:-1.66(3SF)         | 0 | MEF | 7273  |
| 45                                                                                  | 25  | 0 | Inf. Dilution | E:-1.66(3SF)         | 0 | ENS | 7276  |

|    |    |   |               |                      |   |     |       |
|----|----|---|---------------|----------------------|---|-----|-------|
| 46 | 25 | 0 | Inf. Dilution | E:-1.66(3SF)         | 0 | CRV | 7372  |
| 47 | 25 | 0 | Inf. Dilution | E:-1.662(4SF)        | 0 | ENS | 7381  |
| 48 | 25 | 0 | Inf. Dilution | E:-1.677(4SF)        | 0 | CRV | 7761  |
| 49 | 25 | 0 | Inf. Dilution | E:-1.676(4SF)        | 0 | CRV | 22551 |
| 50 | 25 | 0 | Inf. Dilution | dH:127.0(0.1SD)      | 0 | ENS | 802   |
| 51 | 25 | 0 | Inf. Dilution | dH:531.(4.SD)kJ      | 0 | MTD | 2171  |
| 52 | 25 | 0 | Inf. Dilution | dH:537.6(4SF)kJ      | 0 | ENS | 5616  |
| 53 | 25 | 0 | Inf. Dilution | dH:531.(4.0MD)kJ     | 0 | CRV | 5619  |
| 54 | 25 | 0 | Inf. Dilution | dH:126.01(5SF)       | 0 | CRV | 6488  |
| 55 | 25 | 0 | Inf. Dilution | dH:531.(3SF)kJ       | 0 | EFE | 7275  |
| 56 | 25 | 0 | Inf. Dilution | dH:531.4(4SF)kJ      | 0 | CRV | 7761  |
| 57 | 25 | 0 | Inf. Dilution | dH:538.4(1.5MD)kJ    | 0 | CRV | 8746  |
| 58 | 25 | 0 | Inf. Dilution | dH:540.9(4.1MD)kJ    | 0 | CRV | 8775  |
| 59 | 25 | 0 | Inf. Dilution | dH:127.010(6SF)      | 0 | CRV | 8875  |
| 60 | 25 | 0 | Inf. Dilution | dH:126.830(6SF)      | 0 | CRV | 9029  |
| 61 | 25 | 0 | Inf. Dilution | dH:538.40(1.500SD)kJ | 0 | CRV | 14923 |
| 62 | 25 | 0 | Inf. Dilution | dH:530.63(5SF)kJ     | 0 | EOD | 23101 |
| 63 | 25 | 0 | Inf. Dilution | dH:538.400(0.75SD)kJ | 5 | CRV | 27292 |
| 64 | 25 | 0 | Inf. Dilution | dH:126.91(5SF)       | 0 | EOD | 29489 |
| 65 | 25 | 0 | Inf. Dilution | dH:538.35(0.43SD)kJ  | 7 | CRV | 29520 |
| 66 | 25 | 0 | Inf. Dilution | dH:538.4(4SF)kJ      | 5 | EOD | 29520 |
| 67 | 25 | 0 | Inf. Dilution | dH:531.0(4SF)kJ      | 2 | EFE | 32152 |
| 68 | 25 | 0 | GAMSPATH      | dH:531.330(6SF)kJ    | 0 | CRV | 12638 |
| 69 | 25 | 0 | Inf. Dilution | Cp:120.(12.MD)J/K    | 3 | CRV | 8775  |

## Reaction No. 4944

IDA-2 + Al+3 = Al+3\_IDA-2

| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:9.42(3SF)    | 5 | ENS   | 1384   |
| 2  | 25 | 0     | Inf. Dilution | lgK:9.9(2SF)     | 4 | ENS   | 6405   |
| 3  | 25 | 0.1   | NaClO4        | lgK:8.62(3SF)    | 0 | MGL   | 1411   |
| 4  | 25 | 0.5   | NaClO4        | lgK:8.10(0.01SD) | 4 | MGL   | 6953   |
| 5  | 35 | 0.1   | NaClO4        | lgK:8.42(3SF)    | 0 | MGL   | 1411   |
| 6  | 45 | 0.1   | NaClO4        | lgK:8.28(3SF)    | 0 | MGL   | 1411   |

## Reaction No. 4945

| 2<IDA-2> + Al+3 = Al+3_IDA-2 (2) |    |       |               |                   |   |       |        |
|----------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| No                               | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                | 25 | 0     | Inf. Dilution | lgK:17.5 (3SF)    | 4 | ENS   | 6405   |
| 2                                | 25 | 0.1   | NaClO4        | lgK:16.12 (4SF)   | 0 | MGL   | 1411   |
| 3                                | 25 | 0.5   | NaClO4        | lgK:15.1 (0.03SD) | 4 | MGL   | 6953   |
| 4                                | 35 | 0.1   | NaClO4        | lgK:15.64 (4SF)   | 0 | MGL   | 1411   |
| 5                                | 45 | 0.1   | NaClO4        | lgK:15.49 (4SF)   | 0 | MGL   | 1411   |

| Reaction No. 4973                       |    |       |        |                   |   |       |        |
|-----------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| AcHydroxam-1 + Al+3 = Al+3_AcHydroxam-1 |    |       |        |                   |   |       |        |
| No                                      | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                       | 20 | 0.1   | NaNO3  | lgK:7.95 (3SF)    | 5 | MGL   | 3676   |
| 2                                       | 25 | 0.2   | KCl    | lgK:8.15 (0.02SD) | 5 | MGL   | 2353   |

| Reaction No. 4974                              |    |       |               |                    |   |       |        |
|------------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| 2<AcHydroxam-1> + Al+3 = Al+3_AcHydroxam-1 (2) |    |       |               |                    |   |       |        |
| No                                             | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                              | 20 | 0.1   | NaNO3         | lgK:15.29 (4SF)    | 2 | MGL   | 3676   |
| 2                                              | 25 | 0     | Inf. Dilution | lgK:16.5 (3SF)     | 1 | EDH   |        |
| 3                                              | 25 | 0.2   | KCl           | lgK:15.77 (0.02SD) | 0 | MGL   | 2353   |

| Reaction No. 4975                              |    |       |        |                  |   |       |        |
|------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 3<AcHydroxam-1> + Al+3 = Al+3_AcHydroxam-1 (3) |    |       |        |                  |   |       |        |
| No                                             | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                              | 20 | 0.1   | NaNO3  | lgK:21.47 (4SF)  | 6 | MGL   | 3676   |
| 2                                              | 25 | 0.2   | KCl    | lgK:21.5 (0.2SD) | 5 | MGL   | 2353   |

| Reaction No. 5465         |    |       |               |                     |   |       |        |
|---------------------------|----|-------|---------------|---------------------|---|-------|--------|
| Gly-1 + Al+3 = Al+3_Gly-1 |    |       |               |                     |   |       |        |
| No                        | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                         | 25 | 0     | Inf. Dilution | lgK:5.47 (3SF)      | 0 | ENS   | 1384   |
| 2                         | 25 | 0.15  | NaCl          | lgK:7.18 (0.01SD)   | 4 | MGL   | 29906  |
| 3                         | 25 | 0.2   | KCl           | lgK:5.91 (3SF)      | 0 | MGL   | 10434  |
| 4                         | 25 | 0.5   | NaCl          | lgK:7.212 (0.007SD) | 4 | MGL   | 29906  |
| 5                         | 25 | 1     | NaCl          | lgK:7.245 (0.005SD) | 4 | MGL   | 29906  |
| 6                         | 37 | 0.15  | Blood Plasma  | lgK:5. (1SF)        | 0 | ENS   | 94     |

| Reaction No. 5614         |    |       |         |                    |   |       |        |
|---------------------------|----|-------|---------|--------------------|---|-------|--------|
| Cys-2 + Al+3 = Al+3_Cys-2 |    |       |         |                    |   |       |        |
| No                        | t  | I-Str | Medium  | Value (Dev)        | W | Tech. | Ref. # |
| 1                         | 25 | 0.1   | NaClO4  | lgK:6.43 (3SF)     | 0 | MGL   | 3180   |
| 2                         | 25 | 0.15  | NaCl    | lgK:11.91 (0.03SD) | 4 | MGL   | 29906  |
| 3                         | 25 | 0.15  | NaCl    | lgK:11.91 (4SF)    | 3 | MMR   | 29906  |
| 4                         | 25 | 0.15  | NaCl    | lgK:12.20 (0.04SD) | 4 | MSP   | 29906  |
| 5                         | 25 | 0.5   | NaCl    | lgK:11.60 (0.05SD) | 4 | MGL   | 29906  |
| 6                         | 25 | 1     | NaCl    | lgK:11.48 (0.04SD) | 4 | MGL   | 29906  |
| 7                         | 35 | 0.1   | Unknown | lgK:11.60 (4SF)    | 3 | EOD   | 6223   |

| Reaction No. 5645                                    |    |       |        |                 |   |       |        |
|------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Al+3(2)_H+1(-2)_Cat-2(2) + 2<H+1> = Al+3(2)_Cat-2(2) |    |       |        |                 |   |       |        |
| No                                                   | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                    | 25 | 0.2   | KCl    | lgK:-8.35 (3SF) | 6 | MGL   | 2470   |

| Reaction No. 5802                             |    |       |               |                 |   |       |        |
|-----------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Cys-2 + Al+3 + Citric-3 = Al+3_Cys-2_Citric-3 |    |       |               |                 |   |       |        |
| No                                            | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                             | 25 | 0     | Inf. Dilution | lgK:16.7 (3SF)  | 1 | EDH   |        |
| 2                                             | 25 | 0.1   | NaClO4        | lgK:14.90 (4SF) | 2 | CRV   | 905    |
| 3                                             | 25 | 0.1   | NaClO4        | lgK:14.90 (4SF) | 2 | MGL   | 3180   |

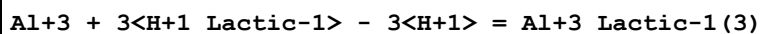
| Reaction No. 5803                       |    |       |               |                 |   |       |        |
|-----------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Cys-2 + Al+3 + NTA-3 = Al+3_Cys-2_NTA-3 |    |       |               |                 |   |       |        |
| No                                      | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                       | 25 | 0     | Inf. Dilution | lgK:20.7 (3SF)  | 1 | EDH   |        |
| 2                                       | 25 | 0.1   | NaClO4        | lgK:18.89 (4SF) | 2 | CRV   | 905    |
| 3                                       | 25 | 0.1   | NaClO4        | lgK:18.89 (4SF) | 2 | MGL   | 3180   |

| Reaction No. 6080                         |    |       |        |                    |   |       |        |
|-------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| Al+3 + H+1_Lactic-1 - H+1 = Al+3_Lactic-1 |    |       |        |                    |   |       |        |
| No                                        | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                         | 25 | 0.6   | NaCl   | lgK:-1.21 (0.02SD) | 3 | MGL   | 916    |

| Reaction No. 6081                                  |  |  |  |  |  |  |  |
|----------------------------------------------------|--|--|--|--|--|--|--|
| Al+3 + 2<H+1_Lactic-1> - 2<H+1> = Al+3_Lactic-1(2) |  |  |  |  |  |  |  |

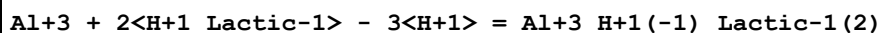
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-2.7(0.03SD) | 4 | MGL   | 916    |

## Reaction No. 6082



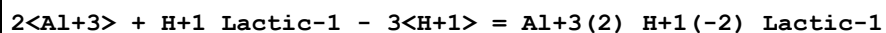
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-4.9(0.1SD) | 4 | MGL   | 916    |

## Reaction No. 6083



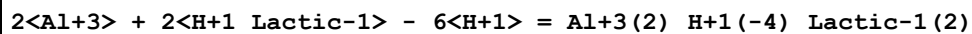
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-6.17(0.01SD) | 6 | MGL   | 916    |

## Reaction No. 6086



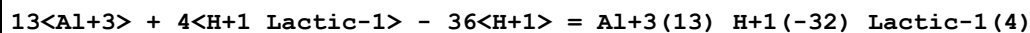
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-6.86(0.044MD) | 4 | MGL   | 910    |

## Reaction No. 6087



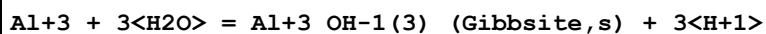
| No | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
|----|----|-------|--------|---------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-16.79(0.055MD) | 4 | MGL   | 910    |

## Reaction No. 6088



| No | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-98.7(3SF)     | 1 | ERG   |        |
| 2  | 25 | 0.6   | NaCl          | lgK:-106.9(0.25MD) | 3 | MGL   | 910    |

## Reaction No. 6090



| No | t | I-Str | Medium          | Value (Dev)       | W | Tech. | Ref. # |
|----|---|-------|-----------------|-------------------|---|-------|--------|
| 1  | 0 | 0     | Inf. Dilution/m | lgK:-9.43(0.09MD) | 4 | MSL   | 5643   |
| 2  | 0 | 0     | NaCl/m          | lgK:-9.43(3SF)    | 0 | MSL   | 7025   |
| 3  | 0 | 0.1   | NaCl/m          | lgK:-9.97(0.09MD) | 0 | MSL   | 5643   |
| 4  | 0 | 0.1   | NaCl/m          | lgK:-9.97(3SF)    | 0 | MSL   | 7025   |

Note (4): Low wgt for internal consistency

|                                                         |    |     |                 |                     |   |     |       |
|---------------------------------------------------------|----|-----|-----------------|---------------------|---|-----|-------|
| 5                                                       | 0  | 0.5 | NaCl/m          | lgK:-10.21 (0.09MD) | 6 | MSL | 5643  |
| 6                                                       | 0  | 0.5 | NaCl/m          | lgK:-10.21 (4SF)    | 6 | MSL | 7025  |
| 7                                                       | 0  | 1   | NaCl/m          | lgK:-10.36 (0.09MD) | 6 | MSL | 5643  |
| 8                                                       | 0  | 1   | NaCl/m          | lgK:-10.36 (4SF)    | 6 | MSL | 7025  |
| 9                                                       | 0  | 3   | NaCl/m          | lgK:-10.9 (0.1MD)   | 6 | MSL | 5643  |
| 10                                                      | 0  | 5   | NaCl/m          | lgK:-11.16 (0.1MD)  | 6 | MSL | 5643  |
| 11                                                      | 0  | 5   | NaCl/m          | lgK:-11.16 (4SF)    | 6 | MSL | 7025  |
| 12                                                      | 20 | 0   | Inf. Dilution   | lgK:-11.40 (4SF)    | 0 | MTN | 5630  |
| 13                                                      | 25 | 0   | Inf. Dilution   | lgK:-9.04 (3SF)     | 0 | MSL | 30    |
| 14                                                      | 25 | 0   | Inf. Dilution   | lgK:-7.961 (4SF)    | 3 | ENS | 440   |
| 15                                                      | 25 | 0   | Inf. Dilution   | lgK:-8.11 (3SF)     | 4 | MSL | 553   |
| 16                                                      | 25 | 0   | Inf. Dilution   | lgK:-8.0 (0.05SD)   | 3 | ENS | 557   |
| 17                                                      | 25 | 0   | Inf. Dilution   | lgK:-8.0 (0.05SD)   | 4 | ENS | 557   |
| 18                                                      | 25 | 0   | Inf. Dilution   | lgK:-7.76 (0.14SD)  | 6 | MSL | 5617  |
| 19                                                      | 25 | 0   | Inf. Dilution   | lgK:-8.071 (4SF)    | 4 | ENS | 5619  |
| 20                                                      | 25 | 0   | Inf. Dilution   | lgK:-7.892 (4SF)    | 3 | ENS | 5627  |
| 21                                                      | 25 | 0   | Inf. Dilution   | lgK:-8.5 (2SF)      | 0 | ENS | 6405  |
| 22                                                      | 25 | 0   | Inf. Dilution   | lgK:-9.218 (4SF)    | 0 | MSL | 6421  |
| 23                                                      | 25 | 0   | Inf. Dilution   | lgK:-8.11 (3SF)     | 4 | ENS | 6496  |
| 24                                                      | 25 | 0   | Inf. Dilution   | lgK:-8.04 (3SF)     | 3 | MGL | 6498  |
| 25                                                      | 25 | 0   | Inf. Dilution   | lgK:-8.205 (4SF)    | 4 | CRV | 8872  |
| 26                                                      | 25 | 0   | Inf. Dilution   | lgK:-8.140 (4SF)    | 4 | EOD | 8872  |
| Note (26): Data of Khodakovsky, 1983                    |    |     |                 |                     |   |     |       |
| 27                                                      | 25 | 0   | Inf. Dilution   | lgK:-8.177 (4SF)    | 4 | EOD | 8872  |
| Note (27): Data of Naumov, Ryzhenko & Khodakovsky, 1971 |    |     |                 |                     |   |     |       |
| 28                                                      | 25 | 0   | Inf. Dilution   | lgK:-8.480 (4SF)    | 0 | ENS | 8873  |
| 29                                                      | 25 | 0   | Inf. Dilution   | lgK:-8.728 (4SF)    | 0 | ENS | 8874  |
| 30                                                      | 25 | 0   | Inf. Dilution   | lgK:-10.38 (4SF)    | 0 | EOD | 23102 |
| 31                                                      | 25 | 0   | Inf. Dilution   | lgK:-8.11 (3SF)     | 3 | EOD | 29419 |
| 32                                                      | 25 | 0   | Inf. Dilution/m | lgK:-7.74 (0.06MD)  | 4 | MSL | 5643  |
| 33                                                      | 25 | 0   | Many Media/m    | lgK:-7.97 (3SF)     | 4 | EOD | 5613  |
| 34                                                      | 25 | 0   | NaCl/m          | lgK:-7.74 (3SF)     | 3 | MSL | 7025  |
| 35                                                      | 25 | 0   | Unknown         | lgK:-8.11 (3SF)     | 0 | ENS | 7373  |
| Note (35): Low wgt for internal consistency             |    |     |                 |                     |   |     |       |
| 36                                                      | 25 | 0.1 | Many Media/m    | lgK:-8.66 (3SF)     | 4 | EOD | 5613  |
| 37                                                      | 25 | 0.1 | NaCl/m          | lgK:-8.31 (0.05MD)  | 6 | MSL | 5643  |



|                                             |    |     |                 |                    |   |     |      |
|---------------------------------------------|----|-----|-----------------|--------------------|---|-----|------|
| 38                                          | 25 | 0.1 | NaCl/m          | lgK:-8.31 (3SF)    | 6 | MSL | 7025 |
| 39                                          | 25 | 0.2 | Many Media/m    | lgK:-8.83 (3SF)    | 4 | EOD | 5613 |
| 40                                          | 25 | 0.3 | Many Media/m    | lgK:-8.92 (3SF)    | 4 | EOD | 5613 |
| 41                                          | 25 | 0.5 | Many Media/m    | lgK:-9.01 (3SF)    | 4 | EOD | 5613 |
| 42                                          | 25 | 0.5 | NaCl/m          | lgK:-8.56 (0.05MD) | 6 | MSL | 5643 |
| 43                                          | 25 | 0.5 | NaCl/m          | lgK:-8.57 (3SF)    | 6 | MSL | 7025 |
| 44                                          | 25 | 1   | Many Media/m    | lgK:-9.05 (3SF)    | 4 | EOD | 5613 |
| 45                                          | 25 | 1   | NaCl/m          | lgK:-8.72 (0.05MD) | 6 | MSL | 5643 |
| 46                                          | 25 | 1   | NaCl/m          | lgK:-8.72 (3SF)    | 6 | MSL | 7025 |
| 47                                          | 25 | 2   | Many Media/m    | lgK:-8.87 (3SF)    | 3 | EOD | 5613 |
| 48                                          | 25 | 3   | Many Media/m    | lgK:-8.56 (3SF)    | 0 | EOD | 5613 |
| Note (48): Low wgt for internal consistency |    |     |                 |                    |   |     |      |
| 49                                          | 25 | 3   | NaCl/m          | lgK:-9.2 (0.07MD)  | 6 | MSL | 5643 |
| 50                                          | 25 | 5   | Many Media/m    | lgK:-7.84 (3SF)    | 0 | EOD | 5613 |
| 51                                          | 25 | 5   | NaCl/m          | lgK:-9.49 (0.07MD) | 6 | MSL | 5643 |
| 52                                          | 25 | 5   | NaCl/m          | lgK:-9.49 (3SF)    | 6 | MSL | 7025 |
| 53                                          | 50 | 0   | Inf. Dilution/m | lgK:-6.31 (0.01MD) | 4 | MSL | 5643 |
| 54                                          | 50 | 0   | NaCl/m          | lgK:-6.31 (3SF)    | 5 | MSL | 7025 |
| 55                                          | 50 | 0.1 | NaCl/m          | lgK:-6.92 (0.04MD) | 6 | MSL | 5643 |
| 56                                          | 50 | 0.1 | NaCl/m          | lgK:-6.92 (3SF)    | 6 | MSL | 7025 |
| 57                                          | 50 | 0.5 | NaCl/m          | lgK:-7.20 (0.04MD) | 6 | MSL | 5643 |
| 58                                          | 50 | 0.5 | NaCl/m          | lgK:-7.20 (3SF)    | 6 | MSL | 7025 |
| 59                                          | 50 | 1   | NaCl/m          | lgK:-7.4 (0.04SD)  | 6 | MSL | 5643 |
| 60                                          | 50 | 1   | NaCl/m          | lgK:-7.37 (3SF)    | 6 | MSL | 7025 |
| 61                                          | 50 | 3   | NaCl/m          | lgK:-7.86 (0.05MD) | 6 | MSL | 5643 |
| 62                                          | 50 | 5   | NaCl/m          | lgK:-8.13 (0.05MD) | 6 | MSL | 5643 |
| 63                                          | 50 | 5   | NaCl/m          | lgK:-8.13 (3SF)    | 6 | MSL | 7025 |
| 64                                          | 75 | 0   | Inf. Dilution/m | lgK:-5.09 (0.07MD) | 4 | MSL | 5643 |
| 65                                          | 75 | 0   | NaCl/m          | lgK:-5.09 (3SF)    | 5 | MSL | 7025 |
| 66                                          | 75 | 0.1 | NaCl/m          | lgK:-5.75 (0.05MD) | 6 | MSL | 5643 |
| 67                                          | 75 | 0.1 | NaCl/m          | lgK:-5.75 (3SF)    | 6 | MSL | 7025 |
| 68                                          | 75 | 0.5 | NaCl/m          | lgK:-6.06 (0.04MD) | 6 | MSL | 5643 |
| 69                                          | 75 | 0.5 | NaCl/m          | lgK:-6.06 (3SF)    | 6 | MSL | 7025 |
| 70                                          | 75 | 1   | NaCl/m          | lgK:-6.24 (0.05MD) | 6 | MSL | 5643 |
| 71                                          | 75 | 1   | NaCl/m          | lgK:-6.24 (3SF)    | 6 | MSL | 7025 |
| 72                                          | 75 | 3   | NaCl/m          | lgK:-6.75 (0.05MD) | 6 | MSL | 5643 |

|                                             |     |     |                 |                    |   |     |      |
|---------------------------------------------|-----|-----|-----------------|--------------------|---|-----|------|
| 73                                          | 75  | 5   | NaCl/m          | lgK:-7.02 (0.07MD) | 6 | MSL | 5643 |
| 74                                          | 75  | 5   | NaCl/m          | lgK:-7.02 (3SF)    | 6 | MSL | 7025 |
| 75                                          | 100 | 0   | Inf. Dilution/m | lgK:-4.04 (0.09MD) | 4 | MSL | 5643 |
| 76                                          | 100 | 0   | NaCl/m          | lgK:-4.04 (3SF)    | 5 | MSL | 7025 |
| 77                                          | 100 | 0.1 | NaCl/m          | lgK:-4.75 (0.07MD) | 6 | MSL | 5643 |
| 78                                          | 100 | 0.1 | NaCl/m          | lgK:-4.75 (3SF)    | 6 | MSL | 7025 |
| 79                                          | 100 | 0.5 | NaCl/m          | lgK:-5.11 (0.07MD) | 6 | MSL | 5643 |
| 80                                          | 100 | 0.5 | NaCl/m          | lgK:-5.11 (3SF)    | 6 | MSL | 7025 |
| 81                                          | 100 | 1   | NaCl/m          | lgK:-5.31 (0.07MD) | 6 | MSL | 5643 |
| 82                                          | 100 | 1   | NaCl/m          | lgK:-5.31 (3SF)    | 6 | MSL | 7025 |
| 83                                          | 100 | 3   | NaCl/m          | lgK:-5.8 (0.07MD)  | 6 | MSL | 5643 |
| 84                                          | 100 | 5   | NaCl/m          | lgK:-6.10 (0.1MD)  | 6 | MSL | 5643 |
| 85                                          | 100 | 5   | NaCl/m          | lgK:-6.10 (3SF)    | 6 | MSL | 7025 |
| 86                                          | 0   | 0   | Inf. Dilution/m | dH:25. (0.7MD)     | 4 | MTD | 5643 |
| 87                                          | 0   | 0.1 | NaCl/m          | dH:25. (0.7MD)     | 5 | MTD | 5643 |
| 88                                          | 0   | 0.5 | NaCl/m          | dH:25. (0.7MD)     | 5 | MTD | 5643 |
| 89                                          | 0   | 1   | NaCl/m          | dH:25. (0.6MD)     | 3 | MTD | 5643 |
| 90                                          | 0   | 3   | NaCl/m          | dH:25. (0.7MD)     | 0 | MTD | 5643 |
| Note (90): Low wgt for internal consistency |     |     |                 |                    |   |     |      |
| 91                                          | 0   | 5   | NaCl/m          | dH:25. (0.9MD)     | 0 | MTD | 5643 |
| Note (91): Low wgt for internal consistency |     |     |                 |                    |   |     |      |
| 92                                          | 25  | 0   | Inf. Dilution   | dH:22.8 (0.1SD)    | 4 | ENS | 766  |
| 93                                          | 25  | 0   | Inf. Dilution   | dH:22.8 (3SF)      | 4 | ENS | 6496 |
| 94                                          | 25  | 0   | Inf. Dilution/m | dH:25. (0.7MD)     | 4 | MTD | 5643 |
| 95                                          | 25  | 0.1 | NaCl/m          | dH:25. (0.7MD)     | 5 | MTD | 5643 |
| 96                                          | 25  | 0.5 | NaCl/m          | dH:24. (0.7MD)     | 5 | MTD | 5643 |
| 97                                          | 25  | 1   | NaCl/m          | dH:24. (0.6MD)     | 5 | MTD | 5643 |
| 98                                          | 25  | 3   | NaCl/m          | dH:24. (0.7MD)     | 3 | MTD | 5643 |
| 99                                          | 25  | 5   | NaCl/m          | dH:24. (0.9MD)     | 3 | MTD | 5643 |
| 100                                         | 50  | 0   | Inf. Dilution/m | dH:25. (0.7MD)     | 4 | MTD | 5643 |
| 101                                         | 50  | 0.1 | NaCl/m          | dH:24. (0.7MD)     | 5 | MTD | 5643 |
| 102                                         | 50  | 0.5 | NaCl/m          | dH:24. (0.7MD)     | 5 | MTD | 5643 |
| 103                                         | 50  | 1   | NaCl/m          | dH:24. (0.6MD)     | 5 | MTD | 5643 |
| 104                                         | 50  | 3   | NaCl/m          | dH:23. (0.7MD)     | 5 | MTD | 5643 |
| 105                                         | 50  | 5   | NaCl/m          | dH:24. (0.9MD)     | 3 | MTD | 5643 |
| 106                                         | 75  | 0   | Inf. Dilution/m | dH:25. (0.7MD)     | 4 | MTD | 5643 |

|     |     |     |                 |                |   |     |      |
|-----|-----|-----|-----------------|----------------|---|-----|------|
| 107 | 75  | 0.1 | NaCl/m          | dH:24. (0.7MD) | 3 | MTD | 5643 |
| 108 | 75  | 0.5 | NaCl/m          | dH:23. (0.7MD) | 3 | MTD | 5643 |
| 109 | 75  | 1   | NaCl/m          | dH:23. (0.6MD) | 3 | MTD | 5643 |
| 110 | 75  | 3   | NaCl/m          | dH:22. (0.7MD) | 3 | MTD | 5643 |
| 111 | 75  | 5   | NaCl/m          | dH:22. (0.9MD) | 5 | MTD | 5643 |
| 112 | 100 | 0   | Inf. Dilution/m | dH:25. (0.7MD) | 4 | MTD | 5643 |
| 113 | 100 | 0.1 | NaCl/m          | dH:23. (0.7MD) | 3 | MTD | 5643 |
| 114 | 100 | 0.5 | NaCl/m          | dH:22. (0.7MD) | 3 | MTD | 5643 |
| 115 | 100 | 1   | NaCl/m          | dH:22. (0.6MD) | 3 | MTD | 5643 |
| 116 | 100 | 3   | NaCl/m          | dH:21. (0.7MD) | 3 | MTD | 5643 |
| 117 | 100 | 5   | NaCl/m          | dH:21. (0.9MD) | 3 | MTD | 5643 |
| 118 | 0   | 0   | Inf. Dilution/m | Cp:-3. (3.MD)  | 4 | MTD | 5643 |
| 119 | 0   | 0.1 | NaCl/m          | Cp:-12. (3.MD) | 5 | MTD | 5643 |
| 120 | 0   | 0.5 | NaCl/m          | Cp:-19. (3.MD) | 5 | MTD | 5643 |
| 121 | 0   | 1   | NaCl/m          | Cp:-23. (3.MD) | 5 | MTD | 5643 |
| 122 | 0   | 3   | NaCl/m          | Cp:-28. (3.MD) | 5 | MTD | 5643 |
| 123 | 0   | 5   | NaCl/m          | Cp:-31. (3.MD) | 5 | MTD | 5643 |
| 124 | 25  | 0   | Inf. Dilution/m | Cp:-3. (3.MD)  | 4 | MTD | 5643 |
| 125 | 25  | 0.1 | NaCl/m          | Cp:-12. (3.MD) | 5 | MTD | 5643 |
| 126 | 25  | 0.5 | NaCl/m          | Cp:-19. (3.MD) | 5 | MTD | 5643 |
| 127 | 25  | 1   | NaCl/m          | Cp:-23. (3.MD) | 5 | MTD | 5643 |
| 128 | 25  | 3   | NaCl/m          | Cp:-30. (3.MD) | 5 | MTD | 5643 |
| 129 | 25  | 5   | NaCl/m          | Cp:-32. (3.MD) | 5 | MTD | 5643 |
| 130 | 50  | 0   | Inf. Dilution/m | Cp:-3. (3.MD)  | 4 | MTD | 5643 |
| 131 | 50  | 0.1 | NaCl/m          | Cp:-14. (3.MD) | 5 | MTD | 5643 |
| 132 | 50  | 0.5 | NaCl/m          | Cp:-24. (3.MD) | 5 | MTD | 5643 |
| 133 | 50  | 1   | NaCl/m          | Cp:-29. (3.MD) | 5 | MTD | 5643 |
| 134 | 50  | 3   | NaCl/m          | Cp:-37. (3.MD) | 5 | MTD | 5643 |
| 135 | 50  | 5   | NaCl/m          | Cp:-41. (3.MD) | 5 | MTD | 5643 |
| 136 | 75  | 0   | Inf. Dilution/m | Cp:-3. (3.MD)  | 4 | MTD | 5643 |
| 137 | 75  | 0.1 | NaCl/m          | Cp:-17. (3.MD) | 5 | MTD | 5643 |
| 138 | 75  | 0.5 | NaCl/m          | Cp:-29. (3.MD) | 5 | MTD | 5643 |
| 139 | 75  | 1   | NaCl/m          | Cp:-35. (3.MD) | 5 | MTD | 5643 |
| 140 | 75  | 3   | NaCl/m          | Cp:-47. (3.MD) | 5 | MTD | 5643 |
| 141 | 75  | 5   | NaCl/m          | Cp:-52. (3.MD) | 5 | MTD | 5643 |
| 142 | 100 | 0   | Inf. Dilution/m | Cp:-3. (3.MD)  | 4 | MTD | 5643 |

|     |     |     |        |                |   |     |      |
|-----|-----|-----|--------|----------------|---|-----|------|
| 143 | 100 | 0.1 | NaCl/m | Cp:-21.(3.MD)  | 5 | MTD | 5643 |
| 144 | 100 | 0.5 | NaCl/m | Cp:-35.(0.7MD) | 5 | MTD | 5643 |
| 145 | 100 | 1   | NaCl/m | Cp:-43.(3.MD)  | 4 | MTD | 5643 |
| 146 | 100 | 3   | NaCl/m | Cp:-58.(3.MD)  | 4 | MTD | 5643 |
| 147 | 100 | 5   | NaCl/m | Cp:-65.(3.MD)  | 4 | MTD | 5643 |

| Reaction No. 6091                                             |    |       |        |                  |   |       |        |
|---------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Al+3 + 3<H+1_Tropolone-1> - 3<H+1> = Al+3_Tropolone-1(3)_ (s) |    |       |        |                  |   |       |        |
| No                                                            | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                             | 25 | 0.6   | NaCl   | lgK:11.2(0.05SD) | 4 | MGL   | 911    |

| Reaction No. 6093                                 |    |       |        |                    |   |       |        |
|---------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| Al+3 + H+1(2)_MeMalon-2 - 2<H+1> = Al+3_MeMalon-2 |    |       |        |                    |   |       |        |
| No                                                | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                 | 25 | 0.6   | NaCl   | lgK:-2.21(0.006SD) | 6 | MGL   | 912    |

| Reaction No. 6094                                       |    |       |        |                   |   |       |        |
|---------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Al+3 + 2<H+1(2)_MeMalon-2> - 4<H+1> = Al+3_MeMalon-2(2) |    |       |        |                   |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.6   | NaCl   | lgK:-5.73(0.01SD) | 6 | MGL   | 912    |

| Reaction No. 6095                                       |    |       |        |                   |   |       |        |
|---------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Al+3 + 3<H+1(2)_MeMalon-2> - 6<H+1> = Al+3_MeMalon-2(3) |    |       |        |                   |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.6   | NaCl   | lgK:-11.2(0.04SD) | 4 | MGL   | 912    |

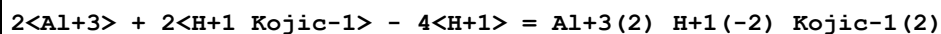
| Reaction No. 6119                       |    |       |        |                    |   |       |        |
|-----------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| Al+3 + H+1_Kojic-1 - H+1 = Al+3_Kojic-1 |    |       |        |                    |   |       |        |
| No                                      | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                       | 25 | 0.6   | NaCl   | lgK:-0.37(0.004SD) | 6 | MGL   | 917    |

| Reaction No. 6120                                |    |       |        |                    |   |       |        |
|--------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| Al+3 + 2<H+1_Kojic-1> - 2<H+1> = Al+3_Kojic-1(2) |    |       |        |                    |   |       |        |
| No                                               | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                | 25 | 0.6   | NaCl   | lgK:-1.50(0.004SD) | 6 | MGL   | 917    |

| Reaction No. 6121                                |  |  |  |  |  |  |  |
|--------------------------------------------------|--|--|--|--|--|--|--|
| Al+3 + 3<H+1_Kojic-1> - 3<H+1> = Al+3_Kojic-1(3) |  |  |  |  |  |  |  |

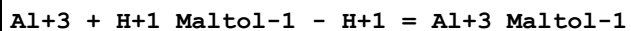
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-3.56(0.007SD) | 6 | MGL   | 917    |

## Reaction No. 6122



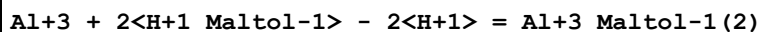
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-7.7(0.02SD) | 4 | MGL   | 917    |

## Reaction No. 6123



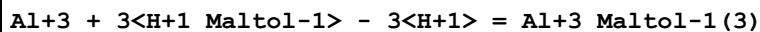
| No | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-0.3(1SF)       | 1 | ELC   |        |
| 2  | 25 | 0.6   | NaCl          | lgK:-0.130(0.002SD) | 6 | MGL   | 917    |

## Reaction No. 6124



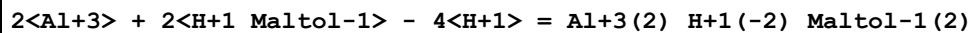
| No | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-1.4(2SF)       | 1 | ELC   |        |
| 2  | 25 | 0.6   | NaCl          | lgK:-0.956(0.002SD) | 3 | MGL   | 917    |

## Reaction No. 6125



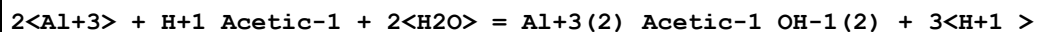
| No | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-3.3(2SF)      | 1 | ELC   |        |
| 2  | 25 | 0.6   | NaCl          | lgK:-2.67(0.005SD) | 4 | MGL   | 917    |

## Reaction No. 6126



| No | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-8.1(2SF)      | 1 | ELC   |        |
| 2  | 25 | 0.6   | NaCl          | lgK:-7.20(0.008SD) | 3 | MGL   | 917    |

## Reaction No. 6127



| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.6   | Na(Cl) | lgK:-7.98(3SF)     | 7 | MGL   | 17107  |
| 2  | 25 | 0.6   | NaCl   | lgK:-7.98(0.006SD) | 4 | MGL   | 918    |

| Reaction No. 6128                                                                                   |    |       |        |                     |   |       |        |
|-----------------------------------------------------------------------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| $2\text{Al}^{3+} + \text{H}_2\text{O} + 2\text{H}_2\text{O} = \text{Al}(\text{OH})_3 + 2\text{H}^+$ |    |       |        |                     |   |       |        |
| No                                                                                                  | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                                                                   | 25 | 0.6   | NaCl   | lgK:-8.038(0.008MD) | 7 | MGL   | 17107  |
| 2                                                                                                   | 25 | 0.6   | NaCl   | lgK:-8.04(0.003SD)  | 4 | MGL   | 918    |

| Reaction No. 6157                                                                                       |    |       |               |                 |   |       |        |
|---------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{Al}^{3+} + 2\text{OH}^- + \text{H}_2\text{PO}_4^- = \text{Al}(\text{OH})_3 + \text{H}_2\text{O}$ |    |       |               |                 |   |       |        |
| No                                                                                                      | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                       | 25 | 0     | Inf. Dilution | lgK:30.5(3SF)   | 4 | ENS   | 5984   |
| 2                                                                                                       | 37 | 0.15  | NaCl          | lgK:27.6(0.1SD) | 0 | MGL   | 922    |

| Reaction No. 6199                                                |    |       |               |                     |   |       |        |
|------------------------------------------------------------------|----|-------|---------------|---------------------|---|-------|--------|
| $3\text{Citric}^{3-} + \text{Al}^{3+} = \text{Al}(\text{Cit})_3$ |    |       |               |                     |   |       |        |
| No                                                               | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                                | 25 | 0     | Inf. Dilution | lgK:14.12(4SF)      | 4 | CRV   | 31301  |
| 2                                                                | 37 | 0.15  | NaCl          | lgK:12.643(0.002SD) | 6 | MGL   | 925    |

| Reaction No. 6343                                                                                        |    |       |        |                    |   |       |        |
|----------------------------------------------------------------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| $\text{Al}^{3+} + \text{H}_2\text{O} + \text{Citric}^{3-} = \text{Al}(\text{OH})_3 + \text{Citric}^{3-}$ |    |       |        |                    |   |       |        |
| No                                                                                                       | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                                                        | 25 | 0.6   | NaCl   | lgK:-4.92(0.008SD) | 4 | MGL   | 1116   |
| 2                                                                                                        | 33 | 0.25  | NaClO4 | lgK:-4.7(2SF)      | 5 | MGL   | 999    |

| Reaction No. 6344                                                                                         |    |       |        |                  |   |       |        |
|-----------------------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $\text{Al}^{3+} + 2\text{H}_2\text{O} + \text{Citric}^{3-} = \text{Al}(\text{OH})_3 + \text{Citric}^{3-}$ |    |       |        |                  |   |       |        |
| No                                                                                                        | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                         | 25 | 0.6   | NaCl   | lgK:-12.5(0.1SD) | 4 | MGL   | 1116   |

| Reaction No. 6345                                                                                          |    |       |        |                    |   |       |        |
|------------------------------------------------------------------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| $3\text{Al}^{3+} + 3\text{H}_2\text{O} + \text{Citric}^{3-} = \text{Al}(\text{OH})_3 + \text{Citric}^{3-}$ |    |       |        |                    |   |       |        |
| No                                                                                                         | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                                                          | 25 | 0.6   | NaCl   | lgK:-21.77(0.02SD) | 4 | MGL   | 1116   |

| Reaction No. 6346                                                                                        |   |       |        |             |   |       |        |
|----------------------------------------------------------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| $\text{Al}^{3+} + \text{H}_2\text{O} + \text{Citric}^{3-} = \text{Al}(\text{OH})_3 + \text{Citric}^{3-}$ |   |       |        |             |   |       |        |
| No                                                                                                       | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

|   |    |     |      |                   |   |     |      |
|---|----|-----|------|-------------------|---|-----|------|
| 1 | 25 | 0.1 | KNO3 | lgK:-2.20 (3SF)   | 5 | ENS | 2171 |
| 2 | 25 | 0.6 | NaCl | lgK:-2.7 (0.02SD) | 4 | MGL | 1116 |

| Reaction No. 6348                                          |    |       |        |                   |   |       |        |
|------------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Al+3 + H+1(3)_Citric-3 + H2O = Al+3_OH-1_Citric-3 + 4<H+1> |    |       |        |                   |   |       |        |
| No                                                         | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                          | 25 | 0.5   | NaNO3  | lgK:-6.6 (0.2SD)  | 0 | MGL   | 2261   |
| 2                                                          | 25 | 0.6   | NaCl   | lgK:-8.5 (0.04SD) | 4 | MGL   | 940    |

| Reaction No. 6349                                                |    |       |        |                    |   |       |        |
|------------------------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| Al+3 + H+1(3)_Citric-3 + 2<H2O> = Al+3_OH-1(2)_Citric-3 + 5<H+1> |    |       |        |                    |   |       |        |
| No                                                               | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                | 25 | 0.6   | NaCl   | lgK:-14.7 (0.05SD) | 4 | MGL   | 940    |

| Reaction No. 6350                                                    |    |       |               |                    |   |       |        |
|----------------------------------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| 3<Al+3> + 3<H+1(3)_Citric-3> = Al+3(3)_H+1(-7)_Citric-3(3) + 16<H+1> |    |       |               |                    |   |       |        |
| No                                                                   | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                    | 25 | 0     | Inf. Dilution | lgK:-53. (2SF)     | 1 | EOR   |        |
| 2                                                                    | 25 | 0.6   | NaCl          | lgK:-47.1 (0.03SD) | 6 | MGL   | 940    |

| Reaction No. 6352                                   |    |       |               |                    |   |       |        |
|-----------------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| Al+3 + H+1(2)_Tartaric-2 = Al+3_Tartaric-2 + 2<H+1> |    |       |               |                    |   |       |        |
| No                                                  | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                   | 25 | 0     | Inf. Dilution | lgK:-0.8 (1SF)     | 1 | ELC   |        |
| 2                                                   | 25 | 0.6   | NaCl          | lgK:-3.44 (0.02SD) | 4 | MGL   | 941    |

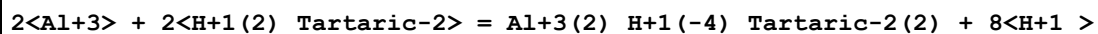
| Reaction No. 6353                                                       |    |       |        |                    |   |       |        |
|-------------------------------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| 2<Al+3> + 2<H+1(2)_Tartaric-2> = Al+3(2)_H+1(-1)_Tartaric-2(2) + 5<H+1> |    |       |        |                    |   |       |        |
| No                                                                      | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                       | 25 | 0.6   | NaCl   | lgK:-6.30 (0.02SD) | 6 | MGL   | 941    |

| Reaction No. 6354                                                       |    |       |        |                     |   |       |        |
|-------------------------------------------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| 2<Al+3> + 2<H+1(2)_Tartaric-2> = Al+3(2)_H+1(-2)_Tartaric-2(2) + 6<H+1> |    |       |        |                     |   |       |        |
| No                                                                      | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                                       | 25 | 0.6   | NaCl   | lgK:-8.91 (0.005SD) | 6 | MGL   | 941    |

| Reaction No. 6355                                                       |  |  |  |  |  |  |  |
|-------------------------------------------------------------------------|--|--|--|--|--|--|--|
| 2<Al+3> + 2<H+1(2)_Tartaric-2> = Al+3(2)_H+1(-3)_Tartaric-2(2) + 7<H+1> |  |  |  |  |  |  |  |

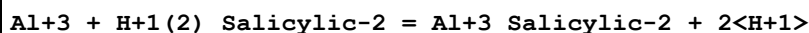
| No | t  | I-Str | Medium | Value (Dev)          | W | Tech. | Ref. # |
|----|----|-------|--------|----------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-13.12 (0.009SD) | 4 | MGL   | 941    |

## Reaction No. 6356



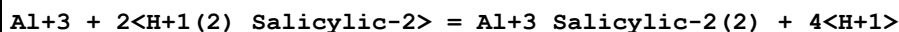
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-19.0 (0.06SD) | 4 | MGL   | 941    |

## Reaction No. 6414



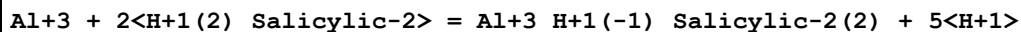
| No | t  | I-Str | Medium | Value (Dev)          | W | Tech. | Ref. # |
|----|----|-------|--------|----------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:-2.97 (3SF)      | 6 | MGL   | 2470   |
| 2  | 25 | 0.6   | NaCl   | lgK:-3.052 (0.005MD) | 6 | MGL   | 950    |

## Reaction No. 6415



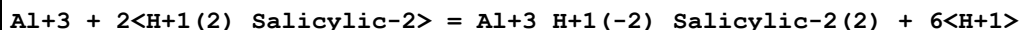
| No | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-8.8 (2SF)     | 1 | ERG   |        |
| 2  | 25 | 0.6   | NaCl          | lgK:-8.39 (0.01MD) | 5 | MGL   | 950    |

## Reaction No. 6416



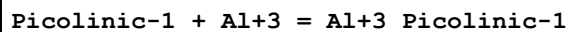
| No | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
|----|----|-------|--------|---------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-15.99 (0.02MD) | 6 | MGL   | 950    |

## Reaction No. 6417



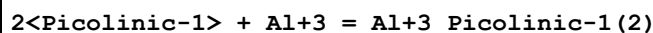
| No | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
|----|----|-------|--------|---------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-25.31 (0.01MD) | 6 | MGL   | 950    |

## Reaction No. 6534



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.15  | KNO3   | lgK:4.49 (0.01SD) | 6 | ENS   | 980    |
| 2  | 25 | 0.5   | NaClO4 | lgK:4.62 (3SF)    | 5 | ENS   | 6953   |

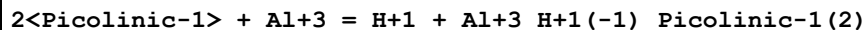
## Reaction No. 6535





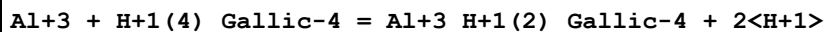
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.15  | KNO3   | lgK:8.42(0.02SD) | 2 | ENS   | 980    |
| 2  | 25 | 0.5   | NaClO4 | lgK:8.60(3SF)    | 2 | ENS   | 6953   |

## Reaction No. 6536



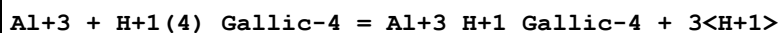
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.15  | KNO3   | lgK:17.59(0.02SD) | 6 | ENS   | 980    |

## Reaction No. 6678



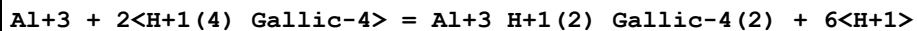
| No | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-4.0(2SF)      | 1 | EDH   |        |
| 2  | 25 | 0.6   | Cl-1 salt     | lgK:-4.3(2SF)      | 0 | ENS   | 2171   |
| 3  | 25 | 0.6   | NaCl          | lgK:-4.93(0.003SD) | 5 | MGL   | 969    |

## Reaction No. 6679



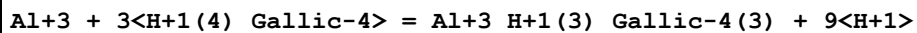
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-9.43(0.006SD) | 6 | MGL   | 969    |

## Reaction No. 6680



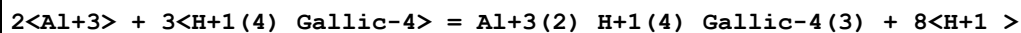
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-21.98(0.01SD) | 4 | MGL   | 969    |

## Reaction No. 6681



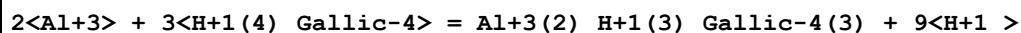
| No | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
|----|----|-------|--------|---------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-37.69(0.005SD) | 6 | MGL   | 969    |

## Reaction No. 6682



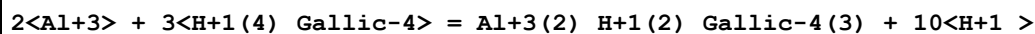
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-22.65(0.01SD) | 6 | MGL   | 969    |

## Reaction No. 6683



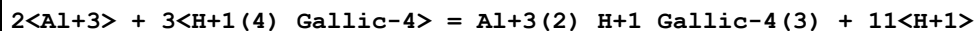
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-27.81(0.02SD) | 4 | MGL   | 969    |

## Reaction No. 6684



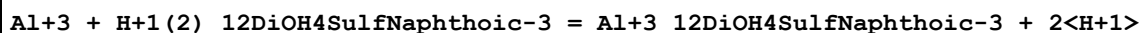
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-32.87(0.01SD) | 4 | MGL   | 969    |

## Reaction No. 6685



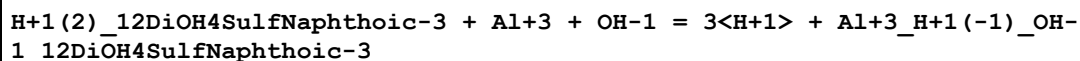
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-39.56(0.02SD) | 6 | MGL   | 969    |

## Reaction No. 6688



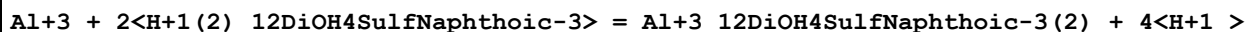
| No | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
|----|----|-------|--------|---------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-5.343(0.002SD) | 6 | MGL   | 966    |

## Reaction No. 6689



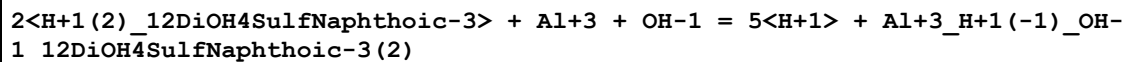
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-11.2(0.08SD) | 4 | MGL   | 966    |

## Reaction No. 6690



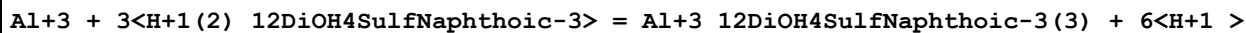
| No | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
|----|----|-------|--------|---------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-13.12(0.003SD) | 6 | MGL   | 966    |

## Reaction No. 6691



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-21.2(0.04SD) | 4 | MGL   | 966    |

## Reaction No. 6692



| No | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
|----|----|-------|--------|---------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-24.47(0.007SD) | 6 | MGL   | 966    |

| Reaction No. 7492                                                                                |    |       |        |                |   |       |        |
|--------------------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Al}+3 + \text{H}+1(2)\text{Oxalic-2} = \text{Al}+3\text{H}+1\text{Oxalic-2} + \text{H}+1$ |    |       |        |                |   |       |        |
| No                                                                                               | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                | 25 | 0.6   | NaCl   | lgK:1.4(0.3MD) | 4 | MGL   | 1044   |

| Reaction No. 7493                                                                       |    |       |        |                |   |       |        |
|-----------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Al}+3 + \text{H}+1(2)\text{Oxalic-2} = \text{Al}+3\text{Oxalic-2} + 2\text{H}+1$ |    |       |        |                |   |       |        |
| No                                                                                      | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                       | 25 | 0.6   | NaCl   | lgK:1.4(0.1MD) | 3 | MGL   | 1044   |

| Reaction No. 7494                                                                           |    |       |        |                  |   |       |        |
|---------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $\text{Al}+3 + 2\text{H}+1(2)\text{Oxalic-2} = \text{Al}+3\text{Oxalic-2}(2) + 4\text{H}+1$ |    |       |        |                  |   |       |        |
| No                                                                                          | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                           | 25 | 0.6   | NaCl   | lgK:1.85(0.04MD) | 4 | MGL   | 1044   |

| Reaction No. 7495                                                                           |    |       |        |                  |   |       |        |
|---------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $\text{Al}+3 + 3\text{H}+1(2)\text{Oxalic-2} = \text{Al}+3\text{Oxalic-2}(3) + 6\text{H}+1$ |    |       |        |                  |   |       |        |
| No                                                                                          | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                           | 25 | 0.6   | NaCl   | lgK:1.26(0.03MD) | 3 | MGL   | 1044   |

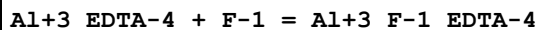
| Reaction No. 7496                                                                                                                   |    |       |        |                   |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| $3\text{Al}+3 + 3\text{H}+1(2)\text{Oxalic-2} + 3\text{H}_2\text{O} = \text{Al}+3(3)\text{OH-1}(3)\text{Oxalic-2}(3) + 9\text{H}+1$ |    |       |        |                   |   |       |        |
| No                                                                                                                                  | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                                   | 25 | 0.6   | NaCl   | lgK:-4.28(0.02MD) | 4 | MGL   | 1044   |

| Reaction No. 7497                                                                                                                    |    |       |        |                   |   |       |        |
|--------------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| $2\text{Al}+3 + 4\text{H}+1(2)\text{Oxalic-2} + 2\text{H}_2\text{O} = \text{Al}+3(2)\text{OH-1}(2)\text{Oxalic-2}(4) + 10\text{H}+1$ |    |       |        |                   |   |       |        |
| No                                                                                                                                   | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                                    | 25 | 0.6   | NaCl   | lgK:-4.62(0.04MD) | 4 | MGL   | 1044   |

| Reaction No. 7617                                                                           |    |       |        |                |   |       |        |
|---------------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Al}+3\text{EDTA-4} + \text{H}_2\text{O} = \text{Al}+3\text{OH-1EDTA-4} + \text{H}+1$ |    |       |        |                |   |       |        |
| No                                                                                          | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                           | 15 | 0.1   | KNO3   | lgK:-5.16(3SF) | 0 | MGL   | 140    |
| Note (1): Low wgt for internal consistency                                                  |    |       |        |                |   |       |        |
| 2                                                                                           | 15 |       |        | lgK:6.16(3SF)  | 0 | MGL   | 903    |
| 3                                                                                           | 20 | 0.1   | KNO3   | lgK:-5.89(3SF) | 6 | MPL   | 140    |

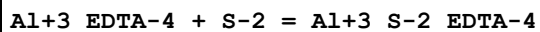
|   |    |     |         |                    |   |     |      |
|---|----|-----|---------|--------------------|---|-----|------|
| 4 | 25 | 0.2 | NaClO4  | lgK:-5.87 (3SF)    | 5 | MGL | 931  |
| 5 | 25 | 0.5 | KNO3    | lgK:-5.81 (0.01SD) | 6 | MSP | 1075 |
| 6 | 20 | 0.1 | Unknown | dH:5.0 (2SF)       | 6 | CRV | 552  |

## Reaction No. 7618



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.5   | KNO3   | lgK:4.8 (0.2SD) | 4 | MSP   | 1075   |

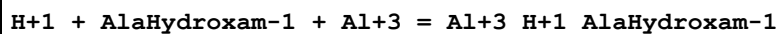
## Reaction No. 7619



Note: The species S-2 does not exist in aqueous solution See reaction: H+1 + S-2 = H+1\_S-2

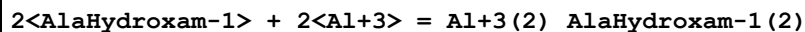
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.5   | KNO3   | lgK:10.3 (3SF) | 0 | MSP   | 1075   |

## Reaction No. 8799



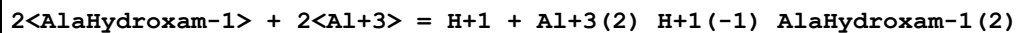
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:14.4 (0.03SD) | 4 | MGL   | 996    |

## Reaction No. 8800



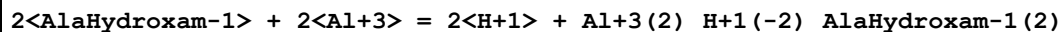
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:22.2 (0.06SD) | 4 | MGL   | 996    |

## Reaction No. 8801



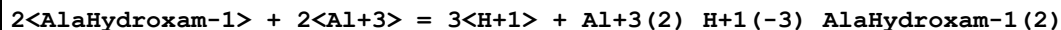
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:17.6 (0.09SD) | 4 | MGL   | 996    |

## Reaction No. 8802



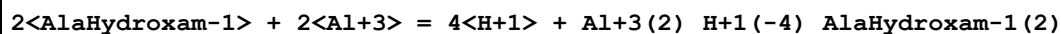
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:12.63 (0.02SD) | 6 | MGL   | 996    |

## Reaction No. 8803



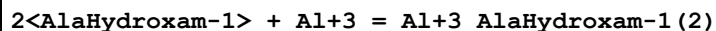
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:5.8(0.03SD) | 4 | MGL   | 996    |

## Reaction No. 8804



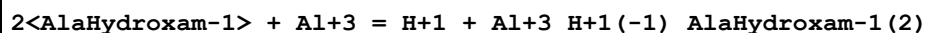
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:-2.(0.3SD) | 4 | MGL   | 996    |

## Reaction No. 8805



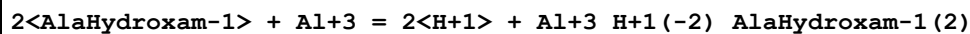
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:16.7(0.1SD) | 4 | MGL   | 996    |

## Reaction No. 8806



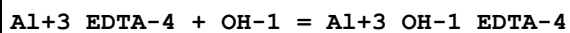
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:9.6(0.05SD) | 3 | MGL   | 996    |

## Reaction No. 8807



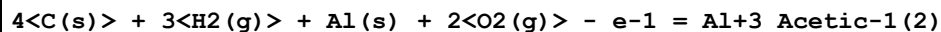
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:-0.2(0.2SD) | 6 | MGL   | 996    |

## Reaction No. 9118



| No | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------|---------------|---|-------|--------|
| 1  | 20 | 0.1   | Unknown | lgK:8.06(3SF) | 4 | ENS   | 773    |
| 2  | 25 | 0.1   | KNO3    | lgK:8.0(2SF)  | 5 | MEF   | 931    |
| 3  | 25 | 0.1   | Unknown | lgK:7.95(3SF) | 4 | ENS   | 773    |

## Reaction No. 9358



| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | dG:-299.36(5SF) | 3 | CRV   | 10174  |
| 2  | 50  | 0     | Inf. Dilution | dG:-299.73(5SF) | 2 | CRV   | 10174  |
| 3  | 75  | 0     | Inf. Dilution | dG:-300.38(5SF) | 2 | CRV   | 10174  |
| 4  | 100 | 0     | Inf. Dilution | dG:-301.25(5SF) | 1 | CRV   | 10174  |

|    |     |   |               |                  |   |     |       |
|----|-----|---|---------------|------------------|---|-----|-------|
| 5  | 125 | 0 | Inf. Dilution | dG:-302.33 (5SF) | 0 | CRV | 10174 |
| 6  | 150 | 0 | Inf. Dilution | dG:-303.58 (5SF) | 0 | CRV | 10174 |
| 7  | 175 | 0 | Inf. Dilution | dG:-304.98 (5SF) | 0 | CRV | 10174 |
| 8  | 200 | 0 | Inf. Dilution | dG:-306.53 (5SF) | 0 | CRV | 10174 |
| 9  | 225 | 0 | Inf. Dilution | dG:-308.22 (5SF) | 0 | CRV | 10174 |
| 10 | 250 | 0 | Inf. Dilution | dG:-310.01 (5SF) | 0 | CRV | 10174 |
| 11 | 300 | 0 | Inf. Dilution | dG:-313.90 (5SF) | 0 | CRV | 10174 |

| Reaction No. 9359                                             |     |       |               |                  |   |       |        |
|---------------------------------------------------------------|-----|-------|---------------|------------------|---|-------|--------|
| 2<C(s)> + 1.5<H2(g)> + Al(s) + O2(g) - 2<e-1> = Al+3_Acetic-1 |     |       |               |                  |   |       |        |
| No                                                            | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                             | 25  | 0     | Inf. Dilution | dG:-208.56 (5SF) | 3 | CRV   | 10174  |
| 2                                                             | 50  | 0     | Inf. Dilution | dG:-207.81 (5SF) | 2 | CRV   | 10174  |
| 3                                                             | 75  | 0     | Inf. Dilution | dG:-207.18 (5SF) | 2 | CRV   | 10174  |
| 4                                                             | 100 | 0     | Inf. Dilution | dG:-206.65 (5SF) | 1 | CRV   | 10174  |
| 5                                                             | 125 | 0     | Inf. Dilution | dG:-206.19 (5SF) | 0 | CRV   | 10174  |
| 6                                                             | 150 | 0     | Inf. Dilution | dG:-205.80 (5SF) | 0 | CRV   | 10174  |
| 7                                                             | 175 | 0     | Inf. Dilution | dG:-205.46 (5SF) | 0 | CRV   | 10174  |
| 8                                                             | 200 | 0     | Inf. Dilution | dG:-205.16 (5SF) | 0 | CRV   | 10174  |
| 9                                                             | 225 | 0     | Inf. Dilution | dG:-204.88 (5SF) | 0 | CRV   | 10174  |
| 10                                                            | 250 | 0     | Inf. Dilution | dG:-204.62 (5SF) | 0 | CRV   | 10174  |
| 11                                                            | 300 | 0     | Inf. Dilution | dG:-204.07 (5SF) | 0 | CRV   | 10174  |

| Reaction No. 9538                                 |    |       |        |                   |   |       |        |
|---------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Al+3 + H+1(2)_Catechin-4 = Al+3_H+1(2)_Catechin-4 |    |       |        |                   |   |       |        |
| No                                                | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                 | 25 | 0.1   | KCl    | lgK:17.1 (0.04SD) | 4 | MGL   | 1148   |

| Reaction No. 9539                                                      |    |       |        |                    |   |       |        |
|------------------------------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| Al+3_H+1(2)_Catechin-4 + H+1(2)_Catechin-4 = Al+3_H+1(4)_Catechin-4(2) |    |       |        |                    |   |       |        |
| No                                                                     | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                      | 25 | 0.1   | KCl    | lgK:13.89 (0.02SD) | 4 | MGL   | 1148   |

| Reaction No. 9540                                                         |    |       |        |                  |   |       |        |
|---------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Al+3_H+1(4)_Catechin-4(2) + H+1(2)_Catechin-4 = Al+3_H+1(6)_Catechin-4(3) |    |       |        |                  |   |       |        |
| No                                                                        | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                         | 25 | 0.1   | KCl    | lgK:9.9 (0.07SD) | 3 | MGL   | 1148   |

| Reaction No. 9541                                                             |    |       |        |                               |   |       |        |
|-------------------------------------------------------------------------------|----|-------|--------|-------------------------------|---|-------|--------|
| $\text{Al+3\_H+1(2)\_Catechin-4} = \text{Al+3\_H+1\_Catechin-4} + \text{H+1}$ |    |       |        |                               |   |       |        |
| No                                                                            | t  | I-Str | Medium | Value (Dev)                   | W | Tech. | Ref. # |
| 1                                                                             | 25 | 0.1   | KCl    | $\lg K: -6.0 (0.09\text{SD})$ | 4 | MGL   | 1148   |

| Reaction No. 9542                                                                |    |       |        |                               |   |       |        |
|----------------------------------------------------------------------------------|----|-------|--------|-------------------------------|---|-------|--------|
| $\text{Al+3\_H+1(2)\_Catechin-4} + \text{H+1} = \text{Al+3\_H+1(3)\_Catechin-4}$ |    |       |        |                               |   |       |        |
| No                                                                               | t  | I-Str | Medium | Value (Dev)                   | W | Tech. | Ref. # |
| 1                                                                                | 25 | 0.1   | KCl    | $\lg K: -8.2 (0.06\text{SD})$ | 4 | MGL   | 1148   |

| Reaction No. 9544                                                                     |    |       |        |                                 |   |       |        |
|---------------------------------------------------------------------------------------|----|-------|--------|---------------------------------|---|-------|--------|
| $\text{Al+3} + \text{H+1(3)\_NTA-3} = \text{Al+3\_NTA-3} + 3\langle\text{H+1}\rangle$ |    |       |        |                                 |   |       |        |
| No                                                                                    | t  | I-Str | Medium | Value (Dev)                     | W | Tech. | Ref. # |
| 1                                                                                     | 25 | 0.6   | NaCl   | $\lg K: -1.85 (0.007\text{MD})$ | 6 | MGL   | 1087   |

| Reaction No. 9545                                                                                                |    |       |               |                                |   |       |        |
|------------------------------------------------------------------------------------------------------------------|----|-------|---------------|--------------------------------|---|-------|--------|
| $\text{Al+3} + \text{H+1(3)\_NTA-3} + \text{H}_2\text{O} = \text{Al+3\_OH-1\_NTA-3} + 4\langle\text{H+1}\rangle$ |    |       |               |                                |   |       |        |
| No                                                                                                               | t  | I-Str | Medium        | Value (Dev)                    | W | Tech. | Ref. # |
| 1                                                                                                                | 25 | 0     | Inf. Dilution | $\lg K: -5.9 (2\text{SF})$     | 1 | ELC   |        |
| 2                                                                                                                | 25 | 0.6   | NaCl          | $\lg K: -7.28 (0.03\text{MD})$ | 6 | MGL   | 1087   |

| Reaction No. 9546                                                                                                                  |    |       |               |                                 |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|---------------------------------|---|-------|--------|
| $\text{Al+3} + \text{H+1(3)\_NTA-3} + 2\langle\text{H}_2\text{O}\rangle = \text{Al+3\_OH-1(2)\_NTA-3} + 5\langle\text{H+1}\rangle$ |    |       |               |                                 |   |       |        |
| No                                                                                                                                 | t  | I-Str | Medium        | Value (Dev)                     | W | Tech. | Ref. # |
| 1                                                                                                                                  | 25 | 0     | Inf. Dilution | $\lg K: -14.7 (3\text{SF})$     | 1 | ELC   |        |
| 2                                                                                                                                  | 25 | 0.6   | NaCl          | $\lg K: -15.47 (0.03\text{MD})$ | 6 | MGL   | 1087   |

| Reaction No. 9547                                                                                                                  |    |       |        |                                |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|--------------------------------|---|-------|--------|
| $2\langle\text{Al+3}\rangle + 2\langle\text{H+1(3)\_NTA-3}\rangle = \text{Al+3(2)\_H+1(-2)\_NTA-3(2)} + 8\langle\text{H+1}\rangle$ |    |       |        |                                |   |       |        |
| No                                                                                                                                 | t  | I-Str | Medium | Value (Dev)                    | W | Tech. | Ref. # |
| 1                                                                                                                                  | 25 | 0.6   | NaCl   | $\lg K: -12.7 (0.04\text{SD})$ | 4 | MGL   | 1087   |

| Reaction No. 9788                                       |    |       |        |                               |   |       |        |
|---------------------------------------------------------|----|-------|--------|-------------------------------|---|-------|--------|
| $\text{Lactic-1} + \text{Al+3} = \text{Al+3\_Lactic-1}$ |    |       |        |                               |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)                   | W | Tech. | Ref. # |
| 1                                                       | 15 | 0.105 | NaCl   | $\lg K: 1.93 (0.04\text{SD})$ | 2 | MGL   | 30072  |
| 2                                                       | 15 | 0.961 | NaCl   | $\lg K: 1.66 (0.06\text{SD})$ | 2 | MGL   | 30072  |

|                                                                              |    |       |               |                       |   |     |       |
|------------------------------------------------------------------------------|----|-------|---------------|-----------------------|---|-----|-------|
| 3                                                                            | 20 | 0.2   | KNO3          | lgK:2.38 (3SF)        | 2 | ENS | 1242  |
| 4                                                                            | 23 |       |               | lgK:0.85 (2SF)        | 0 | MSP | 906   |
| 5                                                                            | 25 | 0.105 | NaCl          | lgK:2.26 (0.01SD)     | 3 | MGL | 30072 |
| 6                                                                            | 25 | 0.2   | KCl           | lgK:2.48 (0.02SD)     | 4 | MGL | 4371  |
| 7                                                                            | 25 | 0.488 | NaCl          | lgK:2.12 (0.02SD)     | 3 | MGL | 30072 |
| 8                                                                            | 25 | 0.6   | NaCl          | lgK:2.36 (3SF)        | 4 | MGL | 4371  |
| Note (8): Cited value from Marklund et al, ACS, 90, 44, 228 with altered NDP |    |       |               |                       |   |     |       |
| 9                                                                            | 25 | 0.6   | Unknown       | lgK:2.36 (3SF)        | 4 | CMN | 1084  |
| 10                                                                           | 25 | 0.962 | NaCl          | lgK:2.07 (0.02SD)     | 3 | MGL | 30072 |
| 11                                                                           | 25 | 1     | NaClO4        | lgK:2.279 (4SF)       | 2 | MGL | 4204  |
| 12                                                                           | 25 | 1     | NaClO4        | lgK:4.255 (4SF)       | 0 | MGL | 4204  |
| 13                                                                           | 45 | 0.107 | NaCl          | lgK:2.51 (0.02SD)     | 3 | MGL | 30072 |
| 14                                                                           | 45 | 0.941 | NaCl          | lgK:2.34 (0.04SD)     | 3 | MGL | 30072 |
| 15                                                                           | 45 | 1     | NaClO4        | lgK:2.398 (4SF)       | 2 | MGL | 4204  |
| 16                                                                           | 25 | 0     | Inf. Dilution | dG:-16.20 (0.56SD) kJ | 3 | EFE | 30072 |
| 17                                                                           | 25 | 0     | Inf. Dilution | dH:28. (4.SD) kJ      | 2 | EFE | 30072 |
| 18                                                                           | 25 | 0.1   | ClO4-1 salt   | dH:7.9 (2SF) kJ       | 4 | MTD | 7703  |

| Reaction No. 9789                                                            |    |       |         |                   |   |       |        |
|------------------------------------------------------------------------------|----|-------|---------|-------------------|---|-------|--------|
| 2<Lactic-1> + Al+3 = Al+3_Lactic-1(2)                                        |    |       |         |                   |   |       |        |
| No                                                                           | t  | I-Str | Medium  | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                            | 15 | 0.105 | NaCl    | lgK:4.85 (0.01SD) | 3 | MGL   | 30072  |
| 2                                                                            | 15 | 0.961 | NaCl    | lgK:4.42 (0.01SD) | 3 | MGL   | 30072  |
| 3                                                                            | 20 | 0.2   | KNO3    | lgK:4.56 (3SF)    | 2 | ENS   | 1242   |
| 4                                                                            | 23 |       |         | lgK:2.92 (3SF)    | 0 | MSP   | 906    |
| 5                                                                            | 25 | 0.105 | NaCl    | lgK:4.88 (0.02SD) | 3 | MGL   | 30072  |
| 6                                                                            | 25 | 0.2   | KCl     | lgK:4.55 (0.06SD) | 3 | MGL   | 4371   |
| 7                                                                            | 25 | 0.488 | NaCl    | lgK:4.66 (0.03SD) | 3 | MGL   | 30072  |
| 8                                                                            | 25 | 0.6   | NaCl    | lgK:4.42 (3SF)    | 3 | MGL   | 4371   |
| Note (8): Cited value from Marklund et al, ACS, 90, 44, 228 with altered NDP |    |       |         |                   |   |       |        |
| 9                                                                            | 25 | 0.6   | Unknown | lgK:4.42 (3SF)    | 4 | CMN   | 1084   |
| 10                                                                           | 25 | 0.962 | NaCl    | lgK:4.60 (0.03SD) | 3 | MGL   | 30072  |
| 11                                                                           | 25 | 1     | NaClO4  | lgK:4.940 (4SF)   | 2 | MGL   | 4204   |
| 12                                                                           | 25 | 1     | NaClO4  | lgK:4.875 (4SF)   | 2 | MGL   | 4204   |
| 13                                                                           | 45 | 0.107 | NaCl    | lgK:5.19 (0.03SD) | 3 | MGL   | 30072  |
| 14                                                                           | 45 | 0.941 | NaCl    | lgK:5.07 (0.03SD) | 3 | MGL   | 30072  |



|    |    |   |                    |                     |   |     |       |
|----|----|---|--------------------|---------------------|---|-----|-------|
| 15 | 45 | 1 | NaClO <sub>4</sub> | lgK:5.255(4SF)      | 2 | MGL | 4204  |
| 16 | 25 | 0 | Inf. Dilution      | dG:-34.18(0.45SD)kJ | 3 | EFE | 30072 |
| 17 | 25 | 0 | Inf. Dilution      | dH:21.(3.SD)kJ      | 3 | EFE | 30072 |

| Reaction No. 9790                                                            |    |       |                  |                  |   |       |        |
|------------------------------------------------------------------------------|----|-------|------------------|------------------|---|-------|--------|
| 3<Lactic-1> + Al+3 = Al+3_Lactic-1(3)                                        |    |       |                  |                  |   |       |        |
| No                                                                           | t  | I-Str | Medium           | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                            | 20 | 0.2   | KNO <sub>3</sub> | lgK:6.66(3SF)    | 0 | ENS   | 1242   |
| 2                                                                            | 23 |       |                  | lgK:3.79(3SF)    | 0 | MSP   | 906    |
| 3                                                                            | 25 | 0.2   | KCl              | lgK:5.95(0.15SD) | 6 | MGL   | 4371   |
| 4                                                                            | 25 | 0.6   | NaCl             | lgK:5.79(3SF)    | 5 | MGL   | 4371   |
| Note (4): Cited value from Marklund et al, ACS, 90, 44, 228 with altered NDP |    |       |                  |                  |   |       |        |
| 5                                                                            | 25 | 0.6   | Unknown          | lgK:5.8(2SF)     | 4 | CMN   | 1084   |

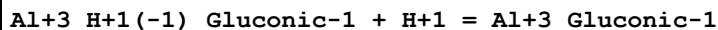
| Reaction No. 9791             |    |       |                    |               |   |       |        |
|-------------------------------|----|-------|--------------------|---------------|---|-------|--------|
| HIMDA-2 + Al+3 = Al+3_HIMDA-2 |    |       |                    |               |   |       |        |
| No                            | t  | I-Str | Medium             | Value (Dev)   | W | Tech. | Ref. # |
| 1                             | 25 | 0.1   | Unknown            | lgK:7.74(3SF) | 6 | CMN   | 1084   |
| 2                             | 25 | 0.5   | NaClO <sub>4</sub> | lgK:7.49(3SF) | 5 | MGL   | 2261   |
| 3                             | 25 | 0.1   | Unknown            | dH:6.2(2SF)   | 5 | CRV   | 552    |

| Reaction No. 9792                       |       |       |                  |               |   |       |        |
|-----------------------------------------|-------|-------|------------------|---------------|---|-------|--------|
| Al+3_Citric-3 + H+1 = Al+3_H+1_Citric-3 |       |       |                  |               |   |       |        |
| No                                      | t     | I-Str | Medium           | Value (Dev)   | W | Tech. | Ref. # |
| 1                                       | 25    | 0.1   | KNO <sub>3</sub> | lgK:2.94(3SF) | 6 | MGL   | 1244   |
| 2                                       | 25    | 0.1   | Unknown          | lgK:2.94(3SF) | 6 | CMN   | 1084   |
| 3                                       | 25:37 | 0.15  | KNO <sub>3</sub> | lgK:2.5(2SF)  | 5 | MPT   | 3607   |

| Reaction No. 9794                   |    |       |                  |                  |   |       |        |
|-------------------------------------|----|-------|------------------|------------------|---|-------|--------|
| Gluconic-1 + Al+3 = Al+3_Gluconic-1 |    |       |                  |                  |   |       |        |
| No                                  | t  | I-Str | Medium           | Value (Dev)      | W | Tech. | Ref. # |
| 1                                   | 25 | 0     | Inf. Dilution    | lgK:3.16(0.08SD) | 3 | EDH   | 30072  |
| 2                                   | 25 | 0.1   | KNO <sub>3</sub> | lgK:1.98(3SF)    | 2 | MGL   | 1244   |
| 3                                   | 25 | 0.1   | NaCl             | lgK:2.01(0.02SD) | 2 | MGL   | 3975   |
| 4                                   | 25 | 0.1   | Unknown          | lgK:1.98(3SF)    | 2 | CMN   | 1084   |
| 5                                   | 25 | 0.107 | NaCl             | lgK:2.49(0.07SD) | 2 | MGL   | 30072  |
| 6                                   | 25 | 0.254 | NaCl             | lgK:2.32(0.40SD) | 3 | MGL   | 30072  |

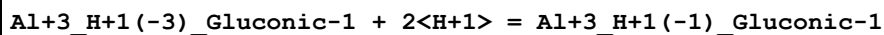
|   |    |       |      |                   |   |     |       |
|---|----|-------|------|-------------------|---|-----|-------|
| 7 | 25 | 0.486 | NaCl | lgK:2.17 (0.16SD) | 4 | MGL | 30072 |
| 8 | 25 | 0.73  | NaCl | lgK:2.04 (0.16SD) | 4 | MGL | 30072 |
| 9 | 25 | 0.963 | NaCl | lgK:1.56 (0.35SD) | 2 | MGL | 30072 |

## Reaction No. 9795



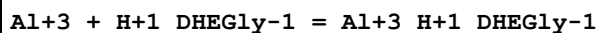
| No | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3    | lgK:2.87 (3SF)   | 6 | MGL   | 1244   |
| 2  | 25 | 0.1   | NaCl    | lgK:2.9 (0.04SD) | 5 | MGL   | 3975   |
| 3  | 25 | 0.1   | Unknown | lgK:2.87 (3SF)   | 6 | CMN   | 1084   |

## Reaction No. 9796



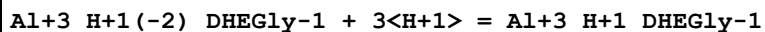
| No | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3    | lgK:9.29 (3SF)   | 6 | MGL   | 1244   |
| 2  | 25 | 0.1   | NaCl    | lgK:9.3 (0.07SD) | 5 | MGL   | 3975   |
| 3  | 25 | 0.1   | Unknown | lgK:9.29 (3SF)   | 6 | CMN   | 1084   |

## Reaction No. 9798



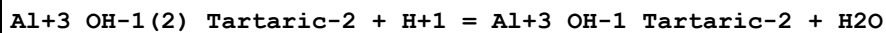
| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | Unknown | lgK:3.38 (3SF) | 6 | CRV   | 552    |
| 2  | 25 | 0.1   | KNO3    | lgK:3.38 (3SF) | 6 | MGL   | 1244   |
| 3  | 25 | 0.1   | Unknown | lgK:3.38 (3SF) | 6 | CMN   | 1084   |

## Reaction No. 9799



| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | Unknown | lgK:13.72 (4SF) | 6 | CRV   | 552    |
| 2  | 25 | 0.1   | KNO3    | lgK:13.72 (4SF) | 6 | MGL   | 1244   |
| 3  | 25 | 0.1   | Unknown | lgK:13.72 (4SF) | 6 | CMN   | 1084   |

## Reaction No. 9800



| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3    | lgK:5.15 (3SF) | 6 | MGL   | 1244   |
| 2  | 25 | 0.1   | Unknown | lgK:5.15 (3SF) | 6 | CMN   | 1084   |

| Reaction No. 9801                                              |    |       |        |                 |   |       |        |
|----------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Al+3_OH-1(4) + BisTris = Al+3_H+1(-2)_BisTris_OH-1(2) + 2<H2O> |    |       |        |                 |   |       |        |
| No                                                             | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                              | 25 | 0.1   | NaCl/m | lgK:2.6(0.03SD) | 4 | MSL   | 1105   |

| Reaction No. 9802                                                |    |       |        |               |   |       |        |
|------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Al+3_OH-1(4) + DGlucose = Al+3_H+1(-2)_DGlucose_OH-1(2) + 2<H2O> |    |       |        |               |   |       |        |
| No                                                               | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                | 25 | 0.1   | NaCl/m | lgK:0.65(2SF) | 6 | MSL   | 1105   |

| Reaction No. 9864                                     |    |       |         |               |   |       |        |
|-------------------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Al+3_H+1(-1)_Tartaric-2(2) + H+1 = Al+3_Tartaric-2(2) |    |       |         |               |   |       |        |
| No                                                    | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                     | 25 | 0.1   | KNO3    | lgK:3.72(3SF) | 6 | MGL   | 1244   |
| 2                                                     | 25 | 0.1   | Unknown | lgK:3.72(3SF) | 6 | CMN   | 1084   |

| Reaction No. 9865                                                |    |       |         |                |   |       |        |
|------------------------------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Al+3_H+1(-3)_Tartaric-2(2) + 2<H+1> = Al+3_H+1(-1)_Tartaric-2(2) |    |       |         |                |   |       |        |
| No                                                               | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                | 25 | 0.1   | KNO3    | lgK:12.67(4SF) | 6 | MGL   | 1244   |
| 2                                                                | 25 | 0.1   | Unknown | lgK:12.67(4SF) | 6 | CMN   | 1084   |

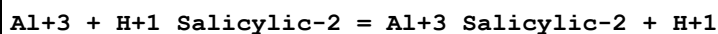
| Reaction No. 9866                                          |    |       |         |                |   |       |        |
|------------------------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Al+3 + 3<Tartaric-2> = Al+3_H+1(-4)_Tartaric-2(3) + 4<H+1> |    |       |         |                |   |       |        |
| No                                                         | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                          | 25 | 0.1   | KNO3    | lgK:15.92(4SF) | 6 | MGL   | 1244   |
| 2                                                          | 25 | 0.1   | Unknown | lgK:15.92(4SF) | 6 | CMN   | 1084   |

| Reaction No. 9867                                             |    |       |         |                |   |       |        |
|---------------------------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Al+3_H+1(-5)_Tartaric-2(3) + H+1 = Al+3_H+1(-4)_Tartaric-2(3) |    |       |         |                |   |       |        |
| No                                                            | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                             | 25 | 0.1   | KNO3    | lgK:10.89(4SF) | 6 | MGL   | 1244   |
| 2                                                             | 25 | 0.1   | Unknown | lgK:10.89(4SF) | 6 | CMN   | 1084   |

| Reaction No. 9868                                             |   |       |        |             |   |       |        |
|---------------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| Al+3_H+1(-6)_Tartaric-2(3) + H+1 = Al+3_H+1(-5)_Tartaric-2(3) |   |       |        |             |   |       |        |
| No                                                            | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

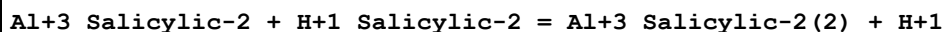
|   |    |     |         |               |   |     |      |
|---|----|-----|---------|---------------|---|-----|------|
| 1 | 25 | 0.1 | KNO3    | lgK:12.7(3SF) | 6 | MGL | 1244 |
| 2 | 25 | 0.1 | Unknown | lgK:12.7(3SF) | 5 | CMN | 1084 |

## Reaction No. 9869



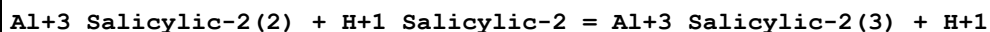
| No | t     | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|-------|-------|---------|----------------|---|-------|--------|
| 1  | 20    | 0.1   | Unknown | lgK:-0.2(1SF)  | 4 | CMN   | 1084   |
| 2  | 25    | 0.1   | Unknown | lgK:0.0(1SF)   | 4 | CRV   | 2556   |
| 3  | 25    | 0.2   | KCl     | lgK:-0.18(2SF) | 6 | MGL   | 2470   |
| 4  | 25    | 0.5   | Unknown | lgK:-0.33(2SF) | 4 | CRV   | 2556   |
| 5  | 26:28 | 0     | NaClO4  | lgK:0.26(2SF)  | 4 | MSP   | 931    |
| 6  | 26:28 | 0.02  | NaClO4  | lgK:0.06(1SF)  | 4 | MSP   | 931    |
| 7  | 26:28 | 0.05  | NaClO4  | lgK:0.01(1SF)  | 4 | MSP   | 931    |
| 8  | 26:28 | 0.1   | NaClO4  | lgK:-0.13(2SF) | 4 | MSP   | 931    |
| 9  | 26:28 | 0.2   | NaClO4  | lgK:-0.18(2SF) | 4 | MSP   | 931    |

## Reaction No. 9870



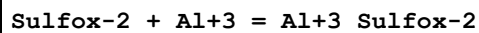
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | Unknown       | lgK:-2.8(2SF)  | 4 | CMN   | 1084   |
| 2  | 25 | 0     | Inf. Dilution | lgK:-3.4(2SF)  | 1 | ERG   |        |
| 3  | 25 | 0.2   | KCl           | lgK:-2.89(3SF) | 4 | MGL   | 2470   |
| 4  | 25 | 0.5   | Unknown       | lgK:-2.61(3SF) | 3 | CRV   | 2556   |

## Reaction No. 9871



| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 20 | 0.1   | Unknown       | lgK:-6.5(2SF) | 6 | CMN   | 1084   |
| 2  | 25 | 0     | Inf. Dilution | lgK:-6.6(2SF) | 1 | ERG   |        |

## Reaction No. 9872



| No | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------|---------------|---|-------|--------|
| 1  | 23 |       |         | lgK:9.69(3SF) | 0 | MSP   | 906    |
| 2  | 25 | 0.1   | Unknown | lgK:9.2(2SF)  | 4 | CMN   | 1084   |
| 3  | 25 | 0.1   | Unknown | lgK:8.7(2SF)  | 4 | ENS   | 6953   |

| Reaction No. 9873                                                                 |    |       |         |               |   |       |        |
|-----------------------------------------------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| $\text{Al}+3 + 2\langle\text{Sulfox-2}\rangle = \text{Al}+3\_ \text{Sulfox-2}(2)$ |    |       |         |               |   |       |        |
| No                                                                                | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                 | 25 | 0.1   | Unknown | lgK:17.6(3SF) | 4 | CMN   | 1084   |
| 2                                                                                 | 25 | 0.1   | Unknown | lgK:16.6(3SF) | 4 | ENS   | 6953   |

| Reaction No. 9874                                                                 |    |       |         |               |   |       |        |
|-----------------------------------------------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| $\text{Al}+3 + 3\langle\text{Sulfox-2}\rangle = \text{Al}+3\_ \text{Sulfox-2}(3)$ |    |       |         |               |   |       |        |
| No                                                                                | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                 | 25 | 0.1   | Unknown | lgK:24.7(3SF) | 4 | CMN   | 1084   |
| 2                                                                                 | 25 | 0.1   | Unknown | lgK:23.2(3SF) | 4 | ENS   | 6953   |

| Reaction No. 9875                                               |    |       |         |               |   |       |        |
|-----------------------------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| $\text{Al}+3 + \text{Maltol-1} = \text{Al}+3\_ \text{Maltol-1}$ |    |       |         |               |   |       |        |
| No                                                              | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                               | 25 | 0.1   | KNO3    | lgK:7.7(2SF)  | 4 | MGL   | 2171   |
| 2                                                               | 25 | 0.6   | Unknown | lgK:8.25(3SF) | 6 | CMN   | 1084   |

| Reaction No. 9876                                                                 |    |       |         |                |   |       |        |
|-----------------------------------------------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| $\text{Al}+3 + 2\langle\text{Maltol-1}\rangle = \text{Al}+3\_ \text{Maltol-1}(2)$ |    |       |         |                |   |       |        |
| No                                                                                | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                 | 25 | 0.1   | KNO3    | lgK:15.25(4SF) | 5 | ENS   | 2171   |
| 2                                                                                 | 25 | 0.6   | Unknown | lgK:14.26(4SF) | 6 | CMN   | 1084   |

| Reaction No. 9877                                                                 |    |       |         |                |   |       |        |
|-----------------------------------------------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| $\text{Al}+3 + 3\langle\text{Maltol-1}\rangle = \text{Al}+3\_ \text{Maltol-1}(3)$ |    |       |         |                |   |       |        |
| No                                                                                | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                 | 25 | 0.1   | KNO3    | lgK:21.90(4SF) | 3 | ENS   | 2171   |
| 2                                                                                 | 25 | 0.6   | Unknown | lgK:20.16(4SF) | 0 | CMN   | 1084   |

| Reaction No. 9878                                                                                     |    |       |         |                   |   |       |        |
|-------------------------------------------------------------------------------------------------------|----|-------|---------|-------------------|---|-------|--------|
| $\text{Al}+3 + \text{H}+1(2)\_ \text{Cat-2} = \text{Al}+3\_ \text{Cat-2} + 2\langle\text{H}+1\rangle$ |    |       |         |                   |   |       |        |
| No                                                                                                    | t  | I-Str | Medium  | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                     | 25 | 0.1   | KNO3    | lgK:-6.08(3SF)    | 6 | MGL   | 1244   |
| 2                                                                                                     | 25 | 0.1   | Unknown | lgK:-6.0(2SF)     | 6 | CMN   | 1084   |
| 3                                                                                                     | 25 | 0.2   | KCl     | lgK:-6.08(3SF)    | 6 | MGL   | 2470   |
| 4                                                                                                     | 25 | 0.6   | NaCl    | lgK:-6.34(0.01SD) | 7 | MGL   | 2357   |

| Reaction No. 9879                                  |    |       |         |                |   |       |        |
|----------------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Al+3_Cat-2 + H+1(2)_Cat-2 = Al+3_Cat-2(2) + 2<H+1> |    |       |         |                |   |       |        |
| No                                                 | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                  | 25 | 0.1   | KNO3    | lgK:-9.18(3SF) | 6 | MGL   | 1244   |
| 2                                                  | 25 | 0.1   | Unknown | lgK:-9.10(3SF) | 6 | CMN   | 1084   |
| 3                                                  | 25 | 0.2   | KCl     | lgK:-9.22(3SF) | 6 | MGL   | 2470   |

| Reaction No. 9880                                     |    |       |         |                 |   |       |        |
|-------------------------------------------------------|----|-------|---------|-----------------|---|-------|--------|
| Al+3_Cat-2(2) + H+1(2)_Cat-2 = Al+3_Cat-2(3) + 2<H+1> |    |       |         |                 |   |       |        |
| No                                                    | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                     | 25 | 0.1   | KNO3    | lgK:-13.52(4SF) | 6 | MGL   | 1244   |
| 2                                                     | 25 | 0.1   | Unknown | lgK:-13.61(4SF) | 6 | CMN   | 1084   |
| 3                                                     | 25 | 0.2   | KCl     | lgK:-13.59(4SF) | 6 | MGL   | 2470   |

| Reaction No. 9881                       |    |       |         |               |   |       |        |
|-----------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Al+3_Cat-2(2) + H+1 = Al+3_H+1_Cat-2(2) |    |       |         |               |   |       |        |
| No                                      | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                       | 25 | 0.1   | KNO3    | lgK:6.03(3SF) | 6 | MGL   | 1244   |
| 2                                       | 25 | 0.1   | Unknown | lgK:6.05(3SF) | 6 | CMN   | 1084   |

| Reaction No. 9882                       |    |       |         |               |   |       |        |
|-----------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Al+3_Cat-2(3) + H+1 = Al+3_H+1_Cat-2(3) |    |       |         |               |   |       |        |
| No                                      | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                       | 25 | 0.1   | KNO3    | lgK:8.01(3SF) | 6 | MGL   | 1244   |
| 2                                       | 25 | 0.1   | Unknown | lgK:8.05(3SF) | 6 | CMN   | 1084   |

| Reaction No. 9884                                 |    |       |               |               |   |       |        |
|---------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Al+3 + H+1(2)_SulfCat-3 = Al+3_SulfCat-3 + 2<H+1> |    |       |               |               |   |       |        |
| No                                                | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                 | 20 | 0.1   | Unknown       | lgK:-4.7(2SF) | 6 | CMN   | 1084   |
| 2                                                 | 25 | 0     | Inf. Dilution | lgK:-4.6(2SF) | 1 | ESC   |        |

| Reaction No. 9885                                              |    |       |         |               |   |       |        |
|----------------------------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Al+3_SulfCat-3 + H+1(2)_SulfCat-3 = Al+3_SulfCat-3(2) + 2<H+1> |    |       |         |               |   |       |        |
| No                                                             | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                              | 20 | 0.1   | Unknown | lgK:-8.0(2SF) | 6 | CMN   | 1084   |

|   |    |   |               |                |   |     |  |
|---|----|---|---------------|----------------|---|-----|--|
| 2 | 25 | 0 | Inf. Dilution | lgK:-7.9 (2SF) | 1 | ESC |  |
|---|----|---|---------------|----------------|---|-----|--|

| Reaction No. 9886                                                 |    |       |               |                 |   |       |        |
|-------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Al+3_SulfCat-3(2) + H+1(2)_SulfCat-3 = Al+3_SulfCat-3(3) + 2<H+1> |    |       |               |                 |   |       |        |
| No                                                                | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                 | 20 | 0.1   | Unknown       | lgK:-12.0 (3SF) | 6 | CMN   | 1084   |
| 2                                                                 | 25 | 0     | Inf. Dilution | lgK:-11.8 (3SF) | 1 | ESC   |        |

| Reaction No. 9887                             |    |       |         |                 |   |       |        |
|-----------------------------------------------|----|-------|---------|-----------------|---|-------|--------|
| Al+3 + H+1(2)_Tiron-4 = Al+3_Tiron-4 + 2<H+1> |    |       |         |                 |   |       |        |
| No                                            | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                                             | 25 | 0.1   | Unknown | lgK:-3.21 (3SF) | 6 | CMN   | 1084   |

| Reaction No. 9888                                        |    |       |         |                |   |       |        |
|----------------------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Al+3_Tiron-4 + H+1(2)_Tiron-4 = Al+3_Tiron-4(2) + 2<H+1> |    |       |         |                |   |       |        |
| No                                                       | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                        | 25 | 0.1   | Unknown | lgK:-6.2 (2SF) | 6 | CMN   | 1084   |

| Reaction No. 9889                                           |    |       |         |                 |   |       |        |
|-------------------------------------------------------------|----|-------|---------|-----------------|---|-------|--------|
| Al+3_Tiron-4(2) + H+1(2)_Tiron-4 = Al+3_Tiron-4(3) + 2<H+1> |    |       |         |                 |   |       |        |
| No                                                          | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                           | 25 | 0.1   | Unknown | lgK:-10.0 (3SF) | 6 | CMN   | 1084   |

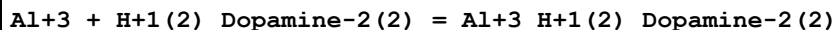
| Reaction No. 9891                                           |    |       |         |                 |   |       |        |
|-------------------------------------------------------------|----|-------|---------|-----------------|---|-------|--------|
| Al+3 + H+1(2)_Chromotropic-4 = Al+3_Chromotropic-4 + 2<H+1> |    |       |         |                 |   |       |        |
| No                                                          | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                           | 20 | 0.1   | Unknown | lgK:-3.82 (3SF) | 6 | CMN   | 1084   |
| 2                                                           | 25 | 0.004 | Unknown | lgK:-3.20 (3SF) | 4 | ENS   | 6953   |
| 3                                                           | 25 | 0.1   | NaClO4  | lgK:-4.70 (3SF) | 4 | ENS   | 6953   |

| Reaction No. 9892                                                             |    |       |               |                 |   |       |        |
|-------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Al+3_Chromotropic-4 + H+1(2)_Chromotropic-4 = Al+3_Chromotropic-4(2) + 2<H+1> |    |       |               |                 |   |       |        |
| No                                                                            | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                             | 20 | 0.1   | Unknown       | lgK:-8.25 (3SF) | 6 | CMN   | 1084   |
| 2                                                                             | 25 | 0     | Inf. Dilution | lgK:-8.1 (2SF)  | 1 | ESC   |        |

| Reaction No. 9896                           |  |  |  |  |  |  |  |
|---------------------------------------------|--|--|--|--|--|--|--|
| Al+3 + H+1_Dopamine-2 = Al+3_H+1_Dopamine-2 |  |  |  |  |  |  |  |

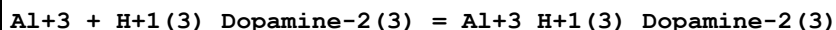
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:13.32 (4SF) | 3 | CRV   | 552    |
| 2  | 25 | 0.1   | Unknown | lgK:13.32 (4SF) | 6 | CMN   | 1084   |

## Reaction No. 9897



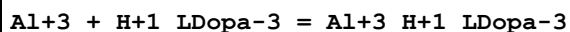
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:23.99 (4SF) | 6 | CMN   | 1084   |

## Reaction No. 9898



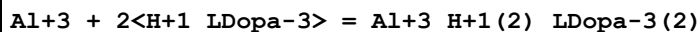
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:30.63 (4SF) | 6 | CMN   | 1084   |

## Reaction No. 9899



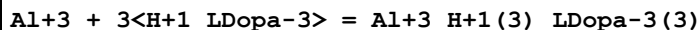
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:12.72 (4SF) | 6 | ENS   | 1084   |

## Reaction No. 9900



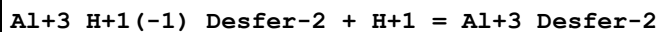
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:22.62 (4SF) | 4 | ENS   | 1084   |

## Reaction No. 9901



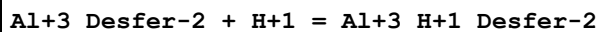
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:28.43 (4SF) | 4 | ENS   | 1084   |

## Reaction No. 9904



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:9.43 (0.01SD) | 6 | MGL   | 1382   |

## Reaction No. 9905



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:1.18 (0.01SD) | 4 | MGL   | 1382   |



| Reaction No. 9920                                                                                                 |    |       |         |                 |   |       |        |
|-------------------------------------------------------------------------------------------------------------------|----|-------|---------|-----------------|---|-------|--------|
| $3\text{Al}^{3+} + 3\text{Citric-3} + \text{H}_2\text{O} = \text{Al}_3(\text{OH})_3\text{Citric-3} + 4\text{H}^+$ |    |       |         |                 |   |       |        |
| No                                                                                                                | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                 | 25 | 0.6   | Unknown | lgK:14.43 (4SF) | 6 | CMN   | 1084   |

| Reaction No. 9921                                                                                                     |    |       |                    |                |   |       |        |
|-----------------------------------------------------------------------------------------------------------------------|----|-------|--------------------|----------------|---|-------|--------|
| $\text{Al}_3(\text{OH})_3\text{Citric-3} + \text{H}^+ = \text{Al}_3(\text{OH})_3\text{Citric-3} + \text{H}_2\text{O}$ |    |       |                    |                |   |       |        |
| No                                                                                                                    | t  | I-Str | Medium             | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                                     | 25 | 0.6   | Unknown            | lgK:6.23 (3SF) | 6 | CMN   | 1084   |
| 2                                                                                                                     | 33 | 0.25  | NaClO <sub>4</sub> | lgK:6.8 (2SF)  | 4 | MGL   | 931    |

| Reaction No. 9992                                                                            |    |       |               |                  |   |       |        |
|----------------------------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| $3\text{Al}^{3+} + 3\text{Citric-3} = \text{Al}_3(\text{OH})_3\text{Citric-3} + 7\text{H}^+$ |    |       |               |                  |   |       |        |
| No                                                                                           | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                            | 25 | 0     | Inf. Dilution | lgK:-10.6 (3SF)  | 1 | ERG   |        |
| 2                                                                                            | 25 | 0.6   | Unknown       | lgK:-10.91 (4SF) | 5 | CMN   | 1084   |

| Reaction No. 9993                                                          |       |       |                  |                   |   |       |        |
|----------------------------------------------------------------------------|-------|-------|------------------|-------------------|---|-------|--------|
| $\text{Al}_3(\text{Cat-2})_2 + \text{Cat-2} = \text{Al}_3(\text{Cat-2})_3$ |       |       |                  |                   |   |       |        |
| No                                                                         | t     | I-Str | Medium           | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                          | 20    | 0.1   | KNO <sub>3</sub> | lgK:8.9 (2SF)     | 4 | MGL   | 3494   |
| 2                                                                          | 20    | 0.1   | Unknown          | lgK:8.9 (2SF)     | 4 | CMN   | 1084   |
| 3                                                                          | 20    | 0.2   | KCl              | lgK:9.00 (3SF)    | 6 | MGL   | 999    |
| 4                                                                          | 22:25 | 0.2   | KNO <sub>3</sub> | lgK:7.74 (0.18SD) | 0 | MGL   | 2349   |
| 5                                                                          | 25    | 0.1   | KCl              | lgK:9.0 (0.08SD)  | 3 | MGL   | 1148   |
| 6                                                                          | 25    | 0.2   | KCl              | lgK:8.69 (3SF)    | 5 | MGL   | 2470   |

| Reaction No. 9994                                                        |       |       |                  |                    |   |       |        |
|--------------------------------------------------------------------------|-------|-------|------------------|--------------------|---|-------|--------|
| $\text{Al}_3(\text{Cat-2}) + \text{Cat-2} = \text{Al}_3(\text{Cat-2})_2$ |       |       |                  |                    |   |       |        |
| No                                                                       | t     | I-Str | Medium           | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                        | 20    | 0.1   | KNO <sub>3</sub> | lgK:13.6 (3SF)     | 4 | MGL   | 3494   |
| 2                                                                        | 20    | 0.1   | Unknown          | lgK:13.6 (3SF)     | 4 | CMN   | 1084   |
| 3                                                                        | 20    | 0.2   | KCl              | lgK:13.48 (4SF)    | 6 | MGL   | 999    |
| 4                                                                        | 22:25 | 0.2   | KNO <sub>3</sub> | lgK:12.36 (0.15SD) | 0 | MGL   | 2349   |
| 5                                                                        | 25    | 0.1   | KCl              | lgK:13.66 (0.01SD) | 6 | MGL   | 1148   |
| 6                                                                        | 25    | 0.2   | KCl              | lgK:13.06 (4SF)    | 5 | MGL   | 2470   |

| Reaction No. 10089                                                                                        |    |       |        |                 |   |       |        |
|-----------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{Al}+3 + \text{H}+1\_ \text{DiPhosGlyceric-5} = \text{Al}+3\_ \text{H}+1\_ \text{DiPhosGlyceric-5}$ |    |       |        |                 |   |       |        |
| No                                                                                                        | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                         | 25 | 0.2   | KCl    | lgK:6.0(0.03SD) | 4 | MGL   | 1085   |

| Reaction No. 10090                                                                                                              |    |       |        |                 |   |       |        |
|---------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{Al}+3\_ \text{H}+1\_ \text{DiPhosGlyceric-5(2)} + \text{H}+1 = \text{Al}+3\_ \text{H}+1(2)\_ \text{DiPhosGlyceric-5(2)}$ |    |       |        |                 |   |       |        |
| No                                                                                                                              | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                               | 25 | 0.2   | KCl    | lgK:5.6(0.03SD) | 4 | MGL   | 1085   |

| Reaction No. 10091                                                                                              |    |       |        |                 |   |       |        |
|-----------------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{Al}+3\_ \text{DiPhosGlyceric-5(2)} + \text{H}+1 = \text{Al}+3\_ \text{H}+1\_ \text{DiPhosGlyceric-5(2)}$ |    |       |        |                 |   |       |        |
| No                                                                                                              | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                               | 25 | 0.2   | KCl    | lgK:6.3(0.03SD) | 4 | MGL   | 1085   |

| Reaction No. 10092                                                                                                                                     |    |       |        |                 |   |       |        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{Al}+3\_ \text{H}+1\_ \text{DiPhosGlyceric-5} + \text{H}+1\_ \text{DiPhosGlyceric-5} = \text{Al}+3\_ \text{H}+1(2)\_ \text{DiPhosGlyceric-5(2)}$ |    |       |        |                 |   |       |        |
| No                                                                                                                                                     | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                                                      | 25 | 0.2   | KCl    | lgK:4.2(0.03SD) | 4 | MGL   | 1085   |

| Reaction No. 10093                                                                                        |    |       |        |                  |   |       |        |
|-----------------------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $\text{H}+1 + \text{DiPhosGlyceric-5} + \text{Al}+3 = \text{Al}+3\_ \text{H}+1\_ \text{DiPhosGlyceric-5}$ |    |       |        |                  |   |       |        |
| No                                                                                                        | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                         | 25 | 0.2   | KCl    | lgK:13.1(0.03SD) | 4 | MGL   | 1085   |

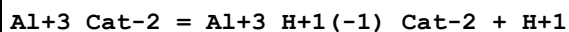
| Reaction No. 10094                                                                                                           |    |       |        |                  |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $2\text{<H}+1\text{>} + 2\text{<DiPhosGlyceric-5>} + \text{Al}+3 = \text{Al}+3\_ \text{H}+1(2)\_ \text{DiPhosGlyceric-5(2)}$ |    |       |        |                  |   |       |        |
| No                                                                                                                           | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                                            | 25 | 0.2   | KCl    | lgK:24.4(0.03SD) | 4 | MGL   | 1085   |

| Reaction No. 10095                                                                                              |    |       |        |                  |   |       |        |
|-----------------------------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $\text{H}+1 + 2\text{<DiPhosGlyceric-5>} + \text{Al}+3 = \text{Al}+3\_ \text{H}+1\_ \text{DiPhosGlyceric-5(2)}$ |    |       |        |                  |   |       |        |
| No                                                                                                              | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                               | 25 | 0.2   | KCl    | lgK:18.8(0.03SD) | 4 | MGL   | 1085   |

| Reaction No. 10096                                                                    |   |       |        |             |   |       |        |
|---------------------------------------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| $2\text{<DiPhosGlyceric-5>} + \text{Al}+3 = \text{Al}+3\_ \text{DiPhosGlyceric-5(2)}$ |   |       |        |             |   |       |        |
| No                                                                                    | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

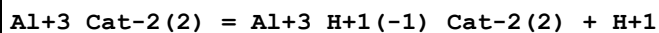
|   |    |     |     |                   |   |     |      |
|---|----|-----|-----|-------------------|---|-----|------|
| 1 | 25 | 0.2 | KCl | lgK:12.5 (0.03SD) | 6 | MGL | 1085 |
|---|----|-----|-----|-------------------|---|-----|------|

## Reaction No. 10097



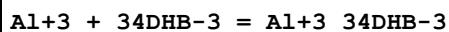
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:-6.1 (0.09SD) | 4 | MGL   | 1148   |

## Reaction No. 10098



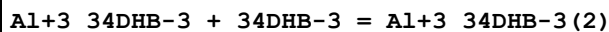
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:-8.10 (0.01SD) | 4 | MGL   | 1148   |

## Reaction No. 10099



| No | t     | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|-------|-------|--------|--------------------|---|-------|--------|
| 1  | 22:25 | 0.2   | KNO3   | lgK:15.03 (0.08SD) | 0 | MGL   | 2349   |
| 2  | 25    | 0.1   | KCl    | lgK:16.87 (0.02SD) | 4 | MGL   | 1148   |
| 3  | 25    | 0.2   | KCl    | lgK:16.47 (0.02SD) | 6 | MGL   | 2470   |
| 4  | 25    | 0.5   | NaClO4 | lgK:16.1 (3SF)     | 5 | ENS   | 6953   |

## Reaction No. 10100



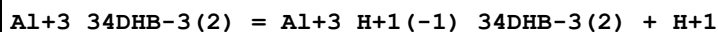
| No | t     | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|-------|-------|--------|--------------------|---|-------|--------|
| 1  | 22:25 | 0.2   | KNO3   | lgK:12.61 (0.12SD) | 2 | MGL   | 2349   |
| 2  | 25    | 0.1   | KCl    | lgK:13.0 (0.05SD)  | 4 | MGL   | 1148   |

## Reaction No. 10101



| No | t     | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|-------|-------|--------|-------------------|---|-------|--------|
| 1  | 22:25 | 0.2   | KNO3   | lgK:9.91 (0.35SD) | 0 | MGL   | 2349   |
| 2  | 25    | 0.1   | KCl    | lgK:8.8 (0.05SD)  | 4 | MGL   | 1148   |

## Reaction No. 10103



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:-8.4 (0.03SD) | 4 | MGL   | 1148   |

## Reaction No. 10104

| Al+3 + H+1(2)_34DHB-3 = Al+3_H+1(2)_34DHB-3 |    |       |        |                |   |       |        |
|---------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| No                                          | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | KCl    | lgK:2.9(0.2SD) | 3 | MGL   | 1148   |

| Reaction No. 10105                    |    |       |        |                |   |       |        |
|---------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Al+3_34DHB-3 + H+1 = Al+3_H+1_34DHB-3 |    |       |        |                |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | KCl    | lgK:4.7(0.1SD) | 4 | MGL   | 1148   |

| Reaction No. 10115                                           |    |       |         |               |   |       |        |
|--------------------------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| 2<Al+3> + 2<Maltol-1> = Al+3(2)_H+1(-2)_Maltol-1(2) + 2<H+1> |    |       |         |               |   |       |        |
| No                                                           | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                            | 25 | 0.1   | Unknown | lgK:8.02(3SF) | 6 | CMN   | 1084   |

| Reaction No. 10122                          |       |       |        |                  |   |       |        |
|---------------------------------------------|-------|-------|--------|------------------|---|-------|--------|
| Al+3_Citric-3 + Citric-3 = Al+3_Citric-3(2) |       |       |        |                  |   |       |        |
| No                                          | t     | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                           | 25    | 0.1   | Inert  | lgK:5.01(0.01SD) | 4 | MGL   | 1151   |
| 2                                           | 25    | 0.1   | KCl    | lgK:4.8(0.07SD)  | 4 | MGL   | 1151   |
| 3                                           | 25:37 | 0.15  | KNO3   | lgK:5.0(2SF)     | 5 | MPT   | 3607   |
| 4                                           | 37    | 0.15  | NaCl   | lgK:4.82(3SF)    | 5 | MGL   | 925    |
| 5                                           | 37    | 0.15  | NaCl   | lgK:5.22(3SF)    | 5 | MGL   | 10029  |

| Reaction No. 10123                                |    |       |        |                   |   |       |        |
|---------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Al+3_Citric-3(2) = Al+3_H+1(-1)_Citric-3(2) + H+1 |    |       |        |                   |   |       |        |
| No                                                | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                 | 25 | 0.1   | Inert  | lgK:-6.07(0.01SD) | 4 | MGL   | 1151   |
| 2                                                 | 25 | 0.1   | KCl    | lgK:-6.1(0.1SD)   | 4 | MGL   | 1151   |

| Reaction No. 10124                                        |    |       |        |                   |   |       |        |
|-----------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Al+3_H+1(-1)_Citric-3(2) = Al+3_H+1(-2)_Citric-3(2) + H+1 |    |       |        |                   |   |       |        |
| No                                                        | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                         | 25 | 0.1   | Inert  | lgK:-7.18(0.02SD) | 4 | MGL   | 1151   |
| 2                                                         | 25 | 0.1   | KCl    | lgK:-7.17(0.02SD) | 4 | MGL   | 1151   |

| Reaction No. 10125                                    |  |  |  |  |  |  |  |
|-------------------------------------------------------|--|--|--|--|--|--|--|
| Al+3 + Citric-3 + H+1_Citric-3 = Al+3_H+1_Citric-3(2) |  |  |  |  |  |  |  |

| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.1   | Inert  | lgK:11.42(0.02SD) | 6 | MGL   | 1151   |
| 2  | 25 | 0.1   | KCl    | lgK:11.1(0.2SD)   | 3 | MGL   | 1151   |

| Reaction No. 10126                              |    |       |        |                  |   |       |        |
|-------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Al+3 + 2<Citric-3> + H+1 = Al+3_H+1_Citric-3(2) |    |       |        |                  |   |       |        |
| No                                              | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                               | 25 | 0.1   | Inert  | lgK:17.3(0.04SD) | 4 | MGL   | 1151   |
| 2                                               | 25 | 0.1   | KCl    | lgK:16.8(0.2SD)  | 4 | MGL   | 1151   |

| Reaction No. 10386                |    |       |         |                 |   |       |        |
|-----------------------------------|----|-------|---------|-----------------|---|-------|--------|
| Squaric-2 + Al+3 = Al+3_Squaric-2 |    |       |         |                 |   |       |        |
| No                                | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 25 | 1     | NH4ClO4 | lgK:1.8(0.03SD) | 4 | MIX   | 906    |
| 2                                 | 25 | 3     | NaClO4  | lgK:2.83(3SF)   | 5 | MGL   | 6953   |

| Reaction No. 10557                  |    |       |        |                  |   |       |        |
|-------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Diglycol-2 + Al+3 = Al+3_Diglycol-2 |    |       |        |                  |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                   | 25 | 0.5   | NaClO4 | lgK:3.16(0.01SD) | 5 | MGL   | 6953   |

| Reaction No. 10558                        |    |       |        |                  |   |       |        |
|-------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 2<Diglycol-2> + Al+3 = Al+3_Diglycol-2(2) |    |       |        |                  |   |       |        |
| No                                        | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                         | 25 | 0.5   | NaClO4 | lgK:5.25(0.02SD) | 4 | MGL   | 6953   |

| Reaction No. 10616                                |    |       |               |               |   |       |        |
|---------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Al+3_Tartaric-2 + Tartaric-2 = Al+3_Tartaric-2(2) |    |       |               |               |   |       |        |
| No                                                | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                 | 25 | 0     | Inf. Dilution | lgK:2.6(2SF)  | 1 | ELC   |        |
| 2                                                 | 25 | 0.1   | NaClO4        | lgK:4.45(3SF) | 0 | MGL   | 906    |
| 3                                                 | 25 | 0.1   | NaClO4        | lgK:4.33(3SF) | 0 | MGL   | 7544   |

| Reaction No. 10929                      |    |       |        |                 |   |       |        |
|-----------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| ThioDiAcet-2 + Al+3 = Al+3_ThioDiAcet-2 |    |       |        |                 |   |       |        |
| No                                      | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                       | 25 | 0.5   | NaClO4 | lgK:1.9(0.08SD) | 3 | MGL   | 6953   |

| Reaction No. 10930                                  |    |       |        |                  |   |       |        |
|-----------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Al+3 + ThioDiAcet-2 + OH-1 = Al+3_OH-1_ThioDiAcet-2 |    |       |        |                  |   |       |        |
| No                                                  | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.5   | NaClO4 | lgK:12.3(0.05SD) | 4 | MGL   | 999    |

| Reaction No. 11405                    |    |       |               |              |   |       |        |
|---------------------------------------|----|-------|---------------|--------------|---|-------|--------|
| Al+3_Acac-1 + Acac-1 = Al+3_Acac-1(2) |    |       |               |              |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
| 1                                     | 25 | 0     | Inf. Dilution | lgK:7.9(2SF) | 1 | ESC   |        |
| 2                                     | 30 | 0     | Inf. Dilution | lgK:7.9(2SF) | 4 | MGL   | 999    |

| Reaction No. 11406                       |    |       |               |              |   |       |        |
|------------------------------------------|----|-------|---------------|--------------|---|-------|--------|
| Al+3_Acac-1(2) + Acac-1 = Al+3_Acac-1(3) |    |       |               |              |   |       |        |
| No                                       | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
| 1                                        | 25 | 0     | Inf. Dilution | lgK:5.9(2SF) | 1 | ESC   |        |
| 2                                        | 30 | 0     | Inf. Dilution | lgK:5.8(2SF) | 4 | MGL   | 999    |

| Reaction No. 11833                          |    |       |        |               |   |       |        |
|---------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Al+3_Maltol-1 + Maltol-1 = Al+3_Maltol-1(2) |    |       |        |               |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | KNO3   | lgK:7.55(3SF) | 6 | MGL   | 999    |

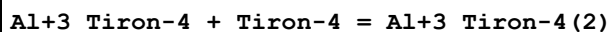
| Reaction No. 11834                             |    |       |        |               |   |       |        |
|------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Al+3_Maltol-1(2) + Maltol-1 = Al+3_Maltol-1(3) |    |       |        |               |   |       |        |
| No                                             | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                              | 25 | 0.1   | KNO3   | lgK:6.65(3SF) | 6 | MGL   | 999    |

| Reaction No. 11907                       |       |       |               |              |   |       |        |
|------------------------------------------|-------|-------|---------------|--------------|---|-------|--------|
| Al+3_Kojic-1 + Kojic-1 = Al+3_Kojic-1(2) |       |       |               |              |   |       |        |
| No                                       | t     | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
| 1                                        | 20:23 | 0.1   | KCl           | lgK:6.5(2SF) | 4 | MHY   | 999    |
| 2                                        | 25    | 0     | Inf. Dilution | lgK:6.7(2SF) | 1 | ESC   |        |

| Reaction No. 11908                          |       |       |        |              |   |       |        |
|---------------------------------------------|-------|-------|--------|--------------|---|-------|--------|
| Al+3_Kojic-1(2) + Kojic-1 = Al+3_Kojic-1(3) |       |       |        |              |   |       |        |
| No                                          | t     | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                           | 20:23 | 0.1   | KCl    | lgK:5.3(2SF) | 4 | MHY   | 999    |

|   |    |   |               |              |   |     |  |
|---|----|---|---------------|--------------|---|-----|--|
| 2 | 25 | 0 | Inf. Dilution | lgK:5.5(2SF) | 1 | ESC |  |
|---|----|---|---------------|--------------|---|-----|--|

## Reaction No. 12465



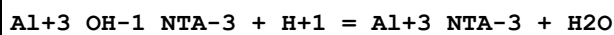
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3          | lgK:13.6(3SF)    | 4 | MGL   | 3494   |
| 2  | 25 | 0     | Inf. Dilution | lgK:12.08(4SF)   | 6 | MGL   | 999    |
| 3  | 25 | 0.2   | KCl           | lgK:13.9(0.03SD) | 4 | MGL   | 1391   |
| 4  | 25 | 0.2   | NaNO3         | lgK:13.60(4SF)   | 6 | MGL   | 999    |

## Reaction No. 12466



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:9.7(2SF)     | 4 | MGL   | 3494   |
| 2  | 25 | 0.2   | KCl    | lgK:10.5(0.03SD) | 6 | MGL   | 1391   |

## Reaction No. 12574

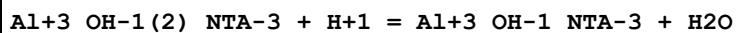


| No | t  | I-Str | Medium  | Value (Dev)  | W | Tech. | Ref. # |
|----|----|-------|---------|--------------|---|-------|--------|
| 1  | 20 | 0.001 | Unknown | lgK:5.8(2SF) | 3 | MGL   | 1118   |
| 2  | 20 | 0.1   | KCl     | lgK:5.8(2SF) | 0 | MGL   | 999    |

Note (2): Low wgt for internal consistency

|   |    |     |          |               |   |     |      |
|---|----|-----|----------|---------------|---|-----|------|
| 3 | 25 | 0.2 | NaClO4   | lgK:5.09(3SF) | 6 | MGL | 999  |
| 4 | 25 | 0.2 | NaClO4   | lgK:5.09(3SF) | 5 | MGL | 1118 |
| 5 | 25 | 0.5 | Unknown  | lgK:5.43(3SF) | 6 | CRV | 552  |
| 6 | 35 | 0.1 | K+1 salt | lgK:6.16(3SF) | 6 | CRV | 552  |

## Reaction No. 12575



| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 20 | 0.1   | KCl           | lgK:8.6(2SF)  | 3 | MGL   | 999    |
| 2  | 25 | 0     | Inf. Dilution | lgK:8.8(2SF)  | 1 | EDH   |        |
| 3  | 25 | 0.2   | NaClO4        | lgK:8.28(3SF) | 6 | MGL   | 999    |
| 4  | 25 | 0.2   | NaClO4        | lgK:8.28(3SF) | 5 | MGL   | 1118   |
| 5  | 25 | 0.5   | Unknown       | lgK:8.19(3SF) | 6 | CRV   | 552    |

## Reaction No. 12875

| H+1 + 5ATP-4 + Al+3 = Al+3_H+1_5ATP-4 |    |       |               |                   |   |       |        |
|---------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| No                                    | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                     | 25 | 0     | Inf. Dilution | lgK:13.8(0.3SD)   | 3 | EDH   | 30067  |
| 2                                     | 25 | 0.1   | NaCl          | lgK:10.84(0.01SD) | 1 | MGL   | 30067  |
| 3                                     | 25 | 0.15  | NaCl          | lgK:12.5(0.03SD)  | 0 | MGL   | 1175   |
| 4                                     | 25 | 0.2   | KCl           | lgK:11.3(0.04SD)  | 2 | MGL   | 1236   |
| 5                                     | 25 | 0.5   | NaCl          | lgK:10.80(0.02SD) | 2 | MGL   | 30067  |
| 6                                     | 25 | 1     | NaCl          | lgK:10.35(0.02SD) | 2 | MGL   | 30067  |

| Reaction No. 12876          |    |       |               |                    |   |       |        |
|-----------------------------|----|-------|---------------|--------------------|---|-------|--------|
| 5ATP-4 + Al+3 = Al+3_5ATP-4 |    |       |               |                    |   |       |        |
| No                          | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                           | 25 | 0     | Inf. Dilution | lgK:10.9(0.2SD)    | 3 | EDH   | 30067  |
| 2                           | 25 | 0.09  | Unknown       | lgK:6.2(2SF)       | 0 | MKN   | 1437   |
| 3                           | 25 | 0.1   | NaCl          | lgK:8.138(0.001SD) | 3 | MGL   | 30067  |
| 4                           | 25 | 0.2   | KCl           | lgK:7.9(0.04SD)    | 3 | MGL   | 1236   |
| 5                           | 25 | 0.5   | NaCl          | lgK:7.72(0.01SD)   | 3 | MGL   | 30067  |
| 6                           | 25 | 0.5   | Unknown       | lgK:7.92(3SF)      | 3 | MMR   | 7163   |
| 7                           | 25 | 1     | NaCl          | lgK:7.38(0.01SD)   | 3 | MGL   | 30067  |
| 8                           | 25 | 0.1   | NaCl          | dH:25.(1.SD)kJ     | 4 | MCL   | 30067  |

| Reaction No. 12877                           |    |       |               |                  |   |       |        |
|----------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Al+3 + 5ATP-4 + H2O = Al+3_OH-1_5ATP-4 + H+1 |    |       |               |                  |   |       |        |
| No                                           | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                            | 25 | 0     | Inf. Dilution | lgK:5.0(0.1SD)   | 3 | EDH   | 30067  |
| 2                                            | 25 | 0.1   | NaCl          | lgK:2.26(0.03SD) | 3 | MGL   | 30067  |
| 3                                            | 25 | 0.2   | KCl           | lgK:2.46(0.07SD) | 2 | MGL   | 1236   |
| 4                                            | 25 | 0.5   | NaCl          | lgK:1.54(0.08SD) | 3 | MGL   | 30067  |
| 5                                            | 25 | 1     | NaCl          | lgK:1.16(0.07SD) | 3 | MGL   | 30067  |

| Reaction No. 12878                |    |       |               |                   |   |       |        |
|-----------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| 2<5ATP-4> + Al+3 = Al+3_5ATP-4(2) |    |       |               |                   |   |       |        |
| No                                | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                 | 25 | 0     | Inf. Dilution | lgK:14.9(0.2SD)   | 3 | EDH   | 30067  |
| 2                                 | 25 | 0.1   | NaCl          | lgK:12.91(0.02SD) | 3 | MGL   | 30067  |
| 3                                 | 25 | 0.2   | KCl           | lgK:12.5(0.04SD)  | 4 | MGL   | 1236   |
| 4                                 | 25 | 0.5   | NaCl          | lgK:12.24(0.01SD) | 3 | MGL   | 30067  |



|   |    |     |      |                    |   |     |       |
|---|----|-----|------|--------------------|---|-----|-------|
| 5 | 25 | 1   | NaCl | lgK:11.54 (0.01SD) | 3 | MGL | 30067 |
| 6 | 25 | 0.1 | NaCl | dH:22. (2.SD) kJ   | 4 | MCL | 30067 |

| Reaction No. 12879                                  |    |       |               |                   |   |       |        |
|-----------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| Al+3 + 2<5ATP-4> + H2O = Al+3_OH-1_5ATP-4 (2) + H+1 |    |       |               |                   |   |       |        |
| No                                                  | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                   | 25 | 0     | Inf. Dilution | lgK:5.7 (0.1SD)   | 3 | EDH   | 30067  |
| 2                                                   | 25 | 0.1   | NaCl          | lgK:5.10 (0.09SD) | 3 | MGL   | 30067  |
| 3                                                   | 25 | 0.2   | KCl           | lgK:4.8 (0.05SD)  | 4 | MGL   | 1236   |
| 4                                                   | 25 | 0.5   | NaCl          | lgK:4.75 (0.04SD) | 3 | MGL   | 30067  |
| 5                                                   | 25 | 1     | NaCl          | lgK:4.18 (0.09SD) | 3 | MGL   | 30067  |

| Reaction No. 12880                    |    |       |        |                    |   |       |        |
|---------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| H+1 + 5ADP-3 + Al+3 = Al+3_H+1_5ADP-3 |    |       |        |                    |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                     | 25 | 0.15  | NaCl   | lgK:12.47 (4SF)    | 0 | MGL   | 1175   |
| 2                                     | 25 | 0.15  | NaCl   | lgK:10.53 (0.06SD) | 2 | MGL   | 30067  |
| 3                                     | 25 | 0.2   | KCl    | lgK:11.0 (0.04SD)  | 2 | MGL   | 1236   |

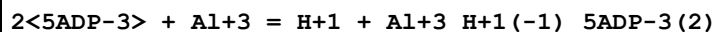
| Reaction No. 12881          |    |       |        |                    |   |       |        |
|-----------------------------|----|-------|--------|--------------------|---|-------|--------|
| 5ADP-3 + Al+3 = Al+3_5ADP-3 |    |       |        |                    |   |       |        |
| No                          | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                           | 25 | 0.15  | NaCl   | lgK:10.03 (0.02SD) | 0 | MGL   | 1175   |
| 2                           | 25 | 0.15  | NaCl   | lgK:7.19 (0.05SD)  | 2 | MGL   | 30067  |
| 3                           | 25 | 0.2   | KCl    | lgK:7.8 (0.03SD)   | 2 | MGL   | 1236   |

| Reaction No. 12882                           |    |       |        |                   |   |       |        |
|----------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Al+3 + 5ADP-3 + H2O = Al+3_OH-1_5ADP-3 + H+1 |    |       |        |                   |   |       |        |
| No                                           | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                            | 25 | 0.15  | NaCl   | lgK:4.2 (0.04SD)  | 0 | MGL   | 1175   |
| 2                                            | 25 | 0.15  | NaCl   | lgK:2.07 (0.09SD) | 2 | MGL   | 30067  |
| 3                                            | 25 | 0.2   | KCl    | lgK:2.9 (0.08SD)  | 0 | MGL   | 1236   |

| Reaction No. 12883                 |    |       |        |                    |   |       |        |
|------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| 2<5ADP-3> + Al+3 = Al+3_5ADP-3 (2) |    |       |        |                    |   |       |        |
| No                                 | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                  | 25 | 0.15  | NaCl   | lgK:11.86 (0.09SD) | 3 | MGL   | 30067  |

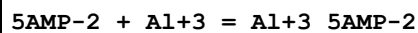
|   |    |     |     |                   |   |     |      |
|---|----|-----|-----|-------------------|---|-----|------|
| 2 | 25 | 0.2 | KCl | lgK:12.2 (0.04SD) | 2 | MGL | 1236 |
|---|----|-----|-----|-------------------|---|-----|------|

## Reaction No. 12884



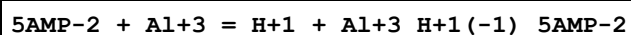
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:5.0 (0.07SD) | 4 | MGL   | 1236   |

## Reaction No. 12885



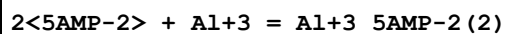
| No | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
|----|----|-------|--------|---------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaCl   | lgK:6.018 (0.004SD) | 5 | MGL   | 30067  |
| 2  | 25 | 0.2   | KCl    | lgK:6.17 (0.01SD)   | 4 | MGL   | 1236   |
| 3  | 25 | 0.2   | KCl    | lgK:6.14 (0.02SD)   | 4 | MGL   | 7048   |

## Reaction No. 12886



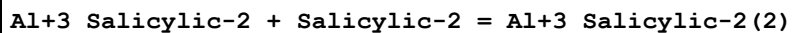
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:2.0 (0.09SD) | 3 | MGL   | 1236   |
| 2  | 25 | 0.2   | KCl    | lgK:1.9 (0.03SD) | 5 | MGL   | 7048   |

## Reaction No. 12887



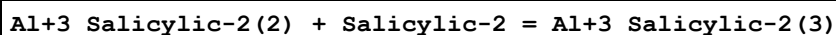
| No | t  | I-Str | Medium | Value (Dev)          | W | Tech. | Ref. # |
|----|----|-------|--------|----------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaCl   | lgK:10.645 (0.007SD) | 5 | MGL   | 30067  |
| 2  | 25 | 0.2   | KCl    | lgK:10.4 (0.1SD)     | 4 | MGL   | 1236   |

## Reaction No. 12983



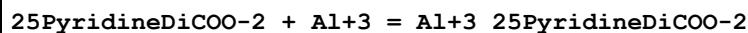
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3          | lgK:10.3 (3SF)  | 1 | MGL   | 3494   |
| 2  | 25 | 0     | Inf. Dilution | lgK:10.4 (3SF)  | 1 | ERG   |        |
| 3  | 25 | 0.12  | NaCl          | lgK:10.7 (3SF)  | 0 | MGL   | 5987   |
| 4  | 25 | 0.12  | NaCl          | lgK:13.1 (3SF)  | 0 | MGL   | 5987   |
| 5  | 25 | 0.2   | KCl           | lgK:10.51 (4SF) | 0 | MGL   | 2470   |

## Reaction No. 12984



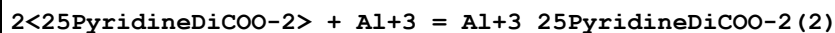
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3          | lgK:6.6 (2SF)  | 0 | MGL   | 3494   |
| 2  | 25 | 0     | Inf. Dilution | lgK:7.2 (2SF)  | 1 | ERG   |        |
| 3  | 25 | 0.1   | KCl           | lgK:8.82 (3SF) | 1 | MGL   | 2470   |
| 4  | 25 | 0.12  | NaCl          | lgK:10.7 (3SF) | 0 | MGL   | 5987   |

## Reaction No. 13238



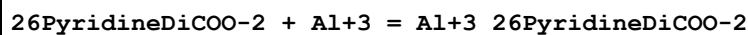
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.5   | NaClO3 | lgK:3.95 (0.02SD) | 4 | MGL   | 6953   |

## Reaction No. 13239



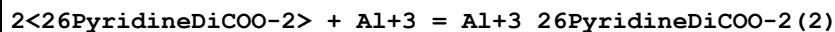
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.5   | NaClO3 | lgK:7.24 (0.02SD) | 4 | MGL   | 6953   |

## Reaction No. 13312



| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 23 | 1     | NaCl          | lgK:11.4 (0.06SD) | 3 | MMX   | 3326   |
| 2  | 25 | 0     | Inf. Dilution | lgK:7.8 (2SF)     | 1 | ELC   |        |
| 3  | 25 | 0.5   | NaClO4        | lgK:4.87 (3SF)    | 0 | MGL   | 6953   |

## Reaction No. 13313



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.5   | NaClO4 | lgK:8.32 (3SF) | 0 | MGL   | 6953   |

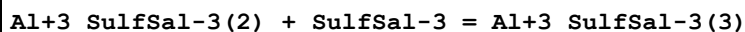
## Reaction No. 13346



| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3          | lgK:9.7 (2SF)   | 4 | MGL   | 3494   |
| 2  | 25 | 0     | Inf. Dilution | lgK:9.9 (2SF)   | 1 | EDH   |        |
| 3  | 25 | 0.1   | NaClO4        | lgK:9.63 (3SF)  | 0 | MGL   | 2122   |
| 4  | 25 | 0.1   | NaClO4        | lgK:10.01 (4SF) | 6 | MSP   | 2122   |
| 5  | 25 | 0.5   | NaClO4        | lgK:9.4 (2SF)   | 0 | MGL   | 5987   |
| 6  | 30 | 0.2   | NaClO4        | lgK:10.00 (4SF) | 6 | MGL   | 2123   |

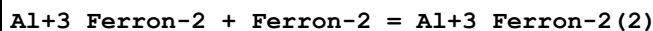
|   |    |     |         |                 |   |     |      |
|---|----|-----|---------|-----------------|---|-----|------|
| 7 | 30 | 0.2 | Unknown | lgK:10.02 (4SF) | 5 | MSC | 2123 |
|---|----|-----|---------|-----------------|---|-----|------|

## Reaction No. 13347



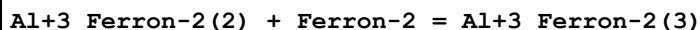
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:5.8 (2SF)  | 4 | MGL   | 3494   |
| 2  | 25 | 0.1   | NaClO4 | lgK:6.06 (3SF) | 6 | MGL   | 2122   |

## Reaction No. 13829



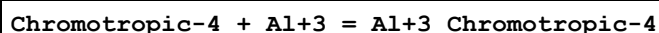
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:7.1 (2SF)   | 4 | MGL   | 999    |
| 2  | 25 | 0.1   | KCl    | lgK:5.8 (2SF) w | 0 | MPH   | 3955   |
| 3  | 25 | 0.1   | KCl    | lgK:7.1 (2SF)   | 5 | MPT   | 31140  |
| 4  | 25 | 0.14  | KCl    | lgK:7.2 (2SF) w | 3 | MPH   | 3955   |
| 5  | 28 | 0.1   | KNO3   | lgK:7.00 (3SF)  | 6 | MGL   | 999    |

## Reaction No. 13832



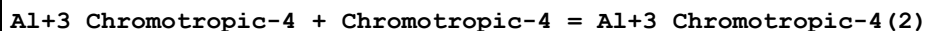
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:5.6 (2SF)    | 3 | MGL   | 999    |
| 2  | 25 | 0.1   | KCl    | lgK:10.8 (3SF) w | 0 | MPH   | 3955   |
| 3  | 25 | 0.1   | KCl    | lgK:5.6 (2SF)    | 5 | MPT   | 31140  |
| 4  | 25 | 0.14  | KCl    | lgK:7.2 (2SF) w  | 0 | MPH   | 3955   |

## Reaction No. 13982



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:17.1 (3SF)  | 4 | MGL   | 3494   |
| 2  | 25 | 0.5   | NaClO4 | lgK:12.9 (3SF)  | 0 | ENS   | 6953   |
| 3  | 30 | 0.2   | NaClO4 | lgK:17.20 (4SF) | 6 | MGL   | 2123   |

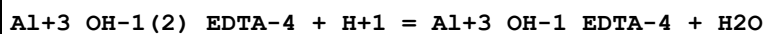
## Reaction No. 13983



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3          | lgK:12.7 (3SF) | 2 | MGL   | 3494   |
| 2  | 25 | 0     | Inf. Dilution | lgK:13.1 (3SF) | 1 | ESC   |        |

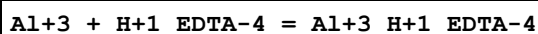
|   |    |     |         |                 |   |     |      |
|---|----|-----|---------|-----------------|---|-----|------|
| 3 | 30 | 0.2 | NaClO4  | lgK:13.25 (4SF) | 6 | MGL | 2123 |
| 4 | 30 | 0.2 | Unknown | lgK:13.25 (4SF) | 5 | MSC | 2123 |

## Reaction No. 14034



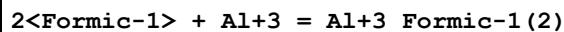
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3    | lgK:9.97 (3SF)  | 5 | MPL   | 140    |
| 2  | 25 | 0.2   | NaClO4  | lgK:10.31 (4SF) | 5 | MGL   | 931    |
| 3  | 25 | 1     | Unknown | lgK:9.97 (3SF)  | 6 | CRV   | 552    |

## Reaction No. 14061



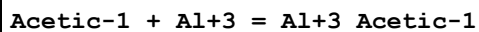
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3          | lgK:8.4 (2SF)  | 4 | MPL   | 140    |
| 2  | 20 | 0.1   | Unknown       | lgK:8.96 (3SF) | 6 | ENS   | 1539   |
| 3  | 25 | 0     | Inf. Dilution | lgK:10.4 (3SF) | 2 | EAE   |        |

## Reaction No. 14071



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 1     | NaClO4 | lgK:2.02 (3SF) | 6 | ENS   | 1241   |
| 2  | 25 | 1     | NaNO3  | lgK:1.76 (3SF) | 6 | MPT   | 2103   |

## Reaction No. 14072



| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 0  | 0     | Inf. Dilution | lgK:2.56 (3SF)    | 4 | CRV   | 8888   |
| 2  | 25 | 0     | Inf. Dilution | lgK:2.41 (3SF)    | 4 | ENS   | 1541   |
| 3  | 25 | 0     | Inf. Dilution | lgK:2.4 (2SF)     | 4 | ENS   | 2152   |
| 4  | 25 | 0     | Inf. Dilution | lgK:2.75 (3SF)    | 4 | CRV   | 8888   |
| 5  | 25 | 0     | Inf. Dilution | lgK:2.668 (4SF)   | 4 | EOD   | 29559  |
| 6  | 25 | 0     | NaCl/m        | lgK:2.75 (0.1MD)  | 7 | MEF   | 4277   |
| 7  | 25 | 0.1   | NaCl/m        | lgK:2.02 (0.1MD)  | 7 | MEF   | 4277   |
| 8  | 25 | 0.2   | NaCl/m        | lgK:1.82 (0.09MD) | 7 | MEF   | 4277   |
| 9  | 25 | 0.5   | NaCl/m        | lgK:1.56 (0.08MD) | 7 | MEF   | 4277   |
| 10 | 25 | 1     | NaCl/m        | lgK:1.43 (0.06MD) | 7 | MEF   | 4277   |
| 11 | 25 | 1     | NaClO4        | lgK:1.51 (3SF)    | 2 | ENS   | 1241   |

|                                                                   |     |      |               |                   |   |     |       |
|-------------------------------------------------------------------|-----|------|---------------|-------------------|---|-----|-------|
| 12                                                                | 25  | 1    | NaNO3         | lgK:1.4 (2SF)     | 5 | MGL | 11516 |
| Note (12): Kreytshuk A et al., Zhur. Neorg. Khim., 1975, 20, 2330 |     |      |               |                   |   |     |       |
| 13                                                                | 50  | 0    | Inf. Dilution | lgK:3.06 (3SF)    | 4 | CRV | 8888  |
| 14                                                                | 50  | 0    | NaCl/m        | lgK:3.06 (0.07MD) | 4 | MEF | 4277  |
| 15                                                                | 50  | 0.1  | NaCl/m        | lgK:2.29 (0.06MD) | 7 | MEF | 4277  |
| 16                                                                | 50  | 0.2  | NaCl/m        | lgK:2.08 (0.06MD) | 7 | MEF | 4277  |
| 17                                                                | 50  | 0.5  | NaCl/m        | lgK:1.80 (0.04MD) | 7 | MEF | 4277  |
| 18                                                                | 50  | 1    | NaCl/m        | lgK:1.65 (0.06MD) | 5 | MEF | 4277  |
| 19                                                                | 60  | 0.02 | NaCl/m        | lgK:2.33 (0.06SD) | 3 | MSL | 6039  |
| 20                                                                | 75  | 0    | Inf. Dilution | lgK:3.42 (3SF)    | 4 | CRV | 8888  |
| 21                                                                | 75  | 0    | NaCl/m        | lgK:3.46 (0.07MD) | 4 | MEF | 4277  |
| 22                                                                | 75  | 0.1  | NaCl/m        | lgK:2.64 (0.07MD) | 7 | MEF | 4277  |
| 23                                                                | 75  | 0.2  | NaCl/m        | lgK:2.42 (0.06MD) | 7 | MEF | 4277  |
| 24                                                                | 75  | 0.5  | NaCl/m        | lgK:2.11 (0.04MD) | 7 | MEF | 4277  |
| 25                                                                | 75  | 1    | NaCl/m        | lgK:1.95 (0.06MD) | 7 | MEF | 4277  |
| 26                                                                | 80  | 0    | Inf. Dilution | lgK:2.9 (2SF)     | 3 | ENS | 6953  |
| 27                                                                | 90  | 0    | Inf. Dilution | lgK:2.405 (4SF)   | 0 | EOD | 29559 |
| 28                                                                | 100 | 0    | Inf. Dilution | lgK:3.81 (3SF)    | 4 | CRV | 8888  |
| 29                                                                | 100 | 0    | NaCl/m        | lgK:3.94 (0.08MD) | 5 | MEF | 4277  |
| 30                                                                | 100 | 0.1  | NaCl/m        | lgK:3.07 (0.07MD) | 7 | MEF | 4277  |
| 31                                                                | 100 | 0.2  | NaCl/m        | lgK:2.83 (0.06MD) | 7 | MEF | 4277  |
| 32                                                                | 100 | 0.5  | NaCl/m        | lgK:2.49 (0.04MD) | 7 | MEF | 4277  |
| 33                                                                | 100 | 1    | NaCl/m        | lgK:2.30 (0.05MD) | 7 | MEF | 4277  |
| 34                                                                | 120 | 0.01 | NaCl/m        | lgK:4.15 (0.07SD) | 3 | MSL | 6039  |
| 35                                                                | 125 | 0    | Inf. Dilution | lgK:4.23 (3SF)    | 4 | CRV | 8888  |
| 36                                                                | 125 | 0    | NaCl/m        | lgK:4.48 (0.11MD) | 7 | MEF | 4277  |
| 37                                                                | 125 | 0.1  | NaCl/m        | lgK:3.55 (0.10MD) | 7 | MEF | 4277  |
| 38                                                                | 125 | 0.2  | NaCl/m        | lgK:3.29 (0.09MD) | 7 | MEF | 4277  |
| 39                                                                | 125 | 0.5  | NaCl/m        | lgK:2.92 (0.06MD) | 4 | MEF | 4277  |
| 40                                                                | 125 | 1    | NaCl/m        | lgK:2.70 (0.05MD) | 7 | MEF | 4277  |
| 41                                                                | 150 | 0    | Inf. Dilution | lgK:4.66 (3SF)    | 4 | CRV | 8888  |
| 42                                                                | 150 | 0    | Inf. Dilution | lgK:2.606 (4SF)   | 0 | EOD | 29559 |
| 43                                                                | 150 | 0    | NaCl/m        | lgK:5.08 (0.16MD) | 7 | MEF | 4277  |
| Note (43): Low wgt for internal consistency                       |     |      |               |                   |   |     |       |
| 44                                                                | 150 | 0.1  | NaCl/m        | lgK:4.07 (0.15MD) | 7 | MEF | 4277  |
| 45                                                                | 150 | 0.2  | NaCl/m        | lgK:3.78 (0.14MD) | 4 | MEF | 4277  |

|    |     |      |               |                      |   |     |       |
|----|-----|------|---------------|----------------------|---|-----|-------|
| 46 | 150 | 0.5  | NaCl/m        | lgK:3.38 (0.12MD)    | 4 | MEF | 4277  |
| 47 | 150 | 1    | NaCl/m        | lgK:3.12 (0.09MD)    | 7 | MEF | 4277  |
| 48 | 170 | 0.01 | NaCl/m        | lgK:5.15 (0.10SD)    | 5 | MSL | 6039  |
| 49 | 175 | 0    | Inf. Dilution | lgK:5.11 (3SF)       | 0 | CRV | 8888  |
| 50 | 200 | 0    | Inf. Dilution | lgK:5.57 (3SF)       | 0 | CRV | 8888  |
| 51 | 200 | 0.01 | NaCl/m        | lgK:6.31 (0.024SD)   | 3 | MSL | 6039  |
| 52 | 210 | 0    | Inf. Dilution | lgK:3.052 (4SF)      | 0 | EOD | 29559 |
| 53 | 225 | 0    | Inf. Dilution | lgK:6.04 (3SF)       | 0 | CRV | 8888  |
| 54 | 250 | 0    | Inf. Dilution | lgK:6.54 (3SF)       | 0 | CRV | 8888  |
| 55 | 275 | 0    | Inf. Dilution | lgK:7.06 (3SF)       | 0 | CRV | 8888  |
| 56 | 300 | 0    | Inf. Dilution | lgK:7.62 (3SF)       | 0 | CRV | 8888  |
| 57 | 325 | 0    | Inf. Dilution | lgK:8.22 (3SF)       | 0 | CRV | 8888  |
| 58 | 350 | 0    | Inf. Dilution | lgK:8.71 (3SF)       | 0 | CRV | 8888  |
| 59 | 25  | 0:1  | NaCl/m        | dH:17. (6.MD) kJ     | 3 | MTD | 4277  |
| 60 | 50  | 0:1  | NaCl/m        | dH:29. (3.MD) kJ     | 3 | MTD | 4277  |
| 61 | 75  | 0:1  | NaCl/m        | dH:41. (2.MD) kJ     | 3 | MTD | 4277  |
| 62 | 100 | 0:1  | NaCl/m        | dH:55. (4.MD) kJ     | 3 | MTD | 4277  |
| 63 | 125 | 0:1  | NaCl/m        | dH:69. (7.MD) kJ     | 3 | MTD | 4277  |
| 64 | 150 | 0:1  | NaCl/m        | dH:85. (10.MD) kJ    | 3 | MTD | 4277  |
| 65 | 25  | 0:1  | NaCl/m        | Cp:450. (100.MD) J/K | 7 | MTD | 4277  |
| 66 | 50  | 0:1  | NaCl/m        | Cp:490. (110.MD) J/K | 7 | MTD | 4277  |
| 67 | 75  | 0:1  | NaCl/m        | Cp:520. (120.MD) J/K | 7 | MTD | 4277  |
| 68 | 100 | 0:1  | NaCl/m        | Cp:560. (130.MD) J/K | 7 | MTD | 4277  |
| 69 | 125 | 0:1  | NaCl/m        | Cp:600. (140.MD) J/K | 7 | MTD | 4277  |
| 70 | 150 | 0:1  | NaCl/m        | Cp:640. (140.MD) J/K | 7 | MTD | 4277  |

| Reaction No. 14136                             |    |       |         |                |   |       |        |
|------------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Al+3_OH-1_HOEDTA-3 + H+1 = Al+3_HOEDTA-3 + H2O |    |       |         |                |   |       |        |
| No                                             | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                              | 20 | 0.1   | Unknown | lgK:5.08 (3SF) | 6 | CRV   | 552    |
| 2                                              | 25 | 0.2   | NaClO4  | lgK:4.89 (3SF) | 6 | MGL   | 999    |

| Reaction No. 14137                                     |    |       |        |                |   |       |        |
|--------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Al+3_OH-1(2)_HOEDTA-3 + H+1 = Al+3_OH-1_HOEDTA-3 + H2O |    |       |        |                |   |       |        |
| No                                                     | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                      | 25 | 0.2   | NaClO4 | lgK:9.19 (3SF) | 6 | MGL   | 999    |

| Reaction No. 14177                           |    |       |                 |                     |   |       |        |
|----------------------------------------------|----|-------|-----------------|---------------------|---|-------|--------|
| Malonic-2 + Al+3 = Al+3_Malonic-2            |    |       |                 |                     |   |       |        |
| No                                           | t  | I-Str | Medium          | Value (Dev)         | W | Tech. | Ref. # |
| 1                                            | 0  | 0     | NaCl/m          | lgK:7.25 (0.19MD)   | 0 | MGL   | 8762   |
| 2                                            | 0  | 0.1   | NaCl/m          | lgK:6.15 (0.11MD)   | 3 | MGL   | 8762   |
| 3                                            | 0  | 0.2   | NaCl/m          | lgK:5.95 (0.08MD)   | 3 | MGL   | 8762   |
| 4                                            | 0  | 0.5   | NaCl/m          | lgK:5.71 (0.06MD)   | 3 | MGL   | 8762   |
| 5                                            | 0  | 1     | NaCl/m          | lgK:5.47 (0.04MD)   | 3 | MGL   | 8762   |
| 6                                            | 25 | 0     | Inf. Dilution   | lgK:7.81 (0.05SD)   | 2 | EDH   | 29190  |
| 7                                            | 25 | 0     | Inf. Dilution/m | lgK:7.49 (3SF)      | 3 | MPT   | 6429   |
| 8                                            | 25 | 0     | NaCl/m          | lgK:7.49 (0.18MD)   | 3 | MGL   | 8762   |
| 9                                            | 25 | 0.1   | KCl             | lgK:6.71 (0.01SD)   | 0 | MGL   | 2356   |
| 10                                           | 25 | 0.1   | NaCl/m          | lgK:6.36 (0.09MD)   | 4 | MGL   | 8762   |
| 11                                           | 25 | 0.15  | NaCl            | lgK:6.40 (0.03SD)   | 3 | MGL   | 29190  |
| 12                                           | 25 | 0.2   | KCl             | lgK:6.23 (0.02SD)   | 3 | MGL   | 4371   |
| 13                                           | 25 | 0.2   | NaCl/m          | lgK:6.16 (0.05MD)   | 6 | MGL   | 8762   |
| 14                                           | 25 | 0.5   | NaCl            | lgK:5.933 (0.009SD) | 5 | MGL   | 29190  |
| 15                                           | 25 | 0.5   | NaCl/m          | lgK:5.91 (0.05MD)   | 6 | MGL   | 8762   |
| 16                                           | 25 | 1     | NaCl            | lgK:5.835 (0.009SD) | 5 | MGL   | 29190  |
| 17                                           | 25 | 1     | NaCl/m          | lgK:5.68 (0.03MD)   | 3 | MGL   | 8762   |
| 18                                           | 25 | 1     | NaClO4          | lgK:3.58 (3SF)      | 0 | MGL   | 1243   |
| 19                                           | 25 | 1     | Unknown/m       | lgK:3.44 (3SF)      | 0 | EOD   | 6366   |
| Note (19): NDP problem (indicated in source) |    |       |                 |                     |   |       |        |
| 20                                           | 30 | 0.1   | NaClO4          | lgK:6.15 (3SF)      | 0 | MGL   | 2356   |
| 21                                           | 30 | 0.2   | NaClO4          | lgK:5.24 (3SF)      | 0 | MGL   | 2123   |
| Note (21): Low wgt for internal consistency  |    |       |                 |                     |   |       |        |
| 22                                           | 31 | 0     | Unknown         | lgK:5.70 (3SF)      | 0 | EOD   | 6004   |
| 23                                           | 35 | 1     | NaClO4          | lgK:4.04 (3SF)      | 0 | MGL   | 1243   |
| Note (23): Low wgt for internal consistency  |    |       |                 |                     |   |       |        |
| 24                                           | 37 | 0.15  | NaCl            | lgK:6.26 (0.02SD)   | 3 | MGL   | 2372   |
| 25                                           | 50 | 0     | NaCl/m          | lgK:7.81 (0.20MD)   | 0 | MGL   | 8762   |
| 26                                           | 50 | 0.1   | NaCl/m          | lgK:6.62 (0.11MD)   | 3 | MGL   | 8762   |
| 27                                           | 50 | 0.2   | NaCl/m          | lgK:6.42 (0.06MD)   | 6 | MGL   | 8762   |
| 28                                           | 50 | 0.5   | NaCl/m          | lgK:6.16 (0.05MD)   | 4 | MGL   | 8762   |
| 29                                           | 50 | 1     | NaCl/m          | lgK:5.92 (0.03MD)   | 3 | MGL   | 8762   |
| 30                                           | 75 | 0     | NaCl/m          | lgK:8.18 (0.24MD)   | 3 | MGL   | 8762   |



|    |    |     |                 |                    |   |     |      |
|----|----|-----|-----------------|--------------------|---|-----|------|
| 31 | 75 | 0.1 | NaCl/m          | lgK:6.93(0.14MD)   | 6 | MGL | 8762 |
| 32 | 75 | 0.2 | NaCl/m          | lgK:6.71(0.09MD)   | 3 | MGL | 8762 |
| 33 | 75 | 0.5 | NaCl/m          | lgK:6.45(0.06MD)   | 0 | MGL | 8762 |
| 34 | 75 | 1   | NaCl/m          | lgK:6.18(0.04MD)   | 6 | MGL | 8762 |
| 35 | 0  | 0   | NaCl/m          | dH:11.(6.MD)kJ     | 0 | MTD | 8762 |
| 36 | 0  | 0.1 | NaCl/m          | dH:13.(5.MD)kJ     | 0 | MTD | 8762 |
| 37 | 0  | 0.2 | NaCl/m          | dH:14.(5.MD)kJ     | 0 | MTD | 8762 |
| 38 | 0  | 0.5 | NaCl/m          | dH:16.(4.MD)kJ     | 0 | MTD | 8762 |
| 39 | 0  | 1   | NaCl/m          | dH:17.(4.MD)kJ     | 0 | MTD | 8762 |
| 40 | 25 | 0   | Inf. Dilution/m | dH:4.59(3SF)       | 0 | MTD | 6429 |
| 41 | 25 | 0   | NaCl/m          | dH:19.(5.MD)kJ     | 0 | MTD | 8762 |
| 42 | 25 | 0.1 | NaCl/m          | dH:21.(4.MD)kJ     | 6 | MTD | 8762 |
| 43 | 25 | 0.2 | NaCl/m          | dH:22.(3.MD)kJ     | 6 | MTD | 8762 |
| 44 | 25 | 0.5 | NaCl/m          | dH:24.(2.MD)kJ     | 3 | MTD | 8762 |
| 45 | 25 | 1   | NaCl/m          | dH:26.(2.MD)kJ     | 0 | MTD | 8762 |
| 46 | 50 | 0   | NaCl/m          | dH:28.(6.MD)kJ     | 4 | MTD | 8762 |
| 47 | 50 | 0.1 | NaCl/m          | dH:30.(5.MD)kJ     | 0 | MTD | 8762 |
| 48 | 50 | 0.2 | NaCl/m          | dH:31.(4.MD)kJ     | 0 | MTD | 8762 |
| 49 | 50 | 0.5 | NaCl/m          | dH:34.(2.MD)kJ     | 0 | MTD | 8762 |
| 50 | 50 | 1   | NaCl/m          | dH:36.(2.MD)kJ     | 0 | MTD | 8762 |
| 51 | 75 | 0   | NaCl/m          | dH:37.(8.MD)kJ     | 0 | MTD | 8762 |
| 52 | 75 | 0.1 | NaCl/m          | dH:39.(7.MD)kJ     | 0 | MTD | 8762 |
| 53 | 75 | 0.2 | NaCl/m          | dH:41.(6.MD)kJ     | 0 | MTD | 8762 |
| 54 | 75 | 0.5 | NaCl/m          | dH:44.(6.MD)kJ     | 0 | MTD | 8762 |
| 55 | 75 | 1   | NaCl/m          | dH:46.(5.MD)kJ     | 0 | MTD | 8762 |
| 56 | 0  | 0   | NaCl/m          | Cp:303.(110.MD)J/K | 4 | MTD | 8762 |
| 57 | 0  | 0.1 | NaCl/m          | Cp:314.(110.MD)J/K | 6 | MTD | 8762 |
| 58 | 0  | 0.2 | NaCl/m          | Cp:321.(110.MD)J/K | 6 | MTD | 8762 |
| 59 | 0  | 0.5 | NaCl/m          | Cp:334.(110.MD)J/K | 6 | MTD | 8762 |
| 60 | 0  | 1   | NaCl/m          | Cp:345.(110.MD)J/K | 6 | MTD | 8762 |
| 61 | 25 | 0   | Inf. Dilution/m | Cp:79.(2SF)        | 5 | MHY | 6429 |
| 62 | 25 | 0   | NaCl/m          | Cp:331.(120.MD)J/K | 4 | MTD | 8762 |
| 63 | 25 | 0.1 | NaCl/m          | Cp:342.(120.MD)J/K | 6 | MTD | 8762 |
| 64 | 25 | 0.2 | NaCl/m          | Cp:350.(120.MD)J/K | 6 | MTD | 8762 |
| 65 | 25 | 0.5 | NaCl/m          | Cp:365.(120.MD)J/K | 6 | MTD | 8762 |
| 66 | 25 | 1   | NaCl/m          | Cp:377.(120.MD)J/K | 6 | MTD | 8762 |

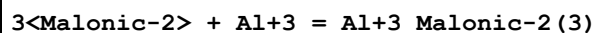
|    |    |     |        |                      |   |     |      |
|----|----|-----|--------|----------------------|---|-----|------|
| 67 | 50 | 0   | NaCl/m | Cp:359. (130.MD) J/K | 4 | MTD | 8762 |
| 68 | 50 | 0.1 | NaCl/m | Cp:371. (130.MD) J/K | 6 | MTD | 8762 |
| 69 | 50 | 0.2 | NaCl/m | Cp:379. (130.MD) J/K | 6 | MTD | 8762 |
| 70 | 50 | 0.5 | NaCl/m | Cp:395. (130.MD) J/K | 6 | MTD | 8762 |
| 71 | 50 | 1   | NaCl/m | Cp:409. (130.MD) J/K | 6 | MTD | 8762 |
| 72 | 75 | 0   | NaCl/m | Cp:387. (140.MD) J/K | 4 | MTD | 8762 |
| 73 | 75 | 0.1 | NaCl/m | Cp:400. (140.MD) J/K | 6 | MTD | 8762 |
| 74 | 75 | 0.2 | NaCl/m | Cp:409. (140.MD) J/K | 6 | MTD | 8762 |
| 75 | 75 | 0.5 | NaCl/m | Cp:426. (140.MD) J/K | 6 | MTD | 8762 |
| 76 | 75 | 1   | NaCl/m | Cp:440. (140.MD) J/K | 6 | MTD | 8762 |

| Reaction No. 14178                      |    |       |               |                      |   |       |        |
|-----------------------------------------|----|-------|---------------|----------------------|---|-------|--------|
| 2<Malonic-2> + Al+3 = Al+3_Malonic-2(2) |    |       |               |                      |   |       |        |
| No                                      | t  | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
| 1                                       | 0  | 0     | NaCl/m        | lgK:12.27 (0.41MD)   | 0 | MGL   | 8762   |
| 2                                       | 0  | 0.1   | NaCl/m        | lgK:10.81 (0.24MD)   | 0 | MGL   | 8762   |
| 3                                       | 0  | 0.2   | NaCl/m        | lgK:10.55 (0.16MD)   | 0 | MGL   | 8762   |
| 4                                       | 0  | 0.5   | NaCl/m        | lgK:10.19 (0.12MD)   | 0 | MGL   | 8762   |
| 5                                       | 0  | 1     | NaCl/m        | lgK:9.78 (0.07MD)    | 0 | MGL   | 8762   |
| 6                                       | 25 | 0     | Inf. Dilution | lgK:13.3 (0.02SD)    | 2 | EDH   | 29190  |
| 7                                       | 25 | 0     | NaCl/m        | lgK:12.62 (0.40MD)   | 0 | MGL   | 8762   |
| 8                                       | 25 | 0.1   | KCl           | lgK:11.53 (4SF)      | 0 | MGL   | 2356   |
| 9                                       | 25 | 0.1   | NaCl/m        | lgK:11.12 (0.21MD)   | 3 | MGL   | 8762   |
| 10                                      | 25 | 0.15  | NaCl          | lgK:11.40 (0.03SD)   | 0 | MGL   | 29190  |
| 11                                      | 25 | 0.2   | KCl           | lgK:10.92 (0.02SD)   | 2 | MGL   | 4371   |
| 12                                      | 25 | 0.2   | NaCl/m        | lgK:10.85 (0.11MD)   | 4 | MGL   | 8762   |
| 13                                      | 25 | 0.5   | NaCl          | lgK:10.278 (0.007SD) | 5 | MGL   | 29190  |
| 14                                      | 25 | 0.5   | NaCl/m        | lgK:10.49 (0.10MD)   | 2 | MGL   | 8762   |
| 15                                      | 25 | 1     | ClO4-1 salt   | lgK:6.0 (2SF)        | 0 | ENS   | 2171   |
| 16                                      | 25 | 1     | NaCl          | lgK:10.152 (0.009SD) | 5 | MGL   | 29190  |
| 17                                      | 25 | 1     | NaCl/m        | lgK:10.08 (0.06MD)   | 0 | MGL   | 8762   |
| 18                                      | 25 | 1     | NaClO4        | lgK:5.99 (3SF)       | 0 | MGL   | 1243   |
| 19                                      | 30 | 0.2   | NaClO4        | lgK:9.40 (3SF)       | 0 | MGL   | 2123   |
| 20                                      | 35 | 0     | Inf. Dilution | lgK:11.3 (0.1SD)     | 0 | MSL   | 7062   |
| 21                                      | 35 | 1     | NaClO4        | lgK:6.32 (3SF)       | 0 | MGL   | 1243   |
| 22                                      | 37 | 0.15  | NaCl          | lgK:11.11 (0.006SD)  | 5 | MGL   | 2372   |

|    |    |     |               |                      |   |     |      |
|----|----|-----|---------------|----------------------|---|-----|------|
| 23 | 50 | 0   | NaCl/m        | lgK:13.12 (0.44MD)   | 0 | MGL | 8762 |
| 24 | 50 | 0.1 | NaCl/m        | lgK:11.55 (0.23MD)   | 3 | MGL | 8762 |
| 25 | 50 | 0.2 | NaCl/m        | lgK:11.27 (0.13MD)   | 6 | MGL | 8762 |
| 26 | 50 | 0.5 | NaCl/m        | lgK:10.90 (0.11MD)   | 3 | MGL | 8762 |
| 27 | 50 | 1   | NaCl/m        | lgK:10.47 (0.06MD)   | 0 | MGL | 8762 |
| 28 | 75 | 0   | NaCl/m        | lgK:13.72 (0.52MD)   | 0 | MGL | 8762 |
| 29 | 75 | 0.1 | NaCl/m        | lgK:12.06 (0.30MD)   | 6 | MGL | 8762 |
| 30 | 75 | 0.2 | NaCl/m        | lgK:11.77 (0.20MD)   | 3 | MGL | 8762 |
| 31 | 75 | 0.5 | NaCl/m        | lgK:11.37 (0.14MD)   | 2 | MGL | 8762 |
| 32 | 75 | 1   | NaCl/m        | lgK:10.91 (0.07MD)   | 0 | MGL | 8762 |
| 33 | 80 | 0   | Inf. Dilution | lgK:14.5 (0.5SD)     | 3 | MSL | 7062 |
| 34 | 0  | 0   | NaCl/m        | dH:15. (11.MD) kJ    | 0 | MTD | 8762 |
| 35 | 0  | 0.1 | NaCl/m        | dH:17. (10.MD) kJ    | 0 | MTD | 8762 |
| 36 | 0  | 0.2 | NaCl/m        | dH:19. (9.MD) kJ     | 0 | MTD | 8762 |
| 37 | 0  | 0.5 | NaCl/m        | dH:21. (8.MD) kJ     | 0 | MTD | 8762 |
| 38 | 0  | 1   | NaCl/m        | dH:22. (8.MD) kJ     | 0 | MTD | 8762 |
| 39 | 25 | 0   | NaCl/m        | dH:29. (10.MD) kJ    | 0 | MTD | 8762 |
| 40 | 25 | 0.1 | NaCl/m        | dH:32. (8.MD) kJ     | 3 | MTD | 8762 |
| 41 | 25 | 0.2 | NaCl/m        | dH:33. (6.MD) kJ     | 3 | MTD | 8762 |
| 42 | 25 | 0.5 | NaCl/m        | dH:36. (4.MD) kJ     | 2 | MTD | 8762 |
| 43 | 25 | 1   | NaCl/m        | dH:38. (3.MD) kJ     | 0 | MTD | 8762 |
| 44 | 50 | 0   | NaCl/m        | dH:44. (12.MD) kJ    | 4 | MTD | 8762 |
| 45 | 50 | 0.1 | NaCl/m        | dH:47. (10.MD) kJ    | 0 | MTD | 8762 |
| 46 | 50 | 0.2 | NaCl/m        | dH:49. (8.MD) kJ     | 0 | MTD | 8762 |
| 47 | 50 | 0.5 | NaCl/m        | dH:52. (5.MD) kJ     | 0 | MTD | 8762 |
| 48 | 50 | 1   | NaCl/m        | dH:54. (4.MD) kJ     | 0 | MTD | 8762 |
| 49 | 75 | 0   | NaCl/m        | dH:60. (17.MD) kJ    | 0 | MTD | 8762 |
| 50 | 75 | 0.1 | NaCl/m        | dH:64. (14.MD) kJ    | 0 | MTD | 8762 |
| 51 | 75 | 0.2 | NaCl/m        | dH:66. (13.MD) kJ    | 0 | MTD | 8762 |
| 52 | 75 | 0.5 | NaCl/m        | dH:69. (10.MD) kJ    | 0 | MTD | 8762 |
| 53 | 75 | 1   | NaCl/m        | dH:72. (10.MD) kJ    | 0 | MTD | 8762 |
| 54 | 0  | 0   | NaCl/m        | Cp:527. (210.MD) J/K | 4 | MTD | 8762 |
| 55 | 0  | 0.1 | NaCl/m        | Cp:541. (210.MD) J/K | 4 | MTD | 8762 |
| 56 | 0  | 0.2 | NaCl/m        | Cp:551. (210.MD) J/K | 6 | MTD | 8762 |
| 57 | 0  | 0.5 | NaCl/m        | Cp:567. (200.MD) J/K | 6 | MTD | 8762 |
| 58 | 0  | 1   | NaCl/m        | Cp:579. (200.MD) J/K | 6 | MTD | 8762 |

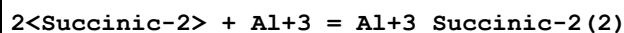
|    |    |     |        |                      |   |     |      |
|----|----|-----|--------|----------------------|---|-----|------|
| 59 | 25 | 0   | NaCl/m | Cp:575. (230.MD) J/K | 4 | MTD | 8762 |
| 60 | 25 | 0.1 | NaCl/m | Cp:591. (230.MD) J/K | 6 | MTD | 8762 |
| 61 | 25 | 0.2 | NaCl/m | Cp:601. (230.MD) J/K | 6 | MTD | 8762 |
| 62 | 25 | 0.5 | NaCl/m | Cp:619. (220.MD) J/K | 6 | MTD | 8762 |
| 63 | 25 | 1   | NaCl/m | Cp:632. (220.MD) J/K | 6 | MTD | 8762 |
| 64 | 50 | 0   | NaCl/m | Cp:623. (250.MD) J/K | 4 | MTD | 8762 |
| 65 | 50 | 0.1 | NaCl/m | Cp:640. (250.MD) J/K | 6 | MTD | 8762 |
| 66 | 50 | 0.2 | NaCl/m | Cp:651. (250.MD) J/K | 6 | MTD | 8762 |
| 67 | 50 | 0.5 | NaCl/m | Cp:671. (240.MD) J/K | 6 | MTD | 8762 |
| 68 | 50 | 1   | NaCl/m | Cp:685. (240.MD) J/K | 6 | MTD | 8762 |
| 69 | 75 | 0   | NaCl/m | Cp:672. (270.MD) J/K | 4 | MTD | 8762 |
| 70 | 75 | 0.1 | NaCl/m | Cp:690. (270.MD) J/K | 6 | MTD | 8762 |
| 71 | 75 | 0.2 | NaCl/m | Cp:702. (260.MD) J/K | 6 | MTD | 8762 |
| 72 | 75 | 0.5 | NaCl/m | Cp:723. (260.MD) J/K | 6 | MTD | 8762 |
| 73 | 75 | 1   | NaCl/m | Cp:738. (260.MD) J/K | 6 | MTD | 8762 |

## Reaction No. 14179



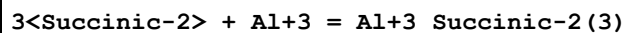
| No | t  | I-Str | Medium      | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|-------------|-------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl         | lgK:13.4 (0.06SD) | 6 | MGL   | 4371   |
| 2  | 25 | 1     | ClO4-1 salt | lgK:8.6 (2SF)     | 0 | ENS   | 2171   |
| 3  | 25 | 1     | NaClO4      | lgK:8.58 (3SF)    | 0 | MGL   | 1243   |
| 4  | 35 | 1     | NaClO4      | lgK:8.98 (3SF)    | 0 | MGL   | 1243   |
| 5  | 37 | 0.15  | NaCl        | lgK:13.3 (0.2SD)  | 6 | MGL   | 2372   |

## Reaction No. 14180



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 1     | NaClO4 | lgK:5.93 (3SF) | 0 | MGL   | 1243   |
| 2  | 35 | 1     | NaClO4 | lgK:6.40 (3SF) | 0 | MGL   | 1243   |

## Reaction No. 14181



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 1     | NaClO4 | lgK:8.98 (3SF) | 0 | MGL   | 1243   |
| 2  | 35 | 1     | NaClO4 | lgK:9.52 (3SF) | 0 | MGL   | 1243   |

| Reaction No. 14209                                                                                                      |    |       |        |                  |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $2\text{Al}^{3+} + 4\text{H}_2\text{O} + \text{Phthalic-2} = \text{Al}_2(\text{OH})_2(\text{Phthalic-2}) + 4\text{H}^+$ |    |       |        |                  |   |       |        |
| No                                                                                                                      | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                                       | 25 | 0.6   | NaCl   | lgK:-8.4(0.08MD) | 6 | MGL   | 1342   |

| Reaction No. 14257                                                     |       |       |                    |               |   |       |        |
|------------------------------------------------------------------------|-------|-------|--------------------|---------------|---|-------|--------|
| $\text{PAR-2} + \text{Al}^{3+} = \text{Al}_2(\text{OH})_2\text{PAR-2}$ |       |       |                    |               |   |       |        |
| No                                                                     | t     | I-Str | Medium             | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                      | 18:22 | 0.1   | NaClO <sub>4</sub> | lgK:11.5(3SF) | 4 | MSP   | 999    |
| 2                                                                      | 25    | 0     | Inf. Dilution      | lgK:11.7(3SF) | 1 | ESC   |        |

| Reaction No. 14326                                                             |    |       |                  |                 |   |       |        |
|--------------------------------------------------------------------------------|----|-------|------------------|-----------------|---|-------|--------|
| $\text{TriMDTA-4} + \text{Al}^{3+} = \text{Al}_2(\text{OH})_2\text{TriMDTA-4}$ |    |       |                  |                 |   |       |        |
| No                                                                             | t  | I-Str | Medium           | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                              | 20 | 0.1   | KNO <sub>3</sub> | lgK:16.33(4SF)w | 6 | MPH   | 2043   |
| 2                                                                              | 20 | 0.1   | KNO <sub>3</sub> | lgK:16.31(4SF)  | 6 | MPL   | 2043   |
| 3                                                                              | 25 | 0     | Inf. Dilution    | lgK:16.2(3SF)   | 2 | EAE   | 1175   |

| Reaction No. 14420                                                                |    |       |                  |                  |   |       |        |
|-----------------------------------------------------------------------------------|----|-------|------------------|------------------|---|-------|--------|
| $2\text{OHTriMDTA-4} + \text{Al}^{3+} = \text{Al}_2(\text{OH})_2\text{TriMDTA-4}$ |    |       |                  |                  |   |       |        |
| No                                                                                | t  | I-Str | Medium           | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                 | 25 | 0.1   | KNO <sub>3</sub> | lgK:15.2(0.03SD) | 4 | MGL   | 999    |

| Reaction No. 14516                                                           |    |       |                  |               |   |       |        |
|------------------------------------------------------------------------------|----|-------|------------------|---------------|---|-------|--------|
| $\text{dlBDTA-4} + \text{Al}^{3+} = \text{Al}_2(\text{OH})_2\text{dlBDTA-4}$ |    |       |                  |               |   |       |        |
| No                                                                           | t  | I-Str | Medium           | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                            | 20 | 0.1   | KNO <sub>3</sub> | lgK:18.5(3SF) | 4 | MSC   | 999    |
| 2                                                                            | 25 | 0     | Inf. Dilution    | lgK:18.8(3SF) | 1 | ESC   |        |

| Reaction No. 14544                                                         |    |       |                  |               |   |       |        |
|----------------------------------------------------------------------------|----|-------|------------------|---------------|---|-------|--------|
| $\text{mBDTA-4} + \text{Al}^{3+} = \text{Al}_2(\text{OH})_2\text{mBDTA-4}$ |    |       |                  |               |   |       |        |
| No                                                                         | t  | I-Str | Medium           | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                          | 20 | 0.1   | KNO <sub>3</sub> | lgK:16.5(3SF) | 4 | MSC   | 999    |
| 2                                                                          | 25 | 0     | Inf. Dilution    | lgK:16.8(3SF) | 1 | ESC   |        |

| Reaction No. 14641                                                       |  |  |  |  |  |  |  |
|--------------------------------------------------------------------------|--|--|--|--|--|--|--|
| $\text{CDTA-4} + \text{Al}^{3+} = \text{Al}_2(\text{OH})_2\text{CDTA-4}$ |  |  |  |  |  |  |  |

| No                                                          | t     | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|-------------------------------------------------------------|-------|-------|---------------|------------------|---|-------|--------|
| 1                                                           | 20    | 0.1   | KNO3          | lgK:18.28(4SF)   | 0 | MGL   | 1963   |
| 2                                                           | 20    | 0.1   | KNO3          | lgK:18.6(0.08SD) | 6 | MEF   | 2095   |
| 3                                                           | 20    | 0.1   | Unknown       | lgK:17.63(4SF)   | 0 | MGL   | 197    |
| 4                                                           | 20    | 0.1   | Unknown       | lgK:19.50(4SF)   | 2 | ENS   | 1539   |
| 5                                                           | 25    | 0     | Inf. Dilution | lgK:22.1(3SF)    | 2 | ENS   | 6405   |
| 6                                                           | 25    | 0.1   | KNO3          | lgK:18.7(0.05SD) | 4 | MEF   | 2095   |
| 7                                                           | 25    | 0.1   | KNO3          | lgK:18.9(3SF)    | 4 | MEF   | 6953   |
| 8                                                           | 25    | 0.1   | Unknown       | lgK:19.6(3SF)    | 0 | ENS   | 2171   |
| 9                                                           | 25    | 0.2   | NaClO4        | lgK:18.50(4SF)   | 2 | MSP   | 6953   |
| 10                                                          | 30    | 0.1   | KNO3          | lgK:19.(0.3SD)   | 6 | MEF   | 2095   |
| 11                                                          | 35    | 0.1   | KNO3          | lgK:19.18(4SF)   | 5 | MGL   | 11516  |
| Note (11): Khan M et al., Indian J. Chem., 1980, 19A, 44,50 |       |       |               |                  |   |       |        |
| 12                                                          | 40    | 0.1   | KNO3          | lgK:19.2(0.06SD) | 6 | MEF   | 2095   |
| 13                                                          | 25    | 0.1   | KNO3          | dG:-25.5(0.07SD) | 4 | MEF   | 2095   |
| 14                                                          | 20:40 | 0.1   | KNO3          | dH:11.(2SF)      | 5 | MEF   | 999    |
| 15                                                          | 25    | 0.1   | KNO3          | dH:10.7(3SF)     | 5 | MCL   | 2095   |
| 16                                                          | 25    | 0.1   | KNO3          | dS:121.7(4SF)    | 5 | EFE   | 2095   |

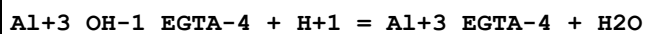
| Reaction No. 14642                         |    |       |         |               |   |       |        |
|--------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Al+3_CDTA-4 + H+1 = Al+3_H+1_CDTA-4        |    |       |         |               |   |       |        |
| No                                         | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                          | 20 | 0.1   | KNO3    | lgK:2.59(3SF) | 6 | MEF   | 2095   |
| 2                                          | 20 | 0.1   | Unknown | lgK:2.0(2SF)  | 4 | ENS   | 773    |
| 3                                          | 25 | 0.1   | KNO3    | lgK:3.4(2SF)  | 0 | ENS   | 6953   |
| Note (3): Low wgt for internal consistency |    |       |         |               |   |       |        |
| 4                                          | 25 | 0.2   | NaClO4  | lgK:2.29(3SF) | 6 | MGL   | 999    |

| Reaction No. 14643                         |    |       |        |               |   |       |        |
|--------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Al+3_OH-1_CDTA-4 + H+1 = Al+3_CDTA-4 + H2O |    |       |        |               |   |       |        |
| No                                         | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                          | 20 | 0.1   | KNO3   | lgK:7.58(3SF) | 6 | MPL   | 999    |
| 2                                          | 25 | 0.2   | NaClO4 | lgK:7.82(3SF) | 6 | MGL   | 999    |

| Reaction No. 14859                  |   |       |        |             |   |       |        |
|-------------------------------------|---|-------|--------|-------------|---|-------|--------|
| Al+3 + H+1_DTPA-5 = Al+3_H+1_DTPA-5 |   |       |        |             |   |       |        |
| No                                  | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

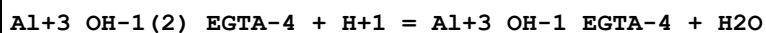
|   |    |     |               |                 |   |     |      |
|---|----|-----|---------------|-----------------|---|-----|------|
| 1 | 20 | 0.1 | Unknown       | lgK:12.70 (4SF) | 6 | ENS | 1539 |
| 2 | 25 | 0   | Inf. Dilution | lgK:15.2 (3SF)  | 2 | EAE |      |

## Reaction No. 14912



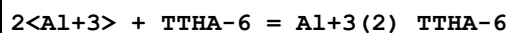
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.2   | NaClO4 | lgK:5.20 (3SF) | 6 | MGL   | 999    |

## Reaction No. 14913



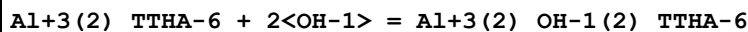
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.2   | NaClO4 | lgK:8.42 (3SF) | 6 | MGL   | 999    |

## Reaction No. 14951



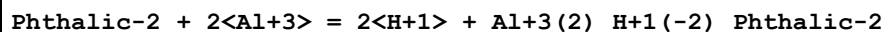
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:28.9 (3SF) | 4 | MRX   | 906    |
| 2  | 25 | 0.1   | KNO3   | lgK:28.6 (3SF) | 4 | MEF   | 999    |

## Reaction No. 14952



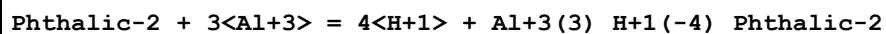
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:15.9 (3SF) | 4 | MGL   | 999    |

## Reaction No. 15578



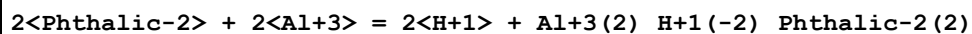
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-2.50 (0.02MD) | 6 | MGL   | 1342   |

## Reaction No. 15579



| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-8.47 (0.03MD) | 6 | MGL   | 1342   |

## Reaction No. 15580



| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:-0.07 (0.06MD) | 6 | MGL   | 1342   |

| Reaction No. 15581                                         |    |       |        |                  |   |       |        |
|------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Phthalic-2 + 2<Al+3> = 4<H+1> + Al+3(2)_H+1(-4)_Phthalic-2 |    |       |        |                  |   |       |        |
| No                                                         | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                          | 25 | 0.6   | NaCl   | lgK:-8.4(0.08MD) | 6 | MGL   | 1342   |

| Reaction No. 15902                          |    |       |               |                   |   |       |        |
|---------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| Malic-2 + Al+3 = H+1 + Al+3_H+1(-1)_Malic-2 |    |       |               |                   |   |       |        |
| No                                          | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                           | 25 | 0     | Inf. Dilution | lgK:2.71(0.06SD)  | 2 | EDH   | 29190  |
| 2                                           | 25 | 0     | UCT_Sea       | lgK:2.26(0.01SD)  | 2 | EDH   | 5390   |
| 3                                           | 25 | 0.1   | UCT_Sea       | lgK:1.20(0.01SD)  | 2 | EDH   | 5390   |
| 4                                           | 25 | 0.1   | Unknown       | lgK:1.2(2SF)      | 4 | ESL   | 1217   |
| 5                                           | 25 | 0.15  | NaCl          | lgK:1.14(0.01SD)  | 4 | MGL   | 29190  |
| 6                                           | 25 | 0.2   | KCl           | lgK:1.3(0.2SD)    | 3 | MGL   | 4371   |
| 7                                           | 25 | 0.2   | UCT_Sea       | lgK:0.99(0.01SD)  | 2 | EDH   | 5390   |
| 8                                           | 25 | 0.3   | UCT_Sea       | lgK:0.87(0.01SD)  | 2 | EDH   | 5390   |
| 9                                           | 25 | 0.4   | UCT_Sea       | lgK:0.79(0.01SD)  | 2 | EDH   | 5390   |
| 10                                          | 25 | 0.5   | NaCl          | lgK:0.46(0.02SD)  | 4 | MGL   | 29190  |
| 11                                          | 25 | 0.5   | UCT_Sea       | lgK:0.73(0.01SD)  | 2 | EDH   | 5390   |
| 12                                          | 25 | 0.6   | UCT_Sea       | lgK:0.68(0.01SD)  | 2 | EDH   | 5390   |
| 13                                          | 25 | 0.7   | UCT_Sea       | lgK:0.64(0.01SD)  | 2 | EDH   | 5390   |
| 14                                          | 25 | 1     | NaCl          | lgK:-0.17(0.02SD) | 3 | MGL   | 29190  |

| Reaction No. 15907              |    |       |               |               |   |       |        |
|---------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Al+3 + Maleic-2 = Al+3_Maleic-2 |    |       |               |               |   |       |        |
| No                              | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                               | 25 | 0     | Inf. Dilution | lgK:5.48(3SF) | 4 | ENS   | 6953   |
| 2                               | 25 | 0.1   | Unknown       | lgK:3.(1SF)   | 3 | ESL   | 1217   |

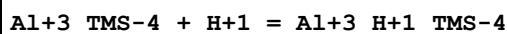
| Reaction No. 15916                |    |       |         |              |   |       |        |
|-----------------------------------|----|-------|---------|--------------|---|-------|--------|
| Al+3 + Fumaric-2 = Al+3_Fumaric-2 |    |       |         |              |   |       |        |
| No                                | t  | I-Str | Medium  | Value (Dev)  | W | Tech. | Ref. # |
| 1                                 | 25 | 0.1   | Unknown | lgK:2.5(2SF) | 4 | ESL   | 1217   |

| Reaction No. 15993        |  |  |  |  |  |  |  |
|---------------------------|--|--|--|--|--|--|--|
| Al+3 + TMS-4 = Al+3_TMS-4 |  |  |  |  |  |  |  |



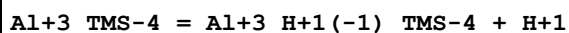
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:7.63 (3SF) | 6 | MGL   | 1217   |

## Reaction No. 15994



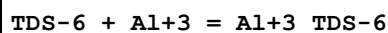
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:2.99 (3SF) | 6 | MGL   | 1217   |

## Reaction No. 15995



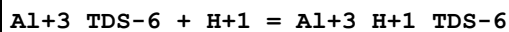
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:-5.03 (3SF) | 6 | MGL   | 1217   |

## Reaction No. 16020



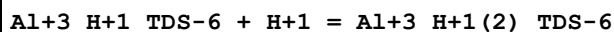
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:8.56 (3SF) | 2 | MGL   | 1217   |
| 2  | 25 | 0.1   | KCl    | lgK:9.04 (3SF) | 2 | MGL   | 1217   |

## Reaction No. 16021



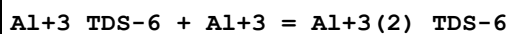
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:5.34 (3SF) | 6 | MGL   | 1217   |

## Reaction No. 16022



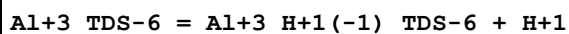
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:3.04 (3SF) | 6 | MGL   | 1217   |

## Reaction No. 16023



| No | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
|----|----|-------|--------|--------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:2. (1SF) | 3 | MGL   | 1217   |

## Reaction No. 16024



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:-6.72 (3SF) | 6 | MGL   | 1217   |

| Reaction No. 16025                                                                                   |    |       |        |               |   |       |        |
|------------------------------------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| $2\text{Al}^{3+} + \text{TDS-6} = \text{Al}^{3+}(2)_{\text{H}+1(-1)}_{\text{TDS-6}} + \text{H}^{+1}$ |    |       |        |               |   |       |        |
| No                                                                                                   | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                                    | 25 | 0.1   | KCl    | lgK:9.36(3SF) | 6 | MGL   | 1217   |

| Reaction No. 16144                                                                                             |    |       |        |               |   |       |        |
|----------------------------------------------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| $\text{Al}^{3+}_{\text{Enterobactin-6}} + \text{H}^{+1} = \text{Al}^{3+}_{\text{H}+1}_{\text{Enterobactin-6}}$ |    |       |        |               |   |       |        |
| No                                                                                                             | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                                              | 25 | 0.1   | KCl    | lgK:5.15(3SF) | 6 | MGL   | 1302   |

| Reaction No. 16145                                                                                                             |    |       |        |              |   |       |        |
|--------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| $\text{Al}^{3+}_{\text{H}+1}_{\text{Enterobactin-6}} + \text{H}^{+1} = \text{Al}^{3+}_{\text{H}+1}(2)_{\text{Enterobactin-6}}$ |    |       |        |              |   |       |        |
| No                                                                                                                             | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                                                                                              | 25 | 0.1   | KCl    | lgK:3.4(2SF) | 4 | MGL   | 1302   |

| Reaction No. 16146                                                                                                                 |    |       |        |              |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| $\text{Al}^{3+}_{\text{H}+1}(2)_{\text{Enterobactin-6}} + 2\text{H}^{+1} = \text{Al}^{3+}_{\text{H}+1}(4)_{\text{Enterobactin-6}}$ |    |       |        |              |   |       |        |
| No                                                                                                                                 | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                                                                                                  | 25 | 0.1   | KCl    | lgK:2.6(2SF) | 4 | MGL   | 1302   |

| Reaction No. 16575                                                                                        |    |       |        |                   |   |       |        |
|-----------------------------------------------------------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| $\text{Al}^{3+} + 3\text{Am}^{3+}_{\text{PhosPr-3}} = \text{Al}^{3+}_{3\text{Am}^{3+}_{\text{PhosPr-3}}}$ |    |       |        |                   |   |       |        |
| No                                                                                                        | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                         | 25 | 1     | KCl    | lgK:10.93(0.02SD) | 4 | MGL   | 1198   |

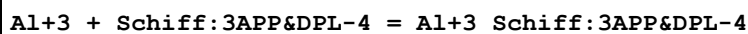
| Reaction No. 16576                                                                                                                     |    |       |        |                  |   |       |        |
|----------------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $\text{Al}^{3+}_{3\text{Am}^{3+}_{\text{PhosPr-3}}} + \text{H}^{+1} = \text{Al}^{3+}_{\text{H}+1}_{3\text{Am}^{3+}_{\text{PhosPr-3}}}$ |    |       |        |                  |   |       |        |
| No                                                                                                                                     | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                                                      | 25 | 1     | KCl    | lgK:4.19(0.02SD) | 4 | MGL   | 1198   |

| Reaction No. 16581                                                                  |    |       |        |                  |   |       |        |
|-------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $\text{Al}^{3+} + 5'\text{DeOxPyridox-1} = \text{Al}^{3+}_{5'\text{DeOxPyridox-1}}$ |    |       |        |                  |   |       |        |
| No                                                                                  | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                   | 25 | 1     | KCl    | lgK:6.15(0.02SD) | 4 | MGL   | 1198   |

| Reaction No. 16582                                                                        |  |  |  |  |  |  |  |
|-------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|
| $\text{Al}^{3+} + 2(5'\text{DeOxPyridox-1}) = \text{Al}^{3+}_{5'\text{DeOxPyridox-1}(2)}$ |  |  |  |  |  |  |  |

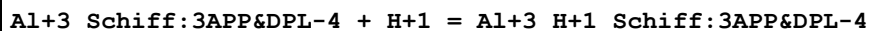
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 1     | KCl    | lgK:12.19 (4SF) | 6 | MGL   | 1198   |

## Reaction No. 16588



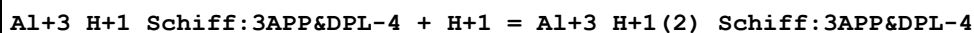
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 1     | KCl    | lgK:18.68 (0.02SD) | 6 | MGL   | 1198   |

## Reaction No. 16589



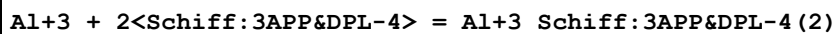
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 1     | KCl    | lgK:4.86 (0.02SD) | 4 | MGL   | 1198   |

## Reaction No. 16590



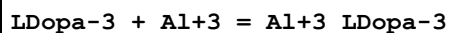
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 1     | KCl    | lgK:3.23 (0.02SD) | 4 | MGL   | 1198   |

## Reaction No. 16591



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 1     | KCl    | lgK:25.10 (4SF) | 6 | MGL   | 1198   |

## Reaction No. 17259

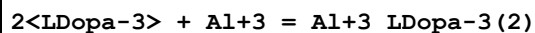


| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.12  | KCl    | lgK:19.60 (4SF) | 0 | ENS   | 6953   |

Note (1): Low wgt for internal consistency

|   |    |     |     |                   |   |     |      |
|---|----|-----|-----|-------------------|---|-----|------|
| 2 | 25 | 0.2 | KCl | lgK:26.1 (0.03SD) | 0 | MGL | 1391 |
| 3 | 25 | 0.2 | KCl | lgK:16.03 (4SF)   | 4 | ENS | 6953 |

## Reaction No. 17260



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:49.4 (0.03SD) | 0 | MGL   | 1391   |
| 2  | 25 | 0.2   | KCl    | lgK:29.24 (4SF)   | 5 | ENS   | 6953   |

## Reaction No. 17261

| 3<LDopa-3> + Al+3 = Al+3_LDopa-3(3) |    |       |        |                  |   |       |        |
|-------------------------------------|----|-------|--------|------------------|---|-------|--------|
| No                                  | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                   | 25 | 0.2   | KCl    | lgK:68.6(0.03SD) | 0 | MGL   | 1391   |
| 2                                   | 25 | 0.2   | KCl    | lgK:38.36(4SF)   | 5 | ENS   | 6953   |

| Reaction No. 17262                                |    |       |        |                  |   |       |        |
|---------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 2<Al+3_H+1_LDopa-3> = Al+3(2)_LDopa-3(2) + 2<H+1> |    |       |        |                  |   |       |        |
| No                                                | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                 | 25 | 0.2   | KCl    | lgK:-7.5(0.03SD) | 6 | MGL   | 1391   |

| Reaction No. 17263                  |    |       |        |                  |   |       |        |
|-------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Al+3 + Dopamine-2 = Al+3_Dopamine-2 |    |       |        |                  |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                   | 25 | 0.2   | KCl    | lgK:26.4(0.03SD) | 4 | MGL   | 1391   |

| Reaction No. 17264                        |    |       |        |                  |   |       |        |
|-------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 2<Dopamine-2> + Al+3 = Al+3_Dopamine-2(2) |    |       |        |                  |   |       |        |
| No                                        | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                         | 25 | 0.2   | KCl    | lgK:50.2(0.03SD) | 4 | MGL   | 1391   |

| Reaction No. 17265                        |    |       |        |                  |   |       |        |
|-------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 3<Dopamine-2> + Al+3 = Al+3_Dopamine-2(3) |    |       |        |                  |   |       |        |
| No                                        | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                         | 25 | 0.2   | KCl    | lgK:69.9(0.03SD) | 4 | MGL   | 1391   |

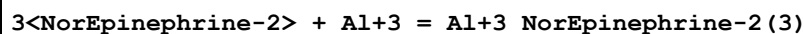
| Reaction No. 17266                                      |    |       |         |                  |   |       |        |
|---------------------------------------------------------|----|-------|---------|------------------|---|-------|--------|
| 2<Al+3_H+1_Dopamine-2> = Al+3(2)_Dopamine-2(2) + 2<H+1> |    |       |         |                  |   |       |        |
| No                                                      | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.1   | Unknown | lgK:-7.82(3SF)   | 3 | CRV   | 552    |
| 2                                                       | 25 | 0.2   | KCl     | lgK:-7.8(0.03SD) | 4 | MGL   | 1391   |

| Reaction No. 17267                              |    |       |        |                  |   |       |        |
|-------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Al+3 + NorEpinephrine-2 = Al+3_NorEpinephrine-2 |    |       |        |                  |   |       |        |
| No                                              | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                               | 25 | 0.2   | KCl    | lgK:25.3(0.03SD) | 4 | MGL   | 1391   |

| Reaction No. 17268                                    |  |  |  |  |  |  |  |
|-------------------------------------------------------|--|--|--|--|--|--|--|
| 2<NorEpinephrine-2> + Al+3 = Al+3_NorEpinephrine-2(2) |  |  |  |  |  |  |  |

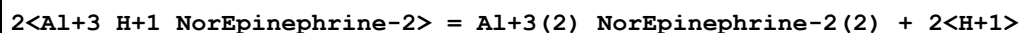
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:48.0(0.03SD) | 4 | MGL   | 1391   |

## Reaction No. 17269



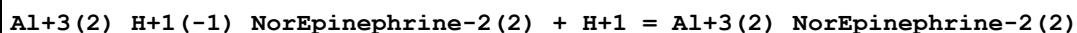
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:67.0(0.03SD) | 4 | MGL   | 1391   |

## Reaction No. 17270



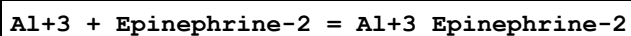
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:-7.6(0.03SD) | 4 | MGL   | 1391   |

## Reaction No. 17271



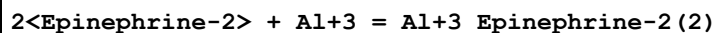
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:6.4(0.03SD) | 4 | MGL   | 1391   |

## Reaction No. 17272



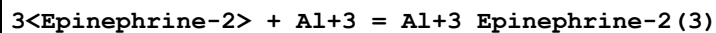
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:25.8(0.03SD) | 4 | MGL   | 1391   |

## Reaction No. 17273



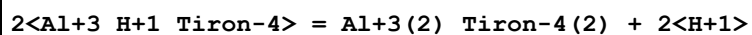
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:49.1(0.03SD) | 4 | MGL   | 1391   |

## Reaction No. 17274



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:68.6(0.03SD) | 4 | MGL   | 1391   |

## Reaction No. 17277



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:-7.7(0.03SD) | 4 | MGL   | 1391   |

| Reaction No. 17278                                                                                      |    |       |        |                 |   |       |        |
|---------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{Al+3(2)}_{\text{H+1(-1)}}_{\text{Tiron-4(2)}} + \text{H+1} = \text{Al+3(2)}_{\text{Tiron-4(2)}}$ |    |       |        |                 |   |       |        |
| No                                                                                                      | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                       | 25 | 0.2   | KCl    | lgK:5.9(0.03SD) | 4 | MGL   | 1391   |

| Reaction No. 17996                                                                    |    |       |               |               |   |       |        |
|---------------------------------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $\text{Al+3} + \text{H+1}_{\text{CDTA-4}} = \text{Al+3}_{\text{H+1}_{\text{CDTA-4}}}$ |    |       |               |               |   |       |        |
| No                                                                                    | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                     | 20 | 0.1   | Unknown       | lgK:9.70(3SF) | 6 | ENS   | 1539   |
| 2                                                                                     | 25 | 0     | Inf. Dilution | lgK:9.9(2SF)  | 1 | ESC   |        |

| Reaction No. 18035                                                                                                 |    |       |        |                    |   |       |        |
|--------------------------------------------------------------------------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| $3\text{Al+3} + \text{H+1(4)}_{\text{Gallic-4}} = \text{Al+3(3)}_{\text{H+1(-1)}}_{\text{Gallic-4}} + 5\text{H+1}$ |    |       |        |                    |   |       |        |
| No                                                                                                                 | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                                                                  | 25 | 0.6   | NaCl   | lgK:-12.52(0.01MD) | 6 | MGL   | 1440   |

| Reaction No. 18036                                                                                                    |    |       |        |                    |   |       |        |
|-----------------------------------------------------------------------------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| $4\text{Al+3} + 3\text{H+1(4)}_{\text{Gallic-4}} = \text{Al+3(4)}_{\text{H+1(3)}}_{\text{Gallic-4(3)}} + 9\text{H+1}$ |    |       |        |                    |   |       |        |
| No                                                                                                                    | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                                                                     | 25 | 0.6   | NaCl   | lgK:-20.25(0.05MD) | 3 | MGL   | 1440   |

| Reaction No. 18037                                                                                                                                                    |    |       |        |                 |   |       |        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{Al+3} + \text{H+1(4)}_{\text{Gallic-4}} + 4\text{H}_2\text{O} = \text{Al+3}_{\text{H+1}_{\text{Gallic-4}}}_{\text{H}_2\text{O(4)}}_{\text{(s)}} + 3\text{H+1}$ |    |       |        |                 |   |       |        |
| No                                                                                                                                                                    | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                                                                     | 25 | 0.6   | NaCl   | lgK:-6.2(0.5MD) | 4 | MGL   | 1440   |

| Reaction No. 18050                                                                                               |    |       |               |                 |   |       |        |
|------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{Al+3}_{\text{EDTA-4}} + 2\text{H}_2\text{O} = \text{Al+3}_{\text{OH-1(2)}}_{\text{EDTA-4}} + 2\text{H+1}$ |    |       |               |                 |   |       |        |
| No                                                                                                               | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                | 20 | 0.1   | Unknown       | lgK:-16.20(4SF) | 6 | ENS   | 1539   |
| 2                                                                                                                | 25 | 0     | Inf. Dilution | lgK:-17.2(3SF)  | 2 | EAE   |        |

| Reaction No. 18085                                                                                          |    |       |               |                |   |       |        |
|-------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $\text{Al+3}_{\text{DTPA-5}} + \text{H}_2\text{O} = \text{Al+3}_{\text{OH-1}_{\text{DTPA-5}}} + \text{H+1}$ |    |       |               |                |   |       |        |
| No                                                                                                          | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                           | 20 | 0.1   | Unknown       | lgK:-7.60(3SF) | 6 | ENS   | 1539   |
| 2                                                                                                           | 25 | 0     | Inf. Dilution | lgK:-8.22(3SF) | 2 | EAE   |        |

| Reaction No. 18162                             |       |       |         |               |   |       |        |
|------------------------------------------------|-------|-------|---------|---------------|---|-------|--------|
| Al+3_OH-1_Citric-3 + H+1 = Al+3_Citric-3 + H2O |       |       |         |               |   |       |        |
| No                                             | t     | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                              | 20    | 0.1   | Unknown | lgK:3.31(3SF) | 6 | CMN   | 1084   |
| 2                                              | 20    | 0.1   | Unknown | lgK:3.50(3SF) | 6 | ENS   | 1539   |
| 3                                              | 25    | 0.1   | KNO3    | lgK:3.31(3SF) | 6 | MGL   | 1244   |
| 4                                              | 25:37 | 0.15  | KNO3    | lgK:3.4(2SF)  | 5 | MPT   | 3607   |
| 5                                              | 33    | 0.25  | NaClO4  | lgK:3.5(2SF)  | 3 | MGL   | 999    |

| Reaction No. 18163                                      |    |       |               |                 |   |       |        |
|---------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Al+3_Citric-3 + 2<H2O> = Al+3_OH-1(2)_Citric-3 + 2<H+1> |    |       |               |                 |   |       |        |
| No                                                      | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                       | 20 | 0.1   | Unknown       | lgK:-10.30(4SF) | 6 | ENS   | 1539   |
| 2                                                       | 25 | 0     | Inf. Dilution | lgK:-10.9(3SF)  | 2 | EAE   |        |

| Reaction No. 18293                          |    |       |               |               |   |       |        |
|---------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Al+3 + H+1_Tartaric-2 = Al+3_H+1_Tartaric-2 |    |       |               |               |   |       |        |
| No                                          | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 20 | 0     | Inf. Dilution | lgK:3.43(3SF) | 6 | MSC   | 999    |
| 2                                           | 20 | 0.1   | Unknown       | lgK:3.43(3SF) | 6 | ENS   | 1539   |

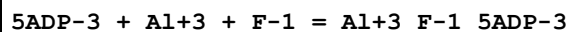
| Reaction No. 18294                                 |    |       |         |                |   |       |        |
|----------------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Al+3_Tartaric-2 + H2O = Al+3_OH-1_Tartaric-2 + H+1 |    |       |         |                |   |       |        |
| No                                                 | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                  | 20 | 0.1   | Unknown | lgK:-2.70(3SF) | 6 | ENS   | 1539   |
| 2                                                  | 25 | 0.1   | Unknown | lgK:1.18(3SF)  | 0 | MGL   | 1217   |

| Reaction No. 18295                                          |    |       |               |                |   |       |        |
|-------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Al+3_Tartaric-2 + 2<H2O> = Al+3_OH-1(2)_Tartaric-2 + 2<H+1> |    |       |               |                |   |       |        |
| No                                                          | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                           | 20 | 0.1   | Unknown       | lgK:-5.41(3SF) | 0 | ENS   | 1539   |
| 2                                                           | 25 | 0     | Inf. Dilution | lgK:-9.6(2SF)  | 1 | EES   |        |

| Reaction No. 18296                                          |    |       |         |               |   |       |        |
|-------------------------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Al+3 + Tartaric-2 + H+1_Tartaric-2 = Al+3_H+1_Tartaric-2(2) |    |       |         |               |   |       |        |
| No                                                          | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                           | 20 | 0.1   | Unknown | lgK:7.60(3SF) | 4 | ENS   | 1539   |

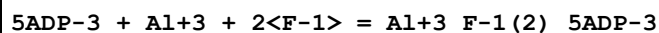
|   |    |   |               |               |   |     |      |
|---|----|---|---------------|---------------|---|-----|------|
| 2 | 25 | 0 | Inf. Dilution | lgK:7.5 (2SF) | 1 | EAE | 1175 |
|---|----|---|---------------|---------------|---|-----|------|

## Reaction No. 18702



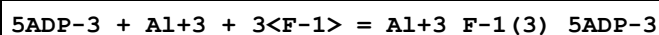
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 21 | 0.01  | Unknown       | lgK:11.9 (3SF) | 4 | MMR   | 1556   |
| 2  | 25 | 0     | Inf. Dilution | lgK:12.0 (3SF) | 1 | ESC   |        |

## Reaction No. 18703



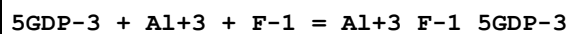
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 21 | 0.01  | Unknown       | lgK:15.0 (3SF) | 4 | MMR   | 1556   |
| 2  | 25 | 0     | Inf. Dilution | lgK:15.1 (3SF) | 1 | ESC   |        |

## Reaction No. 18704



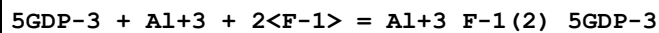
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 21 | 0.01  | Unknown       | lgK:16.8 (3SF) | 4 | MMR   | 1556   |
| 2  | 25 | 0     | Inf. Dilution | lgK:16.9 (3SF) | 1 | ESC   |        |

## Reaction No. 18705



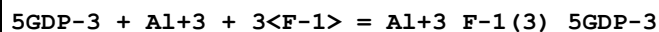
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 21 | 0.01  | Unknown       | lgK:11.9 (3SF) | 4 | MMR   | 1556   |
| 2  | 25 | 0     | Inf. Dilution | lgK:12.1 (3SF) | 1 | ESC   |        |

## Reaction No. 18706



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 21 | 0.01  | Unknown       | lgK:15.0 (3SF) | 4 | MMR   | 1556   |
| 2  | 25 | 0     | Inf. Dilution | lgK:15.1 (3SF) | 1 | ESC   |        |

## Reaction No. 18707



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 21 | 0.01  | Unknown       | lgK:16.8 (3SF) | 4 | MMR   | 1556   |
| 2  | 25 | 0     | Inf. Dilution | lgK:16.9 (3SF) | 1 | ESC   |        |



| Reaction No. 19274              |    |       |        |                 |   |       |        |
|---------------------------------|----|-------|--------|-----------------|---|-------|--------|
| BAMTPH-3 + Al+3 = Al+3_BAMTPH-3 |    |       |        |                 |   |       |        |
| No                              | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                               | 25 | 0.1   | KNO3   | lgK:21.85 (4SF) | 6 | MGL   | 1739   |

| Reaction No. 19794                                     |    |       |        |                |   |       |        |
|--------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Al+3_OH-1_2OHTriMDTA-4 + H+1 = Al+3_2OHTriMDTA-4 + H2O |    |       |        |                |   |       |        |
| No                                                     | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                      | 25 | 0.1   | KNO3   | lgK:6.70 (3SF) | 6 | MGL   | 999    |

| Reaction No. 19795                                             |    |       |        |                |   |       |        |
|----------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Al+3_OH-1(2)_2OHTriMDTA-4 + H+1 = Al+3_OH-1_2OHTriMDTA-4 + H2O |    |       |        |                |   |       |        |
| No                                                             | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                              | 25 | 0.1   | KNO3   | lgK:8.70 (3SF) | 6 | MGL   | 999    |

| Reaction No. 19796                            |    |       |        |                   |   |       |        |
|-----------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| 2<Al+3> + 2OHTriMDTA-4 = Al+3(2)_2OHTriMDTA-4 |    |       |        |                   |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                             | 25 | 0.1   | KNO3   | lgK:16.6 (0.05SD) | 4 | MGL   | 999    |

| Reaction No. 19797                                           |    |       |        |               |   |       |        |
|--------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Al+3(2)_OH-1_2OHTriMDTA-4 + H+1 = Al+3(2)_2OHTriMDTA-4 + H2O |    |       |        |               |   |       |        |
| No                                                           | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                            | 25 | 0.1   | KNO3   | lgK:1.0 (2SF) | 6 | MGL   | 999    |

| Reaction No. 19798                                                   |    |       |        |                |   |       |        |
|----------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Al+3(2)_OH-1(2)_2OHTriMDTA-4 + H+1 = Al+3(2)_OH-1_2OHTriMDTA-4 + H2O |    |       |        |                |   |       |        |
| No                                                                   | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                    | 25 | 0.1   | KNO3   | lgK:3.55 (3SF) | 6 | MGL   | 999    |

| Reaction No. 19799                                                      |    |       |        |                |   |       |        |
|-------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Al+3(2)_OH-1(3)_2OHTriMDTA-4 + H+1 = Al+3(2)_OH-1(2)_2OHTriMDTA-4 + H2O |    |       |        |                |   |       |        |
| No                                                                      | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                       | 25 | 0.1   | KNO3   | lgK:6.20 (3SF) | 6 | MGL   | 999    |

| Reaction No. 19921                             |   |       |        |             |   |       |        |
|------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| Al+3_Malonic-2 + Malonic-2 = Al+3_Malonic-2(2) |   |       |        |             |   |       |        |
| No                                             | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

|                                            |    |      |        |                   |   |     |      |
|--------------------------------------------|----|------|--------|-------------------|---|-----|------|
| 1                                          | 25 | 0.1  | KCl    | lgK:4.82(0.007SD) | 6 | MGL | 2356 |
| 2                                          | 30 | 0.1  | NaClO4 | lgK:3.95(3SF)     | 0 | ENS | 2356 |
| Note (2): Low wgt for internal consistency |    |      |        |                   |   |     |      |
| 3                                          | 30 | 0.2  | NaClO4 | lgK:4.16(3SF)     | 0 | MGL | 2123 |
| 4                                          | 37 | 0.15 | NaCl   | lgK:4.847(4SF)    | 5 | MGL | 2372 |

| Reaction No. 20328          |    |       |        |               |   |       |        |
|-----------------------------|----|-------|--------|---------------|---|-------|--------|
| mODS-4 + Al+3 = Al+3_mODS-4 |    |       |        |               |   |       |        |
| No                          | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                           | 25 | 0.1   | KCl    | lgK:9.17(3SF) | 6 | MGL   | 1747   |

| Reaction No. 20329                  |    |       |        |               |   |       |        |
|-------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Al+3_mODS-4 + H+1 = Al+3_H+1_mODS-4 |    |       |        |               |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | KCl    | lgK:3.37(3SF) | 6 | MGL   | 1747   |

| Reaction No. 20330                         |    |       |        |                |   |       |        |
|--------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Al+3_mODS-4 + H2O = Al+3_OH-1_mODS-4 + H+1 |    |       |        |                |   |       |        |
| No                                         | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                          | 25 | 0.1   | KCl    | lgK:-5.30(3SF) | 6 | MGL   | 1747   |

| Reaction No. 20331                                 |    |       |        |                |   |       |        |
|----------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Al+3_OH-1_mODS-4 + H2O = Al+3_OH-1(2)_mODS-4 + H+1 |    |       |        |                |   |       |        |
| No                                                 | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                  | 25 | 0.1   | KCl    | lgK:-6.96(3SF) | 6 | MGL   | 1747   |

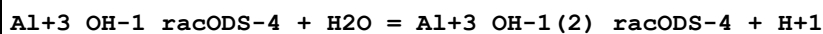
| Reaction No. 20376              |    |       |        |               |   |       |        |
|---------------------------------|----|-------|--------|---------------|---|-------|--------|
| racODS-4 + Al+3 = Al+3_racODS-4 |    |       |        |               |   |       |        |
| No                              | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                               | 25 | 0.1   | KCl    | lgK:8.43(3SF) | 6 | MGL   | 1747   |

| Reaction No. 20377                      |    |       |        |               |   |       |        |
|-----------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Al+3_racODS-4 + H+1 = Al+3_H+1_racODS-4 |    |       |        |               |   |       |        |
| No                                      | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                       | 25 | 0.1   | KCl    | lgK:3.42(3SF) | 6 | MGL   | 1747   |

| Reaction No. 20378                             |  |  |  |  |  |  |  |
|------------------------------------------------|--|--|--|--|--|--|--|
| Al+3_racODS-4 + H2O = Al+3_OH-1_racODS-4 + H+1 |  |  |  |  |  |  |  |

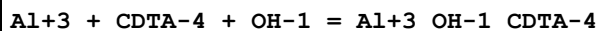
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:-5.31(3SF) | 6 | MGL   | 1747   |

## Reaction No. 20379



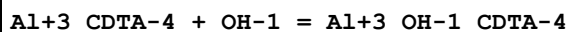
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:-6.43(3SF) | 6 | MGL   | 1747   |

## Reaction No. 20707



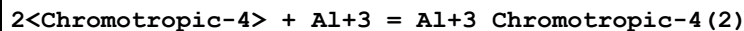
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 20 | 0.1   | KClO4         | lgK:26.6(0.08SD) | 0 | MDS   | 999    |
| 2  | 25 | 0     | Inf. Dilution | lgK:27.2(3SF)    | 1 | ELC   |        |
| 3  | 25 | 0     | Inf. Dilution | lgK:28.1(3SF)    | 0 | ENS   | 6405   |

## Reaction No. 20740



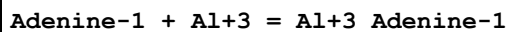
| No | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------|---------------|---|-------|--------|
| 1  | 20 | 0.1   | Unknown | lgK:6.37(3SF) | 6 | ENS   | 773    |
| 2  | 25 | 0.1   | Unknown | lgK:5.96(3SF) | 6 | ENS   | 773    |

## Reaction No. 20768



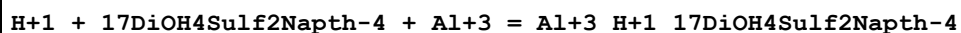
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:29.8(3SF) | 5 | ENS   | 2171   |
| 2  | 25 | 0.5   | NaClO4 | lgK:22.5(3SF) | 0 | ENS   | 6953   |

## Reaction No. 20951



| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:7.8(2SF)  | 1 | ESC   |        |
| 2  | 30 | 0.1   | KNO3          | lgK:7.56(3SF) | 5 | MPT   | 1760   |

## Reaction No. 21085



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.5   | NaClO4 | lgK:20.00(0.02MD) | 3 | MGL   | 1773   |

| Reaction No. 21086                                  |    |       |        |                   |   |       |        |
|-----------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| 17DiOH4Sulf2Naph-4 + Al+3 = Al+3_17DiOH4Sulf2Naph-4 |    |       |        |                   |   |       |        |
| No                                                  | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.5   | NaClO4 | lgK:14.95(0.03MD) | 4 | MGL   | 1773   |

| Reaction No. 21088                                                  |    |       |        |                  |   |       |        |
|---------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| H+1 + 2<17DiOH4Sulf2Naph-4> + Al+3 = Al+3_H+1_17DiOH4Sulf2Naph-4(2) |    |       |        |                  |   |       |        |
| No                                                                  | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                   | 25 | 0.5   | NaClO4 | lgK:30.8(0.08MD) | 4 | MGL   | 1773   |

| Reaction No. 21089                                        |    |       |        |                  |   |       |        |
|-----------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 2<17DiOH4Sulf2Naph-4> + Al+3 = Al+3_17DiOH4Sulf2Naph-4(2) |    |       |        |                  |   |       |        |
| No                                                        | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                         | 25 | 0.5   | NaClO4 | lgK:22.1(0.09MD) | 4 | MGL   | 1773   |

| Reaction No. 21090                                        |    |       |        |                  |   |       |        |
|-----------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 3<17DiOH4Sulf2Naph-4> + Al+3 = Al+3_17DiOH4Sulf2Naph-4(3) |    |       |        |                  |   |       |        |
| No                                                        | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                         | 25 | 0.5   | NaClO4 | lgK:28.2(0.08MD) | 4 | MGL   | 1773   |

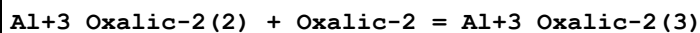
| Reaction No. 21486                             |    |       |               |               |   |       |        |
|------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Al+3_SulfCat-3 + SulfCat-3 = Al+3_SulfCat-3(2) |    |       |               |               |   |       |        |
| No                                             | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                              | 20 | 0.1   | KNO3          | lgK:13.3(3SF) | 4 | MGL   | 3494   |
| 2                                              | 25 | 0     | Inf. Dilution | lgK:13.6(3SF) | 1 | ESC   |        |

| Reaction No. 21487                                |    |       |               |              |   |       |        |
|---------------------------------------------------|----|-------|---------------|--------------|---|-------|--------|
| Al+3_SulfCat-3(2) + SulfCat-3 = Al+3_SulfCat-3(3) |    |       |               |              |   |       |        |
| No                                                | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                 | 20 | 0.1   | KNO3          | lgK:9.3(2SF) | 4 | MGL   | 3494   |
| 2                                                 | 25 | 0     | Inf. Dilution | lgK:9.6(2SF) | 1 | ESC   |        |

| Reaction No. 21653                          |    |       |        |                |   |       |        |
|---------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Al+3_Oxalic-2 + Oxalic-2 = Al+3_Oxalic-2(2) |    |       |        |                |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                           | 20 |       |        | lgK:6.0(2SF)   | 0 | MRM   | 3727   |
| 2                                           | 20 |       |        | lgK:5.0(0.1SD) | 0 | MPT   | 3727   |

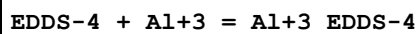
|   |    |      |               |                |   |     |      |
|---|----|------|---------------|----------------|---|-----|------|
| 3 | 25 | 0    | Inf. Dilution | lgK:5.6 (2SF)  | 1 | EOR |      |
| 4 | 25 | 0    | Inf. Dilution | lgK:6.31 (3SF) | 0 | MMR | 3727 |
| 5 | 25 | 1    | NaClO4        | lgK:5.03 (3SF) | 3 | MEF | 9709 |
| 6 | 32 | 1    | KNO3          | lgK:5.45 (3SF) | 0 | MGL | 999  |
| 7 | 37 | 0.15 | NaCl          | lgK:4.31 (3SF) | 3 | MGL | 2372 |

## Reaction No. 21654



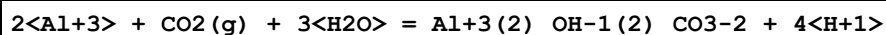
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 20 |       |               | lgK:3.87 (3SF)   | 5 | MRM   | 3727   |
| 2  | 20 |       |               | lgK:3.9 (0.06SD) | 5 | MPT   | 3727   |
| 3  | 25 | 0     | Inf. Dilution | lgK:3.08 (3SF)   | 2 | EAE   |        |
| 4  | 32 | 1     | KNO3          | lgK:3.69 (3SF)   | 6 | MGL   | 999    |

## Reaction No. 24493



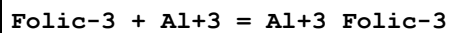
| No | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3          | lgK:14.8 (3SF)     | 0 | MSC   | 2006   |
| 2  | 25 | 0     | Inf. Dilution | lgK:20.6 (0.2SD)   | 2 | EDH   | 29190  |
| 3  | 25 | 0.15  | NaCl          | lgK:17.60 (0.04SD) | 0 | ENS   | 29190  |
| 4  | 25 | 0.5   | NaCl          | lgK:15.75 (0.03SD) | 2 | ENS   | 29190  |
| 5  | 25 | 1     | NaCl          | lgK:15.01 (0.03SD) | 2 | ENS   | 29190  |

## Reaction No. 24575



| No | t  | I-Str | Medium  | Value (Dev)         | W | Tech. | Ref. # |
|----|----|-------|---------|---------------------|---|-------|--------|
| 1  | 25 | 0.1   | Na (Cl) | lgK:-14.95 (0.17SD) | 5 | MGL   | 5494   |
| 2  | 25 | 0.6   | NaCl    | lgK:-20.41 (0.02SD) | 0 | MGL   | 5447   |
| 3  | 25 | 3     | Na (Cl) | lgK:-16.61 (0.12SD) | 5 | MGL   | 5494   |

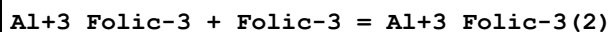
## Reaction No. 24721



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:5.3 (2SF)  | 1 | ESC   |        |
| 2  | 30 | 0     | KNO3          | lgK:5.80 (3SF) | 0 | MGL   | 2171   |
| 3  | 30 | 0.01  | KNO3          | lgK:5.25 (3SF) | 3 | MGL   | 999    |
| 4  | 30 | 0.05  | KNO3          | lgK:4.80 (3SF) | 3 | MGL   | 999    |

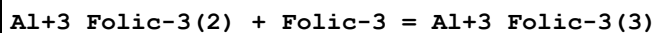
|   |    |     |      |                |   |     |     |
|---|----|-----|------|----------------|---|-----|-----|
| 5 | 30 | 0.1 | KNO3 | lgK:4.65 (3SF) | 3 | MGL | 999 |
|---|----|-----|------|----------------|---|-----|-----|

## Reaction No. 24722



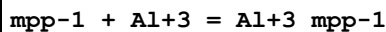
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:4.6 (2SF)  | 1 | ESC   |        |
| 2  | 30 | 0     | KNO3          | lgK:4.70 (3SF) | 2 | MGL   | 999    |
| 3  | 30 | 0.01  | KNO3          | lgK:4.55 (3SF) | 3 | MGL   | 999    |
| 4  | 30 | 0.05  | KNO3          | lgK:4.28 (3SF) | 3 | MGL   | 999    |
| 5  | 30 | 0.1   | KNO3          | lgK:4.20 (3SF) | 3 | MGL   | 999    |

## Reaction No. 24723



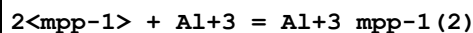
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:4.4 (2SF)  | 1 | ELC   |        |
| 2  | 30 | 0     | KNO3          | lgK:4.65 (3SF) | 0 | MGL   | 999    |
| 3  | 30 | 0.01  | KNO3          | lgK:4.50 (3SF) | 0 | MGL   | 999    |
| 4  | 30 | 0.05  | KNO3          | lgK:4.15 (3SF) | 0 | MGL   | 999    |
| 5  | 30 | 0.1   | KNO3          | lgK:4.10 (3SF) | 0 | MGL   | 999    |

## Reaction No. 25175



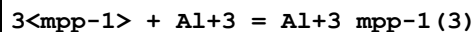
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0     | NaCl   | lgK:12.31 (4SF)    | 6 | MGL   | 2271   |
| 2  | 25 | 0.15  | NaCl   | lgK:11.87 (0.03SD) | 6 | MGL   | 2271   |
| 3  | 25 | 0.6   | NaCl   | lgK:11.43 (0.02MD) | 6 | MGL   | 2271   |

## Reaction No. 25176



| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0     | NaCl   | lgK:23.97 (4SF)    | 6 | MGL   | 2271   |
| 2  | 25 | 0.15  | NaCl   | lgK:22.54 (0.03SD) | 6 | MGL   | 2271   |
| 3  | 25 | 0.6   | NaCl   | lgK:21.73 (0.02MD) | 6 | MGL   | 2271   |

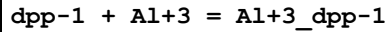
## Reaction No. 25177



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

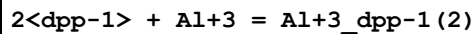
|   |    |      |      |                   |   |     |      |
|---|----|------|------|-------------------|---|-----|------|
| 1 | 25 | 0    | NaCl | lgK:33.98(4SF)    | 6 | MGL | 2271 |
| 2 | 25 | 0.15 | NaCl | lgK:32.05(0.03SD) | 6 | MGL | 2271 |
| 3 | 25 | 0.6  | NaCl | lgK:30.41(0.05MD) | 6 | MGL | 2271 |

## Reaction No. 25178



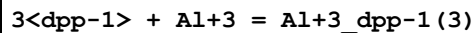
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0     | NaCl   | lgK:12.72(4SF)    | 6 | MGL   | 2271   |
| 2  | 25 | 0.1   | KCl    | lgK:12.2(0.03SD)  | 6 | MGL   | 2512   |
| 3  | 25 | 0.15  | NaCl   | lgK:11.91(0.02SD) | 6 | MGL   | 2271   |
| 4  | 25 | 0.6   | NaCl   | lgK:11.57(0.02MD) | 6 | MGL   | 2271   |

## Reaction No. 25179



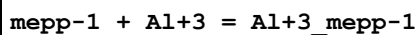
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0     | NaCl   | lgK:24.26(4SF)    | 6 | MGL   | 2271   |
| 2  | 25 | 0.15  | NaCl   | lgK:22.83(0.02SD) | 6 | MGL   | 2271   |
| 3  | 25 | 0.6   | NaCl   | lgK:22.01(0.02MD) | 6 | MGL   | 2271   |

## Reaction No. 25180



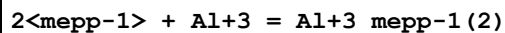
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0     | NaCl   | lgK:34.09(4SF)    | 6 | MGL   | 2271   |
| 2  | 25 | 0.1   | KCl    | lgK:32.6(0.04SD)  | 6 | MGL   | 2512   |
| 3  | 25 | 0.15  | NaCl   | lgK:32.25(0.05SD) | 6 | MGL   | 2271   |
| 4  | 25 | 0.6   | NaCl   | lgK:30.90(0.05MD) | 6 | MGL   | 2271   |

## Reaction No. 25181



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaCl   | lgK:11.75(0.04SD) | 6 | MGL   | 2271   |

## Reaction No. 25182



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaCl   | lgK:22.52(0.05SD) | 6 | MGL   | 2271   |

| Reaction No. 25183                |    |       |        |                   |   |       |        |
|-----------------------------------|----|-------|--------|-------------------|---|-------|--------|
| 3<mepp-1> + Al+3 = Al+3_mepp-1(3) |    |       |        |                   |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                 | 25 | 0.15  | NaCl   | lgK:32.17(0.06SD) | 6 | MGL   | 2271   |

| Reaction No. 25186          |    |       |        |                   |   |       |        |
|-----------------------------|----|-------|--------|-------------------|---|-------|--------|
| mhpp-1 + Al+3 = Al+3_mhpp-1 |    |       |        |                   |   |       |        |
| No                          | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                           | 25 | 0.15  | NaCl   | lgK:11.51(0.01SD) | 6 | MGL   | 2271   |

| Reaction No. 25187                |    |       |        |                   |   |       |        |
|-----------------------------------|----|-------|--------|-------------------|---|-------|--------|
| 2<mhpp-1> + Al+3 = Al+3_mhpp-1(2) |    |       |        |                   |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                 | 25 | 0.15  | NaCl   | lgK:22.49(0.01SD) | 6 | MGL   | 2271   |

| Reaction No. 25188                |    |       |        |                   |   |       |        |
|-----------------------------------|----|-------|--------|-------------------|---|-------|--------|
| 3<mhpp-1> + Al+3 = Al+3_mhpp-1(3) |    |       |        |                   |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                 | 25 | 0.15  | NaCl   | lgK:31.71(0.03SD) | 6 | MGL   | 2271   |

| Reaction No. 25269                                  |    |       |        |               |   |       |        |
|-----------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Al+3_H+1(-1)_Saccharic-2 + H+1 = Al+3 + Saccharic-2 |    |       |        |               |   |       |        |
| No                                                  | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.1   | KNO3   | lgK:1.57(3SF) | 6 | MGL   | 1244   |

| Reaction No. 25270                                        |    |       |        |                |   |       |        |
|-----------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Al+3_H+1(-2)_Saccharic-2 + H+1 = Al+3_H+1(-1)_Saccharic-2 |    |       |        |                |   |       |        |
| No                                                        | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                         | 25 | 0.1   | KNO3   | lgK:-3.76(3SF) | 6 | MGL   | 1244   |

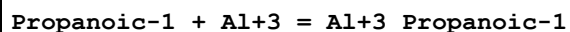
| Reaction No. 25674                              |    |       |        |               |   |       |        |
|-------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| [12]N30:Acet*3-3 + Al+3 = Al+3_[12]N30:Acet*3-3 |    |       |        |               |   |       |        |
| No                                              | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                               | 25 | 0.1   | KCl    | lgK:12.5(3SF) | 6 | MGL   | 2284   |

| Reaction No. 25675                                      |   |       |        |             |   |       |        |
|---------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| Al+3_[12]N30:Acet*3-3 + H+1 = Al+3_H+1_[12]N30:Acet*3-3 |   |       |        |             |   |       |        |
| No                                                      | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |



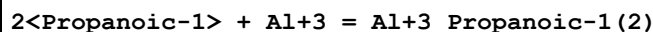
|   |    |     |     |                |   |     |      |
|---|----|-----|-----|----------------|---|-----|------|
| 1 | 25 | 0.1 | KCl | lgK:3.28 (3SF) | 6 | MGL | 2284 |
|---|----|-----|-----|----------------|---|-----|------|

## Reaction No. 26965



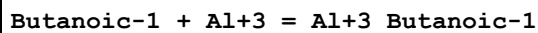
| No | t  | I-Str | Medium    | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|-----------|-------------------|---|-------|--------|
| 1  | 25 | 0     | UCT_Sea   | lgK:2.26 (0.01SD) | 4 | EDH   | 5390   |
| 2  | 25 | 0.1   | UCT_Sea   | lgK:1.81 (0.01SD) | 4 | EDH   | 5390   |
| 3  | 25 | 0.2   | UCT_Sea   | lgK:1.75 (0.01SD) | 4 | EDH   | 5390   |
| 4  | 25 | 0.3   | UCT_Sea   | lgK:1.72 (0.01SD) | 4 | EDH   | 5390   |
| 5  | 25 | 0.4   | UCT_Sea   | lgK:1.70 (0.01SD) | 4 | EDH   | 5390   |
| 6  | 25 | 0.5   | UCT_Sea   | lgK:1.69 (0.01SD) | 4 | EDH   | 5390   |
| 7  | 25 | 0.6   | UCT_Sea   | lgK:1.69 (0.01SD) | 4 | EDH   | 5390   |
| 8  | 25 | 0.7   | UCT_Sea   | lgK:1.69 (0.01SD) | 4 | EDH   | 5390   |
| 9  | 25 | 1     | NaClO4    | lgK:1.69 (3SF)    | 5 | ENS   | 2171   |
| 10 | 25 | 1     | NaClO4    | lgK:1.778 (4SF)   | 5 | MGL   | 4204   |
| 11 | 35 | 1     | NaClO4    | lgK:1.806 (4SF)   | 5 | MGL   | 4204   |
| 12 | 45 | 1     | NaClO4    | lgK:1.892 (4SF)   | 5 | MGL   | 4204   |
| 13 | 25 | 1     | Unknown/m | dH:2.29 (3SF)     | 3 | EOD   | 6366   |

## Reaction No. 26966



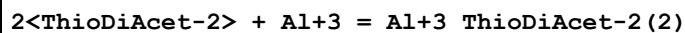
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 1     | NaClO4 | lgK:3.415 (4SF) | 5 | MGL   | 4204   |
| 2  | 35 | 1     | NaClO4 | lgK:4.041 (4SF) | 5 | MGL   | 4204   |
| 3  | 45 | 1     | NaClO4 | lgK:4.380 (4SF) | 5 | MGL   | 4204   |

## Reaction No. 26967



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 1     | KNO3   | lgK:1.58 (3SF) | 5 | ENS   | 2171   |

## Reaction No. 26968



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:6.80 (3SF) | 5 | ENS   | 2171   |

## Reaction No. 26969

| Pyrogallol-3 + Al+3 = Al+3_Pyrogallol-3 |    |       |               |                 |   |       |        |
|-----------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| No                                      | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                       | 22 | 0.2   | KCl           | lgK:24.50 (4SF) | 0 | MGL   | 906    |
| 2                                       | 25 | 0     | Inf. Dilution | lgK:14.34 (4SF) | 3 | ENS   | 6953   |

| Reaction No. 26970                            |    |       |               |                 |   |       |        |
|-----------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| 2<Pyrogallol-3> + Al+3 = Al+3_Pyrogallol-3(2) |    |       |               |                 |   |       |        |
| No                                            | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                             | 22 | 0.2   | NaCl          | lgK:44.55 (4SF) | 0 | ENS   | 2171   |
| 2                                             | 25 | 0     | Inf. Dilution | lgK:34.4 (3SF)  | 1 | ELC   |        |

| Reaction No. 26971                            |    |       |               |                 |   |       |        |
|-----------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| 3<Pyrogallol-3> + Al+3 = Al+3_Pyrogallol-3(3) |    |       |               |                 |   |       |        |
| No                                            | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                             | 22 | 0.2   | NaCl          | lgK:57.95 (4SF) | 0 | ENS   | 2171   |
| 2                                             | 25 | 0     | Inf. Dilution | lgK:39.71 (4SF) | 3 | ENS   | 6953   |

| Reaction No. 26972              |    |       |        |                 |   |       |        |
|---------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Gallic-4 + Al+3 = Al+3_Gallic-4 |    |       |        |                 |   |       |        |
| No                              | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                               | 25 | 0.6   | NaCl   | lgK:14.24 (4SF) | 5 | ENS   | 2171   |

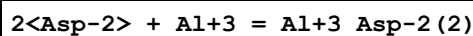
| Reaction No. 26973                    |    |       |        |                 |   |       |        |
|---------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2<Gallic-4> + Al+3 = Al+3_Gallic-4(2) |    |       |        |                 |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                     | 25 | 0.6   | NaCl   | lgK:25.37 (4SF) | 5 | ENS   | 2171   |

| Reaction No. 26974                    |    |       |        |                 |   |       |        |
|---------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 3<Gallic-4> + Al+3 = Al+3_Gallic-4(3) |    |       |        |                 |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                     | 25 | 0.6   | NaCl   | lgK:33.31 (4SF) | 5 | ENS   | 2171   |

| Reaction No. 26976        |    |       |         |                   |   |       |        |
|---------------------------|----|-------|---------|-------------------|---|-------|--------|
| Asp-2 + Al+3 = Al+3_Asp-2 |    |       |         |                   |   |       |        |
| No                        | t  | I-Str | Medium  | Value (Dev)       | W | Tech. | Ref. # |
| 1                         | 25 | 0     | UCT_Sea | lgK:9.26 (0.01SD) | 4 | EDH   | 5390   |
| 2                         | 25 | 0.1   | NaClO4  | lgK:16.29 (4SF)   | 0 | MGL   | 5763   |

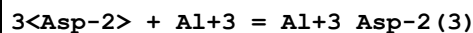
|    |    |      |         |                   |   |     |       |
|----|----|------|---------|-------------------|---|-----|-------|
| 3  | 25 | 0.1  | UCT_Sea | lgK:8.25 (0.01SD) | 4 | EDH | 5390  |
| 4  | 25 | 0.2  | KCl     | lgK:7.87 (3SF)    | 5 | MGL | 10434 |
| 5  | 25 | 0.2  | UCT_Sea | lgK:8.05 (0.01SD) | 4 | EDH | 5390  |
| 6  | 25 | 0.3  | UCT_Sea | lgK:7.95 (0.01SD) | 4 | EDH | 5390  |
| 7  | 25 | 0.4  | UCT_Sea | lgK:7.90 (0.01SD) | 4 | EDH | 5390  |
| 8  | 25 | 0.5  | UCT_Sea | lgK:7.85 (0.01SD) | 4 | EDH | 5390  |
| 9  | 25 | 0.6  | UCT_Sea | lgK:7.83 (0.01SD) | 4 | EDH | 5390  |
| 10 | 25 | 0.7  | UCT_Sea | lgK:7.80 (0.01SD) | 4 | EDH | 5390  |
| 11 | 37 | 0.15 | NaCl    | lgK:7.77 (3SF)    | 6 | MGL | 2737  |

## Reaction No. 26977



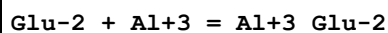
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:30.69 (4SF) | 0 | ENS   | 2171   |

## Reaction No. 26978



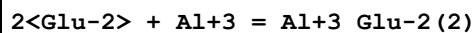
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:42.19 (4SF) | 0 | MGL   | 5763   |

## Reaction No. 26979



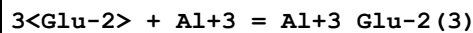
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:15.12 (4SF) | 0 | MGL   | 5763   |
| 2  | 25 | 0.2   | KCl    | lgK:7.29 (3SF)  | 5 | MGL   | 10434  |
| 3  | 37 | 0.15  | NaCl   | lgK:7.69 (3SF)  | 6 | MGL   | 2737   |

## Reaction No. 26980



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:29.40 (4SF) | 0 | ENS   | 2171   |

## Reaction No. 26981



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:38.60 (4SF) | 0 | ENS   | 5763   |

| Reaction No. 26982                  |    |       |                                    |                |   |       |        |
|-------------------------------------|----|-------|------------------------------------|----------------|---|-------|--------|
| 3<Oxine-1> + Al+3 = Al+3_Oxine-1(3) |    |       |                                    |                |   |       |        |
| No                                  | t  | I-Str | Medium                             | Value (Dev)    | W | Tech. | Ref. # |
| 1                                   | 20 | 0.1   | Unknown                            | lgK:30.6(3SF)  | 4 | ENS   | 5971   |
| 2                                   | 25 | 0.1   | Dioxan:Water<br>60:40Vol<br>NaClO4 | lgK:33.75(4SF) | 4 | MGL   | 3082   |
| 3                                   | 25 |       | Ethanol:Water<br>20:80Vol          | lgK:33.42(4SF) | 3 | ENS   | 2171   |

| Reaction No. 26983                      |    |       |        |                |   |       |        |
|-----------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 2Am2PrPhos-2 + Al+3 = Al+3_2Am2PrPhos-2 |    |       |        |                |   |       |        |
| No                                      | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                       | 25 | 0.1   | NaCl   | lgK:11.42(4SF) | 5 | MGL   | 2171   |

| Reaction No. 26984                            |    |       |        |                |   |       |        |
|-----------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 2<2Am2PrPhos-2> + Al+3 = Al+3_2Am2PrPhos-2(2) |    |       |        |                |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                             | 25 | 0.1   | NaCl   | lgK:19.53(4SF) | 5 | MGL   | 2171   |

| Reaction No. 26985                  |    |       |         |               |   |       |        |
|-------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Ascorbic-2 + Al+3 = Al+3_Ascorbic-2 |    |       |         |               |   |       |        |
| No                                  | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | Unknown | lgK:1.89(3SF) | 5 | ENS   | 2171   |

| Reaction No. 26986                        |    |       |         |              |   |       |        |
|-------------------------------------------|----|-------|---------|--------------|---|-------|--------|
| 2<Ascorbic-2> + Al+3 = Al+3_Ascorbic-2(2) |    |       |         |              |   |       |        |
| No                                        | t  | I-Str | Medium  | Value (Dev)  | W | Tech. | Ref. # |
| 1                                         | 25 | 0.1   | Unknown | lgK:3.7(2SF) | 5 | ENS   | 2171   |

| Reaction No. 26987                  |    |       |               |                |   |       |        |
|-------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| 2<Folic-3> + Al+3 = Al+3_Folic-3(2) |    |       |               |                |   |       |        |
| No                                  | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                   | 25 | 0     | Inf. Dilution | lgK:10.7(3SF)  | 1 | ESC   |        |
| 2                                   | 30 | 0     | Inf. Dilution | lgK:10.50(4SF) | 5 | ENS   | 2171   |

| Reaction No. 26988                  |   |       |        |             |   |       |        |
|-------------------------------------|---|-------|--------|-------------|---|-------|--------|
| 3<Folic-3> + Al+3 = Al+3_Folic-3(3) |   |       |        |             |   |       |        |
| No                                  | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

|   |    |   |               |                 |   |     |      |
|---|----|---|---------------|-----------------|---|-----|------|
| 1 | 25 | 0 | Inf. Dilution | lgK:15.4 (3SF)  | 1 | ESC |      |
| 2 | 30 | 0 | Inf. Dilution | lgK:15.15 (4SF) | 5 | ENS | 2171 |

| Reaction No. 26998                                  |    |       |        |                  |   |       |        |
|-----------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Al+3_H+1(-1)_Malonic-2(2) + H+1 = Al+3_Malonic-2(2) |    |       |        |                  |   |       |        |
| No                                                  | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.1   | KCl    | lgK:6.7 (0.08SD) | 6 | MGL   | 2356   |

| Reaction No. 26999                              |    |       |        |                     |   |       |        |
|-------------------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| Al+3 + 2<H+1(2)_Cat-2> = Al+3_Cat-2(2) + 4<H+1> |    |       |        |                     |   |       |        |
| No                                              | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                               | 25 | 0.6   | NaCl   | lgK:-15.44 (0.02SD) | 7 | MGL   | 2357   |

| Reaction No. 27000                              |    |       |        |                    |   |       |        |
|-------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| Al+3 + 3<H+1(2)_Cat-2> = Al+3_Cat-2(3) + 6<H+1> |    |       |        |                    |   |       |        |
| No                                              | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                               | 25 | 0.6   | NaCl   | lgK:-28.6 (0.03SD) | 7 | MGL   | 2357   |

| Reaction No. 27003                  |    |       |        |                 |   |       |        |
|-------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| PyroCatV-4 + Al+3 = Al+3_PyroCatV-4 |    |       |        |                 |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                   | 25 | 0.2   | KCl    | lgK:25.12 (4SF) | 5 | MGL   | 2171   |

| Reaction No. 27004                                |    |       |        |                 |   |       |        |
|---------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Al+3_PyroCatV-4 + PyroCatV-4 = Al+3_PyroCatV-4(2) |    |       |        |                 |   |       |        |
| No                                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                 | 25 | 0.2   | KCl    | lgK:22.27 (4SF) | 5 | MGL   | 999    |

| Reaction No. 27005                                   |    |       |        |                 |   |       |        |
|------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Al+3_PyroCatV-4(2) + PyroCatV-4 = Al+3_PyroCatV-4(3) |    |       |        |                 |   |       |        |
| No                                                   | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                    | 25 | 0.2   | KCl    | lgK:20.74 (4SF) | 5 | MGL   | 999    |

| Reaction No. 27011                              |    |       |        |                 |   |       |        |
|-------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| AmPhenMeDiPhos-4 + Al+3 = Al+3_AmPhenMeDiPhos-4 |    |       |        |                 |   |       |        |
| No                                              | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                               | 25 | 0.1   | KCl    | lgK:18.97 (4SF) | 5 | MGL   | 6953   |

| Reaction No. 27012                                                            |    |       |        |                |   |       |        |
|-------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $2\text{<AmPhenMeDiPhos-4>} + \text{Al+3} = \text{Al+3\_AmPhenMeDiPhos-4(2)}$ |    |       |        |                |   |       |        |
| No                                                                            | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                             | 25 | 0.1   | KCl    | lgK:26.31(4SF) | 5 | MGL   | 6953   |

| Reaction No. 27013                                                                |    |       |        |                |   |       |        |
|-----------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Al+3} + \text{H+1\_AmPhenMeDiPhos-4} = \text{Al+3\_H+1\_AmPhenMeDiPhos-4}$ |    |       |        |                |   |       |        |
| No                                                                                | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                 | 25 | 0.1   | KCl    | lgK:12.50(4SF) | 5 | MGL   | 999    |

| Reaction No. 27014                                                            |    |       |        |                |   |       |        |
|-------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $2\text{<Al+3>} + \text{AmPhenMeDiPhos-4} = \text{Al+3(2)\_AmPhenMeDiPhos-4}$ |    |       |        |                |   |       |        |
| No                                                                            | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                             | 25 | 0.1   | KCl    | lgK:23.09(4SF) | 5 | MGL   | 999    |

| Reaction No. 27015                                                                                             |    |       |        |                |   |       |        |
|----------------------------------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Al+3} + \text{H+1\_AmPhenMeDiPhos-4} + \text{AmPhenMeDiPhos-4} = \text{Al+3\_H+1\_AmPhenMeDiPhos-4(2)}$ |    |       |        |                |   |       |        |
| No                                                                                                             | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                              | 25 | 0.1   | KCl    | lgK:19.68(4SF) | 5 | MGL   | 999    |

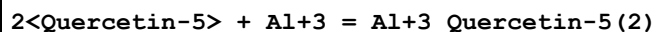
| Reaction No. 27053                                                  |    |       |               |                 |   |       |        |
|---------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{SolochromeVR-3} + \text{Al+3} = \text{Al+3\_SolochromeVR-3}$ |    |       |               |                 |   |       |        |
| No                                                                  | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                   | 25 | 0     | Inf. Dilution | lgK:18.4(0.1SD) | 4 | MSP   | 6845   |
| 2                                                                   | 25 | 0     | Inf. Dilution | lgK:19.5(3SF)   | 4 | EDH   | 7000   |
| 3                                                                   | 25 | 0     | Inf. Dilution | lgK:19.4(3SF)   | 4 | MPS   | 7000   |

| Reaction No. 27054                                                                     |    |       |               |                 |   |       |        |
|----------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{Al+3\_SolochromeVR-3} + \text{SolochromeVR-3} = \text{Al+3\_SolochromeVR-3(2)}$ |    |       |               |                 |   |       |        |
| No                                                                                     | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                      | 25 | 0     | Inf. Dilution | lgK:16.3(3SF)   | 1 | ELC   |        |
| 2                                                                                      | 25 | 0     | Inf. Dilution | lgK:13.2(0.1SD) | 0 | MSP   | 6845   |

| Reaction No. 27082                                                    |    |       |               |                |   |       |        |
|-----------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $3\text{OH2Naphthoic-2} + \text{Al+3} = \text{Al+3\_3OH2Naphthoic-2}$ |    |       |               |                |   |       |        |
| No                                                                    | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                     | 25 | 0     | Inf. Dilution | lgK:13.38(4SF) | 5 | ENS   | 2171   |

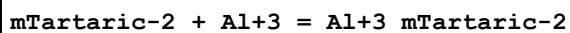
|   |    |     |         |                 |   |     |      |
|---|----|-----|---------|-----------------|---|-----|------|
| 2 | 25 | 0.5 | Unknown | lgK:12.94 (4SF) | 4 | ENS | 5971 |
|---|----|-----|---------|-----------------|---|-----|------|

## Reaction No. 27086



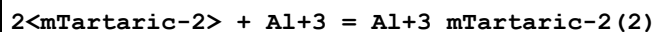
| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:12. (2SF) | 5 | ENS   | 2171   |

## Reaction No. 27089



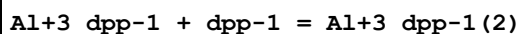
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:5.62 (3SF) | 5 | ENS   | 2171   |

## Reaction No. 27090



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:9.95 (3SF) | 5 | ENS   | 2171   |

## Reaction No. 28114



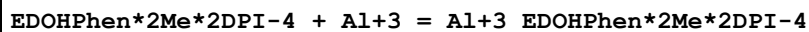
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:11.05 (0.02SD) | 6 | MGL   | 2512   |
| 2  | 25 | 0.15  | NaCl   | lgK:10.92 (0.02SD) | 6 | MGL   | 2512   |

## Reaction No. 28115



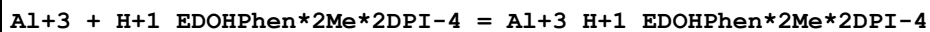
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:9.4 (0.04SD) | 6 | MGL   | 2512   |
| 2  | 25 | 0.15  | NaCl   | lgK:9.4 (0.05SD) | 6 | MGL   | 2512   |

## Reaction No. 28722



| No | t  | I-Str | Medium   | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|----------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | K+1 salt | lgK:20. (2SF) | 4 | CRV   | 2556   |

## Reaction No. 28723



| No | t  | I-Str | Medium   | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|----------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | K+1 salt | lgK:15.36 (4SF) | 5 | CRV   | 2556   |

| Reaction No. 28766                                    |    |       |               |                  |   |       |        |
|-------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 1OH4Sulf2Napthoic-3 + Al+3 = Al+3_1OH4Sulf2Napthoic-3 |    |       |               |                  |   |       |        |
| No                                                    | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                     | 25 | 0     | Inf. Dilution | lgK:14.0 (3SF)   | 1 | EDH   |        |
| 2                                                     | 25 | 0.1   | NaClO4        | lgK:12.102 (5SF) | 3 | MGL   | 5987   |
| 3                                                     | 25 | 0.1   | NaClO4        | lgK:12.64 (4SF)  | 0 | MSP   | 5987   |
| 4                                                     | 25 | 0.5   | NaClO4        | lgK:12.64 (4SF)  | 0 | MSP   | 2261   |

| Reaction No. 28767                                           |    |       |               |                 |   |       |        |
|--------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| 2<1OH4Sulf2Napthoic-3> + Al+3 = Al+3_1OH4Sulf2Napthoic-3 (2) |    |       |               |                 |   |       |        |
| No                                                           | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                            | 25 | 0     | Inf. Dilution | lgK:23.5 (3SF)  | 1 | ELC   |        |
| 2                                                            | 25 | 0.5   | NaClO4        | lgK:21.89 (4SF) | 0 | MGL   | 2261   |

| Reaction No. 28768                                                          |    |       |        |                 |   |       |        |
|-----------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2<1OH4Sulf2Napthoic-3> + Al+3 = H+1 + Al+3_H+1 (-1)_1OH4Sulf2Napthoic-3 (2) |    |       |        |                 |   |       |        |
| No                                                                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                           | 25 | 0.5   | NaClO4 | lgK:13.98 (4SF) | 6 | MGL   | 2261   |

| Reaction No. 28769                                                             |    |       |        |                |   |       |        |
|--------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 2<1OH4Sulf2Napthoic-3> + Al+3 = 2<H+1> + Al+3_H+1 (-2)_1OH4Sulf2Napthoic-3 (2) |    |       |        |                |   |       |        |
| No                                                                             | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                              | 25 | 0.5   | NaClO4 | lgK:5.32 (3SF) | 6 | MGL   | 2261   |

| Reaction No. 28826                                            |    |       |        |                  |   |       |        |
|---------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 1OH47DiSulf2Naphthoic-4 + Al+3 = Al+3_1OH47DiSulf2Naphthoic-4 |    |       |        |                  |   |       |        |
| No                                                            | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                             | 25 | 0.5   | NaClO4 | lgK:12.38 (4SF)  | 5 | MSP   | 2261   |
| 2                                                             | 25 | 0.5   | NaClO4 | lgK:12.379 (5SF) | 6 | MGL   | 5987   |
| 3                                                             | 25 | 0.5   | NaClO4 | lgK:12.38 (4SF)  | 6 | MSP   | 5987   |

| Reaction No. 28827                                                   |    |       |        |                 |   |       |        |
|----------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2<1OH47DiSulf2Naphthoic-4> + Al+3 = Al+3_1OH47DiSulf2Naphthoic-4 (2) |    |       |        |                 |   |       |        |
| No                                                                   | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                    | 25 | 0.5   | NaClO4 | lgK:21.55 (4SF) | 6 | MGL   | 2261   |

| Reaction No. 28828 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|



| 2<1OH47DiSulf2Naphthoic-4> + Al+3 = H+1 + Al+3_H+1(-1)_1OH47DiSulf2Naphthoic-4(2) |    |       |        |                |   |       |        |
|-----------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| No                                                                                | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                 | 25 | 0.5   | NaClO4 | lgK:13.66(4SF) | 6 | MGL   | 2261   |

| Reaction No. 28840                                            |    |       |        |                |   |       |        |
|---------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 3OH57DiSulf2Naphthoic-4 + Al+3 = Al+3_3OH57DiSulf2Naphthoic-4 |    |       |        |                |   |       |        |
| No                                                            | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                             | 25 | 0.5   | NaClO4 | lgK:10.81(4SF) | 6 | MGL   | 2261   |

| Reaction No. 28841                                                  |    |       |        |                |   |       |        |
|---------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 2<3OH57DiSulf2Naphthoic-4> + Al+3 = Al+3_3OH57DiSulf2Naphthoic-4(2) |    |       |        |                |   |       |        |
| No                                                                  | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                   | 25 | 0.5   | NaClO4 | lgK:19.26(4SF) | 6 | MGL   | 2261   |

| Reaction No. 29151                                  |    |       |        |               |   |       |        |
|-----------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Al+3 + H+1_MeThymolBlue-6 = Al+3_H+1_MeThymolBlue-6 |    |       |        |               |   |       |        |
| No                                                  | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.1   | NaClO4 | lgK:16.2(3SF) | 3 | MPT   | 1695   |

| Reaction No. 29152                                |    |       |        |               |   |       |        |
|---------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| 2<Al+3> + MeThymolBlue-6 = Al+3(2)_MeThymolBlue-6 |    |       |        |               |   |       |        |
| No                                                | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                 | 25 | 0.1   | NaClO4 | lgK:26.8(3SF) | 3 | MGL   | 1695   |

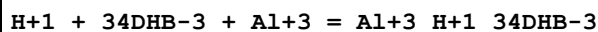
| Reaction No. 29153                                  |    |       |               |              |   |       |        |
|-----------------------------------------------------|----|-------|---------------|--------------|---|-------|--------|
| Al+3_MeThymolBlue-6 + H+1 = Al+3_H+1_MeThymolBlue-6 |    |       |               |              |   |       |        |
| No                                                  | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                   | 25 | 0     | Inf. Dilution | lgK:7.5(2SF) | 1 | EES   |        |
| 2                                                   | 25 | 0.1   | NaClO4        | lgK:6.7(2SF) | 3 | MPT   | 1695   |

| Reaction No. 29154                                                         |    |       |        |               |   |       |        |
|----------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Al+3(2)_OH-1(2)_MeThymolBlue-6 + 2<H+1> = Al+3(2)_MeThymolBlue-6 + 2<H2O > |    |       |        |               |   |       |        |
| No                                                                         | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                          | 25 | 0.1   | NaClO4 | lgK:13.8(3SF) | 3 | MGL   | 1695   |

| Reaction No. 29160                                                |   |       |        |             |   |       |        |
|-------------------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| 8OH76Sulf2NaphQuin5Sulf-3 + Al+3 = Al+3_8OH76Sulf2NaphQuin5Sulf-3 |   |       |        |             |   |       |        |
| No                                                                | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

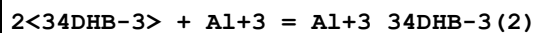
|   |    |     |        |                |   |     |      |
|---|----|-----|--------|----------------|---|-----|------|
| 1 | 25 | 0.1 | NaClO4 | lgK:7.9(0.2SD) | 5 | MGL | 1098 |
|---|----|-----|--------|----------------|---|-----|------|

## Reaction No. 30174



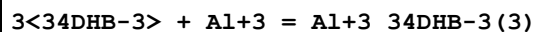
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:21.09(0.01SD) | 6 | MGL   | 2470   |

## Reaction No. 30175



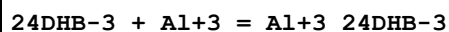
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:29.4(0.04SD) | 6 | MGL   | 2470   |

## Reaction No. 30176



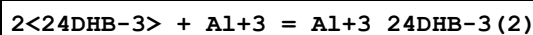
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:38.3(0.05SD) | 6 | MGL   | 2470   |

## Reaction No. 30177



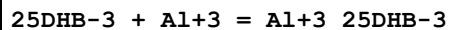
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:8.71(0.01SD) | 6 | MGL   | 2470   |

## Reaction No. 30178



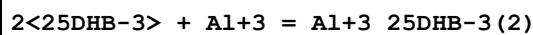
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:15.0(0.03SD) | 6 | MGL   | 2470   |

## Reaction No. 30179



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:9.74(0.01SD) | 7 | MGL   | 2470   |

## Reaction No. 30180



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:17.2(0.04SD) | 7 | MGL   | 2470   |

## Reaction No. 30181

| 26DHB-3 + Al+3 = Al+3_26DHB-3 |    |       |        |                   |   |       |        |
|-------------------------------|----|-------|--------|-------------------|---|-------|--------|
| No                            | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                             | 25 | 0.2   | KCl    | lgK:12.79(0.01SD) | 6 | MGL   | 2470   |

| Reaction No. 30182                  |    |       |        |                  |   |       |        |
|-------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 2<26DHB-3> + Al+3 = Al+3_26DHB-3(2) |    |       |        |                  |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                   | 25 | 0.2   | KCl    | lgK:23.7(0.04SD) | 6 | MGL   | 2470   |

| Reaction No. 30183                                  |    |       |        |                  |   |       |        |
|-----------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Cat-2 + Al+3 + Salicylic-2 = Al+3_Cat-2_Salicylic-2 |    |       |        |                  |   |       |        |
| No                                                  | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.2   | KCl    | lgK:26.9(0.05SD) | 6 | MGL   | 2470   |
| 2                                                   | 25 | 0.2   | KCl    | dH:0.12(2SF)     | 6 | MCL   | 2470   |

| Reaction No. 30184                                        |    |       |        |                  |   |       |        |
|-----------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Cat-2 + Al+3 + 2<Salicylic-2> = Al+3_Cat-2_Salicylic-2(2) |    |       |        |                  |   |       |        |
| No                                                        | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                         | 25 | 0.2   | KCl    | lgK:35.2(0.05SD) | 6 | MGL   | 2470   |

| Reaction No. 30185                                        |    |       |        |                 |   |       |        |
|-----------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2<Cat-2> + Al+3 + Salicylic-2 = Al+3_Cat-2(2)_Salicylic-2 |    |       |        |                 |   |       |        |
| No                                                        | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                         | 25 | 0.2   | KCl    | lgK:37.1(0.2SD) | 6 | MGL   | 2470   |

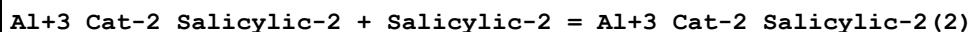
| Reaction No. 30186                                |    |       |        |                |   |       |        |
|---------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Al+3_Cat-2 + Salicylic-2 = Al+3_Cat-2_Salicylic-2 |    |       |        |                |   |       |        |
| No                                                | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                 | 25 | 0.2   | KCl    | lgK:10.72(4SF) | 6 | MGL   | 2470   |

| Reaction No. 30187                                |    |       |        |                |   |       |        |
|---------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Al+3_Salicylic-2 + Cat-2 = Al+3_Cat-2_Salicylic-2 |    |       |        |                |   |       |        |
| No                                                | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                 | 25 | 0.2   | KCl    | lgK:13.70(4SF) | 6 | MGL   | 2470   |

| Reaction No. 30188                                                |   |       |        |             |   |       |        |
|-------------------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| Al+3_Salicylic-2 + H+1(2)_Cat-2 = Al+3_Cat-2_Salicylic-2 + 2<H+1> |   |       |        |             |   |       |        |
| No                                                                | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

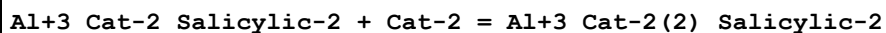
|   |    |     |     |                 |   |     |      |
|---|----|-----|-----|-----------------|---|-----|------|
| 1 | 25 | 0.2 | KCl | lgK:-8.58 (3SF) | 6 | MGL | 2470 |
|---|----|-----|-----|-----------------|---|-----|------|

## Reaction No. 30189



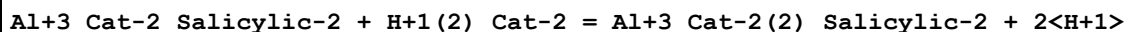
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:8.28 (3SF) | 6 | MGL   | 2470   |

## Reaction No. 30190



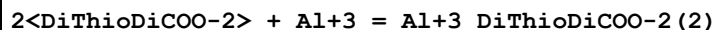
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:10.20 (4SF) | 6 | MGL   | 2470   |

## Reaction No. 30191



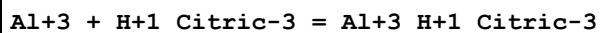
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:-12.08 (4SF) | 6 | MGL   | 2470   |

## Reaction No. 30554



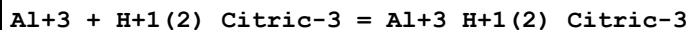
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:6.36 (3SF) | 5 | ENS   | 2171   |

## Reaction No. 30795



| No | t     | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|-------|-------|---------------|----------------|---|-------|--------|
| 1  | 25    | 0     | Inf. Dilution | lgK:6.56 (3SF) | 4 | MSF   | 1717   |
| 2  | 25    | 0     | Inf. Dilution | lgK:6.50 (3SF) | 4 | CRV   | 31301  |
| 3  | 25    | 0.1   | Unknown       | lgK:5.21 (3SF) | 3 | CRV   | 2556   |
| 4  | 25    | 0.5   | Unknown       | lgK:4.17 (3SF) | 4 | CRV   | 2556   |
| 5  | 25:37 | 0.15  | KNO3          | lgK:4.7 (2SF)  | 5 | MPT   | 3607   |
| 6  | 37    | 0.15  | Unknown       | lgK:4.55 (3SF) | 4 | CRV   | 2556   |

## Reaction No. 30796



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:2.91 (3SF) | 4 | MSF   | 1717   |

## Reaction No. 31078

| Al+3_OH-1 + H+1_His-1 = Al+3_His-1 + H2O                   |    |       |               |               |   |       |        |
|------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| No                                                         | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                          | 25 | 0     | Inf. Dilution | lgK:4.1(2SF)  | 1 | ERG   |        |
| 2                                                          | 25 | 0.5   | NaClO4        | lgK:3.62(3SF) | 3 | MGL   | 22560  |
| Note (2): Charlet et al, Bull. Soc. Chim. Fr., 1984, I-222 |    |       |               |               |   |       |        |

| Reaction No. 31079                            |    |       |               |               |   |       |        |
|-----------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Al+3_His-1_OH-1 + H+1 = Al+3_OH-1 + H+1_His-1 |    |       |               |               |   |       |        |
| No                                            | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                             | 25 | 0     | Inf. Dilution | lgK:2.2(2SF)  | 1 | ERG   |        |
| 2                                             | 25 | 0.5   | Unknown       | lgK:4.36(3SF) | 0 | CRV   | 2556   |

| Reaction No. 31462              |    |       |         |               |   |       |        |
|---------------------------------|----|-------|---------|---------------|---|-------|--------|
| 3OHGlu-2 + Al+3 = Al+3_3OHGlu-2 |    |       |         |               |   |       |        |
| No                              | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                               | 25 | 0.1   | Unknown | lgK:4.87(3SF) | 5 | ENS   | 6953   |

| Reaction No. 31468              |    |       |        |               |   |       |        |
|---------------------------------|----|-------|--------|---------------|---|-------|--------|
| PO4Ser-3 + Al+3 = Al+3_PO4Ser-3 |    |       |        |               |   |       |        |
| No                              | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                               | 25 | 0.1   | KNO3   | lgK:4.79(3SF) | 6 | MGL   | 2754   |

| Reaction No. 31469                    |    |       |        |               |   |       |        |
|---------------------------------------|----|-------|--------|---------------|---|-------|--------|
| 2<PO4Ser-3> + Al+3 = Al+3_PO4Ser-3(2) |    |       |        |               |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | KNO3   | lgK:8.81(3SF) | 4 | MGL   | 2754   |

| Reaction No. 31470                    |    |       |        |                |   |       |        |
|---------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 3<PO4Ser-3> + Al+3 = Al+3_PO4Ser-3(3) |    |       |        |                |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | KNO3   | lgK:10.68(4SF) | 4 | MGL   | 2754   |

| Reaction No. 31543                  |    |       |         |                   |   |       |        |
|-------------------------------------|----|-------|---------|-------------------|---|-------|--------|
| H+1 + Asp-2 + Al+3 = Al+3_H+1_Asp-2 |    |       |         |                   |   |       |        |
| No                                  | t  | I-Str | Medium  | Value (Dev)       | W | Tech. | Ref. # |
| 1                                   | 25 | 0     | UCT_Sea | lgK:12.73(0.01SD) | 4 | EDH   | 5390   |
| 2                                   | 25 | 0.1   | UCT_Sea | lgK:11.94(0.01SD) | 4 | EDH   | 5390   |

|   |    |      |         |                    |   |     |      |
|---|----|------|---------|--------------------|---|-----|------|
| 3 | 25 | 0.2  | UCT_Sea | lgK:11.80 (0.01SD) | 4 | EDH | 5390 |
| 4 | 25 | 0.3  | UCT_Sea | lgK:11.73 (0.01SD) | 4 | EDH | 5390 |
| 5 | 25 | 0.4  | UCT_Sea | lgK:11.71 (0.01SD) | 4 | EDH | 5390 |
| 6 | 25 | 0.5  | UCT_Sea | lgK:11.69 (0.01SD) | 4 | EDH | 5390 |
| 7 | 25 | 0.6  | UCT_Sea | lgK:11.69 (0.01SD) | 4 | EDH | 5390 |
| 8 | 25 | 0.7  | UCT_Sea | lgK:11.68 (0.01SD) | 4 | EDH | 5390 |
| 9 | 37 | 0.15 | NaCl    | lgK:11.24 (4SF)    | 6 | MGL | 2737 |

| Reaction No. 31544                        |    |       |               |                 |   |       |        |
|-------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| 2<H+1> + Asp-2 + Al+3 = Al+3_H+1(2)_Asp-2 |    |       |               |                 |   |       |        |
| No                                        | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                         | 25 | 0     | Inf. Dilution | lgK:15.2 (3SF)  | 2 | EAE   |        |
| 2                                         | 37 | 0.15  | NaCl          | lgK:14.48 (4SF) | 6 | MGL   | 2737   |

| Reaction No. 31545                            |    |       |               |                |   |       |        |
|-----------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Asp-2 + 2<Al+3> = H+1 + Al+3(2)_H+1(-1)_Asp-2 |    |       |               |                |   |       |        |
| No                                            | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                             | 25 | 0     | Inf. Dilution | lgK:7.59 (3SF) | 2 | EAE   |        |
| 2                                             | 37 | 0.15  | NaCl          | lgK:6.15 (3SF) | 6 | MGL   | 2737   |

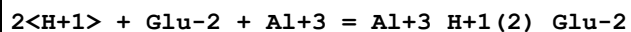
| Reaction No. 31546                               |    |       |               |                |   |       |        |
|--------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Asp-2 + 2<Al+3> = 2<H+1> + Al+3(2)_H+1(-2)_Asp-2 |    |       |               |                |   |       |        |
| No                                               | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                | 25 | 0     | Inf. Dilution | lgK:3.99 (3SF) | 2 | EAE   |        |
| 2                                                | 37 | 0.15  | NaCl          | lgK:2.07 (3SF) | 6 | MGL   | 2737   |

| Reaction No. 31547                               |    |       |               |                 |   |       |        |
|--------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Asp-2 + 2<Al+3> = 4<H+1> + Al+3(2)_H+1(-4)_Asp-2 |    |       |               |                 |   |       |        |
| No                                               | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                | 25 | 0     | Inf. Dilution | lgK:-4.51 (3SF) | 2 | EAE   |        |
| 2                                                | 37 | 0.15  | NaCl          | lgK:-6.67 (3SF) | 6 | MGL   | 2737   |

| Reaction No. 31548                  |    |       |               |                |   |       |        |
|-------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| H+1 + Glu-2 + Al+3 = Al+3_H+1_Glu-2 |    |       |               |                |   |       |        |
| No                                  | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                   | 25 | 0     | Inf. Dilution | lgK:13.2 (3SF) | 1 | ELC   |        |
| 2                                   | 25 | 0     | Inf. Dilution | lgK:12.3 (3SF) | 0 | EAE   |        |

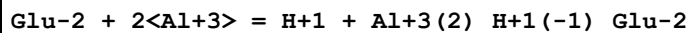
|   |    |      |      |                 |   |     |      |
|---|----|------|------|-----------------|---|-----|------|
| 3 | 37 | 0.15 | NaCl | lgK:11.07 (4SF) | 3 | MGL | 2737 |
|---|----|------|------|-----------------|---|-----|------|

## Reaction No. 31549



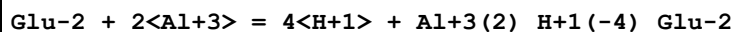
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:15.5 (3SF)  | 2 | EAE   |        |
| 2  | 37 | 0.15  | NaCl          | lgK:14.74 (4SF) | 6 | MGL   | 2737   |

## Reaction No. 31550



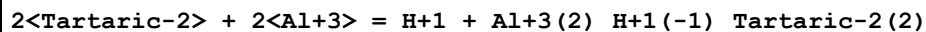
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:7.89 (3SF) | 2 | EAE   |        |
| 2  | 37 | 0.15  | NaCl          | lgK:6.45 (3SF) | 6 | MGL   | 2737   |

## Reaction No. 31551



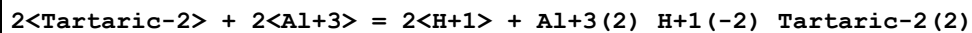
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-4.55 (3SF) | 2 | EAE   |        |
| 2  | 37 | 0.15  | NaCl          | lgK:-6.71 (3SF) | 6 | MGL   | 2737   |

## Reaction No. 31584



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:10.5 (3SF) | 1 | ESC   |        |
| 2  | 37 | 0.15  | NaCl          | lgK:7.93 (3SF) | 4 | MGL   | 5981   |

## Reaction No. 31585



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:7.8 (2SF)  | 1 | ELC   |        |
| 2  | 37 | 0.15  | NaCl          | lgK:5.55 (3SF) | 4 | MGL   | 5981   |

## Reaction No. 31586



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:3.3 (2SF)  | 1 | ESC   |        |
| 2  | 37 | 0.15  | NaCl          | lgK:0.82 (2SF) | 4 | MGL   | 5981   |

| Reaction No. 31587                                                                                               |    |       |               |                |   |       |        |
|------------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $2\text{<Tartaric-2>} + 2\text{<Al+3>} = 4\text{<H+1>} + \text{Al+3(2)}_{\text{H+1(-4)}}_{\text{Tartaric-2(2)}}$ |    |       |               |                |   |       |        |
| No                                                                                                               | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                                | 25 | 0     | Inf. Dilution | lgK:-3.1(2SF)  | 1 | ELC   |        |
| 2                                                                                                                | 37 | 0.15  | NaCl          | lgK:-5.26(3SF) | 4 | MGL   | 5981   |

| Reaction No. 31962                                                      |    |       |        |               |   |       |        |
|-------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| $\text{NPhosMeIDA-4} + \text{Al+3} = \text{Al+3}_{\text{NPhosMeIDA-4}}$ |    |       |        |               |   |       |        |
| No                                                                      | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                       | 25 | 0.1   | KCl    | lgK:14.7(3SF) | 5 | MGL   | 2261   |

| Reaction No. 31963                                                                                |    |       |        |              |   |       |        |
|---------------------------------------------------------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| $\text{Al+3} + \text{H+1}_{\text{NPhosMeIDA-4}} = \text{Al+3}_{\text{H+1}_{\text{NPhosMeIDA-4}}}$ |    |       |        |              |   |       |        |
| No                                                                                                | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                                                                 | 25 | 0.1   | KCl    | lgK:7.3(2SF) | 5 | MGL   | 2261   |

| Reaction No. 32245                                                      |    |       |        |                |   |       |        |
|-------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $2\text{Am3PhosPr-3} + \text{Al+3} = \text{Al+3}_{2\text{Am3PhosPr-3}}$ |    |       |        |                |   |       |        |
| No                                                                      | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                       | 25 | 1     | KCl    | lgK:13.12(4SF) | 5 | MGL   | 1585   |

| Reaction No. 32246                                                            |    |       |        |                |   |       |        |
|-------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Al+3} + 2\text{<2Am3PhosPr-3>} = \text{Al+3}_{2\text{Am3PhosPr-3(2)}}$ |    |       |        |                |   |       |        |
| No                                                                            | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                             | 25 | 1     | KCl    | lgK:21.40(4SF) | 4 | MGL   | 1585   |

| Reaction No. 32247                                                                                |    |       |        |               |   |       |        |
|---------------------------------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| $\text{Al+3}_{2\text{Am3PhosPr-3}} + \text{H+1} = \text{Al+3}_{\text{H+1}_{2\text{Am3PhosPr-3}}}$ |    |       |        |               |   |       |        |
| No                                                                                                | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                                 | 25 | 1     | KCl    | lgK:3.22(3SF) | 5 | MGL   | 1585   |

| Reaction No. 32248                                                                                      |    |       |        |               |   |       |        |
|---------------------------------------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| $\text{Al+3}_{2\text{Am3PhosPr-3(2)}} + \text{H+1} = \text{Al+3}_{\text{H+1}_{2\text{Am3PhosPr-3(2)}}}$ |    |       |        |               |   |       |        |
| No                                                                                                      | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                                       | 25 | 1     | KCl    | lgK:4.80(3SF) | 4 | MGL   | 1585   |

| Reaction No. 32796                                                                                |  |  |  |  |  |  |  |
|---------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|
| $3344\text{TetrOHDiPhenSulfone-4} + \text{Al+3} = \text{Al+3}_{3344\text{TetrOHDiPhenSulfone-4}}$ |  |  |  |  |  |  |  |



| No | t     | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|-------|-------|---------------|-------------------|---|-------|--------|
| 1  | 22:25 | 0.2   | KNO3          | lgK:14.93(0.08SD) | 0 | MGL   | 2349   |
| 2  | 25    | 0     | Inf. Dilution | lgK:15.2(3SF)     | 1 | EES   |        |

## Reaction No. 32797

Al+3\_3344TetrOHDiPhenSulfone-4 + 3344TetrOHDiPhenSulfone-4 =  
Al+3\_3344TetrOHDiPhenSulfone-4(2)

| No | t     | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|-------|-------|---------------|-------------------|---|-------|--------|
| 1  | 22:25 | 0.2   | KNO3          | lgK:12.65(0.32SD) | 0 | MGL   | 2349   |
| 2  | 25    | 0     | Inf. Dilution | lgK:12.8(3SF)     | 1 | EES   |        |

## Reaction No. 32798

Al+3\_3344TetrOHDiPhenSulfone-4(2) + 3344TetrOHDiPhenSulfone-4 =  
Al+3\_3344TetrOHDiPhenSulfone-4(3)

| No | t     | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|-------|-------|---------------|------------------|---|-------|--------|
| 1  | 22:25 | 0.2   | KNO3          | lgK:9.24(0.20SD) | 0 | MGL   | 2349   |
| 2  | 25    | 0     | Inf. Dilution | lgK:9.5(2SF)     | 1 | EES   |        |

## Reaction No. 33037

Al+3 + H+1(2)\_Pyrogallol-3 = Al+3\_Pyrogallol-3 + 2<H+1>

| No | t     | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|-------|-------|---------------|------------------|---|-------|--------|
| 1  | 22:25 | 0.2   | KNO3          | lgK:-6.0(0.03SD) | 0 | MGL   | 2349   |
| 2  | 25    | 0     | Inf. Dilution | lgK:-11.7(3SF)   | 1 | ESC   |        |

## Reaction No. 33038

Al+3\_Pyrogallol-3 + H+1(2)\_Pyrogallol-3 = Al+3\_Pyrogallol-3(2) + 2<H+1>

| No | t     | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|-------|-------|---------------|----------------|---|-------|--------|
| 1  | 22:25 | 0.2   | KNO3          | lgK:-9.(0.4SD) | 0 | MGL   | 2349   |
| 2  | 25    | 0     | Inf. Dilution | lgK:-6.0(2SF)  | 1 | ELC   |        |

## Reaction No. 33039

Al+3\_Pyrogallol-3(2) + H+1(2)\_Pyrogallol-3 = Al+3\_Pyrogallol-3(3) + 2<H+1>

| No | t     | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-------|-------|---------------|-----------------|---|-------|--------|
| 1  | 22:25 | 0.2   | KNO3          | lgK:-15.(0.2SD) | 0 | MGL   | 2349   |
| 2  | 25    | 0     | Inf. Dilution | lgK:-20.8(3SF)  | 1 | ELC   |        |

## Reaction No. 33040

Al+3 + H+1(2)\_334455HexDiPhenSulfone-6 = Al+3\_334455HexDiPhenSulfone-6 + 2<H+1>

| No | t     | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|-------|-------|---------------|----------------|---|-------|--------|
| 1  | 22:25 | 0.2   | KNO3          | lgK:-5.(0.3SD) | 5 | MGL   | 2349   |
| 2  | 25    | 0     | Inf. Dilution | lgK:-5.1(2SF)  | 1 | ESC   |        |

## Reaction No. 33041

Al+3\_334455HexDiPhenSulfone-6 + H+1(2)\_334455HexDiPhenSulfone-6 =  
Al+3\_334455HexDiPhenSulfone-6(2) + 2<H+1>

| No | t     | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|-------|-------|---------------|------------------|---|-------|--------|
| 1  | 22:25 | 0.2   | KNO3          | lgK:-6.6(0.07SD) | 0 | MGL   | 2349   |
| 2  | 25    | 0     | Inf. Dilution | lgK:-6.6(2SF)    | 1 | ESC   |        |

## Reaction No. 33042

Al+3\_334455HexDiPhenSulfone-6(2) + H+1(2)\_334455HexDiPhenSulfone-6 =  
Al+3\_334455HexDiPhenSulfone-6(3) + 2<H+1>

| No | t     | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-------|-------|---------------|-----------------|---|-------|--------|
| 1  | 22:25 | 0.2   | KNO3          | lgK:-10.(0.2SD) | 0 | MGL   | 2349   |
| 2  | 25    | 0     | Inf. Dilution | lgK:-10.2(3SF)  | 1 | ESC   |        |

## Reaction No. 33094

23DiOHTerePhth-2 + Al+3 = Al+3\_23DiOHTerePhth-2

| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:3.9(0.1SD) | 6 | MGL   | 2470   |

## Reaction No. 33728

23DiOH6SulfNaphthoic-3 + Al+3 = Al+3\_23DiOH6SulfNaphthoic-3

| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:16.48(4SF) | 2 | ENS   | 6953   |
| 2  | 25 | 0.5   | NaClO4 | lgK:15.10(4SF) | 2 | MGL   | 3061   |
| 3  | 25 | 0.5   | NaClO4 | lgK:15.7(3SF)  | 2 | ENS   | 6953   |

## Reaction No. 33729

2<23DiOH6SulfNaphthoic-3> + Al+3 = Al+3\_23DiOH6SulfNaphthoic-3(2)

| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:29.82(4SF) | 2 | ENS   | 6953   |
| 2  | 25 | 0.5   | NaClO4 | lgK:27.88(4SF) | 1 | MGL   | 3061   |
| 3  | 25 | 0.5   | NaClO4 | lgK:29.1(3SF)  | 2 | ENS   | 6953   |

## Reaction No. 33730

| 3<23DiOH6SulfNaphthoic-3> + Al+3 = Al+3_23DiOH6SulfNaphthoic-3(3) |    |       |        |                |   |       |        |
|-------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| No                                                                | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                 | 25 | 0.1   | KNO3   | lgK:39.12(4SF) | 0 | ENS   | 6953   |
| 2                                                                 | 25 | 0.5   | NaClO4 | lgK:37.47(4SF) | 4 | MGL   | 3061   |

| Reaction No. 33886                                  |    |       |        |                  |   |       |        |
|-----------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Al+3 + H+1_Tetracycline-2 = Al+3_H+1_Tetracycline-2 |    |       |        |                  |   |       |        |
| No                                                  | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.1   | NaNO3  | lgK:12.5(0.03SD) | 3 | MGL   | 3335   |

| Reaction No. 34151                                                       |    |       |        |                     |   |       |        |
|--------------------------------------------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| 3<Al+3> + 4<H+1(2)_Ascorbic-2> = Al+3(3)_H+1(-1)_Ascorbic-2(4) + 9<H+1 > |    |       |        |                     |   |       |        |
| No                                                                       | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                                        | 25 | 0.6   | NaCl   | lgK:-24.19(0.083SD) | 7 | MGL   | 22912  |

| Reaction No. 34363                                       |    |       |         |              |   |       |        |
|----------------------------------------------------------|----|-------|---------|--------------|---|-------|--------|
| 4<Al+3(2)_EtDiAm_Pr2Ol(6)> = Al+3(8)_EtDiAm(4)_Pr2Ol(24) |    |       |         |              |   |       |        |
| No                                                       | t  | I-Str | Medium  | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                        | 20 |       | Benzene | lgK:6.2(2SF) | 3 | MFP   | 931    |

| Reaction No. 34406                    |    |       |        |                   |   |       |        |
|---------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Al+3 + H+1_23DHB-3 = Al+3_H+1_23DHB-3 |    |       |        |                   |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                     | 25 | 0.2   | KCl    | lgK:10.32(0.01SD) | 6 | MGL   | 2470   |
| 2                                     | 37 | 0.15  | NaCl   | lgK:10.505(5SF)   | 6 | MGL   | 8370   |

| Reaction No. 34407                             |    |       |        |                     |   |       |        |
|------------------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| Al+3 + 2<H+1_23DHB-3> = Al+3_H+1(2)_23DHB-3(2) |    |       |        |                     |   |       |        |
| No                                             | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                              | 25 | 0.2   | KCl    | lgK:18.3(0.06SD)    | 6 | MGL   | 2470   |
| 2                                              | 37 | 0.15  | NaCl   | lgK:18.244(0.032SD) | 6 | MGL   | 8370   |

| Reaction No. 34408                                |    |       |        |                     |   |       |        |
|---------------------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| Al+3 + 2<H+1_23DHB-3> = Al+3_H+1_23DHB-3(2) + H+1 |    |       |        |                     |   |       |        |
| No                                                | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                 | 25 | 0.2   | KCl    | lgK:11.6(0.08SD)    | 6 | MGL   | 2470   |
| 2                                                 | 37 | 0.15  | NaCl   | lgK:11.697(0.067SD) | 6 | MGL   | 8370   |

| Reaction No. 34409                                                                                                     |    |       |        |                 |   |       |        |
|------------------------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{Al}+3 + 2\langle\text{H}+1\_23\text{DHB}-3\rangle = \text{Al}+3\_23\text{DHB}-3(2) + 2\langle\text{H}+1\rangle$ |    |       |        |                 |   |       |        |
| No                                                                                                                     | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                      | 25 | 0.2   | KCl    | lgK:1.7(0.08SD) | 6 | MGL   | 2470   |

| Reaction No. 34410                                                                                                                       |    |       |        |                     |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| $2\langle\text{Al}+3\rangle + 2\langle\text{H}+1\_23\text{DHB}-3\rangle = \text{Al}+3(2)\_23\text{DHB}-3(2) + 2\langle\text{H}+1\rangle$ |    |       |        |                     |   |       |        |
| No                                                                                                                                       | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                                                                                                        | 25 | 0.2   | KCl    | lgK:13.6(0.2SD)     | 4 | MGL   | 2470   |
| 2                                                                                                                                        | 37 | 0.15  | NaCl   | lgK:14.368(0.121SD) | 4 | MGL   | 8370   |

| Reaction No. 34411                                                                                                                                |    |       |        |                    |   |       |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| $2\langle\text{Al}+3\rangle + 2\langle\text{H}+1\_23\text{DHB}-3\rangle = \text{Al}+3(2)\_H+1(-1)\_23\text{DHB}-3(2) + 3\langle\text{H}+1\rangle$ |    |       |        |                    |   |       |        |
| No                                                                                                                                                | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                                                                                                 | 25 | 0.2   | KCl    | lgK:8.9(0.03SD)    | 3 | MGL   | 2470   |
| 2                                                                                                                                                 | 37 | 0.15  | NaCl   | lgK:9.740(0.074SD) | 3 | MGL   | 8370   |

| Reaction No. 34415                                                                    |    |       |        |                  |   |       |        |
|---------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $\text{Al}+3 + \text{H}+1(2)\_234\text{THB}-4 = \text{Al}+3\_H+1(2)\_234\text{THB}-4$ |    |       |        |                  |   |       |        |
| No                                                                                    | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                     | 25 | 0.2   | KCl    | lgK:9.10(0.02SD) | 5 | MGL   | 2470   |

| Reaction No. 34416                                                                                      |    |       |        |                  |   |       |        |
|---------------------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $\text{Al}+3 + 2\langle\text{H}+1(2)\_234\text{THB}-4\rangle = \text{Al}+3\_H+1(4)\_234\text{THB}-4(2)$ |    |       |        |                  |   |       |        |
| No                                                                                                      | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                       | 25 | 0.2   | KCl    | lgK:15.9(0.09SD) | 5 | MGL   | 2470   |

| Reaction No. 34417                                                                                                   |    |       |        |                 |   |       |        |
|----------------------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{Al}+3 + 2\langle\text{H}+1(2)\_234\text{THB}-4\rangle = \text{Al}+3\_H+1(3)\_234\text{THB}-4(2) + \text{H}+1$ |    |       |        |                 |   |       |        |
| No                                                                                                                   | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                    | 25 | 0.2   | KCl    | lgK:10.3(0.1SD) | 5 | MGL   | 2470   |

| Reaction No. 34418                                                                                                                  |    |       |        |               |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| $\text{Al}+3 + 2\langle\text{H}+1(2)\_234\text{THB}-4\rangle = \text{Al}+3\_H+1(2)\_234\text{THB}-4(2) + 2\langle\text{H}+1\rangle$ |    |       |        |               |   |       |        |
| No                                                                                                                                  | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                                                                   | 25 | 0.2   | KCl    | lgK:2.(0.2SD) | 5 | MGL   | 2470   |

| Reaction No. 34419 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| Al+3 + 3<H+1(2)_234THB-4> = Al+3_H+1(4)_234THB-4(3) + 2<H+1> |    |       |        |                |   |       |        |
|--------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| No                                                           | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                            | 25 | 0.2   | KCl    | lgK:6.8(0.1SD) | 5 | MGL   | 2470   |

| Reaction No. 34420                                                 |    |       |        |                 |   |       |        |
|--------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2<Al+3> + 2<H+1(2)_234THB-4> = Al+3(2)_H+1(2)_234THB-4(2) + 2<H+1> |    |       |        |                 |   |       |        |
| No                                                                 | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                  | 25 | 0.2   | KCl    | lgK:11.1(0.2SD) | 5 | MGL   | 2470   |

| Reaction No. 34421                                              |    |       |        |                 |   |       |        |
|-----------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2<Al+3> + 2<H+1(2)_234THB-4> = Al+3(2)_H+1_234THB-4(2) + 3<H+1> |    |       |        |                 |   |       |        |
| No                                                              | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                               | 25 | 0.2   | KCl    | lgK:6.5(0.05SD) | 5 | MGL   | 2470   |

| Reaction No. 34422                                          |    |       |        |                 |   |       |        |
|-------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2<Al+3> + 2<H+1(2)_234THB-4> = Al+3(2)_234THB-4(2) + 4<H+1> |    |       |        |                 |   |       |        |
| No                                                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                           | 25 | 0.2   | KCl    | lgK:-0.7(0.2SD) | 5 | MGL   | 2470   |

| Reaction No. 34423                                            |    |       |        |                |   |       |        |
|---------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 23DiOHTerePhth-2 + Al+3 = H+1 + Al+3_H+1(-1)_23DiOHTerePhth-2 |    |       |        |                |   |       |        |
| No                                                            | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                             | 25 | 0.2   | KCl    | lgK:1.(0.03SD) | 5 | MGL   | 2470   |

| Reaction No. 34424                                                     |    |       |        |                  |   |       |        |
|------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 2<23DiOHTerePhth-2> + Al+3 = 2<H+1> + Al+3_H+1(-2)_23DiOHTerePhth-2(2) |    |       |        |                  |   |       |        |
| No                                                                     | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                      | 25 | 0.2   | KCl    | lgK:-0.4(0.08SD) | 5 | MGL   | 2470   |

| Reaction No. 34425                                                     |    |       |        |                 |   |       |        |
|------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2<23DiOHTerePhth-2> + Al+3 = 3<H+1> + Al+3_H+1(-3)_23DiOHTerePhth-2(2) |    |       |        |                 |   |       |        |
| No                                                                     | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                      | 25 | 0.2   | KCl    | lgK:-8.1(0.2SD) | 5 | MGL   | 2470   |

| Reaction No. 34426                                                           |    |       |        |                 |   |       |        |
|------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2<23DiOHTerePhth-2> + 2<Al+3> = 3<H+1> + Al+3(2)_H+1(-3)_23DiOHTerePhth-2(2) |    |       |        |                 |   |       |        |
| No                                                                           | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                            | 25 | 0.2   | KCl    | lgK:1.4(0.08SD) | 5 | MGL   | 2470   |

| Reaction No. 34427                                                                                                           |    |       |        |                             |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------------------|---|-------|--------|
| $3<23\text{DiOHTerePhth-2}> + 3<\text{Al+3}> = 5<\text{H+1}> + \text{Al+3(3)}_{\text{H+1(-5)}}_{23\text{DiOHTerePhth-2(3)}}$ |    |       |        |                             |   |       |        |
| No                                                                                                                           | t  | I-Str | Medium | Value (Dev)                 | W | Tech. | Ref. # |
| 1                                                                                                                            | 25 | 0.2   | KCl    | $\lg K: 2.3(0.09\text{SD})$ | 5 | MGL   | 2470   |

| Reaction No. 34428                                                                                                           |    |       |        |                             |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------------------|---|-------|--------|
| $3<23\text{DiOHTerePhth-2}> + 3<\text{Al+3}> = 6<\text{H+1}> + \text{Al+3(3)}_{\text{H+1(-6)}}_{23\text{DiOHTerePhth-2(3)}}$ |    |       |        |                             |   |       |        |
| No                                                                                                                           | t  | I-Str | Medium | Value (Dev)                 | W | Tech. | Ref. # |
| 1                                                                                                                            | 25 | 0.2   | KCl    | $\lg K: -2.1(0.1\text{SD})$ | 5 | MGL   | 2470   |

| Reaction No. 34865                                                                                               |    |       |                    |                              |   |       |        |
|------------------------------------------------------------------------------------------------------------------|----|-------|--------------------|------------------------------|---|-------|--------|
| $\text{Al+3} + \text{H+1(3)}_{\text{PyroCatV-4}} = \text{Al+3}_{\text{H+1(2)}}_{\text{PyroCatV-4}} + \text{H+1}$ |    |       |                    |                              |   |       |        |
| No                                                                                                               | t  | I-Str | Medium             | Value (Dev)                  | W | Tech. | Ref. # |
| 1                                                                                                                | 25 | 0.1   | KCl                | $\lg K: -0.2(0.04\text{SD})$ | 5 | MPT   | 349    |
| 2                                                                                                                | 25 | 0.1   | NaClO <sub>4</sub> | $\lg K: 0.1(0.03\text{SD})$  | 4 | MGL   | 3660   |

| Reaction No. 34866                                                                                                                                         |    |       |                    |                              |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|--------------------|------------------------------|---|-------|--------|
| $\text{Al+3}_{\text{H+1(2)}}_{\text{PyroCatV-4}} + \text{H+1(3)}_{\text{PyroCatV-4}} = \text{Al+3}_{\text{H+1(3)}}_{\text{PyroCatV-4(2)}} + 2<\text{H+1}>$ |    |       |                    |                              |   |       |        |
| No                                                                                                                                                         | t  | I-Str | Medium             | Value (Dev)                  | W | Tech. | Ref. # |
| 1                                                                                                                                                          | 25 | 0.1   | NaClO <sub>4</sub> | $\lg K: -6.1(0.09\text{SD})$ | 4 | MGL   | 3660   |

| Reaction No. 34867                                                                                                                                         |    |       |                    |                              |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|--------------------|------------------------------|---|-------|--------|
| $\text{Al+3}_{\text{H+1(3)}}_{\text{PyroCatV-4(2)}} + \text{H+1(2)}_{\text{PyroCatV-4}} = \text{Al+3}_{\text{H+1(4)}}_{\text{PyroCatV-4(3)}} + \text{H+1}$ |    |       |                    |                              |   |       |        |
| No                                                                                                                                                         | t  | I-Str | Medium             | Value (Dev)                  | W | Tech. | Ref. # |
| 1                                                                                                                                                          | 25 | 0     | Inf. Dilution      | $\lg K: -0.8(1\text{SF})$    | 1 | ELC   |        |
| 2                                                                                                                                                          | 25 | 0.1   | NaClO <sub>4</sub> | $\lg K: -1.9(0.05\text{SD})$ | 0 | MGL   | 3660   |

| Reaction No. 34868                                                                                                     |    |       |                    |                              |   |       |        |
|------------------------------------------------------------------------------------------------------------------------|----|-------|--------------------|------------------------------|---|-------|--------|
| $2<\text{Al+3}> + \text{H+1(3)}_{\text{PyroCatV-4}} = \text{Al+3(2)}_{\text{H+1}}_{\text{PyroCatV-4}} + 2<\text{H+1}>$ |    |       |                    |                              |   |       |        |
| No                                                                                                                     | t  | I-Str | Medium             | Value (Dev)                  | W | Tech. | Ref. # |
| 1                                                                                                                      | 25 | 0.1   | NaClO <sub>4</sub> | $\lg K: -0.2(0.04\text{SD})$ | 4 | MGL   | 3660   |

| Reaction No. 34869                                                                                  |    |       |                    |                            |   |       |        |
|-----------------------------------------------------------------------------------------------------|----|-------|--------------------|----------------------------|---|-------|--------|
| $\text{Al+3} + \text{H+1(2)}_{\text{PyroCatV-4}} = \text{Al+3}_{\text{H+1(2)}}_{\text{PyroCatV-4}}$ |    |       |                    |                            |   |       |        |
| No                                                                                                  | t  | I-Str | Medium             | Value (Dev)                | W | Tech. | Ref. # |
| 1                                                                                                   | 25 | 0.1   | NaClO <sub>4</sub> | $\lg K: 7.9(0.1\text{SD})$ | 4 | MGL   | 3660   |

| Reaction No. 34870                                                                                                          |    |       |                    |                 |   |       |        |
|-----------------------------------------------------------------------------------------------------------------------------|----|-------|--------------------|-----------------|---|-------|--------|
| $\text{Al}+3 + \text{H}+1(2)\text{PyroCatV-4} + \text{H}+1\text{PyroCatV-4} = \text{Al}+3\text{H}+1(3)\text{PyroCatV-4}(2)$ |    |       |                    |                 |   |       |        |
| No                                                                                                                          | t  | I-Str | Medium             | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                           | 25 | 0.1   | NaClO <sub>4</sub> | lgK:19.4(0.1SD) | 4 | MGL   | 3660   |

| Reaction No. 34871                                                                                                           |    |       |                    |                 |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------|----|-------|--------------------|-----------------|---|-------|--------|
| $\text{Al}+3 + \text{H}+1(2)\text{PyroCatV-4} + 2\text{H}+1\text{PyroCatV-4} = \text{Al}+3\text{H}+1(4)\text{PyroCatV-4}(3)$ |    |       |                    |                 |   |       |        |
| No                                                                                                                           | t  | I-Str | Medium             | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                            | 25 | 0     | Inf. Dilution      | lgK:31.1(3SF)   | 1 | ELC   |        |
| 2                                                                                                                            | 25 | 0.1   | NaClO <sub>4</sub> | lgK:27.3(0.1SD) | 0 | MGL   | 3660   |

| Reaction No. 34872                                                                       |    |       |                    |                 |   |       |        |
|------------------------------------------------------------------------------------------|----|-------|--------------------|-----------------|---|-------|--------|
| $2\text{Al}+3 + \text{H}+1\text{PyroCatV-4} = \text{Al}+3(2)\text{H}+1\text{PyroCatV-4}$ |    |       |                    |                 |   |       |        |
| No                                                                                       | t  | I-Str | Medium             | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                        | 25 | 0.1   | NaClO <sub>4</sub> | lgK:17.3(0.1SD) | 4 | MGL   | 3660   |

| Reaction No. 35533                                                     |    |       |                                   |               |   |       |        |
|------------------------------------------------------------------------|----|-------|-----------------------------------|---------------|---|-------|--------|
| $\text{Nitroacetic-1} + \text{Al}+3 = \text{Al}+3\text{Nitroacetic-1}$ |    |       |                                   |               |   |       |        |
| No                                                                     | t  | I-Str | Medium                            | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                      | 18 | 0.2   | Ba(NO <sub>3</sub> ) <sub>2</sub> | lgK:0.48(2SF) | 4 | MKN   | 140    |
| 2                                                                      | 25 | 0     | Inf. Dilution                     | lgK:0.4(1SF)  | 1 | ESC   |        |

| Reaction No. 35903                                                                                   |    |       |               |                 |   |       |        |
|------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $13\text{Al}+3 + 4\text{Lactic-1} = \text{Al}+3(13)\text{H}+1(-32)\text{Lactic-1}(4) + 32\text{H}+1$ |    |       |               |                 |   |       |        |
| No                                                                                                   | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                    | 25 | 0     | Inf. Dilution | lgK:-83.5(3SF)  | 1 | ERG   |        |
| 2                                                                                                    | 25 | 0.2   | KCl           | lgK:-96.97(4SF) | 0 | MGL   | 4371   |
| 3                                                                                                    | 25 | 0.6   | NaCl          | lgK:-92.6(3SF)  | 1 | MGL   | 4371   |
| Note (3): Cited value from Marklund et al, ACS, 90, 44, 228 with altered NDP                         |    |       |               |                 |   |       |        |

| Reaction No. 36254                                                                  |    |       |               |                 |   |       |        |
|-------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{H}+1 + \text{Oxalic-2} + \text{Al}+3 = \text{Al}+3\text{H}+1\text{Oxalic-2}$ |    |       |               |                 |   |       |        |
| No                                                                                  | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                   | 25 | 0     | Inf. Dilution | lgK:7.8(3SF)    | 2 | EAE   |        |
| 2                                                                                   | 37 | 0.15  | NaCl          | lgK:6.6(0.08SD) | 5 | MGL   | 2372   |

| Reaction No. 36255 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| Al+3 + 2<Oxalic-2> + H2O = Al+3_OH-1_Oxalic-2(2) + H+1 |    |       |        |                 |   |       |        |
|--------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| No                                                     | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                      | 25 | 0.2   | KCl    | lgK:5.3(0.09SD) | 5 | MGL   | 4371   |
| 2                                                      | 37 | 0.15  | NaCl   | lgK:7.1(0.08SD) | 0 | MGL   | 2372   |

| Reaction No. 36256                                     |    |       |               |                |   |       |        |
|--------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Al+3 + 3<Oxalic-2> + H2O = Al+3_OH-1_Oxalic-2(3) + H+1 |    |       |               |                |   |       |        |
| No                                                     | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                      | 25 | 0     | Inf. Dilution | lgK:10.0(3SF)  | 2 | EAE   |        |
| 2                                                      | 37 | 0.15  | NaCl          | lgK:9.5(0.1SD) | 5 | MGL   | 2372   |

| Reaction No. 36257                                      |    |       |               |                  |   |       |        |
|---------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Succinic-2 + 2<Al+3> = H+1 + Al+3(2)_H+1(-1)_Succinic-2 |    |       |               |                  |   |       |        |
| No                                                      | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                       | 25 | 0     | Inf. Dilution | lgK:4.32(3SF)    | 2 | EAE   |        |
| 2                                                       | 37 | 0.15  | NaCl          | lgK:2.88(0.02SD) | 5 | MGL   | 2372   |

| Reaction No. 36462              |    |       |        |                 |   |       |        |
|---------------------------------|----|-------|--------|-----------------|---|-------|--------|
| NOHIDA-2 + Al+3 = Al+3_NOHIDA-2 |    |       |        |                 |   |       |        |
| No                              | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                               | 25 | 0.1   | KNO3   | lgK:5.1(0.03SD) | 6 | MPT   | 3908   |

| Reaction No. 36556          |    |       |        |                 |   |       |        |
|-----------------------------|----|-------|--------|-----------------|---|-------|--------|
| HIDP-2 + Al+3 = Al+3_HIDP-2 |    |       |        |                 |   |       |        |
| No                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                           | 25 | 0.1   | KNO3   | lgK:5.9(0.03SD) | 6 | MPT   | 3908   |

| Reaction No. 37946                               |    |       |        |                 |   |       |        |
|--------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Al+3 + Oxalic-2 + H2O = Al+3_OH-1_Oxalic-2 + H+1 |    |       |        |                 |   |       |        |
| No                                               | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                | 25 | 0.2   | KCl    | lgK:1.0(0.04SD) | 5 | MGL   | 4371   |

| Reaction No. 37947                                        |    |       |        |                  |   |       |        |
|-----------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Al+3 + Oxalic-2 + 2<H2O> = Al+3_OH-1(2)_Oxalic-2 + 2<H+1> |    |       |        |                  |   |       |        |
| No                                                        | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                         | 25 | 0.2   | KCl    | lgK:-4.4(0.04SD) | 5 | MGL   | 4371   |

| Reaction No. 37948 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|



| Malonic-2 + Al+3 = 2<H+1> + Al+3_H+1(-2)_Malonic-2 |    |       |        |                  |   |       |        |
|----------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| No                                                 | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                  | 25 | 0.2   | KCl    | lgK:-5.3(0.09SD) | 5 | MGL   | 4371   |

| Reaction No. 37949                            |    |       |        |                |   |       |        |
|-----------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Lactic-1 + Al+3 = H+1 + Al+3_H+1(-1)_Lactic-1 |    |       |        |                |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                             | 25 | 0.2   | KCl    | lgK:-1.84(3SF) | 5 | MGL   | 4371   |

| Reaction No. 37950                               |    |       |        |                 |   |       |        |
|--------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Lactic-1 + Al+3 = 3<H+1> + Al+3_H+1(-3)_Lactic-1 |    |       |        |                 |   |       |        |
| No                                               | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                | 25 | 0.2   | KCl    | lgK:-12.39(4SF) | 5 | MGL   | 4371   |

| Reaction No. 37951                                                           |    |       |        |               |   |       |        |
|------------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| 2<Lactic-1> + Al+3 = H+1 + Al+3_H+1(-1)_Lactic-1(2)                          |    |       |        |               |   |       |        |
| No                                                                           | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                            | 25 | 0.2   | KCl    | lgK:1.05(3SF) | 5 | MGL   | 4371   |
| 2                                                                            | 25 | 0.6   | NaCl   | lgK:0.94(2SF) | 5 | MGL   | 4371   |
| Note (2): Cited value from Marklund et al, ACS, 90, 44, 228 with altered NDP |    |       |        |               |   |       |        |

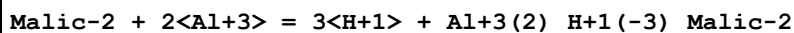
| Reaction No. 37952                                                           |    |       |        |                |   |       |        |
|------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Lactic-1 + 2<Al+3> = 2<H+1> + Al+3(2)_H+1(-2)_Lactic-1                       |    |       |        |                |   |       |        |
| No                                                                           | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                            | 25 | 0.2   | KCl    | lgK:-3.43(3SF) | 5 | MGL   | 4371   |
| 2                                                                            | 25 | 0.6   | NaCl   | lgK:-3.29(3SF) | 5 | MGL   | 4371   |
| Note (2): Cited value from Marklund et al, ACS, 90, 44, 228 with altered NDP |    |       |        |                |   |       |        |

| Reaction No. 37953                                                           |    |       |        |                |   |       |        |
|------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 2<Lactic-1> + 2<Al+3> = 4<H+1> + Al+3(2)_H+1(-4)_Lactic-1(2)                 |    |       |        |                |   |       |        |
| No                                                                           | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                            | 25 | 0.2   | KCl    | lgK:-9.60(3SF) | 5 | MGL   | 4371   |
| 2                                                                            | 25 | 0.6   | NaCl   | lgK:-9.62(3SF) | 5 | MGL   | 4371   |
| Note (2): Cited value from Marklund et al, ACS, 90, 44, 228 with altered NDP |    |       |        |                |   |       |        |

| Reaction No. 37954                             |  |  |  |  |  |  |  |
|------------------------------------------------|--|--|--|--|--|--|--|
| Malic-2 + Al+3 = 2<H+1> + Al+3_H+1(-2)_Malic-2 |  |  |  |  |  |  |  |

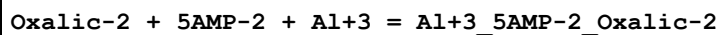
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaCl   | lgK:-4.83(0.05SD) | 2 | MGL   | 29190  |
| 2  | 25 | 0.2   | KCl    | lgK:-3.5(0.03SD)  | 0 | MGL   | 4371   |
| 3  | 25 | 0.5   | NaCl   | lgK:-4.84(0.02SD) | 2 | MGL   | 29190  |
| 4  | 25 | 1     | NaCl   | lgK:-4.48(0.01SD) | 2 | MGL   | 29190  |

## Reaction No. 37955



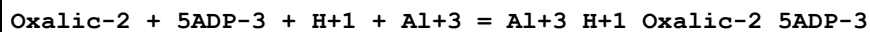
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:-3.8(0.05SD) | 6 | MGL   | 4371   |

## Reaction No. 37956



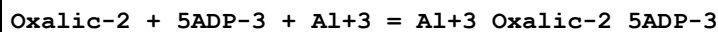
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:12.41(4SF)  | 1 | EOR   |        |
| 2  | 25 | 0.2   | KCl           | lgK:10.4(0.1SD) | 6 | MGL   | 4371   |

## Reaction No. 37957



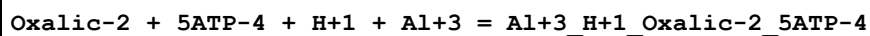
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:18.27(4SF)  | 1 | EOR   |        |
| 2  | 25 | 0.2   | KCl           | lgK:15.5(0.2SD) | 6 | MGL   | 4371   |

## Reaction No. 37958



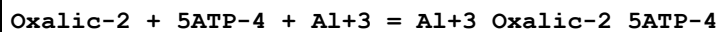
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:14.16(4SF)   | 1 | EOR   |        |
| 2  | 25 | 0.2   | KCl           | lgK:11.9(0.09SD) | 6 | MGL   | 4371   |

## Reaction No. 37959



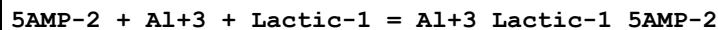
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:19.47(4SF)  | 1 | EOR   |        |
| 2  | 25 | 0.2   | KCl           | lgK:16.2(0.2SD) | 6 | MGL   | 4371   |

## Reaction No. 37960



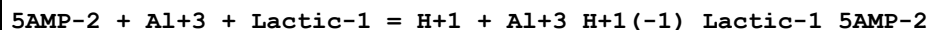
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:15.11(4SF)   | 1 | EOR   |        |
| 2  | 25 | 0.2   | KCl           | lgK:12.6(0.03SD) | 6 | MGL   | 4371   |

## Reaction No. 37961



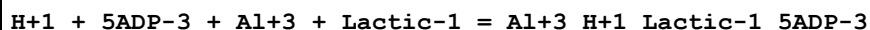
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:8.4(0.03SD) | 6 | MGL   | 4371   |

## Reaction No. 37962



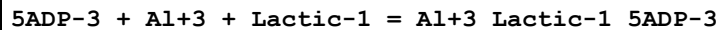
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:3.(0.3SD) | 6 | MGL   | 4371   |

## Reaction No. 37963



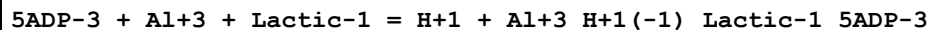
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:13.9(0.04SD) | 6 | MGL   | 4371   |

## Reaction No. 37964



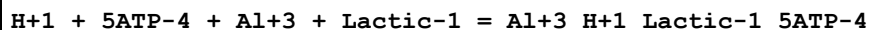
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:10.(0.06SD) | 6 | MGL   | 4371   |

## Reaction No. 37965



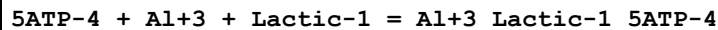
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:4.5(0.2SD) | 6 | MGL   | 4371   |

## Reaction No. 37966



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:14.0(0.09SD) | 6 | MGL   | 4371   |

## Reaction No. 37967



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:10.5(0.08SD) | 6 | MGL   | 4371   |

| Reaction No. 37968                                            |    |       |        |                |   |       |        |
|---------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 5ATP-4 + Al+3 + Lactic-1 = H+1 + Al+3_H+1(-1)_Lactic-1_5ATP-4 |    |       |        |                |   |       |        |
| No                                                            | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                             | 25 | 0.2   | KCl    | lgK:5.0(0.2SD) | 6 | MGL   | 4371   |

| Reaction No. 37969                                      |    |       |        |                  |   |       |        |
|---------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| H+1 + 5AMP-2 + Al+3 + Malic-2 = Al+3_H+1_5AMP-2_Malic-2 |    |       |        |                  |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.2   | KCl    | lgK:13.5(0.05SD) | 6 | MGL   | 4371   |

| Reaction No. 37970                            |    |       |        |                |   |       |        |
|-----------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 5AMP-2 + Al+3 + Malic-2 = Al+3_5AMP-2_Malic-2 |    |       |        |                |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                             | 25 | 0.2   | KCl    | lgK:9.5(0.1SD) | 6 | MGL   | 4371   |

| Reaction No. 37971                                      |    |       |        |                  |   |       |        |
|---------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| H+1 + 5ADP-3 + Al+3 + Malic-2 = Al+3_H+1_Malic-2_5ADP-3 |    |       |        |                  |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.2   | KCl    | lgK:15.1(0.04SD) | 6 | MGL   | 4371   |

| Reaction No. 37972                            |    |       |        |                  |   |       |        |
|-----------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 5ADP-3 + Al+3 + Malic-2 = Al+3_Malic-2_5ADP-3 |    |       |        |                  |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                             | 25 | 0.2   | KCl    | lgK:11.1(0.08SD) | 6 | MGL   | 4371   |

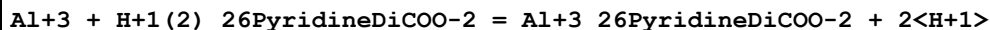
| Reaction No. 37973                                      |    |       |        |                  |   |       |        |
|---------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| H+1 + 5ATP-4 + Al+3 + Malic-2 = Al+3_H+1_Malic-2_5ATP-4 |    |       |        |                  |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.2   | KCl    | lgK:15.5(0.08SD) | 6 | MGL   | 4371   |

| Reaction No. 37974                            |    |       |        |                  |   |       |        |
|-----------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 5ATP-4 + Al+3 + Malic-2 = Al+3_Malic-2_5ATP-4 |    |       |        |                  |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                             | 25 | 0.2   | KCl    | lgK:11.3(0.09SD) | 6 | MGL   | 4371   |

| Reaction No. 38596                                                 |  |  |  |  |  |  |  |
|--------------------------------------------------------------------|--|--|--|--|--|--|--|
| Al+3 + H+1(2)_26PyridineDiCOO-2 = Al+3_H+1_26PyridineDiCOO-2 + H+1 |  |  |  |  |  |  |  |

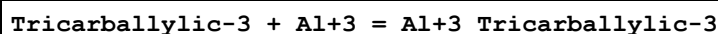
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 23 | 1     | NaCl          | lgK:2.943(4SF) | 5 | MMX   | 3326   |
| 2  | 25 | 0     | Inf. Dilution | lgK:3.0(2SF)   | 1 | ESC   |        |

## Reaction No. 38597



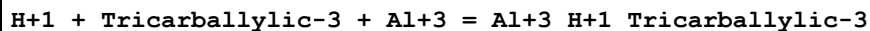
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 23 | 1     | NaCl          | lgK:0.267(3SF) | 5 | MMX   | 3326   |
| 2  | 25 | 0     | Inf. Dilution | lgK:0.3(1SF)   | 1 | ESC   |        |

## Reaction No. 38824



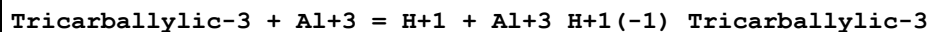
| No | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:6.28(0.05SD)   | 2 | EDH   | 29190  |
| 2  | 25 | 0.15  | NaCl          | lgK:4.765(0.005SD) | 4 | MGL   | 29190  |
| 3  | 25 | 0.5   | NaCl          | lgK:4.124(0.004SD) | 4 | MGL   | 29190  |
| 4  | 25 | 1     | NaCl          | lgK:3.785(0.006SD) | 3 | MGL   | 29190  |
| 5  | 37 | 0.15  | NaCl          | lgK:5.4(0.06SD)    | 0 | MGL   | 3574   |

## Reaction No. 38825



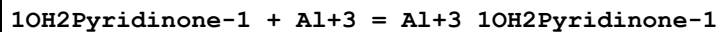
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:9.2(2SF)    | 1 | ESC   |        |
| 2  | 37 | 0.15  | NaCl          | lgK:8.9(0.04SD) | 5 | MGL   | 3574   |

## Reaction No. 38826



| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:2.0(2SF)    | 1 | ESC   |        |
| 2  | 37 | 0.15  | NaCl          | lgK:1.9(0.04SD) | 5 | MGL   | 3574   |

## Reaction No. 39278



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:8.2(0.06SD) | 6 | MGL   | 2686   |

## Reaction No. 39279

| Al+3_1OH2Pyridinone-1 + 1OH2Pyridinone-1 = Al+3_1OH2Pyridinone-1(2) |    |       |        |                 |   |       |        |
|---------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| No                                                                  | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                   | 25 | 0.1   | KCl    | lgK:7.4(0.08SD) | 6 | MGL   | 2686   |

| Reaction No. 39280                                                     |    |       |        |                 |   |       |        |
|------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Al+3_1OH2Pyridinone-1(2) + 1OH2Pyridinone-1 = Al+3_1OH2Pyridinone-1(3) |    |       |        |                 |   |       |        |
| No                                                                     | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                      | 25 | 0.1   | KCl    | lgK:6.0(0.06SD) | 6 | MGL   | 2686   |

| Reaction No. 40280                  |    |       |        |                |   |       |        |
|-------------------------------------|----|-------|--------|----------------|---|-------|--------|
| MeDiPhos-4 + Al+3 = Al+3_MeDiPhos-4 |    |       |        |                |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | KCl    | lgK:14.08(4SF) | 5 | MGL   | 931    |

| Reaction No. 40281                        |    |       |        |                |   |       |        |
|-------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 2<MeDiPhos-4> + Al+3 = Al+3_MeDiPhos-4(2) |    |       |        |                |   |       |        |
| No                                        | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                         | 25 | 0.1   | KCl    | lgK:23.01(4SF) | 5 | MGL   | 931    |

| Reaction No. 40282                          |    |       |        |               |   |       |        |
|---------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Al+3 + H+1_MeDiPhos-4 = Al+3_H+1_MeDiPhos-4 |    |       |        |               |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | KCl    | lgK:9.05(3SF) | 5 | MGL   | 931    |

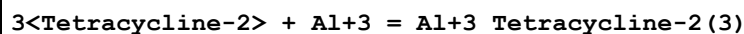
| Reaction No. 40283                                   |    |       |        |                |   |       |        |
|------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Al+3 + 2<H+1_MeDiPhos-4> = Al+3_H+1(2)_MeDiPhos-4(2) |    |       |        |                |   |       |        |
| No                                                   | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                    | 25 | 0.1   | KCl    | lgK:13.66(4SF) | 5 | MGL   | 931    |

| Reaction No. 42438                          |    |       |               |              |   |       |        |
|---------------------------------------------|----|-------|---------------|--------------|---|-------|--------|
| Tetracycline-2 + Al+3 = Al+3_Tetracycline-2 |    |       |               |              |   |       |        |
| No                                          | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
| 1                                           | 20 | 0.01  | Unknown       | lgK:7.4(2SF) | 5 | MGL   | 3682   |
| 2                                           | 25 | 0     | Inf. Dilution | lgK:7.4(2SF) | 1 | ESC   |        |

| Reaction No. 42439                                            |   |       |        |             |   |       |        |
|---------------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| Al+3_Tetracycline-2 + Tetracycline-2 = Al+3_Tetracycline-2(2) |   |       |        |             |   |       |        |
| No                                                            | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

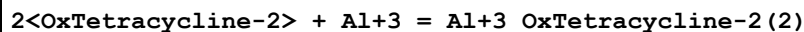
|   |    |      |               |               |   |     |      |
|---|----|------|---------------|---------------|---|-----|------|
| 1 | 20 | 0.01 | Unknown       | lgK:6.4 (2SF) | 5 | MGL | 3682 |
| 2 | 25 | 0    | Inf. Dilution | lgK:6.4 (2SF) | 1 | ESC |      |

## Reaction No. 42440



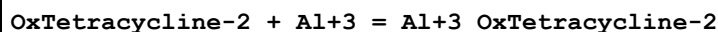
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 20 | 0.01  | Unknown       | lgK:19.2 (3SF) | 5 | MGL   | 3682   |
| 2  | 25 | 0     | Inf. Dilution | lgK:19.2 (3SF) | 1 | ESC   |        |

## Reaction No. 42442



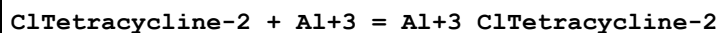
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 20 | 0.01  | Unknown       | lgK:18. (2SF)  | 3 | MGL   | 3682   |
| 2  | 25 | 0     | Inf. Dilution | lgK:18.0 (3SF) | 1 | ESC   |        |

## Reaction No. 42445



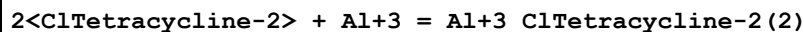
| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 20 | 0.01  | Unknown       | lgK:7.0 (2SF) | 5 | MGL   | 3682   |
| 2  | 25 | 0     | Inf. Dilution | lgK:7.0 (2SF) | 1 | ESC   |        |

## Reaction No. 42459



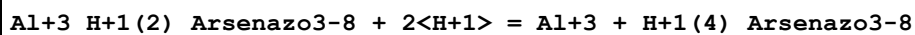
| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 20 | 0.01  | Unknown       | lgK:7.2 (2SF) | 5 | MGL   | 3682   |
| 2  | 25 | 0     | Inf. Dilution | lgK:7.2 (2SF) | 1 | ESC   |        |

## Reaction No. 42460



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 20 | 0.01  | Unknown       | lgK:18. (2SF)  | 3 | MGL   | 3682   |
| 2  | 25 | 0     | Inf. Dilution | lgK:18.0 (3SF) | 1 | ESC   |        |

## Reaction No. 42495



| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:1.173 (4SF) | 5 | MSP   | 4201   |

| Reaction No. 42496                                                                                        |    |       |         |                 |   |       |        |
|-----------------------------------------------------------------------------------------------------------|----|-------|---------|-----------------|---|-------|--------|
| $\text{Al+3\_OH-1} + \text{H+1(5)\_Arsenazo3-8} = \text{Al+3\_H+1(3)\_OH-1\_Arsenazo3-8} + 2\text{<H+1>}$ |    |       |         |                 |   |       |        |
| No                                                                                                        | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                         | 25 | 0.1   | Unknown | lgK:0.468 (3SF) | 5 | MSP   | 4201   |

| Reaction No. 42681                                                                                                                           |    |       |               |                 |   |       |        |
|----------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $2\text{<Al+3>} + 2\text{<H+1(4)\_MeThymolBlue-6>} = \text{Al+3\_H+1(3)\_MeThymolBlue-6} + \text{Al+3\_H+1(4)\_MeThymolBlue-6} + \text{H+1}$ |    |       |               |                 |   |       |        |
| No                                                                                                                                           | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                                            | 22 | 1     | NaClO4        | lgK:3.881 (4SF) | 3 | MGL   | 4364   |
| 2                                                                                                                                            | 25 | 0     | Inf. Dilution | lgK:4.0 (2SF)   | 1 | ESC   |        |

| Reaction No. 42682                                                                                                                                    |    |       |               |                 |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $2\text{<Al+3>} + 4\text{<H+1(4)\_MeThymolBlue-6>} = \text{Al+3\_H+1(6)\_MeThymolBlue-6(2)} + \text{Al+3\_H+1(8)\_MeThymolBlue-6(2)} + 2\text{<H+1>}$ |    |       |               |                 |   |       |        |
| No                                                                                                                                                    | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                                                     | 22 | 1     | NaClO4        | lgK:6.021 (4SF) | 3 | MGL   | 4364   |
| 2                                                                                                                                                     | 25 | 0     | Inf. Dilution | lgK:6.1 (2SF)   | 1 | ESC   |        |

| Reaction No. 42730                                                    |    |       |               |               |   |       |        |
|-----------------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $\text{Al+3\_OH-1(4)} + 5\text{ATP-4} = \text{Al+3\_OH-1(4)\_5ATP-4}$ |    |       |               |               |   |       |        |
| No                                                                    | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                     | 25 | 0     | Inf. Dilution | lgK:3.7 (2SF) | 1 | ESC   |        |
| 2                                                                     | 27 |       |               | lgK:3.7 (2SF) | 0 | MMR   | 4359   |

| Reaction No. 42844                                                        |    |       |               |                |   |       |        |
|---------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $\text{Co+3\_Oxalic-2(3)} + \text{Al+3} = \text{Al+3\_Co+3\_Oxalic-2(3)}$ |    |       |               |                |   |       |        |
| No                                                                        | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                         | 25 | 0     | Inf. Dilution | lgK:3.70 (3SF) | 4 | MPL   | 906    |
| 2                                                                         | 25 | 0.1   | NH4NO3        | lgK:2.07 (3SF) | 4 | MPL   | 906    |

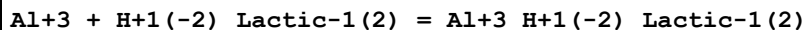
| Reaction No. 42876                                                                                          |    |       |               |                   |   |       |        |
|-------------------------------------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $\text{Al(s)} + 2\text{<B(s)>} + 4\text{<H2(g)>} + 4\text{<O2(g)>} - \text{e-1} = \text{Al+3\_B(OH)4-1(2)}$ |    |       |               |                   |   |       |        |
| No                                                                                                          | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                           | 25 | 0     | Inf. Dilution | dG:-687.910 (6SF) | 5 | CRV   | 406    |

| Reaction No. 43060                                                               |  |  |  |  |  |  |  |
|----------------------------------------------------------------------------------|--|--|--|--|--|--|--|
| $\text{Al+3} + 3\text{<H+1\_3OHPropan-1>} = \text{Al+3\_H+1(3)\_3OHPropan-1(3)}$ |  |  |  |  |  |  |  |



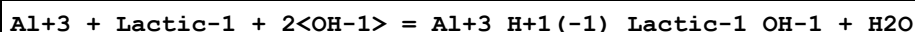
| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 23 |       |               | lgK:7.38(3SF) | 4 | MSP   | 906    |
| 2  | 25 | 0     | Inf. Dilution | lgK:7.5(2SF)  | 1 | ESC   |        |

## Reaction No. 43062



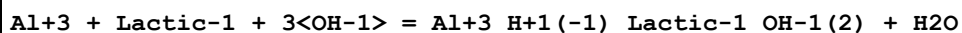
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 23 |       |               | lgK:23.05(4SF) | 3 | MSP   | 906    |
| 2  | 25 | 0     | Inf. Dilution | lgK:23.3(3SF)  | 1 | ESC   |        |

## Reaction No. 43063



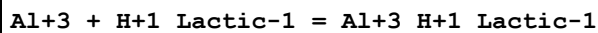
| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 23 |       |               | lgK:8.49(3SF) | 3 | MSP   | 906    |
| 2  | 25 | 0     | Inf. Dilution | lgK:8.6(2SF)  | 1 | ESC   |        |

## Reaction No. 43064



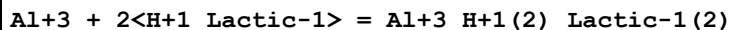
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 23 |       |               | lgK:24.58(4SF) | 3 | MSP   | 906    |
| 2  | 25 | 0     | Inf. Dilution | lgK:24.8(3SF)  | 1 | ESC   |        |

## Reaction No. 43065



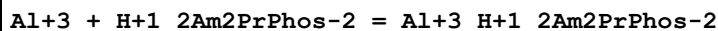
| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 23 |       |               | lgK:4.46(3SF) | 3 | MSP   | 906    |
| 2  | 25 | 0     | Inf. Dilution | lgK:4.6(2SF)  | 1 | ESC   |        |

## Reaction No. 43066



| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 23 |       |               | lgK:6.51(3SF) | 3 | MSP   | 906    |
| 2  | 25 | 0     | Inf. Dilution | lgK:6.6(2SF)  | 1 | ESC   |        |

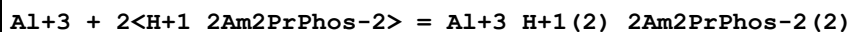
## Reaction No. 43174



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

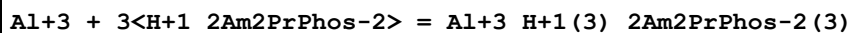
|   |    |     |     |               |   |     |     |
|---|----|-----|-----|---------------|---|-----|-----|
| 1 | 25 | 0.1 | KCl | lgK:5.59(3SF) | 5 | MGL | 906 |
|---|----|-----|-----|---------------|---|-----|-----|

## Reaction No. 43175



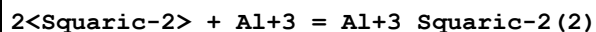
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:9.61(3SF) | 5 | MGL   | 906    |

## Reaction No. 43176



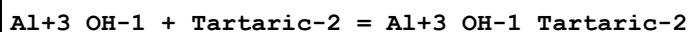
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:13.07(4SF) | 5 | MGL   | 906    |

## Reaction No. 43194



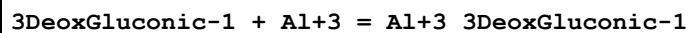
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 1     | NH4ClO4 | lgK:2.7(0.06SD) | 5 | MIX   | 906    |

## Reaction No. 43231



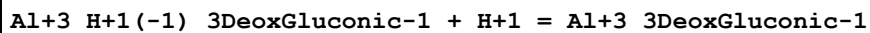
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 20 |       |               | lgK:1.8(0.2SD) | 0 | MSC   | 906    |
| 2  | 25 | 0     | Inf. Dilution | lgK:7.3(2SF)   | 1 | ELC   |        |

## Reaction No. 43251



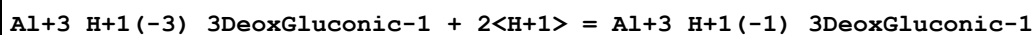
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaCl   | lgK:2.0(0.03SD) | 3 | MGL   | 3975   |

## Reaction No. 43252



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | NaCl   | lgK:-2.8(0.05SD) | 5 | MGL   | 3975   |

## Reaction No. 43253



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | NaCl   | lgK:-9.2(0.08SD) | 5 | MGL   | 3975   |

| Reaction No. 43255                          |    |       |        |                 |   |       |        |
|---------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 25DiOHPentan-1 + Al+3 = Al+3_25DiOHPentan-1 |    |       |        |                 |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | NaCl   | lgK:2.0(0.03SD) | 3 | MGL   | 3975   |

| Reaction No. 43256                                      |    |       |        |                  |   |       |        |
|---------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Al+3_H+1(-1)_25DiOHPentan-1 + H+1 = Al+3_25DiOHPentan-1 |    |       |        |                  |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.1   | NaCl   | lgK:-3.1(0.06SD) | 5 | MGL   | 3975   |

| Reaction No. 43257                                              |    |       |        |                  |   |       |        |
|-----------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Al+3_H+1(-2)_25DiOHPentan-1 + H+1 = Al+3_H+1(-1)_25DiOHPentan-1 |    |       |        |                  |   |       |        |
| No                                                              | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                               | 25 | 0.1   | NaCl   | lgK:-2.3(0.09SD) | 5 | MGL   | 3975   |

| Reaction No. 43615                 |    |       |        |                |   |       |        |
|------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Al+3_Glu-2 + Glu-2 = Al+3_Glu-2(2) |    |       |        |                |   |       |        |
| No                                 | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                  | 25 | 0.1   | NaClO4 | lgK:14.28(4SF) | 0 | MGL   | 5763   |

| Reaction No. 43616                    |    |       |        |               |   |       |        |
|---------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Al+3_Glu-2(2) + Glu-2 = Al+3_Glu-2(3) |    |       |        |               |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | NaClO4 | lgK:9.20(3SF) | 0 | MGL   | 5763   |

| Reaction No. 43666                                      |    |       |               |                |   |       |        |
|---------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Al+3_Pyrogallol-3 + Pyrogallol-3 = Al+3_Pyrogallol-3(2) |    |       |               |                |   |       |        |
| No                                                      | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                       | 22 | 0.2   | KCl           | lgK:20.05(4SF) | 2 | MGL   | 906    |
| 2                                                       | 25 | 0     | Inf. Dilution | lgK:21.5(3SF)  | 1 | EDH   |        |

| Reaction No. 43667                                         |    |       |               |                |   |       |        |
|------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Al+3_Pyrogallol-3(2) + Pyrogallol-3 = Al+3_Pyrogallol-3(3) |    |       |               |                |   |       |        |
| No                                                         | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                          | 22 | 0.2   | KCl           | lgK:13.40(4SF) | 0 | MGL   | 906    |
| 2                                                          | 25 | 0     | Inf. Dilution | lgK:5.3(2SF)   | 1 | ELC   |        |

| Reaction No. 44216 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| Al+3 + H+1(2)_25DHB-3 = Al+3_H+1_25DHB-3 + H+1 |       |       |               |                |   |       |        |
|------------------------------------------------|-------|-------|---------------|----------------|---|-------|--------|
| No                                             | t     | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                              | 20    | 0.01  | NaClO4        | lgK:-3.49(3SF) | 0 | MSP   | 906    |
| 2                                              | 20    | 0.09  | NaClO4        | lgK:-3.96(3SF) | 0 | MSP   | 906    |
| 3                                              | 20    | 0.2   | NaClO4        | lgK:-4.47(3SF) | 0 | MSP   | 906    |
| 4                                              | 25    | 0     | Inf. Dilution | lgK:0.1(1SF)   | 1 | ELC   |        |
| 5                                              | 30    | 0.01  | NaClO4        | lgK:-3.43(3SF) | 0 | MSP   | 906    |
| 6                                              | 40    | 0.01  | NaClO4        | lgK:-3.25(3SF) | 0 | MSP   | 906    |
| 7                                              | 50    | 0.01  | NaClO4        | lgK:-3.11(3SF) | 0 | MSP   | 906    |
| 8                                              | 60    | 0.01  | NaClO4        | lgK:-2.92(3SF) | 0 | MSP   | 906    |
| 9                                              | 20:60 | 0.01  | NaClO4        | dH:5.44(3SF)   | 2 | MTD   | 906    |

| Reaction No. 44480                  |    |       |               |                |   |       |        |
|-------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Mandelic-1 + Al+3 = Al+3_Mandelic-1 |    |       |               |                |   |       |        |
| No                                  | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                   | 23 |       |               | lgK:13.91(4SF) | 4 | MPT   | 906    |
| 2                                   | 25 | 0     | Inf. Dilution | lgK:14.1(3SF)  | 1 | ESC   |        |

| Reaction No. 44481                                |    |       |               |                |   |       |        |
|---------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Al+3_Mandelic-1 + Mandelic-1 = Al+3_Mandelic-1(2) |    |       |               |                |   |       |        |
| No                                                | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                 | 23 |       |               | lgK:12.99(4SF) | 4 | MPT   | 906    |
| 2                                                 | 25 | 0     | Inf. Dilution | lgK:12.2(3SF)  | 1 | ESC   |        |

| Reaction No. 44482                                   |    |       |               |                |   |       |        |
|------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Al+3_Mandelic-1(2) + Mandelic-1 = Al+3_Mandelic-1(3) |    |       |               |                |   |       |        |
| No                                                   | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                    | 23 |       |               | lgK:11.98(4SF) | 4 | MPT   | 906    |
| 2                                                    | 25 | 0     | Inf. Dilution | lgK:12.2(3SF)  | 1 | ESC   |        |

| Reaction No. 44505                          |       |       |                                     |               |   |       |        |
|---------------------------------------------|-------|-------|-------------------------------------|---------------|---|-------|--------|
| NPhenNACOHAm-1 + Al+3 = Al+3_NPhenNACOHAm-1 |       |       |                                     |               |   |       |        |
| No                                          | t     | I-Str | Medium                              | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 29:31 | 0.5   | Acetone:Water<br>50:50Vol<br>NaClO4 | lgK:9.20(3SF) | 5 | MGL   | 906    |

| Reaction No. 44506 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| Al+3_NPhenNacOHAm-1 + NPhenNacOHAm-1 = Al+3_NPhenNacOHAm-1 (2) |       |       |                                     |                |   |       |        |
|----------------------------------------------------------------|-------|-------|-------------------------------------|----------------|---|-------|--------|
| No                                                             | t     | I-Str | Medium                              | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                              | 29:31 | 0.5   | Acetone:Water<br>50:50Vol<br>NaClO4 | lgK:8.10 (3SF) | 5 | MGL   | 906    |

| Reaction No. 44507                                                 |       |       |                                     |                |   |       |        |
|--------------------------------------------------------------------|-------|-------|-------------------------------------|----------------|---|-------|--------|
| Al+3_NPhenNacOHAm-1 (2) + NPhenNacOHAm-1 = Al+3_NPhenNacOHAm-1 (3) |       |       |                                     |                |   |       |        |
| No                                                                 | t     | I-Str | Medium                              | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                  | 29:31 | 0.5   | Acetone:Water<br>50:50Vol<br>NaClO4 | lgK:7.53 (3SF) | 5 | MGL   | 906    |

| Reaction No. 44592                                          |    |       |               |                   |   |       |        |
|-------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| Al(s) + B(s) + 2<H2(g)> + 2<O2(g)> - 2<e-1> = Al+3_B(OH)4-1 |    |       |               |                   |   |       |        |
| No                                                          | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                           | 25 | 0     | Inf. Dilution | dG:-402.850 (6SF) | 5 | CRV   | 406    |

| Reaction No. 44712                           |    |       |               |                |   |       |        |
|----------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Al+3_Sulfox-2 + Sulfox-2 = Al+3_Sulfox-2 (2) |    |       |               |                |   |       |        |
| No                                           | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                            | 23 |       |               | lgK:9.01 (3SF) | 4 | MSP   | 906    |
| 2                                            | 25 | 0     | Inf. Dilution | lgK:9.3 (2SF)  | 1 | ESC   |        |

| Reaction No. 44748                                        |    |       |               |                   |   |       |        |
|-----------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| Al(s) + 3<B(s)> + 6<H2(g)> + 6<O2(g)> = Al+3_B(OH)4-1 (3) |    |       |               |                   |   |       |        |
| No                                                        | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                         | 25 | 0     | Inf. Dilution | dG:-970.740 (6SF) | 5 | CRV   | 406    |

| Reaction No. 45070                                  |    |       |               |                |   |       |        |
|-----------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Al+3 + H+1_N2COOPhenIDA-3 = Al+3_H+1_N2COOPhenIDA-3 |    |       |               |                |   |       |        |
| No                                                  | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                   | 22 |       |               | lgK:4.60 (3SF) | 4 | MSP   | 906    |
| 2                                                   | 25 | 0     | Inf. Dilution | lgK:4.8 (2SF)  | 1 | ESC   |        |

| Reaction No. 45212                                             |    |       |         |                    |   |       |        |
|----------------------------------------------------------------|----|-------|---------|--------------------|---|-------|--------|
| Al+3 + 2<H+1_AlizarinRedS-3> = Al+3_H+1 (2)_AlizarinRedS-3 (2) |    |       |         |                    |   |       |        |
| No                                                             | t  | I-Str | Medium  | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                              | 25 | 0     | Unknown | lgK:12.88 (0.01SD) | 4 | MGL   | 3894   |

|   |    |     |         |                |   |     |     |
|---|----|-----|---------|----------------|---|-----|-----|
| 2 | 25 | 0.1 | Unknown | lgK:12.06(4SF) | 3 | MSP | 906 |
|---|----|-----|---------|----------------|---|-----|-----|

| Reaction No. 45213                                              |    |       |                    |                 |   |       |        |
|-----------------------------------------------------------------|----|-------|--------------------|-----------------|---|-------|--------|
| Al+3 + 2<H+1(2)_AlizarinRedS-3> = Al+3_H+1(4)_AlizarinRedS-3(2) |    |       |                    |                 |   |       |        |
| No                                                              | t  | I-Str | Medium             | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                               | 23 | 0.1   | NaClO <sub>4</sub> | lgK:11.5(0.2SD) | 4 | MSP   | 906    |
| 2                                                               | 25 | 0     | Inf. Dilution      | lgK:11.8(3SF)   | 1 | ESC   |        |

| Reaction No. 45214                          |    |       |                    |                 |   |       |        |
|---------------------------------------------|----|-------|--------------------|-----------------|---|-------|--------|
| AlizarinRedS-3 + Al+3 = Al+3_AlizarinRedS-3 |    |       |                    |                 |   |       |        |
| No                                          | t  | I-Str | Medium             | Value (Dev)     | W | Tech. | Ref. # |
| 1                                           | 23 | 0.1   | Unknown            | lgK:11.31(4SF)w | 0 | MPH   | 906    |
| 2                                           | 25 | 0.1   | NaClO <sub>4</sub> | lgK:14.11(4SF)  | 4 | ENS   | 6953   |

| Reaction No. 45215                                            |    |       |               |                |   |       |        |
|---------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Al+3_AlizarinRedS-3 + AlizarinRedS-3 = Al+3_AlizarinRedS-3(2) |    |       |               |                |   |       |        |
| No                                                            | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                             | 23 | 0.1   | Unknown       | lgK:6.06(3SF)w | 0 | MPH   | 906    |
| 2                                                             | 25 | 0     | Inf. Dilution | lgK:12.8(3SF)  | 1 | ELC   |        |

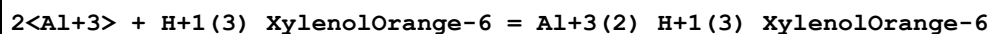
| Reaction No. 45341        |    |       |                                                 |                 |   |       |        |
|---------------------------|----|-------|-------------------------------------------------|-----------------|---|-------|--------|
| PAN-1 + Al+3 = Al+3_PAN-1 |    |       |                                                 |                 |   |       |        |
| No                        | t  | I-Str | Medium                                          | Value (Dev)     | W | Tech. | Ref. # |
| 1                         | 25 | 0.06  | Ethanol:Water<br>50:50Wgt<br>HClO <sub>4</sub>  | lgK:12.9(0.2SD) | 4 | MPL   | 906    |
| 2                         | 25 | 0.06  | Ethanol:Water<br>50:50Wgt<br>NaClO <sub>4</sub> | lgK:12.9(0.2SD) | 4 | MPL   | 906    |

| Reaction No. 45629                                |    |       |               |               |   |       |        |
|---------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Al+3 + H+1(4)_Aluminon-5 = Al+3_H+1(4)_Aluminon-5 |    |       |               |               |   |       |        |
| No                                                | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                 | 23 |       |               | lgK:16.6(3SF) | 4 | MSP   | 906    |
| 2                                                 | 25 | 0     | Inf. Dilution | lgK:16.7(3SF) | 1 | ESC   |        |

| Reaction No. 45667                                          |   |       |        |             |   |       |        |
|-------------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| Al+3 + H+1(2)_XylenolOrange-6 = Al+3_H+1(2)_XylenolOrange-6 |   |       |        |             |   |       |        |
| No                                                          | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

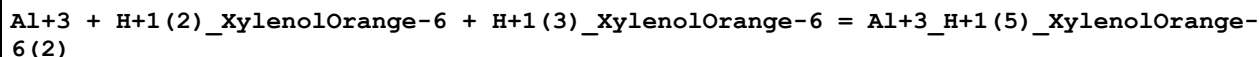
|   |    |   |               |               |   |     |     |
|---|----|---|---------------|---------------|---|-----|-----|
| 1 | 23 |   |               | lgK:6.52(3SF) | 4 | MSP | 906 |
| 2 | 25 | 0 | Inf. Dilution | lgK:6.6(2SF)  | 1 | ESC |     |

## Reaction No. 45668



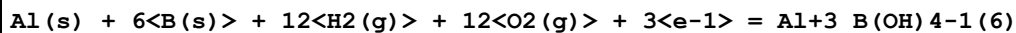
| No | t     | I-Str | Medium             | Value (Dev)    | W | Tech. | Ref. # |
|----|-------|-------|--------------------|----------------|---|-------|--------|
| 1  | 19:21 | 0.1   | NaClO <sub>4</sub> | lgK:13.64(4SF) | 4 | MSP   | 906    |
| 2  | 25    | 0     | Inf. Dilution      | lgK:13.9(3SF)  | 1 | ESC   |        |

## Reaction No. 45669



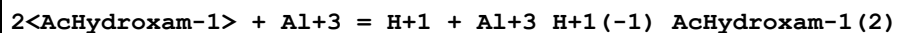
| No | t     | I-Str | Medium             | Value (Dev)    | W | Tech. | Ref. # |
|----|-------|-------|--------------------|----------------|---|-------|--------|
| 1  | 19:21 | 0.1   | NaClO <sub>4</sub> | lgK:11.79(4SF) | 4 | MSP   | 906    |
| 2  | 25    | 0     | Inf. Dilution      | lgK:12.1(3SF)  | 1 | ESC   |        |

## Reaction No. 45841



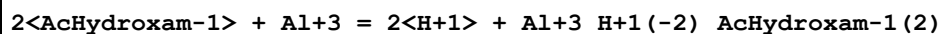
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dG:-1822.50(6SF) | 5 | CRV   | 406    |

## Reaction No. 45904



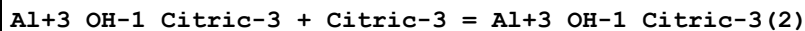
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:10.4(0.03SD) | 5 | MGL   | 2353   |

## Reaction No. 45905



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:1.0(0.05SD) | 5 | MGL   | 2353   |

## Reaction No. 45996



| No | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:1.7(2SF) | 1 | ELC   |        |
| 2  | 37 | 0.15  | Unknown       | lgK:4.2(2SF) | 4 | CRV   | 2556   |

## Reaction No. 46024

| Al+3_H+1_OH-1_Glu-2 + H+1 = Al+3_H+1_Glu-2 + H2O |    |       |         |               |   |       |        |
|--------------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| No                                               | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                | 25 | 0.5   | Unknown | lgK:3.50(3SF) | 4 | CRV   | 2556   |

| Reaction No. 46056                                         |    |       |               |               |   |       |        |
|------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Al+3_OH-1_Salicylic-2(2) + H+1 = Al+3_Salicylic-2(2) + H2O |    |       |               |               |   |       |        |
| No                                                         | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                          | 25 | 0     | Inf. Dilution | lgK:7.1(2SF)  | 1 | ERG   |        |
| 2                                                          | 25 | 0.5   | Unknown       | lgK:7.60(3SF) | 4 | CRV   | 2556   |

| Reaction No. 46106                                      |    |       |        |                  |   |       |        |
|---------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Al+3 + H+1_ChromeAzuroIS-4 = Al+3_ChromeAzuroIS-4 + H+1 |    |       |        |                  |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.1   | KCl    | lgK:2.01(0.02SD) | 2 | MSP   | 4759   |
| 2                                                       | 25 | 0.6   | KCl    | lgK:1.8(0.2SD)   | 2 | MSP   | 4759   |

| Reaction No. 46107                                                      |    |       |        |                  |   |       |        |
|-------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 3<Al+3> + 2<H+1_ChromeAzuroIS-4> = Al+3(3)_ChromeAzuroIS-4(2) + 2<H+1 > |    |       |        |                  |   |       |        |
| No                                                                      | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                       | 25 | 0.1   | KCl    | lgK:12.3(0.08SD) | 0 | MGL   | 4759   |
| 2                                                                       | 25 | 0.6   | KCl    | lgK:12.0(0.05SD) | 5 | MGL   | 4759   |

| Reaction No. 46108                                                      |    |       |        |                  |   |       |        |
|-------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 2<Al+3> + 2<H+1_ChromeAzuroIS-4> = Al+3(2)_H+1_ChromeAzuroIS-4(2) + H+1 |    |       |        |                  |   |       |        |
| No                                                                      | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                       | 25 | 0.1   | KCl    | lgK:12.9(0.07SD) | 4 | MGL   | 4759   |

| Reaction No. 46109                                                              |    |       |        |                 |   |       |        |
|---------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 4<Al+3> + 2<H+1_ChromeAzuroIS-4> = Al+3(4)_H+1(-2)_ChromeAzuroIS-4(2) + 4<H+1 > |    |       |        |                 |   |       |        |
| No                                                                              | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                               | 25 | 0.6   | KCl    | lgK:7.5(0.05SD) | 4 | MGL   | 4759   |

| Reaction No. 46184                                |    |       |               |                  |   |       |        |
|---------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Al+3_H+1(-1)_Gluconic-1 + H+1 = Al+3 + Gluconic-1 |    |       |               |                  |   |       |        |
| No                                                | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                 | 20 | 0.1   | NaNO3         | lgK:0.84(0.02SD) | 5 | MMR   | 4767   |
| 2                                                 | 25 | 0     | Inf. Dilution | lgK:0.9(1SF)     | 1 | ESC   |        |



| Reaction No. 46185                                                                                 |    |       |                   |                   |   |       |        |
|----------------------------------------------------------------------------------------------------|----|-------|-------------------|-------------------|---|-------|--------|
| $\text{Al}+3_{\text{H}+1(-3)}_{\text{Gluconic-1}} + 3\text{H}+1 = \text{Al}+3 + \text{Gluconic-1}$ |    |       |                   |                   |   |       |        |
| No                                                                                                 | t  | I-Str | Medium            | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                  | 20 | 0.1   | NaNO <sub>3</sub> | lgK:10.70(0.02SD) | 0 | MMR   | 4767   |
| 2                                                                                                  | 25 | 0     | Inf. Dilution     | lgK:9.2(2SF)      | 1 | ELC   |        |

| Reaction No. 46193                                                                                       |    |       |                   |                  |   |       |        |
|----------------------------------------------------------------------------------------------------------|----|-------|-------------------|------------------|---|-------|--------|
| $\text{Al}+3_{\text{H}+1(-3)}_{\text{Lactobionic-1}} + 3\text{H}+1 = \text{Al}+3 + \text{Lactobionic-1}$ |    |       |                   |                  |   |       |        |
| No                                                                                                       | t  | I-Str | Medium            | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                        | 20 | 0.1   | NaNO <sub>3</sub> | lgK:12.0(0.03SD) | 5 | MMR   | 4767   |
| 2                                                                                                        | 25 | 0     | Inf. Dilution     | lgK:12.3(3SF)    | 1 | ESC   |        |

| Reaction No. 46353                                                                              |    |       |        |                 |   |       |        |
|-------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{H}+1 + \text{Isocitric-3} + \text{Al}+3 = \text{Al}+3_{\text{H}+1}_{\text{Isocitric-3}}$ |    |       |        |                 |   |       |        |
| No                                                                                              | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                               | 25 | 0.1   | KCl    | lgK:9.6(0.05SD) | 5 | MGL   | 72     |

| Reaction No. 46354                                                    |    |       |        |                  |   |       |        |
|-----------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $\text{Isocitric-3} + \text{Al}+3 = \text{Al}+3_{\text{Isocitric-3}}$ |    |       |        |                  |   |       |        |
| No                                                                    | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                     | 25 | 0.1   | KCl    | lgK:6.90(0.02SD) | 5 | MGL   | 72     |

| Reaction No. 46355                                                                                  |    |       |        |                 |   |       |        |
|-----------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{Isocitric-3} + \text{Al}+3 = \text{H}+1 + \text{Al}+3_{\text{H}+1(-1)}_{\text{Isocitric-3}}$ |    |       |        |                 |   |       |        |
| No                                                                                                  | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                   | 25 | 0.1   | KCl    | lgK:3.1(0.04SD) | 5 | MGL   | 72     |

| Reaction No. 46356                                                                                           |    |       |        |                 |   |       |        |
|--------------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $2\text{Isocitric-3} + 2\text{Al}+3 = 3\text{H}+1 + \text{Al}+3(2)_{\text{H}+1(-3)}_{\text{Isocitric-3}(2)}$ |    |       |        |                 |   |       |        |
| No                                                                                                           | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                            | 25 | 0.1   | KCl    | lgK:4.1(0.04SD) | 5 | MGL   | 72     |

| Reaction No. 46357                                                                                           |    |       |        |                  |   |       |        |
|--------------------------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $3\text{Isocitric-3} + 3\text{Al}+3 = 4\text{H}+1 + \text{Al}+3(3)_{\text{H}+1(-4)}_{\text{Isocitric-3}(3)}$ |    |       |        |                  |   |       |        |
| No                                                                                                           | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                            | 25 | 0.1   | KCl    | lgK:10.4(0.06SD) | 5 | MGL   | 72     |

| Reaction No. 46486 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

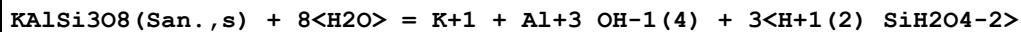
| H+1 + PO4-3 + Al+3 = Al+3_H+1_PO4-3 |    |       |         |                    |   |       |        |
|-------------------------------------|----|-------|---------|--------------------|---|-------|--------|
| No                                  | t  | I-Str | Medium  | Value (Dev)        | W | Tech. | Ref. # |
| 1                                   | 25 | 0     | UCT_Sea | lgK:19.75 (0.01SD) | 2 | EDH   | 5390   |
| 2                                   | 25 | 0.1   | UCT_Sea | lgK:18.04 (0.01SD) | 2 | EDH   | 5390   |
| 3                                   | 25 | 0.15  | NaCl/m  | lgK:17.00 (4SF)    | 3 | MGL   | 29916  |
| 4                                   | 25 | 0.2   | KCl     | lgK:17.60 (4SF)    | 3 | MSC   | 7048   |
| 5                                   | 25 | 0.2   | UCT_Sea | lgK:17.66 (0.01SD) | 2 | EDH   | 5390   |
| 6                                   | 25 | 0.3   | UCT_Sea | lgK:17.45 (0.01SD) | 2 | EDH   | 5390   |
| 7                                   | 25 | 0.4   | UCT_Sea | lgK:17.29 (0.01SD) | 2 | EDH   | 5390   |
| 8                                   | 25 | 0.5   | UCT_Sea | lgK:17.17 (0.01SD) | 2 | EDH   | 5390   |
| 9                                   | 25 | 0.6   | UCT_Sea | lgK:17.08 (0.01SD) | 2 | EDH   | 5390   |
| 10                                  | 25 | 0.7   | UCT_Sea | lgK:17.03 (0.01SD) | 2 | EDH   | 5390   |
| 11                                  | 37 | 0.15  | NaCl    | lgK:19.1 (0.2SD)   | 0 | MGL   | 922    |

| Reaction No. 46487                        |    |       |         |                    |   |       |        |
|-------------------------------------------|----|-------|---------|--------------------|---|-------|--------|
| 2<H+1> + PO4-3 + Al+3 = Al+3_H+1(2)_PO4-3 |    |       |         |                    |   |       |        |
| No                                        | t  | I-Str | Medium  | Value (Dev)        | W | Tech. | Ref. # |
| 1                                         | 25 | 0     | UCT_Sea | lgK:22.65 (0.01SD) | 4 | EDH   | 5390   |
| 2                                         | 25 | 0.1   | UCT_Sea | lgK:21.02 (0.01SD) | 4 | EDH   | 5390   |
| 3                                         | 25 | 0.2   | UCT_Sea | lgK:20.68 (0.01SD) | 4 | EDH   | 5390   |
| 4                                         | 25 | 0.3   | UCT_Sea | lgK:20.47 (0.01SD) | 4 | EDH   | 5390   |
| 5                                         | 25 | 0.4   | UCT_Sea | lgK:20.32 (0.01SD) | 4 | EDH   | 5390   |
| 6                                         | 25 | 0.5   | UCT_Sea | lgK:20.20 (0.01SD) | 4 | EDH   | 5390   |
| 7                                         | 25 | 0.6   | UCT_Sea | lgK:20.11 (0.01SD) | 4 | EDH   | 5390   |
| 8                                         | 25 | 0.7   | UCT_Sea | lgK:20.05 (0.01SD) | 4 | EDH   | 5390   |
| 9                                         | 37 | 0.15  | NaCl    | lgK:22.2 (0.04SD)  | 0 | MGL   | 922    |
| 10                                        | 37 | 0.15  | NaCl    | lgK:21.1 (3SF)     | 6 | MIS   | 5983   |

| Reaction No. 49887                                                                |    |       |               |                   |   |       |        |
|-----------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| CaO.Al2O3.2SiO2(s) + 8<H2O> = Ca+2 + 2<Al+3_OH-1(4)> + 2<H+1(2)_SiH2O4-2 >        |    |       |               |                   |   |       |        |
| No                                                                                | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                 | 5  | 0     | Inf. Dilution | lgK:-21.071 (5SF) | 0 | CRV   | 8798   |
| 2                                                                                 | 25 | 0     | Inf. Dilution | lgK:-19.1 (3SF)   | 1 | ELC   |        |
| 3                                                                                 | 25 | 0     | Inf. Dilution | lgK:-20.484 (5SF) | 0 | CRV   | 8798   |
| 4                                                                                 | 25 | 0     | Inf. Dilution | lgK:-54.579 (5SF) | 0 | CRV   | 25435  |
| Note (4): Given as log K, v. prob. should be ln K but -54.6/2.3 = -23.7 still bad |    |       |               |                   |   |       |        |

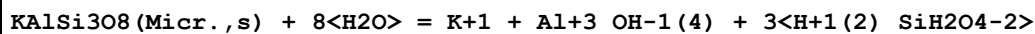
|    |     |   |               |                   |   |     |      |
|----|-----|---|---------------|-------------------|---|-----|------|
| 5  | 50  | 0 | Inf. Dilution | lgK:-19.857 (5SF) | 0 | CRV | 8798 |
| 6  | 100 | 0 | Inf. Dilution | lgK:-18.960 (5SF) | 0 | CRV | 8798 |
| 7  | 150 | 0 | Inf. Dilution | lgK:-18.514 (5SF) | 0 | CRV | 8798 |
| 8  | 200 | 0 | Inf. Dilution | lgK:-18.485 (5SF) | 0 | CRV | 8798 |
| 9  | 250 | 0 | Inf. Dilution | lgK:-18.839 (5SF) | 0 | CRV | 8798 |
| 10 | 300 | 0 | Inf. Dilution | lgK:-19.547 (5SF) | 0 | CRV | 8798 |
| 11 | 350 | 0 | Inf. Dilution | lgK:-20.586 (5SF) | 0 | CRV | 8798 |

## Reaction No. 49888



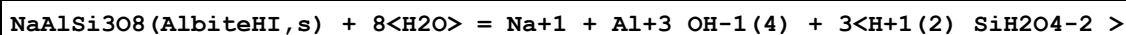
| No | t   | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|-----|-------|---------------|-------------------|---|-------|--------|
| 1  | 5   | 0     | Inf. Dilution | lgK:-23.764 (5SF) | 7 | CRV   | 8798   |
| 2  | 25  | 0     | Inf. Dilution | lgK:-22.338 (5SF) | 7 | CRV   | 8798   |
| 3  | 50  | 0     | Inf. Dilution | lgK:-20.774 (5SF) | 7 | CRV   | 8798   |
| 4  | 100 | 0     | Inf. Dilution | lgK:-18.267 (5SF) | 7 | CRV   | 8798   |
| 5  | 150 | 0     | Inf. Dilution | lgK:-16.441 (5SF) | 7 | CRV   | 8798   |
| 6  | 200 | 0     | Inf. Dilution | lgK:-15.166 (5SF) | 7 | CRV   | 8798   |
| 7  | 250 | 0     | Inf. Dilution | lgK:-14.351 (5SF) | 7 | CRV   | 8798   |
| 8  | 300 | 0     | Inf. Dilution | lgK:-13.930 (5SF) | 7 | CRV   | 8798   |
| 9  | 350 | 0     | Inf. Dilution | lgK:-13.855 (5SF) | 7 | CRV   | 8798   |

## Reaction No. 49901



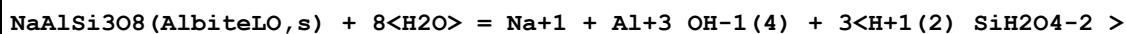
| No | t   | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|-----|-------|---------------|-------------------|---|-------|--------|
| 1  | 5   | 0     | Inf. Dilution | lgK:-24.446 (5SF) | 0 | CRV   | 8798   |
| 2  | 25  | 0     | Inf. Dilution | lgK:-21.6 (3SF)   | 1 | ELC   |        |
| 3  | 25  | 0     | Inf. Dilution | lgK:-22.912 (5SF) | 0 | CRV   | 8798   |
| 4  | 50  | 0     | Inf. Dilution | lgK:-21.231 (5SF) | 0 | CRV   | 8798   |
| 5  | 100 | 0     | Inf. Dilution | lgK:-18.535 (5SF) | 2 | CRV   | 8798   |
| 6  | 150 | 0     | Inf. Dilution | lgK:-16.562 (5SF) | 0 | CRV   | 8798   |
| 7  | 200 | 0     | Inf. Dilution | lgK:-15.170 (5SF) | 0 | CRV   | 8798   |
| 8  | 250 | 0     | Inf. Dilution | lgK:-14.260 (5SF) | 0 | CRV   | 8798   |
| 9  | 300 | 0     | Inf. Dilution | lgK:-13.759 (5SF) | 0 | CRV   | 8798   |
| 10 | 350 | 0     | Inf. Dilution | lgK:-13.616 (5SF) | 0 | CRV   | 8798   |

## Reaction No. 50248



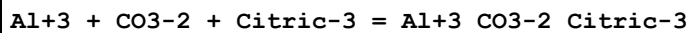
| No | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|-----|-------|---------------|------------------|---|-------|--------|
| 1  | 5   | 0     | Inf. Dilution | lgK:-19.816(5SF) | 7 | CRV   | 8798   |
| 2  | 25  | 0     | Inf. Dilution | lgK:-18.779(5SF) | 7 | CRV   | 8798   |
| 3  | 50  | 0     | Inf. Dilution | lgK:-17.622(5SF) | 7 | CRV   | 8798   |
| 4  | 100 | 0     | Inf. Dilution | lgK:-15.737(5SF) | 7 | CRV   | 8798   |
| 5  | 150 | 0     | Inf. Dilution | lgK:-14.359(5SF) | 7 | CRV   | 8798   |
| 6  | 200 | 0     | Inf. Dilution | lgK:-13.417(5SF) | 7 | CRV   | 8798   |
| 7  | 250 | 0     | Inf. Dilution | lgK:-12.858(5SF) | 7 | CRV   | 8798   |
| 8  | 300 | 0     | Inf. Dilution | lgK:-12.640(5SF) | 7 | CRV   | 8798   |
| 9  | 350 | 0     | Inf. Dilution | lgK:-12.730(5SF) | 7 | CRV   | 8798   |

## Reaction No. 50272



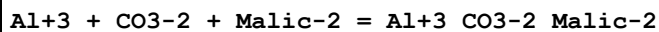
| No | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|-----|-------|---------------|------------------|---|-------|--------|
| 1  | 5   | 0     | Inf. Dilution | lgK:-21.337(5SF) | 7 | CRV   | 8798   |
| 2  | 25  | 0     | Inf. Dilution | lgK:-20.179(5SF) | 7 | CRV   | 8798   |
| 3  | 50  | 0     | Inf. Dilution | lgK:-18.848(5SF) | 7 | CRV   | 8798   |
| 4  | 100 | 0     | Inf. Dilution | lgK:-16.686(5SF) | 7 | CRV   | 8798   |
| 5  | 150 | 0     | Inf. Dilution | lgK:-15.096(5SF) | 7 | CRV   | 8798   |
| 6  | 200 | 0     | Inf. Dilution | lgK:-13.987(5SF) | 7 | CRV   | 8798   |
| 7  | 250 | 0     | Inf. Dilution | lgK:-13.293(5SF) | 7 | CRV   | 8798   |
| 8  | 300 | 0     | Inf. Dilution | lgK:-12.963(5SF) | 7 | CRV   | 8798   |
| 9  | 350 | 0     | Inf. Dilution | lgK:-12.959(5SF) | 7 | CRV   | 8798   |

## Reaction No. 51588



| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:13.2(3SF)     | 2 | EAE   |        |
| 2  | 37 | 0.15  | Blood Plasma  | lgK:11.03(0.01SD) | 3 | ETS   | 94     |

## Reaction No. 51589



| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:10.2(3SF)    | 2 | EAE   |        |
| 2  | 37 | 0.15  | Blood Plasma  | lgK:8.31(0.01SD) | 3 | ETS   | 94     |

## Reaction No. 51590

| Al+3 + Citric-3 + Malic-2 = Al+3_Malic-2_Citric-3 |    |       |               |                   |   |       |        |
|---------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| No                                                | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                 | 25 | 0     | Inf. Dilution | lgK:12.8(3SF)     | 2 | EAE   |        |
| 2                                                 | 37 | 0.15  | Blood Plasma  | lgK:10.64(0.01SD) | 3 | ETS   | 94     |

| Reaction No. 51591                            |    |       |               |                  |   |       |        |
|-----------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Al+3 + CO3-2 + Oxalic-2 = Al+3_CO3-2_Oxalic-2 |    |       |               |                  |   |       |        |
| No                                            | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                             | 25 | 0     | Inf. Dilution | lgK:11.2(3SF)    | 2 | EAE   |        |
| 2                                             | 37 | 0.15  | Blood Plasma  | lgK:9.25(0.01SD) | 3 | ETS   | 94     |

| Reaction No. 51592                                  |    |       |               |                   |   |       |        |
|-----------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| Al+3 + Citric-3 + Oxalic-2 = Al+3_Oxalic-2_Citric-3 |    |       |               |                   |   |       |        |
| No                                                  | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                   | 25 | 0     | Inf. Dilution | lgK:13.7(3SF)     | 2 | EAE   |        |
| 2                                                   | 37 | 0.15  | Blood Plasma  | lgK:11.58(0.01SD) | 3 | ETS   | 94     |

| Reaction No. 51593                                |    |       |               |                  |   |       |        |
|---------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Oxalic-2 + Malic-2 + Al+3 = Al+3_Malic-2_Oxalic-2 |    |       |               |                  |   |       |        |
| No                                                | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                 | 25 | 0     | Inf. Dilution | lgK:10.8(4SF)    | 1 | EOR   |        |
| 2                                                 | 25 | 0     | Inf. Dilution | lgK:10.8(3SF)    | 2 | EAE   |        |
| 3                                                 | 37 | 0.15  | Blood Plasma  | lgK:8.86(0.01SD) | 3 | ETS   | 94     |

| Reaction No. 51594                                  |    |       |               |                   |   |       |        |
|-----------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| Al+3 + CO3-2 + Salicylic-2 = Al+3_CO3-2_Salicylic-2 |    |       |               |                   |   |       |        |
| No                                                  | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                   | 25 | 0     | Inf. Dilution | lgK:17.4(3SF)     | 2 | EAE   |        |
| 2                                                   | 37 | 0.15  | Blood Plasma  | lgK:15.50(0.01SD) | 3 | ETS   | 94     |

| Reaction No. 51595                                        |    |       |               |                   |   |       |        |
|-----------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| Al+3 + Citric-3 + Salicylic-2 = Al+3_Salicylic-2_Citric-3 |    |       |               |                   |   |       |        |
| No                                                        | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                         | 25 | 0     | Inf. Dilution | lgK:20.0(3SF)     | 2 | EAE   |        |
| 2                                                         | 37 | 0.15  | Blood Plasma  | lgK:17.83(0.01SD) | 3 | ETS   | 94     |

| Reaction No. 51596 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| Al+3 + Malic-2 + Salicylic-2 = Al+3_Malic-2_Salicylic-2 |    |       |               |                   |   |       |        |
|---------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| No                                                      | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                       | 25 | 0     | Inf. Dilution | lgK:17.0(3SF)     | 2 | EAE   |        |
| 2                                                       | 37 | 0.15  | Blood Plasma  | lgK:15.11(0.01SD) | 3 | ETS   | 94     |

| Reaction No. 51597                                        |    |       |               |                   |   |       |        |
|-----------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| Al+3 + Oxalic-2 + Salicylic-2 = Al+3_Oxalic-2_Salicylic-2 |    |       |               |                   |   |       |        |
| No                                                        | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                         | 25 | 0     | Inf. Dilution | lgK:18.0(3SF)     | 2 | EAE   |        |
| 2                                                         | 37 | 0.15  | Blood Plasma  | lgK:16.05(0.01SD) | 3 | ETS   | 94     |

| Reaction No. 52285                                 |    |       |         |                   |   |       |        |
|----------------------------------------------------|----|-------|---------|-------------------|---|-------|--------|
| Al+3 + Benzoic-1 + H2O = Al+3_Benzoic-1_OH-1 + H+1 |    |       |         |                   |   |       |        |
| No                                                 | t  | I-Str | Medium  | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                  | 25 | 0     | UCT_Sea | lgK:-0.78(0.01SD) | 4 | EDH   | 5390   |
| 2                                                  | 25 | 0.1   | UCT_Sea | lgK:-1.40(0.01SD) | 4 | EDH   | 5390   |
| 3                                                  | 25 | 0.2   | UCT_Sea | lgK:-1.52(0.01SD) | 4 | EDH   | 5390   |
| 4                                                  | 25 | 0.3   | UCT_Sea | lgK:-1.59(0.01SD) | 4 | EDH   | 5390   |
| 5                                                  | 25 | 0.4   | UCT_Sea | lgK:-1.62(0.01SD) | 4 | EDH   | 5390   |
| 6                                                  | 25 | 0.5   | UCT_Sea | lgK:-1.65(0.01SD) | 4 | EDH   | 5390   |
| 7                                                  | 25 | 0.6   | UCT_Sea | lgK:-1.68(0.01SD) | 4 | EDH   | 5390   |
| 8                                                  | 25 | 0.7   | UCT_Sea | lgK:-1.70(0.01SD) | 4 | EDH   | 5390   |

| Reaction No. 52286                        |    |       |         |                   |   |       |        |
|-------------------------------------------|----|-------|---------|-------------------|---|-------|--------|
| H+1 + Al+3 + 2<Cat-2> = Al+3_H+1_Cat-2(2) |    |       |         |                   |   |       |        |
| No                                        | t  | I-Str | Medium  | Value (Dev)       | W | Tech. | Ref. # |
| 1                                         | 25 | 0     | UCT_Sea | lgK:38.45(0.01SD) | 4 | EDH   | 5390   |
| 2                                         | 25 | 0.1   | UCT_Sea | lgK:36.15(0.01SD) | 4 | EDH   | 5390   |
| 3                                         | 25 | 0.2   | UCT_Sea | lgK:35.68(0.01SD) | 4 | EDH   | 5390   |
| 4                                         | 25 | 0.3   | UCT_Sea | lgK:35.45(0.01SD) | 4 | EDH   | 5390   |
| 5                                         | 25 | 0.4   | UCT_Sea | lgK:35.28(0.01SD) | 4 | EDH   | 5390   |
| 6                                         | 25 | 0.5   | UCT_Sea | lgK:35.20(0.01SD) | 4 | EDH   | 5390   |
| 7                                         | 25 | 0.6   | UCT_Sea | lgK:35.12(0.01SD) | 4 | EDH   | 5390   |
| 8                                         | 25 | 0.7   | UCT_Sea | lgK:35.11(0.01SD) | 4 | EDH   | 5390   |

| Reaction No. 52287                                           |  |  |  |  |  |  |  |
|--------------------------------------------------------------|--|--|--|--|--|--|--|
| Al+3 + 2<Salicylic-2> + H2O = Al+3_OH-1_Salicylic-2(2) + H+1 |  |  |  |  |  |  |  |

| No                                         | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|--------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| 1                                          | 25 | 0     | Inf. Dilution | lgK:17.6(3SF)     | 1 | ERG   |        |
| 2                                          | 25 | 0     | UCT_Sea       | lgK:15.69(0.01SD) | 0 | EDH   | 5390   |
| Note (2): Low wgt for internal consistency |    |       |               |                   |   |       |        |
| 3                                          | 25 | 0.1   | UCT_Sea       | lgK:14.70(0.01SD) | 0 | EDH   | 5390   |
| 4                                          | 25 | 0.2   | KCl           | lgK:16.60(0.02SD) | 5 | MGL   | 2470   |
| 5                                          | 25 | 0.2   | UCT_Sea       | lgK:14.44(0.01SD) | 0 | EDH   | 5390   |
| 6                                          | 25 | 0.3   | UCT_Sea       | lgK:14.32(0.01SD) | 0 | EDH   | 5390   |
| 7                                          | 25 | 0.4   | UCT_Sea       | lgK:14.25(0.01SD) | 0 | EDH   | 5390   |
| 8                                          | 25 | 0.5   | UCT_Sea       | lgK:14.18(0.01SD) | 0 | EDH   | 5390   |
| 9                                          | 25 | 0.6   | UCT_Sea       | lgK:14.12(0.01SD) | 0 | EDH   | 5390   |
| 10                                         | 25 | 0.7   | UCT_Sea       | lgK:14.08(0.01SD) | 0 | EDH   | 5390   |

| Reaction No. 52288                            |    |       |         |                  |   |       |        |
|-----------------------------------------------|----|-------|---------|------------------|---|-------|--------|
| H+1 + Succinic-2 + Al+3 = Al+3_H+1_Succinic-2 |    |       |         |                  |   |       |        |
| No                                            | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
| 1                                             | 25 | 0     | UCT_Sea | lgK:7.62(0.01SD) | 4 | EDH   | 5390   |
| 2                                             | 25 | 0.1   | UCT_Sea | lgK:6.78(0.01SD) | 4 | EDH   | 5390   |
| 3                                             | 25 | 0.2   | UCT_Sea | lgK:6.64(0.01SD) | 4 | EDH   | 5390   |
| 4                                             | 25 | 0.3   | UCT_Sea | lgK:6.57(0.01SD) | 4 | EDH   | 5390   |
| 5                                             | 25 | 0.4   | UCT_Sea | lgK:6.53(0.01SD) | 4 | EDH   | 5390   |
| 6                                             | 25 | 0.5   | NaClO4  | lgK:1.42(3SF)    | 0 | ENS   | 6953   |
| 7                                             | 25 | 0.5   | UCT_Sea | lgK:6.51(0.01SD) | 4 | EDH   | 5390   |
| 8                                             | 25 | 0.6   | UCT_Sea | lgK:6.51(0.01SD) | 4 | EDH   | 5390   |
| 9                                             | 25 | 0.7   | UCT_Sea | lgK:6.50(0.01SD) | 4 | EDH   | 5390   |
| 10                                            | 37 | 0.15  | NaCl    | lgK:7.3(0.08SD)  | 5 | MGL   | 5979   |

| Reaction No. 52579                                               |    |       |        |                 |   |       |        |
|------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2<H+1> + Al+3_H+1(4)_PyroCatV-4(2) = Al+3 + 2<H+1(3)_PyroCatV-4> |    |       |        |                 |   |       |        |
| No                                                               | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                | 25 | 0.1   | KCl    | lgK:1.0(0.03SD) | 5 | MPT   | 349    |

| Reaction No. 52582                                               |    |       |        |                 |   |       |        |
|------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 3<H+1> + Al+3_H+1(6)_PyroCatV-4(3) = Al+3 + 3<H+1(3)_PyroCatV-4> |    |       |        |                 |   |       |        |
| No                                                               | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                | 25 | 0.1   | KCl    | lgK:2.6(0.06SD) | 5 | MPT   | 349    |

| Reaction No. 52583                                                                                |    |       |        |                  |   |       |        |
|---------------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $4\text{<H+1>} + \text{Al+3\_H+1(2)\_PyroCatV-4(2)} = \text{Al+3} + 2\text{<H+1(3)\_PyroCatV-4>}$ |    |       |        |                  |   |       |        |
| No                                                                                                | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                 | 25 | 0.1   | KCl    | lgK:10.2(0.09SD) | 5 | MPT   | 349    |

| Reaction No. 52584                                                                                |    |       |               |                 |   |       |        |
|---------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $5\text{<H+1>} + \text{Al+3\_H+1(4)\_PyroCatV-4(3)} = \text{Al+3} + 3\text{<H+1(3)\_PyroCatV-4>}$ |    |       |               |                 |   |       |        |
| No                                                                                                | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                 | 25 | 0     | Inf. Dilution | lgK:14.9(3SF)   | 1 | EES   |        |
| 2                                                                                                 | 25 | 0.1   | KCl           | lgK:13.0(0.1SD) | 4 | MPT   | 349    |

| Reaction No. 52585                                                                                |    |       |        |                 |   |       |        |
|---------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $6\text{<H+1>} + \text{Al+3\_H+1(3)\_PyroCatV-4(3)} = \text{Al+3} + 3\text{<H+1(3)\_PyroCatV-4>}$ |    |       |        |                 |   |       |        |
| No                                                                                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                 | 25 | 0.1   | KCl    | lgK:21.1(0.1SD) | 4 | MPT   | 349    |

| Reaction No. 52586                                                                                |    |       |        |                |   |       |        |
|---------------------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $7\text{<H+1>} + \text{Al+3\_H+1(2)\_PyroCatV-4(3)} = \text{Al+3} + 3\text{<H+1(3)\_PyroCatV-4>}$ |    |       |        |                |   |       |        |
| No                                                                                                | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                 | 25 | 0.1   | KCl    | lgK:30.(0.3SD) | 4 | MPT   | 349    |

| Reaction No. 52587                                                                             |    |       |        |                |   |       |        |
|------------------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $8\text{<H+1>} + \text{Al+3\_H+1\_PyroCatV-4(3)} = \text{Al+3} + 3\text{<H+1(3)\_PyroCatV-4>}$ |    |       |        |                |   |       |        |
| No                                                                                             | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                              | 25 | 0.1   | KCl    | lgK:41.(0.5SD) | 4 | MPT   | 349    |

| Reaction No. 52588                                                                        |    |       |               |               |   |       |        |
|-------------------------------------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $9\text{<H+1>} + \text{Al+3\_PyroCatV-4(3)} = \text{Al+3} + 3\text{<H+1(3)\_PyroCatV-4>}$ |    |       |               |               |   |       |        |
| No                                                                                        | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                         | 25 | 0     | Inf. Dilution | lgK:30.0(3SF) | 1 | ELC   |        |
| 2                                                                                         | 25 | 0.1   | KCl           | lgK:52.0(3SF) | 0 | MPT   | 349    |
| Note (2): Estimated by authors                                                            |    |       |               |               |   |       |        |

| Reaction No. 52589                                                                                      |    |       |        |                 |   |       |        |
|---------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $6\text{<H+1>} + \text{Al+3(3)\_H+1(3)\_PyroCatV-4(3)} = 3\text{<Al+3>} + 3\text{<H+1(3)\_PyroCatV-4>}$ |    |       |        |                 |   |       |        |
| No                                                                                                      | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                       | 25 | 0.1   | KCl    | lgK:5.1(0.05SD) | 5 | MPT   | 349    |



| Reaction No. 52590                                                                                                                |    |       |        |                 |   |       |        |
|-----------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $16\text{<H+1>} + \text{Al+3(6)}_{\text{H+1(2)}}_{\text{PyroCatV-4(6)}} = 6\text{<Al+3>} + 6\text{<H+1(3)}_{\text{PyroCatV-4}} >$ |    |       |        |                 |   |       |        |
| No                                                                                                                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                                 | 25 | 0.1   | KCl    | lgK:23.2(0.2SD) | 5 | MPT   | 349    |

| Reaction No. 52591                                            |    |       |        |                  |   |       |        |
|---------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $2\text{<Tris>} + \text{Al+3} = \text{Al+3}_{\text{Tris(2)}}$ |    |       |        |                  |   |       |        |
| No                                                            | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                             | 25 | 0.5   | NaClO4 | lgK:11.0(0.07SD) | 5 | MPT   | 510    |
| 2                                                             | 25 | 1     | NaClO4 | lgK:10.7(0.05SD) | 5 | MPT   | 510    |
| 3                                                             | 25 | 2     | NaClO4 | lgK:11.1(0.05SD) | 5 | MPT   | 510    |
| 4                                                             | 25 | 3     | NaClO4 | lgK:11.8(0.05SD) | 5 | MPT   | 510    |

| Reaction No. 52592                                                                        |    |       |               |                  |   |       |        |
|-------------------------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| $\text{Al+3} + 2\text{<Tris>} + \text{OH-1} = \text{Al+3}_{\text{Tris(2)}}_{\text{OH-1}}$ |    |       |               |                  |   |       |        |
| No                                                                                        | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                         | 25 | 0     | Inf. Dilution | lgK:20.4(3SF)    | 1 | EDH   |        |
| 2                                                                                         | 25 | 0.5   | NaClO4        | lgK:20.0(0.04SD) | 4 | MPT   | 510    |
| 3                                                                                         | 25 | 1     | NaClO4        | lgK:19.8(0.03SD) | 2 | MPT   | 510    |
| 4                                                                                         | 25 | 2     | NaClO4        | lgK:20.3(0.03SD) | 0 | MPT   | 510    |
| 5                                                                                         | 25 | 3     | NaClO4        | lgK:21.6(0.08SD) | 5 | MPT   | 510    |

| Reaction No. 52593                                                                              |    |       |               |                  |   |       |        |
|-------------------------------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| $\text{Al+3} + 2\text{<Tris>} + 2\text{<OH-1>} = \text{Al+3}_{\text{Tris(2)}}_{\text{OH-1(2)}}$ |    |       |               |                  |   |       |        |
| No                                                                                              | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                               | 25 | 0     | Inf. Dilution | lgK:26.9(3SF)    | 1 | EDH   |        |
| 2                                                                                               | 25 | 0.5   | NaClO4        | lgK:25.9(0.04SD) | 4 | MPT   | 510    |
| 3                                                                                               | 25 | 1     | NaClO4        | lgK:25.7(0.04SD) | 3 | MPT   | 510    |
| 4                                                                                               | 25 | 2     | NaClO4        | lgK:26.1(0.04SD) | 0 | MPT   | 510    |
| 5                                                                                               | 25 | 3     | NaClO4        | lgK:27.4(0.08SD) | 5 | MPT   | 510    |

| Reaction No. 52679                                                                                                                |    |       |        |                |   |       |        |
|-----------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Al+3(2)}_{\text{Br-1(2)}}_{\text{Cl-1(6)}} = \text{Al+3}_{\text{Cl-1(4)}} + \text{Al+3}_{\text{Br-1(2)}}_{\text{Cl-1(2)}}$ |    |       |        |                |   |       |        |
| No                                                                                                                                | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                                                 | 26 |       |        | lgK:-0.5(1SF)  | 0 | MMR   | 5398   |
| 2                                                                                                                                 | 26 |       | MeCN   | lgK:-0.35(2SF) | 4 | MMR   | 6765   |
| 3                                                                                                                                 | 26 |       | MeCN   | dH:0.04(1SF)   | 4 | MTD   | 6765   |

| Reaction No. 52680                                                                                                                           |    |       |        |                 |   |       |        |
|----------------------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{Al}+3(2)_{\text{Br}-1(4)}_{\text{Cl}-1(4)} = \text{Al}+3_{\text{Br}-1}_{\text{Cl}-1(3)} + \text{Al}+3_{\text{Br}-1(3)}_{\text{Cl}-1}$ |    |       |        |                 |   |       |        |
| No                                                                                                                                           | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                                            | 26 |       | MeCN   | lgK:-0.8(0.1SD) | 4 | MMR   | 6765   |
| 2                                                                                                                                            | 26 |       | MeCN   | dH:-0.35(2SF)   | 4 | MTD   | 6765   |

| Reaction No. 52681                                                                                                                |    |       |        |                  |   |       |        |
|-----------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $\text{Al}+3(2)_{\text{Br}-1(6)}_{\text{Cl}-1(2)} = \text{Al}+3_{\text{Br}-1(2)}_{\text{Cl}-1(2)} + \text{Al}+3_{\text{Br}-1(4)}$ |    |       |        |                  |   |       |        |
| No                                                                                                                                | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                                                 | 26 |       | MeCN   | lgK:-0.4(0.04SD) | 4 | MMR   | 6765   |
| 2                                                                                                                                 | 26 |       | MeCN   | dH:-0.07(1SF)    | 4 | MTD   | 6765   |

| Reaction No. 52682                                                                                             |     |       |        |               |   |       |        |
|----------------------------------------------------------------------------------------------------------------|-----|-------|--------|---------------|---|-------|--------|
| $2<\text{Al}+3_{\text{Br}-1(4)}> + \text{F}-1 = \text{Al}+3(2)_{\text{Br}-1(6)}_{\text{F}-1} + 2<\text{Br}-1>$ |     |       |        |               |   |       |        |
| No                                                                                                             | t   | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                                              | 210 |       |        | lgK:-4.0(2SF) | 0 | MEF   | 5398   |

| Reaction No. 52683                                                                                             |     |       |        |               |   |       |        |
|----------------------------------------------------------------------------------------------------------------|-----|-------|--------|---------------|---|-------|--------|
| $2<\text{Al}+3_{\text{Cl}-1(4)}> + \text{F}-1 = \text{Al}+3(2)_{\text{Cl}-1(6)}_{\text{F}-1} + 2<\text{Cl}-1>$ |     |       |        |               |   |       |        |
| No                                                                                                             | t   | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                                              | 210 |       |        | lgK:-3.5(2SF) | 0 | MEF   | 5398   |

| Reaction No. 52779                                                                  |    |       |               |               |   |       |        |
|-------------------------------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $\text{Al}+3 + \text{H}+1_{\text{SO4}-2} = \text{Al}+3_{\text{H}+1_{\text{SO4}-2}}$ |    |       |               |               |   |       |        |
| No                                                                                  | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                   | 25 | 0     | Inf. Dilution | lgK:0.46(2SF) | 4 | ENS   | 6496   |
| 2                                                                                   | 25 |       |               | lgK:0.5(1SF)  | 0 | MMR   | 5398   |

| Reaction No. 52902                                                                |         |       |        |                |   |       |        |
|-----------------------------------------------------------------------------------|---------|-------|--------|----------------|---|-------|--------|
| $2<\text{Al}+3_{\text{Br}-1(4)}> = \text{Al}+3(2)_{\text{Br}-1(7)} + \text{Br}-1$ |         |       |        |                |   |       |        |
| No                                                                                | t       | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                 | 210     |       |        | lgK:-4.40(3SF) | 0 | MEF   | 5398   |
| 2                                                                                 | 225     |       |        | lgK:-4.30(3SF) | 0 | MEF   | 5398   |
| 3                                                                                 | 240     |       |        | lgK:-4.15(3SF) | 0 | MEF   | 5398   |
| 4                                                                                 | 210:240 |       |        | dH:4.8(2SF)    | 0 | MTD   | 5398   |

| Reaction No. 53076 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| Al+3_C1-1 + Cl-1 = Al+3_C1-1(2) |    |       |        |              |   |       |        |
|---------------------------------|----|-------|--------|--------------|---|-------|--------|
| No                              | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                               | 23 |       |        | lgK:0.5(1SF) | 0 | MSC   | 5398   |

| Reaction No. 53077                 |    |       |        |               |   |       |        |
|------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Al+3_C1-1(2) + Cl-1 = Al+3_C1-1(3) |    |       |        |               |   |       |        |
| No                                 | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                  | 23 |       |        | lgK:-2.7(2SF) | 0 | MSC   | 5398   |

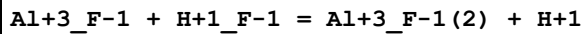
| Reaction No. 53078                       |         |       |        |                |   |       |        |
|------------------------------------------|---------|-------|--------|----------------|---|-------|--------|
| 2<Al+3_C1-1(4)> = Al+3(2)_C1-1(7) + Cl-1 |         |       |        |                |   |       |        |
| No                                       | t       | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                        | 175     |       |        | lgK:-7.1(2SF)  | 0 | MEF   | 5398   |
| 2                                        | 175     |       |        | lgK:-5.50(3SF) | 0 | MEF   | 5398   |
| 3                                        | 190     |       |        | lgK:-5.32(3SF) | 0 | MEF   | 5398   |
| 4                                        | 210     |       |        | lgK:-5.00(3SF) | 0 | MEF   | 5398   |
| 5                                        | 225     |       |        | lgK:-4.87(3SF) | 0 | MEF   | 5398   |
| 6                                        | 240     |       |        | lgK:-4.75(3SF) | 0 | MEF   | 5398   |
| 7                                        | 250     |       |        | lgK:-6.3(2SF)  | 0 | MEF   | 5398   |
| 8                                        | 289     |       |        | lgK:-3.52(3SF) | 0 | MGC   | 5398   |
| 9                                        | 300     |       |        | lgK:-5.7(2SF)  | 0 | MEF   | 5398   |
| 10                                       | 300     |       |        | lgK:-7.83(3SF) | 0 | MEF   | 5398   |
| 11                                       | 350     |       |        | lgK:-5.3(2SF)  | 0 | MEF   | 5398   |
| 12                                       | 400     |       |        | lgK:-5.0(2SF)  | 0 | MEF   | 5398   |
| 13                                       | 450     |       |        | lgK:-6.12(3SF) | 0 | MEF   | 5398   |
| 14                                       | 450     |       |        | lgK:-6.24(3SF) | 0 | MEF   | 5398   |
| 15                                       | 450     |       |        | lgK:-6.54(3SF) | 0 | MEF   | 5398   |
| 16                                       | 175:240 |       |        | dH:6.7(2SF)    | 0 | MTD   | 5398   |

| Reaction No. 53079                                     |         |       |        |              |   |       |        |
|--------------------------------------------------------|---------|-------|--------|--------------|---|-------|--------|
| Al+3(2)_C1-1(6) + 2<Al+3_C1-1(4)> = 2<Al+3(2)_C1-1(7)> |         |       |        |              |   |       |        |
| No                                                     | t       | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                      | 170:240 |       |        | lgK:2.1(2SF) | 0 | MRM   | 5398   |

| Reaction No. 53207              |   |       |        |             |   |       |        |
|---------------------------------|---|-------|--------|-------------|---|-------|--------|
| Al+3 + H+1_F-1 = Al+3_F-1 + H+1 |   |       |        |             |   |       |        |
| No                              | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

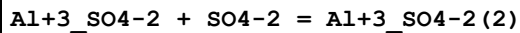
|   |    |   |        |                |   |     |      |
|---|----|---|--------|----------------|---|-----|------|
| 1 | 25 | 1 | NaClO4 | lgK:3.15 (3SF) | 5 | MQH | 5398 |
| 2 | 25 | 1 | NaClO4 | dH:0.74 (2SF)  | 4 | MCL | 5398 |

## Reaction No. 53208



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 1     | NaClO4 | lgK:1.99 (3SF) | 4 | MQH   | 5398   |

## Reaction No. 53347



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 15 | 0     | Inf. Dilution | lgK:0.50 (2SF) | 0 | MEF   | 5398   |
| 2  | 15 | 0     | Inf. Dilution | lgK:0.50 (2SF) | 0 | MEF   | 8374   |
| 3  | 25 | 0     | Inf. Dilution | lgK:1.3 (2SF)  | 1 | ELC   |        |
| 4  | 25 | 0     | Inf. Dilution | lgK:1.9 (2SF)  | 0 | MEF   | 140    |
| 5  | 25 | 0     | Inf. Dilution | lgK:1.89 (3SF) | 0 | MCL   | 5398   |
| 6  | 25 | 0     | Inf. Dilution | lgK:0.80 (2SF) | 0 | MEF   | 5398   |

Note (6): Low wgt for internal consistency

|    |    |     |               |                   |   |     |      |
|----|----|-----|---------------|-------------------|---|-----|------|
| 7  | 25 | 0   | Inf. Dilution | lgK:1.90 (0.1SD)  | 0 | ENS | 5573 |
| 8  | 25 | 0   | Inf. Dilution | lgK:0.80 (2SF)    | 0 | MEF | 8374 |
| 9  | 25 | 0   | Inf. Dilution | lgK:1.89 (0.10SD) | 0 | MCL | 8402 |
| 10 | 25 | 0.6 | NaClO4        | lgK:1.0 (2SF)     | 0 | MEF | 621  |
| 11 | 25 | 1   | NaClO4        | lgK:0.305 (3SF)   | 0 | MCL | 8684 |

Note (11): Suspect other reaction

|    |       |   |               |                  |   |     |      |
|----|-------|---|---------------|------------------|---|-----|------|
| 12 | 35    | 0 | Inf. Dilution | lgK:1.05 (3SF)   | 0 | MEF | 5398 |
| 13 | 35    | 0 | Inf. Dilution | lgK:1.05 (3SF)   | 0 | MEF | 8374 |
| 14 | 35    | 1 | NaClO4        | lgK:0.382 (3SF)  | 3 | MCL | 8684 |
| 15 | 50    | 1 | NaClO4        | lgK:0.556 (3SF)  | 0 | MCL | 8684 |
| 16 | 70    | 1 | NaClO4        | lgK:0.908 (3SF)  | 0 | MCL | 8684 |
| 17 | 15:35 | 0 | Inf. Dilution | dH:11. (2SF)     | 0 | MCL | 5398 |
| 18 | 25    | 0 | Inf. Dilution | dH:0.78 (2SF)    | 3 | MCL | 5398 |
| 19 | 25    | 0 | Inf. Dilution | dH:0.78 (0.02SD) | 3 | MCL | 8402 |
| 20 | 25    | 1 | NaClO4        | dH:2.46 (3SF)    | 0 | MCL | 8684 |
| 21 | 35    | 1 | NaClO4        | dH:3.99 (3SF)    | 0 | MCL | 8684 |
| 22 | 50    | 1 | NaClO4        | dH:6.75 (3SF)    | 0 | MCL | 8684 |
| 23 | 70    | 1 | NaClO4        | dH:11.3 (3SF)    | 0 | MCL | 8684 |

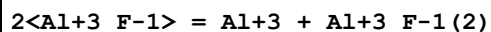
| Reaction No. 53729          |     |       |               |                   |   |       |        |
|-----------------------------|-----|-------|---------------|-------------------|---|-------|--------|
| 5<F-1> + Al+3 = Al+3_F-1(5) |     |       |               |                   |   |       |        |
| No                          | t   | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                           | 25  | 0     | Inf. Dilution | lgK:21.(0.5SD)    | 3 | CMN   | 2171   |
| 2                           | 25  | 0     | Inf. Dilution | lgK:20.91(4SF)    | 4 | EMU   | 5398   |
| 3                           | 25  | 0     | Inf. Dilution | lgK:20.6(3SF)     | 4 | ENS   | 6496   |
| 4                           | 25  | 0     | Many Media    | lgK:21.(0.5SD)    | 7 | EOD   | 6953   |
| 5                           | 25  | 0     | SOLVEQ        | lgK:20.97(4SF)    | 4 | CMN   | 5416   |
| 6                           | 25  | 0     | UCT_Sea       | lgK:20.59(0.01SD) | 4 | EDH   | 5390   |
| 7                           | 25  | 0     | WATCH         | lgK:20.35(4SF)    | 4 | CMN   | 5416   |
| 8                           | 25  | 0.1   | UCT_Sea       | lgK:19.70(0.01SD) | 4 | EDH   | 5390   |
| 9                           | 25  | 0.2   | UCT_Sea       | lgK:19.54(0.01SD) | 4 | EDH   | 5390   |
| 10                          | 25  | 0.3   | UCT_Sea       | lgK:19.46(0.01SD) | 4 | EDH   | 5390   |
| 11                          | 25  | 0.4   | UCT_Sea       | lgK:19.42(0.01SD) | 4 | EDH   | 5390   |
| 12                          | 25  | 0.5   | UCT_Sea       | lgK:19.40(0.01SD) | 4 | EDH   | 5390   |
| 13                          | 25  | 0.6   | UCT_Sea       | lgK:19.39(0.01SD) | 4 | EDH   | 5390   |
| 14                          | 25  | 0.7   | UCT_Sea       | lgK:19.39(0.01SD) | 4 | EDH   | 5390   |
| 15                          | 50  | 0     | SOLVEQ        | lgK:21.16(4SF)    | 4 | CMN   | 5416   |
| 16                          | 50  | 0     | WATCH         | lgK:20.57(4SF)    | 4 | CMN   | 5416   |
| 17                          | 75  | 0     | WATCH         | lgK:20.97(4SF)    | 4 | CMN   | 5416   |
| 18                          | 100 | 0     | SOLVEQ        | lgK:22.10(4SF)    | 4 | CMN   | 5416   |
| 19                          | 100 | 0     | WATCH         | lgK:21.53(4SF)    | 4 | CMN   | 5416   |
| 20                          | 125 | 0     | WATCH         | lgK:22.24(4SF)    | 4 | CMN   | 5416   |
| 21                          | 150 | 0     | SOLVEQ        | lgK:23.76(4SF)    | 4 | CMN   | 5416   |
| 22                          | 150 | 0     | WATCH         | lgK:23.09(4SF)    | 4 | CMN   | 5416   |
| 23                          | 175 | 0     | WATCH         | lgK:24.06(4SF)    | 3 | CMN   | 5416   |
| 24                          | 200 | 0     | SOLVEQ        | lgK:26.24(4SF)    | 3 | CMN   | 5416   |
| 25                          | 200 | 0     | WATCH         | lgK:25.15(4SF)    | 3 | CMN   | 5416   |
| 26                          | 225 | 0     | WATCH         | lgK:26.35(4SF)    | 3 | CMN   | 5416   |
| 27                          | 250 | 0     | SOLVEQ        | lgK:30.01(4SF)    | 0 | CMN   | 5416   |
| 28                          | 250 | 0     | WATCH         | lgK:27.65(4SF)    | 0 | CMN   | 5416   |
| 29                          | 275 | 0     | WATCH         | lgK:29.06(4SF)    | 0 | CMN   | 5416   |
| 30                          | 300 | 0     | SOLVEQ        | lgK:36.50(4SF)    | 0 | CMN   | 5416   |
| 31                          | 300 | 0     | WATCH         | lgK:30.57(4SF)    | 0 | CMN   | 5416   |
| 32                          | 325 | 0     | WATCH         | lgK:32.17(4SF)    | 0 | CMN   | 5416   |
| 33                          | 350 | 0     | WATCH         | lgK:33.87(4SF)    | 0 | CMN   | 5416   |

|    |    |   |               |                  |   |     |      |
|----|----|---|---------------|------------------|---|-----|------|
| 34 | 25 | 0 | Inf. Dilution | dH:1.84 (0.02SD) | 4 | ENS | 2171 |
| 35 | 25 | 0 | Inf. Dilution | dH:1.84 (3SF)    | 4 | ENS | 6496 |

| Reaction No. 53730           |     |       |               |                    |   |       |        |
|------------------------------|-----|-------|---------------|--------------------|---|-------|--------|
| 6<F-1> + Al+3 = Al+3_F-1 (6) |     |       |               |                    |   |       |        |
| No                           | t   | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                            | 18  |       |               | lgK:27. (2SF)      | 0 | MGL   | 140    |
| 2                            | 25  | 0     | Inf. Dilution | lgK:21. (0.9SD)    | 3 | CMN   | 2171   |
| 3                            | 25  | 0     | Inf. Dilution | lgK:20.86 (4SF)    | 4 | EMU   | 5398   |
| 4                            | 25  | 0     | Inf. Dilution | lgK:20.6 (3SF)     | 4 | ENS   | 6496   |
| 5                            | 25  | 0     | Inf. Dilution | lgK:19.842 (5SF)   | 0 | CRV   | 22551  |
| 6                            | 25  | 0     | Many Media    | lgK:21. (1.SD)     | 5 | EOD   | 6953   |
| 7                            | 25  | 0     | UCT_Sea       | lgK:20.61 (0.01SD) | 4 | EDH   | 5390   |
| 8                            | 25  | 0     | WATCH         | lgK:21.01 (4SF)    | 4 | CMN   | 5416   |
| 9                            | 25  | 0.1   | UCT_Sea       | lgK:20.00 (0.01SD) | 4 | EDH   | 5390   |
| 10                           | 25  | 0.2   | UCT_Sea       | lgK:19.88 (0.01SD) | 4 | EDH   | 5390   |
| 11                           | 25  | 0.3   | UCT_Sea       | lgK:19.83 (0.01SD) | 4 | EDH   | 5390   |
| 12                           | 25  | 0.4   | UCT_Sea       | lgK:19.81 (0.01SD) | 4 | EDH   | 5390   |
| 13                           | 25  | 0.5   | UCT_Sea       | lgK:19.80 (0.01SD) | 4 | EDH   | 5390   |
| 14                           | 25  | 0.6   | UCT_Sea       | lgK:19.81 (0.01SD) | 4 | EDH   | 5390   |
| 15                           | 25  | 0.7   | UCT_Sea       | lgK:19.82 (0.01SD) | 4 | EDH   | 5390   |
| 16                           | 50  | 0     | WATCH         | lgK:21.17 (4SF)    | 4 | CMN   | 5416   |
| 17                           | 75  | 0     | WATCH         | lgK:21.52 (4SF)    | 4 | CMN   | 5416   |
| 18                           | 100 | 0     | WATCH         | lgK:22.04 (4SF)    | 3 | CMN   | 5416   |
| 19                           | 125 | 0     | WATCH         | lgK:22.71 (4SF)    | 3 | CMN   | 5416   |
| 20                           | 150 | 0     | WATCH         | lgK:23.52 (4SF)    | 3 | CMN   | 5416   |
| 21                           | 175 | 0     | WATCH         | lgK:24.46 (4SF)    | 3 | CMN   | 5416   |
| 22                           | 200 | 0     | WATCH         | lgK:25.52 (4SF)    | 0 | CMN   | 5416   |
| 23                           | 225 | 0     | WATCH         | lgK:26.70 (4SF)    | 0 | CMN   | 5416   |
| 24                           | 250 | 0     | WATCH         | lgK:27.98 (4SF)    | 0 | CMN   | 5416   |
| 25                           | 275 | 0     | WATCH         | lgK:29.37 (4SF)    | 0 | CMN   | 5416   |
| 26                           | 300 | 0     | WATCH         | lgK:30.87 (4SF)    | 0 | CMN   | 5416   |
| 27                           | 325 | 0     | WATCH         | lgK:32.46 (4SF)    | 0 | CMN   | 5416   |
| 28                           | 350 | 0     | WATCH         | lgK:34.14 (4SF)    | 0 | CMN   | 5416   |
| 29                           | 25  | 0     | Inf. Dilution | dH:7.00 (3SF) kJ   | 4 | CRV   | 2171   |
| 30                           | 25  | 0     | Inf. Dilution | dH:1.67 (3SF)      | 3 | ENS   | 6496   |

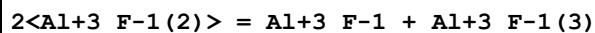
Note (30): Sign changed

## Reaction No. 53731



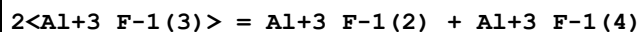
| No | t   | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|-----|-------|--------|---------------|---|-------|--------|
| 1  | -15 |       |        | lgK:-0.8(1SF) | 0 | MMR   | 5398   |

## Reaction No. 53732



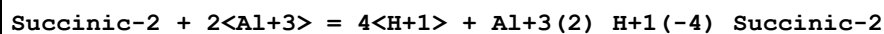
| No | t   | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|-----|-------|--------|---------------|---|-------|--------|
| 1  | -15 |       |        | lgK:-0.9(1SF) | 0 | MMR   | 5398   |

## Reaction No. 53733



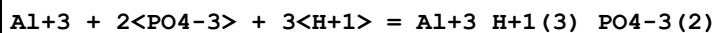
| No | t   | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|-----|-------|--------|---------------|---|-------|--------|
| 1  | -15 |       |        | lgK:-1.4(2SF) | 0 | MMR   | 5398   |

## Reaction No. 53998



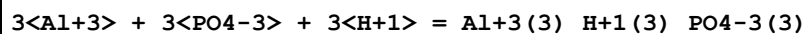
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-7.64(3SF)  | 2 | EAE   |        |
| 2  | 37 | 0.15  | NaCl          | lgK:-9.8(0.1SD) | 4 | MGL   | 5979   |

## Reaction No. 53999



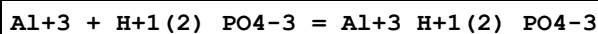
| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:42.2(3SF) | 2 | EAE   |        |
| 2  | 37 | 0.15  | NaCl          | lgK:38.6(3SF) | 6 | MIS   | 5983   |

## Reaction No. 54443



| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:62.2(2SF) | 1 | EAE   |        |
| 2  | 37 | 0.15  | NaCl          | lgK:61.1(3SF) | 0 | MIS   | 5983   |

## Reaction No. 54582



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

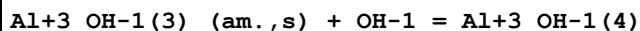
|   |    |      |               |               |   |     |      |
|---|----|------|---------------|---------------|---|-----|------|
| 1 | 18 | 0.1  | Unknown       | lgK:3.(0.5SD) | 0 | ENS | 2171 |
| 2 | 25 | 0    | Inf. Dilution | lgK:3.3(2SF)  | 1 | ERG |      |
| 3 | 25 |      |               | lgK:3.1(2SF)  | 0 | EES | 6953 |
| 4 | 37 | 0.15 | Unknown       | lgK:4.25(3SF) | 6 | CRV | 552  |

## Reaction No. 54584



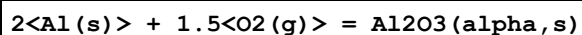
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-10.8(3SF)   | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:-9.7(0.05SD) | 0 | ENS   | 557    |
| 3  | 25 | 0     | Inf. Dilution | lgK:-10.39(4SF)  | 0 | EOD   | 5629   |
| 4  | 25 | 0     | Inf. Dilution | lgK:-10.8(3SF)   | 0 | ENS   | 6496   |
| 5  | 25 | 0     | Inf. Dilution | lgK:-9.66(3SF)   | 0 | MGL   | 6498   |
| 6  | 25 | 0     | Inf. Dilution | lgK:-10.359(5SF) | 0 | CRV   | 32224  |
| 7  | 25 | 0     | Inf. Dilution | dH:27.0(0.1SD)   | 1 | ENS   | 766    |
| 8  | 25 | 0     | Inf. Dilution | dH:26.5(3SF)     | 1 | ENS   | 6496   |

## Reaction No. 54585



| No | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:2.0(2SF) | 1 | ERG   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:1.3(2SF) | 0 | MSL   | 30     |

## Reaction No. 54587



| No | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|-----|-------|---------------|------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:278.9(4SF)   | 1 | EOR   |        |
| 2  | 25  | 0     | Inf. Dilution | lgK:278.6(4SF)   | 1 | ERG   |        |
| 3  | 25  | 0     | Inf. Dilution | lgK:277.210(6SF) | 0 | ENS   | 6422   |
| 4  | 25  | 0     | Inf. Dilution | lgK:277.20(5SF)  | 0 | ENS   | 6494   |
| 5  | 27  | 0     | Inf. Dilution | lgK:275.4(4SF)   | 0 | ENS   | 6422   |
| 6  | 27  | 0     | Inf. Dilution | lgK:275.4(4SF)   | 0 | ENS   | 6494   |
| 7  | 127 | 0     | Inf. Dilution | lgK:202.4(4SF)   | 0 | ENS   | 6422   |
| 8  | 127 | 0     | Inf. Dilution | lgK:202.4(4SF)   | 0 | ENS   | 6494   |
| 9  | 227 | 0     | Inf. Dilution | lgK:158.7(4SF)   | 0 | ENS   | 6422   |
| 10 | 227 | 0     | Inf. Dilution | lgK:158.7(4SF)   | 0 | ENS   | 6494   |
| 11 | 327 | 0     | Inf. Dilution | lgK:129.5(4SF)   | 0 | ENS   | 6422   |



|                                                                                    |     |   |               |                         |   |     |       |
|------------------------------------------------------------------------------------|-----|---|---------------|-------------------------|---|-----|-------|
| 12                                                                                 | 327 | 0 | Inf. Dilution | lgK:129.5 (4SF)         | 0 | ENS | 6494  |
| 13                                                                                 | 25  | 0 | Inf. Dilution | dG:-1568.30 (6SF) kJ    | 0 | CRV | 440   |
| 14                                                                                 | 25  | 0 | Inf. Dilution | dG:-378. (0.3SD)        | 0 | ENS | 2171  |
| 15                                                                                 | 25  | 0 | Inf. Dilution | dG:-1582.3 (5SF) kJ     | 0 | EOD | 5616  |
| 16                                                                                 | 25  | 0 | Inf. Dilution | dG:-1587.4 (2.1SD) kJ   | 0 | EFE | 5617  |
| 17                                                                                 | 25  | 0 | Inf. Dilution | dG:-1582.20 (6SF) kJ    | 0 | CRV | 5619  |
| 18                                                                                 | 25  | 0 | Inf. Dilution | dG:-1582.40 (6SF) kJ    | 0 | CRV | 5627  |
| 19                                                                                 | 25  | 0 | Inf. Dilution | dG:-371. (3SF)          | 0 | CRV | 5642  |
| 20                                                                                 | 25  | 0 | Inf. Dilution | dG:-378.1 (4SF)         | 0 | CRV | 5642  |
| 21                                                                                 | 25  | 0 | Inf. Dilution | dG:-1582.3 (5SF) kJ     | 0 | CRV | 6330  |
| 22                                                                                 | 25  | 0 | Inf. Dilution | dG:-1581.90 (6SF) kJ    | 0 | CRV | 6421  |
| 23                                                                                 | 25  | 0 | Inf. Dilution | dG:-378.15 (5SF)        | 0 | EFE | 6482  |
| 24                                                                                 | 25  | 0 | Inf. Dilution | dG:-378.18 (5SF)        | 0 | EFE | 6494  |
| 25                                                                                 | 25  | 0 | Inf. Dilution | dG:-1582.3 (5SF) kJ     | 0 | EFE | 7275  |
| 26                                                                                 | 25  | 0 | Inf. Dilution | dG:-378.2 (4SF)         | 0 | CRV | 7382  |
| 27                                                                                 | 25  | 0 | Inf. Dilution | dG:-1582. (4SF) kJ      | 0 | CRV | 8746  |
| 28                                                                                 | 25  | 0 | Inf. Dilution | dG:-1580.4 (5SF) kJ     | 0 | CRV | 8872  |
| 29                                                                                 | 25  | 0 | Inf. Dilution | dG:-1557.30 (6SF) kJ    | 0 | EOD | 8872  |
| Note (29): Data from Naumov, Ryzhenko & Khodakovsky, 1971; low wgt for consistency |     |   |               |                         |   |     |       |
| 30                                                                                 | 25  | 0 | Inf. Dilution | dG:-1581.10 (6SF) kJ    | 0 | EOD | 8872  |
| Note (30): Data from Khodakovsky, 1983                                             |     |   |               |                         |   |     |       |
| 31                                                                                 | 25  | 0 | Inf. Dilution | dG:-1582.00 (6SF) kJ    | 0 | CRV | 8873  |
| 32                                                                                 | 25  | 0 | Inf. Dilution | dG:-378.17 (5SF)        | 0 | CRV | 10174 |
| 33                                                                                 | 25  | 0 | Inf. Dilution | dG:-1675.7 (5SF) kJ     | 0 | CRV | 10802 |
| 34                                                                                 | 25  | 0 | Inf. Dilution | dG:-1582.3 (1.302SD) kJ | 0 | CRV | 14923 |
| 35                                                                                 | 25  | 0 | Inf. Dilution | dG:-1582.30 (0.65SD) kJ | 0 | CRV | 27292 |
| 36                                                                                 | 25  | 0 | Inf. Dilution | dG:-1582.3 (5SF) kJ     | 2 | EOD | 29520 |
| 37                                                                                 | 25  | 0 | GAMSPATH      | dG:-1568. (4SF) kJ      | 0 | CRV | 12638 |
| 38                                                                                 | 50  | 0 | Inf. Dilution | dG:-1574.4 (5SF) kJ     | 0 | EOD | 5616  |
| 39                                                                                 | 50  | 0 | Inf. Dilution | dG:-378.49 (5SF)        | 0 | CRV | 10174 |
| 40                                                                                 | 75  | 0 | Inf. Dilution | dG:-378.86 (5SF)        | 0 | CRV | 10174 |
| 41                                                                                 | 100 | 0 | Inf. Dilution | dG:-1558.7 (5SF) kJ     | 0 | EOD | 5616  |
| 42                                                                                 | 100 | 0 | Inf. Dilution | dG:-379.26 (5SF)        | 0 | CRV | 10174 |
| 43                                                                                 | 125 | 0 | Inf. Dilution | dG:-379.69 (5SF)        | 0 | CRV | 10174 |
| 44                                                                                 | 150 | 0 | Inf. Dilution | dG:-1542.9 (5SF) kJ     | 0 | MSL | 5616  |
| 45                                                                                 | 150 | 0 | Inf. Dilution | dG:-380.17 (5SF)        | 0 | CRV | 10174 |

|    |     |   |               |                         |   |     |       |
|----|-----|---|---------------|-------------------------|---|-----|-------|
| 46 | 175 | 0 | Inf. Dilution | dG:-380.67 (5SF)        | 0 | CRV | 10174 |
| 47 | 200 | 0 | Inf. Dilution | dG:-1527.2 (5SF) kJ     | 0 | MSL | 5616  |
| 48 | 200 | 0 | Inf. Dilution | dG:-381.21 (5SF)        | 0 | CRV | 10174 |
| 49 | 225 | 0 | Inf. Dilution | dG:-381.78 (5SF)        | 0 | CRV | 10174 |
| 50 | 250 | 0 | Inf. Dilution | dG:-1511.5 (5SF) kJ     | 0 | MSL | 5616  |
| 51 | 250 | 0 | Inf. Dilution | dG:-382.38 (5SF)        | 0 | CRV | 10174 |
| 52 | 300 | 0 | Inf. Dilution | dG:-1495.8 (5SF) kJ     | 0 | MSL | 5616  |
| 53 | 300 | 0 | Inf. Dilution | dG:-383.66 (5SF)        | 0 | CRV | 10174 |
| 54 | 25  | 0 | Inf. Dilution | dH:-400. (0.3SD)        | 0 | ENS | 2171  |
| 55 | 25  | 0 | Inf. Dilution | dH:-1675.7 (1.3SD) kJ   | 0 | EOD | 5616  |
| 56 | 25  | 0 | Inf. Dilution | dH:-1675.7 (1.3MD) kJ   | 0 | CRV | 5619  |
| 57 | 25  | 0 | Inf. Dilution | dH:-1675.7 (5SF) kJ     | 0 | CRV | 6330  |
| 58 | 25  | 0 | Inf. Dilution | dH:-400.5 (4SF)         | 0 | ENS | 6422  |
| 59 | 25  | 0 | Inf. Dilution | dH:-400.50 (5SF)        | 0 | EFE | 6482  |
| 60 | 25  | 0 | Inf. Dilution | dH:-400.5 (4SF)         | 0 | MTD | 6494  |
| 61 | 25  | 0 | Inf. Dilution | dH:-1675.7 (5SF) kJ     | 0 | EFE | 7275  |
| 62 | 25  | 0 | Inf. Dilution | dH:-400.5 (4SF)         | 0 | CRV | 7382  |
| 63 | 25  | 0 | Inf. Dilution | dH:-1675.7 (1.3MD) kJ   | 0 | CRV | 8746  |
| 64 | 25  | 0 | Inf. Dilution | dH:-1582.3 (5SF) kJ     | 0 | CRV | 10802 |
| 65 | 25  | 0 | Inf. Dilution | dH:-1675.7 (1.300SD) kJ | 0 | CRV | 14923 |
| 66 | 25  | 0 | Inf. Dilution | dH:-1675.70 (0.65SD) kJ | 0 | CRV | 27292 |
| 67 | 25  | 0 | Inf. Dilution | dH:-1675.7 (5SF) kJ     | 2 | EOD | 29520 |
| 68 | 25  | 0 | GAMSPATH      | dH:-1661.70 (6SF) kJ    | 0 | CRV | 12638 |
| 69 | 127 | 0 | Inf. Dilution | dH:-400.6 (4SF)         | 0 | ENS | 6422  |
| 70 | 127 | 0 | Inf. Dilution | dH:-400.6 (4SF)         | 0 | MTD | 6494  |
| 71 | 227 | 0 | Inf. Dilution | dH:-400.6 (4SF)         | 0 | ENS | 6422  |
| 72 | 227 | 0 | Inf. Dilution | dH:-400.6 (4SF)         | 0 | MTD | 6494  |
| 73 | 327 | 0 | Inf. Dilution | dH:-400.4 (4SF)         | 0 | ENS | 6422  |
| 74 | 327 | 0 | Inf. Dilution | dH:-400.4 (4SF)         | 0 | MTD | 6494  |
| 75 | 25  | 0 | Inf. Dilution | Cp:18.88 (4SF)          | 0 | ENS | 6333  |
| 76 | 25  | 0 | Inf. Dilution | Cp:79.033 (0.1SD) J/K   | 0 | CRV | 27292 |

## Reaction No. 54588



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.2   | NH4NO3 | lgK:-2.0 (0.05SD) | 5 | MIX   | 5451   |

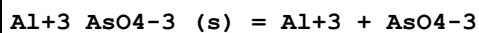
| Reaction No. 54618                                                                                           |     |       |               |                   |   |       |        |
|--------------------------------------------------------------------------------------------------------------|-----|-------|---------------|-------------------|---|-------|--------|
| $\text{Ca(s)} + 3.5\text{O}_2(\text{g}) + 4\text{Al(s)} = \text{CaO} \cdot 2\text{Al}_2\text{O}_3(\text{s})$ |     |       |               |                   |   |       |        |
| No                                                                                                           | t   | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                            | 25  | 0     | Inf. Dilution | lgK:669.0(4SF)    | 4 | ENS   | 6422   |
| 2                                                                                                            | 27  | 0     | Inf. Dilution | lgK:664.7(4SF)    | 4 | ENS   | 6422   |
| 3                                                                                                            | 127 | 0     | Inf. Dilution | lgK:489.4(4SF)    | 4 | ENS   | 6422   |
| 4                                                                                                            | 227 | 0     | Inf. Dilution | lgK:384.220(6SF)  | 4 | ENS   | 6422   |
| 5                                                                                                            | 327 | 0     | Inf. Dilution | lgK:314.1(4SF)    | 4 | ENS   | 6422   |
| 6                                                                                                            | 25  | 0     | Inf. Dilution | dG:-901.2(0.05SD) | 0 | ENS   | 802    |
| Note (6): Low wgt for internal consistency                                                                   |     |       |               |                   |   |       |        |
| 7                                                                                                            | 25  | 0     | Inf. Dilution | dG:-912.7(4SF)    | 4 | ENS   | 6422   |
| 8                                                                                                            | 25  | 0     | Inf. Dilution | dH:-950.7(0.1SD)  | 0 | ENS   | 802    |
| 9                                                                                                            | 25  | 0     | Inf. Dilution | dH:-962.2(4SF)    | 4 | MTD   | 6422   |
| 10                                                                                                           | 127 | 0     | Inf. Dilution | dH:-962.5(4SF)    | 4 | MTD   | 6422   |
| 11                                                                                                           | 227 | 0     | Inf. Dilution | dH:-962.4(4SF)    | 4 | MTD   | 6422   |
| 12                                                                                                           | 327 | 0     | Inf. Dilution | dH:-962.1(4SF)    | 4 | MTD   | 6422   |

| Reaction No. 54652                                                                                          |     |       |               |                   |   |       |        |
|-------------------------------------------------------------------------------------------------------------|-----|-------|---------------|-------------------|---|-------|--------|
| $3\text{Ca(s)} + 3\text{O}_2(\text{g}) + 2\text{Al(s)} = 3\text{CaO} \cdot \text{Al}_2\text{O}_3(\text{s})$ |     |       |               |                   |   |       |        |
| No                                                                                                          | t   | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                           | 25  | 0     | Inf. Dilution | lgK:597.7(4SF)    | 4 | ENS   | 6422   |
| 2                                                                                                           | 27  | 0     | Inf. Dilution | lgK:593.9(4SF)    | 4 | ENS   | 6422   |
| 3                                                                                                           | 127 | 0     | Inf. Dilution | lgK:437.7(4SF)    | 4 | ENS   | 6422   |
| 4                                                                                                           | 227 | 0     | Inf. Dilution | lgK:344.0(4SF)    | 4 | ENS   | 6422   |
| 5                                                                                                           | 327 | 0     | Inf. Dilution | lgK:281.6(4SF)    | 4 | ENS   | 6422   |
| 6                                                                                                           | 25  | 0     | Inf. Dilution | dG:-815.4(0.05SD) | 4 | ENS   | 802    |
| 7                                                                                                           | 25  | 0     | Inf. Dilution | dG:-815.4(4SF)    | 4 | ENS   | 6422   |
| 8                                                                                                           | 25  | 0     | Inf. Dilution | dH:-857.5(0.1SD)  | 4 | ENS   | 802    |
| 9                                                                                                           | 25  | 0     | Inf. Dilution | dH:-857.5(4SF)    | 4 | MTD   | 6422   |
| 10                                                                                                          | 127 | 0     | Inf. Dilution | dH:-857.3(4SF)    | 4 | MTD   | 6422   |
| 11                                                                                                          | 227 | 0     | Inf. Dilution | dH:-856.8(4SF)    | 4 | MTD   | 6422   |
| 12                                                                                                          | 327 | 0     | Inf. Dilution | dH:-856.3(4SF)    | 4 | MTD   | 6422   |

| Reaction No. 54681                                                                                        |   |       |        |             |   |       |        |
|-----------------------------------------------------------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| $\text{Ca(s)} + 2\text{O}_2(\text{g}) + 2\text{Al(s)} = \text{CaO} \cdot \text{Al}_2\text{O}_3(\text{s})$ |   |       |        |             |   |       |        |
| No                                                                                                        | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

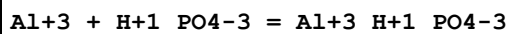
|    |     |   |               |                   |   |     |      |
|----|-----|---|---------------|-------------------|---|-----|------|
| 1  | 25  | 0 | Inf. Dilution | lgK:387.0(4SF)    | 4 | ENS | 6422 |
| 2  | 27  | 0 | Inf. Dilution | lgK:384.5(4SF)    | 4 | ENS | 6422 |
| 3  | 127 | 0 | Inf. Dilution | lgK:283.2(4SF)    | 4 | ENS | 6422 |
| 4  | 227 | 0 | Inf. Dilution | lgK:222.4(4SF)    | 4 | ENS | 6422 |
| 5  | 327 | 0 | Inf. Dilution | lgK:181.9(4SF)    | 4 | ENS | 6422 |
| 6  | 25  | 0 | Inf. Dilution | dG:-527.9(0.05SD) | 3 | ENS | 802  |
| 7  | 25  | 0 | Inf. Dilution | dG:-527.9(4SF)    | 4 | ENS | 6422 |
| 8  | 25  | 0 | Inf. Dilution | dH:-556.0(0.1SD)  | 3 | ENS | 802  |
| 9  | 25  | 0 | Inf. Dilution | dH:-556.0(4SF)    | 4 | MTD | 6422 |
| 10 | 127 | 0 | Inf. Dilution | dH:-556.1(4SF)    | 4 | MTD | 6422 |
| 11 | 227 | 0 | Inf. Dilution | dH:-555.9(4SF)    | 4 | MTD | 6422 |
| 12 | 327 | 0 | Inf. Dilution | dH:-555.8(4SF)    | 4 | MTD | 6422 |

## Reaction No. 54714



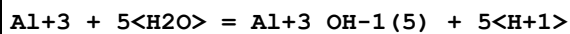
| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 20 | 0     | Inf. Dilution | lgK:-15.8(0.05SD) | 0 | ENS   | 5468   |
| 2  | 25 | 0     | Inf. Dilution | lgK:-34.6(3SF)    | 1 | ELC   |        |

## Reaction No. 54754



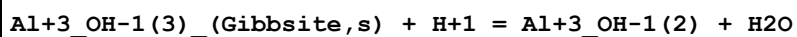
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 18 | 0.1   | Unknown       | lgK:7.(0.5SD)  | 0 | ENS   | 2171   |
| 2  | 25 | 0     | Inf. Dilution | lgK:7.6(2SF)   | 1 | ERG   |        |
| 3  | 25 | 0.1   | NaClO4        | lgK:9.19(3SF)w | 0 | MPH   | 3180   |
| 4  | 25 | 0.15  | NaCl          | lgK:11.4(3SF)  | 0 | MGL   | 3614   |
| 5  | 25 |       |               | lgK:7.4(2SF)   | 0 | EES   | 6953   |
| 6  | 37 | 0.15  | Unknown       | lgK:7.7(2SF)   | 1 | CRV   | 552    |

## Reaction No. 54783



| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-43.1(0.05SD) | 0 | ENS   | 557    |

## Reaction No. 54784



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

|                        |     |     |               |                 |   |     |      |
|------------------------|-----|-----|---------------|-----------------|---|-----|------|
| 1                      | 0   | 0   | NaCl/m        | lgK:-2.8 (2SF)  | 0 | MSL | 7025 |
| 2                      | 0   | 0.1 | NaCl/m        | lgK:-2.8 (2SF)  | 0 | MSL | 7025 |
| 3                      | 0   | 0.5 | NaCl/m        | lgK:-2.8 (2SF)  | 0 | MSL | 7025 |
| 4                      | 0   | 1   | NaCl/m        | lgK:-2.8 (2SF)  | 0 | MSL | 7025 |
| 5                      | 0   | 5   | NaCl/m        | lgK:-2.7 (2SF)  | 0 | MSL | 7025 |
| 6                      | 20  | 0   | Inf. Dilution | lgK:-2.59 (3SF) | 0 | MTN | 5630 |
| Note (6): Sign changed |     |     |               |                 |   |     |      |
| 7                      | 25  | 0   | Inf. Dilution | lgK:-2.9 (2SF)  | 1 | ELC |      |
| 8                      | 25  | 0   | Inf. Dilution | lgK:0.40 (2SF)  | 0 | MSL | 30   |
| 9                      | 25  | 0   | Inf. Dilution | lgK:-2.02 (3SF) | 0 | MSL | 553  |
| 10                     | 25  | 0   | NaCl/m        | lgK:-2.8 (2SF)  | 0 | MSL | 7025 |
| 11                     | 25  | 0.1 | NaCl/m        | lgK:-2.8 (2SF)  | 0 | MSL | 7025 |
| 12                     | 25  | 0.5 | NaCl/m        | lgK:-2.8 (2SF)  | 0 | MSL | 7025 |
| 13                     | 25  | 1   | NaCl/m        | lgK:-2.8 (2SF)  | 0 | MSL | 7025 |
| 14                     | 25  | 5   | NaCl/m        | lgK:-2.7 (2SF)  | 0 | MSL | 7025 |
| 15                     | 50  | 0   | NaCl/m        | lgK:-2.9 (2SF)  | 0 | MSL | 7025 |
| 16                     | 50  | 0.1 | NaCl/m        | lgK:-2.9 (2SF)  | 0 | MSL | 7025 |
| 17                     | 50  | 0.5 | NaCl/m        | lgK:-2.9 (2SF)  | 0 | MSL | 7025 |
| 18                     | 50  | 1   | NaCl/m        | lgK:-2.9 (2SF)  | 0 | MSL | 7025 |
| 19                     | 50  | 5   | NaCl/m        | lgK:-2.8 (2SF)  | 0 | MSL | 7025 |
| 20                     | 75  | 0   | NaCl/m        | lgK:-2.9 (2SF)  | 0 | MSL | 7025 |
| 21                     | 75  | 0.1 | NaCl/m        | lgK:-2.9 (2SF)  | 0 | MSL | 7025 |
| 22                     | 75  | 0.5 | NaCl/m        | lgK:-2.9 (2SF)  | 0 | MSL | 7025 |
| 23                     | 75  | 1   | NaCl/m        | lgK:-2.9 (2SF)  | 0 | MSL | 7025 |
| 24                     | 75  | 5   | NaCl/m        | lgK:-2.8 (2SF)  | 0 | MSL | 7025 |
| 25                     | 100 | 0   | NaCl/m        | lgK:-2.9 (2SF)  | 0 | MSL | 7025 |
| 26                     | 100 | 0.1 | NaCl/m        | lgK:-2.9 (2SF)  | 0 | MSL | 7025 |
| 27                     | 100 | 0.5 | NaCl/m        | lgK:-2.9 (2SF)  | 0 | MSL | 7025 |
| 28                     | 100 | 1   | NaCl/m        | lgK:-2.9 (2SF)  | 0 | MSL | 7025 |
| 29                     | 100 | 5   | NaCl/m        | lgK:-2.9 (2SF)  | 0 | MSL | 7025 |

## Reaction No. 54785

$$\text{Al+3\_OH-1(3)\_(am.,s)} = \text{Al+3} + 3\text{<OH-1>}$$

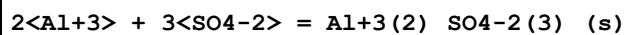
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-31.1 (3SF) | 1 | EOR   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:-31.2 (3SF) | 3 | MSL   | 30     |

|                                                                    |     |     |               |                  |   |     |       |
|--------------------------------------------------------------------|-----|-----|---------------|------------------|---|-----|-------|
| 3                                                                  | 25  | 0   | Inf. Dilution | lgK:-32.34 (4SF) | 0 | ENS | 773   |
| 4                                                                  | 25  | 0   | Inf. Dilution | lgK:-31.80 (4SF) | 0 | MSL | 8620  |
| 5                                                                  | 25  | 0   | Inf. Dilution | lgK:-31.66 (4SF) | 0 | CRV | 22551 |
| 6                                                                  | 25  | 0   | Ca(OH)2       | lgK:-31.00 (4SF) | 0 | MGL | 5632  |
| 7                                                                  | 25  | 0   | KOH           | lgK:-30.36 (4SF) | 0 | MGL | 5632  |
| 8                                                                  | 25  | 0   | LiOH          | lgK:-33.15 (4SF) | 0 | MGL | 5632  |
| 9                                                                  | 25  | 0   | NH4OH         | lgK:-30.55 (4SF) | 0 | MGL | 5632  |
| 10                                                                 | 25  | 0   | NaOH          | lgK:-30.75 (4SF) | 0 | MGL | 5632  |
| 11                                                                 | 25  | 0.1 | Unknown       | lgK:-31.6 (3SF)  | 0 | ENS | 773   |
| 12                                                                 | 25  | 3   | Unknown       | lgK:-33. (2SF)   | 3 | EES |       |
| Note (12): Based on behaviour of Gibbsite (needed for consistency) |     |     |               |                  |   |     |       |
| 13                                                                 | 100 | 0   | Inf. Dilution | lgK:-30.8 (3SF)  | 1 | EOR |       |

| Reaction No. 54786                               |       |       |               |                 |   |       |        |
|--------------------------------------------------|-------|-------|---------------|-----------------|---|-------|--------|
| Al+3_OH-1(4) = Al+3_OH-1(3)_ (Gibbsite,s) + OH-1 |       |       |               |                 |   |       |        |
| No                                               | t     | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                | 21    | 0     | Inf. Dilution | lgK:1.264 (4SF) | 6 | MSL   | 6028   |
| 2                                                | 25    | 0     | Inf. Dilution | lgK:1.2 (3SF)   | 2 | EAE   |        |
| 3                                                | 25    | 0     | Inf. Dilution | lgK:0.57 (2SF)  | 0 | CRV   | 30     |
| 4                                                | 40    | 2     | NaClO4        | lgK:1.0 (0.2SD) | 5 | MHY   | 12539  |
| 5                                                | 50    | 0     | Inf. Dilution | lgK:0.759 (3SF) | 6 | MSL   | 6028   |
| 6                                                | 70    | 0     | Inf. Dilution | lgK:0.461 (3SF) | 6 | MSL   | 6028   |
| 7                                                | 100.1 | 0     | Inf. Dilution | dH:-7.187 (4SF) | 4 | MCL   | 5626   |
| 8                                                | 100.1 | 1     | NaOH          | dH:-7.201 (4SF) | 5 | MCL   | 5626   |
| 9                                                | 100.1 | 3     | NaOH          | dH:-7.239 (4SF) | 5 | MCL   | 5626   |
| 10                                               | 100.1 | 5     | NaOH          | dH:-7.232 (4SF) | 5 | MCL   | 5626   |
| 11                                               | 112.9 | 0     | Inf. Dilution | dH:-7.543 (4SF) | 4 | MCL   | 5626   |
| 12                                               | 112.9 | 1     | NaOH          | dH:-7.562 (4SF) | 6 | MCL   | 5626   |
| 13                                               | 112.9 | 3     | NaOH          | dH:-7.581 (4SF) | 6 | MCL   | 5626   |
| 14                                               | 112.9 | 5     | NaOH          | dH:-7.526 (4SF) | 6 | MCL   | 5626   |
| 15                                               | 125.7 | 0     | Inf. Dilution | dH:-7.902 (4SF) | 4 | MCL   | 5626   |
| 16                                               | 125.7 | 1     | NaOH          | dH:-7.928 (4SF) | 6 | MCL   | 5626   |
| 17                                               | 125.7 | 3     | NaOH          | dH:-7.933 (4SF) | 6 | MCL   | 5626   |
| 18                                               | 125.7 | 5     | NaOH          | dH:-7.823 (4SF) | 6 | MCL   | 5626   |
| 19                                               | 138.6 | 0     | Inf. Dilution | dH:-8.267 (4SF) | 4 | MCL   | 5626   |
| 20                                               | 138.6 | 1     | NaOH          | dH:-8.243 (4SF) | 6 | MCL   | 5626   |

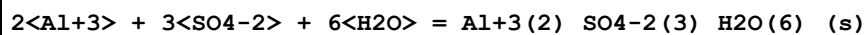
|    |       |   |               |                |   |     |      |
|----|-------|---|---------------|----------------|---|-----|------|
| 21 | 138.6 | 3 | NaOH          | dH:-8.267(4SF) | 6 | MCL | 5626 |
| 22 | 138.6 | 5 | NaOH          | dH:-8.164(4SF) | 6 | MCL | 5626 |
| 23 | 150.3 | 0 | Inf. Dilution | dH:-8.597(4SF) | 4 | MCL | 5626 |
| 24 | 150.3 | 1 | NaOH          | dH:-8.592(4SF) | 4 | MCL | 5626 |
| 25 | 150.3 | 3 | NaOH          | dH:-8.566(4SF) | 4 | MCL | 5626 |
| 26 | 150.3 | 5 | NaOH          | dH:-8.444(4SF) | 4 | MCL | 5626 |
| 27 | 100.1 | 0 | Inf. Dilution | Cp:-27.77(4SF) | 4 | MCL | 5626 |
| 28 | 112.9 | 0 | Inf. Dilution | Cp:-27.94(4SF) | 4 | MCL | 5626 |
| 29 | 125.7 | 0 | Inf. Dilution | Cp:-28.11(4SF) | 4 | MCL | 5626 |
| 30 | 138.6 | 0 | Inf. Dilution | Cp:-28.27(4SF) | 4 | MCL | 5626 |
| 31 | 150.3 | 0 | Inf. Dilution | Cp:-28.42(4SF) | 3 | MCL | 5626 |

## Reaction No. 54787



| No                                                                                                                                          | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|---------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| 1                                                                                                                                           | 25 | 0     | Inf. Dilution | lgK:-20.2(3SF)     | 1 | ELC   |        |
| 2                                                                                                                                           | 25 | 0     | Inf. Dilution | lgK:-1.84(3SF)     | 0 | CRV   |        |
| Note (2): <a href="http://www.saltlakemetals.com/SolubilityProducts.htm">http://www.saltlakemetals.com/SolubilityProducts.htm</a> (15/7/14) |    |       |               |                    |   |       |        |
| 3                                                                                                                                           | 25 | 0     | Inf. Dilution | lgK:-21.(0.5SD)    | 0 | ENS   | 557    |
| 4                                                                                                                                           | 25 | 0     | Inf. Dilution | lgK:-19.817(5SF)   | 0 | CRV   | 32224  |
| 5                                                                                                                                           | 25 | 0     | CHEMVAL1      | lgK:-19.82(0.01SD) | 0 | ENS   | 5156   |
| 6                                                                                                                                           | 25 | 0     | CHEMVAL2      | lgK:-19.82(0.01SD) | 0 | ENS   | 5156   |
| 7                                                                                                                                           | 25 | 0     | CHEMVAL3      | lgK:-19.82(0.01SD) | 0 | ENS   | 5156   |
| 8                                                                                                                                           | 25 | 0     | CHEMVAL4      | lgK:-19.80(0.01SD) | 0 | ENS   | 5156   |

## Reaction No. 54788



| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-2.35(3SF)    | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:-3.(0.5SD)    | 0 | ENS   | 557    |
| 3  | 25 | 0     | Inf. Dilution | lgK:-2.317(4SF)   | 4 | CRV   | 32224  |
| 4  | 25 | 0     | CHEMVAL1      | lgK:-2.32(0.01SD) | 0 | ENS   | 5156   |
| 5  | 25 | 0     | CHEMVAL2      | lgK:-2.32(0.01SD) | 0 | ENS   | 5156   |
| 6  | 25 | 0     | CHEMVAL3      | lgK:-2.32(0.01SD) | 1 | ENS   | 5156   |
| 7  | 25 | 0     | CHEMVAL4      | lgK:-0.75(0.01SD) | 0 | ENS   | 5156   |

## Reaction No. 54789

| 3<Al+3> + K+1 + 2<SO4-2> + 6<H2O> = Al+3(3)_K+1_OH-1(6)_SO4-2(2)_(s) + 6<H+1> |    |       |               |                   |   |       |        |
|-------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| No                                                                            | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                             | 25 | 0     | Inf. Dilution | lgK:-1.4(2SF)     | 1 | ERG   |        |
| Note (1): Was lgK=-1.5,-1.8 prev iter                                         |    |       |               |                   |   |       |        |
| 2                                                                             | 25 | 0     | Inf. Dilution | lgK:-3.04(3SF)    | 0 | ENS   | 557    |
| 3                                                                             | 25 | 0     | Inf. Dilution | lgK:1.3(2SF)      | 0 | ENS   | 766    |
| 4                                                                             | 25 | 0     | Inf. Dilution | lgK:1.4(2SF)      | 0 | ENS   | 6496   |
| 5                                                                             | 25 | 0     | Inf. Dilution | lgK:-0.78(2SF)    | 4 | CRV   | 32224  |
| 6                                                                             | 25 | 0     | CHEMVAL1      | lgK:-3.03(3SF)    | 0 | ENS   | 5156   |
| 7                                                                             | 25 | 0     | CHEMVAL2      | lgK:-3.03(3SF)    | 0 | ENS   | 5156   |
| 8                                                                             | 25 | 0     | CHEMVAL3      | lgK:-3.03(3SF)    | 0 | ENS   | 5156   |
| 9                                                                             | 25 | 0     | CHEMVAL4      | lgK:-3.03(0.01SD) | 0 | ENS   | 5156   |
| 10                                                                            | 25 | 0     | Inf. Dilution | dH:-3.9(2SF)      | 0 | ENS   | 766    |
| 11                                                                            | 25 | 0     | Inf. Dilution | dH:50.25(4SF)     | 1 | ENS   | 6496   |

| Reaction No. 54790                                          |    |       |               |                 |   |       |        |
|-------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| 2<Al+3> + 3<SO4-2> + 17<H2O> = Al+3(2)_SO4-2(3)_H2O(17)_(s) |    |       |               |                 |   |       |        |
| No                                                          | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                           | 25 | 0     | Inf. Dilution | lgK:7.0(0.05SD) | 4 | ENS   | 2171   |

| Reaction No. 54791                                                      |    |       |               |                   |   |       |        |
|-------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| 4<Al+3> + SO4-2 + 15<H2O> = Al+3(4)_OH-1(10)_SO4-2_H2O(5)_(s) + 10<H+1> |    |       |               |                   |   |       |        |
| No                                                                      | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                       | 25 | 0     | Inf. Dilution | lgK:-22.3(0.05SD) | 4 | ENS   | 2171   |

| Reaction No. 54792                                       |    |       |               |                 |   |       |        |
|----------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Al+3 + SO4-2 + 6<H2O> = Al+3_OH-1_SO4-2_H2O(5)_(s) + H+1 |    |       |               |                 |   |       |        |
| No                                                       | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                        | 25 | 0     | Inf. Dilution | lgK:3.8(0.05SD) | 0 | ENS   | 2171   |
| 2                                                        | 25 | 0     | Inf. Dilution | lgK:8.04(3SF)   | 4 | MGL   | 6498   |

| Reaction No. 54793                                                          |    |       |               |                   |   |       |        |
|-----------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| 4<Al+3> + SO4-2 + 15<H2O> = Al+3(4)_OH-1(10)_SO4-2_H2O(5)_(am.,s) + 10<H+1> |    |       |               |                   |   |       |        |
| No                                                                          | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                           | 25 | 0     | Inf. Dilution | lgK:-24.0(0.05SD) | 4 | ENS   | 2171   |

| Reaction No. 54794 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|



| 3<F-1> + Al+3 = Al+3_F-1(3)_(s) |    |       |               |                   |   |       |        |
|---------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| No                              | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                               | 25 | 0     | Inf. Dilution | lgK:18.(0.5SD)    | 2 | CMN   | 5458   |
| 2                               | 25 | 0     | Inf. Dilution | lgK:17.25(4SF)    | 2 | CRV   | 32224  |
| 3                               | 25 | 0     | CHEMVAL1      | lgK:18.16(0.01SD) | 0 | ENS   | 5156   |
| 4                               | 25 | 0     | CHEMVAL2      | lgK:18.16(0.01SD) | 0 | ENS   | 5156   |
| 5                               | 25 | 0     | CHEMVAL3      | lgK:18.16(0.01SD) | 0 | ENS   | 5156   |
| 6                               | 25 | 0     | CHEMVAL4      | lgK:16.90(0.01SD) | 2 | ENS   | 5156   |

| Reaction No. 54795                       |     |       |               |                    |   |       |        |
|------------------------------------------|-----|-------|---------------|--------------------|---|-------|--------|
| Al(s) + 2<O2(g)> + P(s) = Al+3_PO4-3_(s) |     |       |               |                    |   |       |        |
| No                                       | t   | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                        | 25  | 0     | Inf. Dilution | lgK:283.4(4SF)     | 0 | ENS   | 6422   |
| 2                                        | 25  | 0     | Inf. Dilution | lgK:283.4(4SF)     | 0 | ENS   | 6494   |
| 3                                        | 27  | 0     | Inf. Dilution | lgK:281.5(4SF)     | 0 | ENS   | 6422   |
| 4                                        | 27  | 0     | Inf. Dilution | lgK:281.6(4SF)     | 0 | ENS   | 6494   |
| 5                                        | 127 | 0     | Inf. Dilution | lgK:206.0(4SF)     | 1 | ENS   | 6422   |
| 6                                        | 127 | 0     | Inf. Dilution | lgK:206.1(4SF)     | 1 | ENS   | 6494   |
| 7                                        | 227 | 0     | Inf. Dilution | lgK:160.7(4SF)     | 1 | ENS   | 6422   |
| 8                                        | 227 | 0     | Inf. Dilution | lgK:160.7(4SF)     | 1 | ENS   | 6494   |
| 9                                        | 327 | 0     | Inf. Dilution | lgK:130.5(4SF)     | 0 | ENS   | 6422   |
| 10                                       | 327 | 0     | Inf. Dilution | lgK:130.5(4SF)     | 0 | ENS   | 6494   |
| 11                                       | 25  | 0     | Inf. Dilution | dG:-387.(0.5SD)    | 0 | ENS   | 802    |
| 12                                       | 25  | 0     | Inf. Dilution | dG:-1617.9(5SF) kJ | 0 | CRV   | 6330   |
| 13                                       | 25  | 0     | Inf. Dilution | dG:-386.6(4SF)     | 0 | EFE   | 6422   |
| 14                                       | 25  | 0     | Inf. Dilution | dG:-386.7(4SF)     | 0 | EFE   | 6494   |
| 15                                       | 25  | 0     | Inf. Dilution | dG:-1617.9(5SF) kJ | 0 | CRV   | 9015   |
| 16                                       | 25  | 0     | Inf. Dilution | dG:-382.7(4SF)     | 0 | CRV   | 12011  |
| 17                                       | 25  | 0     | Inf. Dilution | dH:-414.(0.5SD)    | 2 | ENS   | 802    |
| 18                                       | 25  | 0     | Inf. Dilution | dH:-1733.8(5SF) kJ | 2 | CRV   | 6330   |
| 19                                       | 25  | 0     | Inf. Dilution | dH:-414.3(4SF)     | 2 | MTD   | 6422   |
| 20                                       | 25  | 0     | Inf. Dilution | dH:-414.4(4SF)     | 2 | MTD   | 6494   |
| 21                                       | 25  | 0     | Inf. Dilution | dH:-1733.8(5SF) kJ | 2 | CRV   | 9015   |
| 22                                       | 25  | 0     | Inf. Dilution | dH:-414.40(5SF)    | 2 | CRV   | 12011  |
| 23                                       | 127 | 0     | Inf. Dilution | dH:-414.7(4SF)     | 1 | MTD   | 6422   |
| 24                                       | 127 | 0     | Inf. Dilution | dH:-414.7(4SF)     | 1 | MTD   | 6494   |

|    |     |   |               |                 |   |     |      |
|----|-----|---|---------------|-----------------|---|-----|------|
| 25 | 227 | 0 | Inf. Dilution | dH:-414.7 (4SF) | 0 | MTD | 6422 |
| 26 | 227 | 0 | Inf. Dilution | dH:-414.7 (4SF) | 0 | MTD | 6494 |
| 27 | 327 | 0 | Inf. Dilution | dH:-414.4 (4SF) | 0 | MTD | 6422 |
| 28 | 327 | 0 | Inf. Dilution | dH:-414.4 (4SF) | 0 | MTD | 6494 |

| Reaction No. 54838                                                    |     |       |               |                     |   |       |        |
|-----------------------------------------------------------------------|-----|-------|---------------|---------------------|---|-------|--------|
| 2<Al(s)> + 2<H2(g)> + 4.5<O2(g)> + 2<Si(s)> = Al2Si2O5(OH)4 (Hall.,s) |     |       |               |                     |   |       |        |
| No                                                                    | t   | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                                     | 25  | 0     | Inf. Dilution | lgK:662.3 (4SF)     | 4 | ENS   | 6422   |
| 2                                                                     | 27  | 0     | Inf. Dilution | lgK:657.9 (4SF)     | 0 | ENS   | 6422   |
| 3                                                                     | 127 | 0     | Inf. Dilution | lgK:479.3 (4SF)     | 3 | ENS   | 6422   |
| 4                                                                     | 227 | 0     | Inf. Dilution | lgK:372.2 (4SF)     | 2 | ENS   | 6422   |
| 5                                                                     | 327 | 0     | Inf. Dilution | lgK:300.8 (4SF)     | 0 | ENS   | 6422   |
| 6                                                                     | 25  | 0     | Inf. Dilution | dG:-903. (0.4SD)    | 2 | ENS   | 2171   |
| 7                                                                     | 25  | 0     | Inf. Dilution | dG:-903.6 (4SF)     | 4 | EFE   | 6422   |
| 8                                                                     | 25  | 0     | Inf. Dilution | dG:-898.5 (4SF)     | 0 | CRV   | 7382   |
| 9                                                                     | 25  | 0     | Inf. Dilution | dG:-3780.5 (5SF) kJ | 4 | CRV   | 9015   |
| 10                                                                    | 25  | 0     | Inf. Dilution | dG:-898.5 (4SF)     | 0 | CRV   | 12011  |
| 11                                                                    | 25  | 0     | Inf. Dilution | dH:-980. (0.3SD)    | 4 | ENS   | 2171   |
| 12                                                                    | 25  | 0     | Inf. Dilution | dH:-980.2 (4SF)     | 4 | MTD   | 6422   |
| 13                                                                    | 25  | 0     | Inf. Dilution | dH:-975.0 (4SF)     | 0 | CRV   | 7382   |
| 14                                                                    | 25  | 0     | Inf. Dilution | dH:-4101.2 (5SF) kJ | 4 | CRV   | 9015   |
| 15                                                                    | 25  | 0     | Inf. Dilution | dH:-975.1 (4SF)     | 0 | CRV   | 12011  |
| 16                                                                    | 127 | 0     | Inf. Dilution | dH:-980.5 (4SF)     | 3 | MTD   | 6422   |
| 17                                                                    | 227 | 0     | Inf. Dilution | dH:-980.2 (4SF)     | 2 | MTD   | 6422   |
| 18                                                                    | 327 | 0     | Inf. Dilution | dH:-979.6 (4SF)     | 0 | MTD   | 6422   |

| Reaction No. 54839                                           |    |       |               |                      |   |       |        |
|--------------------------------------------------------------|----|-------|---------------|----------------------|---|-------|--------|
| 2<Al(s)> + Si(s) + 3.5<O2(g)> + 2<H2(g)> = Al2O3SiOH(OH)3(s) |    |       |               |                      |   |       |        |
| No                                                           | t  | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
| 1                                                            | 25 | 0     | Inf. Dilution | dG:-700. (0.5SD)     | 2 | MCO   | 5479   |
| 2                                                            | 25 | 0     | Inf. Dilution | dG:-2921.80 (6SF) kJ | 3 | CRV   | 25435  |
| 3                                                            | 25 | 0     | Inf. Dilution | dH:-763. (0.5SD)     | 0 | MCO   | 5479   |
| 4                                                            | 25 | 0     | Inf. Dilution | dH:-3170.40 (6SF) kJ | 3 | CRV   | 25435  |

| Reaction No. 54840                                                    |  |  |  |  |  |  |  |
|-----------------------------------------------------------------------|--|--|--|--|--|--|--|
| 2<Al(s)> + 2<H2(g)> + 4.5<O2(g)> + 2<Si(s)> = Al2Si2O5(OH)4 (Kaol.,s) |  |  |  |  |  |  |  |

| No                                                                                 | t   | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
|------------------------------------------------------------------------------------|-----|-------|---------------|----------------------|---|-------|--------|
| 1                                                                                  | 25  | 0     | Inf. Dilution | lgK:665.0 (4SF)      | 1 | ELC   |        |
| 2                                                                                  | 25  | 0     | Inf. Dilution | lgK:665.7 (4SF)      | 0 | ENS   | 6422   |
| Note (2): Changed from 655.7                                                       |     |       |               |                      |   |       |        |
| 3                                                                                  | 25  | 0     | Inf. Dilution | lgK:665.3 (4SF)      | 0 | ENS   | 6494   |
| 4                                                                                  | 27  | 0     | Inf. Dilution | lgK:661.2 (4SF)      | 0 | ENS   | 6422   |
| 5                                                                                  | 27  | 0     | Inf. Dilution | lgK:660.8 (4SF)      | 0 | ENS   | 6494   |
| 6                                                                                  | 127 | 0     | Inf. Dilution | lgK:481.8 (4SF)      | 0 | ENS   | 6422   |
| 7                                                                                  | 127 | 0     | Inf. Dilution | lgK:481.5 (4SF)      | 0 | ENS   | 6494   |
| 8                                                                                  | 227 | 0     | Inf. Dilution | lgK:374.2 (4SF)      | 0 | ENS   | 6422   |
| 9                                                                                  | 227 | 0     | Inf. Dilution | lgK:373.9 (4SF)      | 0 | ENS   | 6494   |
| 10                                                                                 | 327 | 0     | Inf. Dilution | lgK:302.5 (4SF)      | 0 | ENS   | 6422   |
| 11                                                                                 | 327 | 0     | Inf. Dilution | lgK:302.2 (4SF)      | 0 | ENS   | 6494   |
| 12                                                                                 | 25  | 0     | Inf. Dilution | dG:-3789.10 (6SF) kJ | 0 | CRV   | 440    |
| 13                                                                                 | 25  | 0     | Inf. Dilution | dG:-908.1 (0.2SD)    | 0 | ENS   | 2171   |
| 14                                                                                 | 25  | 0     | Inf. Dilution | dG:-3799.40 (6SF) kJ | 0 | CRV   | 5619   |
| 15                                                                                 | 25  | 0     | Inf. Dilution | dG:-3777.60 (6SF) kJ | 0 | CRV   | 6421   |
| 16                                                                                 | 25  | 0     | Inf. Dilution | dG:-908.1 (4SF)      | 0 | EFE   | 6422   |
| 17                                                                                 | 25  | 0     | Inf. Dilution | dG:-908.17 (5SF)     | 0 | EFE   | 6482   |
| 18                                                                                 | 25  | 0     | Inf. Dilution | dG:-907.6 (4SF)      | 0 | EFE   | 6494   |
| 19                                                                                 | 25  | 0     | Inf. Dilution | dG:-3789.5 (5SF) kJ  | 0 | MSL   | 7380   |
| 20                                                                                 | 25  | 0     | Inf. Dilution | dG:-903.0 (4SF)      | 0 | CRV   | 7382   |
| 21                                                                                 | 25  | 0     | Inf. Dilution | dG:-3789.40 (6SF) kJ | 0 | CRV   | 8872   |
| 22                                                                                 | 25  | 0     | Inf. Dilution | dG:-3778.20 (6SF) kJ | 0 | EOD   | 8872   |
| Note (22): Data from Naumov, Ryzhenko & Khodakovsky, 1971; low wgt for consistency |     |       |               |                      |   |       |        |
| 23                                                                                 | 25  | 0     | Inf. Dilution | dG:-3776.90 (6SF) kJ | 0 | CRV   | 8873   |
| 24                                                                                 | 25  | 0     | Inf. Dilution | dG:-3780.00 (6SF) kJ | 0 | CRV   | 8874   |
| 25                                                                                 | 25  | 0     | Inf. Dilution | dG:-3799.7 (5SF) kJ  | 0 | CRV   | 9015   |
| 26                                                                                 | 25  | 0     | Inf. Dilution | dG:-908.50 (5SF)     | 0 | CRV   | 10174  |
| 27                                                                                 | 25  | 0     | Inf. Dilution | dG:-903.0 (4SF)      | 0 | CRV   | 12011  |
| 28                                                                                 | 25  | 0     | Inf. Dilution | dG:-3801.70 (6SF) kJ | 0 | CRV   | 25435  |
| 29                                                                                 | 25  | 0     | Inf. Dilution | dG:-907.88 (5SF)     | 5 | CRV   | 29553  |
| 30                                                                                 | 25  | 0     | GAMSPATH      | dG:-3789. (4SF) kJ   | 0 | CRV   | 12638  |
| 31                                                                                 | 50  | 0     | Inf. Dilution | dG:-909.76 (5SF)     | 0 | CRV   | 10174  |
| 32                                                                                 | 75  | 0     | Inf. Dilution | dG:-911.15 (5SF)     | 0 | CRV   | 10174  |
| 33                                                                                 | 100 | 0     | Inf. Dilution | dG:-912.64 (5SF)     | 0 | CRV   | 10174  |

|    |     |   |               |                      |   |     |       |
|----|-----|---|---------------|----------------------|---|-----|-------|
| 34 | 125 | 0 | Inf. Dilution | dG:-914.26 (5SF)     | 0 | CRV | 10174 |
| 35 | 150 | 0 | Inf. Dilution | dG:-915.97 (5SF)     | 0 | CRV | 10174 |
| 36 | 175 | 0 | Inf. Dilution | dG:-917.80 (5SF)     | 0 | CRV | 10174 |
| 37 | 200 | 0 | Inf. Dilution | dG:-919.72 (5SF)     | 0 | CRV | 10174 |
| 38 | 225 | 0 | Inf. Dilution | dG:-921.73 (5SF)     | 0 | CRV | 10174 |
| 39 | 250 | 0 | Inf. Dilution | dG:-923.83 (5SF)     | 0 | CRV | 10174 |
| 40 | 300 | 0 | Inf. Dilution | dG:-928.28 (5SF)     | 0 | CRV | 10174 |
| 41 | 25  | 0 | Inf. Dilution | dH:-985. (0.3SD)     | 0 | ENS | 2171  |
| 42 | 25  | 0 | Inf. Dilution | dH:-984.6 (4SF)      | 0 | MTD | 6422  |
| 43 | 25  | 0 | Inf. Dilution | dH:-984.8 (4SF)      | 0 | ENS | 6437  |
| 44 | 25  | 0 | Inf. Dilution | dH:-984.78 (5SF)     | 0 | EFE | 6482  |
| 45 | 25  | 0 | Inf. Dilution | dH:-984.5 (4SF)      | 0 | MTD | 6494  |
| 46 | 25  | 0 | Inf. Dilution | dH:-979.6 (4SF)      | 0 | CRV | 7382  |
| 47 | 25  | 0 | Inf. Dilution | dH:-4119.6 (5SF) kJ  | 0 | CRV | 9015  |
| 48 | 25  | 0 | Inf. Dilution | dH:-976.6 (4SF)      | 0 | CRV | 12011 |
| 49 | 25  | 0 | Inf. Dilution | dH:-4122.30 (6SF) kJ | 0 | CRV | 25435 |
| 50 | 25  | 0 | Inf. Dilution | dH:-984.53 (5SF)     | 5 | CRV | 29553 |
| 51 | 25  | 0 | GAMSPATH      | dH:-4109.60 (6SF) kJ | 0 | CRV | 12638 |
| 52 | 127 | 0 | Inf. Dilution | dH:-984.9 (4SF)      | 0 | MTD | 6422  |
| 53 | 127 | 0 | Inf. Dilution | dH:-984.8 (4SF)      | 0 | MTD | 6494  |
| 54 | 227 | 0 | Inf. Dilution | dH:-984.6 (4SF)      | 0 | MTD | 6422  |
| 55 | 227 | 0 | Inf. Dilution | dH:-984.6 (4SF)      | 0 | MTD | 6494  |
| 56 | 327 | 0 | Inf. Dilution | dH:-983.9 (4SF)      | 0 | MTD | 6422  |
| 57 | 327 | 0 | Inf. Dilution | dH:-984.1 (4SF)      | 0 | MTD | 6494  |

## Reaction No. 54851

PO4-3 + Al+3 = Al+3\_PO4-3

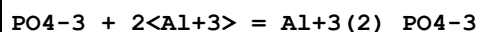
| No | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:16.0 (3SF)     | 1 | EES   |        |
| 2  | 25 | 0.15  | NaCl/m        | lgK:14.07 (0.01SD) | 3 | MGL   | 29916  |
| 3  | 37 | 0.15  | NaCl          | lgK:16. (0.4SD)    | 0 | MGL   | 922    |
| 4  | 37 | 0.15  | NaCl          | lgK:15.32 (4SF)    | 2 | MGL   | 10028  |
| 5  | 37 | 0.15  | Unknown       | lgK:15.9 (3SF)     | 0 | CRV   | 2556   |

## Reaction No. 54852

Al+3(2)\_OH-1(2)\_PO4-3 + 2&lt;H+1&gt; = Al+3(2)\_PO4-3 + 2&lt;H2O&gt;

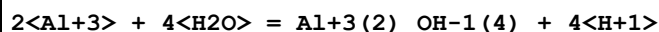
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:7.0(2SF)     | 2 | EES   |        |
| 2  | 25 | 0.2   | KCl           | lgK:7.42(0.04SD) | 2 | MGL   | 7048   |
| 3  | 37 | 0.15  | Na+1 salt     | lgK:6.14(3SF)    | 2 | CRV   | 7271   |

## Reaction No. 54854



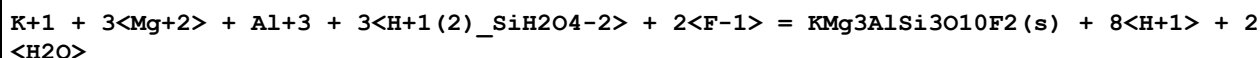
| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:16.8(3SF)     | 1 | ELC   |        |
| 2  | 25 | 0.2   | KCl           | lgK:16.65(0.18SD) | 0 | MGL   | 7048   |
| 3  | 37 | 0.15  | NaCl          | lgK:20.9(0.1SD)   | 0 | MGL   | 922    |
| 4  | 37 | 0.15  | NaCl          | lgK:18.7(3SF)     | 0 | MIS   | 5983   |

## Reaction No. 54883



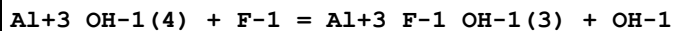
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 3     | NaCl   | lgK:-17.(0.3SD)  | 0 | MGL   | 5061   |
| 2  | 25 | 3     | NaClO4 | lgK:-15.5(0.2SD) | 0 | MGL   | 5485   |
| 3  | 25 | 3     | NaNO3  | lgK:-16.4(0.1SD) | 4 | MGL   | 5060   |

## Reaction No. 55374



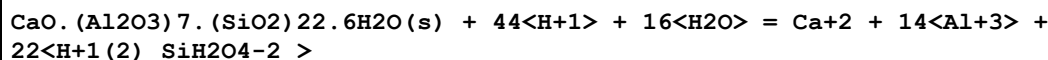
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-21.7(3SF)   | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:-7.8(0.05SD) | 0 | ENS   | 557    |

## Reaction No. 55731



| No                                                 | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| 1                                                  | 25 | 0.1   | KNO3   | lgK:-3.73(3SF)    | 0 | ENS   | 6953   |
| Note (1): Reaction as stated in ref. is unbalanced |    |       |        |                   |   |       |        |
| 2                                                  | 25 | 0.5   | KNO3   | lgK:-2.20(0.13SD) | 4 | MPT   | 5222   |
| 3                                                  | 25 | 0.5   | KNO3   | lgK:-2.20(3SF)    | 0 | ENS   | 6953   |

## Reaction No. 56091



| No | t  | I-Str | Medium   | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|----------|-------------------|---|-------|--------|
| 1  | 25 | 0     | CHEMVAL1 | lgK:36.84(0.01SD) | 0 | ENS   | 5156   |
| 2  | 25 | 0     | CHEMVAL2 | lgK:36.84(0.01SD) | 0 | ENS   | 5156   |
| 3  | 25 | 0     | CHEMVAL3 | lgK:36.84(0.01SD) | 0 | ENS   | 5156   |
| 4  | 25 | 0     | CHEMVAL4 | lgK:36.80(0.01SD) | 4 | ENS   | 5156   |

| Reaction No. 56155                                                                                                                                                                                                                                            |    |       |          |                   |   |       |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|----------|-------------------|---|-------|--------|
| $\text{Ca}_2\text{Al}_2\text{Si}_4\text{O}_{12} \cdot 4\text{H}_2\text{O}(\text{s}) + 8\langle\text{H}+1\rangle - 2\langle\text{e}-1\rangle = 2\langle\text{Ca}+2\rangle + 2\langle\text{Al}+3\rangle + 4\langle\text{H}+1(2)\text{SiH}_2\text{O}_4-2\rangle$ |    |       |          |                   |   |       |        |
| No                                                                                                                                                                                                                                                            | t  | I-Str | Medium   | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                                                                                                                                                             | 25 | 0     | CHEMVAL1 | lgK:11.85(0.01SD) | 0 | ENS   | 5156   |
| 2                                                                                                                                                                                                                                                             | 25 | 0     | CHEMVAL2 | lgK:11.85(0.01SD) | 0 | ENS   | 5156   |
| 3                                                                                                                                                                                                                                                             | 25 | 0     | CHEMVAL3 | lgK:11.85(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                                                                                                                                                                                                             | 25 | 0     | CHEMVAL4 | lgK:11.80(0.01SD) | 1 | ENS   | 5156   |
| Note (4): query mineral "laumontite"                                                                                                                                                                                                                          |    |       |          |                   |   |       |        |

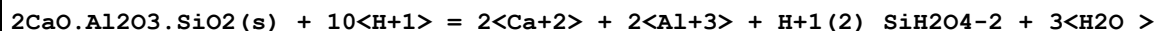
| Reaction No. 56156                                                                                                                                                                                   |    |       |               |                   |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2(\text{s}) + 8\langle\text{H}+1\rangle = \text{Ca}+2 + 2\langle\text{Al}+3\rangle + 2\langle\text{H}+1(2)\text{SiH}_2\text{O}_4-2\rangle$ |    |       |               |                   |   |       |        |
| No                                                                                                                                                                                                   | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                                                                                                    | 25 | 0     | Inf. Dilution | lgK:26.2(3SF)     | 1 | ELC   |        |
| 2                                                                                                                                                                                                    | 25 | 0     | Inf. Dilution | lgK:27.19(4SF)    | 0 | CRV   | 32224  |
| 3                                                                                                                                                                                                    | 25 | 0     | CHEMVAL1      | lgK:27.06(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                                                                                                                                                    | 25 | 0     | CHEMVAL2      | lgK:27.06(0.01SD) | 0 | ENS   | 5156   |
| 5                                                                                                                                                                                                    | 25 | 0     | CHEMVAL3      | lgK:27.06(0.01SD) | 0 | ENS   | 5156   |
| 6                                                                                                                                                                                                    | 25 | 0     | CHEMVAL4      | lgK:23.00(0.01SD) | 0 | ENS   | 5156   |

| Reaction No. 56157                                                                                                                                                                                                                                                |    |       |          |                   |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|----------|-------------------|---|-------|--------|
| $\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 4\text{SiO}_2 \cdot \text{H}_2\text{O}(\text{s}) + 8\langle\text{H}+1\rangle + 3\langle\text{H}_2\text{O}\rangle = \text{Ca}+2 + 2\langle\text{Al}+3\rangle + 4\langle\text{H}+1(2)\text{SiH}_2\text{O}_4-2\rangle$ |    |       |          |                   |   |       |        |
| No                                                                                                                                                                                                                                                                | t  | I-Str | Medium   | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                                                                                                                                                                 | 25 | 0     | CHEMVAL1 | lgK:17.46(0.01SD) | 0 | ENS   | 5156   |
| 2                                                                                                                                                                                                                                                                 | 25 | 0     | CHEMVAL2 | lgK:17.46(0.01SD) | 0 | ENS   | 5156   |
| 3                                                                                                                                                                                                                                                                 | 25 | 0     | CHEMVAL3 | lgK:17.46(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                                                                                                                                                                                                                 | 25 | 0     | CHEMVAL4 | lgK:17.50(0.01SD) | 3 | ENS   | 5156   |

| Reaction No. 56158                                                                                                                                                                                                                                                 |    |       |               |               |   |       |        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2 \cdot 2\text{H}_2\text{O}(\text{s}) + 8\langle\text{H}+1\rangle = \text{Ca}+2 + 2\langle\text{Al}+3\rangle + 2\langle\text{H}+1(2)\text{SiH}_2\text{O}_4-2\rangle + 2\langle\text{H}_2\text{O}\rangle$ |    |       |               |               |   |       |        |
| No                                                                                                                                                                                                                                                                 | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                                                                                                                                                                                                  | 25 | 0     | Inf. Dilution | lgK:21.0(3SF) | 1 | ELC   |        |

|   |    |   |               |                   |   |     |       |
|---|----|---|---------------|-------------------|---|-----|-------|
| 2 | 25 | 0 | Inf. Dilution | lgK:22.81(4SF)    | 0 | CRV | 32224 |
| 3 | 25 | 0 | CHEMVAL1      | lgK:22.06(0.01SD) | 0 | ENS | 5156  |
| 4 | 25 | 0 | CHEMVAL2      | lgK:22.06(0.01SD) | 0 | ENS | 5156  |
| 5 | 25 | 0 | CHEMVAL3      | lgK:22.06(0.01SD) | 0 | ENS | 5156  |
| 6 | 25 | 0 | CHEMVAL4      | lgK:17.00(0.01SD) | 0 | ENS | 5156  |

## Reaction No. 56159



| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:57.0(3SF)     | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:56.88(4SF)    | 4 | CRV   | 32224  |
| 3  | 25 | 0     | CHEMVAL1      | lgK:56.42(0.01SD) | 0 | ENS   | 5156   |
| 4  | 25 | 0     | CHEMVAL2      | lgK:56.42(0.01SD) | 0 | ENS   | 5156   |
| 5  | 25 | 0     | CHEMVAL3      | lgK:56.42(0.01SD) | 0 | ENS   | 5156   |
| 6  | 25 | 0     | CHEMVAL4      | lgK:52.00(0.01SD) | 0 | ENS   | 5156   |

## Reaction No. 56160



| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:34.6(3SF)     | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:33.51(4SF)    | 0 | CRV   | 32224  |
| 3  | 25 | 0     | CHEMVAL1      | lgK:32.41(0.01SD) | 0 | ENS   | 5156   |
| 4  | 25 | 0     | CHEMVAL2      | lgK:32.41(0.01SD) | 0 | ENS   | 5156   |
| 5  | 25 | 0     | CHEMVAL3      | lgK:32.41(0.01SD) | 0 | ENS   | 5156   |
| 6  | 25 | 0     | CHEMVAL4      | lgK:32.40(0.01SD) | 0 | ENS   | 5156   |

## Reaction No. 56161



| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:41.2(3SF)     | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:44.20(4SF)    | 0 | CRV   | 32224  |
| 3  | 25 | 0     | CHEMVAL1      | lgK:39.83(0.01SD) | 0 | ENS   | 5156   |
| 4  | 25 | 0     | CHEMVAL2      | lgK:39.83(0.01SD) | 0 | ENS   | 5156   |
| 5  | 25 | 0     | CHEMVAL3      | lgK:39.83(0.01SD) | 0 | ENS   | 5156   |
| 6  | 25 | 0     | CHEMVAL4      | lgK:39.80(0.01SD) | 0 | ENS   | 5156   |

## Reaction No. 56162

| Ca <sub>2</sub> Al <sub>4</sub> Si <sub>8</sub> O <sub>24</sub> ·7H <sub>2</sub> O(s) + 16<H+1> + H <sub>2</sub> O = 2<Ca+2> + 4<Al+3> + 8<H+1(2)_SiH <sub>2</sub> O <sub>4</sub> -2 > |    |       |          |                   |   |       |        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|----------|-------------------|---|-------|--------|
| No                                                                                                                                                                                     | t  | I-Str | Medium   | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                                                                                      | 25 | 0     | CHEMVAL1 | lgK:25.06(0.01SD) | 0 | ENS   | 5156   |
| 2                                                                                                                                                                                      | 25 | 0     | CHEMVAL2 | lgK:25.06(0.01SD) | 0 | ENS   | 5156   |
| 3                                                                                                                                                                                      | 25 | 0     | CHEMVAL3 | lgK:25.06(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                                                                                                                                      | 25 | 0     | CHEMVAL4 | lgK:16.60(0.01SD) | 3 | ENS   | 5156   |

| Reaction No. 56164                                                                                                                       |    |       |               |                   |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| 3CaO·Al <sub>2</sub> O <sub>3</sub> ·3SiO <sub>2</sub> (s) + 12<H+1> = 3<Ca+2> + 2<Al+3> + 3<H+1(2)_SiH <sub>2</sub> O <sub>4</sub> -2 > |    |       |               |                   |   |       |        |
| No                                                                                                                                       | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                                        | 25 | 0     | Inf. Dilution | lgK:51.4(3SF)     | 1 | ELC   |        |
| 2                                                                                                                                        | 25 | 0     | Inf. Dilution | lgK:52.47(4SF)    | 4 | CRV   | 32224  |
| 3                                                                                                                                        | 25 | 0     | CHEMVAL1      | lgK:57.21(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                                                                                        | 25 | 0     | CHEMVAL2      | lgK:57.21(0.01SD) | 0 | ENS   | 5156   |
| 5                                                                                                                                        | 25 | 0     | CHEMVAL3      | lgK:57.21(0.01SD) | 0 | ENS   | 5156   |
| 6                                                                                                                                        | 25 | 0     | CHEMVAL4      | lgK:52.60(0.01SD) | 0 | ENS   | 5156   |

| Reaction No. 56170                                                                                                                                    |    |       |               |                   |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| CaO·Al <sub>2</sub> O <sub>3</sub> ·SiO <sub>2</sub> (s) + 8<H+1> = Ca+2 + 2<Al+3> + H+1(2)_SiH <sub>2</sub> O <sub>4</sub> -2 + 2<H <sub>2</sub> O > |    |       |               |                   |   |       |        |
| No                                                                                                                                                    | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                                                     | 25 | 0     | Inf. Dilution | lgK:34.2(3SF)     | 1 | ELC   |        |
| 2                                                                                                                                                     | 25 | 0     | Inf. Dilution | lgK:36.59(4SF)    | 4 | CRV   | 32224  |
| 3                                                                                                                                                     | 25 | 0     | CHEMVAL1      | lgK:33.71(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                                                                                                     | 25 | 0     | CHEMVAL2      | lgK:33.71(0.01SD) | 0 | ENS   | 5156   |
| 5                                                                                                                                                     | 25 | 0     | CHEMVAL3      | lgK:33.71(0.01SD) | 0 | ENS   | 5156   |
| 6                                                                                                                                                     | 25 | 0     | CHEMVAL4      | lgK:33.60(0.01SD) | 0 | ENS   | 5156   |

| Reaction No. 56173                                                                     |    |       |               |                   |   |       |        |
|----------------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| MgO·Al <sub>2</sub> O <sub>3</sub> (s) + 8<H+1> = Mg+2 + 2<Al+3> + 4<H <sub>2</sub> O> |    |       |               |                   |   |       |        |
| No                                                                                     | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                      | 25 | 0     | Inf. Dilution | lgK:36.9(3SF)     | 1 | ELC   |        |
| 2                                                                                      | 25 | 0     | Inf. Dilution | lgK:38.42(4SF)    | 0 | CRV   | 32224  |
| 3                                                                                      | 25 | 0     | CHEMVAL1      | lgK:37.02(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                                      | 25 | 0     | CHEMVAL2      | lgK:37.02(0.01SD) | 0 | ENS   | 5156   |
| 5                                                                                      | 25 | 0     | CHEMVAL3      | lgK:37.02(0.01SD) | 0 | ENS   | 5156   |
| 6                                                                                      | 25 | 0     | CHEMVAL4      | lgK:36.30(0.01SD) | 0 | ENS   | 5156   |



| Reaction No. 56187                                                                                                                                                                                                                            |    |       |          |                   |   |       |        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|----------|-------------------|---|-------|--------|
| $\text{MgAl}_{14}\text{Si}_{22}\text{O}_{72}\text{H}_{12}(\text{s}) + 16\langle\text{H}_2\text{O}\rangle + 44\langle\text{H}+1\rangle = \text{Mg}+2 + 14\langle\text{Al}+3\rangle + 22\langle\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}\rangle$ |    |       |          |                   |   |       |        |
| No                                                                                                                                                                                                                                            | t  | I-Str | Medium   | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                                                                                                                                             | 25 | 0     | CHEMVAL1 | lgK:36.24(0.01SD) | 0 | ENS   | 5156   |
| 2                                                                                                                                                                                                                                             | 25 | 0     | CHEMVAL2 | lgK:36.24(0.01SD) | 0 | ENS   | 5156   |
| 3                                                                                                                                                                                                                                             | 25 | 0     | CHEMVAL3 | lgK:36.24(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                                                                                                                                                                                             | 25 | 0     | CHEMVAL4 | lgK:36.20(0.01SD) | 4 | ENS   | 5156   |

| Reaction No. 56193                                                                                                                                                                                                                                            |    |       |               |                   |   |       |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $2\text{MgO} \cdot 2\text{Al}_2\text{O}_3 \cdot 5\text{SiO}_2(\text{s}) + 16\langle\text{H}+1\rangle + 2\langle\text{H}_2\text{O}\rangle = 2\langle\text{Mg}+2\rangle + 4\langle\text{Al}+3\rangle + 5\langle\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}\rangle$ |    |       |               |                   |   |       |        |
| No                                                                                                                                                                                                                                                            | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                                                                                                                                                             | 25 | 0     | Inf. Dilution | lgK:50.6(3SF)     | 1 | ELC   |        |
| 2                                                                                                                                                                                                                                                             | 25 | 0     | Inf. Dilution | lgK:45.5(0.05SD)  | 0 | ENS   | 557    |
| 3                                                                                                                                                                                                                                                             | 25 | 0     | Inf. Dilution | lgK:53.88(4SF)    | 0 | CRV   | 32224  |
| 4                                                                                                                                                                                                                                                             | 25 | 0     | CHEMVAL1      | lgK:53.77(0.01SD) | 0 | ENS   | 5156   |
| 5                                                                                                                                                                                                                                                             | 25 | 0     | CHEMVAL2      | lgK:53.77(0.01SD) | 0 | ENS   | 5156   |
| 6                                                                                                                                                                                                                                                             | 25 | 0     | CHEMVAL3      | lgK:53.77(0.01SD) | 0 | ENS   | 5156   |
| 7                                                                                                                                                                                                                                                             | 25 | 0     | CHEMVAL4      | lgK:49.40(0.01SD) | 0 | ENS   | 5156   |

| Reaction No. 56197                                                                                                                                                                 |    |       |               |                   |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $\text{Al}+3_{\text{K}+1_{\text{SO}_4-2(2)_{\text{H}_2\text{O}(12)}_{\text{(s)}}}} = \text{K}+1 + \text{Al}+3 + 2\langle\text{SO}_4-2\rangle + 12\langle\text{H}_2\text{O}\rangle$ |    |       |               |                   |   |       |        |
| No                                                                                                                                                                                 | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                                                                                  | 25 | 0     | Inf. Dilution | lgK:-5.17(3SF)    | 0 | CRV   | 32224  |
| 2                                                                                                                                                                                  | 25 | 0     | CHEMVAL1      | lgK:-5.17(0.01SD) | 0 | ENS   | 5156   |
| 3                                                                                                                                                                                  | 25 | 0     | CHEMVAL2      | lgK:-5.17(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                                                                                                                                  | 25 | 0     | CHEMVAL3      | lgK:-5.17(0.01SD) | 0 | ENS   | 5156   |
| 5                                                                                                                                                                                  | 25 | 0     | CHEMVAL4      | lgK:-6.68(0.01SD) | 4 | ENS   | 5156   |

| Reaction No. 56198                                                                                                                                                                                     |    |       |               |                  |   |       |        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| $\text{KAlSi}_3\text{O}_8(\text{Micr.},\text{s}) + 4\langle\text{H}_2\text{O}\rangle + 4\langle\text{H}+1\rangle = \text{K}+1 + \text{Al}+3 + 3\langle\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}\rangle$ |    |       |               |                  |   |       |        |
| No                                                                                                                                                                                                     | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                                                                                                                      | 25 | 0     | Inf. Dilution | lgK:1.2(2SF)     | 1 | ELC   |        |
| 2                                                                                                                                                                                                      | 25 | 0     | CHEMVAL1      | lgK:1.29(0.01SD) | 0 | ENS   | 5156   |
| 3                                                                                                                                                                                                      | 25 | 0     | CHEMVAL2      | lgK:1.29(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                                                                                                                                                      | 25 | 0     | CHEMVAL3      | lgK:1.29(0.01SD) | 0 | ENS   | 5156   |
| 5                                                                                                                                                                                                      | 25 | 0     | CHEMVAL4      | lgK:0.88(0.01SD) | 0 | ENS   | 5156   |

| Reaction No. 56199                                                                                                                                                                                                                     |    |       |          |                   |   |       |        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|----------|-------------------|---|-------|--------|
| $\text{KAl}_7\text{Si}_{11}\text{O}_{30}(\text{OH})_6(\text{s}) + 8\langle\text{H}_2\text{O}\rangle + 22\langle\text{H}+1\rangle = \text{K}+1 + 7\langle\text{Al}+3\rangle + 11\langle\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}\rangle$ |    |       |          |                   |   |       |        |
| No                                                                                                                                                                                                                                     | t  | I-Str | Medium   | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                                                                                                                                      | 25 | 0     | CHEMVAL1 | lgK:17.58(0.01SD) | 0 | ENS   | 5156   |
| 2                                                                                                                                                                                                                                      | 25 | 0     | CHEMVAL2 | lgK:17.58(0.01SD) | 0 | ENS   | 5156   |
| 3                                                                                                                                                                                                                                      | 25 | 0     | CHEMVAL3 | lgK:17.58(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                                                                                                                                                                                      | 25 | 0     | CHEMVAL4 | lgK:17.60(0.01SD) | 4 | ENS   | 5156   |

| Reaction No. 56200                                                                                                   |    |       |               |                  |   |       |        |
|----------------------------------------------------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| $\text{Al}+3_{\text{K}+1_{\text{SO}_4-2(2)}_{\text{(s)}}} = \text{K}+1 + \text{Al}+3 + 2\langle\text{SO}_4-2\rangle$ |    |       |               |                  |   |       |        |
| No                                                                                                                   | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                                    | 25 | 0     | Inf. Dilution | lgK:3.77(3SF)    | 4 | CRV   | 32224  |
| 2                                                                                                                    | 25 | 0     | CHEMVAL1      | lgK:3.77(0.01SD) | 0 | ENS   | 5156   |
| 3                                                                                                                    | 25 | 0     | CHEMVAL2      | lgK:3.77(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                                                                    | 25 | 0     | CHEMVAL3      | lgK:3.77(0.01SD) | 0 | ENS   | 5156   |
| 5                                                                                                                    | 25 | 0     | CHEMVAL4      | lgK:3.74(0.01SD) | 4 | ENS   | 5156   |

| Reaction No. 56201                                                                                                                                                                        |    |       |               |                  |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| $\text{KAlSi}_2\text{O}_6(\text{s}) + 4\langle\text{H}+1\rangle + 2\langle\text{H}_2\text{O}\rangle = 2\langle\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}\rangle + \text{Al}+3 + \text{K}+1$ |    |       |               |                  |   |       |        |
| No                                                                                                                                                                                        | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                                                                                                         | 25 | 0     | Inf. Dilution | lgK:7.2(2SF)     | 1 | ELC   |        |
| 2                                                                                                                                                                                         | 25 | 0     | Inf. Dilution | lgK:6.42(3SF)    | 0 | EOD   | 23102  |
| 3                                                                                                                                                                                         | 25 | 0     | CHEMVAL1      | lgK:9.78(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                                                                                                                                         | 25 | 0     | CHEMVAL2      | lgK:9.78(0.01SD) | 0 | ENS   | 5156   |
| 5                                                                                                                                                                                         | 25 | 0     | CHEMVAL3      | lgK:9.78(0.01SD) | 0 | ENS   | 5156   |
| 6                                                                                                                                                                                         | 25 | 0     | CHEMVAL4      | lgK:6.61(0.01SD) | 0 | ENS   | 5156   |

| Reaction No. 56202                                                                                                            |    |       |               |                   |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $\text{KAlSiO}_4(\text{s}) + 4\langle\text{H}+1\rangle = \text{K}+1 + \text{Al}+3 + \text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}$ |    |       |               |                   |   |       |        |
| No                                                                                                                            | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                             | 25 | 0     | Inf. Dilution | lgK:13.4(3SF)     | 1 | ELC   |        |
| 2                                                                                                                             | 25 | 0     | Inf. Dilution | lgK:11.23(4SF)    | 0 | CRV   | 32224  |
| 3                                                                                                                             | 25 | 0     | CHEMVAL1      | lgK:12.79(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                                                                             | 25 | 0     | CHEMVAL2      | lgK:12.79(0.01SD) | 0 | ENS   | 5156   |
| 5                                                                                                                             | 25 | 0     | CHEMVAL3      | lgK:12.79(0.01SD) | 0 | ENS   | 5156   |
| 6                                                                                                                             | 25 | 0     | CHEMVAL4      | lgK:13.00(0.01SD) | 0 | ENS   | 5156   |

| Reaction No. 56203                                                |    |       |               |                   |   |       |        |
|-------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| KAl3Si3O10(OH)2(s) + 10<H+1> = K+1 + 3<Al+3> + 3<H+1(2)_SiH2O4-2> |    |       |               |                   |   |       |        |
| No                                                                | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                 | 25 | 0     | Inf. Dilution | lgK:13.8(3SF)     | 1 | ELC   |        |
| 2                                                                 | 25 | 0     | Inf. Dilution | lgK:14.56(4SF)    | 0 | CRV   | 32224  |
| 3                                                                 | 25 | 0     | CHEMVAL1      | lgK:13.96(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                 | 25 | 0     | CHEMVAL2      | lgK:13.96(0.01SD) | 0 | ENS   | 5156   |
| 5                                                                 | 25 | 0     | CHEMVAL3      | lgK:13.96(0.01SD) | 0 | ENS   | 5156   |
| 6                                                                 | 25 | 0     | CHEMVAL4      | lgK:13.00(0.01SD) | 0 | ENS   | 5156   |

| Reaction No. 56208                                                  |    |       |               |                   |   |       |        |
|---------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| Al+3_Na+1_OH-1(2)_CO3-2_(s) + 2<H+1> = Na+1 + Al+3 + CO3-2 + 2<H2O> |    |       |               |                   |   |       |        |
| No                                                                  | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                   | 25 | 0     | Inf. Dilution | lgK:-4.8(2SF)     | 1 | ELC   |        |
| 2                                                                   | 25 | 0     | Inf. Dilution | lgK:-5.67(3SF)    | 0 | CRV   | 32224  |
| 3                                                                   | 25 | 0     | CHEMVAL1      | lgK:-5.67(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                   | 25 | 0     | CHEMVAL2      | lgK:-5.67(0.01SD) | 0 | ENS   | 5156   |
| 5                                                                   | 25 | 0     | CHEMVAL3      | lgK:-5.67(0.01SD) | 0 | ENS   | 5156   |
| 6                                                                   | 25 | 0     | CHEMVAL4      | lgK:-5.70(0.01SD) | 0 | ENS   | 5156   |

| Reaction No. 56214                                                          |    |       |               |                  |   |       |        |
|-----------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| NaAlSi3O8(AlbiteHI,s) + 4<H+1> + 4<H2O> = Na+1 + Al+3 + 3<H+1(2)_SiH2O4-2 > |    |       |               |                  |   |       |        |
| No                                                                          | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                           | 25 | 0     | Inf. Dilution | lgK:4.0(2SF)     | 1 | ELC   |        |
| 2                                                                           | 25 | 0     | Inf. Dilution | lgK:4.41(3SF)    | 4 | CRV   | 32224  |
| 3                                                                           | 25 | 0     | CHEMVAL1      | lgK:7.02(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                           | 25 | 0     | CHEMVAL2      | lgK:7.02(0.01SD) | 0 | ENS   | 5156   |
| 5                                                                           | 25 | 0     | CHEMVAL3      | lgK:7.02(0.01SD) | 0 | ENS   | 5156   |
| 6                                                                           | 25 | 0     | CHEMVAL4      | lgK:7.02(0.01SD) | 0 | ENS   | 5156   |

| Reaction No. 56215                                                              |    |       |          |                   |   |       |        |
|---------------------------------------------------------------------------------|----|-------|----------|-------------------|---|-------|--------|
| NaAl7Si11O30(OH)6(s) + 8<H2O> + 22<H+1> = Na+1 + 7<Al+3> + 11<H+1(2)_SiH2O4-2 > |    |       |          |                   |   |       |        |
| No                                                                              | t  | I-Str | Medium   | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                               | 25 | 0     | CHEMVAL1 | lgK:18.36(0.01SD) | 0 | ENS   | 5156   |
| 2                                                                               | 25 | 0     | CHEMVAL2 | lgK:18.36(0.01SD) | 0 | ENS   | 5156   |
| 3                                                                               | 25 | 0     | CHEMVAL3 | lgK:18.36(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                               | 25 | 0     | CHEMVAL4 | lgK:18.70(0.01SD) | 4 | ENS   | 5156   |

| Reaction No. 56216                                                                                         |    |       |               |                    |   |       |        |
|------------------------------------------------------------------------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| $\text{Al}+3\_ \text{Na}+1(3)\_ \text{F}-1(6)\_ (\text{s}) = 3<\text{Na}+1> + \text{Al}+3 + 6<\text{F}-1>$ |    |       |               |                    |   |       |        |
| No                                                                                                         | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                                                          | 25 | 0     | Inf. Dilution | lgK:-33.7(3SF)     | 1 | ELC   |        |
| 2                                                                                                          | 25 | 0     | Inf. Dilution | lgK:-33.8(0.1SD)   | 0 | ENS   | 5331   |
| 3                                                                                                          | 25 | 0     | Inf. Dilution | lgK:-33.84(4SF)    | 3 | ENS   | 6496   |
| 4                                                                                                          | 25 | 0     | Inf. Dilution | lgK:-32.14(4SF)    | 0 | CRV   | 32224  |
| 5                                                                                                          | 25 | 0     | CHEMVAL1      | lgK:-32.14(0.01SD) | 0 | ENS   | 5156   |
| 6                                                                                                          | 25 | 0     | CHEMVAL2      | lgK:-32.14(0.01SD) | 0 | ENS   | 5156   |
| 7                                                                                                          | 25 | 0     | CHEMVAL3      | lgK:-32.14(0.01SD) | 0 | ENS   | 5156   |
| 8                                                                                                          | 25 | 0     | CHEMVAL4      | lgK:-31.50(0.01SD) | 0 | ENS   | 5156   |
| 9                                                                                                          | 25 | 0     | Inf. Dilution | dH:9.09(3SF)       | 2 | ENS   | 6496   |

| Reaction No. 56217                                                                                                                                                     |    |       |               |                   |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $\text{NaAlSi}_2\text{O}_6(\text{Dehyd. Anal., s}) + 2<\text{H}_2\text{O}> + 4<\text{H}+1> = \text{Na}+1 + \text{Al}+3 + 2<\text{H}+1(2)\_ \text{SiH}_2\text{O}_4-2 >$ |    |       |               |                   |   |       |        |
| No                                                                                                                                                                     | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                                                                      | 25 | 0     | Inf. Dilution | lgK:13.6(3SF)     | 1 | ELC   |        |
| 2                                                                                                                                                                      | 25 | 0     | CHEMVAL1      | lgK:15.20(0.01SD) | 0 | ENS   | 5156   |
| 3                                                                                                                                                                      | 25 | 0     | CHEMVAL2      | lgK:15.20(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                                                                                                                      | 25 | 0     | CHEMVAL3      | lgK:15.20(0.01SD) | 0 | ENS   | 5156   |
| 5                                                                                                                                                                      | 25 | 0     | CHEMVAL4      | lgK:15.20(0.01SD) | 0 | ENS   | 5156   |

| Reaction No. 56218                                                                                                                                                      |    |       |               |                   |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $\text{NaAlSi}_2\text{O}_6.\text{H}_2\text{O}(\text{s}) + 4<\text{H}+1> + \text{H}_2\text{O} = \text{Na}+1 + \text{Al}+3 + 2<\text{H}+1(2)\_ \text{SiH}_2\text{O}_4-2>$ |    |       |               |                   |   |       |        |
| No                                                                                                                                                                      | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                                                                       | 25 | 0     | Inf. Dilution | lgK:9.0(2SF)      | 1 | ELC   |        |
| 2                                                                                                                                                                       | 25 | 0     | Inf. Dilution | lgK:6.64(3SF)     | 0 | CRV   | 27383  |
| 3                                                                                                                                                                       | 25 | 0     | Inf. Dilution | lgK:7.27(3SF)     | 0 | CRV   | 32224  |
| 4                                                                                                                                                                       | 25 | 0     | Inf. Dilution | lgK:11.98(4SF)    | 0 | CRV   | 32224  |
| 5                                                                                                                                                                       | 25 | 0     | CHEMVAL1      | lgK:10.58(0.01SD) | 0 | ENS   | 5156   |
| 6                                                                                                                                                                       | 25 | 0     | CHEMVAL2      | lgK:10.58(0.01SD) | 0 | ENS   | 5156   |
| 7                                                                                                                                                                       | 25 | 0     | CHEMVAL3      | lgK:10.58(0.01SD) | 0 | ENS   | 5156   |
| 8                                                                                                                                                                       | 25 | 0     | CHEMVAL4      | lgK:10.60(0.01SD) | 0 | ENS   | 5156   |

| Reaction No. 56219                                                                                                         |  |  |  |  |  |  |  |
|----------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|
| $\text{NaAlSiO}_4(\text{Neph., s}) + 4<\text{H}+1> = \text{Na}+1 + \text{Al}+3 + \text{H}+1(2)\_ \text{SiH}_2\text{O}_4-2$ |  |  |  |  |  |  |  |

| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:13.8(3SF)     | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:14.13(4SF)    | 2 | CRV   | 32224  |
| 3  | 25 | 0     | CHEMVAL1      | lgK:14.96(0.01SD) | 0 | ENS   | 5156   |
| 4  | 25 | 0     | CHEMVAL2      | lgK:14.96(0.01SD) | 0 | ENS   | 5156   |
| 5  | 25 | 0     | CHEMVAL3      | lgK:14.96(0.01SD) | 0 | ENS   | 5156   |
| 6  | 25 | 0     | CHEMVAL4      | lgK:15.00(0.01SD) | 0 | ENS   | 5156   |

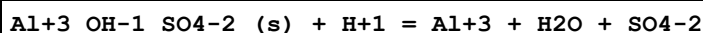
| Reaction No. 56220                                                                                                                                                                               |    |       |               |                   |   |       |        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $\text{NaAl}_3\text{Si}_3\text{O}_{10}(\text{OH})_2(\text{s}) + 10\langle\text{H}+1\rangle = \text{Na}+1 + 3\langle\text{Al}+3\rangle + 3\langle\text{H}+1(2)_{\text{SiH}_2\text{O}_4-2}\rangle$ |    |       |               |                   |   |       |        |
| No                                                                                                                                                                                               | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                                                                                                | 25 | 0     | Inf. Dilution | lgK:15.6(3SF)     | 1 | ELC   |        |
| 2                                                                                                                                                                                                | 25 | 0     | Inf. Dilution | lgK:18.49(4SF)    | 0 | CRV   | 32224  |
| 3                                                                                                                                                                                                | 25 | 0     | CHEMVAL1      | lgK:23.23(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                                                                                                                                                | 25 | 0     | CHEMVAL2      | lgK:23.23(0.01SD) | 0 | ENS   | 5156   |
| 5                                                                                                                                                                                                | 25 | 0     | CHEMVAL3      | lgK:23.23(0.01SD) | 0 | ENS   | 5156   |
| 6                                                                                                                                                                                                | 25 | 0     | CHEMVAL4      | lgK:23.20(0.01SD) | 0 | ENS   | 5156   |

| Reaction No. 56225                                                                                                                                      |    |       |               |                   |   |       |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $\text{FeO}.\text{Al}_2\text{O}_3(\text{s}) + 8\langle\text{H}+1\rangle = \text{Fe}+2 + 2\langle\text{Al}+3\rangle + 4\langle\text{H}_2\text{O}\rangle$ |    |       |               |                   |   |       |        |
| No                                                                                                                                                      | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                                                       | 25 | 0     | Inf. Dilution | lgK:24.7(3SF)     | 1 | ELC   |        |
| 2                                                                                                                                                       | 25 | 0     | Inf. Dilution | lgK:27.28(4SF)    | 0 | CRV   | 32224  |
| 3                                                                                                                                                       | 25 | 0     | CHEMVAL1      | lgK:27.28(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                                                                                                       | 25 | 0     | CHEMVAL2      | lgK:27.28(0.01SD) | 0 | ENS   | 5156   |
| 5                                                                                                                                                       | 25 | 0     | CHEMVAL3      | lgK:27.28(0.01SD) | 0 | ENS   | 5156   |
| 6                                                                                                                                                       | 25 | 0     | CHEMVAL4      | lgK:27.20(0.01SD) | 0 | ENS   | 5156   |

| Reaction No. 56231                                                                              |    |       |               |                    |   |       |        |
|-------------------------------------------------------------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| $\text{Al}+3_{\text{OH}-1(3)}_{\text{(Gibbsite,s)}} = \text{Al}+3 + 3\langle\text{OH}-1\rangle$ |    |       |               |                    |   |       |        |
| No                                                                                              | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                                               | 15 | 0     | Inf. Dilution | lgK:-34.45(0.02SD) | 0 | MSL   | 5618   |
| 2                                                                                               | 23 | 0     | Inf. Dilution | lgK:-32.3(3SF)     | 0 | ENS   | 7276   |
| 3                                                                                               | 25 | 0     | Inf. Dilution | lgK:-34.5(3SF)     | 0 | ELC   |        |
| 4                                                                                               | 25 | 0     | Inf. Dilution | lgK:-33.0(3SF)     | 0 | MSL   | 30     |
| 5                                                                                               | 25 | 0     | Inf. Dilution | lgK:-32.96(4SF)    | 0 | MSL   | 560    |
| 6                                                                                               | 25 | 0     | Inf. Dilution | lgK:-33.5(3SF)     | 0 | ENS   | 773    |

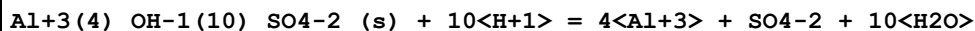
|    |       |   |               |                     |   |     |       |
|----|-------|---|---------------|---------------------|---|-----|-------|
| 7  | 25    | 0 | Inf. Dilution | lgK:-33.7 (3SF)     | 0 | CRV | 2556  |
| 8  | 25    | 0 | Inf. Dilution | lgK:-33.96 (0.03SD) | 0 | MSL | 5618  |
| 9  | 25    | 0 | Inf. Dilution | lgK:-33.5 (3SF)     | 0 | CRV | 6405  |
| 10 | 25    | 0 | Inf. Dilution | lgK:-33.80 (4SF)    | 0 | CRV | 28992 |
| 11 | 25    | 1 | Unknown       | lgK:-31.79 (4SF)    | 0 | ENS | 773   |
| 12 | 25    | 3 | Unknown       | lgK:-35. (2SF)      | 0 | ELC |       |
| 13 | 35    | 0 | Inf. Dilution | lgK:-33.52 (0.02SD) | 0 | MSL | 5618  |
| 14 | 50    | 0 | Inf. Dilution | lgK:-32.85 (4SF)    | 5 | CRV | 28992 |
| 15 | 100   | 0 | Inf. Dilution | lgK:-31.68 (4SF)    | 5 | CRV | 28992 |
| 16 | 150   | 0 | Inf. Dilution | lgK:-31.23 (4SF)    | 5 | CRV | 28992 |
| 17 | 200   | 0 | Inf. Dilution | lgK:-31.24 (4SF)    | 5 | CRV | 28992 |
| 18 | 10:40 | 0 | Inf. Dilution | dH:18.94 (4SF)      | 0 | MSL | 5618  |

## Reaction No. 56232



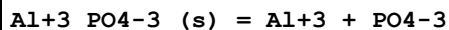
| No | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-1.903 (4SF)   | 0 | CRV   | 22551  |
| 2  | 25 | 0     | Inf. Dilution | lgK:-3.23 (3SF)    | 5 | MSL   | 23912  |
| 3  | 25 | 0     | Inf. Dilution | lgK:-3.23 (3SF)    | 4 | CRV   | 32224  |
| 4  | 25 | 0     | CHEMVAL1      | lgK:-3.23 (0.01SD) | 0 | ENS   | 5156   |
| 5  | 25 | 0     | CHEMVAL2      | lgK:-3.23 (0.01SD) | 0 | ENS   | 5156   |
| 6  | 25 | 0     | CHEMVAL3      | lgK:-3.23 (0.01SD) | 0 | ENS   | 5156   |
| 7  | 25 | 0     | CHEMVAL4      | lgK:-3.23 (0.01SD) | 3 | ENS   | 5156   |

## Reaction No. 56233



| No | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:22.7 (3SF)     | 4 | CRV   | 32224  |
| 2  | 25 | 0     | CHEMVAL1      | lgK:22.70 (0.01SD) | 0 | ENS   | 5156   |
| 3  | 25 | 0     | CHEMVAL2      | lgK:22.70 (0.01SD) | 0 | ENS   | 5156   |
| 4  | 25 | 0     | CHEMVAL3      | lgK:22.70 (0.01SD) | 0 | ENS   | 5156   |
| 5  | 25 | 0     | CHEMVAL4      | lgK:22.70 (0.01SD) | 4 | ENS   | 5156   |

## Reaction No. 56235



| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 19 | 0     | Inf. Dilution | lgK:-18.24 (4SF) | 0 | MSL   |        |

|                                                                                                                                             |    |      |               |                    |   |     |       |
|---------------------------------------------------------------------------------------------------------------------------------------------|----|------|---------------|--------------------|---|-----|-------|
| Note (1): Zharovskii FG, Trudy Kom. Anal. Khim., Akad. Nauk SSSR, 1951, 3, 101                                                              |    |      |               |                    |   |     |       |
| 2                                                                                                                                           | 25 | 0    | Inf. Dilution | lgK:-20.1(3SF)     | 1 | EDH |       |
| 3                                                                                                                                           | 25 | 0    | Inf. Dilution | lgK:-18.2(3SF)     | 0 | CRV |       |
| Note (3): <a href="http://www.saltlakemetals.com/SolubilityProducts.htm">http://www.saltlakemetals.com/SolubilityProducts.htm</a> (15/7/14) |    |      |               |                    |   |     |       |
| 4                                                                                                                                           | 25 | 0    | Inf. Dilution | lgK:-18.2(3SF)     | 0 | ENS | 5648  |
| 5                                                                                                                                           | 25 | 0    | Inf. Dilution | lgK:-20.01(4SF)    | 3 | CRV | 6330  |
| 6                                                                                                                                           | 25 | 0    | Inf. Dilution | lgK:-20.6(3SF)     | 3 | CRV | 6405  |
| 7                                                                                                                                           | 25 | 0    | Inf. Dilution | lgK:-19.89(4SF)    | 3 | EOD | 23430 |
| Note (7): Zharovskii, Trudy Komissii Anal. Khim, 1951, 3, 101                                                                               |    |      |               |                    |   |     |       |
| 8                                                                                                                                           | 25 | 0    | CHEMVAL2      | lgK:-22.84(0.01SD) | 0 | ENS | 5156  |
| 9                                                                                                                                           | 25 | 0    | CHEMVAL3      | lgK:-22.84(0.01SD) | 0 | ENS | 5156  |
| 10                                                                                                                                          | 25 | 0    | CHEMVAL4      | lgK:-20.10(0.01SD) | 2 | ENS | 5156  |
| 11                                                                                                                                          | 37 | 0.15 | Na+1 salt     | lgK:-18.34(4SF)    | 2 | CRV | 7271  |
| 12                                                                                                                                          | 37 | 0.15 | NaCl          | lgK:-18.6(3SF)     | 2 | MIS | 5983  |
| 13                                                                                                                                          | 37 | 0.15 | Unknown       | lgK:-19.1(3SF)     | 0 | CRV | 2556  |

| Reaction No. 56236                                                                                                           |    |       |               |                   |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| Al <sub>2</sub> SiO <sub>5</sub> (Kyan.,s) + 6<H+1> = 2<Al+3> + H+1(2)_SiH <sub>2</sub> O <sub>4</sub> -2 + H <sub>2</sub> O |    |       |               |                   |   |       |        |
| No                                                                                                                           | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                            | 25 | 0     | Inf. Dilution | lgK:15.1(3SF)     | 1 | ELC   |        |
| 2                                                                                                                            | 25 | 0     | Inf. Dilution | lgK:16.326(5SF)   | 0 | CRV   | 32224  |
| 3                                                                                                                            | 25 | 0     | CHEMVAL1      | lgK:13.99(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                                                                            | 25 | 0     | CHEMVAL2      | lgK:13.99(0.01SD) | 0 | ENS   | 5156   |
| 5                                                                                                                            | 25 | 0     | CHEMVAL3      | lgK:13.99(0.01SD) | 0 | ENS   | 5156   |
| 6                                                                                                                            | 25 | 0     | CHEMVAL4      | lgK:14.00(0.01SD) | 0 | ENS   | 5156   |

| Reaction No. 56237                                                                                                                                            |    |       |               |                  |   |       |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Al <sub>2</sub> Si <sub>4</sub> O <sub>10</sub> (OH) <sub>2</sub> (s) + 4<H <sub>2</sub> O> = 2<Al+3> + 4<H+1(2)_SiH <sub>2</sub> O <sub>4</sub> -2> - 6<H+1> |    |       |               |                  |   |       |        |
| No                                                                                                                                                            | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                                                                             | 25 | 0     | Inf. Dilution | lgK:0.8(1SF)     | 1 | ELC   |        |
| 2                                                                                                                                                             | 25 | 0     | Inf. Dilution | lgK:1.081(4SF)   | 0 | CRV   | 32224  |
| 3                                                                                                                                                             | 25 | 0     | CHEMVAL1      | lgK:3.46(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                                                                                                             | 25 | 0     | CHEMVAL2      | lgK:3.46(0.01SD) | 0 | ENS   | 5156   |
| 5                                                                                                                                                             | 25 | 0     | CHEMVAL3      | lgK:3.46(0.01SD) | 0 | ENS   | 5156   |
| 6                                                                                                                                                             | 25 | 0     | CHEMVAL4      | lgK:3.42(0.01SD) | 0 | ENS   | 5156   |

|                    |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
| Reaction No. 56331 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|

| Mg <sub>5</sub> Al <sub>2</sub> Si <sub>3</sub> O <sub>10</sub> (OH) <sub>8</sub> (s) + 16<H+1> = 5<Mg+2> + 2<Al+3> + 3<H+1(2)_SiH <sub>2</sub> O <sub>4</sub> -2> + 6<H <sub>2</sub> O > |    |       |               |                   |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| No                                                                                                                                                                                        | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                                                                                         | 25 | 0     | Inf. Dilution | lgK:61.6(3SF)     | 1 | ELC   |        |
| 2                                                                                                                                                                                         | 25 | 0     | Inf. Dilution | lgK:71.92(4SF)    | 0 | CRV   | 32224  |
| 3                                                                                                                                                                                         | 25 | 0     | CHEMVAL1      | lgK:57.02(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                                                                                                                                         | 25 | 0     | CHEMVAL2      | lgK:57.02(0.01SD) | 0 | ENS   | 5156   |
| 5                                                                                                                                                                                         | 25 | 0     | CHEMVAL3      | lgK:57.02(0.01SD) | 0 | ENS   | 5156   |

| Reaction No. 56334                                                                                                                                              |    |       |               |                  |   |       |        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Al <sub>2</sub> Si <sub>2</sub> O <sub>7</sub> .2H <sub>2</sub> O(Dick.,s) + 6<H+1> = 2<Al+3> + 2<H+1(2)_SiH <sub>2</sub> O <sub>4</sub> -2> + H <sub>2</sub> O |    |       |               |                  |   |       |        |
| No                                                                                                                                                              | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                                                                               | 25 | 0     | Inf. Dilution | lgK:7.8(2SF)     | 1 | ELC   |        |
| 2                                                                                                                                                               | 25 | 0     | Inf. Dilution | lgK:5.95(3SF)    | 0 | MGL   | 6498   |
| 3                                                                                                                                                               | 25 | 0     | CHEMVAL1      | lgK:7.78(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                                                                                                               | 25 | 0     | CHEMVAL2      | lgK:7.78(0.01SD) | 0 | ENS   | 5156   |
| 5                                                                                                                                                               | 25 | 0     | CHEMVAL3      | lgK:7.78(0.01SD) | 0 | ENS   | 5156   |
| 6                                                                                                                                                               | 25 | 0     | CHEMVAL4      | lgK:7.74(0.01SD) | 0 | ENS   | 5156   |

| Reaction No. 56342                                                                               |     |           |                    |                 |   |       |        |
|--------------------------------------------------------------------------------------------------|-----|-----------|--------------------|-----------------|---|-------|--------|
| Al+3(3)_K+1_OH-1(6)_SO <sub>4</sub> -2(2)_(s) = K+1 + 3<Al+3> + 6<OH-1> + 2<SO <sub>4</sub> -2 > |     |           |                    |                 |   |       |        |
| No                                                                                               | t   | I-Str     | Medium             | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                | 25  | 0.001:0.3 | NaClO <sub>4</sub> | lgK:-85.(2.SD)  | 0 | MSL   | 556    |
| Note (1): Low wgt for inter-reaction consistency                                                 |     |           |                    |                 |   |       |        |
| 2                                                                                                | 25  | 0         | Inf. Dilution      | lgK:-82.3(3SF)  | 1 | ERG   |        |
| 3                                                                                                | 25  | 0         | Inf. Dilution      | lgK:-82.56(4SF) | 3 | CRV   | 28992  |
| 4                                                                                                | 50  | 0         | Inf. Dilution      | lgK:-80.8(3SF)  | 3 | CRV   | 28992  |
| 5                                                                                                | 100 | 0         | Inf. Dilution      | lgK:-79.5(3SF)  | 3 | CRV   | 28992  |
| 6                                                                                                | 150 | 0         | Inf. Dilution      | lgK:-80.1(3SF)  | 2 | CRV   | 28992  |
| 7                                                                                                | 200 | 0         | Inf. Dilution      | lgK:-82.0(3SF)  | 2 | CRV   | 28992  |

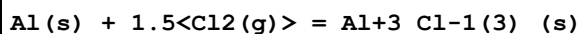
| Reaction No. 56343                                                                                 |    |           |                    |                 |   |       |        |
|----------------------------------------------------------------------------------------------------|----|-----------|--------------------|-----------------|---|-------|--------|
| Al+3(3)_Na+1_OH-1(6)_SO <sub>4</sub> -2(2)_(s) = Na+1 + 3<Al+3> + 6<OH-1> + 2<SO <sub>4</sub> -2 > |    |           |                    |                 |   |       |        |
| No                                                                                                 | t  | I-Str     | Medium             | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                  | 25 | 0.001:0.1 | NaClO <sub>4</sub> | lgK:-80.(0.7SD) | 4 | MSL   | 556    |

| Reaction No. 56355                                                                                     |  |  |  |  |  |  |  |
|--------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|
| 6<Al(s)> + 6.5<O <sub>2</sub> (g)> + 2<Si(s)> = 3Al <sub>2</sub> O <sub>3</sub> .2SiO <sub>2</sub> (s) |  |  |  |  |  |  |  |



| No | t   | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|-----|-------|---------------|--------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:1129.(4SF)     | 4 | ENS   | 6422   |
| 2  | 25  | 0     | Inf. Dilution | lgK:1129.(4SF)     | 4 | ENS   | 6494   |
| 3  | 27  | 0     | Inf. Dilution | lgK:1121.(4SF)     | 3 | ENS   | 6422   |
| 4  | 27  | 0     | Inf. Dilution | lgK:1121.(4SF)     | 0 | ENS   | 6494   |
| 5  | 127 | 0     | Inf. Dilution | lgK:824.5(4SF)     | 2 | ENS   | 6422   |
| 6  | 127 | 0     | Inf. Dilution | lgK:824.3(4SF)     | 2 | ENS   | 6494   |
| 7  | 227 | 0     | Inf. Dilution | lgK:646.3(4SF)     | 1 | ENS   | 6422   |
| 8  | 227 | 0     | Inf. Dilution | lgK:646.1(4SF)     | 1 | ENS   | 6494   |
| 9  | 327 | 0     | Inf. Dilution | lgK:527.5(4SF)     | 0 | ENS   | 6422   |
| 10 | 327 | 0     | Inf. Dilution | lgK:527.4(4SF)     | 0 | ENS   | 6494   |
| 11 | 25  | 0     | Inf. Dilution | dG:-1541.2(0.05SD) | 0 | ENS   | 802    |
| 12 | 25  | 0     | Inf. Dilution | dG:-1540.(4SF)     | 3 | EFE   | 6422   |
| 13 | 25  | 0     | Inf. Dilution | dG:-1540.(4SF)     | 4 | EFE   | 6494   |
| 14 | 25  | 0     | Inf. Dilution | dG:-6432.7(5SF)kJ  | 0 | CRV   | 9015   |
| 15 | 25  | 0     | Inf. Dilution | dG:-1811.(4SF)     | 0 | CRV   | 12011  |
| 16 | 25  | 0     | Inf. Dilution | dH:-1632.8(0.1SD)  | 4 | ENS   | 802    |
| 17 | 25  | 0     | Inf. Dilution | dH:-1637.(4SF)     | 2 | MTD   | 6422   |
| 18 | 25  | 0     | Inf. Dilution | dH:-1630.(4SF)     | 2 | MTD   | 6494   |
| 19 | 25  | 0     | Inf. Dilution | dH:-6816.2(5SF)kJ  | 0 | CRV   | 9015   |
| 20 | 25  | 0     | Inf. Dilution | dH:-1903.(4SF)     | 0 | CRV   | 12011  |
| 21 | 127 | 0     | Inf. Dilution | dH:-1631.(4SF)     | 1 | MTD   | 6422   |
| 22 | 127 | 0     | Inf. Dilution | dH:-1630.(4SF)     | 1 | MTD   | 6494   |
| 23 | 227 | 0     | Inf. Dilution | dH:-1630.(4SF)     | 0 | MTD   | 6422   |
| 24 | 227 | 0     | Inf. Dilution | dH:-1630.(4SF)     | 0 | MTD   | 6494   |
| 25 | 327 | 0     | Inf. Dilution | dH:-1630.(4SF)     | 0 | MTD   | 6422   |
| 26 | 327 | 0     | Inf. Dilution | dH:-1629.(4SF)     | 0 | MTD   | 6494   |

## Reaction No. 56356



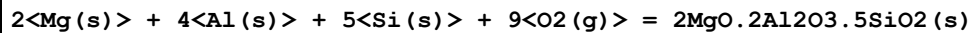
| No | t     | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|-------|-------|---------------|-------------------|---|-------|--------|
| 1  | 25    | 0     | Inf. Dilution | lgK:110.4(4SF)    | 2 | ENS   | 6422   |
| 2  | 27    | 0     | Inf. Dilution | lgK:109.6(4SF)    | 0 | ENS   | 6422   |
| 3  | 127   | 0     | Inf. Dilution | lgK:78.93(4SF)    | 3 | ENS   | 6422   |
| 4  | 192.7 | 0     | Inf. Dilution | lgK:65.98(4SF)    | 2 | ENS   | 6422   |
| 5  | 25    | 0     | Inf. Dilution | dG:-150.3(0.05SD) | 4 | ENS   | 802    |

|    |       |   |               |                    |   |     |       |
|----|-------|---|---------------|--------------------|---|-----|-------|
| 6  | 25    | 0 | Inf. Dilution | dG:-628.8(4SF)kJ   | 4 | CRV | 6330  |
| 7  | 25    | 0 | Inf. Dilution | dG:-150.6(4SF)     | 2 | EFE | 6422  |
| 8  | 25    | 0 | Inf. Dilution | dG:-628.8(4SF)kJ   | 5 | EFE | 7275  |
| 9  | 25    | 0 | Inf. Dilution | dG:-628.8(4SF)kJ   | 4 | CRV | 9015  |
| 10 | 25    | 0 | Inf. Dilution | dG:-628.8(4SF)kJ   | 4 | CRV | 10802 |
| 11 | 25    | 0 | Inf. Dilution | dH:-168.3(0.1SD)   | 4 | ENS | 802   |
| 12 | 25    | 0 | Inf. Dilution | dH:-704.2(4SF)kJ   | 4 | CRV | 6330  |
| 13 | 25    | 0 | Inf. Dilution | dH:-168.6(4SF)     | 4 | MTD | 6422  |
| 14 | 25    | 0 | Inf. Dilution | dH:-704.2(4SF)kJ   | 5 | EFE | 7275  |
| 15 | 25    | 0 | Inf. Dilution | dH:-704.2(4SF)kJ   | 4 | CRV | 9015  |
| 16 | 25    | 0 | Inf. Dilution | dH:-704.2(4SF)kJ   | 4 | CRV | 10802 |
| 17 | 25    | 0 | Inf. Dilution | dH:-168.80(0.16SD) | 3 | MCL | 29546 |
| 18 | 127   | 0 | Inf. Dilution | dH:-168.2(4SF)     | 3 | MTD | 6422  |
| 19 | 192.7 | 0 | Inf. Dilution | dH:-167.8(4SF)     | 2 | MTD | 6422  |

| Reaction No. 56370                         |     |       |               |                   |   |       |        |
|--------------------------------------------|-----|-------|---------------|-------------------|---|-------|--------|
| Fe(s) + 2<Al(s)> + 2<O2(g)> = FeO.Al2O3(s) |     |       |               |                   |   |       |        |
| No                                         | t   | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                          | 25  | 0     | Inf. Dilution | lgK:329.3(4SF)    | 4 | ENS   | 6422   |
| 2                                          | 25  | 0     | Inf. Dilution | lgK:322.0(4SF)    | 0 | ENS   | 6494   |
| 3                                          | 27  | 0     | Inf. Dilution | lgK:327.2(4SF)    | 0 | ENS   | 6422   |
| 4                                          | 27  | 0     | Inf. Dilution | lgK:319.9(4SF)    | 0 | ENS   | 6494   |
| 5                                          | 127 | 0     | Inf. Dilution | lgK:240.3(4SF)    | 3 | ENS   | 6422   |
| 6                                          | 127 | 0     | Inf. Dilution | lgK:235.0(4SF)    | 0 | ENS   | 6494   |
| 7                                          | 227 | 0     | Inf. Dilution | lgK:188.2(4SF)    | 2 | ENS   | 6422   |
| 8                                          | 227 | 0     | Inf. Dilution | lgK:184.1(4SF)    | 0 | ENS   | 6494   |
| 9                                          | 327 | 0     | Inf. Dilution | lgK:153.4(4SF)    | 0 | ENS   | 6422   |
| 10                                         | 327 | 0     | Inf. Dilution | lgK:150.2(4SF)    | 0 | ENS   | 6494   |
| 11                                         | 25  | 0     | Inf. Dilution | dG:-442.0(0.05SD) | 0 | ENS   | 802    |
| 12                                         | 25  | 0     | Inf. Dilution | dG:-449.2(4SF)    | 4 | EFE   | 6422   |
| 13                                         | 25  | 0     | Inf. Dilution | dG:-439.3(4SF)    | 0 | EFE   | 6494   |
| 14                                         | 25  | 0     | Inf. Dilution | dH:-470.0(0.1SD)  | 3 | ENS   | 802    |
| 15                                         | 25  | 0     | Inf. Dilution | dH:-476.9(4SF)    | 0 | MTD   | 6422   |
| 16                                         | 25  | 0     | Inf. Dilution | dH:-468.0(4SF)    | 0 | ENS   | 6437   |
| 17                                         | 25  | 0     | Inf. Dilution | dH:-466.2(4SF)    | 0 | MTD   | 6494   |
| 18                                         | 127 | 0     | Inf. Dilution | dH:-476.9(4SF)    | 2 | MTD   | 6422   |

|    |     |   |               |                 |   |     |      |
|----|-----|---|---------------|-----------------|---|-----|------|
| 19 | 127 | 0 | Inf. Dilution | dH:-466.1 (4SF) | 0 | MTD | 6494 |
| 20 | 227 | 0 | Inf. Dilution | dH:-476.8 (4SF) | 1 | MTD | 6422 |
| 21 | 227 | 0 | Inf. Dilution | dH:-465.8 (4SF) | 0 | MTD | 6494 |
| 22 | 327 | 0 | Inf. Dilution | dH:-476.7 (4SF) | 0 | MTD | 6422 |
| 23 | 327 | 0 | Inf. Dilution | dH:-465.5 (4SF) | 0 | MTD | 6494 |

## Reaction No. 56374



| No                                          | t   | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|---------------------------------------------|-----|-------|---------------|--------------------|---|-------|--------|
| 1                                           | 25  | 0     | Inf. Dilution | lgK:1516. (4SF)    | 5 | ENS   | 6422   |
| 2                                           | 25  | 0     | Inf. Dilution | lgK:1516. (4SF)    | 4 | ENS   | 6494   |
| 3                                           | 27  | 0     | Inf. Dilution | lgK:1506. (4SF)    | 0 | ENS   | 6422   |
| 4                                           | 27  | 0     | Inf. Dilution | lgK:1506. (4SF)    | 0 | ENS   | 6494   |
| 5                                           | 127 | 0     | Inf. Dilution | lgK:1107. (4SF)    | 5 | ENS   | 6422   |
| 6                                           | 127 | 0     | Inf. Dilution | lgK:1107. (4SF)    | 4 | ENS   | 6494   |
| 7                                           | 227 | 0     | Inf. Dilution | lgK:867.7 (4SF)    | 5 | ENS   | 6422   |
| 8                                           | 227 | 0     | Inf. Dilution | lgK:867.5 (4SF)    | 4 | ENS   | 6494   |
| 9                                           | 327 | 0     | Inf. Dilution | lgK:708.3 (4SF)    | 5 | ENS   | 6422   |
| 10                                          | 327 | 0     | Inf. Dilution | lgK:707.9 (4SF)    | 4 | ENS   | 6494   |
| 11                                          | 25  | 0     | Inf. Dilution | dG:-2055.0 (0.1SD) | 0 | ENS   | 802    |
| 12                                          | 25  | 0     | Inf. Dilution | dG:-2068. (4SF)    | 5 | EFE   | 6422   |
| 13                                          | 25  | 0     | Inf. Dilution | dG:-2067.8 (5SF)   | 4 | EFE   | 6482   |
| 14                                          | 25  | 0     | Inf. Dilution | dG:-2068. (4SF)    | 4 | EFE   | 6494   |
| 15                                          | 25  | 0     | Inf. Dilution | dH:-2177.0 (0.1SD) | 0 | ENS   | 802    |
| Note (15): Low wgt for internal consistency |     |       |               |                    |   |       |        |
| 16                                          | 25  | 0     | Inf. Dilution | dH:-2190. (4SF)    | 5 | MTD   | 6422   |
| 17                                          | 25  | 0     | Inf. Dilution | dH:-2190. (4SF)    | 4 | ENS   | 6437   |
| 18                                          | 25  | 0     | Inf. Dilution | dH:-2189.0 (5SF)   | 4 | EFE   | 6482   |
| 19                                          | 25  | 0     | Inf. Dilution | dH:-2190. (4SF)    | 4 | MTD   | 6494   |
| 20                                          | 127 | 0     | Inf. Dilution | dH:-2190. (4SF)    | 5 | MTD   | 6422   |
| 21                                          | 127 | 0     | Inf. Dilution | dH:-2190. (4SF)    | 4 | MTD   | 6494   |
| 22                                          | 227 | 0     | Inf. Dilution | dH:-2189. (4SF)    | 5 | MTD   | 6422   |
| 23                                          | 227 | 0     | Inf. Dilution | dH:-2190. (4SF)    | 4 | MTD   | 6494   |
| 24                                          | 327 | 0     | Inf. Dilution | dH:-2187. (4SF)    | 5 | MTD   | 6422   |
| 25                                          | 327 | 0     | Inf. Dilution | dH:-2190. (4SF)    | 4 | MTD   | 6494   |

| Reaction No. 56814          |    |       |               |               |   |       |        |
|-----------------------------|----|-------|---------------|---------------|---|-------|--------|
| $V(s) + 3<Al(s)> = VA13(s)$ |    |       |               |               |   |       |        |
| No                          | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                           | 25 | 0     | Inf. Dilution | dH:-26.(1.SD) | 4 | ENS   | 3006   |

| Reaction No. 56815              |    |       |               |               |   |       |        |
|---------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $5<V(s)> + 8<Al(s)> = V5Al8(s)$ |    |       |               |               |   |       |        |
| No                              | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                               | 25 | 0     | Inf. Dilution | dH:-70.(1.SD) | 4 | ENS   | 3006   |

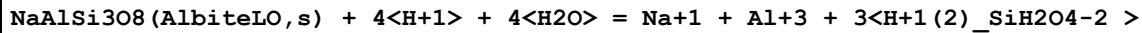
| Reaction No. 57064                                                        |    |       |               |                  |   |       |        |
|---------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| $KAlSi3O8(Adul.,s) + 4<H2O> + 4<H+1> = K+1 + Al+3 + 3<H+1(2)_{SiH2O4-2}>$ |    |       |               |                  |   |       |        |
| No                                                                        | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                         | 25 | 0     | Inf. Dilution | lgK:3.4(2SF)     | 1 | ELC   |        |
| 2                                                                         | 25 | 0     | CHEMVAL1      | lgK:1.26(0.01SD) | 0 | ENS   | 5156   |
| 3                                                                         | 25 | 0     | CHEMVAL2      | lgK:1.26(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                         | 25 | 0     | CHEMVAL3      | lgK:1.26(0.01SD) | 0 | ENS   | 5156   |
| 5                                                                         | 25 | 0     | CHEMVAL4      | lgK:1.26(0.01SD) | 0 | ENS   | 5156   |

| Reaction No. 57065                                                       |    |       |               |                  |   |       |        |
|--------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| $KAlSi3O8(San.,s) + 4<H2O> + 4<H+1> = K+1 + Al+3 + 3<H+1(2)_{SiH2O4-2}>$ |    |       |               |                  |   |       |        |
| No                                                                       | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                        | 25 | 0     | Inf. Dilution | lgK:0.5(1SF)     | 1 | ELC   |        |
| 2                                                                        | 25 | 0     | Inf. Dilution | lgK:1.25(3SF)    | 0 | CRV   | 32224  |
| 3                                                                        | 25 | 0     | CHEMVAL1      | lgK:5.20(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                        | 25 | 0     | CHEMVAL2      | lgK:5.20(0.01SD) | 0 | ENS   | 5156   |
| 5                                                                        | 25 | 0     | CHEMVAL3      | lgK:5.20(0.01SD) | 0 | ENS   | 5156   |
| 6                                                                        | 25 | 0     | CHEMVAL4      | lgK:5.19(0.01SD) | 0 | ENS   | 5156   |

| Reaction No. 57066                                                          |    |       |               |                  |   |       |        |
|-----------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| $NaAlSi2O6(Jade.,s) + 2<H2O> + 4<H+1> = Na+1 + Al+3 + 2<H+1(2)_{SiH2O4-2}>$ |    |       |               |                  |   |       |        |
| No                                                                          | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                           | 25 | 0     | Inf. Dilution | lgK:7.7(2SF)     | 1 | ELC   |        |
| 2                                                                           | 25 | 0     | Inf. Dilution | lgK:8.71(3SF)    | 0 | CRV   | 32224  |
| 3                                                                           | 25 | 0     | CHEMVAL1      | lgK:9.42(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                           | 25 | 0     | CHEMVAL2      | lgK:9.42(0.01SD) | 0 | ENS   | 5156   |
| 5                                                                           | 25 | 0     | CHEMVAL3      | lgK:9.42(0.01SD) | 0 | ENS   | 5156   |

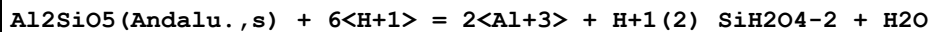
|   |    |   |          |                  |   |     |      |
|---|----|---|----------|------------------|---|-----|------|
| 6 | 25 | 0 | CHEMVAL4 | lgK:9.41(0.01SD) | 0 | ENS | 5156 |
|---|----|---|----------|------------------|---|-----|------|

## Reaction No. 57067



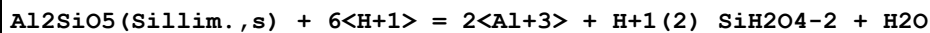
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:2.6(2SF)     | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:3.09(3SF)    | 0 | CRV   | 32224  |
| 3  | 25 | 0     | CHEMVAL1      | lgK:6.43(0.01SD) | 0 | ENS   | 5156   |
| 4  | 25 | 0     | CHEMVAL2      | lgK:6.43(0.01SD) | 0 | ENS   | 5156   |
| 5  | 25 | 0     | CHEMVAL3      | lgK:6.43(0.01SD) | 0 | ENS   | 5156   |
| 6  | 25 | 0     | CHEMVAL4      | lgK:6.43(0.01SD) | 0 | ENS   | 5156   |

## Reaction No. 57068



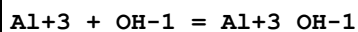
| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:15.4(3SF)     | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:16.596(5SF)   | 0 | CRV   | 32224  |
| 3  | 25 | 0     | CHEMVAL1      | lgK:14.20(0.01SD) | 0 | ENS   | 5156   |
| 4  | 25 | 0     | CHEMVAL2      | lgK:14.20(0.01SD) | 0 | ENS   | 5156   |
| 5  | 25 | 0     | CHEMVAL3      | lgK:14.20(0.01SD) | 0 | ENS   | 5156   |
| 6  | 25 | 0     | CHEMVAL4      | lgK:14.20(0.01SD) | 0 | ENS   | 5156   |

## Reaction No. 57069



| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:15.2(3SF)     | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:16.960(5SF)   | 0 | CRV   | 32224  |
| 3  | 25 | 0     | CHEMVAL1      | lgK:14.50(0.01SD) | 0 | ENS   | 5156   |
| 4  | 25 | 0     | CHEMVAL2      | lgK:14.50(0.01SD) | 0 | ENS   | 5156   |
| 5  | 25 | 0     | CHEMVAL3      | lgK:14.50(0.01SD) | 0 | ENS   | 5156   |
| 6  | 25 | 0     | CHEMVAL4      | lgK:14.50(0.01SD) | 0 | ENS   | 5156   |

## Reaction No. 57276



| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 20 | 0     | Inf. Dilution | lgK:8.99(3SF)     | 5 | CRV   | 5627   |
| 2  | 20 | 0.1   | K+1 salt      | lgK:12.30(0.3SD)w | 0 | MPH   | 5639   |

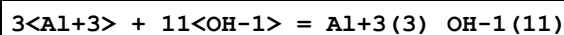
|                                            |       |      |                    |                     |   |     |      |
|--------------------------------------------|-------|------|--------------------|---------------------|---|-----|------|
| 3                                          | 25    | 0    | Inf. Dilution      | lgK:9.0 (0.1SD)     | 4 | ENS | 5573 |
| 4                                          | 25    | 0    | Inf. Dilution      | lgK:9.0 (2SF)       | 5 | CRV | 6405 |
| 5                                          | 25    | 0.1  | NaClO <sub>4</sub> | lgK:9.10 (3SF)      | 0 | MLC | 5620 |
| Note (5): Low wgt for internal consistency |       |      |                    |                     |   |     |      |
| 6                                          | 25    | 0.1  | Unknown            | lgK:8.45 (0.01SD)   | 4 | CRV | 2556 |
| 7                                          | 25    | 0.3  | NaClO <sub>4</sub> | lgK:9.40 (3SF)      | 0 | MLC | 5620 |
| 8                                          | 25    | 0.5  | NaClO <sub>4</sub> | lgK:9.70 (3SF)      | 0 | MLC | 5620 |
| 9                                          | 25    | 0.5  | Unknown            | lgK:8.15 (3SF)      | 4 | CRV | 2556 |
| 10                                         | 25    | 1    | NaCl               | lgK:8.27 (3SF)      | 5 | CRV | 7271 |
| 11                                         | 25    | 1    | NaClO <sub>4</sub> | lgK:10.40 (4SF)     | 0 | MLC | 5620 |
| 12                                         | 25    | 3    | NaCl               | lgK:8.50 (3SF)      | 4 | CRV | 2556 |
| 13                                         | 37    | 0.15 | NaCl               | lgK:8.577 (0.020SD) | 0 | MGL | 3086 |
| 14                                         | 100   | 0    | Inf. Dilution      | lgK:9.3 (2SF)       | 1 | ELC |      |
| 15                                         | 10:40 | 0    | Inf. Dilution      | dH:-2. (1SF)        | 4 | CRV | 2556 |

| Reaction No. 57277             |    |       |                    |                      |   |       |        |
|--------------------------------|----|-------|--------------------|----------------------|---|-------|--------|
| Al+3 + 3<OH-1> = Al+3_OH-1 (3) |    |       |                    |                      |   |       |        |
| No                             | t  | I-Str | Medium             | Value (Dev)          | W | Tech. | Ref. # |
| 1                              | 20 | 0     | Inf. Dilution      | lgK:26.8 (0.1SD)     | 0 | CRV   | 5627   |
| 2                              | 20 | 0.1   | K+1 salt           | lgK:31.98 (0.3SD) w  | 0 | MPH   | 5639   |
| 3                              | 25 | 0     | Inf. Dilution      | lgK:24.8 (3SF)       | 1 | ELC   |        |
| 4                              | 25 | 0     | Inf. Dilution      | lgK:25.2 (3SF)       | 0 | CRV   | 2556   |
| 5                              | 25 | 0     | Inf. Dilution      | lgK:26.99 (0.1SD)    | 0 | ENS   | 5573   |
| 6                              | 25 | 0     | Inf. Dilution      | lgK:27.0 (3SF)       | 0 | CRV   | 6405   |
| 7                              | 25 | 0.1   | NaClO <sub>4</sub> | lgK:25.75 (4SF)      | 0 | MLC   | 5620   |
| 8                              | 25 | 0.3   | NaClO <sub>4</sub> | lgK:26.75 (4SF)      | 0 | MLC   | 5620   |
| 9                              | 25 | 0.5   | NaClO <sub>4</sub> | lgK:27.62 (4SF)      | 0 | MLC   | 5620   |
| 10                             | 25 | 1     | NaClO <sub>4</sub> | lgK:30.16 (4SF)      | 0 | MLC   | 5620   |
| 11                             | 37 | 0.15  | NaCl               | lgK:26.138 (0.035SD) | 0 | MGL   | 3086   |

| Reaction No. 57278             |    |       |               |                     |   |       |        |
|--------------------------------|----|-------|---------------|---------------------|---|-------|--------|
| Al+3 + 4<OH-1> = Al+3_OH-1 (4) |    |       |               |                     |   |       |        |
| No                             | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                              | 20 | 0     | Inf. Dilution | lgK:32.70 (0.02SD)  | 0 | CRV   | 5627   |
| 2                              | 20 | 0.1   | K+1 salt      | lgK:36.58 (0.3SD) w | 0 | MPH   | 5639   |
| 3                              | 25 | 0     | Inf. Dilution | lgK:33.2 (3SF)      | 1 | ERG   |        |

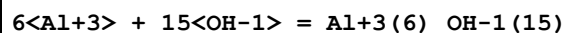
|    |       |      |               |                      |   |     |       |
|----|-------|------|---------------|----------------------|---|-----|-------|
| 4  | 25    | 0    | Inf. Dilution | lgK:33. (0.5SD)      | 0 | CRV | 2556  |
| 5  | 25    | 0    | Inf. Dilution | lgK:33.0 (0.1SD)     | 0 | ENS | 5573  |
| 6  | 25    | 0    | Inf. Dilution | lgK:33.0 (3SF)       | 0 | CRV | 6405  |
| 7  | 25    | 0    | Inf. Dilution | lgK:32.17 (4SF)      | 0 | CRV | 22551 |
| 8  | 25    | 0    | Inf. Dilution | lgK:34.05 (0.05SD)   | 4 | EOD | 30201 |
| 9  | 25    | 0.5  | Unknown       | lgK:31.5 (0.1SD)     | 0 | CRV | 2556  |
| 10 | 37    | 0.15 | NaCl          | lgK:29.044 (0.120SD) | 0 | MGL | 3086  |
| 11 | 50    | 0    | Inf. Dilution | lgK:33.44 (0.05SD)   | 4 | EOD | 30201 |
| 12 | 75    | 0    | Inf. Dilution | lgK:33.11 (0.05SD)   | 4 | EOD | 30201 |
| 13 | 100   | 0    | Inf. Dilution | lgK:32.99 (0.05SD)   | 4 | EOD | 30201 |
| 14 | 20:70 | 0    | Inf. Dilution | dH:-12. (2SF)        | 0 | CRV | 2556  |
| 15 | 25    | 0    | Inf. Dilution | Cp:157.7 (4SF)       | 0 | MTD | 6439  |
| 16 | 25    | 0    | Inf. Dilution | dS:104. (3SF)        | 0 | EFE | 6439  |

## Reaction No. 57279



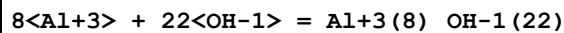
| No | t  | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:95.1 (3SF)       | 2 | EAE   |        |
| 2  | 37 | 0.15  | NaCl          | lgK:91.025 (0.099SD) | 1 | MGL   | 3086   |

## Reaction No. 57280



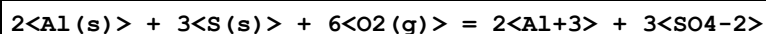
| No | t  | I-Str | Medium        | Value (Dev)           | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:159.3 (4SF)       | 1 | ELC   |        |
| 2  | 37 | 0.15  | NaCl          | lgK:149.280 (0.015SD) | 3 | MGL   | 3086   |

## Reaction No. 57281



| No | t  | I-Str | Medium        | Value (Dev)           | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:226.0 (3SF)       | 2 | EAE   |        |
| 2  | 37 | 0.15  | NaCl          | lgK:214.970 (0.184SD) | 3 | MGL   | 3086   |

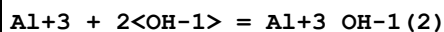
## Reaction No. 57550



| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:563.4 (4SF)  | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | dG:-766. (0.5SD) | 0 | ENS   | 802    |

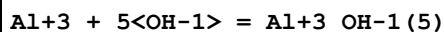
|   |    |   |               |                 |   |     |     |
|---|----|---|---------------|-----------------|---|-----|-----|
| 3 | 25 | 0 | Inf. Dilution | dH:-906.(0.5SD) | 2 | ENS | 802 |
|---|----|---|---------------|-----------------|---|-----|-----|

## Reaction No. 57614



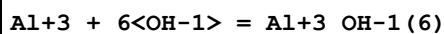
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 20 | 0     | Inf. Dilution | lgK:19.3(0.1SD)  | 0 | CRV   | 5627   |
| 2  | 25 | 0     | Inf. Dilution | lgK:17.5(3SF)    | 1 | ELC   |        |
| 3  | 25 | 0     | Inf. Dilution | lgK:17.9(3SF)    | 0 | CRV   | 2556   |
| 4  | 25 | 0     | Inf. Dilution | lgK:18.69(0.1SD) | 0 | ENS   | 5573   |
| 5  | 25 | 0     | Inf. Dilution | lgK:18.7(3SF)    | 0 | CRV   | 6405   |
| 6  | 25 | 0.1   | NaClO4        | lgK:17.65(4SF)   | 0 | MLC   | 5620   |
| 7  | 25 | 0.1   | Unknown       | lgK:16.7(3SF)    | 0 | CRV   | 2556   |
| 8  | 25 | 0.3   | NaClO4        | lgK:18.27(4SF)   | 0 | MLC   | 5620   |
| 9  | 25 | 0.5   | NaClO4        | lgK:18.88(4SF)   | 0 | MLC   | 5620   |
| 10 | 25 | 1     | NaClO4        | lgK:20.42(4SF)   | 0 | MLC   | 5620   |

## Reaction No. 57615



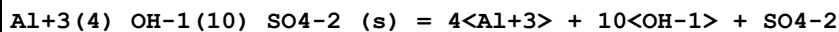
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:26.9(0.1SD) | 0 | ENS   | 5573   |

## Reaction No. 57616



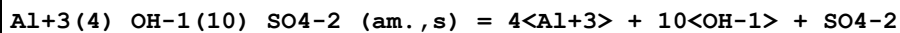
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:15.65(0.1SD) | 0 | ENS   | 5573   |

## Reaction No. 57919



| No | t  | I-Str     | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-----------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0.001:0.1 | NaClO4        | lgK:-118.(0.5SD) | 0 | MSL   | 556    |
| 2  | 25 | 0         | Inf. Dilution | lgK:-117.3(4SF)  | 1 | ELC   |        |

## Reaction No. 57920



| No | t  | I-Str     | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-----------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.001:0.1 | NaClO4 | lgK:-116.(0.5SD) | 4 | MSL   | 556    |



| Reaction No. 57942                               |     |       |               |                         |   |       |        |
|--------------------------------------------------|-----|-------|---------------|-------------------------|---|-------|--------|
| Al(s) + 2<H2(g)> + 2<O2(g)> + e-1 = Al+3_OH-1(4) |     |       |               |                         |   |       |        |
| No                                               | t   | I-Str | Medium        | Value (Dev)             | W | Tech. | Ref. # |
| 1                                                | 25  | 0     | Inf. Dilution | lgK:229.5 (4SF)         | 1 | ELC   |        |
| 2                                                | 0   | 0     | Inf. Dilution | dG:-315.850 (6SF)       | 0 | EFE   | 5626   |
| 3                                                | 10  | 0     | Inf. Dilution | dG:-314.270 (6SF)       | 0 | EFE   | 5626   |
| 4                                                | 25  | 0     | Inf. Dilution | dG:-313.5 (4SF)         | 0 | MSL   | 553    |
| 5                                                | 25  | 0     | Inf. Dilution | dG:-1307. (10.SD) kJ    | 0 | EFE   | 2171   |
| 6                                                | 25  | 0     | Inf. Dilution | dG:-1305.0 (5SF) kJ     | 0 | EOD   | 5616   |
| 7                                                | 25  | 0     | Inf. Dilution | dG:-1305. (1.MD) kJ     | 0 | CRV   | 5619   |
| 8                                                | 25  | 0     | Inf. Dilution | dG:-1305.2 (5SF) kJ     | 0 | EFE   | 5621   |
| 9                                                | 25  | 0     | Inf. Dilution | dG:-311.900 (6SF)       | 0 | EFE   | 5626   |
| 10                                               | 25  | 0     | Inf. Dilution | dG:-1305.40 (1.59SD) kJ | 0 | EOD   | 6028   |
| 11                                               | 25  | 0     | Inf. Dilution | dG:-312.160 (6SF)       | 0 | ENS   | 6333   |
| 12                                               | 25  | 0     | Inf. Dilution | dG:-1305. (4SF) kJ      | 0 | CRV   | 7421   |
| 13                                               | 25  | 0     | Inf. Dilution | dG:-1305. (2.0MD) kJ    | 0 | CRV   | 8775   |
| 14                                               | 25  | 0     | Inf. Dilution | dG:-1305.60 (6SF) kJ    | 0 | CRV   | 8798   |
| 15                                               | 25  | 0     | Inf. Dilution | dG:-1305.7 (5SF) kJ     | 0 | EOD   | 26787  |
| 16                                               | 25  | 0     | Inf. Dilution | dG:1305.1 (5SF) kJ      | 0 | EFE   | 28693  |
| 17                                               | 25  | 0     | Inf. Dilution | dG:-1305.8 (5SF) kJ     | 0 | CRV   | 29482  |
| 18                                               | 25  | 0     | Inf. Dilution | dG:-312.09 (5SF)        | 0 | CRV   | 29553  |
| 19                                               | 25  | 0     | GAMSPATH      | dG:-1301. (4SF) kJ      | 0 | CRV   | 12638  |
| 20                                               | 40  | 0     | Inf. Dilution | dG:-309.520 (6SF)       | 0 | EFE   | 5626   |
| 21                                               | 50  | 0     | Inf. Dilution | dG:-1288.7 (5SF) kJ     | 0 | EOD   | 5616   |
| 22                                               | 60  | 0     | Inf. Dilution | dG:-1309.2 (5SF) kJ     | 0 | EFE   | 5621   |
| 23                                               | 60  | 0     | Inf. Dilution | dG:-306.330 (6SF)       | 0 | EFE   | 5626   |
| 24                                               | 80  | 0     | Inf. Dilution | dG:-303.130 (6SF)       | 0 | EFE   | 5626   |
| 25                                               | 100 | 0     | Inf. Dilution | dG:-1255.8 (5SF) kJ     | 0 | EOD   | 5616   |
| 26                                               | 100 | 0     | Inf. Dilution | dG:-1314.2 (5SF) kJ     | 0 | EFE   | 5621   |
| 27                                               | 100 | 0     | Inf. Dilution | dG:-299.920 (6SF)       | 0 | EFE   | 5626   |
| 28                                               | 120 | 0     | Inf. Dilution | dG:-296.710 (6SF)       | 0 | EFE   | 5626   |
| 29                                               | 140 | 0     | Inf. Dilution | dG:-293.490 (6SF)       | 0 | EFE   | 5626   |
| 30                                               | 150 | 0     | Inf. Dilution | dG:-1222.9 (5SF) kJ     | 0 | MSL   | 5616   |
| 31                                               | 150 | 0     | Inf. Dilution | dG:-1320.7 (5SF) kJ     | 0 | EFE   | 5621   |
| 32                                               | 160 | 0     | Inf. Dilution | dG:-290.260 (6SF)       | 0 | EFE   | 5626   |
| 33                                               | 200 | 0     | Inf. Dilution | dG:-1189.3 (5SF) kJ     | 0 | MSL   | 5616   |

|    |     |   |               |                       |   |     |       |
|----|-----|---|---------------|-----------------------|---|-----|-------|
| 34 | 200 | 0 | Inf. Dilution | dG:-1326.8 (5SF) kJ   | 0 | EFE | 5621  |
| 35 | 250 | 0 | Inf. Dilution | dG:-1155.3 (5SF) kJ   | 0 | MSL | 5616  |
| 36 | 250 | 0 | Inf. Dilution | dG:-1332.6 (5SF) kJ   | 0 | EFE | 5621  |
| 37 | 300 | 0 | Inf. Dilution | dG:-1120.8 (5SF) kJ   | 0 | MSL | 5616  |
| 38 | 300 | 0 | Inf. Dilution | dG:-1338.7 (5SF) kJ   | 0 | EFE | 5621  |
| 39 | 0   | 0 | Inf. Dilution | dH:-358.760 (6SF)     | 0 | MCL | 5626  |
| 40 | 10  | 0 | Inf. Dilution | dH:-358.910 (6SF)     | 0 | MCL | 5626  |
| 41 | 25  | 0 | Inf. Dilution | dH:-1478. (4SF) kJ    | 0 | MTD | 2171  |
| 42 | 25  | 0 | Inf. Dilution | dH:-1499.8 (5SF) kJ   | 0 | EOD | 5616  |
| 43 | 25  | 0 | Inf. Dilution | dH:-359.170 (6SF)     | 0 | MCL | 5626  |
| 44 | 25  | 0 | Inf. Dilution | dH:-1491.50 (6SF) kJ  | 0 | MTD | 6028  |
| 45 | 25  | 0 | Inf. Dilution | dH:-359.110 (6SF)     | 0 | ENS | 6333  |
| 46 | 25  | 0 | Inf. Dilution | dH:-1502.8 (1.6MD) kJ | 0 | CRV | 8775  |
| 47 | 25  | 0 | Inf. Dilution | dH:-1500.70 (6SF) kJ  | 0 | CRV | 8798  |
| 48 | 25  | 0 | Inf. Dilution | dH:-1503.0 (5SF) kJ   | 0 | EOD | 26787 |
| 49 | 25  | 0 | Inf. Dilution | dH:-1483.6 (5SF) kJ   | 0 | CRV | 29482 |
| 50 | 25  | 0 | Inf. Dilution | dH:-359.24 (5SF)      | 3 | CRV | 29553 |
| 51 | 25  | 0 | GAMSPATH      | dH:-1494.40 (6SF) kJ  | 0 | CRV | 12638 |
| 52 | 40  | 0 | Inf. Dilution | dH:-359.380 (6SF)     | 0 | MCL | 5626  |
| 53 | 60  | 0 | Inf. Dilution | dH:-359.570 (6SF)     | 0 | MCL | 5626  |
| 54 | 80  | 0 | Inf. Dilution | dH:-359.710 (6SF)     | 0 | MCL | 5626  |
| 55 | 100 | 0 | Inf. Dilution | dH:-359.820 (6SF)     | 0 | MCL | 5626  |
| 56 | 120 | 0 | Inf. Dilution | dH:-359.940 (6SF)     | 0 | MCL | 5626  |
| 57 | 140 | 0 | Inf. Dilution | dH:-360.090 (6SF)     | 0 | MCL | 5626  |
| 58 | 160 | 0 | Inf. Dilution | dH:-360.330 (6SF)     | 0 | MCL | 5626  |
| 59 | 25  | 0 | Inf. Dilution | Cp:23.06 (4SF)        | 0 | ENS | 6333  |
| 60 | 25  | 0 | Inf. Dilution | Cp:89.6 (12.MD) J/K   | 0 | CRV | 8775  |
| 61 | 25  | 0 | Inf. Dilution | Cp:96.5 (3SF) J/K     | 3 | EOD | 26787 |

## Reaction No. 57943

Al+3(3)\_OH-1(3)\_Oxalic-2(3) + 4<H2O> = Al+3(3)\_OH-1(7)\_Oxalic-2\_(s) + 2<Oxalic-2> + 4<H+1 >

| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:0.0 (1SF) | 1 | ENS   |        |

Note (1): Very rough guess to link reactions

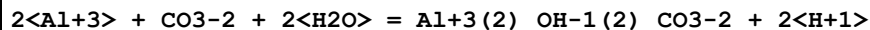
## Reaction No. 57986

| Al(s) = Al(g) |    |       |               |                        |   |       |        |
|---------------|----|-------|---------------|------------------------|---|-------|--------|
| No            | t  | I-Str | Medium        | Value (Dev)            | W | Tech. | Ref. # |
| 1             | 25 | 0     | Inf. Dilution | dG:289.4 (4SF) kJ      | 4 | CRV   | 6330   |
| 2             | 25 | 0     | Inf. Dilution | dG:285.7 (4SF) kJ      | 0 | EFE   | 7275   |
| 3             | 25 | 0     | Inf. Dilution | dG:289.4 (4SF) kJ      | 6 | CRV   | 8746   |
| 4             | 25 | 0     | Inf. Dilution | dG:285.7 (4SF) kJ      | 0 | CRV   | 10802  |
| 5             | 25 | 0     | Inf. Dilution | dG:289.38 (4.000SD) kJ | 6 | CRV   | 14923  |
| 6             | 25 | 0     | Inf. Dilution | dH:79. (1.SD)          | 6 | CRV   | 5605   |
| 7             | 25 | 0     | Inf. Dilution | dH:330.0 (4SF) kJ      | 4 | CRV   | 6330   |
| 8             | 25 | 0     | Inf. Dilution | dH:326.4 (4SF) kJ      | 5 | EFE   | 7275   |
| 9             | 25 | 0     | Inf. Dilution | dH:330.0 (4.MD) kJ     | 6 | CRV   | 8746   |
| 10            | 25 | 0     | Inf. Dilution | dH:326.4 (4SF) kJ      | 0 | CRV   | 10802  |
| 11            | 25 | 0     | Inf. Dilution | dH:330.00 (4.000SD) kJ | 6 | CRV   | 14923  |

| Reaction No. 57987                   |     |       |               |                         |   |       |        |
|--------------------------------------|-----|-------|---------------|-------------------------|---|-------|--------|
| Al(s) + 1.5<F2(g)> = Al+3_F-1(3)_(s) |     |       |               |                         |   |       |        |
| No                                   | t   | I-Str | Medium        | Value (Dev)             | W | Tech. | Ref. # |
| 1                                    | 25  | 0     | Inf. Dilution | lgK:250.7 (4SF)         | 4 | ENS   | 6422   |
| 2                                    | 27  | 0     | Inf. Dilution | lgK:249.1 (4SF)         | 0 | ENS   | 6422   |
| 3                                    | 127 | 0     | Inf. Dilution | lgK:183.4 (4SF)         | 2 | ENS   | 6422   |
| 4                                    | 227 | 0     | Inf. Dilution | lgK:143.9 (4SF)         | 1 | ENS   | 6422   |
| 5                                    | 327 | 0     | Inf. Dilution | lgK:117.7 (4SF)         | 0 | ENS   | 6422   |
| 6                                    | 25  | 0     | Inf. Dilution | dG:-1431.1 (5SF) kJ     | 4 | CRV   | 6330   |
| 7                                    | 25  | 0     | Inf. Dilution | dG:-342.0 (4SF)         | 4 | EFE   | 6422   |
| 8                                    | 25  | 0     | Inf. Dilution | dG:-340.6 (4SF)         | 0 | CRV   | 7382   |
| 9                                    | 25  | 0     | Inf. Dilution | dG:-1431. (4SF) kJ      | 6 | CRV   | 8746   |
| 10                                   | 25  | 0     | Inf. Dilution | dG:-1425.0 (5SF) kJ     | 0 | CRV   | 9015   |
| 11                                   | 25  | 0     | Inf. Dilution | dG:-1431.1 (1.309SD) kJ | 6 | CRV   | 14923  |
| 12                                   | 25  | 0     | Inf. Dilution | dH:-361. (0.3SD)        | 7 | CRV   | 5605   |
| 13                                   | 25  | 0     | Inf. Dilution | dH:-1510.4 (5SF) kJ     | 4 | CRV   | 6330   |
| 14                                   | 25  | 0     | Inf. Dilution | dH:-361.0 (4SF)         | 4 | MTD   | 6422   |
| 15                                   | 25  | 0     | Inf. Dilution | dH:-359.5 (4SF)         | 0 | CRV   | 7382   |
| 16                                   | 25  | 0     | Inf. Dilution | dH:-1510.4 (1.3MD) kJ   | 6 | CRV   | 8746   |
| 17                                   | 25  | 0     | Inf. Dilution | dH:-1504.1 (5SF) kJ     | 0 | CRV   | 9015   |
| 18                                   | 25  | 0     | Inf. Dilution | dH:-1510.4 (1.300SD) kJ | 6 | CRV   | 14923  |
| 19                                   | 127 | 0     | Inf. Dilution | dH:-360.8 (4SF)         | 1 | MTD   | 6422   |

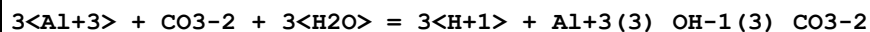
|    |     |   |               |                       |   |     |       |
|----|-----|---|---------------|-----------------------|---|-----|-------|
| 20 | 227 | 0 | Inf. Dilution | dH:-360.5 (4SF)       | 1 | MTD | 6422  |
| 21 | 327 | 0 | Inf. Dilution | dH:-360.1 (4SF)       | 0 | MTD | 6422  |
| 22 | 25  | 0 | Inf. Dilution | Cp:75.122 (0.4SD) J/K | 5 | CRV | 27292 |

## Reaction No. 58056



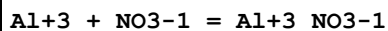
| No | t  | I-Str | Medium  | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------|-------------------|---|-------|--------|
| 1  | 25 | 0     | UCT_Sea | lgK:3.74 (0.01SD) | 0 | EDH   | 5390   |
| 2  | 25 | 0.1   | UCT_Sea | lgK:2.38 (0.01SD) | 0 | EDH   | 5390   |
| 3  | 25 | 0.2   | UCT_Sea | lgK:2.13 (0.01SD) | 0 | EDH   | 5390   |
| 4  | 25 | 0.3   | UCT_Sea | lgK:1.96 (0.01SD) | 0 | EDH   | 5390   |
| 5  | 25 | 0.4   | UCT_Sea | lgK:1.85 (0.01SD) | 0 | EDH   | 5390   |
| 6  | 25 | 0.5   | UCT_Sea | lgK:1.75 (0.01SD) | 0 | EDH   | 5390   |
| 7  | 25 | 0.6   | UCT_Sea | lgK:1.68 (0.01SD) | 0 | EDH   | 5390   |
| 8  | 25 | 0.7   | UCT_Sea | lgK:1.62 (0.01SD) | 0 | EDH   | 5390   |

## Reaction No. 58057



| No | t  | I-Str | Medium  | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------|--------------------|---|-------|--------|
| 1  | 25 | 0     | UCT_Sea | lgK:1.16 (0.01SD)  | 0 | EDH   | 5390   |
| 2  | 25 | 0.1   | UCT_Sea | lgK:0.02 (0.01SD)  | 0 | EDH   | 5390   |
| 3  | 25 | 0.2   | UCT_Sea | lgK:-0.22 (0.01SD) | 0 | EDH   | 5390   |
| 4  | 25 | 0.3   | UCT_Sea | lgK:-0.39 (0.01SD) | 0 | EDH   | 5390   |
| 5  | 25 | 0.4   | UCT_Sea | lgK:-0.49 (0.01SD) | 0 | EDH   | 5390   |
| 6  | 25 | 0.5   | UCT_Sea | lgK:-0.58 (0.01SD) | 0 | EDH   | 5390   |
| 7  | 25 | 0.6   | UCT_Sea | lgK:-0.64 (0.01SD) | 0 | EDH   | 5390   |
| 8  | 25 | 0.7   | UCT_Sea | lgK:-0.69 (0.01SD) | 0 | EDH   | 5390   |

## Reaction No. 58059



| No | t  | I-Str | Medium  | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------|--------------------|---|-------|--------|
| 1  | 25 | 0     | UCT_Sea | lgK:0.12 (0.01SD)  | 4 | EDH   | 5390   |
| 2  | 25 | 0.1   | UCT_Sea | lgK:-0.35 (0.01SD) | 4 | EDH   | 5390   |
| 3  | 25 | 0.2   | UCT_Sea | lgK:-0.42 (0.01SD) | 4 | EDH   | 5390   |
| 4  | 25 | 0.3   | UCT_Sea | lgK:-0.45 (0.01SD) | 4 | EDH   | 5390   |
| 5  | 25 | 0.4   | UCT_Sea | lgK:-0.48 (0.01SD) | 4 | EDH   | 5390   |
| 6  | 25 | 0.5   | UCT_Sea | lgK:-0.49 (0.01SD) | 4 | EDH   | 5390   |

|   |    |     |         |                    |   |     |      |
|---|----|-----|---------|--------------------|---|-----|------|
| 7 | 25 | 0.6 | UCT_Sea | lgK:-0.50 (0.01SD) | 4 | EDH | 5390 |
| 8 | 25 | 0.7 | UCT_Sea | lgK:-0.50 (0.01SD) | 4 | EDH | 5390 |

## Reaction No. 58082

$\text{Ca}_4\text{SO}_4\text{AlSiF}_{13}\cdot 10\text{H}_2\text{O}(\text{s}) = 4\langle\text{Ca}+2\rangle + \text{Al}+3 + 4\langle\text{H}+1\rangle + \text{H}+1(2)\text{SiH}_2\text{O}_4-2 + 6\langle\text{H}_2\text{O}\rangle + \text{SO}_4-2 + 13\langle\text{F}-1\rangle$

| No                                                                                                      | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|---------------------------------------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 1                                                                                                       | 23 | 0     | Inf. Dilution | lgK:-78.77 (4SF) | 3 | ENS   |        |
| Note (1): <a href="http://www.fipr.state.fl.us/fipr100.htm">http://www.fipr.state.fl.us/fipr100.htm</a> |    |       |               |                  |   |       |        |
| 2                                                                                                       | 25 | 0     | Inf. Dilution | lgK:-78.8 (3SF)  | 2 | EAE   |        |

## Reaction No. 58222

$\text{AlOOH}(\text{Boehmite},\text{s}) + \text{H}_2\text{O} = \text{Al}+3 + 3\langle\text{OH}-1\rangle$

| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-33.9 (3SF)  | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:-31.7 (3SF)  | 0 | MSL   | 30     |
| 3  | 25 | 0     | Inf. Dilution | lgK:-34.02 (4SF) | 0 | ENS   | 773    |

## Reaction No. 58223

$\text{AlOOH}(\text{Boehmite},\text{s}) + \text{OH}-1 + \text{H}_2\text{O} = \text{Al}+3\text{OH}-1(4)$

| No | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|-----|-------|---------------|------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:-0.5 (1SF)   | 1 | ELC   |        |
| 2  | 25  | 0     | Inf. Dilution | lgK:0.8 (1SF)    | 0 | MSL   | 30     |
| 3  | 70  | 0     | Inf. Dilution | lgK:-0.812 (3SF) | 0 | MSL   | 6028   |
| 4  | 90  | 0     | Inf. Dilution | lgK:-0.662 (3SF) | 0 | MSL   | 6028   |
| 5  | 110 | 0     | Inf. Dilution | lgK:-0.525 (3SF) | 0 | MSL   | 6028   |
| 6  | 150 | 0     | Inf. Dilution | lgK:-0.279 (3SF) | 2 | MSL   | 6028   |
| 7  | 200 | 0     | Inf. Dilution | lgK:-0.017 (2SF) | 0 | MSL   | 6028   |
| 8  | 250 | 0     | Inf. Dilution | lgK:0.207 (3SF)  | 0 | MSL   | 6028   |
| 9  | 300 | 0     | Inf. Dilution | lgK:0.403 (3SF)  | 0 | MSL   | 6028   |

## Reaction No. 58224

$\text{Al}+3\text{OH}-1(3)\text{_(Bayerite},\text{s}) = \text{Al}+3 + 3\langle\text{OH}-1\rangle$

| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-33.8 (3SF) | 0 | MSL   | 30     |

## Reaction No. 58225

$\text{Al}+3\text{OH}-1(3)\text{_(Bayerite},\text{s}) + \text{OH}-1 = \text{Al}+3\text{OH}-1(4)$

| No                                             | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|------------------------------------------------|-----|-------|---------------|------------------|---|-------|--------|
| 1                                              | 21  | 0     | Inf. Dilution | lgK:-0.355 (3SF) | 6 | MSL   | 6028   |
| 2                                              | 25  | 0     | Inf. Dilution | lgK:-1.3 (2SF)   | 0 | MSL   | 30     |
| 3                                              | 25  | 3     | Unknown       | lgK:-5.8 (2SF)   | 1 | ELC   |        |
| Note (3): Caught between devil & deep blue sea |     |       |               |                  |   |       |        |
| 4                                              | 50  | 0     | Inf. Dilution | lgK:-0.017 (2SF) | 6 | MSL   | 6028   |
| 5                                              | 65  | 0     | Inf. Dilution | lgK:0.136 (3SF)  | 6 | MSL   | 6028   |
| 6                                              | 100 | 0     | Inf. Dilution | lgK:1.5 (2SF)    | 1 | ELC   |        |

| Reaction No. 58226                                      |     |       |               |                |   |       |        |
|---------------------------------------------------------|-----|-------|---------------|----------------|---|-------|--------|
| Al+3_OH-1(3)_(Gibbsite,s) + 2<H+1> = Al+3_OH-1 + 2<H2O> |     |       |               |                |   |       |        |
| No                                                      | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                       | 0   | 0     | NaCl/m        | lgK:3.58 (3SF) | 5 | MSL   | 7025   |
| 2                                                       | 0   | 0.1   | NaCl/m        | lgK:3.81 (3SF) | 6 | MSL   | 7025   |
| 3                                                       | 0   | 0.5   | NaCl/m        | lgK:3.98 (3SF) | 6 | MSL   | 7025   |
| 4                                                       | 0   | 1     | NaCl/m        | lgK:4.08 (3SF) | 6 | MSL   | 7025   |
| 5                                                       | 0   | 5     | NaCl/m        | lgK:4.80 (3SF) | 6 | MSL   | 7025   |
| 6                                                       | 25  | 0     | Inf. Dilution | lgK:3.12 (3SF) | 3 | MSL   | 553    |
| 7                                                       | 25  | 0     | NaCl/m        | lgK:2.78 (3SF) | 5 | MSL   | 7025   |
| 8                                                       | 25  | 0.1   | NaCl/m        | lgK:3.00 (3SF) | 6 | MSL   | 7025   |
| 9                                                       | 25  | 0.5   | NaCl/m        | lgK:3.13 (3SF) | 6 | MSL   | 7025   |
| 10                                                      | 25  | 1     | NaCl/m        | lgK:3.21 (3SF) | 6 | MSL   | 7025   |
| 11                                                      | 25  | 5     | NaCl/m        | lgK:3.91 (3SF) | 6 | MSL   | 7025   |
| 12                                                      | 50  | 0     | NaCl/m        | lgK:2.11 (3SF) | 5 | MSL   | 7025   |
| 13                                                      | 50  | 0.1   | NaCl/m        | lgK:2.32 (3SF) | 6 | MSL   | 7025   |
| 14                                                      | 50  | 0.5   | NaCl/m        | lgK:2.43 (3SF) | 6 | MSL   | 7025   |
| 15                                                      | 50  | 1     | NaCl/m        | lgK:2.49 (3SF) | 6 | MSL   | 7025   |
| 16                                                      | 50  | 5     | NaCl/m        | lgK:3.19 (3SF) | 6 | MSL   | 7025   |
| 17                                                      | 75  | 0     | NaCl/m        | lgK:1.54 (3SF) | 5 | MSL   | 7025   |
| 18                                                      | 75  | 0.1   | NaCl/m        | lgK:1.75 (3SF) | 6 | MSL   | 7025   |
| 19                                                      | 75  | 0.5   | NaCl/m        | lgK:1.84 (3SF) | 6 | MSL   | 7025   |
| 20                                                      | 75  | 1     | NaCl/m        | lgK:1.89 (3SF) | 6 | MSL   | 7025   |
| 21                                                      | 75  | 5     | NaCl/m        | lgK:2.60 (3SF) | 6 | MSL   | 7025   |
| 22                                                      | 100 | 0     | NaCl/m        | lgK:1.04 (3SF) | 5 | MSL   | 7025   |
| 23                                                      | 100 | 0.1   | NaCl/m        | lgK:1.26 (3SF) | 6 | MSL   | 7025   |
| 24                                                      | 100 | 0.5   | NaCl/m        | lgK:1.34 (3SF) | 6 | MSL   | 7025   |

|    |     |   |        |                |   |     |      |
|----|-----|---|--------|----------------|---|-----|------|
| 25 | 100 | 1 | NaCl/m | lgK:1.38 (3SF) | 6 | MSL | 7025 |
| 26 | 100 | 5 | NaCl/m | lgK:2.13 (3SF) | 6 | MSL | 7025 |

| Reaction No. 58227                                   |     |       |               |                   |   |       |        |
|------------------------------------------------------|-----|-------|---------------|-------------------|---|-------|--------|
| Al+3_OH-1(3)_(Gibbsite,s) + H2O = Al+3_OH-1(4) + H+1 |     |       |               |                   |   |       |        |
| No                                                   | t   | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                    | 0   | 0     | Inf. Dilution | lgK:-16.41 (4SF)  | 3 | MSL   | 7025   |
| 2                                                    | 0   | 0.1   | NaCl/m        | lgK:-16.21 (4SF)  | 6 | MSL   | 7025   |
| 3                                                    | 0   | 0.5   | NaCl/m        | lgK:-16.12 (4SF)  | 6 | MSL   | 7025   |
| 4                                                    | 0   | 1     | NaCl/m        | lgK:-16.12 (4SF)  | 6 | MSL   | 7025   |
| 5                                                    | 0   | 5     | NaCl/m        | lgK:-16.51 (4SF)  | 6 | MSL   | 7025   |
| 6                                                    | 20  | 0     | Inf. Dilution | lgK:-12.62 (4SF)  | 0 | MTN   | 5630   |
| 7                                                    | 25  | 0     | Inf. Dilution | lgK:-14.05 (4SF)  | 0 | MSL   | 553    |
| 8                                                    | 25  | 0     | Inf. Dilution | lgK:-15.25 (4SF)  | 4 | EOD   | 5628   |
| 9                                                    | 25  | 0     | Inf. Dilution | lgK:-15.13 (4SF)  | 3 | MSL   | 7025   |
| 10                                                   | 25  | 0.1   | NaCl/m        | lgK:-14.92 (4SF)  | 6 | MSL   | 7025   |
| 11                                                   | 25  | 0.5   | NaCl/m        | lgK:-14.83 (4SF)  | 6 | MSL   | 7025   |
| 12                                                   | 25  | 1     | NaCl/m        | lgK:-14.83 (4SF)  | 6 | MSL   | 7025   |
| 13                                                   | 25  | 5     | NaCl/m        | lgK:-15.22 (4SF)  | 6 | MSL   | 7025   |
| 14                                                   | 40  | 2     | NaClO4        | lgK:-12.5 (0.2SD) | 0 | MHY   | 12539  |
| 15                                                   | 50  | 0     | Inf. Dilution | lgK:-14.08 (4SF)  | 3 | MSL   | 7025   |
| 16                                                   | 50  | 0.1   | NaCl/m        | lgK:-13.86 (4SF)  | 6 | MSL   | 7025   |
| 17                                                   | 50  | 0.5   | NaCl/m        | lgK:-13.75 (4SF)  | 6 | MSL   | 7025   |
| 18                                                   | 50  | 1     | NaCl/m        | lgK:-13.75 (4SF)  | 6 | MSL   | 7025   |
| 19                                                   | 50  | 5     | NaCl/m        | lgK:-14.12 (4SF)  | 6 | MSL   | 7025   |
| 20                                                   | 75  | 0     | Inf. Dilution | lgK:-13.20 (4SF)  | 3 | MSL   | 7025   |
| 21                                                   | 75  | 0.1   | NaCl/m        | lgK:-12.96 (4SF)  | 6 | MSL   | 7025   |
| 22                                                   | 75  | 0.5   | NaCl/m        | lgK:-12.84 (4SF)  | 6 | MSL   | 7025   |
| 23                                                   | 75  | 1     | NaCl/m        | lgK:-12.83 (4SF)  | 6 | MSL   | 7025   |
| 24                                                   | 75  | 5     | NaCl/m        | lgK:-13.15 (4SF)  | 6 | MSL   | 7025   |
| 25                                                   | 100 | 0     | Inf. Dilution | lgK:-12.45 (4SF)  | 3 | MSL   | 7025   |
| 26                                                   | 100 | 0.1   | NaCl/m        | lgK:-12.19 (4SF)  | 6 | MSL   | 7025   |
| 27                                                   | 100 | 0.5   | NaCl/m        | lgK:-12.05 (4SF)  | 6 | MSL   | 7025   |
| 28                                                   | 100 | 1     | NaCl/m        | lgK:-12.02 (4SF)  | 6 | MSL   | 7025   |
| 29                                                   | 100 | 5     | NaCl/m        | lgK:-12.28 (4SF)  | 6 | MSL   | 7025   |

| Reaction No. 58228                                                                 |    |       |        |                     |   |       |        |
|------------------------------------------------------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| $5\text{Al}^{+3} + 12\text{H}_2\text{O} = \text{Al}_3(\text{OH})_5 + 12\text{H}^+$ |    |       |        |                     |   |       |        |
| No                                                                                 | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                                                  | 25 | 0.1   | NaNO3  | lgK:-43.41 (0.03SD) | 0 | MGL   | 4878   |

| Reaction No. 58229                                                                 |    |       |               |                     |   |       |        |
|------------------------------------------------------------------------------------|----|-------|---------------|---------------------|---|-------|--------|
| $6\text{Al}^{+3} + 15\text{H}_2\text{O} = \text{Al}_3(\text{OH})_6 + 15\text{H}^+$ |    |       |               |                     |   |       |        |
| No                                                                                 | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                                                  | 25 | 0     | Inf. Dilution | lgK:-52. (2SF)      | 1 | EOR   |        |
| 2                                                                                  | 25 | 0.1   | NaNO3         | lgK:-53.78 (0.03SD) | 4 | MGL   | 4878   |
| 3                                                                                  | 40 | 2     | NaClO4        | lgK:-47.00 (4SF)    | 0 | CRV   | 5627   |

| Reaction No. 58230                                                                 |    |       |        |                     |   |       |        |
|------------------------------------------------------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| $7\text{Al}^{+3} + 17\text{H}_2\text{O} = \text{Al}_3(\text{OH})_7 + 17\text{H}^+$ |    |       |        |                     |   |       |        |
| No                                                                                 | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                                                  | 25 | 0.1   | NaNO3  | lgK:-59.48 (0.03SD) | 3 | MGL   | 4878   |
| 2                                                                                  | 50 | 3     | NaClO4 | lgK:-48.8 (3SF)     | 5 | CRV   | 5627   |

| Reaction No. 58231                                                                 |    |       |        |                     |   |       |        |
|------------------------------------------------------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| $8\text{Al}^{+3} + 20\text{H}_2\text{O} = \text{Al}_3(\text{OH})_8 + 20\text{H}^+$ |    |       |        |                     |   |       |        |
| No                                                                                 | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                                                  | 25 | 0.1   | NaNO3  | lgK:-69.82 (0.03SD) | 3 | MGL   | 4878   |

| Reaction No. 58232                                                                 |    |       |        |                     |   |       |        |
|------------------------------------------------------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| $9\text{Al}^{+3} + 23\text{H}_2\text{O} = \text{Al}_3(\text{OH})_9 + 23\text{H}^+$ |    |       |        |                     |   |       |        |
| No                                                                                 | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                                                  | 25 | 0.1   | NaNO3  | lgK:-80.14 (0.03SD) | 3 | MGL   | 4878   |

| Reaction No. 58233                                                                     |    |       |        |                     |   |       |        |
|----------------------------------------------------------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| $10\text{Al}^{+3} + 24\text{H}_2\text{O} = \text{Al}_3(\text{OH})_{10} + 24\text{H}^+$ |    |       |        |                     |   |       |        |
| No                                                                                     | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                                                      | 25 | 0.1   | NaNO3  | lgK:-81.18 (0.03SD) | 3 | MGL   | 4878   |

| Reaction No. 58234                                                                     |    |       |        |                      |   |       |        |
|----------------------------------------------------------------------------------------|----|-------|--------|----------------------|---|-------|--------|
| $13\text{Al}^{+3} + 32\text{H}_2\text{O} = \text{Al}_3(\text{OH})_{13} + 32\text{H}^+$ |    |       |        |                      |   |       |        |
| No                                                                                     | t  | I-Str | Medium | Value (Dev)          | W | Tech. | Ref. # |
| 1                                                                                      | 10 | 1     | NaNO3  | lgK:-120.94 (0.10SD) | 1 | MGL   | 30097  |



|    |    |      |               |                     |   |     |       |
|----|----|------|---------------|---------------------|---|-----|-------|
| 2  | 25 | 0    | Inf. Dilution | lgK:-99.5(1.2SD)    | 2 | EDH | 30072 |
| 3  | 25 | 0.06 | Cl-1 salt     | lgK:-103.4(4SF)     | 0 | MLC | 5588  |
| 4  | 25 | 0.1  | NaCl          | lgK:-103.61(5SF)    | 3 | MGL | 29939 |
| 5  | 25 | 0.1  | NaNO3         | lgK:-107.47(0.04SD) | 0 | MGL | 4878  |
| 6  | 25 | 0.25 | NaCl          | lgK:-105.13(5SF)    | 3 | MGL | 29939 |
| 7  | 25 | 0.5  | NaCl          | lgK:-106.54(5SF)    | 3 | MGL | 29939 |
| 8  | 25 | 0.6  | NaCl          | lgK:-105.5(0.20MD)  | 0 | MGL | 910   |
| 9  | 25 | 0.7  | NaCl          | lgK:-107.34(5SF)    | 3 | MGL | 29939 |
| 10 | 25 | 1    | NaClO4        | lgK:-104.5(0.06SD)  | 0 | CRV | 5627  |
| 11 | 25 | 1    | NaNO3         | lgK:-109.13(5SF)    | 1 | MGL | 30097 |
| 12 | 25 | 3    | Na(Cl)        | lgK:-113.35(0.13SD) | 0 | MGL | 5494  |
| 13 | 50 | 1    | NaNO3         | lgK:-91.73(0.10SD)  | 1 | MGL | 30097 |
| 14 | 70 | 1    | NaNO3         | lgK:-79.96(0.12SD)  | 1 | MGL | 30097 |
| 15 | 25 | 0    | Inf. Dilution | dH:1283.3(14.0SD)kJ | 2 | EDH | 30072 |

| Reaction No. 58235                               |    |       |        |                     |   |       |        |
|--------------------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| 14<Al+3> + 34<H2O> = Al+3(14)_OH-1(34) + 34<H+1> |    |       |        |                     |   |       |        |
| No                                               | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                | 25 | 0.1   | NaNO3  | lgK:-113.14(0.04SD) | 3 | MGL   | 4878   |

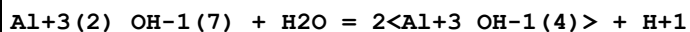
| Reaction No. 58236                          |    |       |               |                 |   |       |        |
|---------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| 2<Al+3> + 3<H2O> = Al+3(2)_OH-1(3) + 3<H+1> |    |       |               |                 |   |       |        |
| No                                          | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                           | 25 | 0     | Inf. Dilution | lgK:-9.5(3SF)   | 2 | EAE   |        |
| 2                                           | 40 | 2     | NaClO4        | lgK:-10.30(4SF) | 5 | MHY   | 5624   |

| Reaction No. 58237                                         |    |       |               |                |   |       |        |
|------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Al+3(2)_OH-1(3) + Al+3 + 3<H2O> = Al+3(3)_OH-1(6) + 3<H+1> |    |       |               |                |   |       |        |
| No                                                         | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                          | 25 | 0     | Inf. Dilution | lgK:-12.(2SF)  | 0 | EES   |        |
| 2                                                          | 40 | 2     | NaClO4        | lgK:-9.22(3SF) | 0 | MHY   | 5624   |
| Note (2): Alone and discredited                            |    |       |               |                |   |       |        |

| Reaction No. 58238                                            |    |       |               |                |   |       |        |
|---------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Al+3(3)_OH-1(10) + H2O = Al+3(2)_OH-1(7) + Al+3_OH-1(4) + H+1 |    |       |               |                |   |       |        |
| No                                                            | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                             | 25 | 0     | Inf. Dilution | lgK:-10.9(3SF) | 2 | EAE   |        |

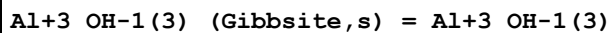
|   |    |   |        |                  |   |     |      |
|---|----|---|--------|------------------|---|-----|------|
| 2 | 40 | 2 | NaClO4 | lgK:-10.70 (4SF) | 1 | MHY | 5624 |
|---|----|---|--------|------------------|---|-----|------|

## Reaction No. 58239



| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-12.9 (3SF)  | 2 | EAE   |        |
| 2  | 40 | 2     | NaClO4        | lgK:-12.64 (4SF) | 2 | MHY   | 5624   |

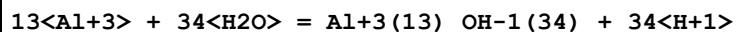
## Reaction No. 58240



| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 0   | 0     | NaCl/m        | lgK:-10.8 (3SF) | 4 | MSL   | 7025   |
| 2  | 0   | 0.1   | NaCl/m        | lgK:-10.8 (3SF) | 5 | MSL   | 7025   |
| 3  | 0   | 0.5   | NaCl/m        | lgK:-10.8 (3SF) | 5 | MSL   | 7025   |
| 4  | 0   | 1     | NaCl/m        | lgK:-10.8 (3SF) | 5 | MSL   | 7025   |
| 5  | 0   | 5     | NaCl/m        | lgK:-10.8 (3SF) | 5 | MSL   | 7025   |
| 6  | 20  | 0     | Inf. Dilution | lgK:-3.92 (3SF) | 0 | MTN   | 5630   |
| 7  | 25  | 0     | NaCl/m        | lgK:-9.6 (2SF)  | 4 | MSL   | 7025   |
| 8  | 25  | 0.1   | NaCl/m        | lgK:-9.6 (2SF)  | 5 | MSL   | 7025   |
| 9  | 25  | 0.5   | NaCl/m        | lgK:-9.6 (2SF)  | 5 | MSL   | 7025   |
| 10 | 25  | 1     | NaCl/m        | lgK:-9.6 (2SF)  | 5 | MSL   | 7025   |
| 11 | 25  | 5     | NaCl/m        | lgK:-9.6 (2SF)  | 5 | MSL   | 7025   |
| 12 | 50  | 0     | NaCl/m        | lgK:-8.6 (2SF)  | 4 | MSL   | 7025   |
| 13 | 50  | 0.1   | NaCl/m        | lgK:-8.6 (2SF)  | 5 | MSL   | 7025   |
| 14 | 50  | 0.5   | NaCl/m        | lgK:-8.6 (2SF)  | 5 | MSL   | 7025   |
| 15 | 50  | 1     | NaCl/m        | lgK:-8.6 (2SF)  | 5 | MSL   | 7025   |
| 16 | 50  | 5     | NaCl/m        | lgK:-8.6 (2SF)  | 5 | MSL   | 7025   |
| 17 | 75  | 0     | NaCl/m        | lgK:-7.9 (2SF)  | 4 | MSL   | 7025   |
| 18 | 75  | 0.1   | NaCl/m        | lgK:-7.9 (2SF)  | 5 | MSL   | 7025   |
| 19 | 75  | 0.5   | NaCl/m        | lgK:-7.9 (2SF)  | 5 | MSL   | 7025   |
| 20 | 75  | 1     | NaCl/m        | lgK:-7.9 (2SF)  | 5 | MSL   | 7025   |
| 21 | 75  | 5     | NaCl/m        | lgK:-7.9 (2SF)  | 5 | MSL   | 7025   |
| 22 | 100 | 0     | NaCl/m        | lgK:-7.2 (2SF)  | 4 | MSL   | 7025   |
| 23 | 100 | 0.1   | NaCl/m        | lgK:-7.2 (2SF)  | 5 | MSL   | 7025   |
| 24 | 100 | 0.5   | NaCl/m        | lgK:-7.2 (2SF)  | 5 | MSL   | 7025   |
| 25 | 100 | 1     | NaCl/m        | lgK:-7.2 (2SF)  | 5 | MSL   | 7025   |

|    |     |   |        |                |   |     |      |
|----|-----|---|--------|----------------|---|-----|------|
| 26 | 100 | 5 | NaCl/m | lgK:-7.2 (2SF) | 5 | MSL | 7025 |
|----|-----|---|--------|----------------|---|-----|------|

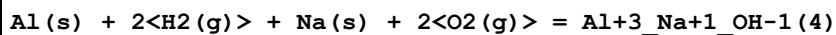
## Reaction No. 58241



| No | t  | I-Str | Medium             | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution      | lgK:-110. (3SF) | 0 | EES   |        |
| 2  | 25 | 0     | Inf. Dilution      | lgK:-97.6 (3SF) | 0 | CRV   | 5629   |
| 3  | 50 | 3     | NaClO <sub>4</sub> | lgK:-97.6 (3SF) | 0 | MGL   | 140    |
| 4  | 50 | 3     | NaClO <sub>4</sub> | lgK:-97.6 (3SF) | 0 | CRV   | 5627   |

Note (4): Error comedy. Cites Biedermann 1962. Actually unpubl. see [64SiM\_140]

## Reaction No. 58242



| No | t   | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
|----|-----|-------|---------------|---------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:275.3 (4SF)     | 1 | ELC   |        |
| 2  | 0   | 0     | Inf. Dilution | dG:-378.030 (6SF)   | 0 | EFE   | 5626   |
| 3  | 10  | 0     | Inf. Dilution | dG:-376.620 (6SF)   | 0 | EFE   | 5626   |
| 4  | 25  | 0     | Inf. Dilution | dG:-374.510 (6SF)   | 0 | EFE   | 5626   |
| 5  | 25  | 0     | Inf. Dilution | dG:-374.610 (6SF)   | 0 | ENS   | 6333   |
| 6  | 25  | 0     | Inf. Dilution | dG:-1562.1 (5SF) kJ | 0 | EFE   | 28693  |
| 7  | 40  | 0     | Inf. Dilution | dG:-372.390 (6SF)   | 0 | EFE   | 5626   |
| 8  | 60  | 0     | Inf. Dilution | dG:-369.550 (6SF)   | 0 | EFE   | 5626   |
| 9  | 80  | 0     | Inf. Dilution | dG:-366.720 (6SF)   | 0 | EFE   | 5626   |
| 10 | 100 | 0     | Inf. Dilution | dG:-363.890 (6SF)   | 0 | EFE   | 5626   |
| 11 | 120 | 0     | Inf. Dilution | dG:-361.030 (6SF)   | 0 | EFE   | 5626   |
| 12 | 140 | 0     | Inf. Dilution | dG:-358.170 (6SF)   | 0 | EFE   | 5626   |
| 13 | 160 | 0     | Inf. Dilution | dG:-355.300 (6SF)   | 0 | EFE   | 5626   |
| 14 | 200 | 0     | Inf. Dilution | dG:-349.530 (6SF)   | 0 | EFE   | 5626   |
| 15 | 250 | 0     | Inf. Dilution | dG:-342.150 (6SF)   | 0 | EFE   | 5626   |
| 16 | 0   | 0     | Inf. Dilution | dH:-416.260 (6SF)   | 0 | MCL   | 5626   |
| 17 | 10  | 0     | Inf. Dilution | dH:-416.440 (6SF)   | 0 | MCL   | 5626   |
| 18 | 25  | 0     | Inf. Dilution | dH:-416.610 (6SF)   | 0 | MCL   | 5626   |
| 19 | 25  | 0     | Inf. Dilution | dH:-413.600 (6SF)   | 0 | ENS   | 6333   |
| 20 | 40  | 0     | Inf. Dilution | dH:-416.710 (6SF)   | 0 | MCL   | 5626   |
| 21 | 60  | 0     | Inf. Dilution | dH:-416.740 (6SF)   | 0 | MCL   | 5626   |
| 22 | 80  | 0     | Inf. Dilution | dH:-416.700 (6SF)   | 0 | MCL   | 5626   |

|    |     |   |               |                   |   |     |      |
|----|-----|---|---------------|-------------------|---|-----|------|
| 23 | 100 | 0 | Inf. Dilution | dH:-417.270 (6SF) | 0 | MCL | 5626 |
| 24 | 120 | 0 | Inf. Dilution | dH:-417.240 (6SF) | 0 | MCL | 5626 |
| 25 | 140 | 0 | Inf. Dilution | dH:-417.290 (6SF) | 0 | MCL | 5626 |
| 26 | 160 | 0 | Inf. Dilution | dH:-417.460 (6SF) | 0 | MCL | 5626 |
| 27 | 200 | 0 | Inf. Dilution | dH:-418.330 (6SF) | 0 | MCL | 5626 |
| 28 | 250 | 0 | Inf. Dilution | dH:-420.930 (6SF) | 0 | MCL | 5626 |
| 29 | 25  | 0 | Inf. Dilution | Cp:32.2 (3SF)     | 0 | ENS | 6333 |

| Reaction No. 58243                                          |     |       |               |                      |   |       |        |
|-------------------------------------------------------------|-----|-------|---------------|----------------------|---|-------|--------|
| Al(s) + 1.5<H2(g)> + 1.5<O2(g)> = Al+3_OH-1(3)_(Gibbsite,s) |     |       |               |                      |   |       |        |
| No                                                          | t   | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
| 1                                                           | 25  | 0     | Inf. Dilution | lgK:203.1 (4SF)      | 1 | ELC   |        |
| 2                                                           | 25  | 0     | Inf. Dilution | lgK:202.4 (4SF)      | 0 | ENS   | 6422   |
| 3                                                           | 27  | 0     | Inf. Dilution | lgK:200.9 (4SF)      | 0 | ENS   | 6422   |
| 4                                                           | 127 | 0     | Inf. Dilution | lgK:144.6 (4SF)      | 0 | ENS   | 6422   |
| 5                                                           | 227 | 0     | Inf. Dilution | lgK:110.8 (4SF)      | 0 | ENS   | 6422   |
| 6                                                           | 25  | 0     | Inf. Dilution | dG:-1155.50 (6SF) kJ | 0 | CRV   | 440    |
| 7                                                           | 25  | 0     | Inf. Dilution | dG:-275.2 (0.2SD)    | 0 | MSL   | 5615   |
| 8                                                           | 25  | 0     | Inf. Dilution | dG:-1155.0 (5SF) kJ  | 0 | EOD   | 5616   |
| 9                                                           | 25  | 0     | Inf. Dilution | dG:-1156.7 (5SF) kJ  | 0 | EFE   | 5617   |
| 10                                                          | 25  | 0     | Inf. Dilution | dG:-1154.90 (6SF) kJ | 0 | CRV   | 5619   |
| 11                                                          | 25  | 0     | Inf. Dilution | dG:-1154.9 (5SF) kJ  | 0 | EFE   | 5621   |
| 12                                                          | 25  | 0     | Inf. Dilution | dG:-1151.90 (6SF) kJ | 0 | CRV   | 5627   |
| 13                                                          | 25  | 0     | Inf. Dilution | dG:-273.49 (0.31MD)  | 0 | CRV   | 5633   |
| 14                                                          | 25  | 0     | Inf. Dilution | dG:-1144.30 (6SF) kJ | 0 | CRV   | 6421   |
| 15                                                          | 25  | 0     | Inf. Dilution | dG:-276.1 (4SF)      | 0 | EFE   | 6422   |
| 16                                                          | 25  | 0     | Inf. Dilution | dG:-273.4 (4SF)      | 0 | CRV   | 7382   |
| 17                                                          | 25  | 0     | Inf. Dilution | dG:-1154.1 (6SF) kJ  | 0 | CRV   | 8872   |
| 18                                                          | 25  | 0     | Inf. Dilution | dG:-1154.50 (6SF) kJ | 0 | EOD   | 8872   |
| Note (18): Data from Khodakovsky, 1983                      |     |       |               |                      |   |       |        |
| 19                                                          | 25  | 0     | Inf. Dilution | dG:-1156.90 (6SF) kJ | 0 | EOD   | 8872   |
| Note (19): Data from Naumov, Ryzhenko & Khodakovsky, 1971   |     |       |               |                      |   |       |        |
| 20                                                          | 25  | 0     | Inf. Dilution | dG:-1144.30 (6SF) kJ | 0 | CRV   | 8873   |
| 21                                                          | 25  | 0     | Inf. Dilution | dG:-1147.10 (6SF) kJ | 0 | CRV   | 8874   |
| 22                                                          | 25  | 0     | Inf. Dilution | dG:-276.03 (5SF)     | 0 | CRV   | 10174  |
| 23                                                          | 25  | 0     | Inf. Dilution | dG:-1155.1 (5SF) kJ  | 0 | EOD   | 29520  |

|    |     |   |               |                       |   |     |       |
|----|-----|---|---------------|-----------------------|---|-----|-------|
| 24 | 25  | 0 | Inf. Dilution | dG:-276.03 (5SF)      | 0 | CRV | 29553 |
| 25 | 25  | 0 | Inf. Dilution | dG:-1151.0 (5SF) kJ   | 0 | CRV | 31308 |
| 26 | 25  | 0 | GAMSPATH      | dG:-1155. (4SF) kJ    | 0 | CRV | 12638 |
| 27 | 50  | 0 | Inf. Dilution | dG:-1143.4 (5SF) kJ   | 0 | EOD | 5616  |
| 28 | 50  | 0 | Inf. Dilution | dG:-276.46 (5SF)      | 0 | CRV | 10174 |
| 29 | 60  | 0 | Inf. Dilution | dG:-1157.6 (5SF) kJ   | 0 | EFE | 5621  |
| 30 | 75  | 0 | Inf. Dilution | dG:-276.93 (5SF)      | 0 | CRV | 10174 |
| 31 | 100 | 0 | Inf. Dilution | dG:-1120.2 (5SF) kJ   | 0 | EOD | 5616  |
| 32 | 100 | 0 | Inf. Dilution | dG:-1160.9 (5SF) kJ   | 0 | EFE | 5621  |
| 33 | 100 | 0 | Inf. Dilution | dG:-277.46 (5SF)      | 0 | CRV | 10174 |
| 34 | 125 | 0 | Inf. Dilution | dG:-278.02 (5SF)      | 0 | CRV | 10174 |
| 35 | 150 | 0 | Inf. Dilution | dG:-1096.9 (5SF) kJ   | 0 | MSL | 5616  |
| 36 | 150 | 0 | Inf. Dilution | dG:-1165.8 (5SF) kJ   | 0 | EFE | 5621  |
| 37 | 150 | 0 | Inf. Dilution | dG:-278.63 (5SF)      | 0 | CRV | 10174 |
| 38 | 175 | 0 | Inf. Dilution | dG:-279.27 (5SF)      | 0 | CRV | 10174 |
| 39 | 200 | 0 | Inf. Dilution | dG:-1073.6 (5SF) kJ   | 0 | MSL | 5616  |
| 40 | 200 | 0 | Inf. Dilution | dG:-279.96 (5SF)      | 0 | CRV | 10174 |
| 41 | 225 | 0 | Inf. Dilution | dG:-280.69 (5SF)      | 0 | CRV | 10174 |
| 42 | 250 | 0 | Inf. Dilution | dG:-1050.4 (5SF) kJ   | 0 | MSL | 5616  |
| 43 | 250 | 0 | Inf. Dilution | dG:-281.45 (5SF)      | 0 | CRV | 10174 |
| 44 | 300 | 0 | Inf. Dilution | dG:-1027.3 (5SF) kJ   | 0 | MSL | 5616  |
| 45 | 300 | 0 | Inf. Dilution | dG:-283.08 (5SF)      | 0 | CRV | 10174 |
| 46 | 25  | 0 | Inf. Dilution | dH:-1293.2 (0.4SD) kJ | 0 | EOD | 5616  |
| 47 | 25  | 0 | Inf. Dilution | dH:-1293.1 (1.2MD) kJ | 0 | CRV | 5619  |
| 48 | 25  | 0 | Inf. Dilution | dH:-309.1 (4SF)       | 0 | MTD | 6422  |
| 49 | 25  | 0 | Inf. Dilution | dH:-306.3 (4SF)       | 0 | CRV | 7382  |
| 50 | 25  | 0 | Inf. Dilution | dH:-1293.3 (5SF) kJ   | 2 | EOD | 29520 |
| 51 | 25  | 0 | Inf. Dilution | dH:-309.06 (5SF)      | 2 | CRV | 29553 |
| 52 | 25  | 0 | Inf. Dilution | dH:-1288.7 (5SF) kJ   | 2 | CRV | 31308 |
| 53 | 25  | 0 | GAMSPATH      | dH:-1293.10 (6SF) kJ  | 0 | CRV | 12638 |
| 54 | 127 | 0 | Inf. Dilution | dH:-309.4 (4SF)       | 0 | MTD | 6422  |
| 55 | 227 | 0 | Inf. Dilution | dH:-309.3 (4SF)       | 0 | MTD | 6422  |

## Reaction No. 58244

$$\text{Al(s)} + 0.5\text{H}_2\text{(g)} + \text{O}_2\text{(g)} = \text{AlOOH(Boehmite,s)}$$

Note: In the exceptional case of Boehmite, the conventional formation reaction is considered a better anchor than the solubility product at 25C

| No                                     | t   | I-Str | Medium        | Value (Dev)            | W | Tech. | Ref. # |
|----------------------------------------|-----|-------|---------------|------------------------|---|-------|--------|
| 1                                      | 25  | 0     | Inf. Dilution | lgK:160.4 (4SF)        | 0 | ENS   | 6422   |
| 2                                      | 25  | 0     | Inf. Dilution | lgK:160.9 (4SF)        | 2 | ENS   | 6494   |
| 3                                      | 27  | 0     | Inf. Dilution | lgK:159.4 (4SF)        | 0 | ENS   | 6422   |
| 4                                      | 27  | 0     | Inf. Dilution | lgK:159.8 (4SF)        | 0 | ENS   | 6494   |
| 5                                      | 127 | 0     | Inf. Dilution | lgK:116.3 (4SF)        | 0 | ENS   | 6422   |
| 6                                      | 127 | 0     | Inf. Dilution | lgK:116.4 (4SF)        | 3 | ENS   | 6494   |
| 7                                      | 227 | 0     | Inf. Dilution | lgK:90.37 (4SF)        | 2 | ENS   | 6422   |
| 8                                      | 227 | 0     | Inf. Dilution | lgK:90.38 (4SF)        | 2 | ENS   | 6494   |
| 9                                      | 327 | 0     | Inf. Dilution | lgK:73.02 (4SF)        | 0 | ENS   | 6494   |
| 10                                     | 25  | 0     | Inf. Dilution | dG:-217.6 (0.2SD)      | 0 | MSL   | 5615   |
| 11                                     | 25  | 0     | Inf. Dilution | dG:-915.9 (4SF) kJ     | 0 | EOD   | 5616   |
| 12                                     | 25  | 0     | Inf. Dilution | dG:-921.0 (1.5SD) kJ   | 0 | EFE   | 5617   |
| 13                                     | 25  | 0     | Inf. Dilution | dG:-918.40 (5SF) kJ    | 3 | CRV   | 5619   |
| 14                                     | 25  | 0     | Inf. Dilution | dG:-915.0 (4SF) kJ     | 0 | EFE   | 5621   |
| 15                                     | 25  | 0     | Inf. Dilution | dG:-915.04 (5SF) kJ    | 0 | CRV   | 5627   |
| 16                                     | 25  | 0     | Inf. Dilution | dG:-218. (4.MD)        | 0 | CRV   | 5631   |
| 17                                     | 25  | 0     | Inf. Dilution | dG:-917.82 (1.93SD) kJ | 0 | MSL   | 6028   |
| 18                                     | 25  | 0     | Inf. Dilution | dG:-219.500 (6SF)      | 3 | ENS   | 6333   |
| 19                                     | 25  | 0     | Inf. Dilution | dG:-218.9 (4SF)        | 0 | EFE   | 6422   |
| 20                                     | 25  | 0     | Inf. Dilution | dG:-219.5 (4SF)        | 3 | EFE   | 6494   |
| 21                                     | 25  | 0     | Inf. Dilution | dG:-218.2 (4SF)        | 0 | CRV   | 7382   |
| 22                                     | 25  | 0     | Inf. Dilution | dG:-916.150 (6SF) kJ   | 0 | CRV   | 8872   |
| 23                                     | 25  | 0     | Inf. Dilution | dG:-915.30 (5SF) kJ    | 0 | EOD   | 8872   |
| Note (23): Data from Khodakovsky, 1983 |     |       |               |                        |   |       |        |
| 24                                     | 25  | 0     | Inf. Dilution | dG:-907.30 (5SF) kJ    | 0 | CRV   | 8874   |
| 25                                     | 25  | 0     | Inf. Dilution | dG:-219.60 (5SF)       | 3 | CRV   | 10174  |
| 26                                     | 25  | 0     | Inf. Dilution | dG:-219.50 (5SF)       | 5 | CRV   | 29553  |
| 27                                     | 25  | 0     | GAMSPATH      | dG:-909. (3SF) kJ      | 0 | CRV   | 12638  |
| 28                                     | 50  | 0     | Inf. Dilution | dG:-909.5 (4SF) kJ     | 0 | EOD   | 5616   |
| 29                                     | 50  | 0     | Inf. Dilution | dG:-219.84 (5SF)       | 0 | CRV   | 10174  |
| 30                                     | 60  | 0     | Inf. Dilution | dG:-916.8 (4SF) kJ     | 0 | EFE   | 5621   |
| 31                                     | 75  | 0     | Inf. Dilution | dG:-220.10 (5SF)       | 0 | CRV   | 10174  |
| 32                                     | 100 | 0     | Inf. Dilution | dG:-897.0 (4SF) kJ     | 0 | EOD   | 5616   |

|    |     |   |               |                      |   |     |       |
|----|-----|---|---------------|----------------------|---|-----|-------|
| 33 | 100 | 0 | Inf. Dilution | dG:-919.2 (4SF) kJ   | 0 | EFE | 5621  |
| 34 | 100 | 0 | Inf. Dilution | dG:-220.39 (5SF)     | 0 | CRV | 10174 |
| 35 | 125 | 0 | Inf. Dilution | dG:-220.70 (5SF)     | 0 | CRV | 10174 |
| 36 | 150 | 0 | Inf. Dilution | dG:-884.6 (4SF) kJ   | 0 | MSL | 5616  |
| 37 | 150 | 0 | Inf. Dilution | dG:-922.7 (4SF) kJ   | 0 | EFE | 5621  |
| 38 | 150 | 0 | Inf. Dilution | dG:-221.04 (5SF)     | 0 | CRV | 10174 |
| 39 | 175 | 0 | Inf. Dilution | dG:-221.40 (5SF)     | 0 | CRV | 10174 |
| 40 | 200 | 0 | Inf. Dilution | dG:-872.2 (4SF) kJ   | 3 | MSL | 5616  |
| 41 | 200 | 0 | Inf. Dilution | dG:-926.7 (4SF) kJ   | 0 | EFE | 5621  |
| 42 | 200 | 0 | Inf. Dilution | dG:-221.78 (5SF)     | 0 | CRV | 10174 |
| 43 | 225 | 0 | Inf. Dilution | dG:-222.19 (5SF)     | 0 | CRV | 10174 |
| 44 | 250 | 0 | Inf. Dilution | dG:-859.9 (4SF) kJ   | 0 | MSL | 5616  |
| 45 | 250 | 0 | Inf. Dilution | dG:-931.1 (4SF) kJ   | 0 | EFE | 5621  |
| 46 | 250 | 0 | Inf. Dilution | dG:-222.61 (5SF)     | 0 | CRV | 10174 |
| 47 | 300 | 0 | Inf. Dilution | dG:-847.7 (4SF) kJ   | 0 | MSL | 5616  |
| 48 | 300 | 0 | Inf. Dilution | dG:-936.0 (4SF) kJ   | 0 | EFE | 5621  |
| 49 | 300 | 0 | Inf. Dilution | dG:-223.53 (5SF)     | 0 | CRV | 10174 |
| 50 | 25  | 0 | Inf. Dilution | dH:-990.4 (1.3SD) kJ | 0 | EOD | 5616  |
| 51 | 25  | 0 | Inf. Dilution | dH:-993.1 (2.2MD) kJ | 0 | CRV | 5619  |
| 52 | 25  | 0 | Inf. Dilution | dH:-992.19 (5SF) kJ  | 0 | MTD | 6028  |
| 53 | 25  | 0 | Inf. Dilution | dH:-238.140 (6SF)    | 2 | ENS | 6333  |
| 54 | 25  | 0 | Inf. Dilution | dH:-236.7 (4SF)      | 0 | MTD | 6422  |
| 55 | 25  | 0 | Inf. Dilution | dH:-238.1 (4SF)      | 2 | MTD | 6494  |
| 56 | 25  | 0 | Inf. Dilution | dH:-236.0 (4SF)      | 0 | CRV | 7382  |
| 57 | 25  | 0 | Inf. Dilution | dH:-238.15 (5SF)     | 5 | CRV | 29553 |
| 58 | 25  | 0 | GAMSPATH      | dH:-983.570 (6SF) kJ | 0 | CRV | 12638 |
| 59 | 127 | 0 | Inf. Dilution | dH:-236.8 (4SF)      | 0 | MTD | 6422  |
| 60 | 127 | 0 | Inf. Dilution | dH:-238.4 (4SF)      | 0 | MTD | 6494  |
| 61 | 227 | 0 | Inf. Dilution | dH:-236.8 (4SF)      | 0 | MTD | 6422  |
| 62 | 227 | 0 | Inf. Dilution | dH:-238.4 (4SF)      | 0 | MTD | 6494  |
| 63 | 327 | 0 | Inf. Dilution | dH:-238.3 (4SF)      | 0 | MTD | 6494  |
| 64 | 25  | 0 | Inf. Dilution | Cp:12.96 (4SF)       | 2 | ENS | 6333  |

Reaction No. 58245

$$\text{AlOOH(Boehmite,s)} + 2\text{<H2O>} = \text{Al+3\_OH-1(4)} + \text{H+1}$$

Note: In the exceptional case of Boehmite, the conventional formation reaction is considered a better anchor than the solubility product at 25C

| No | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|-----|-------|---------------|------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:-14.5 (3SF)  | 1 | ELC   |        |
| 2  | 25  | 0     | Inf. Dilution | lgK:-15.19 (4SF) | 0 | ENS   | 6333   |
| 3  | 50  | 0     | Inf. Dilution | lgK:-14.32 (4SF) | 0 | ENS   | 6333   |
| 4  | 60  | 0     | Inf. Dilution | lgK:-14.00 (4SF) | 0 | ENS   | 6333   |
| 5  | 70  | 0     | Inf. Dilution | lgK:-13.37 (4SF) | 0 | ENS   | 6333   |
| 6  | 75  | 0     | Inf. Dilution | lgK:-13.56 (4SF) | 0 | ENS   | 6333   |
| 7  | 80  | 0     | Inf. Dilution | lgK:-13.35 (4SF) | 0 | EDT   | 5622   |
| 8  | 80  | 0     | Inf. Dilution | lgK:-13.42 (4SF) | 0 | ENS   | 6333   |
| 9  | 90  | 0     | Inf. Dilution | lgK:-13.17 (4SF) | 0 | ENS   | 6333   |
| 10 | 100 | 0     | Inf. Dilution | lgK:-12.79 (4SF) | 0 | EDT   | 5622   |
| 11 | 100 | 0     | Inf. Dilution | lgK:-12.96 (4SF) | 0 | ENS   | 6333   |
| 12 | 120 | 0     | Inf. Dilution | lgK:-12.44 (4SF) | 0 | EDT   | 5622   |
| 13 | 125 | 0     | Inf. Dilution | lgK:-12.48 (4SF) | 0 | ENS   | 6333   |
| 14 | 130 | 0     | Inf. Dilution | lgK:-12.34 (4SF) | 0 | EDT   | 5622   |
| 15 | 150 | 0     | Inf. Dilution | lgK:-11.99 (4SF) | 0 | EDT   | 5622   |
| 16 | 150 | 0     | Inf. Dilution | lgK:-12.06 (4SF) | 0 | ENS   | 6333   |
| 17 | 150 |       |               | lgK:-12.03 (4SF) | 0 | MSL   | 4281   |
| 18 | 170 | 0     | Inf. Dilution | lgK:-11.68 (4SF) | 0 | EDT   | 5622   |
| 19 | 170 | 0     | Inf. Dilution | lgK:-11.75 (4SF) | 0 | ENS   | 6333   |
| 20 | 200 | 0     | Inf. Dilution | lgK:-11.39 (4SF) | 0 | ENS   | 6333   |
| 21 | 200 |       |               | lgK:-11.41 (4SF) | 0 | MSL   | 4281   |
| 22 | 250 | 0     | Inf. Dilution | lgK:-10.75 (4SF) | 0 | EDT   | 5622   |
| 23 | 250 | 0     | Inf. Dilution | lgK:-10.99 (4SF) | 0 | ENS   | 6333   |
| 24 | 250 |       |               | lgK:-11.13 (4SF) | 0 | MSL   | 4281   |
| 25 | 300 | 0     | Inf. Dilution | lgK:-10.53 (4SF) | 0 | EDT   | 5622   |
| 26 | 300 | 0     | Inf. Dilution | lgK:-10.74 (4SF) | 0 | ENS   | 6333   |
| 27 | 325 | 0     | Inf. Dilution | lgK:-10.76 (4SF) | 0 | ENS   | 6333   |
| 28 | 350 | 0     | Inf. Dilution | lgK:-10.77 (4SF) | 0 | ENS   | 6333   |

## Reaction No. 58246

$$0.5\text{Al}_2\text{O}_3(\text{alpha},\text{s}) + 3\text{H}^+ = \text{Al}^{3+} + 1.5\text{H}_2\text{O}$$

| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:9.0 (2SF)     | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:9.01 (0.05SD) | 0 | MSL   | 5617   |



| Reaction No. 58247                                                          |       |       |        |                    |   |       |        |
|-----------------------------------------------------------------------------|-------|-------|--------|--------------------|---|-------|--------|
| $\text{Al}+3_{\text{OH}-1(4)} + \text{OH}-1 = \text{Al}+3_{\text{OH}-1(5)}$ |       |       |        |                    |   |       |        |
| No                                                                          | t     | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                           | 15    | 1     | NaOH   | lgR:0.158(0.15MD)  | 3 | MCN   | 5636   |
| 2                                                                           | 15    | 2     | NaOH   | lgR:0.228(0.18MD)  | 3 | MCN   | 5636   |
| 3                                                                           | 15    | 4     | NaOH   | lgR:0.480(0.18MD)  | 3 | MCN   | 5636   |
| 4                                                                           | 15    | 6     | NaOH   | lgR:0.885(0.32MD)  | 3 | MCN   | 5636   |
| 5                                                                           | 25    | 1     | NaOH   | lgR:-0.119(0.09MD) | 3 | MCN   | 5636   |
| 6                                                                           | 25    | 2     | NaOH   | lgR:-0.046(0.11MD) | 3 | MCN   | 5636   |
| 7                                                                           | 25    | 4     | NaOH   | lgR:0.193(0.12MD)  | 3 | MCN   | 5636   |
| 8                                                                           | 25    | 6     | NaOH   | lgR:0.589(0.18MD)  | 3 | MCN   | 5636   |
| 9                                                                           | 35    | 1     | NaOH   | lgR:-0.284(0.08MD) | 3 | MCN   | 5636   |
| 10                                                                          | 35    | 2     | NaOH   | lgR:-0.260(0.09MD) | 3 | MCN   | 5636   |
| 11                                                                          | 35    | 4     | NaOH   | lgR:-0.125(0.09MD) | 3 | MCN   | 5636   |
| 12                                                                          | 35    | 6     | NaOH   | lgR:0.207(0.15MD)  | 3 | MCN   | 5636   |
| 13                                                                          | 15:35 | 1     | NaOH   | dH:-37.5(0.1MD) kJ | 0 | MTD   | 5636   |
| 14                                                                          | 15:35 | 2     | NaOH   | dH:-40.8(0.1MD) kJ | 0 | MTD   | 5636   |
| 15                                                                          | 15:35 | 4     | NaOH   | dH:-50.4(0.1MD) kJ | 0 | MTD   | 5636   |
| 16                                                                          | 15:35 | 6     | NaOH   | dH:-57.5(0.1MD) kJ | 0 | MTD   | 5636   |

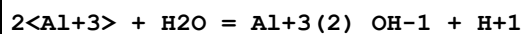
| Reaction No. 58248                                                                                                    |    |       |               |                |   |       |        |
|-----------------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $\text{Al}+3 + 2\langle\text{H}_2\text{O}\rangle = \text{AlOOH}(\text{Pseudoboehmite},s) + 3\langle\text{H}+1\rangle$ |    |       |               |                |   |       |        |
| No                                                                                                                    | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                                     | 25 | 0     | Inf. Dilution | lgK:-9.80(3SF) | 3 | EOD   | 5629   |

| Reaction No. 58249                                                                                                             |    |       |               |                 |   |       |        |
|--------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $4\langle\text{Al}+3\rangle + 8\langle\text{H}_2\text{O}\rangle = \text{Al}+3(4)_{\text{OH}-1(8)} + 8\langle\text{H}+1\rangle$ |    |       |               |                 |   |       |        |
| No                                                                                                                             | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                              | 25 | 0     | Inf. Dilution | lgK:-23.9(3SF)  | 2 | EAE   |        |
| 2                                                                                                                              | 30 | 2     | NaNO3         | lgK:-27.0(3SF)w | 0 | MPH   | 5640   |
| 3                                                                                                                              | 40 | 2     | NaNO3         | lgK:-25.5(3SF)w | 3 | MPH   | 5640   |

| Reaction No. 58250                                                                                                                |    |       |               |                |   |       |        |
|-----------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $7\langle\text{Al}+3\rangle + 16\langle\text{H}_2\text{O}\rangle = \text{Al}+3(7)_{\text{OH}-1(16)} + 16\langle\text{H}+1\rangle$ |    |       |               |                |   |       |        |
| No                                                                                                                                | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                                                 | 25 | 0     | Inf. Dilution | lgK:-52.2(3SF) | 2 | EAE   |        |

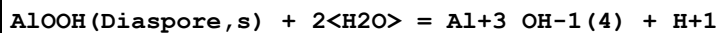
|   |    |   |       |                  |   |     |      |
|---|----|---|-------|------------------|---|-----|------|
| 2 | 30 | 2 | NaNO3 | lgK:-52.7 (3SF)w | 3 | MPH | 5640 |
| 3 | 40 | 2 | NaNO3 | lgK:-48.5 (3SF)w | 3 | MPH | 5640 |

## Reaction No. 58251



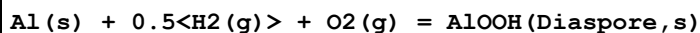
| No | t  | I-Str | Medium    | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|-----------|----------------|---|-------|--------|
| 1  | 25 | 0.06  | Cl-1 salt | lgK:-3.1 (2SF) | 5 | MLC   | 5588   |

## Reaction No. 58252



| No | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|-----|-------|---------------|------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:-14.7 (3SF)  | 1 | ELC   |        |
| 2  | 280 | 0     | NaOH/m        | lgK:-11.15 (4SF) | 0 | MSL   | 5631   |
| 3  | 280 | 0.1   | NaOH/m        | lgK:-11.14 (4SF) | 4 | MSL   | 5631   |
| 4  | 280 | 0.5   | NaOH/m        | lgK:-11.11 (4SF) | 4 | MSL   | 5631   |
| 5  | 280 | 1     | NaOH/m        | lgK:-11.06 (4SF) | 4 | MSL   | 5631   |
| 6  | 280 | 2     | NaOH/m        | lgK:-10.98 (4SF) | 3 | MSL   | 5631   |
| 7  | 280 | 5     | NaOH/m        | lgK:-10.70 (4SF) | 3 | MSL   | 5631   |
| 8  | 300 | 0     | NaOH/m        | lgK:-11.06 (4SF) | 0 | MSL   | 5631   |
| 9  | 300 | 0.1   | NaOH/m        | lgK:-11.05 (4SF) | 0 | MSL   | 5631   |
| 10 | 300 | 0.5   | NaOH/m        | lgK:-11.02 (4SF) | 0 | MSL   | 5631   |
| 11 | 300 | 1     | NaOH/m        | lgK:-10.98 (4SF) | 0 | MSL   | 5631   |
| 12 | 300 | 2     | NaOH/m        | lgK:-10.89 (4SF) | 0 | MSL   | 5631   |
| 13 | 300 | 5     | NaOH/m        | lgK:-10.49 (4SF) | 0 | MSL   | 5631   |
| 14 | 325 | 0     | NaOH/m        | lgK:-10.96 (4SF) | 0 | MSL   | 5631   |
| 15 | 325 | 0.1   | NaOH/m        | lgK:-10.95 (4SF) | 0 | MSL   | 5631   |
| 16 | 325 | 0.5   | NaOH/m        | lgK:-10.92 (4SF) | 0 | MSL   | 5631   |
| 17 | 325 | 1     | NaOH/m        | lgK:-10.88 (4SF) | 0 | MSL   | 5631   |
| 18 | 325 | 2     | NaOH/m        | lgK:-10.77 (4SF) | 0 | MSL   | 5631   |
| 19 | 325 | 5     | NaOH/m        | lgK:-10.23 (4SF) | 0 | MSL   | 5631   |

## Reaction No. 58253

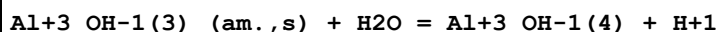


| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:162.4 (4SF) | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:161.4 (4SF) | 0 | ENS   | 6422   |
| 3  | 27 | 0     | Inf. Dilution | lgK:160.3 (4SF) | 0 | ENS   | 6422   |

|                                                           |     |   |               |                       |   |     |       |
|-----------------------------------------------------------|-----|---|---------------|-----------------------|---|-----|-------|
| 4                                                         | 127 | 0 | Inf. Dilution | lgK:116.7 (4SF)       | 0 | ENS | 6422  |
| 5                                                         | 227 | 0 | Inf. Dilution | lgK:90.57 (4SF)       | 0 | ENS | 6422  |
| 6                                                         | 25  | 0 | Inf. Dilution | dG:-913.78 (5SF) kJ   | 0 | CRV | 440   |
| 7                                                         | 25  | 0 | Inf. Dilution | dG:-219. (0.4SD)      | 0 | MSL | 5615  |
| 8                                                         | 25  | 0 | Inf. Dilution | dG:-921.3 (4SF) kJ    | 0 | EOD | 5616  |
| 9                                                         | 25  | 0 | Inf. Dilution | dG:-925.1 (2.0SD) kJ  | 0 | EFE | 5617  |
| 10                                                        | 25  | 0 | Inf. Dilution | dG:-922.00 (5SF) kJ   | 0 | CRV | 5619  |
| 11                                                        | 25  | 0 | Inf. Dilution | dG:-918.39 (5SF) kJ   | 0 | CRV | 5627  |
| 12                                                        | 25  | 0 | Inf. Dilution | dG:-216. (3SF)        | 0 | CRV | 5642  |
| 13                                                        | 25  | 0 | Inf. Dilution | dG:-219.9 (4SF)       | 0 | CRV | 5642  |
| 14                                                        | 25  | 0 | Inf. Dilution | dG:-924.59 (3.5SD) kJ | 0 | MSL | 6028  |
| 15                                                        | 25  | 0 | Inf. Dilution | dG:-220.1 (4SF)       | 0 | EFE | 6422  |
| 16                                                        | 25  | 0 | Inf. Dilution | dG:-220.08 (5SF)      | 0 | EFE | 6482  |
| 17                                                        | 25  | 0 | Inf. Dilution | dG:-220. (3SF)        | 0 | CRV | 7382  |
| 18                                                        | 25  | 0 | Inf. Dilution | dG:-912.700 (6SF) kJ  | 0 | CRV | 8872  |
| 19                                                        | 25  | 0 | Inf. Dilution | dG:-920.80 (5SF) kJ   | 0 | EOD | 8872  |
| Note (19): Data from Khodakovsky, 1983                    |     |   |               |                       |   |     |       |
| 20                                                        | 25  | 0 | Inf. Dilution | dG:-921.32 (5SF) kJ   | 0 | EOD | 8872  |
| Note (20): Data from Naumov, Ryzhenko & Khodakovsky, 1971 |     |   |               |                       |   |     |       |
| 21                                                        | 25  | 0 | Inf. Dilution | dG:-220.15 (5SF)      | 0 | CRV | 10174 |
| 22                                                        | 25  | 0 | Inf. Dilution | dG:-219.68 (5SF)      | 5 | CRV | 29553 |
| 23                                                        | 50  | 0 | Inf. Dilution | dG:-914.7 (4SF) kJ    | 0 | EOD | 5616  |
| 24                                                        | 50  | 0 | Inf. Dilution | dG:-220.37 (5SF)      | 0 | CRV | 10174 |
| 25                                                        | 75  | 0 | Inf. Dilution | dG:-220.63 (5SF)      | 0 | CRV | 10174 |
| 26                                                        | 100 | 0 | Inf. Dilution | dG:-901.4 (4SF) kJ    | 0 | EOD | 5616  |
| 27                                                        | 100 | 0 | Inf. Dilution | dG:-220.90 (5SF)      | 0 | CRV | 10174 |
| 28                                                        | 125 | 0 | Inf. Dilution | dG:-221.20 (5SF)      | 0 | CRV | 10174 |
| 29                                                        | 150 | 0 | Inf. Dilution | dG:-888.1 (4SF) kJ    | 0 | MSL | 5616  |
| 30                                                        | 150 | 0 | Inf. Dilution | dG:-221.53 (5SF)      | 0 | CRV | 10174 |
| 31                                                        | 175 | 0 | Inf. Dilution | dG:-221.88 (5SF)      | 0 | CRV | 10174 |
| 32                                                        | 200 | 0 | Inf. Dilution | dG:-874.7 (4SF) kJ    | 0 | MSL | 5616  |
| 33                                                        | 200 | 0 | Inf. Dilution | dG:-222.25 (5SF)      | 0 | CRV | 10174 |
| 34                                                        | 225 | 0 | Inf. Dilution | dG:-222.64 (5SF)      | 0 | CRV | 10174 |
| 35                                                        | 250 | 0 | Inf. Dilution | dG:-861.4 (4SF) kJ    | 0 | MSL | 5616  |
| 36                                                        | 250 | 0 | Inf. Dilution | dG:-223.06 (5SF)      | 0 | CRV | 10174 |
| 37                                                        | 300 | 0 | Inf. Dilution | dG:-848.1 (4SF) kJ    | 0 | MSL | 5616  |

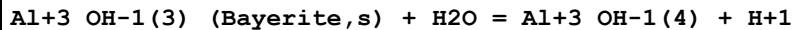
|    |     |   |               |                        |   |     |       |
|----|-----|---|---------------|------------------------|---|-----|-------|
| 38 | 300 | 0 | Inf. Dilution | dG:-223.94 (5SF)       | 0 | CRV | 10174 |
| 39 | 25  | 0 | Inf. Dilution | dH:-999.8 (0.4SD) kJ   | 0 | EOD | 5616  |
| 40 | 25  | 0 | Inf. Dilution | dH:-1000.6 (5.0MD) kJ  | 0 | CRV | 5619  |
| 41 | 25  | 0 | Inf. Dilution | dH:-1003.20 (3.5SD) kJ | 0 | MTD | 6028  |
| 42 | 25  | 0 | Inf. Dilution | dH:-238.9 (4SF)        | 0 | MTD | 6422  |
| 43 | 25  | 0 | Inf. Dilution | dH:-238.86 (5SF)       | 0 | EFE | 6482  |
| 44 | 25  | 0 | Inf. Dilution | dH:-239. (3SF)         | 0 | ENS | 7382  |
| 45 | 25  | 0 | Inf. Dilution | dH:-238.46 (5SF)       | 5 | CRV | 29553 |
| 46 | 127 | 0 | Inf. Dilution | dH:-239.3 (4SF)        | 0 | MTD | 6422  |
| 47 | 227 | 0 | Inf. Dilution | dH:-239.7 (4SF)        | 0 | MTD | 6422  |

## Reaction No. 58254



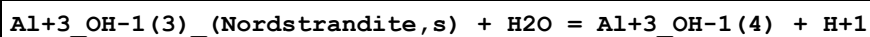
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 20 | 0     | Inf. Dilution | lgK:-12.62 (4SF) | 0 | MTN   | 5630   |
| 2  | 25 | 0     | Inf. Dilution | lgK:-11.95 (4SF) | 1 | ELC   |        |
| 3  | 25 | 0     | Inf. Dilution | lgK:-12.71 (4SF) | 0 | EOD   | 5628   |

## Reaction No. 58255



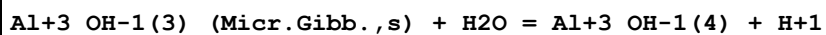
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-14.22 (4SF) | 0 | EOD   | 5628   |

## Reaction No. 58256



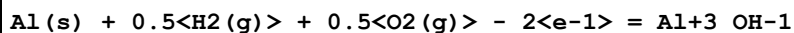
| No | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|-----|-------|---------------|------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:-14.66 (4SF) | 4 | EOD   | 5628   |
| 2  | 100 | 0     | Inf. Dilution | lgK:-12. (2SF)   | 1 | ELC   |        |

## Reaction No. 58257



| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-13.4 (3SF)  | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:-13.36 (4SF) | 0 | EOD   | 5628   |

## Reaction No. 58258



| No | t   | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
|----|-----|-------|---------------|----------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:122.8 (4SF)      | 1 | ELC   |        |
| 2  | 25  | 0     | Inf. Dilution | dG:-696.5 (2.SD) kJ  | 0 | EFE   | 2171   |
| 3  | 25  | 0     | Inf. Dilution | dG:-698.5 (4SF) kJ   | 0 | EFE   | 5616   |
| 4  | 25  | 0     | Inf. Dilution | dG:-165.9 (4SF)      | 0 | CRV   | 7382   |
| 5  | 25  | 0     | Inf. Dilution | dG:-696.5 (4SF) kJ   | 0 | CRV   | 8775   |
| 6  | 25  | 0     | Inf. Dilution | dG:-166.47 (5SF)     | 0 | CRV   | 10174  |
| 7  | 25  | 0     | Inf. Dilution | dG:-692.60 (5SF) kJ  | 0 | EOD   | 23101  |
| 8  | 25  | 0     | Inf. Dilution | dG:-695.57 (5SF) kJ  | 0 | EFE   | 28693  |
| 9  | 25  | 0     | GAMSPATH      | dG:-695.5 (4SF) kJ   | 0 | CRV   | 12638  |
| 10 | 50  | 0     | Inf. Dilution | dG:-692.4 (4SF) kJ   | 0 | EFE   | 5616   |
| 11 | 50  | 0     | Inf. Dilution | dG:-165.31 (5SF)     | 0 | CRV   | 10174  |
| 12 | 75  | 0     | Inf. Dilution | dG:-164.19 (5SF)     | 0 | CRV   | 10174  |
| 13 | 100 | 0     | Inf. Dilution | dG:-680.0 (4SF) kJ   | 0 | EFE   | 5616   |
| 14 | 100 | 0     | Inf. Dilution | dG:-163.10 (5SF)     | 0 | CRV   | 10174  |
| 15 | 125 | 0     | Inf. Dilution | dG:-162.03 (5SF)     | 0 | CRV   | 10174  |
| 16 | 150 | 0     | Inf. Dilution | dG:-667.2 (4SF) kJ   | 0 | MSL   | 5616   |
| 17 | 150 | 0     | Inf. Dilution | dG:-160.97 (5SF)     | 0 | CRV   | 10174  |
| 18 | 175 | 0     | Inf. Dilution | dG:-159.91 (5SF)     | 0 | CRV   | 10174  |
| 19 | 200 | 0     | Inf. Dilution | dG:-654.4 (4SF) kJ   | 0 | MSL   | 5616   |
| 20 | 200 | 0     | Inf. Dilution | dG:-158.85 (5SF)     | 0 | CRV   | 10174  |
| 21 | 225 | 0     | Inf. Dilution | dG:-157.78 (5SF)     | 0 | CRV   | 10174  |
| 22 | 250 | 0     | Inf. Dilution | dG:-641.5 (4SF) kJ   | 0 | MSL   | 5616   |
| 23 | 250 | 0     | Inf. Dilution | dG:-156.68 (5SF)     | 0 | CRV   | 10174  |
| 24 | 300 | 0     | Inf. Dilution | dG:-628.7 (4SF) kJ   | 0 | MSL   | 5616   |
| 25 | 300 | 0     | Inf. Dilution | dG:-154.35 (5SF)     | 0 | CRV   | 10174  |
| 26 | 25  | 0     | Inf. Dilution | dH:-769. (4.SD) kJ   | 0 | MTD   | 2171   |
| 27 | 25  | 0     | Inf. Dilution | dH:-771.7 (4SF) kJ   | 0 | EOD   | 5616   |
| 28 | 25  | 0     | Inf. Dilution | dH:-771. (5.0MD) kJ  | 0 | CRV   | 8775   |
| 29 | 25  | 0     | Inf. Dilution | dH:-767.27 (5SF) kJ  | 0 | EOD   | 23101  |
| 30 | 25  | 0     | GAMSPATH      | dH:-753.040 (6SF) kJ | 0 | CRV   | 12638  |
| 31 | 25  | 0     | Inf. Dilution | Cp:-44. (2SF) J/K    | 0 | CRV   | 8775   |

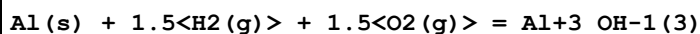
Reaction No. 58259

Al(s) + H2(g) + O2(g) - e-1 = Al+3\_OH-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

|    |     |   |               |                      |   |     |       |
|----|-----|---|---------------|----------------------|---|-----|-------|
| 1  | 25  | 0 | Inf. Dilution | lgK:158.7 (4SF)      | 1 | ELC |       |
| 2  | 25  | 0 | Inf. Dilution | dG:-904.3 (4SF) kJ   | 0 | EFE | 2171  |
| 3  | 25  | 0 | Inf. Dilution | dG:-904.7 (4SF) kJ   | 0 | EOD | 5616  |
| 4  | 25  | 0 | Inf. Dilution | dG:-914.2 (2.1MD) kJ | 0 | CRV | 5619  |
| 5  | 25  | 0 | Inf. Dilution | dG:-904.3 (4SF) kJ   | 0 | CRV | 8775  |
| 6  | 25  | 0 | Inf. Dilution | dG:-215.47 (5SF)     | 0 | CRV | 10174 |
| 7  | 25  | 0 | Inf. Dilution | dG:-898.30 (5SF) kJ  | 0 | EFE | 28693 |
| 8  | 50  | 0 | Inf. Dilution | dG:-896.5 (4SF) kJ   | 0 | EOD | 5616  |
| 9  | 50  | 0 | Inf. Dilution | dG:-215.08 (5SF)     | 0 | CRV | 10174 |
| 10 | 75  | 0 | Inf. Dilution | dG:-214.76 (5SF)     | 0 | CRV | 10174 |
| 11 | 100 | 0 | Inf. Dilution | dG:-880.1 (4SF) kJ   | 0 | EOD | 5616  |
| 12 | 100 | 0 | Inf. Dilution | dG:-214.51 (5SF)     | 0 | CRV | 10174 |
| 13 | 125 | 0 | Inf. Dilution | dG:-214.32 (5SF)     | 0 | CRV | 10174 |
| 14 | 150 | 0 | Inf. Dilution | dG:-863.7 (4SF) kJ   | 0 | MSL | 5616  |
| 15 | 150 | 0 | Inf. Dilution | dG:-214.17 (5SF)     | 0 | CRV | 10174 |
| 16 | 175 | 0 | Inf. Dilution | dG:-214.07 (5SF)     | 0 | CRV | 10174 |
| 17 | 200 | 0 | Inf. Dilution | dG:-847.4 (4SF) kJ   | 0 | MSL | 5616  |
| 18 | 200 | 0 | Inf. Dilution | dG:-214.01 (5SF)     | 0 | CRV | 10174 |
| 19 | 225 | 0 | Inf. Dilution | dG:-213.98 (5SF)     | 0 | CRV | 10174 |
| 20 | 250 | 0 | Inf. Dilution | dG:-831.2 (4SF) kJ   | 0 | MSL | 5616  |
| 21 | 250 | 0 | Inf. Dilution | dG:-213.97 (5SF)     | 0 | CRV | 10174 |
| 22 | 300 | 0 | Inf. Dilution | dG:-815.0 (4SF) kJ   | 0 | MSL | 5616  |
| 23 | 300 | 0 | Inf. Dilution | dG:-213.99 (5SF)     | 0 | CRV | 10174 |
| 24 | 25  | 0 | Inf. Dilution | dH:-990. (3SF) kJ    | 0 | MTD | 2171  |
| 25 | 25  | 0 | Inf. Dilution | dH:-1002.8 (5SF) kJ  | 0 | EOD | 5616  |
| 26 | 25  | 0 | Inf. Dilution | dH:-990. (3SF) kJ    | 0 | CRV | 8775  |
| 27 | 25  | 0 | Inf. Dilution | Cp:-194. (3SF) J/K   | 0 | CRV | 8775  |

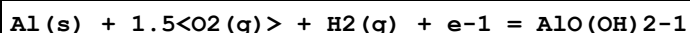
## Reaction No. 58260



| No | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:193.5 (4SF)     | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | dG:-1104. (4SF) kJ  | 0 | EFE   | 2171   |
| 3  | 25 | 0     | Inf. Dilution | dG:-1106.8 (5SF) kJ | 0 | EOD   | 5616   |
| 4  | 25 | 0     | Inf. Dilution | dG:-265.250 (6SF)   | 0 | ENS   | 6333   |
| 5  | 25 | 0     | Inf. Dilution | dG:-1104. (4SF) kJ  | 0 | MSL   | 8775   |

|    |     |   |               |                     |   |     |       |
|----|-----|---|---------------|---------------------|---|-----|-------|
| 6  | 25  | 0 | Inf. Dilution | dG:1105.8 (5SF) kJ  | 0 | EFE | 28693 |
| 7  | 50  | 0 | Inf. Dilution | dG:-1095.2 (5SF) kJ | 0 | EOD | 5616  |
| 8  | 100 | 0 | Inf. Dilution | dG:-1072.2 (5SF) kJ | 0 | EOD | 5616  |
| 9  | 150 | 0 | Inf. Dilution | dG:-1049.4 (5SF) kJ | 0 | MSL | 5616  |
| 10 | 200 | 0 | Inf. Dilution | dG:-1026.9 (5SF) kJ | 0 | MSL | 5616  |
| 11 | 250 | 0 | Inf. Dilution | dG:-1004.7 (5SF) kJ | 0 | MSL | 5616  |
| 12 | 300 | 0 | Inf. Dilution | dG:-982.7 (4SF) kJ  | 0 | MSL | 5616  |
| 13 | 25  | 0 | Inf. Dilution | dH:-1222. (4SF) kJ  | 0 | MTD | 2171  |
| 14 | 25  | 0 | Inf. Dilution | dH:-1245.6 (5SF) kJ | 0 | EOD | 5616  |
| 15 | 25  | 0 | Inf. Dilution | dH:-300.550 (6SF)   | 0 | ENS | 6333  |
| 16 | 25  | 0 | Inf. Dilution | dH:-1222. (4SF) kJ  | 2 | CRV | 8775  |
| 17 | 25  | 0 | Inf. Dilution | dH:-1276. (4SF) kJ  | 0 | CRV | 10802 |
| 18 | 25  | 0 | Inf. Dilution | Cp:24.3 (3SF)       | 2 | ENS | 6333  |

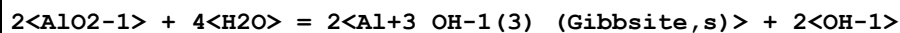
## Reaction No. 58261



Note: The species AlO<sub>2</sub><sup>-</sup>, corresponding to AlO(OH)<sub>2</sub><sup>-1</sup>, was proposed by Russell et al., [55RET\_18518]. It has also appeared on different occasions in the Russian literature. We found no sign of such under 'normal' conditions. I doubt it ever exists but the matter awaits definitive resolution when Al solubilities at higher temperatures and (very high Na concs.) are fully understood. PMM (31/5/2013)

| No | t   | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
|----|-----|-------|---------------|---------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | dG:-1067.8 (5SF) kJ | 0 | EOD   | 5616   |
| 2  | 50  | 0     | Inf. Dilution | dG:-1055.5 (5SF) kJ | 0 | EOD   | 5616   |
| 3  | 100 | 0     | Inf. Dilution | dG:-1030.6 (5SF) kJ | 0 | EOD   | 5616   |
| 4  | 150 | 0     | Inf. Dilution | dG:-1005.2 (5SF) kJ | 0 | MSL   | 5616   |
| 5  | 200 | 0     | Inf. Dilution | dG:-979.3 (4SF) kJ  | 0 | MSL   | 5616   |
| 6  | 250 | 0     | Inf. Dilution | dG:-952.6 (4SF) kJ  | 0 | MSL   | 5616   |
| 7  | 300 | 0     | Inf. Dilution | dG:-925.3 (4SF) kJ  | 0 | MSL   | 5616   |
| 8  | 25  | 0     | Inf. Dilution | dH:-1213.9 (5SF) kJ | 0 | EOD   | 5616   |

## Reaction No. 58262



Note: The species AlO<sub>2</sub><sup>-1</sup> is an artefact; it should be Al+3\_OH<sup>-1(4)</sup>

| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 40 | 0     | Inf. Dilution | lgK:2.01 (3SF) | 0 | MSL   | 5614   |
| 2  | 40 | 0.5   | Inf. Dilution | lgK:1.97 (3SF) | 0 | MSL   | 5614   |
| 3  | 50 | 0     | Inf. Dilution | lgK:1.69 (3SF) | 0 | MSL   | 5614   |

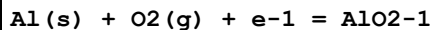
|    |     |     |               |                   |   |     |      |
|----|-----|-----|---------------|-------------------|---|-----|------|
| 4  | 50  | 0.5 | Inf. Dilution | lgK:1.66 (3SF)    | 0 | MSL | 5614 |
| 5  | 60  | 0   | Inf. Dilution | lgK:1.38 (3SF)    | 0 | MSL | 5614 |
| 6  | 60  | 0.5 | Inf. Dilution | lgK:1.36 (3SF)    | 0 | MSL | 5614 |
| 7  | 70  | 0   | Inf. Dilution | lgK:1.10 (3SF)    | 0 | MSL | 5614 |
| 8  | 70  | 0.5 | Inf. Dilution | lgK:1.08 (3SF)    | 0 | MSL | 5614 |
| 9  | 80  | 0   | Inf. Dilution | lgK:0.84 (2SF)    | 0 | MSL | 5614 |
| 10 | 80  | 0.5 | Inf. Dilution | lgK:0.819 (3SF)   | 0 | MSL | 5614 |
| 11 | 90  | 0   | Inf. Dilution | lgK:0.60 (2SF)    | 0 | MSL | 5614 |
| 12 | 90  | 0.5 | Inf. Dilution | lgK:0.575 (3SF)   | 0 | MSL | 5614 |
| 13 | 100 | 0   | Inf. Dilution | lgK:0.37 (2SF)    | 0 | MSL | 5614 |
| 14 | 100 | 0.5 | Inf. Dilution | lgK:0.347 (3SF)   | 0 | MSL | 5614 |
| 15 | 110 | 0   | Inf. Dilution | lgK:0.160 (3SF)   | 0 | MSL | 5614 |
| 16 | 110 | 0.5 | Inf. Dilution | lgK:0.133 (3SF)   | 0 | MSL | 5614 |
| 17 | 120 | 0   | Inf. Dilution | lgK:-0.0448 (3SF) | 0 | MSL | 5614 |
| 18 | 120 | 0.5 | Inf. Dilution | lgK:-0.0680 (3SF) | 0 | MSL | 5614 |
| 19 | 40  | 1   | NaOH          | lgR:1.96 (3SF)    | 3 | MSL | 5614 |
| 20 | 40  | 2   | NaOH          | lgR:1.92 (3SF)    | 3 | MSL | 5614 |
| 21 | 40  | 3   | NaOH          | lgR:1.87 (3SF)    | 3 | MSL | 5614 |
| 22 | 40  | 4   | NaOH          | lgR:1.82 (3SF)    | 3 | MSL | 5614 |
| 23 | 40  | 5   | NaOH          | lgR:1.78 (3SF)    | 3 | MSL | 5614 |
| 24 | 40  | 6   | NaOH          | lgR:1.73 (3SF)    | 3 | MSL | 5614 |
| 25 | 40  | 7   | NaOH          | lgR:1.68 (3SF)    | 3 | MSL | 5614 |
| 26 | 50  | 1   | NaOH          | lgR:1.64 (3SF)    | 3 | MSL | 5614 |
| 27 | 50  | 2   | NaOH          | lgR:1.59 (3SF)    | 3 | MSL | 5614 |
| 28 | 50  | 3   | NaOH          | lgR:1.55 (3SF)    | 3 | MSL | 5614 |
| 29 | 50  | 4   | NaOH          | lgR:1.50 (3SF)    | 3 | MSL | 5614 |
| 30 | 50  | 5   | NaOH          | lgR:1.45 (3SF)    | 3 | MSL | 5614 |
| 31 | 50  | 6   | NaOH          | lgR:1.41 (3SF)    | 3 | MSL | 5614 |
| 32 | 50  | 7   | NaOH          | lgR:1.36 (3SF)    | 3 | MSL | 5614 |
| 33 | 60  | 1   | NaOH          | lgR:1.34 (3SF)    | 3 | MSL | 5614 |
| 34 | 60  | 2   | NaOH          | lgR:1.29 (3SF)    | 3 | MSL | 5614 |
| 35 | 60  | 3   | NaOH          | lgR:1.24 (3SF)    | 3 | MSL | 5614 |
| 36 | 60  | 4   | NaOH          | lgR:1.20 (3SF)    | 3 | MSL | 5614 |
| 37 | 60  | 5   | NaOH          | lgR:1.15 (3SF)    | 3 | MSL | 5614 |
| 38 | 60  | 6   | NaOH          | lgR:1.105 (4SF)   | 3 | MSL | 5614 |
| 39 | 60  | 7   | NaOH          | lgR:1.058 (4SF)   | 3 | MSL | 5614 |



|    |     |   |      |                   |   |     |      |
|----|-----|---|------|-------------------|---|-----|------|
| 40 | 70  | 1 | NaOH | lgR:1.06 (3SF)    | 3 | MSL | 5614 |
| 41 | 70  | 2 | NaOH | lgR:1.01 (3SF)    | 3 | MSL | 5614 |
| 42 | 70  | 3 | NaOH | lgR:0.964 (3SF)   | 3 | MSL | 5614 |
| 43 | 70  | 4 | NaOH | lgR:0.918 (3SF)   | 3 | MSL | 5614 |
| 44 | 70  | 5 | NaOH | lgR:0.871 (3SF)   | 3 | MSL | 5614 |
| 45 | 70  | 6 | NaOH | lgR:0.825 (3SF)   | 3 | MSL | 5614 |
| 46 | 70  | 7 | NaOH | lgR:0.778 (3SF)   | 3 | MSL | 5614 |
| 47 | 80  | 1 | NaOH | lgR:0.796 (3SF)   | 3 | MSL | 5614 |
| 48 | 80  | 2 | NaOH | lgR:0.750 (3SF)   | 3 | MSL | 5614 |
| 49 | 80  | 3 | NaOH | lgR:0.703 (3SF)   | 3 | MSL | 5614 |
| 50 | 80  | 4 | NaOH | lgR:0.656 (3SF)   | 3 | MSL | 5614 |
| 51 | 80  | 5 | NaOH | lgR:0.610 (3SF)   | 3 | MSL | 5614 |
| 52 | 80  | 6 | NaOH | lgR:0.563 (3SF)   | 3 | MSL | 5614 |
| 53 | 80  | 7 | NaOH | lgR:0.516 (3SF)   | 3 | MSL | 5614 |
| 54 | 90  | 1 | NaOH | lgR:0.552 (3SF)   | 3 | MSL | 5614 |
| 55 | 90  | 2 | NaOH | lgR:0.505 (3SF)   | 3 | MSL | 5614 |
| 56 | 90  | 3 | NaOH | lgR:0.457 (3SF)   | 3 | MSL | 5614 |
| 57 | 90  | 4 | NaOH | lgR:0.412 (3SF)   | 3 | MSL | 5614 |
| 58 | 90  | 5 | NaOH | lgR:0.365 (3SF)   | 3 | MSL | 5614 |
| 59 | 90  | 6 | NaOH | lgR:0.319 (3SF)   | 3 | MSL | 5614 |
| 60 | 90  | 7 | NaOH | lgR:0.272 (3SF)   | 3 | MSL | 5614 |
| 61 | 100 | 1 | NaOH | lgR:0.323 (3SF)   | 3 | MSL | 5614 |
| 62 | 100 | 2 | NaOH | lgR:0.277 (3SF)   | 3 | MSL | 5614 |
| 63 | 100 | 3 | NaOH | lgR:0.230 (3SF)   | 3 | MSL | 5614 |
| 64 | 100 | 4 | NaOH | lgR:0.184 (3SF)   | 3 | MSL | 5614 |
| 65 | 100 | 5 | NaOH | lgR:0.137 (3SF)   | 3 | MSL | 5614 |
| 66 | 100 | 6 | NaOH | lgR:0.0903 (3SF)  | 3 | MSL | 5614 |
| 67 | 100 | 7 | NaOH | lgR:0.0438 (3SF)  | 3 | MSL | 5614 |
| 68 | 110 | 1 | NaOH | lgR:0.110 (3SF)   | 3 | MSL | 5614 |
| 69 | 110 | 2 | NaOH | lgR:0.0629 (3SF)  | 3 | MSL | 5614 |
| 70 | 110 | 3 | NaOH | lgR:0.0162 (3SF)  | 3 | MSL | 5614 |
| 71 | 110 | 4 | NaOH | lgR:-0.0306 (3SF) | 3 | MSL | 5614 |
| 72 | 110 | 5 | NaOH | lgR:-0.0773 (3SF) | 3 | MSL | 5614 |
| 73 | 110 | 6 | NaOH | lgR:-0.124 (3SF)  | 3 | MSL | 5614 |
| 74 | 110 | 7 | NaOH | lgR:-0.170 (3SF)  | 3 | MSL | 5614 |
| 75 | 120 | 1 | NaOH | lgR:-0.0910 (3SF) | 3 | MSL | 5614 |

|    |     |   |      |                  |   |     |      |
|----|-----|---|------|------------------|---|-----|------|
| 76 | 120 | 2 | NaOH | lgR:-0.138 (3SF) | 3 | MSL | 5614 |
| 77 | 120 | 3 | NaOH | lgR:-0.184 (3SF) | 3 | MSL | 5614 |
| 78 | 120 | 4 | NaOH | lgR:-0.231 (3SF) | 3 | MSL | 5614 |
| 79 | 120 | 5 | NaOH | lgR:-0.277 (3SF) | 3 | MSL | 5614 |
| 80 | 120 | 6 | NaOH | lgR:-0.324 (3SF) | 3 | MSL | 5614 |
| 81 | 120 | 7 | NaOH | lgR:-0.371 (3SF) | 3 | MSL | 5614 |

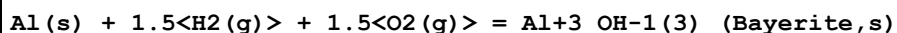
## Reaction No. 58263



Note: The species AlO<sub>2</sub><sup>-</sup> is an artefact; it should be Al<sub>3</sub>OH<sup>-</sup>(4)

| No | t   | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|-----|-------|---------------|--------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | dG:-830.9 (4SF) kJ | 0 | EOD   | 5619   |
| 2  | 25  | 0     | Inf. Dilution | dG:-196.7 (4SF)    | 0 | CRV   | 7382   |
| 3  | 25  | 0     | Inf. Dilution | dG:-198.70 (5SF)   | 0 | CRV   | 10174  |
| 4  | 50  | 0     | Inf. Dilution | dG:-198.52 (5SF)   | 0 | CRV   | 10174  |
| 5  | 75  | 0     | Inf. Dilution | dG:-198.32 (5SF)   | 0 | CRV   | 10174  |
| 6  | 100 | 0     | Inf. Dilution | dG:-198.12 (5SF)   | 0 | CRV   | 10174  |
| 7  | 125 | 0     | Inf. Dilution | dG:-197.90 (5SF)   | 0 | CRV   | 10174  |
| 8  | 150 | 0     | Inf. Dilution | dG:-197.66 (5SF)   | 0 | CRV   | 10174  |
| 9  | 175 | 0     | Inf. Dilution | dG:-197.39 (5SF)   | 0 | CRV   | 10174  |
| 10 | 200 | 0     | Inf. Dilution | dG:-197.07 (5SF)   | 0 | CRV   | 10174  |
| 11 | 225 | 0     | Inf. Dilution | dG:-196.71 (5SF)   | 0 | CRV   | 10174  |
| 12 | 250 | 0     | Inf. Dilution | dG:-196.27 (5SF)   | 0 | CRV   | 10174  |
| 13 | 300 | 0     | Inf. Dilution | dG:-195.01 (5SF)   | 0 | CRV   | 10174  |
| 14 | 25  | 0     | Inf. Dilution | dH:-219.6 (4SF)    | 0 | CRV   | 7382   |

## Reaction No. 58264



| No | t  | I-Str | Medium        | Value (Dev)             | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:202.4 (4SF)         | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | dG:-1153. (2.MD) kJ     | 0 | CRV   | 5619   |
| 3  | 25 | 0     | Inf. Dilution | dG:-1149.80 (1.87SD) kJ | 0 | MSL   | 6028   |
| 4  | 25 | 0     | Inf. Dilution | dG:-275.53 (5SF)        | 0 | EFE   | 6215   |
| 5  | 25 | 0     | Inf. Dilution | dH:-1282.70 (6SF) kJ    | 0 | MTD   | 6028   |
| 6  | 25 | 0     | Inf. Dilution | dH:-305.1 (4SF)         | 0 | CRV   | 7382   |

## Reaction No. 58331

| 3<Al+3> + 3<PO4-3> = Al+3(3)_PO4-3(3) |    |       |               |               |   |       |        |
|---------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| No                                    | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                     | 25 | 0     | Inf. Dilution | lgK:59.1(3SF) | 1 | EAE   |        |
| 2                                     | 37 | 0.15  | NaCl          | lgK:52.6(3SF) | 2 | MIS   | 5983   |

| Reaction No. 58680                        |    |       |        |                |   |       |        |
|-------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Hex16DiPhos-4 + Al+3 = Al+3_Hex16DiPhos-4 |    |       |        |                |   |       |        |
| No                                        | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                         | 25 | 0.1   | KCl    | lgK:14.66(4SF) | 4 | MGL   | 931    |

| Reaction No. 58728                                         |    |       |        |                    |   |       |        |
|------------------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| 3<Al+3> + CO2(g) + 4<H2O> = Al+3(3)_OH-1(3)_CO3-2 + 5<H+1> |    |       |        |                    |   |       |        |
| No                                                         | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                          | 25 | 0.1   | Na(Cl) | lgK:-17.2(0.34SD)  | 5 | MGL   | 5494   |
| 2                                                          | 25 | 0.6   | NaCl   | lgK:-22.74(4SF)    | 0 | MGL   | 5447   |
| 3                                                          | 25 | 3     | Na(Cl) | lgK:-18.39(0.07SD) | 5 | MGL   | 5494   |

| Reaction No. 59036                          |    |       |         |                |   |       |        |
|---------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Al+3_OH-1 + Benzoic-1 = Al+3_Benzoic-1_OH-1 |    |       |         |                |   |       |        |
| No                                          | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                           | 25 | 0.5   | Unknown | lgK:12.09(4SF) | 0 | ENS   | 5971   |

| Reaction No. 59037                                    |    |       |               |              |   |       |        |
|-------------------------------------------------------|----|-------|---------------|--------------|---|-------|--------|
| Al+3 + H+1_Chromotropic-4 = Al+3_Chromotropic-4 + H+1 |    |       |               |              |   |       |        |
| No                                                    | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                     | 20 | 0.1   | Unknown       | lgK:1.5(2SF) | 4 | ENS   | 5971   |
| 2                                                     | 25 | 0     | Inf. Dilution | lgK:1.5(2SF) | 1 | ESC   |        |

| Reaction No. 59038                  |    |       |        |                 |   |       |        |
|-------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2OxButan-1 + Al+3 = Al+3_2OxButan-1 |    |       |        |                 |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | KCl    | lgK:2.1(0.03SD) | 5 | MGL   | 6523   |

| Reaction No. 59039                    |    |       |         |               |   |       |        |
|---------------------------------------|----|-------|---------|---------------|---|-------|--------|
| PhenylSer-1 + Al+3 = Al+3_PhenylSer-1 |    |       |         |               |   |       |        |
| No                                    | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | Unknown | lgK:3.98(3SF) | 4 | ENS   | 5971   |

| Reaction No. 59040          |    |       |         |               |   |       |        |
|-----------------------------|----|-------|---------|---------------|---|-------|--------|
| MIDA-2 + Al+3 = Al+3_MIDA-2 |    |       |         |               |   |       |        |
| No                          | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                           | 25 | 0.5   | NaClO4  | lgK:7.6(2SF)  | 5 | CRV   | 13242  |
| 2                           | 25 | 0.5   | Unknown | lgK:7.55(3SF) | 4 | ENS   | 5971   |

| Reaction No. 59041                           |    |       |         |                 |   |       |        |
|----------------------------------------------|----|-------|---------|-----------------|---|-------|--------|
| Al+3_OH-1_34DHB-3 + H+1 = Al+3_34DHB-3 + H2O |    |       |         |                 |   |       |        |
| No                                           | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                                            | 25 | 0.1   | KCl     | lgK:5.8(0.07SD) | 4 | MGL   | 1148   |
| 2                                            | 25 | 0.1   | Unknown | lgK:5.77(3SF)   | 4 | ENS   | 5971   |

| Reaction No. 59042                                          |    |       |               |               |   |       |        |
|-------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| 12DiOH4SulfNaphthoic-3 + Al+3 = Al+3_12DiOH4SulfNaphthoic-3 |    |       |               |               |   |       |        |
| No                                                          | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                           | 25 | 0     | Inf. Dilution | lgK:17.5(3SF) | 1 | ELC   |        |
| 2                                                           | 25 | 0.6   | Unknown       | lgK:5.9(2SF)  | 0 | ENS   | 5971   |
| Note (2): Likely to be an NDP problem                       |    |       |               |               |   |       |        |

| Reaction No. 59043                        |    |       |               |               |   |       |        |
|-------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Al+3 + H+1_DHEGly-1 = Al+3_DHEGly-1 + H+1 |    |       |               |               |   |       |        |
| No                                        | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                         | 20 | 0.1   | Unknown       | lgK:3.38(3SF) | 4 | ENS   | 5971   |
| 2                                         | 25 | 0     | Inf. Dilution | lgK:3.6(2SF)  | 1 | ESC   |        |

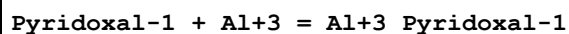
| Reaction No. 59044                            |    |       |        |                 |   |       |        |
|-----------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| aKetoglutaric-2 + Al+3 = Al+3_aKetoglutaric-2 |    |       |        |                 |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                             | 25 | 0.1   | KCl    | lgK:6.2(0.07SD) | 4 | ENS   | 5971   |

| Reaction No. 59045                |    |       |         |               |   |       |        |
|-----------------------------------|----|-------|---------|---------------|---|-------|--------|
| Al+3 + H+1_Asp-2 = Al+3_H+1_Asp-2 |    |       |         |               |   |       |        |
| No                                | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                 | 25 | 0.5   | Unknown | lgK:2.17(3SF) | 4 | ENS   | 5971   |

| Reaction No. 59046                |  |  |  |  |  |  |  |
|-----------------------------------|--|--|--|--|--|--|--|
| Al+3 + H+1_Glu-2 = Al+3_H+1_Glu-2 |  |  |  |  |  |  |  |

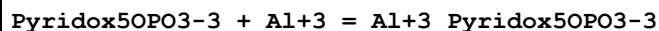
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.5   | NaClO4 | lgK:2.30 (3SF) | 4 | ENS   | 6953   |

## Reaction No. 59047



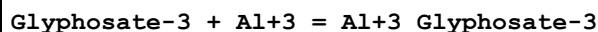
| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:2.45 (3SF) | 4 | ENS   | 5971   |

## Reaction No. 59048



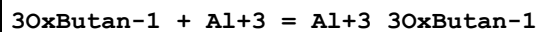
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 22 | 0.1   | Unknown | lgK:10.48 (4SF) | 4 | ENS   | 5971   |
| 2  | 25 | 1     | KCl     | lgK:10.48 (4SF) | 5 | MGL   | 1585   |

## Reaction No. 59049



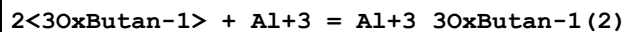
| No | t  | I-Str | Medium  | Value (Dev)         | W | Tech. | Ref. # |
|----|----|-------|---------|---------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3    | lgK:13.70 (0.003SD) | 4 | MGL   | 6861   |
| 2  | 25 | 0.1   | Unknown | lgK:13.7 (0.1SD)    | 6 | CIU   | 7242   |

## Reaction No. 59054



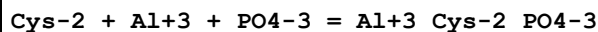
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:1.2 (0.04SD) | 5 | MGL   | 6523   |

## Reaction No. 59055



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:1.8 (0.04SD) | 5 | MGL   | 6523   |

## Reaction No. 59315



| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:18. (2SF) | 1 | EES   |        |

Note (1): Minimum value given m+l+xh is 15.7 from [3180]

|   |    |     |        |                 |   |     |     |
|---|----|-----|--------|-----------------|---|-----|-----|
| 2 | 25 | 0.1 | NaClO4 | lgK:15.66 (4SF) | 0 | CRV | 905 |
|---|----|-----|--------|-----------------|---|-----|-----|

Note (2): NDP error

| Reaction No. 59329                              |    |       |               |                 |   |       |        |
|-------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Al+3 + H+1_PO4-3 + Cys-2 = Al+3_H+1_Cys-2_PO4-3 |    |       |               |                 |   |       |        |
| No                                              | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                               | 25 | 0     | Inf. Dilution | lgK:17.5 (3SF)  | 1 | EDH   |        |
| 2                                               | 25 | 0.1   | NaClO4        | lgK:15.66 (4SF) | 3 | MGL   | 3180   |

| Reaction No. 59330                              |    |       |        |                 |   |       |        |
|-------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Al+3 + NTA-3 + H+1_PO4-3 = Al+3_H+1_NTA-3_PO4-3 |    |       |        |                 |   |       |        |
| No                                              | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                               | 25 | 0.1   | NaClO4 | lgK:23.89 (4SF) | 2 | MGL   | 3180   |

| Reaction No. 59331                                    |    |       |        |                 |   |       |        |
|-------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Al+3 + Citric-3 + H+1_PO4-3 = Al+3_H+1_Citric-3_PO4-3 |    |       |        |                 |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                     | 25 | 0.1   | NaClO4 | lgK:19.29 (4SF) | 2 | MGL   | 3180   |

| Reaction No. 59412        |    |       |        |                |   |       |        |
|---------------------------|----|-------|--------|----------------|---|-------|--------|
| Ser-1 + Al+3 = Al+3_Ser-1 |    |       |        |                |   |       |        |
| No                        | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                         | 25 | 0.2   | KCl    | lgK:5.66 (3SF) | 5 | MGL   | 10434  |
| 2                         | 37 | 0.15  | NaCl   | lgK:5.97 (3SF) | 4 | MGL   | 905    |

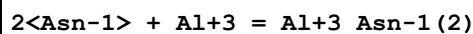
| Reaction No. 59499        |    |       |        |                |   |       |        |
|---------------------------|----|-------|--------|----------------|---|-------|--------|
| Thr-1 + Al+3 = Al+3_Thr-1 |    |       |        |                |   |       |        |
| No                        | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                         | 25 | 0.2   | KCl    | lgK:5.51 (3SF) | 5 | MGL   | 10434  |
| 2                         | 37 | 0.15  | NaCl   | lgK:5.71 (3SF) | 4 | MGL   | 905    |

| Reaction No. 59509                          |    |       |               |                |   |       |        |
|---------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Thr-1 + Al+3 + P3O10-5 = Al+3_Thr-1_P3O10-5 |    |       |               |                |   |       |        |
| No                                          | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                           | 25 | 0     | Inf. Dilution | lgK:7.9 (2SF)  | 1 | ESC   |        |
| 2                                           | 35 | 0.1   | KNO3          | lgK:7.55 (3SF) | 4 | MGL   | 905    |

| Reaction No. 59547        |   |       |        |             |   |       |        |
|---------------------------|---|-------|--------|-------------|---|-------|--------|
| Asn-1 + Al+3 = Al+3_Asn-1 |   |       |        |             |   |       |        |
| No                        | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

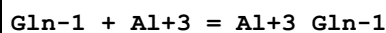
|   |    |     |        |                |   |     |       |
|---|----|-----|--------|----------------|---|-----|-------|
| 1 | 25 | 0.1 | NaClO4 | lgK:6.65 (3SF) | 0 | MGL | 905   |
| 2 | 25 | 0.2 | KCl    | lgK:5.50 (3SF) | 5 | MGL | 10434 |

## Reaction No. 59548



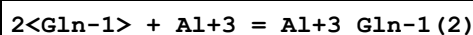
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:10.30 (4SF) | 5 | MGL   | 905    |

## Reaction No. 59596



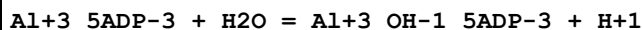
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:6.59 (3SF) | 0 | MGL   | 905    |
| 2  | 25 | 0.2   | KCl    | lgK:5.61 (3SF) | 3 | MGL   | 10434  |

## Reaction No. 59597



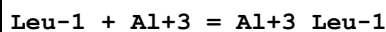
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:10.25 (4SF) | 4 | MGL   | 905    |

## Reaction No. 59973



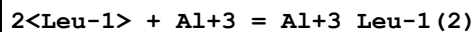
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.15  | NaCl   | lgK:-5.85 (3SF) | 4 | MGL   | 1437   |

## Reaction No. 60015



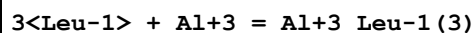
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:8.7 (2SF)  | 1 | ESC   |        |
| 2  | 35 | 0.1   | NaClO4        | lgK:8.50 (3SF) | 0 | MEP   | 2392   |

## Reaction No. 60016



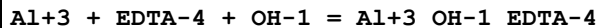
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:15.5 (3SF)  | 1 | ESC   |        |
| 2  | 35 | 0.1   | NaClO4        | lgK:15.20 (4SF) | 0 | MEP   | 2392   |

## Reaction No. 60019



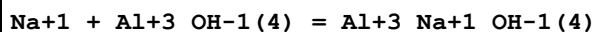
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:19.2 (3SF)  | 1 | ESC   |        |
| 2  | 35 | 0.1   | NaClO4        | lgK:18.90 (4SF) | 0 | MGL   | 2392   |

## Reaction No. 60196



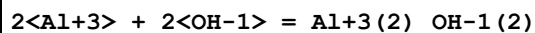
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | KClO4         | lgK:25.04 (4SF) | 3 | MDS   | 903    |
| 2  | 25 | 0     | Inf. Dilution | lgK:26.6 (3SF)  | 4 | ENS   | 6405   |

## Reaction No. 60311



| No | t   | I-Str | Medium                 | Value (Dev)       | W | Tech. | Ref. # |
|----|-----|-------|------------------------|-------------------|---|-------|--------|
| 1  | 25  | 0     | P=500<br>Inf. Dilution | lgK:-0.17 (2SF)   | 4 | EIE   | 6333   |
| 2  | 50  | 0     | P=500<br>Inf. Dilution | lgK:-0.01 (1SF)   | 4 | EIE   | 6333   |
| 3  | 100 | 0     | P=500<br>Inf. Dilution | lgK:0.29 (2SF)    | 3 | EIE   | 6333   |
| 4  | 150 | 0     | P=500<br>Inf. Dilution | lgK:0.54 (2SF)    | 2 | EIE   | 6333   |
| 5  | 200 | 0     | P=500<br>Inf. Dilution | lgK:0.77 (2SF)    | 1 | EIE   | 6333   |
| 6  | 250 | 0     | P=500<br>Inf. Dilution | lgK:1.02 (3SF)    | 0 | EIE   | 6333   |
| 7  | 300 | 0     | P=500<br>Inf. Dilution | lgK:1.31 (3SF)    | 0 | EIE   | 6333   |
| 8  | 25  | 0     | Inf. Dilution          | lgK:-0.10 (2SF)   | 5 | EIE   | 6333   |
| 9  | 25  | 5     | Me4NC1                 | lgK:0.30 (0.04SD) | 6 | MHG   | 7693   |
| 10 | 50  | 0     | Inf. Dilution          | lgK:0.07 (1SF)    | 5 | EIE   | 6333   |
| 11 | 100 | 0     | Inf. Dilution          | lgK:0.35 (2SF)    | 5 | EIE   | 6333   |
| 12 | 150 | 0     | Inf. Dilution          | lgK:0.61 (2SF)    | 5 | EIE   | 6333   |
| 13 | 200 | 0     | Inf. Dilution          | lgK:0.87 (2SF)    | 5 | EIE   | 6333   |
| 14 | 250 | 0     | Inf. Dilution          | lgK:1.17 (3SF)    | 4 | EIE   | 6333   |
| 15 | 300 | 0     | Inf. Dilution          | lgK:1.58 (3SF)    | 0 | EIE   | 6333   |
| 16 | 350 | 0     | Inf. Dilution          | lgK:2.32 (3SF)    | 0 | EIE   | 6333   |

## Reaction No. 60566



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|



|   |    |   |               |               |   |     |      |
|---|----|---|---------------|---------------|---|-----|------|
| 1 | 25 | 0 | Inf. Dilution | lgK:20.3(3SF) | 4 | CRV | 2556 |
| 2 | 25 | 0 | Inf. Dilution | lgK:25.0(3SF) | 0 | ENS | 6405 |
| 3 | 25 | 0 | Inf. Dilution | lgK:20.3(3SF) | 5 | CRV | 6405 |
| 4 | 25 | 1 | Unknown       | lgK:19.6(3SF) | 5 | CRV | 2556 |
| 5 | 25 | 1 | Unknown       | dH:-9.(1SF)   | 5 | CRV | 552  |

| Reaction No. 60567                                                      |     |       |               |               |   |       |        |
|-------------------------------------------------------------------------|-----|-------|---------------|---------------|---|-------|--------|
| 3<Al+3> + 4<OH-1> = Al+3(3)_OH-1(4)                                     |     |       |               |               |   |       |        |
| No                                                                      | t   | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                       | 25  | 0     | Inf. Dilution | lgK:42.1(3SF) | 4 | CRV   | 2556   |
| 2                                                                       | 25  | 0     | Inf. Dilution | lgK:42.1(3SF) | 5 | CRV   | 6405   |
| 3                                                                       | 25  | 0.1   | Unknown       | lgK:41.4(3SF) | 0 | CRV   | 2556   |
| 4                                                                       | 25  | 0.5   | Unknown       | lgK:41.3(3SF) | 0 | CRV   | 2556   |
| 5                                                                       | 25  | 1     | Unknown       | lgK:41.7(3SF) | 0 | CRV   | 2556   |
| Note (5): Low wgt for inter-reaction consistency                        |     |       |               |               |   |       |        |
| 6                                                                       | 25  | 3     | NaCl          | lgK:42.1(3SF) | 0 | CRV   | 2556   |
| Note (6): Very sensitive to pKw; query its dependence on ionic strength |     |       |               |               |   |       |        |
| 7                                                                       | 25  | 3     | Unknown       | lgK:45.6(3SF) | 1 | EOR   |        |
| 8                                                                       | 100 | 0     | Inf. Dilution | lgK:41.9(3SF) | 1 | EOR   |        |
| 9                                                                       | 25  | 1     | Unknown       | dH:-20.(2SF)  | 3 | CRV   | 552    |

| Reaction No. 60568                                                       |       |       |               |                |   |       |        |
|--------------------------------------------------------------------------|-------|-------|---------------|----------------|---|-------|--------|
| 13<Al+3> + 32<OH-1> = Al13O4OH24+7 + 4<H2O>                              |       |       |               |                |   |       |        |
| No                                                                       | t     | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                        | 25    | 0     | Inf. Dilution | lgK:348.3(4SF) | 0 | ELC   |        |
| Note (1): Best not use this rxn; v. small error in pKw = v. large effect |       |       |               |                |   |       |        |
| 2                                                                        | 25    | 0     | Inf. Dilution | lgK:349.3(4SF) | 0 | CRV   | 2556   |
| Note (2): Best not use this rxn; v. small error in pKw = v. large effect |       |       |               |                |   |       |        |
| 3                                                                        | 25    | 0.1   | Unknown       | lgK:333.3(4SF) | 0 | CRV   | 2556   |
| Note (3): Low wgt for internal consistency                               |       |       |               |                |   |       |        |
| 4                                                                        | 25    | 0.5   | Unknown       | lgK:333.8(4SF) | 0 | CRV   | 2556   |
| 5                                                                        | 25    | 1     | Unknown       | lgK:336.2(4SF) | 0 | CRV   | 2556   |
| Note (5): Low wgt for inter-reaction consistency                         |       |       |               |                |   |       |        |
| 6                                                                        | 25    | 3     | NaCl          | lgK:335.3(4SF) | 0 | CRV   | 2556   |
| 7                                                                        | 10:40 | 1     | Unknown       | dH:-160.(3SF)  | 0 | CRV   | 2556   |

| Reaction No. 60656 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| Al+3_PO4-3 + Al+3 = Al+3(2)_PO4-3 |    |       |               |              |   |       |        |
|-----------------------------------|----|-------|---------------|--------------|---|-------|--------|
| No                                | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
| 1                                 | 25 | 0     | Inf. Dilution | lgK:1.0(2SF) | 1 | EES   |        |
| 2                                 | 37 | 0.15  | Unknown       | lgK:5.2(2SF) | 0 | CRV   | 2556   |

| Reaction No. 60657                                        |    |       |               |                  |   |       |        |
|-----------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 2<Al+3> + PO4-3 + 2<H2O> = Al+3(2)_OH-1(2)_PO4-3 + 2<H+1> |    |       |               |                  |   |       |        |
| No                                                        | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                         | 25 | 0     | Inf. Dilution | lgK:10.1(3SF)    | 1 | EES   |        |
| 2                                                         | 37 | 0.15  | NaCl          | lgK:15.8(0.06SD) | 0 | MGL   | 922    |
| 3                                                         | 37 | 0.15  | Unknown       | lgK:15.80(4SF)   | 0 | CRV   | 2556   |

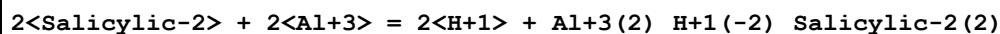
| Reaction No. 60658                                        |    |       |               |                |   |       |        |
|-----------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| 2<Al+3> + PO4-3 + 3<H2O> = Al+3(2)_OH-1(3)_PO4-3 + 3<H+1> |    |       |               |                |   |       |        |
| No                                                        | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                         | 25 | 0     | Inf. Dilution | lgK:7.5(2SF)   | 1 | EES   |        |
| 2                                                         | 37 | 0.15  | NaCl          | lgK:6.7(0.2SD) | 1 | MGL   | 922    |
| 3                                                         | 37 | 0.15  | Unknown       | lgK:6.7(2SF)   | 0 | CRV   | 2556   |

| Reaction No. 60697                       |    |       |               |                    |   |       |        |
|------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| Al+3_F-1(2)_OH-1(s) = Al+3_OH-1 + 2<F-1> |    |       |               |                    |   |       |        |
| No                                       | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                        | 25 | 0     | Inf. Dilution | lgK:-13.59(4SF)    | 4 | CRV   | 2556   |
| 2                                        | 25 | 0     | Inf. Dilution | lgK:-13.59(0.15SD) | 3 | EDH   | 30072  |
| 3                                        | 25 | 0.15  | NaCl          | lgK:-12.88(0.04SD) | 3 | MGL   | 30072  |
| 4                                        | 25 | 1     | NaCl          | lgK:-12.66(0.04SD) | 3 | MGL   | 30072  |

| Reaction No. 60769                                          |    |       |               |                |   |       |        |
|-------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Al+3_OH-1(3)_(Bayerite,s) + 3<H+1> = Al+3 + 3<H2O>          |    |       |               |                |   |       |        |
| No                                                          | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                           | 25 | 0     | Inf. Dilution | lgK:8.2(2SF)   | 1 | EOR   |        |
| 2                                                           | 25 | 0     | Inf. Dilution | lgK:7.00(3SF)  | 0 | EDH   | 6215   |
| 3                                                           | 25 | 0     | Inf. Dilution | lgK:8.5(0.1SD) | 4 | CRV   | 6478   |
| Note (3): Solid phase not specified, assumed to be Bayerite |    |       |               |                |   |       |        |
| 4                                                           | 25 | 0     | Inf. Dilution | lgK:8.51(3SF)  | 3 | MGL   | 6498   |
| 5                                                           | 25 | 3     | Unknown       | lgK:10.1(3SF)  | 1 | EOR   |        |
| 6                                                           | 90 | 1     | (H,Na)ClO4    | lgK:5.43(3SF)  | 4 | MSL   | 6215   |

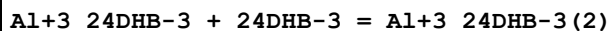
|   |     |   |               |                |   |     |      |
|---|-----|---|---------------|----------------|---|-----|------|
| 7 | 100 | 0 | Inf. Dilution | lgK:6.6(2SF)   | 1 | EOR |      |
| 8 | 25  | 0 | Inf. Dilution | dH:-23.5(1.SD) | 4 | CRV | 6478 |

## Reaction No. 60789



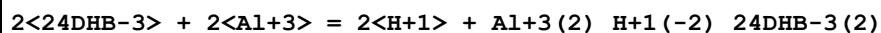
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:17.9(0.2SD) | 7 | MGL   | 2470   |

## Reaction No. 60872



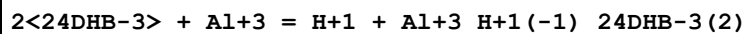
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:6.32(3SF) | 6 | MGL   | 5987   |

## Reaction No. 60873



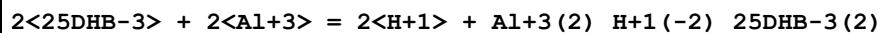
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:9.(0.3SD) | 6 | MGL   | 2470   |

## Reaction No. 60874



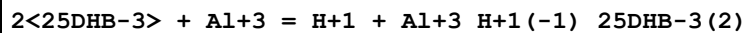
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:7.2(0.1SD) | 6 | MGL   | 2470   |

## Reaction No. 60875



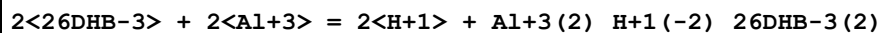
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:11.5(0.2SD) | 7 | MGL   | 2470   |

## Reaction No. 60876



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:10.(0.08SD) | 7 | MGL   | 2470   |

## Reaction No. 60877



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:17.(0.3SD) | 6 | MGL   | 2470   |

| Reaction No. 60878                                      |    |       |        |                   |   |       |        |
|---------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| $2<26DHB-3> + Al+3 = H+1 + Al+3_{H+1(-1)}_{26DHB-3(2)}$ |    |       |        |                   |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.2   | KCl    | lgK:16.5 (0.09SD) | 6 | MGL   | 2470   |

| Reaction No. 60920                            |    |       |               |                |   |       |        |
|-----------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $Al+3 + H+1_{25DHB-3} = Al+3_{H+1}_{25DHB-3}$ |    |       |               |                |   |       |        |
| No                                            | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                             | 25 | 0     | Inf. Dilution | lgK:10.7 (3SF) | 1 | EES   |        |
| 2                                             | 25 | 0.5   | NaClO4        | lgK:10.4 (3SF) | 0 | MGL   | 5987   |

| Reaction No. 60921                                        |    |       |        |                 |   |       |        |
|-----------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $2<H+1> + 2<25DHB-3> + Al+3 = Al+3_{H+1(2)}_{25DHB-3(2)}$ |    |       |        |                 |   |       |        |
| No                                                        | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                         | 25 | 0.5   | NaClO4 | lgK:18.15 (4SF) | 6 | MGL   | 5987   |

| Reaction No. 60922                             |    |       |        |                |   |       |        |
|------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $Al+3_{25DHB-3} + 25DHB-3 = Al+3_{25DHB-3(2)}$ |    |       |        |                |   |       |        |
| No                                             | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                              | 25 | 0.5   | NaClO4 | lgK:7.43 (3SF) | 7 | MGL   | 5987   |

| Reaction No. 61053                                  |    |       |         |                |   |       |        |
|-----------------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| $Al+3 + H+1_{5NitrSal-2} = Al+3_{5NitrSal-2} + H+1$ |    |       |         |                |   |       |        |
| No                                                  | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.1   | Unknown | lgK:1.11 (3SF) | 6 | MSP   | 3229   |

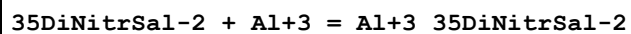
| Reaction No. 61054                      |    |       |        |                 |   |       |        |
|-----------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $5NitrSal-2 + Al+3 = Al+3_{5NitrSal-2}$ |    |       |        |                 |   |       |        |
| No                                      | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                       | 25 | 0.1   | NaClO4 | lgK:11.11 (4SF) | 7 | MSP   | 5987   |

| Reaction No. 61055                                      |    |       |        |                |   |       |        |
|---------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $Al+3_{5NitrSal-2} + 5NitrSal-2 = Al+3_{5NitrSal-2(2)}$ |    |       |        |                |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.1   | NaClO4 | lgK:8.62 (3SF) | 7 | MGL   | 5987   |

| Reaction No. 61056                                         |  |  |  |  |  |  |  |
|------------------------------------------------------------|--|--|--|--|--|--|--|
| $Al+3_{5NitrSal-2(2)} + 5NitrSal-2 = Al+3_{5NitrSal-2(3)}$ |  |  |  |  |  |  |  |

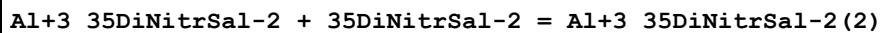
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:6.13 (3SF) | 7 | MGL   | 5987   |

## Reaction No. 61063



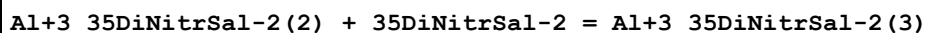
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:8.81 (3SF) | 6 | MSP   | 5987   |

## Reaction No. 61064



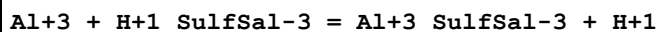
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:6.58 (3SF) | 6 | MGL   | 5987   |

## Reaction No. 61065



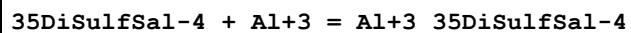
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:4.34 (3SF) | 6 | MGL   | 5987   |

## Reaction No. 61074



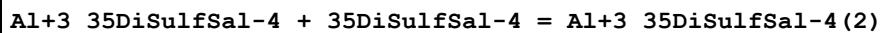
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:0.88 (2SF) | 6 | MSP   | 5987   |

## Reaction No. 61097



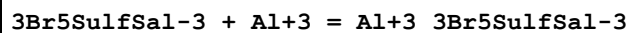
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.5   | NaClO4 | lgK:11.507 (5SF) | 6 | MGL   | 5987   |

## Reaction No. 61098



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.5   | NaClO4 | lgK:8.68 (3SF) | 6 | MGL   | 5987   |

## Reaction No. 61122



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:11.32 (4SF) | 6 | MGL   | 5987   |

| Reaction No. 61123                                                                  |    |       |        |                |   |       |        |
|-------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Al+3\_3Br5SulfSal-3} + 3\text{Br5SulfSal-3} = \text{Al+3\_3Br5SulfSal-3(2)}$ |    |       |        |                |   |       |        |
| No                                                                                  | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                   | 25 | 0.1   | NaClO4 | lgK:8.78 (3SF) | 6 | MGL   | 5987   |

| Reaction No. 61124                                                                     |    |       |        |                |   |       |        |
|----------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Al+3\_3Br5SulfSal-3(2)} + 3\text{Br5SulfSal-3} = \text{Al+3\_3Br5SulfSal-3(3)}$ |    |       |        |                |   |       |        |
| No                                                                                     | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                      | 25 | 0.1   | NaClO4 | lgK:5.43 (3SF) | 6 | MGL   | 5987   |

| Reaction No. 61265                                                    |    |       |               |                 |   |       |        |
|-----------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $2\text{OH3Naphthoic-2} + \text{Al+3} = \text{Al+3\_2OH3Naphthoic-2}$ |    |       |               |                 |   |       |        |
| No                                                                    | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                     | 23 |       |               | lgK:13.19 (4SF) | 0 | MSP   | 5987   |
| 2                                                                     | 25 | 0     | Inf. Dilution | lgK:13.1 (3SF)  | 1 | EAE   | 1175   |

| Reaction No. 61271                                                                                       |    |       |        |                 |   |       |        |
|----------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{Al+3\_1OH4Sulf2Naphthoic-3} + 1\text{OH4Sulf2Naphthoic-3} = \text{Al+3\_1OH4Sulf2Naphthoic-3(2)}$ |    |       |        |                 |   |       |        |
| No                                                                                                       | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                        | 25 | 0.1   | NaClO4 | lgK:9.25 (3SF)  | 6 | MGL   | 5987   |
| 2                                                                                                        | 25 | 0.1   | NaClO4 | lgK:9.259 (4SF) | 6 | MGL   | 5987   |

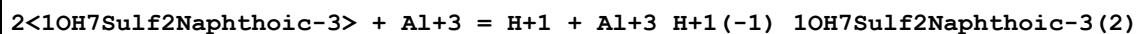
| Reaction No. 61272                                                                                          |    |       |        |                 |   |       |        |
|-------------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{Al+3\_1OH4Sulf2Naphthoic-3(2)} + 1\text{OH4Sulf2Naphthoic-3} = \text{Al+3\_1OH4Sulf2Naphthoic-3(3)}$ |    |       |        |                 |   |       |        |
| No                                                                                                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                           | 25 | 0.1   | NaClO4 | lgK:7.699 (4SF) | 6 | MGL   | 5987   |

| Reaction No. 61303                                                              |    |       |        |                  |   |       |        |
|---------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $1\text{OH7Sulf2Naphthoic-3} + \text{Al+3} = \text{Al+3\_1OH7Sulf2Naphthoic-3}$ |    |       |        |                  |   |       |        |
| No                                                                              | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                               | 25 | 0.1   | NaClO4 | lgK:13.884 (5SF) | 6 | MGL   | 5987   |
| 2                                                                               | 25 | 0.1   | NaClO4 | lgK:13.88 (4SF)  | 6 | MSP   | 5987   |
| 3                                                                               | 25 | 0.1   | NaClO4 | lgK:13.884 (5SF) | 6 | MSP   | 5987   |

| Reaction No. 61304                                                                                       |   |       |        |             |   |       |        |
|----------------------------------------------------------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| $\text{Al+3\_1OH7Sulf2Naphthoic-3} + 1\text{OH7Sulf2Naphthoic-3} = \text{Al+3\_1OH7Sulf2Naphthoic-3(2)}$ |   |       |        |             |   |       |        |
| No                                                                                                       | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

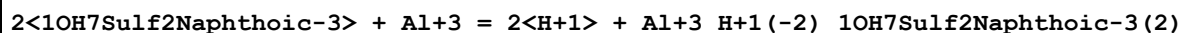
|   |    |     |        |                  |   |     |      |
|---|----|-----|--------|------------------|---|-----|------|
| 1 | 25 | 0.1 | NaClO4 | lgK:11.625 (5SF) | 0 | MGL | 5987 |
| 2 | 25 | 0.1 | NaClO4 | lgK:9.64 (3SF)   | 6 | MGL | 5987 |

## Reaction No. 61305



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:15.22 (4SF) | 6 | MGL   | 5987   |

## Reaction No. 61306



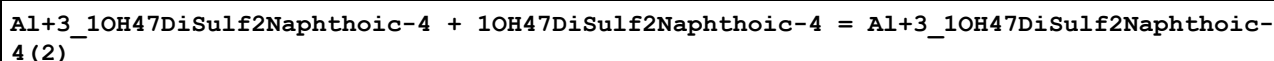
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:6.54 (3SF) | 6 | MGL   | 5987   |

## Reaction No. 61307



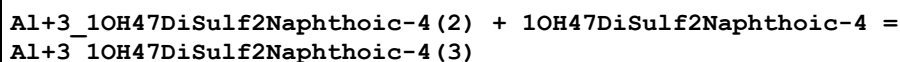
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:5.040 (4SF) | 0 | MGL   | 5987   |

## Reaction No. 61317



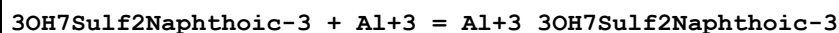
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.5   | NaClO4 | lgK:8.600 (4SF) | 0 | MGL   | 5987   |
| 2  | 25 | 0.5   | NaClO4 | lgK:9.17 (3SF)  | 2 | MGL   | 5987   |

## Reaction No. 61318



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.5   | NaClO4 | lgK:5.793 (4SF) | 6 | MGL   | 5987   |

## Reaction No. 61352



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:11.316 (5SF) | 1 | MSP   | 5987   |
| 2  | 25 | 0.1   | NaClO4 | lgK:11.894 (5SF) | 3 | MGL   | 5987   |
| 3  | 25 | 0.1   | NaClO4 | lgK:11.932 (5SF) | 3 | MSP   | 5987   |

## Reaction No. 61353

| Al+3_3OH7Sulf2Naphthoic-3 + 3OH7Sulf2Naphthoic-3 = Al+3_3OH7Sulf2Naphthoic-3(2) |    |       |        |                |   |       |        |
|---------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| No                                                                              | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                               | 25 | 0.1   | NaClO4 | lgK:9.24 (3SF) | 6 | MGL   | 5987   |

| Reaction No. 61354                                                                 |    |       |        |                |   |       |        |
|------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Al+3_3OH7Sulf2Naphthoic-3(2) + 3OH7Sulf2Naphthoic-3 = Al+3_3OH7Sulf2Naphthoic-3(3) |    |       |        |                |   |       |        |
| No                                                                                 | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                  | 25 | 0.1   | NaClO4 | lgK:7.92 (3SF) | 6 | MGL   | 5987   |

| Reaction No. 61376                                                                       |    |       |        |                |   |       |        |
|------------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Al+3_3OH57DiSulf2Naphthoic-4 + 3OH57DiSulf2Naphthoic-4 = Al+3_3OH57DiSulf2Naphthoic-4(2) |    |       |        |                |   |       |        |
| No                                                                                       | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                        | 25 | 0.5   | NaClO4 | lgK:8.45 (3SF) | 6 | MGL   | 5987   |

| Reaction No. 61388                                                        |    |       |        |                 |   |       |        |
|---------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Al+3_17DiOH4Sulf2Naph-4 + 17DiOH4Sulf2Naph-4 = Al+3_17DiOH4Sulf2Naph-4(2) |    |       |        |                 |   |       |        |
| No                                                                        | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                         | 25 | 0.5   | NaClO4 | lgK:7.188 (4SF) | 4 | MGL   | 5987   |

| Reaction No. 61389                                                           |    |       |        |                |   |       |        |
|------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Al+3_17DiOH4Sulf2Naph-4(2) + 17DiOH4Sulf2Naph-4 = Al+3_17DiOH4Sulf2Naph-4(3) |    |       |        |                |   |       |        |
| No                                                                           | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                            | 25 | 0.5   | NaClO4 | lgK:6.07 (3SF) | 4 | MGL   | 5987   |

| Reaction No. 61431                                                |    |       |        |                      |   |       |        |
|-------------------------------------------------------------------|----|-------|--------|----------------------|---|-------|--------|
| 3<Al+3> + H+1(2)_Ascorbic-2 = Al+3(3)_H+1(-4)_Ascorbic-2 + 6<H+1> |    |       |        |                      |   |       |        |
| No                                                                | t  | I-Str | Medium | Value (Dev)          | W | Tech. | Ref. # |
| 1                                                                 | 25 | 0.6   | NaCl   | lgK:-18.38 (0.016SD) | 7 | MGL   | 22912  |

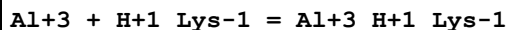
| Reaction No. 61509        |    |       |        |                |   |       |        |
|---------------------------|----|-------|--------|----------------|---|-------|--------|
| Arg-1 + Al+3 = Al+3_Arg-1 |    |       |        |                |   |       |        |
| No                        | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                         | 17 |       |        | lgK:6.67 (3SF) | 0 | MGL   | 6000   |
| 2                         | 25 |       |        | lgK:6.63 (3SF) | 0 | MGL   | 6000   |

| Reaction No. 61510                 |  |  |  |  |  |  |  |
|------------------------------------|--|--|--|--|--|--|--|
| Al+3_Arg-1 + Arg-1 = Al+3_Arg-1(2) |  |  |  |  |  |  |  |



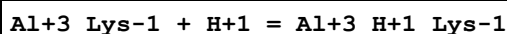
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 17 |       |        | lgK:6.38 (3SF) | 0 | MGL   | 6000   |
| 2  | 25 |       |        | lgK:6.23 (3SF) | 0 | MGL   | 6000   |

## Reaction No. 61531



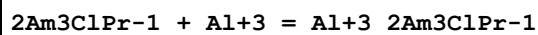
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:5.0 (2SF)    | 1 | ERG   |        |
| 2  | 35 | 0.1   | KNO3          | lgK:6.1 (0.03SD) | 2 | MGL   | 6000   |

## Reaction No. 61532



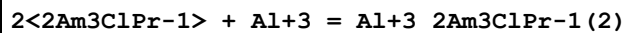
| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:8.1 (2SF)     | 1 | ERG   |        |
| 2  | 35 | 0.1   | KNO3          | lgK:7.44 (0.01SD) | 2 | MGL   | 6000   |

## Reaction No. 61659



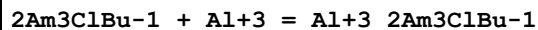
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:3.98 (3SF) | 6 | CRV   | 552    |
| 2  | 25 | 0.1   | KNO3   | lgK:4.05 (3SF) | 6 | MGL   | 2754   |

## Reaction No. 61660



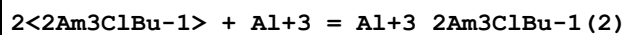
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:6.0 (2SF)  | 6 | CRV   | 552    |
| 2  | 25 | 0.1   | KNO3   | lgK:6.05 (3SF) | 6 | MGL   | 2754   |

## Reaction No. 61666



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:4.13 (3SF) | 6 | MGL   | 2754   |

## Reaction No. 61667



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:6.33 (3SF) | 6 | MGL   | 2754   |

| Reaction No. 61700                                                                  |    |       |         |                |   |       |        |
|-------------------------------------------------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| $\text{Al+3\_H+1\_OH-1\_Asp-2} + \text{H+1} = \text{Al+3\_H+1\_Asp-2} + \text{H2O}$ |    |       |         |                |   |       |        |
| No                                                                                  | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                   | 25 | 0.5   | Unknown | lgK:3.84 (3SF) | 6 | CRV   | 552    |

| Reaction No. 62137                                                     |    |       |         |                    |   |       |        |
|------------------------------------------------------------------------|----|-------|---------|--------------------|---|-------|--------|
| $2<\text{Glyphosate-3}> + \text{Al+3} = \text{Al+3\_Glyphosate-3 (2)}$ |    |       |         |                    |   |       |        |
| No                                                                     | t  | I-Str | Medium  | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                      | 25 | 0.1   | KNO3    | lgK:22.05 (0.02SD) | 4 | MGL   | 6861   |
| 2                                                                      | 25 | 0.1   | Unknown | lgK:22.05 (4SF)    | 6 | CRV   | 552    |
| 3                                                                      | 25 | 0.1   | Unknown | lgK:22.1 (0.2SD)   | 6 | CIU   | 7242   |

| Reaction No. 62138                                                        |    |       |         |                |   |       |        |
|---------------------------------------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| $\text{Al+3\_Glyphosate-3} + \text{H+1} = \text{Al+3\_H+1\_Glyphosate-3}$ |    |       |         |                |   |       |        |
| No                                                                        | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                         | 25 | 0.1   | Unknown | lgK:2.48 (3SF) | 6 | CRV   | 552    |

| Reaction No. 62139                                                                |    |       |         |                |   |       |        |
|-----------------------------------------------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| $\text{Al+3\_Glyphosate-3 (2)} + \text{H+1} = \text{Al+3\_H+1\_Glyphosate-3 (2)}$ |    |       |         |                |   |       |        |
| No                                                                                | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                 | 25 | 0.1   | Unknown | lgK:5.71 (3SF) | 6 | CRV   | 552    |

| Reaction No. 62712                                                   |    |       |         |                |   |       |        |
|----------------------------------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| $2<\text{Al+3\_OH-1\_NTA-3}> = \text{Al+3 (2)\_OH-1 (2)\_NTA-3 (2)}$ |    |       |         |                |   |       |        |
| No                                                                   | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                    | 25 | 0.5   | Unknown | lgK:1.82 (3SF) | 6 | CRV   | 552    |

| Reaction No. 65530                            |    |       |               |                |   |       |        |
|-----------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $\text{I-1} + \text{Al+3} = \text{Al+3\_I-1}$ |    |       |               |                |   |       |        |
| No                                            | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                             | 25 | 0     | Inf. Dilution | lgK:0.38 (2SF) | 4 | EBU   | 6372   |

| Reaction No. 65531                                   |    |       |               |                |   |       |        |
|------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $2<\text{I-1}> + \text{Al+3} = \text{Al+3\_I-1 (2)}$ |    |       |               |                |   |       |        |
| No                                                   | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                    | 25 | 0     | Inf. Dilution | lgK:0.47 (2SF) | 4 | EBU   | 6372   |

| Reaction No. 65532 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| 3<I-1> + Al+3 = Al+3_I-1(3) |    |       |               |               |   |       |        |
|-----------------------------|----|-------|---------------|---------------|---|-------|--------|
| No                          | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                           | 25 | 0     | Inf. Dilution | lgK:0.38(2SF) | 4 | EBU   | 6372   |

| Reaction No. 65533          |    |       |               |               |   |       |        |
|-----------------------------|----|-------|---------------|---------------|---|-------|--------|
| 4<I-1> + Al+3 = Al+3_I-1(4) |    |       |               |               |   |       |        |
| No                          | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                           | 25 | 0     | Inf. Dilution | lgK:0.16(2SF) | 4 | EBU   | 6372   |

| Reaction No. 65534          |    |       |               |                |   |       |        |
|-----------------------------|----|-------|---------------|----------------|---|-------|--------|
| 5<I-1> + Al+3 = Al+3_I-1(5) |    |       |               |                |   |       |        |
| No                          | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                           | 25 | 0     | Inf. Dilution | lgK:-0.15(2SF) | 4 | EBU   | 6372   |

| Reaction No. 65535          |    |       |               |                |   |       |        |
|-----------------------------|----|-------|---------------|----------------|---|-------|--------|
| 6<I-1> + Al+3 = Al+3_I-1(6) |    |       |               |                |   |       |        |
| No                          | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                           | 25 | 0     | Inf. Dilution | lgK:-0.55(2SF) | 4 | EBU   | 6372   |

| Reaction No. 65536        |    |       |               |               |   |       |        |
|---------------------------|----|-------|---------------|---------------|---|-------|--------|
| IO3-1 + Al+3 = Al+3_IO3-1 |    |       |               |               |   |       |        |
| No                        | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                         | 25 | 0     | Inf. Dilution | lgK:2.46(3SF) | 4 | EBU   | 6372   |

| Reaction No. 65537              |    |       |               |               |   |       |        |
|---------------------------------|----|-------|---------------|---------------|---|-------|--------|
| 2<IO3-1> + Al+3 = Al+3_IO3-1(2) |    |       |               |               |   |       |        |
| No                              | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                               | 25 | 0     | Inf. Dilution | lgK:4.30(3SF) | 4 | EBU   | 6372   |

| Reaction No. 65538              |    |       |               |               |   |       |        |
|---------------------------------|----|-------|---------------|---------------|---|-------|--------|
| 3<IO3-1> + Al+3 = Al+3_IO3-1(3) |    |       |               |               |   |       |        |
| No                              | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                               | 25 | 0     | Inf. Dilution | lgK:5.75(3SF) | 4 | EBU   | 6372   |

| Reaction No. 65539              |    |       |               |               |   |       |        |
|---------------------------------|----|-------|---------------|---------------|---|-------|--------|
| 4<IO3-1> + Al+3 = Al+3_IO3-1(4) |    |       |               |               |   |       |        |
| No                              | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                               | 25 | 0     | Inf. Dilution | lgK:6.93(3SF) | 4 | EBU   | 6372   |

| Reaction No. 65540              |    |       |               |                |   |       |        |
|---------------------------------|----|-------|---------------|----------------|---|-------|--------|
| 5<IO3-1> + Al+3 = Al+3_IO3-1(5) |    |       |               |                |   |       |        |
| No                              | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                               | 25 | 0     | Inf. Dilution | lgK:7.90 (3SF) | 4 | EBU   | 6372   |

| Reaction No. 65541              |    |       |               |                |   |       |        |
|---------------------------------|----|-------|---------------|----------------|---|-------|--------|
| 6<IO3-1> + Al+3 = Al+3_IO3-1(6) |    |       |               |                |   |       |        |
| No                              | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                               | 25 | 0     | Inf. Dilution | lgK:8.67 (3SF) | 4 | EBU   | 6372   |

| Reaction No. 65721        |    |       |                 |                   |   |       |        |
|---------------------------|----|-------|-----------------|-------------------|---|-------|--------|
| SCN-1 + Al+3 = Al+3_SCN-1 |    |       |                 |                   |   |       |        |
| No                        | t  | I-Str | Medium          | Value (Dev)       | W | Tech. | Ref. # |
| 1                         | 22 | 0     | Inf. Dilution   | lgK:0.42 (2SF)    | 4 | CRV   | 552    |
| 2                         | 25 | 0     | Inf. Dilution   | lgK:0.42 (3SF)    | 2 | EAE   |        |
| 3                         | 25 | 0.2   | DMF<br>Et4NClO4 | lgK:2.1 (0.1MD)   | 5 | MCL   | 6237   |
| 4                         | 25 | 0.4   | DMF<br>Et4NClO4 | lgK:1.7 (0.1MD)   | 5 | MCL   | 6237   |
| 5                         | 25 | 0.2   | DMF<br>Et4NClO4 | dH:9.0 (0.9MD) kJ | 5 | MCL   | 6237   |
| 6                         | 25 | 0.4   | DMF<br>Et4NClO4 | dH:8.7 (0.9MD) kJ | 5 | MCL   | 6237   |

| Reaction No. 65763                                            |    |       |               |                 |   |       |        |
|---------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Al+3(2)_OH-1(2)_CO3-2 + 3<H+1> = 2<Al+3> + H+1_CO3-2 + 2<H2O> |    |       |               |                 |   |       |        |
| No                                                            | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                             | 25 | 0     | Inf. Dilution | lgK:6.1 (2SF)   | 1 | ELC   |        |
| 2                                                             | 25 | 0.1   | Unknown       | lgK:-7.3 (2SF)  | 0 | CRV   | 552    |
| 3                                                             | 25 | 0.5   | Unknown       | lgK:-7.87 (3SF) | 0 | CRV   | 552    |
| 4                                                             | 25 | 3     | NaCl          | lgK:-9.0 (2SF)  | 0 | CRV   | 552    |

| Reaction No. 65764                                                |    |       |         |                  |   |       |        |
|-------------------------------------------------------------------|----|-------|---------|------------------|---|-------|--------|
| 3<Al+3> + H+1_CO3-2 + 4<H2O> = Al+3(3)_H+1_OH-1(4)_CO3-2 + 4<H+1> |    |       |         |                  |   |       |        |
| No                                                                | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                 | 25 | 0.1   | Unknown | lgK:-9.4 (2SF)   | 6 | CRV   | 552    |
| 2                                                                 | 25 | 0.5   | Unknown | lgK:-10.20 (4SF) | 6 | CRV   | 552    |
| 3                                                                 | 25 | 3     | NaCl    | lgK:-10.57 (4SF) | 6 | CRV   | 552    |

| Reaction No. 65960                                                         |     |       |               |                                       |   |       |        |
|----------------------------------------------------------------------------|-----|-------|---------------|---------------------------------------|---|-------|--------|
| $\text{Al(s)} + 1.5\text{F}_2\text{(g)} = \text{Al}_3\text{F}_4\text{(s)}$ |     |       |               |                                       |   |       |        |
| No                                                                         | t   | I-Str | Medium        | Value (Dev)                           | W | Tech. | Ref. # |
| 1                                                                          | 25  | 0     | Inf. Dilution | $\lg K: 249.9(4\text{SF})$            | 1 | ELC   |        |
| 2                                                                          | 25  | 0     | Inf. Dilution | $dG: -338.(3\text{SF})$               | 0 | CRV   | 7382   |
| 3                                                                          | 25  | 0     | GAMSPATH      | $dG: -1417.(4\text{SF}) \text{ kJ}$   | 0 | CRV   | 12638  |
| 4                                                                          | 300 | 0     | Inf. Dilution | $dG: -340.(3\text{SF})$               | 0 | EFE   | 6427   |
| 5                                                                          | 25  | 0     | Inf. Dilution | $dH: -363.(3\text{SF})$               | 4 | CRV   | 7382   |
| 6                                                                          | 25  | 0     | GAMSPATH      | $dH: -1520.50(6\text{SF}) \text{ kJ}$ | 2 | CRV   | 12638  |

| Reaction No. 65961                                                                                                                          |     |       |               |                            |   |       |        |
|---------------------------------------------------------------------------------------------------------------------------------------------|-----|-------|---------------|----------------------------|---|-------|--------|
| $\text{Al(s)} + 0.5\text{O}_2\text{(g)} + 0.5\text{H}_2\text{(g)} + \text{F}_2\text{(g)} = \text{Al}_3\text{F}_4\text{(s)} \cdot \text{OH}$ |     |       |               |                            |   |       |        |
| No                                                                                                                                          | t   | I-Str | Medium        | Value (Dev)                | W | Tech. | Ref. # |
| 1                                                                                                                                           | 25  | 0     | Inf. Dilution | $\lg K: 231.6(4\text{SF})$ | 1 | EES   | 6953   |
| 2                                                                                                                                           | 300 | 0     | Inf. Dilution | $dG: -0.321(3\text{SF})$   | 0 | EFE   | 6427   |

| Reaction No. 65962                                                                                                                       |     |       |               |                             |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------------|-----|-------|---------------|-----------------------------|---|-------|--------|
| $\text{Al(s)} + \text{O}_2\text{(g)} + \text{H}_2\text{(g)} + 0.5\text{F}_2\text{(g)} = \text{Al}_3\text{F}_4\text{(s)} \cdot \text{OH}$ |     |       |               |                             |   |       |        |
| No                                                                                                                                       | t   | I-Str | Medium        | Value (Dev)                 | W | Tech. | Ref. # |
| 1                                                                                                                                        | 25  | 0     | Inf. Dilution | $\lg K: -0.248(3\text{SF})$ | 2 | EAE   |        |
| 2                                                                                                                                        | 300 | 0     | Inf. Dilution | $dG: -0.305(3\text{SF})$    | 4 | EFE   | 6427   |

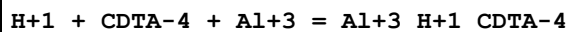
| Reaction No. 66076                                                    |    |       |               |                           |   |       |        |
|-----------------------------------------------------------------------|----|-------|---------------|---------------------------|---|-------|--------|
| $\text{NTA-3} + \text{Al}_3 + \text{OH} = \text{Al}_3\text{OH-NTA-3}$ |    |       |               |                           |   |       |        |
| No                                                                    | t  | I-Str | Medium        | Value (Dev)               | W | Tech. | Ref. # |
| 1                                                                     | 25 | 0     | Inf. Dilution | $\lg K: 23.5(3\text{SF})$ | 1 | ELC   |        |
| 2                                                                     | 25 | 0     | Inf. Dilution | $\lg K: 22.1(3\text{SF})$ | 0 | ENS   | 6405   |

| Reaction No. 66082                                                                                                                                                                                            |    |       |               |                            |   |       |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------------------|---|-------|--------|
| $\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2\text{(s)} + 2\text{H}^+ + 6\text{H}_2\text{O} = 2\text{Al}_3\text{OH-1(3)}_{\text{(Gibbsite,s)}} + 2\text{H}_2\text{SiO}_4^{2-} + \text{Ca}^{2+}$ |    |       |               |                            |   |       |        |
| No                                                                                                                                                                                                            | t  | I-Str | Medium        | Value (Dev)                | W | Tech. | Ref. # |
| 1                                                                                                                                                                                                             | 25 | 0     | Inf. Dilution | $\lg K: 12.1(3\text{SF})$  | 1 | ELC   |        |
| 2                                                                                                                                                                                                             | 25 | 0     | Inf. Dilution | $\lg K: 11.16(4\text{SF})$ | 0 | ENS   | 6405   |

| Reaction No. 66089                                                                     |  |  |  |  |  |  |  |
|----------------------------------------------------------------------------------------|--|--|--|--|--|--|--|
| $\text{EDTA-4} + \text{Al}_3 + 2\text{OH} = \text{Al}_3\text{OH-1(2)}_{\text{EDTA-4}}$ |  |  |  |  |  |  |  |

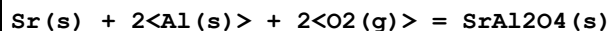
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:30.0 (3SF) | 4 | ENS   | 6405   |

## Reaction No. 66099



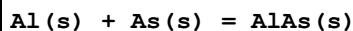
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:24.3 (3SF) | 4 | ENS   | 6405   |

## Reaction No. 66393



| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:385.5 (4SF) | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:383.0 (4SF) | 4 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:281.8 (4SF) | 4 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:221.1 (4SF) | 4 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:180.7 (4SF) | 4 | ENS   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dG:-526.0 (4SF) | 4 | EFE   | 6422   |
| 7  | 25  | 0     | Inf. Dilution | dH:-555.6 (4SF) | 4 | MTD   | 6422   |
| 8  | 127 | 0     | Inf. Dilution | dH:-555.7 (4SF) | 4 | MTD   | 6422   |
| 9  | 227 | 0     | Inf. Dilution | dH:-555.4 (4SF) | 4 | MTD   | 6422   |
| 10 | 327 | 0     | Inf. Dilution | dH:-555.1 (4SF) | 4 | MTD   | 6422   |

## Reaction No. 66464



| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:20.18 (4SF) | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:20.06 (4SF) | 0 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:14.99 (4SF) | 2 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:11.93 (4SF) | 1 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:9.885 (4SF) | 0 | ENS   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dG:-27.53 (4SF) | 4 | EFE   | 6422   |
| 7  | 25  | 0     | Inf. Dilution | dH:-27.8 (3SF)  | 4 | MTD   | 6422   |
| 8  | 127 | 0     | Inf. Dilution | dH:-27.9 (3SF)  | 1 | MTD   | 6422   |
| 9  | 227 | 0     | Inf. Dilution | dH:-28.0 (3SF)  | 0 | MTD   | 6422   |
| 10 | 327 | 0     | Inf. Dilution | dH:-28.2 (3SF)  | 0 | MTD   | 6422   |

## Reaction No. 66465

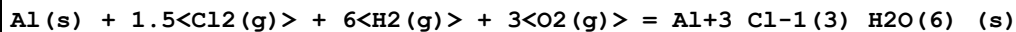
| Al(s) + As(s) + 2<O2(g)> = Al+3_AsO4-3_(s) |     |       |               |                |   |       |        |
|--------------------------------------------|-----|-------|---------------|----------------|---|-------|--------|
| No                                         | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                          | 25  | 0     | Inf. Dilution | lgK:233.5(4SF) | 4 | ENS   | 6422   |
| 2                                          | 27  | 0     | Inf. Dilution | lgK:232.0(4SF) | 0 | ENS   | 6422   |
| 3                                          | 127 | 0     | Inf. Dilution | lgK:169.7(4SF) | 2 | ENS   | 6422   |
| 4                                          | 227 | 0     | Inf. Dilution | lgK:132.4(4SF) | 1 | ENS   | 6422   |
| 5                                          | 327 | 0     | Inf. Dilution | lgK:107.6(4SF) | 0 | ENS   | 6422   |
| 6                                          | 25  | 0     | Inf. Dilution | dG:-318.6(4SF) | 4 | EFE   | 6422   |
| 7                                          | 25  | 0     | Inf. Dilution | dH:-342.0(4SF) | 4 | MTD   | 6422   |
| 8                                          | 127 | 0     | Inf. Dilution | dH:-341.6(4SF) | 1 | MTD   | 6422   |
| 9                                          | 227 | 0     | Inf. Dilution | dH:-341.1(4SF) | 0 | MTD   | 6422   |
| 10                                         | 327 | 0     | Inf. Dilution | dH:-340.5(4SF) | 0 | MTD   | 6422   |

| Reaction No. 66466                     |      |       |               |                   |   |       |        |
|----------------------------------------|------|-------|---------------|-------------------|---|-------|--------|
| Al(s) + 1.5<Br2(l)> = Al+3_Br-1(3)_(s) |      |       |               |                   |   |       |        |
| No                                     | t    | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                      | 25   | 0     | Inf. Dilution | lgK:85.59(4SF)    | 4 | ENS   | 6422   |
| 2                                      | 27   | 0     | Inf. Dilution | lgK:85.03(4SF)    | 4 | ENS   | 6422   |
| 3                                      | 97.6 | 0     | Inf. Dilution | lgK:67.34(4SF)    | 4 | ENS   | 6422   |
| 4                                      | 25   | 0     | Inf. Dilution | dG:-116.8(4SF)    | 4 | EFE   | 6422   |
| 5                                      | 25   | 0     | Inf. Dilution | dH:-527.2(4SF) kJ | 0 | CRV   | 6330   |
| 6                                      | 25   | 0     | Inf. Dilution | dH:-122.2(4SF)    | 4 | MTD   | 6422   |
| 7                                      | 97.6 | 0     | Inf. Dilution | dH:-132.8(4SF)    | 0 | MTD   | 6422   |

| Reaction No. 66467            |     |       |               |                |   |       |        |
|-------------------------------|-----|-------|---------------|----------------|---|-------|--------|
| 3<C(s)> + 4<Al(s)> = Al4C3(s) |     |       |               |                |   |       |        |
| No                            | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                             | 25  | 0     | Inf. Dilution | lgK:34.42(4SF) | 3 | ENS   | 6422   |
| 2                             | 27  | 0     | Inf. Dilution | lgK:34.19(4SF) | 0 | ENS   | 6422   |
| 3                             | 127 | 0     | Inf. Dilution | lgK:25.10(4SF) | 2 | ENS   | 6422   |
| 4                             | 227 | 0     | Inf. Dilution | lgK:19.64(4SF) | 1 | ENS   | 6422   |
| 5                             | 327 | 0     | Inf. Dilution | lgK:16.01(4SF) | 0 | ENS   | 6422   |
| 6                             | 25  | 0     | Inf. Dilution | dG:-46.96(4SF) | 4 | EFE   | 6422   |
| 7                             | 25  | 0     | Inf. Dilution | dG:-46.9(3SF)  | 4 | CRV   | 12011  |
| 8                             | 25  | 0     | Inf. Dilution | dH:-49.9(3SF)  | 4 | MTD   | 6422   |
| 9                             | 25  | 0     | Inf. Dilution | dH:-49.9(3SF)  | 4 | CRV   | 12011  |

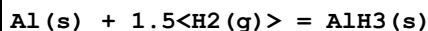
|    |     |   |               |                |   |     |      |
|----|-----|---|---------------|----------------|---|-----|------|
| 10 | 127 | 0 | Inf. Dilution | dH:-49.9 (3SF) | 1 | MTD | 6422 |
| 11 | 227 | 0 | Inf. Dilution | dH:-49.9 (3SF) | 0 | MTD | 6422 |
| 12 | 327 | 0 | Inf. Dilution | dH:-49.9 (3SF) | 0 | MTD | 6422 |

## Reaction No. 66468



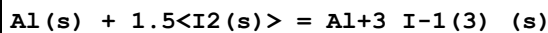
| No | t   | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
|----|-----|-------|---------------|---------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:396.1 (4SF)     | 2 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:393.2 (4SF)     | 0 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:276.0 (4SF)     | 2 | ENS   | 6422   |
| 4  | 25  | 0     | Inf. Dilution | dG:-540.4 (4SF)     | 2 | EFE   | 6422   |
| 5  | 25  | 0     | Inf. Dilution | dG:-2262.5 (5SF) kJ | 5 | EOD   | 29520  |
| 6  | 25  | 0     | Inf. Dilution | dH:-643.3 (4SF)     | 2 | MTD   | 6422   |
| 7  | 25  | 0     | Inf. Dilution | dH:-643.3 (4SF)     | 3 | CRV   | 7382   |
| 8  | 25  | 0     | Inf. Dilution | dH:-2693.1 (5SF) kJ | 5 | EOD   | 29520  |
| 9  | 127 | 0     | Inf. Dilution | dH:-644.3 (4SF)     | 2 | MTD   | 6422   |

## Reaction No. 66469



| No | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|-----|-------|---------------|------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:-8.149 (4SF) | 3 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:-8.162 (4SF) | 0 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:-8.710 (4SF) | 1 | ENS   | 6422   |
| 4  | 25  | 0     | Inf. Dilution | dG:11.12 (4SF)   | 3 | EFE   | 6422   |
| 5  | 25  | 0     | Inf. Dilution | dH:-2.72 (3SF)   | 3 | MTD   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dH:-11. (2SF)    | 0 | CRV   | 7382   |
| 7  | 127 | 0     | Inf. Dilution | dH:-3.32 (3SF)   | 0 | MTD   | 6422   |

## Reaction No. 66470



| No | t     | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|-------|-------|---------------|--------------------|---|-------|--------|
| 1  | 25    | 0     | Inf. Dilution | lgK:52.42 (4SF)    | 4 | ENS   | 6422   |
| 2  | 27    | 0     | Inf. Dilution | lgK:52.09 (4SF)    | 4 | ENS   | 6422   |
| 3  | 127   | 0     | Inf. Dilution | lgK:38.79 (4SF)    | 4 | ENS   | 6422   |
| 4  | 191.2 | 0     | Inf. Dilution | lgK:32.76 (4SF)    | 4 | ENS   | 6422   |
| 5  | 25    | 0     | Inf. Dilution | dG:-300.8 (4SF) kJ | 4 | CRV   | 6330   |
| 6  | 25    | 0     | Inf. Dilution | dG:-71.51 (4SF)    | 4 | EFE   | 6422   |



|    |       |   |               |                    |   |     |      |
|----|-------|---|---------------|--------------------|---|-----|------|
| 7  | 25    | 0 | Inf. Dilution | dH:-313.8 (4SF) kJ | 4 | CRV | 6330 |
| 8  | 25    | 0 | Inf. Dilution | dH:-72.4 (3SF)     | 4 | MTD | 6422 |
| 9  | 127   | 0 | Inf. Dilution | dH:-78.3 (3SF)     | 4 | MTD | 6422 |
| 10 | 191.2 | 0 | Inf. Dilution | dH:-93.8 (3SF)     | 4 | MTD | 6422 |

| Reaction No. 66471          |     |       |               |                    |   |       |        |
|-----------------------------|-----|-------|---------------|--------------------|---|-------|--------|
| Al(s) + 0.5<N2(g)> = AlN(s) |     |       |               |                    |   |       |        |
| No                          | t   | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                           | 25  | 0     | Inf. Dilution | lgK:50.28 (4SF)    | 4 | ENS   | 6422   |
| 2                           | 27  | 0     | Inf. Dilution | lgK:49.94 (4SF)    | 0 | ENS   | 6422   |
| 3                           | 127 | 0     | Inf. Dilution | lgK:36.08 (4SF)    | 2 | ENS   | 6422   |
| 4                           | 227 | 0     | Inf. Dilution | lgK:27.76 (4SF)    | 1 | ENS   | 6422   |
| 5                           | 327 | 0     | Inf. Dilution | lgK:22.20 (4SF)    | 0 | ENS   | 6422   |
| 6                           | 25  | 0     | Inf. Dilution | dG:-287.0 (4SF) kJ | 4 | CRV   | 6330   |
| 7                           | 25  | 0     | Inf. Dilution | dG:-68.59 (4SF)    | 4 | EFE   | 6422   |
| 8                           | 25  | 0     | Inf. Dilution | dG:-68.6 (3SF)     | 4 | CRV   | 7382   |
| 9                           | 25  | 0     | Inf. Dilution | dG:-68.6 (3SF)     | 4 | CRV   | 12011  |
| 10                          | 25  | 0     | Inf. Dilution | dH:-318.0 (4SF) kJ | 4 | CRV   | 6330   |
| 11                          | 25  | 0     | Inf. Dilution | dH:-76.0 (3SF)     | 4 | MTD   | 6422   |
| 12                          | 25  | 0     | Inf. Dilution | dH:-76.0 (3SF)     | 4 | CRV   | 7382   |
| 13                          | 25  | 0     | Inf. Dilution | dH:-76.0 (3SF)     | 4 | CRV   | 12011  |
| 14                          | 127 | 0     | Inf. Dilution | dH:-76.1 (3SF)     | 1 | MTD   | 6422   |
| 15                          | 227 | 0     | Inf. Dilution | dH:-76.2 (3SF)     | 0 | MTD   | 6422   |
| 16                          | 327 | 0     | Inf. Dilution | dH:-76.2 (3SF)     | 0 | MTD   | 6422   |

| Reaction No. 66472                     |     |       |               |                 |   |       |        |
|----------------------------------------|-----|-------|---------------|-----------------|---|-------|--------|
| 2<Al(s)> + 1.5<O2(g)> = Al2O3(gamma,s) |     |       |               |                 |   |       |        |
| No                                     | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                      | 25  | 0     | Inf. Dilution | lgK:273.8 (4SF) | 1 | ELC   |        |
| 2                                      | 25  | 0     | Inf. Dilution | lgK:274.0 (4SF) | 0 | ENS   | 6422   |
| 3                                      | 27  | 0     | Inf. Dilution | lgK:272.2 (4SF) | 0 | ENS   | 6422   |
| 4                                      | 127 | 0     | Inf. Dilution | lgK:200.1 (4SF) | 0 | ENS   | 6422   |
| 5                                      | 227 | 0     | Inf. Dilution | lgK:156.8 (4SF) | 0 | ENS   | 6422   |
| 6                                      | 327 | 0     | Inf. Dilution | lgK:128.0 (4SF) | 0 | ENS   | 6422   |
| 7                                      | 25  | 0     | Inf. Dilution | dG:-373.8 (4SF) | 0 | EFE   | 6422   |
| 8                                      | 25  | 0     | Inf. Dilution | dH:-396.0 (4SF) | 0 | MTD   | 6422   |

|    |     |   |               |                |   |     |      |
|----|-----|---|---------------|----------------|---|-----|------|
| 9  | 25  | 0 | Inf. Dilution | dH:-395.(3SF)  | 0 | CRV | 7382 |
| 10 | 127 | 0 | Inf. Dilution | dH:-396.0(4SF) | 0 | MTD | 6422 |
| 11 | 227 | 0 | Inf. Dilution | dH:-395.9(4SF) | 0 | MTD | 6422 |
| 12 | 327 | 0 | Inf. Dilution | dH:-395.6(4SF) | 0 | MTD | 6422 |

| Reaction No. 66473                     |     |       |               |                |   |       |        |
|----------------------------------------|-----|-------|---------------|----------------|---|-------|--------|
| 2<Al(s)> + 1.5<O2(g)> = Al2O3(delta,s) |     |       |               |                |   |       |        |
| No                                     | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                      | 25  | 0     | Inf. Dilution | lgK:275.6(4SF) | 1 | ELC   |        |
| 2                                      | 25  | 0     | Inf. Dilution | lgK:275.6(4SF) | 0 | ENS   | 6422   |
| 3                                      | 27  | 0     | Inf. Dilution | lgK:273.8(4SF) | 0 | ENS   | 6422   |
| 4                                      | 127 | 0     | Inf. Dilution | lgK:201.2(4SF) | 0 | ENS   | 6422   |
| 5                                      | 227 | 0     | Inf. Dilution | lgK:157.7(4SF) | 0 | ENS   | 6422   |
| 6                                      | 327 | 0     | Inf. Dilution | lgK:128.7(4SF) | 0 | ENS   | 6422   |
| 7                                      | 25  | 0     | Inf. Dilution | dG:-376.0(4SF) | 0 | EFE   | 6422   |
| 8                                      | 25  | 0     | Inf. Dilution | dH:-398.3(4SF) | 0 | MTD   | 6422   |
| 9                                      | 25  | 0     | Inf. Dilution | dH:-398.(3SF)  | 0 | CRV   | 7382   |
| 10                                     | 127 | 0     | Inf. Dilution | dH:-398.4(4SF) | 0 | MTD   | 6422   |
| 11                                     | 227 | 0     | Inf. Dilution | dH:-398.2(4SF) | 0 | MTD   | 6422   |
| 12                                     | 327 | 0     | Inf. Dilution | dH:-398.0(4SF) | 0 | MTD   | 6422   |

| Reaction No. 66474                     |     |       |               |                |   |       |        |
|----------------------------------------|-----|-------|---------------|----------------|---|-------|--------|
| 2<Al(s)> + 1.5<O2(g)> = Al2O3(kappa,s) |     |       |               |                |   |       |        |
| No                                     | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                      | 25  | 0     | Inf. Dilution | lgK:275.6(4SF) | 1 | ELC   |        |
| 2                                      | 25  | 0     | Inf. Dilution | lgK:275.7(4SF) | 0 | ENS   | 6422   |
| 3                                      | 27  | 0     | Inf. Dilution | lgK:273.9(4SF) | 0 | ENS   | 6422   |
| 4                                      | 127 | 0     | Inf. Dilution | lgK:201.4(4SF) | 0 | ENS   | 6422   |
| 5                                      | 227 | 0     | Inf. Dilution | lgK:157.8(4SF) | 0 | ENS   | 6422   |
| 6                                      | 327 | 0     | Inf. Dilution | lgK:128.8(4SF) | 0 | ENS   | 6422   |
| 7                                      | 25  | 0     | Inf. Dilution | dG:-376.2(4SF) | 0 | EFE   | 6422   |
| 8                                      | 25  | 0     | Inf. Dilution | dH:-398.3(4SF) | 0 | MTD   | 6422   |
| 9                                      | 25  | 0     | Inf. Dilution | dH:-397.(3SF)  | 0 | CRV   | 7382   |
| 10                                     | 127 | 0     | Inf. Dilution | dH:-398.4(4SF) | 0 | MTD   | 6422   |
| 11                                     | 227 | 0     | Inf. Dilution | dH:-398.3(4SF) | 0 | MTD   | 6422   |
| 12                                     | 327 | 0     | Inf. Dilution | dH:-398.1(4SF) | 0 | MTD   | 6422   |

| Reaction No. 66475                          |     |       |               |                |   |       |        |
|---------------------------------------------|-----|-------|---------------|----------------|---|-------|--------|
| Al(s) + 0.5<O2(g)> + 0.5<Cl2(g)> = AlOCl(s) |     |       |               |                |   |       |        |
| No                                          | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                           | 25  | 0     | Inf. Dilution | lgK:129.2(4SF) | 4 | ENS   | 6422   |
| 2                                           | 27  | 0     | Inf. Dilution | lgK:128.3(4SF) | 0 | ENS   | 6422   |
| 3                                           | 127 | 0     | Inf. Dilution | lgK:93.78(4SF) | 2 | ENS   | 6422   |
| 4                                           | 227 | 0     | Inf. Dilution | lgK:73.08(4SF) | 1 | ENS   | 6422   |
| 5                                           | 327 | 0     | Inf. Dilution | lgK:59.29(4SF) | 0 | ENS   | 6422   |
| 6                                           | 25  | 0     | Inf. Dilution | dG:-176.2(4SF) | 4 | EFE   | 6422   |
| 7                                           | 25  | 0     | Inf. Dilution | dH:-189.6(4SF) | 2 | MTD   | 6422   |
| 8                                           | 127 | 0     | Inf. Dilution | dH:-189.5(4SF) | 1 | MTD   | 6422   |
| 9                                           | 227 | 0     | Inf. Dilution | dH:-189.3(4SF) | 0 | MTD   | 6422   |
| 10                                          | 327 | 0     | Inf. Dilution | dH:-189.1(4SF) | 0 | MTD   | 6422   |

| Reaction No. 66476                                     |     |       |               |                    |   |       |        |
|--------------------------------------------------------|-----|-------|---------------|--------------------|---|-------|--------|
| Al(s) + 1.5<H2(g)> + 1.5<O2(g)> = Al+3_OH-1(3)_(am.,s) |     |       |               |                    |   |       |        |
| No                                                     | t   | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                      | 25  | 0     | Inf. Dilution | lgK:199.9(4SF)     | 1 | ELC   |        |
| 2                                                      | 25  | 0     | Inf. Dilution | lgK:199.5(4SF)     | 0 | ENS   | 6422   |
| 3                                                      | 27  | 0     | Inf. Dilution | lgK:198.1(4SF)     | 0 | ENS   | 6422   |
| 4                                                      | 127 | 0     | Inf. Dilution | lgK:142.5(4SF)     | 0 | ENS   | 6422   |
| 5                                                      | 227 | 0     | Inf. Dilution | lgK:109.2(4SF)     | 0 | ENS   | 6422   |
| 6                                                      | 327 | 0     | Inf. Dilution | lgK:87.00(4SF)     | 0 | ENS   | 6422   |
| 7                                                      | 25  | 0     | Inf. Dilution | dG:-272.2(4SF)     | 0 | EFE   | 6422   |
| 8                                                      | 25  | 0     | Inf. Dilution | dG:-1142.20(6SF)kJ | 0 | CRV   | 25435  |
| 9                                                      | 25  | 0     | Inf. Dilution | dG:-1143.2(5SF)kJ  | 5 | CRV   | 31308  |
| 10                                                     | 25  | 0     | Inf. Dilution | dH:-305.0(4SF)     | 0 | MTD   | 6422   |
| 11                                                     | 25  | 0     | Inf. Dilution | dH:-305.(3SF)      | 0 | CRV   | 7382   |
| 12                                                     | 25  | 0     | Inf. Dilution | dH:-1276.10(6SF)kJ | 0 | CRV   | 25435  |
| 13                                                     | 127 | 0     | Inf. Dilution | dH:-305.2(4SF)     | 0 | MTD   | 6422   |
| 14                                                     | 227 | 0     | Inf. Dilution | dH:-305.0(4SF)     | 0 | MTD   | 6422   |
| 15                                                     | 327 | 0     | Inf. Dilution | dH:-304.4(4SF)     | 0 | MTD   | 6422   |

| Reaction No. 66477                               |   |       |        |             |   |       |        |
|--------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| 2<Al(s)> + 2.5<O2(g)> + Si(s) = Al2SiO5(Kyan.,s) |   |       |        |             |   |       |        |
| No                                               | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

|    |     |   |               |                      |   |     |       |
|----|-----|---|---------------|----------------------|---|-----|-------|
| 1  | 25  | 0 | Inf. Dilution | lgK:427.3 (4SF)      | 1 | ELC |       |
| 2  | 25  | 0 | Inf. Dilution | lgK:428.2 (4SF)      | 2 | ENS | 6422  |
| 3  | 25  | 0 | Inf. Dilution | lgK:428.0 (4SF)      | 2 | ENS | 6494  |
| 4  | 27  | 0 | Inf. Dilution | lgK:425.4 (4SF)      | 0 | ENS | 6422  |
| 5  | 27  | 0 | Inf. Dilution | lgK:425.2 (4SF)      | 0 | ENS | 6494  |
| 6  | 127 | 0 | Inf. Dilution | lgK:312.4 (4SF)      | 1 | ENS | 6422  |
| 7  | 127 | 0 | Inf. Dilution | lgK:312.3 (4SF)      | 1 | ENS | 6494  |
| 8  | 227 | 0 | Inf. Dilution | lgK:244.6 (4SF)      | 0 | ENS | 6422  |
| 9  | 227 | 0 | Inf. Dilution | lgK:244.5 (4SF)      | 0 | ENS | 6494  |
| 10 | 327 | 0 | Inf. Dilution | lgK:199.5 (4SF)      | 0 | ENS | 6422  |
| 11 | 327 | 0 | Inf. Dilution | lgK:199.4 (4SF)      | 0 | ENS | 6494  |
| 12 | 25  | 0 | Inf. Dilution | dG:-584.1 (4SF)      | 0 | EFE | 6422  |
| 13 | 25  | 0 | Inf. Dilution | dG:-583.98 (5SF)     | 0 | EFE | 6482  |
| 14 | 25  | 0 | Inf. Dilution | dG:-583.9 (4SF)      | 0 | EFE | 6494  |
| 15 | 25  | 0 | Inf. Dilution | dG:-2443.90 (6SF) kJ | 0 | CRV | 9015  |
| 16 | 25  | 0 | Inf. Dilution | dG:-583.88 (5SF)     | 0 | CRV | 10174 |
| 17 | 25  | 0 | Inf. Dilution | dG:-620.5 (4SF)      | 0 | CRV | 12011 |
| 18 | 50  | 0 | Inf. Dilution | dG:-584.41 (5SF)     | 0 | CRV | 10174 |
| 19 | 75  | 0 | Inf. Dilution | dG:-585.00 (5SF)     | 0 | CRV | 10174 |
| 20 | 100 | 0 | Inf. Dilution | dG:-585.65 (5SF)     | 0 | CRV | 10174 |
| 21 | 125 | 0 | Inf. Dilution | dG:-586.36 (5SF)     | 0 | CRV | 10174 |
| 22 | 150 | 0 | Inf. Dilution | dG:-587.12 (5SF)     | 0 | CRV | 10174 |
| 23 | 175 | 0 | Inf. Dilution | dG:-587.94 (5SF)     | 0 | CRV | 10174 |
| 24 | 200 | 0 | Inf. Dilution | dG:-588.80 (5SF)     | 0 | CRV | 10174 |
| 25 | 225 | 0 | Inf. Dilution | dG:-589.72 (5SF)     | 0 | CRV | 10174 |
| 26 | 250 | 0 | Inf. Dilution | dG:-590.67 (5SF)     | 0 | CRV | 10174 |
| 27 | 300 | 0 | Inf. Dilution | dG:-592.72 (5SF)     | 0 | CRV | 10174 |
| 28 | 25  | 0 | Inf. Dilution | dH:-620.0 (4SF)      | 0 | MTD | 6422  |
| 29 | 25  | 0 | Inf. Dilution | dH:-620.5 (4SF)      | 0 | ENS | 6437  |
| 30 | 25  | 0 | Inf. Dilution | dH:-620.03 (5SF)     | 0 | EFE | 6482  |
| 31 | 25  | 0 | Inf. Dilution | dH:-619.9 (4SF)      | 0 | MTD | 6494  |
| 32 | 25  | 0 | Inf. Dilution | dH:-2594.30 (6SF) kJ | 0 | CRV | 9015  |
| 33 | 25  | 0 | Inf. Dilution | dH:-656.4 (4SF)      | 0 | CRV | 12011 |
| 34 | 127 | 0 | Inf. Dilution | dH:-620.2 (4SF)      | 0 | MTD | 6422  |
| 35 | 127 | 0 | Inf. Dilution | dH:-620.1 (4SF)      | 0 | MTD | 6494  |
| 36 | 227 | 0 | Inf. Dilution | dH:-620.1 (4SF)      | 0 | MTD | 6422  |

|    |     |   |               |                 |   |     |      |
|----|-----|---|---------------|-----------------|---|-----|------|
| 37 | 227 | 0 | Inf. Dilution | dH:-620.0 (4SF) | 0 | MTD | 6494 |
| 38 | 327 | 0 | Inf. Dilution | dH:-619.8 (4SF) | 0 | MTD | 6422 |
| 39 | 327 | 0 | Inf. Dilution | dH:-619.7 (4SF) | 0 | MTD | 6494 |

| Reaction No. 66478                                 |     |       |               |                      |   |       |        |
|----------------------------------------------------|-----|-------|---------------|----------------------|---|-------|--------|
| 2<Al(s)> + 2.5<O2(g)> + Si(s) = Al2SiO5(Andalu.,s) |     |       |               |                      |   |       |        |
| No                                                 | t   | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
| 1                                                  | 25  | 0     | Inf. Dilution | lgK:427.(3SF)        | 1 | ELC   |        |
| 2                                                  | 25  | 0     | Inf. Dilution | lgK:428.0 (4SF)      | 0 | ENS   | 6422   |
| 3                                                  | 25  | 0     | Inf. Dilution | lgK:427.8 (4SF)      | 2 | ENS   | 6494   |
| 4                                                  | 27  | 0     | Inf. Dilution | lgK:425.2 (4SF)      | 0 | ENS   | 6422   |
| 5                                                  | 27  | 0     | Inf. Dilution | lgK:425.0 (4SF)      | 0 | ENS   | 6494   |
| 6                                                  | 127 | 0     | Inf. Dilution | lgK:312.4 (4SF)      | 1 | ENS   | 6422   |
| 7                                                  | 127 | 0     | Inf. Dilution | lgK:312.2 (4SF)      | 1 | ENS   | 6494   |
| 8                                                  | 227 | 0     | Inf. Dilution | lgK:244.7 (4SF)      | 0 | ENS   | 6422   |
| 9                                                  | 227 | 0     | Inf. Dilution | lgK:244.6 (4SF)      | 0 | ENS   | 6494   |
| 10                                                 | 327 | 0     | Inf. Dilution | lgK:199.6 (4SF)      | 0 | ENS   | 6422   |
| 11                                                 | 327 | 0     | Inf. Dilution | lgK:199.5 (4SF)      | 0 | ENS   | 6494   |
| 12                                                 | 25  | 0     | Inf. Dilution | dG:-583.9 (4SF)      | 0 | EFE   | 6422   |
| 13                                                 | 25  | 0     | Inf. Dilution | dG:-583.60 (5SF)     | 0 | EFE   | 6482   |
| 14                                                 | 25  | 0     | Inf. Dilution | dG:-583.6 (4SF)      | 0 | EFE   | 6494   |
| 15                                                 | 25  | 0     | Inf. Dilution | dG:-2442.70 (6SF) kJ | 0 | CRV   | 9015   |
| 16                                                 | 25  | 0     | Inf. Dilution | dG:-583.51 (5SF)     | 0 | CRV   | 10174  |
| 17                                                 | 25  | 0     | Inf. Dilution | dG:-620.8 (4SF)      | 0 | CRV   | 12011  |
| 18                                                 | 25  | 0     | Inf. Dilution | dG:-583.16 (5SF)     | 5 | CRV   | 29553  |
| 19                                                 | 50  | 0     | Inf. Dilution | dG:-584.10 (5SF)     | 0 | CRV   | 10174  |
| 20                                                 | 75  | 0     | Inf. Dilution | dG:-584.75 (5SF)     | 0 | CRV   | 10174  |
| 21                                                 | 100 | 0     | Inf. Dilution | dG:-585.45 (5SF)     | 0 | CRV   | 10174  |
| 22                                                 | 125 | 0     | Inf. Dilution | dG:-586.21 (5SF)     | 0 | CRV   | 10174  |
| 23                                                 | 150 | 0     | Inf. Dilution | dG:-587.03 (5SF)     | 0 | CRV   | 10174  |
| 24                                                 | 175 | 0     | Inf. Dilution | dG:-587.90 (5SF)     | 0 | CRV   | 10174  |
| 25                                                 | 200 | 0     | Inf. Dilution | dG:-588.82 (5SF)     | 0 | CRV   | 10174  |
| 26                                                 | 225 | 0     | Inf. Dilution | dG:-589.79 (5SF)     | 0 | CRV   | 10174  |
| 27                                                 | 250 | 0     | Inf. Dilution | dG:-590.80 (5SF)     | 0 | CRV   | 10174  |
| 28                                                 | 300 | 0     | Inf. Dilution | dG:-592.95 (5SF)     | 0 | CRV   | 10174  |
| 29                                                 | 25  | 0     | Inf. Dilution | dH:-619.1 (4SF)      | 0 | MTD   | 6422   |

|    |     |   |               |                      |   |     |       |
|----|-----|---|---------------|----------------------|---|-----|-------|
| 30 | 25  | 0 | Inf. Dilution | dH:-619.4 (4SF)      | 0 | ENS | 6437  |
| 31 | 25  | 0 | Inf. Dilution | dH:-619.02 (5SF)     | 0 | EFE | 6482  |
| 32 | 25  | 0 | Inf. Dilution | dH:-619.0 (4SF)      | 0 | MTD | 6494  |
| 33 | 25  | 0 | Inf. Dilution | dH:-2590.30 (6SF) kJ | 0 | CRV | 9015  |
| 34 | 25  | 0 | Inf. Dilution | dH:-655.9 (4SF)      | 0 | CRV | 12011 |
| 35 | 25  | 0 | Inf. Dilution | dH:-618.57 (5SF)     | 5 | CRV | 29553 |
| 36 | 127 | 0 | Inf. Dilution | dH:-619.3 (4SF)      | 0 | MTD | 6422  |
| 37 | 127 | 0 | Inf. Dilution | dH:-619.2 (4SF)      | 0 | MTD | 6494  |
| 38 | 227 | 0 | Inf. Dilution | dH:-619.2 (4SF)      | 0 | MTD | 6422  |
| 39 | 227 | 0 | Inf. Dilution | dH:-619.0 (4SF)      | 0 | MTD | 6494  |
| 40 | 327 | 0 | Inf. Dilution | dH:-618.9 (4SF)      | 0 | MTD | 6422  |
| 41 | 327 | 0 | Inf. Dilution | dH:-618.7 (4SF)      | 0 | MTD | 6494  |

| Reaction No. 66479                                 |     |       |               |                      |   |       |        |
|----------------------------------------------------|-----|-------|---------------|----------------------|---|-------|--------|
| 2<Al(s)> + 2.5<O2(g)> + Si(s) = Al2SiO5(Sillim.,s) |     |       |               |                      |   |       |        |
| No                                                 | t   | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
| 1                                                  | 25  | 0     | Inf. Dilution | lgK:427.2 (4SF)      | 1 | ELC   |        |
| 2                                                  | 25  | 0     | Inf. Dilution | lgK:427.7 (4SF)      | 3 | ENS   | 6422   |
| 3                                                  | 25  | 0     | Inf. Dilution | lgK:427.3 (4SF)      | 3 | ENS   | 6494   |
| 4                                                  | 27  | 0     | Inf. Dilution | lgK:424.9 (4SF)      | 0 | ENS   | 6422   |
| 5                                                  | 27  | 0     | Inf. Dilution | lgK:424.5 (4SF)      | 0 | ENS   | 6494   |
| 6                                                  | 127 | 0     | Inf. Dilution | lgK:312.2 (4SF)      | 1 | ENS   | 6422   |
| 7                                                  | 127 | 0     | Inf. Dilution | lgK:311.9 (4SF)      | 1 | ENS   | 6494   |
| 8                                                  | 227 | 0     | Inf. Dilution | lgK:244.6 (4SF)      | 0 | ENS   | 6422   |
| 9                                                  | 227 | 0     | Inf. Dilution | lgK:244.4 (4SF)      | 0 | ENS   | 6494   |
| 10                                                 | 327 | 0     | Inf. Dilution | lgK:199.5 (4SF)      | 0 | ENS   | 6422   |
| 11                                                 | 327 | 0     | Inf. Dilution | lgK:199.3 (4SF)      | 0 | ENS   | 6494   |
| 12                                                 | 25  | 0     | Inf. Dilution | dG:-583.4 (4SF)      | 0 | EFE   | 6422   |
| 13                                                 | 25  | 0     | Inf. Dilution | dG:-583.00 (5SF)     | 0 | EFE   | 6482   |
| 14                                                 | 25  | 0     | Inf. Dilution | dG:-582.9 (4SF)      | 0 | EFE   | 6494   |
| 15                                                 | 25  | 0     | Inf. Dilution | dG:-2441.00 (6SF) kJ | 0 | CRV   | 9015   |
| 16                                                 | 25  | 0     | Inf. Dilution | dG:-583.02 (5SF)     | 0 | CRV   | 10174  |
| 17                                                 | 25  | 0     | Inf. Dilution | dG:-627.6 (4SF)      | 0 | CRV   | 12011  |
| 18                                                 | 50  | 0     | Inf. Dilution | dG:-583.63 (5SF)     | 0 | CRV   | 10174  |
| 19                                                 | 75  | 0     | Inf. Dilution | dG:-584.30 (5SF)     | 0 | CRV   | 10174  |
| 20                                                 | 100 | 0     | Inf. Dilution | dG:-585.02 (5SF)     | 0 | CRV   | 10174  |

|    |     |   |               |                      |   |     |       |
|----|-----|---|---------------|----------------------|---|-----|-------|
| 21 | 125 | 0 | Inf. Dilution | dG:-585.81 (5SF)     | 0 | CRV | 10174 |
| 22 | 150 | 0 | Inf. Dilution | dG:-586.65 (5SF)     | 0 | CRV | 10174 |
| 23 | 175 | 0 | Inf. Dilution | dG:-587.54 (5SF)     | 0 | CRV | 10174 |
| 24 | 200 | 0 | Inf. Dilution | dG:-588.48 (5SF)     | 0 | CRV | 10174 |
| 25 | 225 | 0 | Inf. Dilution | dG:-589.46 (5SF)     | 0 | CRV | 10174 |
| 26 | 250 | 0 | Inf. Dilution | dG:-590.49 (5SF)     | 0 | CRV | 10174 |
| 27 | 300 | 0 | Inf. Dilution | dG:-592.68 (5SF)     | 0 | CRV | 10174 |
| 28 | 25  | 0 | Inf. Dilution | dH:-618.5 (4SF)      | 0 | MTD | 6422  |
| 29 | 25  | 0 | Inf. Dilution | dH:-618.5 (4SF)      | 0 | ENS | 6437  |
| 30 | 25  | 0 | Inf. Dilution | dH:-618.09 (5SF)     | 0 | EFE | 6482  |
| 31 | 25  | 0 | Inf. Dilution | dH:-618.1 (4SF)      | 0 | MTD | 6494  |
| 32 | 25  | 0 | Inf. Dilution | dH:-2587.80 (6SF) kJ | 0 | CRV | 9015  |
| 33 | 25  | 0 | Inf. Dilution | dH:-662.6 (4SF)      | 0 | CRV | 12011 |
| 34 | 127 | 0 | Inf. Dilution | dH:-618.7 (4SF)      | 0 | MTD | 6422  |
| 35 | 127 | 0 | Inf. Dilution | dH:-618.3 (4SF)      | 0 | MTD | 6494  |
| 36 | 227 | 0 | Inf. Dilution | dH:-618.6 (4SF)      | 0 | MTD | 6422  |
| 37 | 227 | 0 | Inf. Dilution | dH:-618.2 (4SF)      | 0 | MTD | 6494  |
| 38 | 327 | 0 | Inf. Dilution | dH:-618.4 (4SF)      | 0 | MTD | 6422  |
| 39 | 327 | 0 | Inf. Dilution | dH:-617.9 (4SF)      | 0 | MTD | 6494  |

| Reaction No. 66480                                                   |     |       |               |                     |   |       |        |
|----------------------------------------------------------------------|-----|-------|---------------|---------------------|---|-------|--------|
| 2<Al(s)> + 2<H2(g)> + 4.5<O2(g)> + 2<Si(s)> = Al2Si2O7.2H2O(Dick.,s) |     |       |               |                     |   |       |        |
| No                                                                   | t   | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                                    | 25  | 0     | Inf. Dilution | lgK:665.0 (4SF)     | 4 | ENS   | 6422   |
| 2                                                                    | 25  | 0     | Inf. Dilution | lgK:665.0 (4SF)     | 4 | ENS   | 6494   |
| 3                                                                    | 27  | 0     | Inf. Dilution | lgK:660.6 (4SF)     | 0 | ENS   | 6422   |
| 4                                                                    | 27  | 0     | Inf. Dilution | lgK:660.6 (4SF)     | 0 | ENS   | 6494   |
| 5                                                                    | 127 | 0     | Inf. Dilution | lgK:481.2 (4SF)     | 3 | ENS   | 6422   |
| 6                                                                    | 127 | 0     | Inf. Dilution | lgK:481.3 (4SF)     | 3 | ENS   | 6494   |
| 7                                                                    | 227 | 0     | Inf. Dilution | lgK:373.6 (4SF)     | 2 | ENS   | 6422   |
| 8                                                                    | 227 | 0     | Inf. Dilution | lgK:373.7 (4SF)     | 2 | ENS   | 6494   |
| 9                                                                    | 327 | 0     | Inf. Dilution | lgK:301.9 (4SF)     | 0 | ENS   | 6422   |
| 10                                                                   | 327 | 0     | Inf. Dilution | lgK:302.0 (4SF)     | 0 | ENS   | 6494   |
| 11                                                                   | 25  | 0     | Inf. Dilution | dG:-907.2 (4SF)     | 4 | EFE   | 6422   |
| 12                                                                   | 25  | 0     | Inf. Dilution | dG:-907.3 (4SF)     | 4 | EFE   | 6494   |
| 13                                                                   | 25  | 0     | Inf. Dilution | dG:-3795.9 (5SF) kJ | 4 | CRV   | 9015   |

|    |     |   |               |                   |   |     |       |
|----|-----|---|---------------|-------------------|---|-----|-------|
| 14 | 25  | 0 | GAMSPATH      | dG:-3795.(4SF)kJ  | 3 | CRV | 12638 |
| 15 | 25  | 0 | Inf. Dilution | dH:-984.3(4SF)    | 4 | MTD | 6422  |
| 16 | 25  | 0 | Inf. Dilution | dH:-984.3(4SF)    | 4 | MTD | 6494  |
| 17 | 25  | 0 | Inf. Dilution | dH:-4118.3(5SF)kJ | 4 | CRV | 9015  |
| 18 | 27  | 0 | Inf. Dilution | dH:-984.4(4SF)    | 0 | MTD | 6494  |
| 19 | 127 | 0 | Inf. Dilution | dH:-984.8(4SF)    | 3 | MTD | 6422  |
| 20 | 127 | 0 | Inf. Dilution | dH:-984.7(4SF)    | 3 | MTD | 6494  |
| 21 | 227 | 0 | Inf. Dilution | dH:-984.7(4SF)    | 2 | MTD | 6422  |
| 22 | 227 | 0 | Inf. Dilution | dH:-984.6(4SF)    | 2 | MTD | 6494  |
| 23 | 327 | 0 | Inf. Dilution | dH:-984.2(4SF)    | 0 | MTD | 6422  |
| 24 | 327 | 0 | Inf. Dilution | dH:-984.0(4SF)    | 0 | MTD | 6494  |

| Reaction No. 66481                         |     |       |               |                |   |       |        |
|--------------------------------------------|-----|-------|---------------|----------------|---|-------|--------|
| 2<Al(s)> + Ti(s) + 2.5<O2(g)> = Al2TiO5(s) |     |       |               |                |   |       |        |
| No                                         | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                          | 25  | 0     | Inf. Dilution | lgK:434.9(4SF) | 4 | ENS   | 6422   |
| 2                                          | 27  | 0     | Inf. Dilution | lgK:432.1(4SF) | 4 | ENS   | 6422   |
| 3                                          | 127 | 0     | Inf. Dilution | lgK:317.7(4SF) | 4 | ENS   | 6422   |
| 4                                          | 227 | 0     | Inf. Dilution | lgK:249.0(4SF) | 4 | ENS   | 6422   |
| 5                                          | 327 | 0     | Inf. Dilution | lgK:203.3(4SF) | 4 | ENS   | 6422   |
| 6                                          | 25  | 0     | Inf. Dilution | dG:-593.3(4SF) | 4 | EFE   | 6422   |
| 7                                          | 25  | 0     | Inf. Dilution | dH:-628.3(4SF) | 4 | MTD   | 6422   |
| 8                                          | 127 | 0     | Inf. Dilution | dH:-628.3(4SF) | 4 | MTD   | 6422   |
| 9                                          | 227 | 0     | Inf. Dilution | dH:-628.0(4SF) | 4 | MTD   | 6422   |
| 10                                         | 327 | 0     | Inf. Dilution | dH:-627.6(4SF) | 4 | MTD   | 6422   |

| Reaction No. 66482                      |     |       |               |                  |   |       |        |
|-----------------------------------------|-----|-------|---------------|------------------|---|-------|--------|
| 2<Al(s)> + 3<S(s)> = Al+3(2)_S-2(3)_(s) |     |       |               |                  |   |       |        |
| No                                      | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                       | 25  | 0     | Inf. Dilution | lgK:125.0(4SF)   | 4 | ENS   | 6422   |
| 2                                       | 27  | 0     | Inf. Dilution | lgK:124.2(4SF)   | 4 | ENS   | 6422   |
| 3                                       | 127 | 0     | Inf. Dilution | lgK:92.63(4SF)   | 4 | ENS   | 6422   |
| 4                                       | 227 | 0     | Inf. Dilution | lgK:73.47(4SF)   | 4 | ENS   | 6422   |
| 5                                       | 327 | 0     | Inf. Dilution | lgK:60.61(4SF)   | 4 | ENS   | 6422   |
| 6                                       | 25  | 0     | Inf. Dilution | dG:-170.5(4SF)   | 4 | EFE   | 6422   |
| 7                                       | 25  | 0     | Inf. Dilution | dH:-724.0(4SF)kJ | 4 | CRV   | 6330   |



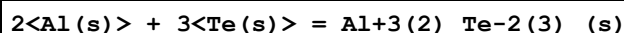
|    |     |   |               |                 |   |     |      |
|----|-----|---|---------------|-----------------|---|-----|------|
| 8  | 25  | 0 | Inf. Dilution | dH:-173.0 (4SF) | 4 | MTD | 6422 |
| 9  | 25  | 0 | Inf. Dilution | dH:-173. (3SF)  | 4 | CRV | 7382 |
| 10 | 127 | 0 | Inf. Dilution | dH:-174.8 (4SF) | 4 | MTD | 6422 |
| 11 | 227 | 0 | Inf. Dilution | dH:-176.0 (4SF) | 4 | MTD | 6422 |
| 12 | 327 | 0 | Inf. Dilution | dH:-177.0 (4SF) | 4 | MTD | 6422 |

| Reaction No. 66483                                   |     |       |               |                  |   |       |        |
|------------------------------------------------------|-----|-------|---------------|------------------|---|-------|--------|
| 2<Al(s)> + 6<O2(g)> + 3<S(s)> = Al+3(2)_SO4-2(3)_(s) |     |       |               |                  |   |       |        |
| No                                                   | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                    | 25  | 0     | Inf. Dilution | lgK:543.0 (4SF)  | 3 | ENS   | 6422   |
| 2                                                    | 25  | 0     | Inf. Dilution | lgK:543.2 (4SF)  | 3 | ENS   | 6494   |
| 3                                                    | 27  | 0     | Inf. Dilution | lgK:539.3 (4SF)  | 0 | ENS   | 6422   |
| 4                                                    | 27  | 0     | Inf. Dilution | lgK:539.5 (4SF)  | 0 | ENS   | 6494   |
| 5                                                    | 127 | 0     | Inf. Dilution | lgK:389.5 (4SF)  | 2 | ENS   | 6422   |
| 6                                                    | 127 | 0     | Inf. Dilution | lgK:389.6 (4SF)  | 2 | ENS   | 6494   |
| 7                                                    | 227 | 0     | Inf. Dilution | lgK:299.4 (4SF)  | 1 | ENS   | 6422   |
| 8                                                    | 227 | 0     | Inf. Dilution | lgK:299.5 (4SF)  | 1 | ENS   | 6494   |
| 9                                                    | 327 | 0     | Inf. Dilution | lgK:239.4 (4SF)  | 0 | ENS   | 6422   |
| 10                                                   | 327 | 0     | Inf. Dilution | lgK:239.4 (4SF)  | 0 | ENS   | 6494   |
| 11                                                   | 25  | 0     | Inf. Dilution | dG:-740.8 (4SF)  | 3 | EFE   | 6422   |
| 12                                                   | 25  | 0     | Inf. Dilution | dG:-741.1 (4SF)  | 3 | EFE   | 6494   |
| 13                                                   | 25  | 0     | Inf. Dilution | dG:-740.95 (5SF) | 3 | CRV   | 7382   |
| 14                                                   | 25  | 0     | Inf. Dilution | dG:-740.95 (5SF) | 4 | CRV   | 12011  |
| 15                                                   | 25  | 0     | Inf. Dilution | dH:-822.4 (4SF)  | 4 | MTD   | 6422   |
| 16                                                   | 25  | 0     | Inf. Dilution | dH:-822.6 (4SF)  | 4 | MTD   | 6494   |
| 17                                                   | 25  | 0     | Inf. Dilution | dH:-822.38 (5SF) | 4 | CRV   | 7382   |
| 18                                                   | 25  | 0     | Inf. Dilution | dH:-822.38 (5SF) | 4 | CRV   | 12011  |
| 19                                                   | 127 | 0     | Inf. Dilution | dH:-824.1 (4SF)  | 1 | MTD   | 6422   |
| 20                                                   | 127 | 0     | Inf. Dilution | dH:-824.4 (4SF)  | 1 | MTD   | 6494   |
| 21                                                   | 227 | 0     | Inf. Dilution | dH:-824.4 (4SF)  | 4 | MTD   | 6422   |
| 22                                                   | 227 | 0     | Inf. Dilution | dH:-824.8 (4SF)  | 0 | MTD   | 6494   |
| 23                                                   | 327 | 0     | Inf. Dilution | dH:-824.1 (4SF)  | 0 | MTD   | 6422   |
| 24                                                   | 327 | 0     | Inf. Dilution | dH:-824.5 (4SF)  | 0 | MTD   | 6494   |

| Reaction No. 66484                        |  |  |  |  |  |  |  |
|-------------------------------------------|--|--|--|--|--|--|--|
| 2<Al(s)> + 3<Se(s)> = Al+3(2)_Se-2(3)_(s) |  |  |  |  |  |  |  |

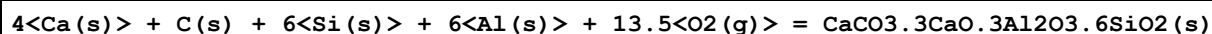
| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:97.83 (4SF) | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:97.22 (4SF) | 4 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:72.53 (4SF) | 4 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:57.65 (4SF) | 4 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:47.39 (4SF) | 4 | ENS   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dG:-133.5 (4SF) | 4 | EFE   | 6422   |
| 7  | 25  | 0     | Inf. Dilution | dH:-135.5 (4SF) | 4 | MTD   | 6422   |
| 8  | 25  | 0     | Inf. Dilution | dH:-135. (3SF)  | 4 | CRV   | 7382   |
| 9  | 127 | 0     | Inf. Dilution | dH:-135.7 (4SF) | 4 | MTD   | 6422   |
| 10 | 227 | 0     | Inf. Dilution | dH:-140.4 (4SF) | 4 | MTD   | 6422   |
| 11 | 327 | 0     | Inf. Dilution | dH:-141.2 (4SF) | 4 | MTD   | 6422   |

## Reaction No. 66485



| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:54.33 (4SF) | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:53.89 (4SF) | 4 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:40.09 (4SF) | 4 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:31.73 (4SF) | 4 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:26.12 (4SF) | 4 | ENS   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dG:-74.11 (4SF) | 4 | EFE   | 6422   |
| 7  | 25  | 0     | Inf. Dilution | dH:-76.2 (3SF)  | 4 | MTD   | 6422   |
| 8  | 25  | 0     | Inf. Dilution | dH:-78. (2SF)   | 4 | CRV   | 7382   |
| 9  | 127 | 0     | Inf. Dilution | dH:-76.4 (3SF)  | 4 | MTD   | 6422   |
| 10 | 227 | 0     | Inf. Dilution | dH:-76.7 (3SF)  | 4 | MTD   | 6422   |
| 11 | 327 | 0     | Inf. Dilution | dH:-77.2 (3SF)  | 4 | MTD   | 6422   |

## Reaction No. 66498



| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:2301. (4SF) | 4 | ENS   | 6494   |
| 2  | 27  | 0     | Inf. Dilution | lgK:2286. (4SF) | 0 | ENS   | 6494   |
| 3  | 127 | 0     | Inf. Dilution | lgK:1681. (4SF) | 4 | ENS   | 6494   |
| 4  | 227 | 0     | Inf. Dilution | lgK:1319. (4SF) | 4 | ENS   | 6494   |
| 5  | 327 | 0     | Inf. Dilution | lgK:1077. (4SF) | 4 | ENS   | 6494   |

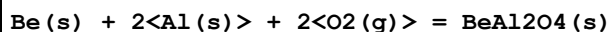
|                                            |     |   |               |                  |   |     |      |
|--------------------------------------------|-----|---|---------------|------------------|---|-----|------|
| 6                                          | 25  | 0 | Inf. Dilution | dG:-3132.4 (5SF) | 0 | EFE | 6482 |
| Note (6): Low wgt for internal consistency |     |   |               |                  |   |     |      |
| 7                                          | 25  | 0 | Inf. Dilution | dG:-3139. (4SF)  | 4 | EFE | 6494 |
| 8                                          | 25  | 0 | Inf. Dilution | dH:-3315.3 (5SF) | 4 | ENS | 6437 |
| 9                                          | 25  | 0 | Inf. Dilution | dH:-3310.2 (5SF) | 0 | EFE | 6482 |
| 10                                         | 25  | 0 | Inf. Dilution | dH:-3318. (4SF)  | 4 | MTD | 6494 |
| 11                                         | 127 | 0 | Inf. Dilution | dH:-3318. (4SF)  | 4 | MTD | 6494 |
| 12                                         | 227 | 0 | Inf. Dilution | dH:-3317. (4SF)  | 4 | MTD | 6494 |
| 13                                         | 327 | 0 | Inf. Dilution | dH:-3315. (4SF)  | 4 | MTD | 6494 |

| Reaction No. 66513                       |     |       |               |                 |   |       |        |
|------------------------------------------|-----|-------|---------------|-----------------|---|-------|--------|
| Ba(s) + 2<Al(s)> + 2<O2(g)> = BaAl2O4(s) |     |       |               |                 |   |       |        |
| No                                       | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                        | 25  | 0     | Inf. Dilution | lgK:387.6 (4SF) | 4 | ENS   | 6422   |
| 2                                        | 27  | 0     | Inf. Dilution | lgK:385.1 (4SF) | 4 | ENS   | 6422   |
| 3                                        | 127 | 0     | Inf. Dilution | lgK:283.8 (4SF) | 4 | ENS   | 6422   |
| 4                                        | 227 | 0     | Inf. Dilution | lgK:223.1 (4SF) | 4 | ENS   | 6422   |
| 5                                        | 327 | 0     | Inf. Dilution | lgK:182.6 (4SF) | 4 | ENS   | 6422   |
| 6                                        | 25  | 0     | Inf. Dilution | dG:-528.8 (4SF) | 4 | EFE   | 6422   |
| 7                                        | 25  | 0     | Inf. Dilution | dH:-555.9 (4SF) | 4 | MTD   | 6422   |
| 8                                        | 127 | 0     | Inf. Dilution | dH:-556.0 (4SF) | 4 | MTD   | 6422   |
| 9                                        | 227 | 0     | Inf. Dilution | dH:-556.0 (4SF) | 4 | MTD   | 6422   |
| 10                                       | 327 | 0     | Inf. Dilution | dH:-556.1 (4SF) | 4 | MTD   | 6422   |

| Reaction No. 66514                           |     |       |               |                 |   |       |        |
|----------------------------------------------|-----|-------|---------------|-----------------|---|-------|--------|
| 3<Ba(s)> + 2<Al(s)> + 3<O2(g)> = Ba3Al2O6(s) |     |       |               |                 |   |       |        |
| No                                           | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                            | 25  | 0     | Inf. Dilution | lgK:587.4 (4SF) | 4 | ENS   | 6422   |
| 2                                            | 27  | 0     | Inf. Dilution | lgK:583.6 (4SF) | 4 | ENS   | 6422   |
| 3                                            | 127 | 0     | Inf. Dilution | lgK:430.4 (4SF) | 4 | ENS   | 6422   |
| 4                                            | 227 | 0     | Inf. Dilution | lgK:338.5 (4SF) | 4 | ENS   | 6422   |
| 5                                            | 327 | 0     | Inf. Dilution | lgK:277.2 (4SF) | 4 | ENS   | 6422   |
| 6                                            | 25  | 0     | Inf. Dilution | dG:-801.3 (4SF) | 4 | EFE   | 6422   |
| 7                                            | 25  | 0     | Inf. Dilution | dH:-841.1 (4SF) | 4 | MTD   | 6422   |
| 8                                            | 127 | 0     | Inf. Dilution | dH:-841.0 (4SF) | 4 | MTD   | 6422   |
| 9                                            | 227 | 0     | Inf. Dilution | dH:-841.1 (4SF) | 4 | MTD   | 6422   |

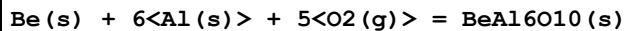
|    |     |   |               |                 |   |     |      |
|----|-----|---|---------------|-----------------|---|-----|------|
| 10 | 327 | 0 | Inf. Dilution | dH:-841.8 (4SF) | 4 | MTD | 6422 |
|----|-----|---|---------------|-----------------|---|-----|------|

## Reaction No. 66526



| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:381.7 (4SF) | 4 | ENS   | 6422   |
| 2  | 25  | 0     | Inf. Dilution | lgK:381.3 (4SF) | 4 | ENS   | 6494   |
| 3  | 27  | 0     | Inf. Dilution | lgK:379.3 (4SF) | 4 | ENS   | 6422   |
| 4  | 27  | 0     | Inf. Dilution | lgK:378.8 (4SF) | 4 | ENS   | 6494   |
| 5  | 127 | 0     | Inf. Dilution | lgK:279.1 (4SF) | 4 | ENS   | 6422   |
| 6  | 127 | 0     | Inf. Dilution | lgK:278.7 (4SF) | 4 | ENS   | 6494   |
| 7  | 227 | 0     | Inf. Dilution | lgK:218.9 (4SF) | 4 | MTD   | 6422   |
| 8  | 227 | 0     | Inf. Dilution | lgK:218.7 (4SF) | 4 | ENS   | 6494   |
| 9  | 327 | 0     | Inf. Dilution | lgK:178.9 (4SF) | 4 | ENS   | 6422   |
| 10 | 327 | 0     | Inf. Dilution | lgK:178.6 (4SF) | 4 | ENS   | 6494   |
| 11 | 25  | 0     | Inf. Dilution | dG:-520.8 (4SF) | 4 | EFE   | 6422   |
| 12 | 25  | 0     | Inf. Dilution | dG:-520.1 (4SF) | 4 | EFE   | 6494   |
| 13 | 25  | 0     | Inf. Dilution | dH:-550.0 (4SF) | 4 | MTD   | 6422   |
| 14 | 25  | 0     | Inf. Dilution | dH:-549.4 (4SF) | 4 | MTD   | 6494   |
| 15 | 127 | 0     | Inf. Dilution | dH:-550.2 (4SF) | 4 | MTD   | 6422   |
| 16 | 127 | 0     | Inf. Dilution | dH:-549.5 (4SF) | 4 | MTD   | 6494   |
| 17 | 227 | 0     | Inf. Dilution | dH:-550.1 (4SF) | 4 | MTD   | 6422   |
| 18 | 227 | 0     | Inf. Dilution | dH:-549.5 (4SF) | 4 | MTD   | 6494   |
| 19 | 327 | 0     | Inf. Dilution | dH:-549.9 (4SF) | 4 | MTD   | 6422   |
| 20 | 327 | 0     | Inf. Dilution | dH:-549.2 (4SF) | 4 | MTD   | 6494   |

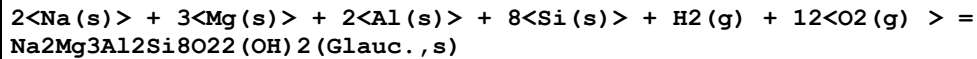
## Reaction No. 66527



| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:931.6 (4SF) | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:925.5 (4SF) | 0 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:680.6 (4SF) | 4 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:533.7 (4SF) | 4 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:435.8 (4SF) | 4 | ENS   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dG:-1271. (4SF) | 4 | EFE   | 6422   |
| 7  | 25  | 0     | Inf. Dilution | dH:-1344. (4SF) | 4 | MTD   | 6422   |

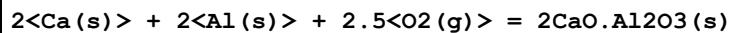
|    |     |   |               |                |   |     |      |
|----|-----|---|---------------|----------------|---|-----|------|
| 8  | 127 | 0 | Inf. Dilution | dH:-1345.(4SF) | 4 | MTD | 6422 |
| 9  | 227 | 0 | Inf. Dilution | dH:-1344.(4SF) | 4 | MTD | 6422 |
| 10 | 327 | 0 | Inf. Dilution | dH:-1344.(4SF) | 4 | MTD | 6422 |

## Reaction No. 66535



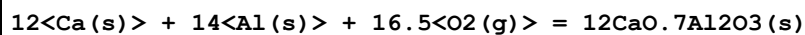
| No | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|-----|-------|---------------|----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:1968.(4SF) | 4 | ENS   | 6494   |
| 2  | 27  | 0     | Inf. Dilution | lgK:1955.(4SF) | 0 | ENS   | 6494   |
| 3  | 127 | 0     | Inf. Dilution | lgK:1434.(4SF) | 4 | ENS   | 6494   |
| 4  | 227 | 0     | Inf. Dilution | lgK:1121.(4SF) | 4 | ENS   | 6494   |
| 5  | 327 | 0     | Inf. Dilution | lgK:912.6(4SF) | 4 | ENS   | 6494   |
| 6  | 25  | 0     | Inf. Dilution | dG:-2684.(4SF) | 4 | EFE   | 6494   |
| 7  | 25  | 0     | Inf. Dilution | dH:-2857.(4SF) | 4 | ENS   | 6437   |
| 8  | 25  | 0     | Inf. Dilution | dH:-2860.(4SF) | 4 | MTD   | 6494   |
| 9  | 127 | 0     | Inf. Dilution | dH:-2862.(4SF) | 4 | MTD   | 6494   |
| 10 | 227 | 0     | Inf. Dilution | dH:-2861.(4SF) | 4 | MTD   | 6494   |
| 11 | 327 | 0     | Inf. Dilution | dH:-2860.(4SF) | 4 | MTD   | 6494   |

## Reaction No. 66554



| No | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|-----|-------|---------------|----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:490.8(4SF) | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:487.6(4SF) | 4 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:358.8(4SF) | 4 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:281.6(4SF) | 4 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:230.2(4SF) | 4 | ENS   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dG:-669.6(4SF) | 4 | ENS   | 6422   |
| 7  | 25  | 0     | Inf. Dilution | dH:-707.0(4SF) | 4 | MTD   | 6422   |
| 8  | 127 | 0     | Inf. Dilution | dH:-706.9(4SF) | 4 | MTD   | 6422   |
| 9  | 227 | 0     | Inf. Dilution | dH:-706.5(4SF) | 4 | MTD   | 6422   |
| 10 | 327 | 0     | Inf. Dilution | dH:-706.1(4SF) | 4 | MTD   | 6422   |

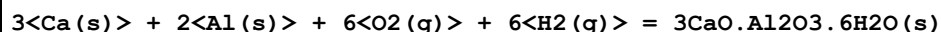
## Reaction No. 66555



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

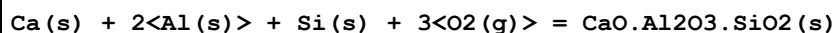
|    |     |   |               |                 |   |     |      |
|----|-----|---|---------------|-----------------|---|-----|------|
| 1  | 25  | 0 | Inf. Dilution | lgK:3235. (4SF) | 4 | ENS | 6422 |
| 2  | 27  | 0 | Inf. Dilution | lgK:3214. (4SF) | 0 | ENS | 6422 |
| 3  | 127 | 0 | Inf. Dilution | lgK:2369. (4SF) | 1 | ENS | 6422 |
| 4  | 227 | 0 | Inf. Dilution | lgK:1861. (4SF) | 1 | ENS | 6422 |
| 5  | 327 | 0 | Inf. Dilution | lgK:1523. (4SF) | 0 | ENS | 6422 |
| 6  | 25  | 0 | Inf. Dilution | dG:-4414. (4SF) | 4 | EFE | 6422 |
| 7  | 25  | 0 | Inf. Dilution | dH:-4644. (4SF) | 4 | MTD | 6422 |
| 8  | 127 | 0 | Inf. Dilution | dH:-4643. (4SF) | 1 | MTD | 6422 |
| 9  | 227 | 0 | Inf. Dilution | dH:-4642. (4SF) | 0 | MTD | 6422 |
| 10 | 327 | 0 | Inf. Dilution | dH:-4640. (4SF) | 0 | MTD | 6422 |

## Reaction No. 66556



| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:877.0 (4SF) | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:871.0 (4SF) | 4 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:629.4 (4SF) | 4 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:484.5 (4SF) | 4 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:388.0 (4SF) | 4 | ENS   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dG:-1196. (4SF) | 4 | EFE   | 6422   |
| 7  | 25  | 0     | Inf. Dilution | dH:-1326. (4SF) | 4 | MTD   | 6422   |
| 8  | 127 | 0     | Inf. Dilution | dH:-1326. (4SF) | 4 | MTD   | 6422   |
| 9  | 227 | 0     | Inf. Dilution | dH:-1326. (4SF) | 4 | MTD   | 6422   |
| 10 | 327 | 0     | Inf. Dilution | dH:-1324. (4SF) | 4 | MTD   | 6422   |

## Reaction No. 66557



| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:544.7 (4SF) | 1 | ELC   |        |
| 2  | 25  | 0     | Inf. Dilution | lgK:547.0 (4SF) | 0 | ENS   | 6422   |
| 3  | 25  | 0     | Inf. Dilution | lgK:548.3 (4SF) | 0 | ENS   | 6494   |
| 4  | 27  | 0     | Inf. Dilution | lgK:543.4 (4SF) | 0 | ENS   | 6422   |
| 5  | 27  | 0     | Inf. Dilution | lgK:544.7 (4SF) | 0 | ENS   | 6494   |
| 6  | 127 | 0     | Inf. Dilution | lgK:399.8 (4SF) | 0 | ENS   | 6422   |
| 7  | 127 | 0     | Inf. Dilution | lgK:400.8 (4SF) | 0 | ENS   | 6494   |
| 8  | 227 | 0     | Inf. Dilution | lgK:313.7 (4SF) | 0 | ENS   | 6422   |

|    |     |   |               |                  |   |     |      |
|----|-----|---|---------------|------------------|---|-----|------|
| 9  | 227 | 0 | Inf. Dilution | lgK:314.4 (4SF)  | 0 | ENS | 6494 |
| 10 | 327 | 0 | Inf. Dilution | lgK:256.4 (4SF)  | 0 | ENS | 6422 |
| 11 | 327 | 0 | Inf. Dilution | lgK:256.9 (4SF)  | 0 | ENS | 6494 |
| 12 | 25  | 0 | Inf. Dilution | dG:-746.2 (4SF)  | 0 | EFE | 6422 |
| 13 | 25  | 0 | Inf. Dilution | dG:-746.25 (5SF) | 0 | EFE | 6482 |
| 14 | 25  | 0 | Inf. Dilution | dG:-748.0 (4SF)  | 0 | EFE | 6494 |
| 15 | 25  | 0 | Inf. Dilution | dH:-788.3 (4SF)  | 2 | MTD | 6422 |
| 16 | 25  | 0 | Inf. Dilution | dH:-788.42 (5SF) | 0 | EFE | 6482 |
| 17 | 25  | 0 | Inf. Dilution | dH:-790.2 (4SF)  | 0 | MTD | 6494 |
| 18 | 127 | 0 | Inf. Dilution | dH:-788.2 (4SF)  | 0 | MTD | 6422 |
| 19 | 127 | 0 | Inf. Dilution | dH:-790.4 (4SF)  | 0 | MTD | 6494 |
| 20 | 227 | 0 | Inf. Dilution | dH:-787.8 (4SF)  | 0 | MTD | 6422 |
| 21 | 227 | 0 | Inf. Dilution | dH:-790.2 (4SF)  | 0 | MTD | 6494 |
| 22 | 327 | 0 | Inf. Dilution | dH:-787.1 (4SF)  | 0 | MTD | 6422 |
| 23 | 327 | 0 | Inf. Dilution | dH:-789.7 (4SF)  | 0 | MTD | 6494 |

| Reaction No. 66558                                                               |     |       |               |                      |   |       |        |
|----------------------------------------------------------------------------------|-----|-------|---------------|----------------------|---|-------|--------|
| 2<Al(s)> + Ca(s) + 4<O2(g)> + 2<Si(s)> = CaO.Al2O3.2SiO2(s)                      |     |       |               |                      |   |       |        |
| No                                                                               | t   | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
| 1                                                                                | 25  | 0     | Inf. Dilution | lgK:701.2 (4SF)      | 0 | ENS   | 6422   |
| 2                                                                                | 25  | 0     | Inf. Dilution | lgK:702.2 (4SF)      | 0 | ENS   | 6494   |
| 3                                                                                | 27  | 0     | Inf. Dilution | lgK:696.6 (4SF)      | 0 | ENS   | 6422   |
| 4                                                                                | 27  | 0     | Inf. Dilution | lgK:697.6 (4SF)      | 0 | ENS   | 6494   |
| 5                                                                                | 127 | 0     | Inf. Dilution | lgK:512.6 (4SF)      | 3 | ENS   | 6422   |
| 6                                                                                | 127 | 0     | Inf. Dilution | lgK:513.2 (4SF)      | 0 | ENS   | 6494   |
| 7                                                                                | 227 | 0     | Inf. Dilution | lgK:402.1 (4SF)      | 0 | ENS   | 6422   |
| 8                                                                                | 227 | 0     | Inf. Dilution | lgK:402.7 (4SF)      | 1 | ENS   | 6494   |
| 9                                                                                | 327 | 0     | Inf. Dilution | lgK:328.6 (4SF)      | 0 | ENS   | 6422   |
| 10                                                                               | 327 | 0     | Inf. Dilution | lgK:329.0 (4SF)      | 0 | ENS   | 6494   |
| 11                                                                               | 25  | 0     | Inf. Dilution | dG:-956.6 (4SF)      | 0 | EFE   | 6422   |
| 12                                                                               | 25  | 0     | Inf. Dilution | dG:-956.8 (4SF)      | 0 | EFE   | 6482   |
| 13                                                                               | 25  | 0     | Inf. Dilution | dG:-957.9 (4SF)      | 0 | EFE   | 6494   |
| 14                                                                               | 25  | 0     | Inf. Dilution | dG:-4002.10 (6SF) kJ | 6 | EFE   | 8798   |
| Note (14): To avoid confusion note source units (J/mol) & value c.a.= -4 million |     |       |               |                      |   |       |        |
| 15                                                                               | 25  | 0     | Inf. Dilution | dG:-4002.1 (5SF) kJ  | 4 | MTD   | 12843  |
| 16                                                                               | 25  | 0     | Inf. Dilution | dH:-1010. (4SF)      | 4 | MTD   | 6422   |

|    |     |   |               |                      |   |     |       |
|----|-----|---|---------------|----------------------|---|-----|-------|
| 17 | 25  | 0 | Inf. Dilution | dH:-1011.6 (5SF)     | 0 | ENS | 6437  |
| 18 | 25  | 0 | Inf. Dilution | dH:-1010.7 (5SF)     | 0 | EFE | 6482  |
| 19 | 25  | 0 | Inf. Dilution | dH:-1012. (4SF)      | 0 | MTD | 6494  |
| 20 | 25  | 0 | Inf. Dilution | dH:-4227.80 (6SF) kJ | 6 | EFE | 8798  |
| 21 | 25  | 0 | Inf. Dilution | dH:-4227.8 (5SF) kJ  | 4 | MTD | 12843 |
| 22 | 25  | 0 | GAMSPATH      | dH:-4216.5 (5SF) kJ  | 0 | CRV | 12638 |
| 23 | 127 | 0 | Inf. Dilution | dH:-1011. (4SF)      | 2 | MTD | 6422  |
| 24 | 127 | 0 | Inf. Dilution | dH:-1012. (4SF)      | 2 | MTD | 6494  |
| 25 | 227 | 0 | Inf. Dilution | dH:-1010. (4SF)      | 1 | MTD | 6422  |
| 26 | 227 | 0 | Inf. Dilution | dH:-1012. (4SF)      | 1 | MTD | 6494  |
| 27 | 327 | 0 | Inf. Dilution | dH:-1010. (4SF)      | 0 | MTD | 6422  |
| 28 | 327 | 0 | Inf. Dilution | dH:-1011. (4SF)      | 0 | MTD | 6494  |

| Reaction No. 66559                                            |     |       |               |                  |   |       |        |
|---------------------------------------------------------------|-----|-------|---------------|------------------|---|-------|--------|
| 2<Ca(s)> + 2<Al(s)> + Si(s) + 3.5<O2(g)> = 2CaO.Al2O3.SiO2(s) |     |       |               |                  |   |       |        |
| No                                                            | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                             | 25  | 0     | Inf. Dilution | lgK:662.7 (4SF)  | 4 | ENS   | 6422   |
| 2                                                             | 25  | 0     | Inf. Dilution | lgK:663.2 (4SF)  | 4 | ENS   | 6494   |
| 3                                                             | 27  | 0     | Inf. Dilution | lgK:658.4 (4SF)  | 4 | ENS   | 6422   |
| 4                                                             | 27  | 0     | Inf. Dilution | lgK:658.9 (4SF)  | 4 | ENS   | 6494   |
| 5                                                             | 127 | 0     | Inf. Dilution | lgK:485.1 (4SF)  | 4 | ENS   | 6422   |
| 6                                                             | 127 | 0     | Inf. Dilution | lgK:485.4 (4SF)  | 4 | ENS   | 6494   |
| 7                                                             | 227 | 0     | Inf. Dilution | lgK:381.2 (4SF)  | 4 | ENS   | 6422   |
| 8                                                             | 227 | 0     | Inf. Dilution | lgK:381.3 (4SF)  | 4 | ENS   | 6494   |
| 9                                                             | 327 | 0     | Inf. Dilution | lgK:311.9 (4SF)  | 4 | ENS   | 6422   |
| 10                                                            | 327 | 0     | Inf. Dilution | lgK:312.0 (4SF)  | 4 | ENS   | 6494   |
| 11                                                            | 25  | 0     | Inf. Dilution | dG:-904.1 (4SF)  | 4 | EFE   | 6422   |
| 12                                                            | 25  | 0     | Inf. Dilution | dG:-904.86 (5SF) | 4 | EFE   | 6482   |
| 13                                                            | 25  | 0     | Inf. Dilution | dG:-904.8 (4SF)  | 4 | EFE   | 6494   |
| 14                                                            | 25  | 0     | Inf. Dilution | dH:-951.6 (4SF)  | 4 | MTD   | 6422   |
| 15                                                            | 25  | 0     | Inf. Dilution | dH:-950.7 (4SF)  | 4 | ENS   | 6437   |
| 16                                                            | 25  | 0     | Inf. Dilution | dH:-953.19 (5SF) | 4 | EFE   | 6482   |
| 17                                                            | 25  | 0     | Inf. Dilution | dH:-952.4 (4SF)  | 4 | MTD   | 6494   |
| 18                                                            | 127 | 0     | Inf. Dilution | dH:-951.5 (4SF)  | 4 | MTD   | 6422   |
| 19                                                            | 127 | 0     | Inf. Dilution | dH:-952.6 (4SF)  | 4 | MTD   | 6494   |
| 20                                                            | 227 | 0     | Inf. Dilution | dH:-951.0 (4SF)  | 4 | MTD   | 6422   |



|    |     |   |               |                 |   |     |      |
|----|-----|---|---------------|-----------------|---|-----|------|
| 21 | 227 | 0 | Inf. Dilution | dH:-952.4 (4SF) | 4 | MTD | 6494 |
| 22 | 327 | 0 | Inf. Dilution | dH:-950.4 (4SF) | 4 | MTD | 6422 |
| 23 | 327 | 0 | Inf. Dilution | dH:-952.0 (4SF) | 4 | MTD | 6494 |

| Reaction No. 66560                                              |     |       |               |                  |   |       |        |
|-----------------------------------------------------------------|-----|-------|---------------|------------------|---|-------|--------|
| 3<Ca(s)> + 2<Al(s)> + 3<Si(s)> + 6<O2(g)> = 3CaO.Al2O3.3SiO2(s) |     |       |               |                  |   |       |        |
| No                                                              | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                               | 25  | 0     | Inf. Dilution | lgK:1098.0 (5SF) | 1 | ELC   |        |
| 2                                                               | 25  | 0     | Inf. Dilution | lgK:1100. (4SF)  | 0 | ENS   | 6422   |
| 3                                                               | 25  | 0     | Inf. Dilution | lgK:1100. (4SF)  | 0 | ENS   | 6494   |
| 4                                                               | 27  | 0     | Inf. Dilution | lgK:1093. (4SF)  | 0 | ENS   | 6422   |
| 5                                                               | 27  | 0     | Inf. Dilution | lgK:1093. (4SF)  | 0 | ENS   | 6494   |
| 6                                                               | 127 | 0     | Inf. Dilution | lgK:803.8 (4SF)  | 0 | ENS   | 6422   |
| 7                                                               | 127 | 0     | Inf. Dilution | lgK:803.7 (4SF)  | 0 | ENS   | 6494   |
| 8                                                               | 227 | 0     | Inf. Dilution | lgK:630.2 (4SF)  | 0 | ENS   | 6422   |
| 9                                                               | 227 | 0     | Inf. Dilution | lgK:630.3 (4SF)  | 0 | ENS   | 6494   |
| 10                                                              | 327 | 0     | Inf. Dilution | lgK:514.6 (4SF)  | 0 | ENS   | 6422   |
| 11                                                              | 327 | 0     | Inf. Dilution | lgK:514.7 (4SF)  | 0 | ENS   | 6494   |
| 12                                                              | 25  | 0     | Inf. Dilution | dG:-1501. (4SF)  | 0 | EFE   | 6422   |
| 13                                                              | 25  | 0     | Inf. Dilution | dG:-1498.8 (5SF) | 0 | EFE   | 6482   |
| 14                                                              | 25  | 0     | Inf. Dilution | dG:-1501. (4SF)  | 0 | EFE   | 6494   |
| 15                                                              | 25  | 0     | Inf. Dilution | dH:-1588. (4SF)  | 2 | MTD   | 6422   |
| 16                                                              | 25  | 0     | Inf. Dilution | dH:-1585.6 (5SF) | 0 | ENS   | 6437   |
| 17                                                              | 25  | 0     | Inf. Dilution | dH:-1585.3 (5SF) | 0 | EFE   | 6482   |
| 18                                                              | 25  | 0     | Inf. Dilution | dH:-1587. (4SF)  | 0 | MTD   | 6494   |
| 19                                                              | 127 | 0     | Inf. Dilution | dH:-1589. (4SF)  | 0 | MTD   | 6422   |
| 20                                                              | 127 | 0     | Inf. Dilution | dH:-1587. (4SF)  | 0 | MTD   | 6494   |
| 21                                                              | 227 | 0     | Inf. Dilution | dH:-1588.0 (5SF) | 0 | MTD   | 6422   |
| 22                                                              | 227 | 0     | Inf. Dilution | dH:-1587. (4SF)  | 0 | MTD   | 6494   |
| 23                                                              | 327 | 0     | Inf. Dilution | dH:-1587. (4SF)  | 0 | MTD   | 6422   |
| 24                                                              | 327 | 0     | Inf. Dilution | dH:-1586. (4SF)  | 0 | MTD   | 6494   |

| Reaction No. 66561                                                          |    |       |               |                 |   |       |        |
|-----------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Ca(s) + 2<Al(s)> + 2<Si(s)> + 5<O2(g)> + 2<H2(g)> = CaO.Al2O3.2SiO2.2H2O(s) |    |       |               |                 |   |       |        |
| No                                                                          | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                           | 25 | 0     | Inf. Dilution | lgK:789.3 (4SF) | 4 | ENS   | 6422   |

|    |     |   |               |                  |   |     |      |
|----|-----|---|---------------|------------------|---|-----|------|
| 2  | 25  | 0 | Inf. Dilution | lgK:790.6(4SF)   | 4 | ENS | 6494 |
| 3  | 27  | 0 | Inf. Dilution | lgK:784.0(4SF)   | 0 | ENS | 6422 |
| 4  | 27  | 0 | Inf. Dilution | lgK:785.4(4SF)   | 0 | ENS | 6494 |
| 5  | 127 | 0 | Inf. Dilution | lgK:572.4(4SF)   | 2 | ENS | 6422 |
| 6  | 127 | 0 | Inf. Dilution | lgK:573.4(4SF)   | 2 | ENS | 6494 |
| 7  | 227 | 0 | Inf. Dilution | lgK:445.4(4SF)   | 1 | ENS | 6422 |
| 8  | 227 | 0 | Inf. Dilution | lgK:446.2(4SF)   | 1 | ENS | 6494 |
| 9  | 327 | 0 | Inf. Dilution | lgK:360.7(4SF)   | 0 | ENS | 6422 |
| 10 | 327 | 0 | Inf. Dilution | lgK:361.4(4SF)   | 0 | ENS | 6494 |
| 11 | 25  | 0 | Inf. Dilution | dG:-1077.(4SF)   | 4 | EFE | 6422 |
| 12 | 25  | 0 | Inf. Dilution | dG:-1077.90(6SF) | 4 | EFE | 6482 |
| 13 | 25  | 0 | Inf. Dilution | dG:-1079.(4SF)   | 4 | EFE | 6494 |
| 14 | 25  | 0 | Inf. Dilution | dH:-1161.2(5SF)  | 4 | MTD | 6422 |
| 15 | 25  | 0 | Inf. Dilution | dH:-1163.5(5SF)  | 4 | ENS | 6437 |
| 16 | 25  | 0 | Inf. Dilution | dH:-1162.9(5SF)  | 4 | EFE | 6482 |
| 17 | 25  | 0 | Inf. Dilution | dH:-1164.(4SF)   | 4 | MTD | 6494 |
| 18 | 127 | 0 | Inf. Dilution | dH:-1162.(4SF)   | 1 | MTD | 6422 |
| 19 | 127 | 0 | Inf. Dilution | dH:-1164.(4SF)   | 1 | MTD | 6494 |
| 20 | 227 | 0 | Inf. Dilution | dH:-1163.(4SF)   | 0 | MTD | 6422 |
| 21 | 227 | 0 | Inf. Dilution | dH:-1164.(4SF)   | 0 | MTD | 6494 |
| 22 | 327 | 0 | Inf. Dilution | dH:-1163.(4SF)   | 0 | MTD | 6422 |
| 23 | 327 | 0 | Inf. Dilution | dH:-1163.(4SF)   | 0 | MTD | 6494 |

| Reaction No. 66608                         |     |       |               |                |   |       |        |
|--------------------------------------------|-----|-------|---------------|----------------|---|-------|--------|
| Cd(s) + 2<Al(s)> + 2<O2(g)> = CdO.Al2O3(s) |     |       |               |                |   |       |        |
| No                                         | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                          | 25  | 0     | Inf. Dilution | lgK:315.6(4SF) | 4 | ENS   | 6422   |
| 2                                          | 27  | 0     | Inf. Dilution | lgK:313.5(4SF) | 4 | ENS   | 6422   |
| 3                                          | 127 | 0     | Inf. Dilution | lgK:230.0(4SF) | 4 | ENS   | 6422   |
| 4                                          | 227 | 0     | Inf. Dilution | lgK:179.9(4SF) | 4 | ENS   | 6422   |
| 5                                          | 327 | 0     | Inf. Dilution | lgK:146.6(4SF) | 4 | ENS   | 6422   |
| 6                                          | 25  | 0     | Inf. Dilution | dG:-430.6(4SF) | 4 | EFE   | 6422   |
| 7                                          | 25  | 0     | Inf. Dilution | dH:-458.6(4SF) | 4 | MTD   | 6422   |
| 8                                          | 127 | 0     | Inf. Dilution | dH:-458.5(4SF) | 4 | MTD   | 6422   |
| 9                                          | 227 | 0     | Inf. Dilution | dH:-458.2(4SF) | 4 | MTD   | 6422   |
| 10                                         | 327 | 0     | Inf. Dilution | dH:-459.3(4SF) | 4 | MTD   | 6422   |

| Reaction No. 66629                                                                 |     |       |               |                |   |       |        |
|------------------------------------------------------------------------------------|-----|-------|---------------|----------------|---|-------|--------|
| $\text{Ce(s)} + \text{Al(s)} + 1.5\text{O}_2\text{(g)} = \text{CeAlO}_3\text{(s)}$ |     |       |               |                |   |       |        |
| No                                                                                 | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                  | 25  | 0     | Inf. Dilution | lgK:291.8(4SF) | 4 | ENS   | 6422   |
| 2                                                                                  | 27  | 0     | Inf. Dilution | lgK:289.9(4SF) | 4 | ENS   | 6422   |
| 3                                                                                  | 127 | 0     | Inf. Dilution | lgK:213.5(4SF) | 4 | ENS   | 6422   |
| 4                                                                                  | 227 | 0     | Inf. Dilution | lgK:167.8(4SF) | 4 | ENS   | 6422   |
| 5                                                                                  | 327 | 0     | Inf. Dilution | lgK:137.3(4SF) | 4 | ENS   | 6422   |
| 6                                                                                  | 25  | 0     | Inf. Dilution | dG:-398.0(4SF) | 4 | EFE   | 6422   |
| 7                                                                                  | 25  | 0     | Inf. Dilution | dH:-419.1(4SF) | 4 | MTD   | 6422   |
| 8                                                                                  | 127 | 0     | Inf. Dilution | dH:-419.0(4SF) | 4 | MTD   | 6422   |
| 9                                                                                  | 227 | 0     | Inf. Dilution | dH:-418.8(4SF) | 4 | MTD   | 6422   |
| 10                                                                                 | 327 | 0     | Inf. Dilution | dH:-418.5(4SF) | 4 | MTD   | 6422   |

| Reaction No. 66776                                                                                                                              |     |       |               |                 |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------|---------------|-----------------|---|-------|--------|
| $3\text{Fe(s)} + 2\text{Al(s)} + 3\text{Si(s)} + 6\text{O}_2\text{(g)} = 3\text{FeO} \cdot \text{Al}_2\text{O}_3 \cdot 3\text{SiO}_2\text{(s)}$ |     |       |               |                 |   |       |        |
| No                                                                                                                                              | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                                               | 25  | 0     | Inf. Dilution | lgK:865.8(4SF)  | 4 | ENS   | 6494   |
| 2                                                                                                                                               | 27  | 0     | Inf. Dilution | lgK:860.1(4SF)  | 4 | ENS   | 6494   |
| 3                                                                                                                                               | 127 | 0     | Inf. Dilution | lgK:630.9(4SF)  | 4 | ENS   | 6494   |
| 4                                                                                                                                               | 227 | 0     | Inf. Dilution | lgK:493.5(4SF)  | 4 | ENS   | 6494   |
| 5                                                                                                                                               | 327 | 0     | Inf. Dilution | lgK:401.9(4SF)  | 4 | ENS   | 6494   |
| 6                                                                                                                                               | 25  | 0     | Inf. Dilution | dG:-1181.1(5SF) | 4 | EFE   | 6482   |
| 7                                                                                                                                               | 25  | 0     | Inf. Dilution | dG:-1181.(4SF)  | 4 | EFE   | 6494   |
| 8                                                                                                                                               | 25  | 0     | Inf. Dilution | dH:-1261.1(5SF) | 4 | ENS   | 6437   |
| 9                                                                                                                                               | 25  | 0     | Inf. Dilution | dH:-1258.5(5SF) | 4 | MTD   | 6482   |
| 10                                                                                                                                              | 25  | 0     | Inf. Dilution | dH:-1258.(4SF)  | 4 | MTD   | 6494   |
| 11                                                                                                                                              | 127 | 0     | Inf. Dilution | dH:-1258.(4SF)  | 4 | MTD   | 6494   |
| 12                                                                                                                                              | 227 | 0     | Inf. Dilution | dH:-1258.(4SF)  | 4 | MTD   | 6494   |
| 13                                                                                                                                              | 327 | 0     | Inf. Dilution | dH:-1257.(4SF)  | 4 | MTD   | 6494   |

| Reaction No. 66777                                                                                                                              |    |       |               |                |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $3\text{Mg(s)} + 2\text{Al(s)} + 3\text{Si(s)} + 6\text{O}_2\text{(g)} = 3\text{MgO} \cdot \text{Al}_2\text{O}_3 \cdot 3\text{SiO}_2\text{(s)}$ |    |       |               |                |   |       |        |
| No                                                                                                                                              | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                                                               | 25 | 0     | Inf. Dilution | lgK:1040.(4SF) | 4 | ENS   | 6494   |
| 2                                                                                                                                               | 27 | 0     | Inf. Dilution | lgK:1033.(4SF) | 0 | ENS   | 6494   |

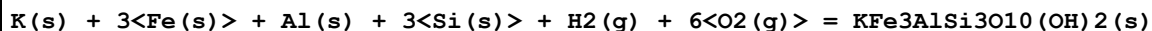
|    |     |   |               |                 |   |     |      |
|----|-----|---|---------------|-----------------|---|-----|------|
| 3  | 127 | 0 | Inf. Dilution | lgK:759.3(4SF)  | 4 | ENS | 6494 |
| 4  | 227 | 0 | Inf. Dilution | lgK:595.1(4SF)  | 4 | ENS | 6494 |
| 5  | 327 | 0 | Inf. Dilution | lgK:485.7(4SF)  | 4 | ENS | 6494 |
| 6  | 25  | 0 | Inf. Dilution | dG:-1418.7(5SF) | 4 | EFE | 6482 |
| 7  | 25  | 0 | Inf. Dilution | dG:-1418.(4SF)  | 4 | EFE | 6494 |
| 8  | 25  | 0 | Inf. Dilution | dH:-1503.7(5SF) | 4 | ENS | 6437 |
| 9  | 25  | 0 | Inf. Dilution | dH:-1502.5(5SF) | 4 | EFE | 6482 |
| 10 | 25  | 0 | Inf. Dilution | dH:-1502.(4SF)  | 4 | MTD | 6494 |
| 11 | 127 | 0 | Inf. Dilution | dH:-1502.(4SF)  | 4 | MTD | 6494 |
| 12 | 227 | 0 | Inf. Dilution | dH:-1502.(4SF)  | 4 | MTD | 6494 |
| 13 | 327 | 0 | Inf. Dilution | dH:-1501.(4SF)  | 4 | MTD | 6494 |

## Reaction No. 66778



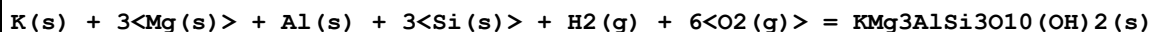
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dH:-2018.9(5SF) | 4 | ENS   | 6437   |

## Reaction No. 66779



| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:847.9(4SF)  | 1 | ELC   |        |
| 2  | 25  | 0     | Inf. Dilution | lgK:840.6(4SF)  | 0 | ENS   | 6494   |
| 3  | 27  | 0     | Inf. Dilution | lgK:835.1(4SF)  | 0 | ENS   | 6494   |
| 4  | 127 | 0     | Inf. Dilution | lgK:610.9(4SF)  | 0 | ENS   | 6494   |
| 5  | 227 | 0     | Inf. Dilution | lgK:476.5(4SF)  | 0 | ENS   | 6494   |
| 6  | 327 | 0     | Inf. Dilution | lgK:386.9(4SF)  | 0 | ENS   | 6494   |
| 7  | 25  | 0     | Inf. Dilution | dG:-1147.(4SF)  | 0 | EFE   | 6494   |
| 8  | 25  | 0     | Inf. Dilution | dH:-1225.0(5SF) | 0 | ENS   | 6437   |
| 9  | 25  | 0     | Inf. Dilution | dH:-1231.(4SF)  | 1 | MTD   | 6494   |
| 10 | 127 | 0     | Inf. Dilution | dH:-1231.(4SF)  | 0 | MTD   | 6494   |
| 11 | 227 | 0     | Inf. Dilution | dH:-1230.(4SF)  | 0 | MTD   | 6494   |
| 12 | 327 | 0     | Inf. Dilution | dH:-1228.(4SF)  | 0 | MTD   | 6494   |

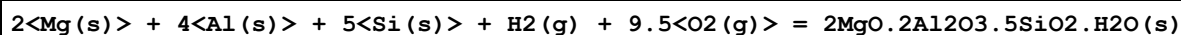
## Reaction No. 66780



| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:1024.0(5SF) | 1 | ELC   |        |

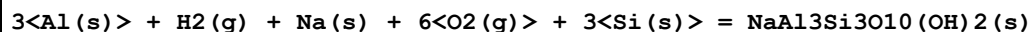
|    |     |   |               |                 |   |     |      |
|----|-----|---|---------------|-----------------|---|-----|------|
| 2  | 25  | 0 | Inf. Dilution | lgK:1027.(4SF)  | 0 | ENS | 6494 |
| 3  | 27  | 0 | Inf. Dilution | lgK:1020.(4SF)  | 0 | ENS | 6494 |
| 4  | 127 | 0 | Inf. Dilution | lgK:748.0(4SF)  | 0 | ENS | 6494 |
| 5  | 227 | 0 | Inf. Dilution | lgK:584.8(4SF)  | 0 | ENS | 6494 |
| 6  | 327 | 0 | Inf. Dilution | lgK:476.0(4SF)  | 0 | ENS | 6494 |
| 7  | 25  | 0 | Inf. Dilution | dG:-1392.7(5SF) | 0 | EFE | 6482 |
| 8  | 25  | 0 | Inf. Dilution | dG:-1401.(4SF)  | 0 | EFE | 6494 |
| 9  | 25  | 0 | Inf. Dilution | dH:-1482.8(5SF) | 2 | ENS | 6437 |
| 10 | 25  | 0 | Inf. Dilution | dH:-1483.6(5SF) | 2 | EFE | 6482 |
| 11 | 25  | 0 | Inf. Dilution | dH:-1493.(4SF)  | 0 | MTD | 6494 |
| 12 | 127 | 0 | Inf. Dilution | dH:-1494.(4SF)  | 0 | MTD | 6494 |
| 13 | 227 | 0 | Inf. Dilution | dH:-1493.(4SF)  | 0 | MTD | 6494 |
| 14 | 327 | 0 | Inf. Dilution | dH:-1493.(4SF)  | 0 | MTD | 6494 |

## Reaction No. 66781



| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dH:-2253.9(5SF) | 4 | ENS   | 6437   |

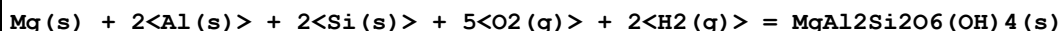
## Reaction No. 66783



| No | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|-----|-------|---------------|------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:975.6(4SF)   | 2 | ENS   | 6494   |
| 2  | 27  | 0     | Inf. Dilution | lgK:969.1(4SF)   | 0 | ENS   | 6494   |
| 3  | 127 | 0     | Inf. Dilution | lgK:710.1(4SF)   | 0 | ENS   | 6494   |
| 4  | 227 | 0     | Inf. Dilution | lgK:554.6(4SF)   | 0 | ENS   | 6494   |
| 5  | 327 | 0     | Inf. Dilution | lgK:450.9(4SF)   | 0 | ENS   | 6494   |
| 6  | 25  | 0     | Inf. Dilution | dG:-1329.7(5SF)  | 0 | EFE   | 6482   |
| 7  | 25  | 0     | Inf. Dilution | dG:-1331.(4SF)   | 2 | EFE   | 6494   |
| 8  | 25  | 0     | Inf. Dilution | dG:-1331.70(6SF) | 0 | CRV   | 10174  |
| 9  | 25  | 0     | Inf. Dilution | dG:-1332.5(5SF)  | 5 | CRV   | 29553  |
| 10 | 50  | 0     | Inf. Dilution | dG:-1333.40(6SF) | 0 | CRV   | 10174  |
| 11 | 75  | 0     | Inf. Dilution | dG:-1335.30(6SF) | 0 | CRV   | 10174  |
| 12 | 100 | 0     | Inf. Dilution | dG:-1337.30(6SF) | 0 | CRV   | 10174  |
| 13 | 125 | 0     | Inf. Dilution | dG:-1339.50(6SF) | 0 | CRV   | 10174  |
| 14 | 150 | 0     | Inf. Dilution | dG:-1341.80(6SF) | 0 | CRV   | 10174  |

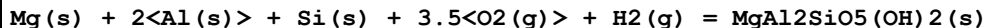
|    |     |   |               |                   |   |     |       |
|----|-----|---|---------------|-------------------|---|-----|-------|
| 15 | 175 | 0 | Inf. Dilution | dG:-1344.30 (6SF) | 0 | CRV | 10174 |
| 16 | 200 | 0 | Inf. Dilution | dG:-1346.90 (6SF) | 0 | CRV | 10174 |
| 17 | 225 | 0 | Inf. Dilution | dG:-1349.60 (6SF) | 0 | CRV | 10174 |
| 18 | 250 | 0 | Inf. Dilution | dG:-1352.40 (6SF) | 0 | CRV | 10174 |
| 19 | 300 | 0 | Inf. Dilution | dG:-1358.30 (6SF) | 0 | CRV | 10174 |
| 20 | 25  | 0 | Inf. Dilution | dH:-1422.3 (5SF)  | 4 | ENS | 6437  |
| 21 | 25  | 0 | Inf. Dilution | dH:-1420.7 (5SF)  | 0 | EFE | 6482  |
| 22 | 25  | 0 | Inf. Dilution | dH:-1422. (4SF)   | 4 | MTD | 6494  |
| 23 | 25  | 0 | Inf. Dilution | dH:-1423.60 (6SF) | 5 | CRV | 29553 |
| 24 | 127 | 0 | Inf. Dilution | dH:-1423. (4SF)   | 0 | MTD | 6494  |
| 25 | 227 | 0 | Inf. Dilution | dH:-1423. (4SF)   | 0 | MTD | 6494  |
| 26 | 327 | 0 | Inf. Dilution | dH:-1422. (4SF)   | 0 | MTD | 6494  |

## Reaction No. 66787



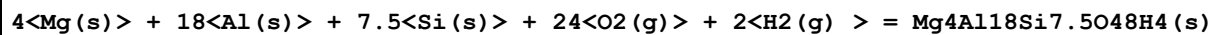
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dH:-1146. (4SF) | 4 | ENS   | 6437   |

## Reaction No. 66788



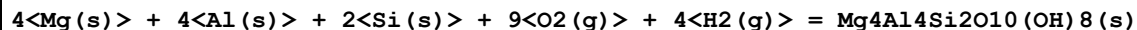
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dH:-850.4 (4SF) | 4 | ENS   | 6437   |

## Reaction No. 66789



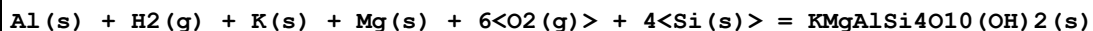
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dH:-6003.5 (5SF) | 4 | ENS   | 6437   |

## Reaction No. 66791



| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dH:-2162. (4SF) | 4 | ENS   | 6437   |

## Reaction No. 66792



| No | t  | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dG:-5464.10 (6SF) kJ | 3 | CRV   | 25435  |

|   |    |   |               |                    |   |     |       |
|---|----|---|---------------|--------------------|---|-----|-------|
| 2 | 25 | 0 | GAMSPATH      | dG:-7971.(4SF)kJ   | 0 | CRV | 12638 |
| 3 | 25 | 0 | Inf. Dilution | dH:-1394.4(5SF)    | 2 | ENS | 6437  |
| 4 | 25 | 0 | Inf. Dilution | dH:-5844.50(6SF)kJ | 3 | CRV | 25435 |

| Reaction No. 66793                                                                                                                                                      |    |       |               |                    |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| $2\text{Al(s)} + 5\text{Fe(s)} + 4\text{H}_2\text{(g)} + 9\text{O}_2\text{(g)} + 3\text{Si(s)} = \text{Fe}_5\text{Al}_2\text{Si}_3\text{O}_{10}(\text{OH})_8\text{(s)}$ |    |       |               |                    |   |       |        |
| No                                                                                                                                                                      | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                                                                                                                       | 25 | 0     | Inf. Dilution | dG:-6535.60(6SF)kJ | 3 | CRV   | 25435  |
| 2                                                                                                                                                                       | 25 | 0     | Inf. Dilution | dH:-1708.5(5SF)    | 2 | ENS   | 6437   |
| 3                                                                                                                                                                       | 25 | 0     | Inf. Dilution | dH:-7154.00(6SF)kJ | 3 | CRV   | 25435  |

| Reaction No. 66795                                                                                                                                                                    |    |       |               |                 |   |       |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{K(s)} + 2\text{Mg(s)} + 3\text{Al(s)} + 2\text{Si(s)} + 6\text{O}_2\text{(g)} + \text{H}_2\text{(g)} = \text{KMg}_2\text{Al}_3\text{Si}_2\text{O}_{10}(\text{OH})_2\text{(s)}$ |    |       |               |                 |   |       |        |
| No                                                                                                                                                                                    | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                                                                                     | 25 | 0     | Inf. Dilution | dH:-1514.5(5SF) | 4 | ENS   | 6437   |

| Reaction No. 66796                                                                                                                                                                                        |    |       |               |                 |   |       |        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{Na(s)} + 2\text{Ca(s)} + 5\text{Mg(s)} + 7\text{Si(s)} + \text{Al(s)} + 12\text{O}_2\text{(g)} + \text{H}_2\text{(g)} = \text{NaCa}_2\text{Mg}_5\text{Si}_7\text{AlO}_{22}(\text{OH})_2\text{(s)}$ |    |       |               |                 |   |       |        |
| No                                                                                                                                                                                                        | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                                                                                                         | 25 | 0     | Inf. Dilution | dH:-3006.8(5SF) | 4 | ENS   | 6437   |

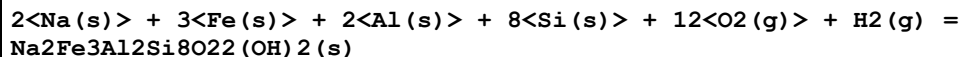
| Reaction No. 66797                                                                                                                                                      |    |       |               |                   |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $4\text{Al(s)} + 4\text{Fe(s)} + 4\text{H}_2\text{(g)} + 9\text{O}_2\text{(g)} + 2\text{Si(s)} = \text{Fe}_4\text{Al}_4\text{Si}_2\text{O}_{10}(\text{OH})_8\text{(s)}$ |    |       |               |                   |   |       |        |
| No                                                                                                                                                                      | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                                                                       | 25 | 0     | GAMSPATH      | dG:-7167.(4SF)kJ  | 3 | CRV   | 12638  |
| 2                                                                                                                                                                       | 25 | 0     | Inf. Dilution | dH:-1823.5(5SF)   | 4 | ENS   | 6437   |
| 3                                                                                                                                                                       | 25 | 0     | GAMSPATH      | dH:-7784.4(5SF)kJ | 0 | CRV   | 12638  |
| Note (3): Called '14A-Chamosite'                                                                                                                                        |    |       |               |                   |   |       |        |

| Reaction No. 66799                                                                                                                                         |    |       |               |                 |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{Fe(s)} + 2\text{Al(s)} + 2\text{Si(s)} + 5\text{O}_2\text{(g)} + 2\text{H}_2\text{(g)} = \text{FeAl}_2\text{Si}_2\text{O}_6(\text{OH})_4\text{(s)}$ |    |       |               |                 |   |       |        |
| No                                                                                                                                                         | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                                                          | 25 | 0     | Inf. Dilution | dH:-1060.9(5SF) | 4 | ENS   | 6437   |

| Reaction No. 66800                                                                                                                                                |   |       |        |             |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| $\text{K(s)} + \text{Fe(s)} + \text{Al(s)} + 4\text{Si(s)} + 6\text{O}_2\text{(g)} + \text{H}_2\text{(g)} = \text{KFeAlSi}_4\text{O}_{10}(\text{OH})_2\text{(s)}$ |   |       |        |             |   |       |        |
| No                                                                                                                                                                | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

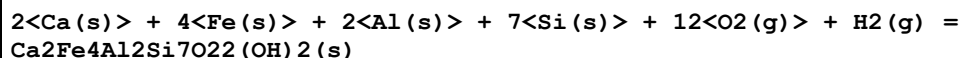
|   |    |   |               |                      |   |     |       |
|---|----|---|---------------|----------------------|---|-----|-------|
| 1 | 25 | 0 | Inf. Dilution | dG:-5127.80 (6SF) kJ | 3 | CRV | 25435 |
| 2 | 25 | 0 | Inf. Dilution | dH:-1310.9 (5SF)     | 2 | ENS | 6437  |
| 3 | 25 | 0 | Inf. Dilution | dH:-5497.30 (6SF) kJ | 3 | CRV | 25435 |

## Reaction No. 66801



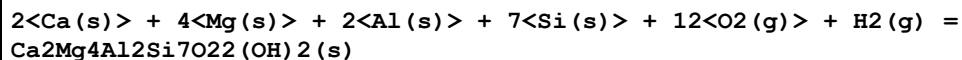
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dH:-2605.4 (5SF) | 4 | ENS   | 6437   |

## Reaction No. 66802



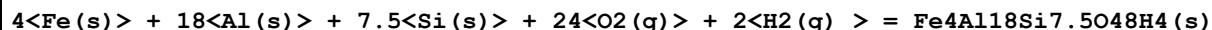
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dH:-2629. (4SF) | 4 | ENS   | 6437   |

## Reaction No. 66803



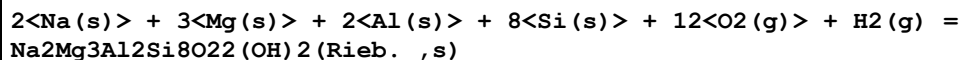
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dH:-2968.5 (5SF) | 4 | ENS   | 6437   |

## Reaction No. 66804



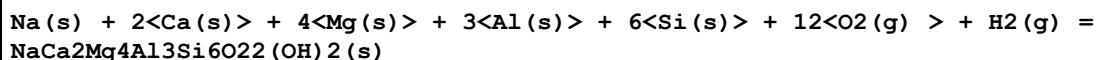
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dH:-5675.2 (5SF) | 4 | ENS   | 6437   |

## Reaction No. 66806



| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dH:-2650. (4SF) | 4 | ENS   | 6437   |

## Reaction No. 66807



| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dH:-3040.1 (5SF) | 4 | ENS   | 6437   |

## Reaction No. 66808



| K(s) + 2<Fe(s)> + 3<Al(s)> + 2<Si(s)> + 6<O2(g)> + H2(g) = KFe2Al3Si2O10(OH)2(s) |    |       |               |                 |   |       |        |
|----------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| No                                                                               | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                | 25 | 0     | Inf. Dilution | dH:-1345.2(5SF) | 4 | ENS   | 6437   |

| Reaction No. 66809                                                   |     |       |               |                  |   |       |        |
|----------------------------------------------------------------------|-----|-------|---------------|------------------|---|-------|--------|
| Ca(s) + 4<Al(s)> + 2<Si(s)> + 6<O2(g)> + H2(g) = CaAl4Si2O10(OH)2(s) |     |       |               |                  |   |       |        |
| No                                                                   | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                    | 25  | 0     | Inf. Dilution | lgK:1026.(4SF)   | 1 | ENS   | 6494   |
| 2                                                                    | 27  | 0     | Inf. Dilution | lgK:1020.(4SF)   | 0 | ENS   | 6494   |
| 3                                                                    | 127 | 0     | Inf. Dilution | lgK:747.8(4SF)   | 0 | ENS   | 6494   |
| 4                                                                    | 227 | 0     | Inf. Dilution | lgK:584.7(4SF)   | 0 | ENS   | 6494   |
| 5                                                                    | 327 | 0     | Inf. Dilution | lgK:476.0(4SF)   | 0 | ENS   | 6494   |
| 6                                                                    | 25  | 0     | Inf. Dilution | dG:-1398.70(6SF) | 1 | EFE   | 6482   |
| 7                                                                    | 25  | 0     | Inf. Dilution | dG:-1400.(4SF)   | 4 | EFE   | 6494   |
| 8                                                                    | 25  | 0     | Inf. Dilution | dH:-1492.5(5SF)  | 4 | ENS   | 6437   |
| 9                                                                    | 25  | 0     | Inf. Dilution | dH:-1490.6(5SF)  | 4 | EFE   | 6482   |
| 10                                                                   | 25  | 0     | Inf. Dilution | dH:-1492.(4SF)   | 1 | MTD   | 6494   |
| 11                                                                   | 127 | 0     | Inf. Dilution | dH:-1493.(4SF)   | 0 | MTD   | 6494   |
| 12                                                                   | 227 | 0     | Inf. Dilution | dH:-1493.(4SF)   | 0 | MTD   | 6494   |
| 13                                                                   | 327 | 0     | Inf. Dilution | dH:-1492.(4SF)   | 0 | MTD   | 6494   |

| Reaction No. 66810                                                                       |    |       |               |                    |   |       |        |
|------------------------------------------------------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| 2<Al(s)> + 2<Ca(s)> + Fe(s) + 0.5<H2(g)> + 6.5<O2(g)> + 3<Si(s)> = Ca2FeAl2Si3O12(OH)(s) |    |       |               |                    |   |       |        |
| No                                                                                       | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                                        | 25 | 0     | GAMSPATH      | dG:-6072.(4SF)kJ   | 1 | CRV   | 12638  |
| 2                                                                                        | 25 | 0     | Inf. Dilution | dH:-1544.5(5SF)    | 3 | ENS   | 6437   |
| 3                                                                                        | 25 | 0     | GAMSPATH      | dH:-6461.90(6SF)kJ | 2 | CRV   | 12638  |
| Note (3): c.f. dG = -6072 & dH = -6462 kJ for 'Ordered Epidote'                          |    |       |               |                    |   |       |        |

| Reaction No. 66812                                                |    |       |               |                 |   |       |        |
|-------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Fe(s) + 2<Al(s)> + Si(s) + 3.5<O2(g)> + H2(g) = FeAl2SiO5(OH)2(s) |    |       |               |                 |   |       |        |
| No                                                                | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                 | 25 | 0     | Inf. Dilution | dH:-767.54(5SF) | 4 | ENS   | 6437   |

| Reaction No. 66815                               |  |  |  |  |  |  |  |
|--------------------------------------------------|--|--|--|--|--|--|--|
| K(s) + Al(s) + 2<Si(s)> + 3<O2(g)> = KAlSi2O6(s) |  |  |  |  |  |  |  |

| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:503.1 (4SF) | 4 | ENS   | 6422   |
| 2  | 25  | 0     | Inf. Dilution | lgK:503.7 (4SF) | 4 | ENS   | 6494   |
| 3  | 27  | 0     | Inf. Dilution | lgK:499.8 (4SF) | 0 | ENS   | 6422   |
| 4  | 27  | 0     | Inf. Dilution | lgK:500.4 (4SF) | 0 | ENS   | 6494   |
| 5  | 127 | 0     | Inf. Dilution | lgK:367.6 (4SF) | 3 | ENS   | 6422   |
| 6  | 127 | 0     | Inf. Dilution | lgK:368.1 (4SF) | 3 | ENS   | 6494   |
| 7  | 227 | 0     | Inf. Dilution | lgK:288.2 (4SF) | 2 | ENS   | 6422   |
| 8  | 227 | 0     | Inf. Dilution | lgK:288.7 (4SF) | 2 | ENS   | 6494   |
| 9  | 327 | 0     | Inf. Dilution | lgK:235.3 (4SF) | 0 | ENS   | 6422   |
| 10 | 327 | 0     | Inf. Dilution | lgK:235.7 (4SF) | 0 | ENS   | 6494   |
| 11 | 25  | 0     | Inf. Dilution | dG:-686.3 (4SF) | 4 | EFE   | 6422   |
| 12 | 25  | 0     | Inf. Dilution | dG:-687.2 (4SF) | 4 | EFE   | 6494   |
| 13 | 25  | 0     | Inf. Dilution | dH:-725.2 (4SF) | 4 | MTD   | 6422   |
| 14 | 25  | 0     | Inf. Dilution | dH:-722. (3SF)  | 4 | ENS   | 6437   |
| 15 | 25  | 0     | Inf. Dilution | dH:-726.1 (4SF) | 4 | MTD   | 6494   |
| 16 | 127 | 0     | Inf. Dilution | dH:-726.2 (4SF) | 3 | MTD   | 6422   |
| 17 | 127 | 0     | Inf. Dilution | dH:-726.9 (4SF) | 3 | MTD   | 6494   |
| 18 | 227 | 0     | Inf. Dilution | dH:-726.6 (4SF) | 2 | MTD   | 6422   |
| 19 | 227 | 0     | Inf. Dilution | dH:-726.8 (4SF) | 2 | MTD   | 6494   |
| 20 | 327 | 0     | Inf. Dilution | dH:-726.8 (4SF) | 0 | MTD   | 6422   |
| 21 | 327 | 0     | Inf. Dilution | dH:-726.4 (4SF) | 0 | MTD   | 6494   |

| Reaction No. 66816                                                                                                                                                      |    |       |               |                      |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------------|---|-------|--------|
| $2\text{Al(s)} + 4\text{H}_2\text{(g)} + 5\text{Mg(s)} + 9\text{O}_2\text{(g)} + 3\text{Si(s)} = \text{Mg}_5\text{Al}_2\text{Si}_3\text{O}_{10}(\text{OH})_8\text{(s)}$ |    |       |               |                      |   |       |        |
| No                                                                                                                                                                      | t  | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
| 1                                                                                                                                                                       | 25 | 0     | Inf. Dilution | lgK:1446.3 (5SF)     | 3 | CRV   | 6494   |
| 2                                                                                                                                                                       | 25 | 0     | Inf. Dilution | dG:-1971.9 (5SF)     | 3 | EFE   | 6482   |
| 3                                                                                                                                                                       | 25 | 0     | Inf. Dilution | dG:-8263.40 (6SF) kJ | 1 | CRV   | 25435  |
| 4                                                                                                                                                                       | 25 | 0     | GAMSPATH      | dG:-8186. (4SF) kJ   | 0 | CRV   | 12638  |
| Note (4): Low wgt for internal consistency                                                                                                                              |    |       |               |                      |   |       |        |
| 5                                                                                                                                                                       | 25 | 0     | Inf. Dilution | dH:-2133.9 (5SF)     | 1 | ENS   | 6437   |
| 6                                                                                                                                                                       | 25 | 0     | Inf. Dilution | dH:-2129.4 (5SF)     | 3 | EFE   | 6482   |
| 7                                                                                                                                                                       | 25 | 0     | Inf. Dilution | dH:-8929.90 (6SF) kJ | 1 | CRV   | 25435  |
| 8                                                                                                                                                                       | 25 | 0     | GAMSPATH      | dH:-8841.6 (5SF) kJ  | 0 | CRV   | 12638  |

| Reaction No. 66817                                                                   |     |       |               |                     |   |       |        |
|--------------------------------------------------------------------------------------|-----|-------|---------------|---------------------|---|-------|--------|
| 3<Al(s)> + 2<Ca(s)> + 0.5<H2(g)> + 6.5<O2(g)> + 3<Si(s)> = Ca2Al3Si3O12(OH)(ortho,s) |     |       |               |                     |   |       |        |
| No                                                                                   | t   | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                                                    | 25  | 0     | Inf. Dilution | lgK:1140.(4SF)      | 1 | EDH   |        |
| 2                                                                                    | 25  | 0     | Inf. Dilution | lgK:1140.(4SF)      | 4 | ENS   | 6494   |
| 3                                                                                    | 27  | 0     | Inf. Dilution | lgK:1132.(4SF)      | 0 | ENS   | 6494   |
| 4                                                                                    | 127 | 0     | Inf. Dilution | lgK:831.7(4SF)      | 2 | ENS   | 6494   |
| 5                                                                                    | 227 | 0     | Inf. Dilution | lgK:651.4(4SF)      | 1 | ENS   | 6494   |
| 6                                                                                    | 327 | 0     | Inf. Dilution | lgK:531.3(4SF)      | 0 | ENS   | 6494   |
| 7                                                                                    | 25  | 0     | Inf. Dilution | dG:-1552.1(5SF)     | 0 | EFE   | 6482   |
| 8                                                                                    | 25  | 0     | Inf. Dilution | dG:-1555.(4SF)      | 4 | EFE   | 6494   |
| 9                                                                                    | 25  | 0     | GAMSPATH      | dG:-6502.(4SF) kJ   | 0 | CRV   | 12638  |
| 10                                                                                   | 25  | 0     | Inf. Dilution | dH:-1648.9(5SF)     | 2 | ENS   | 6437   |
| 11                                                                                   | 25  | 0     | Inf. Dilution | dH:-1646.6(5SF)     | 0 | EFE   | 6482   |
| 12                                                                                   | 25  | 0     | Inf. Dilution | dH:-1649.(4SF)      | 4 | MTD   | 6494   |
| 13                                                                                   | 25  | 0     | GAMSPATH      | dH:-6879.00(6SF) kJ | 0 | CRV   | 12638  |
| 14                                                                                   | 127 | 0     | Inf. Dilution | dH:-1650.(4SF)      | 1 | MTD   | 6494   |
| 15                                                                                   | 227 | 0     | Inf. Dilution | dH:-1649.(4SF)      | 0 | MTD   | 6494   |
| 16                                                                                   | 327 | 0     | Inf. Dilution | dH:-1649.(4SF)      | 0 | MTD   | 6494   |

| Reaction No. 66819                                                       |     |       |               |                   |   |       |        |
|--------------------------------------------------------------------------|-----|-------|---------------|-------------------|---|-------|--------|
| 2<Al(s)> + 2<Ca(s)> + H2(g) + 6<O2(g)> + 3<Si(s)> = Ca2Al2Si3O10(OH)2(s) |     |       |               |                   |   |       |        |
| No                                                                       | t   | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                        | 25  | 0     | Inf. Dilution | lgK:1020.(4SF)    | 3 | ENS   | 6494   |
| 2                                                                        | 27  | 0     | Inf. Dilution | lgK:1014.(4SF)    | 0 | ENS   | 6494   |
| 3                                                                        | 127 | 0     | Inf. Dilution | lgK:743.7(4SF)    | 2 | ENS   | 6494   |
| 4                                                                        | 227 | 0     | Inf. Dilution | lgK:581.7(4SF)    | 1 | ENS   | 6494   |
| 5                                                                        | 327 | 0     | Inf. Dilution | lgK:473.8(4SF)    | 0 | ENS   | 6494   |
| 6                                                                        | 25  | 0     | Inf. Dilution | dG:-1391.1(5SF)   | 5 | EFE   | 6482   |
| 7                                                                        | 25  | 0     | Inf. Dilution | dG:-1392.(4SF)    | 3 | EFE   | 6494   |
| 8                                                                        | 25  | 0     | GAMSPATH      | dG:-5813.(4SF) kJ | 2 | CRV   | 12638  |
| 9                                                                        | 25  | 0     | Inf. Dilution | dH:-1481.8(5SF)   | 3 | ENS   | 6437   |
| 10                                                                       | 25  | 0     | Inf. Dilution | dH:-1481.5(5SF)   | 5 | EFE   | 6482   |
| 11                                                                       | 25  | 0     | Inf. Dilution | dH:-1483.(4SF)    | 3 | MTD   | 6494   |
| 12                                                                       | 127 | 0     | Inf. Dilution | dH:-1483.(4SF)    | 1 | MTD   | 6494   |
| 13                                                                       | 227 | 0     | Inf. Dilution | dH:-1482.(4SF)    | 0 | MTD   | 6494   |

|    |     |   |               |                |   |     |      |
|----|-----|---|---------------|----------------|---|-----|------|
| 14 | 327 | 0 | Inf. Dilution | dH:-1481.(4SF) | 0 | MTD | 6494 |
|----|-----|---|---------------|----------------|---|-----|------|

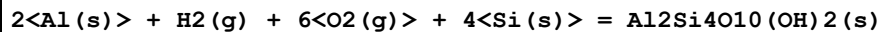
| Reaction No. 66820                                   |     |       |               |                 |   |       |        |
|------------------------------------------------------|-----|-------|---------------|-----------------|---|-------|--------|
| Na(s) + Al(s) + Si(s) + 2<O2(g)> = NaAlSiO4(Neph.,s) |     |       |               |                 |   |       |        |
| No                                                   | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                    | 25  | 0     | Inf. Dilution | lgK:347.2(4SF)  | 1 | ELC   |        |
| 2                                                    | 25  | 0     | Inf. Dilution | lgK:346.9(4SF)  | 3 | ENS   | 6422   |
| 3                                                    | 25  | 0     | Inf. Dilution | lgK:346.1(4SF)  | 2 | ENS   | 6494   |
| 4                                                    | 27  | 0     | Inf. Dilution | lgK:344.6(4SF)  | 0 | ENS   | 6422   |
| 5                                                    | 27  | 0     | Inf. Dilution | lgK:343.9(4SF)  | 0 | ENS   | 6494   |
| 6                                                    | 127 | 0     | Inf. Dilution | lgK:253.4(4SF)  | 1 | ENS   | 6422   |
| 7                                                    | 127 | 0     | Inf. Dilution | lgK:252.8(4SF)  | 1 | ENS   | 6494   |
| 8                                                    | 194 | 0     | Inf. Dilution | lgK:214.1(4SF)  | 0 | ENS   | 6422   |
| 9                                                    | 227 | 0     | Inf. Dilution | lgK:198.4(4SF)  | 0 | ENS   | 6494   |
| 10                                                   | 327 | 0     | Inf. Dilution | lgK:161.8(4SF)  | 0 | ENS   | 6494   |
| 11                                                   | 25  | 0     | Inf. Dilution | dG:-473.2(4SF)  | 0 | EFE   | 6422   |
| 12                                                   | 25  | 0     | Inf. Dilution | dG:-472.2(4SF)  | 0 | EFE   | 6494   |
| 13                                                   | 25  | 0     | Inf. Dilution | dG:-475.50(5SF) | 0 | CRV   | 10174  |
| 14                                                   | 50  | 0     | Inf. Dilution | dG:-476.27(5SF) | 0 | CRV   | 10174  |
| 15                                                   | 75  | 0     | Inf. Dilution | dG:-477.11(5SF) | 0 | CRV   | 10174  |
| 16                                                   | 100 | 0     | Inf. Dilution | dG:-478.00(5SF) | 0 | CRV   | 10174  |
| 17                                                   | 125 | 0     | Inf. Dilution | dG:-478.94(5SF) | 0 | CRV   | 10174  |
| 18                                                   | 150 | 0     | Inf. Dilution | dG:-479.94(5SF) | 0 | CRV   | 10174  |
| 19                                                   | 175 | 0     | Inf. Dilution | dG:-480.98(5SF) | 0 | CRV   | 10174  |
| 20                                                   | 200 | 0     | Inf. Dilution | dG:-482.07(5SF) | 0 | CRV   | 10174  |
| 21                                                   | 225 | 0     | Inf. Dilution | dG:-483.21(5SF) | 0 | CRV   | 10174  |
| 22                                                   | 250 | 0     | Inf. Dilution | dG:-484.38(5SF) | 0 | CRV   | 10174  |
| 23                                                   | 300 | 0     | Inf. Dilution | dG:-486.84(5SF) | 0 | CRV   | 10174  |
| 24                                                   | 25  | 0     | Inf. Dilution | dH:-500.6(4SF)  | 0 | MTD   | 6422   |
| 25                                                   | 25  | 0     | Inf. Dilution | dH:-503.21(5SF) | 0 | ENS   | 6437   |
| 26                                                   | 25  | 0     | Inf. Dilution | dH:-499.6(4SF)  | 0 | MTD   | 6494   |
| 27                                                   | 127 | 0     | Inf. Dilution | dH:-501.4(4SF)  | 0 | MTD   | 6422   |
| 28                                                   | 127 | 0     | Inf. Dilution | dH:-500.4(4SF)  | 0 | MTD   | 6494   |
| 29                                                   | 194 | 0     | Inf. Dilution | dH:-501.1(4SF)  | 0 | MTD   | 6422   |
| 30                                                   | 227 | 0     | Inf. Dilution | dH:-500.6(4SF)  | 0 | MTD   | 6494   |
| 31                                                   | 327 | 0     | Inf. Dilution | dH:-499.9(4SF)  | 0 | MTD   | 6494   |

| Reaction No. 66821                         |     |       |               |                 |   |       |        |
|--------------------------------------------|-----|-------|---------------|-----------------|---|-------|--------|
| Mg(s) + 2<Al(s)> + 2<O2(g)> = MgO.Al2O3(s) |     |       |               |                 |   |       |        |
| No                                         | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                          | 25  | 0     | Inf. Dilution | lgK:381.1(4SF)  | 4 | ENS   | 6422   |
| 2                                          | 25  | 0     | Inf. Dilution | lgK:381.3(4SF)  | 4 | ENS   | 6494   |
| 3                                          | 27  | 0     | Inf. Dilution | lgK:378.6(4SF)  | 4 | ENS   | 6422   |
| 4                                          | 27  | 0     | Inf. Dilution | lgK:378.8(4SF)  | 4 | ENS   | 6494   |
| 5                                          | 127 | 0     | Inf. Dilution | lgK:278.4(4SF)  | 4 | ENS   | 6422   |
| 6                                          | 127 | 0     | Inf. Dilution | lgK:278.7(4SF)  | 4 | ENS   | 6494   |
| 7                                          | 227 | 0     | Inf. Dilution | lgK:218.4(4SF)  | 4 | ENS   | 6422   |
| 8                                          | 227 | 0     | Inf. Dilution | lgK:218.7(4SF)  | 4 | ENS   | 6494   |
| 9                                          | 327 | 0     | Inf. Dilution | lgK:178.3(4SF)  | 4 | ENS   | 6422   |
| 10                                         | 327 | 0     | Inf. Dilution | lgK:178.7(4SF)  | 4 | ENS   | 6494   |
| 11                                         | 25  | 0     | Inf. Dilution | dG:-519.8(4SF)  | 4 | EFE   | 6422   |
| 12                                         | 25  | 0     | Inf. Dilution | dG:-520.20(5SF) | 5 | EFE   | 6482   |
| 13                                         | 25  | 0     | Inf. Dilution | dG:-520.2(4SF)  | 4 | EFE   | 6494   |
| 14                                         | 25  | 0     | Inf. Dilution | dH:-549.7(4SF)  | 4 | MTD   | 6422   |
| 15                                         | 25  | 0     | Inf. Dilution | dH:-551.(3SF)   | 4 | ENS   | 6437   |
| 16                                         | 25  | 0     | Inf. Dilution | dH:-549.79(5SF) | 5 | EFE   | 6482   |
| 17                                         | 25  | 0     | Inf. Dilution | dH:-549.5(4SF)  | 4 | MTD   | 6494   |
| 18                                         | 127 | 0     | Inf. Dilution | dH:-549.8(4SF)  | 4 | MTD   | 6422   |
| 19                                         | 127 | 0     | Inf. Dilution | dH:-549.7(4SF)  | 4 | MTD   | 6494   |
| 20                                         | 227 | 0     | Inf. Dilution | dH:-549.7(4SF)  | 4 | MTD   | 6422   |
| 21                                         | 227 | 0     | Inf. Dilution | dH:-549.6(4SF)  | 4 | MTD   | 6494   |
| 22                                         | 327 | 0     | Inf. Dilution | dH:-549.5(4SF)  | 4 | MTD   | 6422   |
| 23                                         | 327 | 0     | Inf. Dilution | dH:-549.4(4SF)  | 4 | MTD   | 6494   |

| Reaction No. 66822                           |    |       |               |                |   |       |        |
|----------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| K(s) + Al(s) + Si(s) + 2<O2(g)> = KAlSiO4(s) |    |       |               |                |   |       |        |
| No                                           | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                            | 25 | 0     | Inf. Dilution | lgK:351.3(4SF) | 1 | ELC   |        |
| 2                                            | 25 | 0     | Inf. Dilution | lgK:351.3(4SF) | 3 | ENS   | 6422   |
| 3                                            | 25 | 0     | Inf. Dilution | lgK:351.9(4SF) | 3 | ENS   | 6494   |
| 4                                            | 27 | 0     | Inf. Dilution | lgK:349.0(4SF) | 0 | ENS   | 6422   |
| 5                                            | 27 | 0     | Inf. Dilution | lgK:349.6(4SF) | 0 | ENS   | 6494   |

|    |     |   |               |                  |   |     |       |
|----|-----|---|---------------|------------------|---|-----|-------|
| 6  | 127 | 0 | Inf. Dilution | lgK:256.6 (4SF)  | 1 | ENS | 6422  |
| 7  | 127 | 0 | Inf. Dilution | lgK:257.1 (4SF)  | 1 | ENS | 6494  |
| 8  | 227 | 0 | Inf. Dilution | lgK:201.1 (4SF)  | 0 | ENS | 6422  |
| 9  | 227 | 0 | Inf. Dilution | lgK:201.5 (4SF)  | 0 | ENS | 6494  |
| 10 | 327 | 0 | Inf. Dilution | lgK:164.1 (4SF)  | 0 | ENS | 6422  |
| 11 | 327 | 0 | Inf. Dilution | lgK:164.5 (4SF)  | 0 | ENS | 6494  |
| 12 | 25  | 0 | Inf. Dilution | dG:-479.3 (4SF)  | 0 | EFE | 6422  |
| 13 | 25  | 0 | Inf. Dilution | dG:-480.1 (4SF)  | 0 | EFE | 6494  |
| 14 | 25  | 0 | Inf. Dilution | dG:-482.98 (5SF) | 0 | CRV | 10174 |
| 15 | 50  | 0 | Inf. Dilution | dG:-483.81 (5SF) | 0 | CRV | 10174 |
| 16 | 75  | 0 | Inf. Dilution | dG:-484.69 (5SF) | 0 | CRV | 10174 |
| 17 | 100 | 0 | Inf. Dilution | dG:-485.63 (5SF) | 0 | CRV | 10174 |
| 18 | 125 | 0 | Inf. Dilution | dG:-486.62 (5SF) | 0 | CRV | 10174 |
| 19 | 150 | 0 | Inf. Dilution | dG:-487.66 (5SF) | 0 | CRV | 10174 |
| 20 | 175 | 0 | Inf. Dilution | dG:-488.75 (5SF) | 0 | CRV | 10174 |
| 21 | 200 | 0 | Inf. Dilution | dG:-489.89 (5SF) | 0 | CRV | 10174 |
| 22 | 225 | 0 | Inf. Dilution | dG:-491.06 (5SF) | 0 | CRV | 10174 |
| 23 | 250 | 0 | Inf. Dilution | dG:-492.28 (5SF) | 0 | CRV | 10174 |
| 24 | 300 | 0 | Inf. Dilution | dG:-494.81 (5SF) | 0 | CRV | 10174 |
| 25 | 25  | 0 | Inf. Dilution | dH:-507.0 (4SF)  | 0 | MTD | 6422  |
| 26 | 25  | 0 | Inf. Dilution | dH:-505.4 (4SF)  | 0 | ENS | 6437  |
| 27 | 25  | 0 | Inf. Dilution | dH:-507.8 (4SF)  | 0 | MTD | 6494  |
| 28 | 127 | 0 | Inf. Dilution | dH:-507.9 (4SF)  | 0 | MTD | 6422  |
| 29 | 127 | 0 | Inf. Dilution | dH:-508.6 (4SF)  | 0 | MTD | 6494  |
| 30 | 227 | 0 | Inf. Dilution | dH:-508.2 (4SF)  | 0 | MTD | 6422  |
| 31 | 227 | 0 | Inf. Dilution | dH:-508.5 (4SF)  | 0 | MTD | 6494  |
| 32 | 327 | 0 | Inf. Dilution | dH:-508.4 (4SF)  | 0 | MTD | 6422  |
| 33 | 327 | 0 | Inf. Dilution | dH:-508.2 (4SF)  | 0 | MTD | 6494  |

## Reaction No. 66823



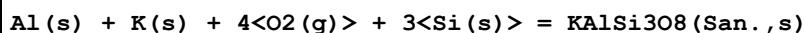
| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:853.0 (4SF) | 0 | ELC   |        |
| 2  | 25  | 0     | Inf. Dilution | lgK:922.6 (4SF) | 2 | ENS   | 6494   |
| 3  | 27  | 0     | Inf. Dilution | lgK:916.5 (4SF) | 0 | ENS   | 6494   |
| 4  | 127 | 0     | Inf. Dilution | lgK:670.9 (4SF) | 0 | ENS   | 6494   |

|    |     |   |               |                      |   |     |       |
|----|-----|---|---------------|----------------------|---|-----|-------|
| 5  | 227 | 0 | Inf. Dilution | lgK:523.6 (4SF)      | 0 | ENS | 6494  |
| 6  | 327 | 0 | Inf. Dilution | lgK:425.4 (4SF)      | 0 | ENS | 6494  |
| 7  | 25  | 0 | Inf. Dilution | dG:-1258.8 (5SF)     | 0 | EFE | 6482  |
| 8  | 25  | 0 | Inf. Dilution | dG:-1259. (4SF)      | 2 | EFE | 6494  |
| 9  | 25  | 0 | Inf. Dilution | dG:-1259.00 (6SF)    | 0 | CRV | 10174 |
| 10 | 25  | 0 | Inf. Dilution | dG:-1258.4 (5SF)     | 3 | CRV | 29553 |
| 11 | 25  | 0 | GAMSPATH      | dG:-5253. (4SF) kJ   | 0 | CRV | 12638 |
| 12 | 50  | 0 | Inf. Dilution | dG:-1260.50 (6SF)    | 0 | CRV | 10174 |
| 13 | 75  | 0 | Inf. Dilution | dG:-1262.20 (6SF)    | 0 | CRV | 10174 |
| 14 | 100 | 0 | Inf. Dilution | dG:-1264.00 (6SF)    | 0 | CRV | 10174 |
| 15 | 125 | 0 | Inf. Dilution | dG:-1265.90 (6SF)    | 0 | CRV | 10174 |
| 16 | 150 | 0 | Inf. Dilution | dG:-1267.90 (6SF)    | 0 | CRV | 10174 |
| 17 | 175 | 0 | Inf. Dilution | dG:-1270.10 (6SF)    | 0 | CRV | 10174 |
| 18 | 200 | 0 | Inf. Dilution | dG:-1272.40 (6SF)    | 0 | CRV | 10174 |
| 19 | 225 | 0 | Inf. Dilution | dG:-1274.80 (6SF)    | 0 | CRV | 10174 |
| 20 | 250 | 0 | Inf. Dilution | dG:-1277.30 (6SF)    | 0 | CRV | 10174 |
| 21 | 300 | 0 | Inf. Dilution | dG:-1282.50 (6SF)    | 0 | CRV | 10174 |
| 22 | 25  | 0 | Inf. Dilution | dH:-1348.2 (5SF)     | 0 | ENS | 6437  |
| 23 | 25  | 0 | Inf. Dilution | dH:-1348.2 (5SF)     | 0 | EFE | 6482  |
| 24 | 25  | 0 | Inf. Dilution | dH:-1348. (4SF)      | 0 | MTD | 6494  |
| 25 | 25  | 0 | Inf. Dilution | dH:-1347.7 (5SF)     | 3 | CRV | 29553 |
| 26 | 25  | 0 | GAMSPATH      | dH:-5626.60 (6SF) kJ | 0 | CRV | 12638 |
| 27 | 127 | 0 | Inf. Dilution | dH:-1348. (4SF)      | 0 | MTD | 6494  |
| 28 | 227 | 0 | Inf. Dilution | dH:-1348. (4SF)      | 0 | MTD | 6494  |
| 29 | 327 | 0 | Inf. Dilution | dH:-1347. (4SF)      | 0 | MTD | 6494  |

| Reaction No. 66826                                                 |     |       |               |                 |   |       |        |
|--------------------------------------------------------------------|-----|-------|---------------|-----------------|---|-------|--------|
| 3<Al(s)> + H2(g) + K(s) + 6<O2(g)> + 3<Si(s)> = KAl3Si3O10(OH)2(s) |     |       |               |                 |   |       |        |
| No                                                                 | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                  | 25  | 0     | Inf. Dilution | lgK:983. (3SF)  | 1 | ELC   |        |
| 2                                                                  | 25  | 0     | Inf. Dilution | lgK:982.6 (4SF) | 0 | ENS   | 6422   |
| 3                                                                  | 25  | 0     | Inf. Dilution | lgK:982.6 (4SF) | 0 | ENS   | 6494   |
| 4                                                                  | 27  | 0     | Inf. Dilution | lgK:976.1 (4SF) | 0 | ENS   | 6422   |
| 5                                                                  | 27  | 0     | Inf. Dilution | lgK:976.1 (4SF) | 0 | ENS   | 6494   |
| 6                                                                  | 127 | 0     | Inf. Dilution | lgK:715.5 (4SF) | 0 | ENS   | 6422   |
| 7                                                                  | 127 | 0     | Inf. Dilution | lgK:715.2 (4SF) | 0 | ENS   | 6494   |

|                                             |     |   |               |                      |   |     |       |
|---------------------------------------------|-----|---|---------------|----------------------|---|-----|-------|
| 8                                           | 227 | 0 | Inf. Dilution | lgK:559.1 (4SF)      | 0 | ENS | 6422  |
| 9                                           | 227 | 0 | Inf. Dilution | lgK:558.7 (4SF)      | 0 | ENS | 6494  |
| 10                                          | 327 | 0 | Inf. Dilution | lgK:454.8 (4SF)      | 0 | ENS | 6422  |
| 11                                          | 327 | 0 | Inf. Dilution | lgK:454.4 (4SF)      | 0 | ENS | 6494  |
| 12                                          | 25  | 0 | Inf. Dilution | dG:-1340. (4SF)      | 0 | EFE | 6422  |
| 13                                          | 25  | 0 | Inf. Dilution | dG:-1337.6 (5SF)     | 0 | EFE | 6482  |
| 14                                          | 25  | 0 | Inf. Dilution | dG:-1340. (4SF)      | 0 | EFE | 6494  |
| 15                                          | 25  | 0 | Inf. Dilution | dG:-1340.30 (6SF)    | 0 | CRV | 10174 |
| 16                                          | 25  | 0 | Inf. Dilution | dG:-1339.0 (5SF)     | 5 | CRV | 29553 |
| 17                                          | 25  | 0 | GAMSPATH      | dG:-5588. (4SF) kJ   | 0 | CRV | 12638 |
| 18                                          | 50  | 0 | Inf. Dilution | dG:-1342.10 (6SF)    | 0 | CRV | 10174 |
| 19                                          | 75  | 0 | Inf. Dilution | dG:-1344.10 (6SF)    | 0 | CRV | 10174 |
| 20                                          | 100 | 0 | Inf. Dilution | dG:-1346.20 (6SF)    | 0 | CRV | 10174 |
| 21                                          | 125 | 0 | Inf. Dilution | dG:-1348.50 (6SF)    | 0 | CRV | 10174 |
| 22                                          | 150 | 0 | Inf. Dilution | dG:-1350.90 (6SF)    | 0 | CRV | 10174 |
| 23                                          | 175 | 0 | Inf. Dilution | dG:-1353.40 (6SF)    | 0 | CRV | 10174 |
| 24                                          | 200 | 0 | Inf. Dilution | dG:-1356.00 (6SF)    | 0 | CRV | 10174 |
| 25                                          | 225 | 0 | Inf. Dilution | dG:-1358.80 (6SF)    | 0 | CRV | 10174 |
| 26                                          | 250 | 0 | Inf. Dilution | dG:-1361.70 (6SF)    | 0 | CRV | 10174 |
| 27                                          | 300 | 0 | Inf. Dilution | dG:-1367.80 (6SF)    | 0 | CRV | 10174 |
| 28                                          | 25  | 0 | Inf. Dilution | dH:-1430. (4SF)      | 0 | MTD | 6422  |
| 29                                          | 25  | 0 | Inf. Dilution | dH:-1428.4 (5SF)     | 0 | ENS | 6437  |
| 30                                          | 25  | 0 | Inf. Dilution | dH:-1428.5 (5SF)     | 0 | EFE | 6482  |
| 31                                          | 25  | 0 | Inf. Dilution | dH:-1432. (4SF)      | 0 | MTD | 6494  |
| 32                                          | 25  | 0 | Inf. Dilution | dH:-1429.9 (5SF)     | 5 | CRV | 29553 |
| 33                                          | 25  | 0 | GAMSPATH      | dH:-5969.40 (6SF) kJ | 0 | CRV | 12638 |
| 34                                          | 127 | 0 | Inf. Dilution | dH:-1431. (4SF)      | 0 | MTD | 6422  |
| Note (34): Low Wgt for internal consistency |     |   |               |                      |   |     |       |
| 35                                          | 127 | 0 | Inf. Dilution | dH:-1433. (4SF)      | 0 | MTD | 6494  |
| 36                                          | 227 | 0 | Inf. Dilution | dH:-1431. (4SF)      | 0 | MTD | 6422  |
| 37                                          | 227 | 0 | Inf. Dilution | dH:-1432. (4SF)      | 0 | MTD | 6494  |
| 38                                          | 327 | 0 | Inf. Dilution | dH:-1430. (4SF)      | 0 | MTD | 6422  |
| 39                                          | 327 | 0 | Inf. Dilution | dH:-1431. (4SF)      | 0 | MTD | 6494  |

Reaction No. 66828





| No | t   | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
|----|-----|-------|---------------|----------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:657.4 (4SF)      | 1 | ELC   |        |
| 2  | 25  | 0     | Inf. Dilution | lgK:655.2 (4SF)      | 0 | ENS   | 6422   |
| 3  | 27  | 0     | Inf. Dilution | lgK:650.9 (4SF)      | 0 | ENS   | 6422   |
| 4  | 127 | 0     | Inf. Dilution | lgK:478.5 (4SF)      | 0 | ENS   | 6422   |
| 5  | 227 | 0     | Inf. Dilution | lgK:375.0 (4SF)      | 0 | ENS   | 6422   |
| 6  | 327 | 0     | Inf. Dilution | lgK:306.1 (4SF)      | 0 | ENS   | 6422   |
| 7  | 25  | 0     | Inf. Dilution | dG:-893.9 (4SF)      | 0 | EFE   | 6422   |
| 8  | 25  | 0     | Inf. Dilution | dG:-893.60 (5SF)     | 0 | EFE   | 6482   |
| 9  | 25  | 0     | Inf. Dilution | dG:-3746.00 (6SF) kJ | 0 | EFE   | 8798   |
| 10 | 25  | 0     | Inf. Dilution | dG:-895.43 (5SF)     | 0 | CRV   | 10174  |
| 11 | 25  | 0     | GAMSPATH      | dG:-3739. (4SF) kJ   | 0 | CRV   | 12638  |
| 12 | 50  | 0     | Inf. Dilution | dG:-896.85 (5SF)     | 0 | CRV   | 10174  |
| 13 | 75  | 0     | Inf. Dilution | dG:-898.36 (5SF)     | 0 | CRV   | 10174  |
| 14 | 100 | 0     | Inf. Dilution | dG:-899.97 (5SF)     | 0 | CRV   | 10174  |
| 15 | 125 | 0     | Inf. Dilution | dG:-901.67 (5SF)     | 0 | CRV   | 10174  |
| 16 | 150 | 0     | Inf. Dilution | dG:-903.46 (5SF)     | 0 | CRV   | 10174  |
| 17 | 175 | 0     | Inf. Dilution | dG:-905.33 (5SF)     | 0 | CRV   | 10174  |
| 18 | 200 | 0     | Inf. Dilution | dG:-907.29 (5SF)     | 0 | CRV   | 10174  |
| 19 | 225 | 0     | Inf. Dilution | dG:-909.31 (5SF)     | 0 | CRV   | 10174  |
| 20 | 250 | 0     | Inf. Dilution | dG:-911.41 (5SF)     | 0 | CRV   | 10174  |
| 21 | 300 | 0     | Inf. Dilution | dG:-915.78 (5SF)     | 0 | CRV   | 10174  |
| 22 | 25  | 0     | Inf. Dilution | dH:-946.4 (4SF)      | 0 | MTD   | 6422   |
| 23 | 25  | 0     | Inf. Dilution | dH:-946.4 (4SF)      | 0 | ENS   | 6437   |
| 24 | 25  | 0     | Inf. Dilution | dH:-945.91 (5SF)     | 2 | EFE   | 6482   |
| 25 | 25  | 0     | Inf. Dilution | dH:-3965.70 (6SF) kJ | 2 | EFE   | 8798   |
| 26 | 25  | 0     | GAMSPATH      | dH:-3960.30 (6SF) kJ | 0 | CRV   | 12638  |
| 27 | 127 | 0     | Inf. Dilution | dH:-947.1 (4SF)      | 0 | MTD   | 6422   |
| 28 | 227 | 0     | Inf. Dilution | dH:-946.6 (4SF)      | 0 | MTD   | 6422   |
| 29 | 327 | 0     | Inf. Dilution | dH:-945.7 (4SF)      | 0 | MTD   | 6422   |

## Reaction No. 66830

$$\text{Na(s)} + \text{Al(s)} + 2\text{Si(s)} + 3\text{O}_2\text{(g)} = \text{NaAlSi}_2\text{O}_6\text{(Jade.,s)}$$

| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:499.6 (4SF) | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:500.0 (4SF) | 3 | ENS   | 6422   |

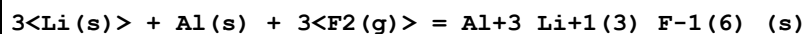
|    |     |   |               |                  |   |     |       |
|----|-----|---|---------------|------------------|---|-----|-------|
| 3  | 25  | 0 | Inf. Dilution | lgK:499.4 (4SF)  | 3 | ENS | 6494  |
| 4  | 27  | 0 | Inf. Dilution | lgK:496.7 (4SF)  | 0 | ENS | 6422  |
| 5  | 27  | 0 | Inf. Dilution | lgK:496.1 (4SF)  | 0 | ENS | 6494  |
| 6  | 127 | 0 | Inf. Dilution | lgK:364.7 (4SF)  | 1 | ENS | 6422  |
| 7  | 127 | 0 | Inf. Dilution | lgK:364.2 (4SF)  | 1 | ENS | 6494  |
| 8  | 227 | 0 | Inf. Dilution | lgK:285.4 (4SF)  | 0 | ENS | 6422  |
| 9  | 227 | 0 | Inf. Dilution | lgK:285.0 (4SF)  | 0 | ENS | 6494  |
| 10 | 327 | 0 | Inf. Dilution | lgK:232.5 (4SF)  | 0 | ENS | 6422  |
| 11 | 327 | 0 | Inf. Dilution | lgK:232.2 (4SF)  | 0 | ENS | 6494  |
| 12 | 25  | 0 | Inf. Dilution | dG:-682.1 (4SF)  | 0 | EFE | 6422  |
| 13 | 25  | 0 | Inf. Dilution | dG:-680.33 (5SF) | 0 | EFE | 6482  |
| 14 | 25  | 0 | Inf. Dilution | dG:-681.3 (4SF)  | 0 | EFE | 6494  |
| 15 | 25  | 0 | Inf. Dilution | dG:-682.07 (5SF) | 0 | CRV | 10174 |
| 16 | 50  | 0 | Inf. Dilution | dG:-682.91 (5SF) | 0 | CRV | 10174 |
| 17 | 75  | 0 | Inf. Dilution | dG:-683.82 (5SF) | 0 | CRV | 10174 |
| 18 | 100 | 0 | Inf. Dilution | dG:-684.82 (5SF) | 0 | CRV | 10174 |
| 19 | 125 | 0 | Inf. Dilution | dG:-685.88 (5SF) | 0 | CRV | 10174 |
| 20 | 150 | 0 | Inf. Dilution | dG:-687.01 (5SF) | 0 | CRV | 10174 |
| 21 | 175 | 0 | Inf. Dilution | dG:-688.21 (5SF) | 0 | CRV | 10174 |
| 22 | 200 | 0 | Inf. Dilution | dG:-689.47 (5SF) | 0 | CRV | 10174 |
| 23 | 225 | 0 | Inf. Dilution | dG:-690.79 (5SF) | 0 | CRV | 10174 |
| 24 | 250 | 0 | Inf. Dilution | dG:-692.17 (5SF) | 0 | CRV | 10174 |
| 25 | 300 | 0 | Inf. Dilution | dG:-695.07 (5SF) | 0 | CRV | 10174 |
| 26 | 25  | 0 | Inf. Dilution | dH:-725.8 (4SF)  | 0 | MTD | 6422  |
| 27 | 25  | 0 | Inf. Dilution | dH:-724. (3SF)   | 0 | ENS | 6437  |
| 28 | 25  | 0 | Inf. Dilution | dH:-723.02 (5SF) | 0 | EFE | 6482  |
| 29 | 25  | 0 | Inf. Dilution | dH:-724.0 (4SF)  | 0 | MTD | 6494  |
| 30 | 127 | 0 | Inf. Dilution | dH:-725.7 (4SF)  | 0 | MTD | 6422  |
| 31 | 127 | 0 | Inf. Dilution | dH:-724.9 (4SF)  | 0 | MTD | 6494  |
| 32 | 227 | 0 | Inf. Dilution | dH:-725.6 (4SF)  | 0 | MTD | 6422  |
| 33 | 227 | 0 | Inf. Dilution | dH:-724.8 (4SF)  | 0 | MTD | 6494  |
| 34 | 327 | 0 | Inf. Dilution | dH:-725.4 (4SF)  | 0 | MTD | 6422  |
| 35 | 327 | 0 | Inf. Dilution | dH:-724.6 (4SF)  | 0 | MTD | 6494  |

Reaction No. 66901

$$\text{Al}+3\text{OH-1(3)}_{\text{(Hydrag.,s)}} = \text{Al}+3 + 3\text{OH-1}$$

| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-36.30(4SF) | 3 | ENS   | 773    |

## Reaction No. 66932



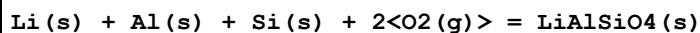
| No | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|-----|-------|---------------|----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:564.8(4SF) | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:561.1(4SF) | 4 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:413.9(4SF) | 4 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:325.5(4SF) | 4 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:266.5(4SF) | 4 | ENS   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dG:-770.5(4SF) | 4 | EFE   | 6422   |
| 7  | 25  | 0     | Inf. Dilution | dH:-808.7(4SF) | 4 | MTD   | 6422   |
| 8  | 127 | 0     | Inf. Dilution | dH:-808.3(4SF) | 4 | MTD   | 6422   |
| 9  | 227 | 0     | Inf. Dilution | dH:-809.9(4SF) | 4 | MTD   | 6422   |
| 10 | 327 | 0     | Inf. Dilution | dH:-809.0(4SF) | 4 | MTD   | 6422   |

## Reaction No. 66947



| No | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|-----|-------|---------------|----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:197.3(4SF) | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:196.0(4SF) | 4 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:144.3(4SF) | 4 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:113.2(4SF) | 4 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:92.43(4SF) | 4 | ENS   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dG:-269.2(4SF) | 4 | EFE   | 6422   |
| 7  | 25  | 0     | Inf. Dilution | dH:-284.1(4SF) | 4 | MTD   | 6422   |
| 8  | 127 | 0     | Inf. Dilution | dH:-284.2(4SF) | 4 | MTD   | 6422   |
| 9  | 227 | 0     | Inf. Dilution | dH:-285.0(4SF) | 4 | MTD   | 6422   |
| 10 | 327 | 0     | Inf. Dilution | dH:-284.9(4SF) | 4 | MTD   | 6422   |

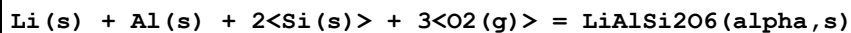
## Reaction No. 66948



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:352.2(4SF) | 4 | ENS   | 6422   |
| 2  | 25 | 0     | Inf. Dilution | lgK:352.0(4SF) | 4 | ENS   | 6494   |
| 3  | 27 | 0     | Inf. Dilution | lgK:349.9(4SF) | 0 | ENS   | 6422   |

|    |     |   |               |                 |   |     |      |
|----|-----|---|---------------|-----------------|---|-----|------|
| 4  | 27  | 0 | Inf. Dilution | lgK:349.7 (4SF) | 0 | ENS | 6494 |
| 5  | 127 | 0 | Inf. Dilution | lgK:257.4 (4SF) | 3 | ENS | 6422 |
| 6  | 127 | 0 | Inf. Dilution | lgK:257.3 (4SF) | 3 | ENS | 6494 |
| 7  | 227 | 0 | Inf. Dilution | lgK:201.9 (4SF) | 2 | ENS | 6422 |
| 8  | 227 | 0 | Inf. Dilution | lgK:201.8 (4SF) | 2 | ENS | 6494 |
| 9  | 327 | 0 | Inf. Dilution | lgK:164.8 (4SF) | 0 | ENS | 6422 |
| 10 | 327 | 0 | Inf. Dilution | lgK:164.8 (4SF) | 0 | ENS | 6494 |
| 11 | 25  | 0 | Inf. Dilution | dG:-480.4 (4SF) | 4 | EFE | 6422 |
| 12 | 25  | 0 | Inf. Dilution | dG:-480.2 (4SF) | 4 | EFE | 6494 |
| 13 | 25  | 0 | Inf. Dilution | dH:-507.7 (4SF) | 4 | MTD | 6422 |
| 14 | 25  | 0 | Inf. Dilution | dH:-507.5 (4SF) | 4 | MTD | 6494 |
| 15 | 127 | 0 | Inf. Dilution | dH:-507.8 (4SF) | 3 | MTD | 6422 |
| 16 | 127 | 0 | Inf. Dilution | dH:-507.6 (4SF) | 3 | MTD | 6494 |
| 17 | 227 | 0 | Inf. Dilution | dH:-508.4 (4SF) | 2 | MTD | 6422 |
| 18 | 227 | 0 | Inf. Dilution | dH:-508.2 (4SF) | 2 | MTD | 6494 |
| 19 | 327 | 0 | Inf. Dilution | dH:-508.1 (4SF) | 0 | MTD | 6422 |
| 20 | 327 | 0 | Inf. Dilution | dH:-508.0 (4SF) | 0 | MTD | 6494 |

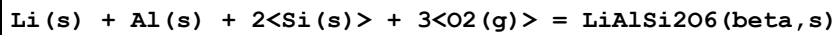
## Reaction No. 66949



| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:504.8 (4SF) | 4 | ENS   | 6422   |
| 2  | 25  | 0     | Inf. Dilution | lgK:504.6 (4SF) | 4 | ENS   | 6494   |
| 3  | 27  | 0     | Inf. Dilution | lgK:501.5 (4SF) | 0 | ENS   | 6422   |
| 4  | 27  | 0     | Inf. Dilution | lgK:501.3 (4SF) | 0 | ENS   | 6494   |
| 5  | 127 | 0     | Inf. Dilution | lgK:368.5 (4SF) | 3 | ENS   | 6422   |
| 6  | 127 | 0     | Inf. Dilution | lgK:368.4 (4SF) | 3 | ENS   | 6494   |
| 7  | 227 | 0     | Inf. Dilution | lgK:288.7 (4SF) | 2 | ENS   | 6422   |
| 8  | 227 | 0     | Inf. Dilution | lgK:288.6 (4SF) | 2 | ENS   | 6494   |
| 9  | 327 | 0     | Inf. Dilution | lgK:235.4 (4SF) | 0 | ENS   | 6422   |
| 10 | 327 | 0     | Inf. Dilution | lgK:235.3 (4SF) | 0 | ENS   | 6494   |
| 11 | 25  | 0     | Inf. Dilution | dG:-688.7 (4SF) | 4 | EFE   | 6422   |
| 12 | 25  | 0     | Inf. Dilution | dG:-688.4 (4SF) | 4 | EFE   | 6494   |
| 13 | 25  | 0     | Inf. Dilution | dH:-730.1 (4SF) | 3 | MTD   | 6422   |
| 14 | 25  | 0     | Inf. Dilution | dH:-729.8 (4SF) | 3 | MTD   | 6494   |
| 15 | 127 | 0     | Inf. Dilution | dH:-730.3 (4SF) | 1 | MTD   | 6422   |

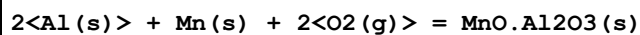
|    |     |   |               |                 |   |     |      |
|----|-----|---|---------------|-----------------|---|-----|------|
| 16 | 127 | 0 | Inf. Dilution | dH:-730.0 (4SF) | 1 | MTD | 6494 |
| 17 | 227 | 0 | Inf. Dilution | dH:-730.9 (4SF) | 0 | MTD | 6422 |
| 18 | 227 | 0 | Inf. Dilution | dH:-730.7 (4SF) | 0 | MTD | 6494 |
| 19 | 327 | 0 | Inf. Dilution | dH:-730.7 (4SF) | 0 | MTD | 6422 |
| 20 | 327 | 0 | Inf. Dilution | dH:-730.5 (4SF) | 0 | MTD | 6494 |

## Reaction No. 66950



| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:501.2 (4SF) | 4 | ENS   | 6422   |
| 2  | 25  | 0     | Inf. Dilution | lgK:501.0 (4SF) | 4 | ENS   | 6494   |
| 3  | 27  | 0     | Inf. Dilution | lgK:498.0 (4SF) | 0 | ENS   | 6422   |
| 4  | 27  | 0     | Inf. Dilution | lgK:497.7 (4SF) | 0 | ENS   | 6494   |
| 5  | 127 | 0     | Inf. Dilution | lgK:366.2 (4SF) | 3 | ENS   | 6422   |
| 6  | 127 | 0     | Inf. Dilution | lgK:366.0 (4SF) | 3 | ENS   | 6494   |
| 7  | 227 | 0     | Inf. Dilution | lgK:287.1 (4SF) | 2 | ENS   | 6422   |
| 8  | 227 | 0     | Inf. Dilution | lgK:286.9 (4SF) | 2 | ENS   | 6494   |
| 9  | 327 | 0     | Inf. Dilution | lgK:234.4 (4SF) | 0 | ENS   | 6422   |
| 10 | 327 | 0     | Inf. Dilution | lgK:234.2 (4SF) | 0 | ENS   | 6494   |
| 11 | 25  | 0     | Inf. Dilution | dG:-683.8 (4SF) | 4 | EFE   | 6422   |
| 12 | 25  | 0     | Inf. Dilution | dG:-683.4 (4SF) | 4 | EFE   | 6494   |
| 13 | 25  | 0     | Inf. Dilution | dH:-723.4 (4SF) | 3 | MTD   | 6422   |
| 14 | 25  | 0     | Inf. Dilution | dH:-723.1 (4SF) | 3 | MTD   | 6494   |
| 15 | 127 | 0     | Inf. Dilution | dH:-723.5 (4SF) | 1 | MTD   | 6422   |
| 16 | 127 | 0     | Inf. Dilution | dH:-723.2 (4SF) | 1 | MTD   | 6494   |
| 17 | 227 | 0     | Inf. Dilution | dH:-724.0 (4SF) | 0 | MTD   | 6422   |
| 18 | 227 | 0     | Inf. Dilution | dH:-723.8 (4SF) | 0 | MTD   | 6494   |
| 19 | 327 | 0     | Inf. Dilution | dH:-723.6 (4SF) | 0 | MTD   | 6422   |
| 20 | 327 | 0     | Inf. Dilution | dH:-732.4 (4SF) | 0 | MTD   | 6494   |

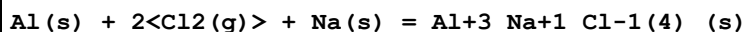
## Reaction No. 66996



| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:348.4 (4SF) | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:346.1 (4SF) | 4 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:254.8 (4SF) | 4 | ENS   | 6422   |

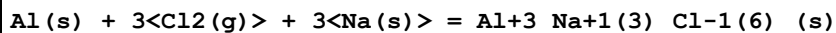
|    |     |   |               |                 |   |     |      |
|----|-----|---|---------------|-----------------|---|-----|------|
| 4  | 227 | 0 | Inf. Dilution | lgK:200.0 (4SF) | 4 | ENS | 6422 |
| 5  | 327 | 0 | Inf. Dilution | lgK:163.4 (4SF) | 4 | ENS | 6422 |
| 6  | 25  | 0 | Inf. Dilution | dG:-475.3 (4SF) | 4 | EFE | 6422 |
| 7  | 25  | 0 | Inf. Dilution | dH:-501.6 (4SF) | 4 | MTD | 6422 |
| 8  | 127 | 0 | Inf. Dilution | dH:-501.6 (4SF) | 4 | MTD | 6422 |
| 9  | 227 | 0 | Inf. Dilution | dH:-501.4 (4SF) | 4 | MTD | 6422 |
| 10 | 327 | 0 | Inf. Dilution | dH:-501.2 (4SF) | 4 | MTD | 6422 |

## Reaction No. 67023



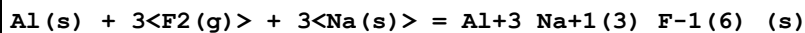
| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:182.5 (4SF) | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:181.2 (4SF) | 0 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:131.6 (4SF) | 2 | ENS   | 6422   |
| 4  | 151 | 0     | Inf. Dilution | lgK:123.1 (4SF) | 1 | ENS   | 6422   |
| 5  | 25  | 0     | Inf. Dilution | dG:-248.9 (4SF) | 4 | EFE   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dH:-273.0 (4SF) | 2 | MTD   | 6422   |
| 7  | 127 | 0     | Inf. Dilution | dH:-272.8 (4SF) | 1 | MTD   | 6422   |
| 8  | 151 | 0     | Inf. Dilution | dH:-272.5 (4SF) | 0 | MTD   | 6422   |

## Reaction No. 67024



| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:320.4 (4SF) | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:318.2 (4SF) | 0 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:232.1 (4SF) | 2 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:180.3 (4SF) | 1 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:145.9 (4SF) | 0 | ENS   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dG:-437.1 (4SF) | 4 | EFE   | 6422   |
| 7  | 25  | 0     | Inf. Dilution | dH:-473.0 (4SF) | 2 | MTD   | 6422   |
| 8  | 127 | 0     | Inf. Dilution | dH:-474.1 (4SF) | 1 | MTD   | 6422   |
| 9  | 227 | 0     | Inf. Dilution | dH:-473.4 (4SF) | 0 | MTD   | 6422   |
| 10 | 327 | 0     | Inf. Dilution | dH:-472.4 (4SF) | 0 | MTD   | 6422   |

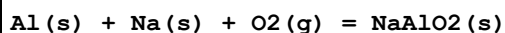
## Reaction No. 67025



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

|    |     |   |               |                 |   |     |      |
|----|-----|---|---------------|-----------------|---|-----|------|
| 1  | 25  | 0 | Inf. Dilution | lgK:551.0 (4SF) | 2 | ENS | 6422 |
| 2  | 25  | 0 | Inf. Dilution | lgK:552.2 (4SF) | 4 | ENS | 6494 |
| 3  | 27  | 0 | Inf. Dilution | lgK:547.4 (4SF) | 0 | ENS | 6422 |
| 4  | 27  | 0 | Inf. Dilution | lgK:548.6 (4SF) | 0 | ENS | 6494 |
| 5  | 127 | 0 | Inf. Dilution | lgK:403.3 (4SF) | 3 | ENS | 6422 |
| 6  | 127 | 0 | Inf. Dilution | lgK:404.2 (4SF) | 3 | ENS | 6494 |
| 7  | 227 | 0 | Inf. Dilution | lgK:316.7 (4SF) | 2 | ENS | 6422 |
| 8  | 227 | 0 | Inf. Dilution | lgK:317.5 (4SF) | 2 | ENS | 6494 |
| 9  | 327 | 0 | Inf. Dilution | lgK:259.0 (4SF) | 0 | ENS | 6422 |
| 10 | 327 | 0 | Inf. Dilution | lgK:259.7 (4SF) | 0 | ENS | 6494 |
| 11 | 25  | 0 | Inf. Dilution | dG:-751.6 (4SF) | 2 | EFE | 6422 |
| 12 | 25  | 0 | Inf. Dilution | dG:-753.4 (4SF) | 3 | EFE | 6494 |
| 13 | 25  | 0 | Inf. Dilution | dH:-791.0 (4SF) | 4 | MTD | 6422 |
| 14 | 25  | 0 | Inf. Dilution | dH:-792.7 (4SF) | 4 | MTD | 6494 |
| 15 | 127 | 0 | Inf. Dilution | dH:-792.5 (4SF) | 2 | MTD | 6422 |
| 16 | 127 | 0 | Inf. Dilution | dH:-794.2 (4SF) | 2 | MTD | 6494 |
| 17 | 227 | 0 | Inf. Dilution | dH:-792.0 (4SF) | 1 | MTD | 6422 |
| 18 | 227 | 0 | Inf. Dilution | dH:-793.6 (4SF) | 1 | MTD | 6494 |
| 19 | 327 | 0 | Inf. Dilution | dH:-791.2 (4SF) | 0 | MTD | 6422 |
| 20 | 327 | 0 | Inf. Dilution | dH:-792.8 (4SF) | 0 | MTD | 6494 |

## Reaction No. 67041



| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:187.3 (4SF) | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:186.1 (4SF) | 4 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:136.7 (4SF) | 4 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:107.1 (4SF) | 4 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:87.27 (4SF) | 4 | ENS   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dG:-255.6 (4SF) | 4 | EFE   | 6422   |
| 7  | 25  | 0     | Inf. Dilution | dH:-270.8 (4SF) | 4 | MTD   | 6422   |
| 8  | 127 | 0     | Inf. Dilution | dH:-271.6 (4SF) | 4 | MTD   | 6422   |
| 9  | 227 | 0     | Inf. Dilution | dH:-271.6 (4SF) | 4 | MTD   | 6422   |
| 10 | 327 | 0     | Inf. Dilution | dH:-271.6 (4SF) | 4 | MTD   | 6422   |

## Reaction No. 67042

| Al(s) + Na(s) + 3<O2(g)> + 2<Si(s)> = NaAlSi2O6(Dehyd.Anal.,s) |     |       |               |                |   |       |        |
|----------------------------------------------------------------|-----|-------|---------------|----------------|---|-------|--------|
| No                                                             | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                              | 25  | 0     | Inf. Dilution | lgK:493.9(4SF) | 4 | ENS   | 6422   |
| 2                                                              | 25  | 0     | Inf. Dilution | lgK:493.3(4SF) | 4 | ENS   | 6494   |
| 3                                                              | 27  | 0     | Inf. Dilution | lgK:490.7(4SF) | 0 | ENS   | 6422   |
| 4                                                              | 27  | 0     | Inf. Dilution | lgK:490.1(4SF) | 0 | ENS   | 6494   |
| 5                                                              | 127 | 0     | Inf. Dilution | lgK:360.7(4SF) | 3 | ENS   | 6422   |
| 6                                                              | 127 | 0     | Inf. Dilution | lgK:360.2(4SF) | 3 | ENS   | 6494   |
| 7                                                              | 227 | 0     | Inf. Dilution | lgK:282.6(4SF) | 2 | ENS   | 6422   |
| 8                                                              | 227 | 0     | Inf. Dilution | lgK:282.3(4SF) | 2 | ENS   | 6494   |
| 9                                                              | 327 | 0     | Inf. Dilution | lgK:230.6(4SF) | 0 | ENS   | 6422   |
| 10                                                             | 327 | 0     | Inf. Dilution | lgK:230.3(4SF) | 0 | ENS   | 6494   |
| 11                                                             | 25  | 0     | Inf. Dilution | dG:-673.8(4SF) | 4 | EFE   | 6422   |
| 12                                                             | 25  | 0     | Inf. Dilution | dG:-673.0(4SF) | 4 | EFE   | 6494   |
| 13                                                             | 25  | 0     | Inf. Dilution | dH:-713.5(4SF) | 4 | MTD   | 6422   |
| 14                                                             | 25  | 0     | Inf. Dilution | dH:-713.0(4SF) | 4 | MTD   | 6494   |
| 15                                                             | 127 | 0     | Inf. Dilution | dH:-714.3(4SF) | 3 | MTD   | 6422   |
| 16                                                             | 127 | 0     | Inf. Dilution | dH:-713.7(4SF) | 3 | MTD   | 6494   |
| 17                                                             | 227 | 0     | Inf. Dilution | dH:-714.0(4SF) | 2 | MTD   | 6422   |
| 18                                                             | 227 | 0     | Inf. Dilution | dH:-713.5(4SF) | 2 | MTD   | 6494   |
| 19                                                             | 327 | 0     | Inf. Dilution | dH:-713.7(4SF) | 0 | MTD   | 6422   |
| 20                                                             | 327 | 0     | Inf. Dilution | dH:-713.0(4SF) | 0 | MTD   | 6494   |

| Reaction No. 67043                                          |     |       |               |                     |   |       |        |
|-------------------------------------------------------------|-----|-------|---------------|---------------------|---|-------|--------|
| Al(s) + Na(s) + 4<O2(g)> + 3<Si(s)> = NaAlSi3O8(AlbiteLO,s) |     |       |               |                     |   |       |        |
| No                                                          | t   | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                           | 25  | 0     | Inf. Dilution | lgK:651.1(4SF)      | 1 | ELC   |        |
| 2                                                           | 25  | 0     | Inf. Dilution | lgK:650.0(4SF)      | 0 | ENS   | 6422   |
| 3                                                           | 27  | 0     | Inf. Dilution | lgK:645.7(4SF)      | 0 | ENS   | 6422   |
| 4                                                           | 127 | 0     | Inf. Dilution | lgK:474.7(4SF)      | 0 | ENS   | 6422   |
| Note (4): Low Wgt for internal consistency                  |     |       |               |                     |   |       |        |
| 5                                                           | 227 | 0     | Inf. Dilution | lgK:372.1(4SF)      | 0 | ENS   | 6422   |
| 6                                                           | 327 | 0     | Inf. Dilution | lgK:303.7(4SF)      | 0 | ENS   | 6422   |
| 7                                                           | 25  | 0     | Inf. Dilution | dG:-886.7(4SF)      | 0 | EFE   | 6422   |
| 8                                                           | 25  | 0     | Inf. Dilution | dG:-887.12(5SF)     | 0 | EFE   | 6482   |
| 9                                                           | 25  | 0     | Inf. Dilution | dG:-3713.00(6SF) kJ | 0 | EFE   | 8798   |



|                                             |     |   |               |                      |   |     |       |
|---------------------------------------------|-----|---|---------------|----------------------|---|-----|-------|
| 10                                          | 25  | 0 | Inf. Dilution | dG:-888.94 (5SF)     | 0 | CRV | 10174 |
| 11                                          | 25  | 0 | Inf. Dilution | dG:-887.37 (5SF)     | 5 | CRV | 29553 |
| 12                                          | 50  | 0 | Inf. Dilution | dG:-890.22 (5SF)     | 0 | CRV | 10174 |
| 13                                          | 75  | 0 | Inf. Dilution | dG:-891.61 (5SF)     | 0 | CRV | 10174 |
| 14                                          | 100 | 0 | Inf. Dilution | dG:-893.10 (5SF)     | 0 | CRV | 10174 |
| 15                                          | 125 | 0 | Inf. Dilution | dG:-894.67 (5SF)     | 0 | CRV | 10174 |
| 16                                          | 150 | 0 | Inf. Dilution | dG:-896.34 (5SF)     | 0 | CRV | 10174 |
| 17                                          | 175 | 0 | Inf. Dilution | dG:-898.09 (5SF)     | 0 | CRV | 10174 |
| 18                                          | 200 | 0 | Inf. Dilution | dG:-899.91 (5SF)     | 0 | CRV | 10174 |
| 19                                          | 225 | 0 | Inf. Dilution | dG:-901.81 (5SF)     | 0 | CRV | 10174 |
| 20                                          | 250 | 0 | Inf. Dilution | dG:-903.78 (5SF)     | 0 | CRV | 10174 |
| 21                                          | 300 | 0 | Inf. Dilution | dG:-907.90 (5SF)     | 0 | CRV | 10174 |
| 22                                          | 25  | 0 | Inf. Dilution | dH:-938.7 (4SF)      | 0 | MTD | 6422  |
| 23                                          | 25  | 0 | Inf. Dilution | dH:-940.51 (5SF)     | 0 | MTD | 6482  |
| 24                                          | 25  | 0 | Inf. Dilution | dH:-3936.20 (6SF) kJ | 0 | EFE | 8798  |
| 25                                          | 25  | 0 | Inf. Dilution | dH:-940.77 (5SF)     | 5 | CRV | 29553 |
| 26                                          | 127 | 0 | Inf. Dilution | dH:-939.5 (4SF)      | 0 | MTD | 6422  |
| Note (26): Low Wgt for internal consistency |     |   |               |                      |   |     |       |
| 27                                          | 227 | 0 | Inf. Dilution | dH:-939.1 (4SF)      | 0 | MTD | 6422  |
| 28                                          | 327 | 0 | Inf. Dilution | dH:-938.4 (4SF)      | 0 | MTD | 6422  |

| Reaction No. 67044                                          |     |       |               |                     |   |       |        |
|-------------------------------------------------------------|-----|-------|---------------|---------------------|---|-------|--------|
| Al(s) + Na(s) + 4<O2(g)> + 3<Si(s)> = NaAlSi3O8(AlbiteHI,s) |     |       |               |                     |   |       |        |
| No                                                          | t   | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                           | 25  | 0     | Inf. Dilution | lgK:649.(3SF)       | 1 | ELC   |        |
| 2                                                           | 25  | 0     | Inf. Dilution | lgK:650.6(4SF)      | 0 | ENS   | 6422   |
| 3                                                           | 27  | 0     | Inf. Dilution | lgK:646.3(4SF)      | 0 | ENS   | 6422   |
| 4                                                           | 127 | 0     | Inf. Dilution | lgK:474.9(4SF)      | 0 | ENS   | 6422   |
| 5                                                           | 227 | 0     | Inf. Dilution | lgK:372.0(4SF)      | 0 | ENS   | 6422   |
| 6                                                           | 327 | 0     | Inf. Dilution | lgK:303.4(4SF)      | 0 | ENS   | 6422   |
| 7                                                           | 25  | 0     | Inf. Dilution | dG:-887.6(4SF)      | 0 | EFE   | 6422   |
| 8                                                           | 25  | 0     | Inf. Dilution | dG:-885.11(5SF)     | 0 | EFE   | 6482   |
| 9                                                           | 25  | 0     | Inf. Dilution | dG:-3705.10(6SF) kJ | 0 | EFE   | 8798   |
| 10                                                          | 25  | 0     | Inf. Dilution | dG:-887.14(5SF)     | 0 | CRV   | 10174  |
| 11                                                          | 25  | 0     | Inf. Dilution | dG:-3705.1(5SF) kJ  | 3 | EOD   | 12843  |
| 12                                                          | 50  | 0     | Inf. Dilution | dG:-888.50(5SF)     | 0 | CRV   | 10174  |

|                                             |     |   |               |                      |   |     |       |
|---------------------------------------------|-----|---|---------------|----------------------|---|-----|-------|
| 13                                          | 75  | 0 | Inf. Dilution | dG:-889.95 (5SF)     | 0 | CRV | 10174 |
| 14                                          | 100 | 0 | Inf. Dilution | dG:-891.51 (5SF)     | 0 | CRV | 10174 |
| 15                                          | 125 | 0 | Inf. Dilution | dG:-893.15 (5SF)     | 0 | CRV | 10174 |
| 16                                          | 150 | 0 | Inf. Dilution | dG:-894.89 (5SF)     | 0 | CRV | 10174 |
| 17                                          | 175 | 0 | Inf. Dilution | dG:-896.71 (5SF)     | 0 | CRV | 10174 |
| 18                                          | 200 | 0 | Inf. Dilution | dG:-898.60 (5SF)     | 0 | CRV | 10174 |
| 19                                          | 225 | 0 | Inf. Dilution | dG:-900.57 (5SF)     | 0 | CRV | 10174 |
| 20                                          | 250 | 0 | Inf. Dilution | dG:-902.61 (5SF)     | 0 | CRV | 10174 |
| 21                                          | 300 | 0 | Inf. Dilution | dG:-906.87 (5SF)     | 0 | CRV | 10174 |
| 22                                          | 25  | 0 | Inf. Dilution | dH:-940.9 (4SF)      | 0 | MTD | 6422  |
| 23                                          | 25  | 0 | Inf. Dilution | dH:-937.29 (5SF)     | 0 | MTD | 6482  |
| 24                                          | 25  | 0 | Inf. Dilution | dH:-3923.40 (6SF) kJ | 0 | EFE | 8798  |
| 25                                          | 25  | 0 | Inf. Dilution | dH:-3923.4 (5SF) kJ  | 3 | EOD | 12843 |
| 26                                          | 127 | 0 | Inf. Dilution | dH:-941.8 (4SF)      | 0 | MTD | 6422  |
| Note (26): Low Wgt for internal consistency |     |   |               |                      |   |     |       |
| 27                                          | 227 | 0 | Inf. Dilution | dH:-941.7 (4SF)      | 0 | MTD | 6422  |
| 28                                          | 327 | 0 | Inf. Dilution | dH:-941.4 (4SF)      | 0 | MTD | 6422  |

| Reaction No. 67045                                               |     |       |               |                      |   |       |        |
|------------------------------------------------------------------|-----|-------|---------------|----------------------|---|-------|--------|
| Al(s) + H2(g) + Na(s) + 3.5<O2(g)> + 2<Si(s)> = NaAlSi2O6.H2O(s) |     |       |               |                      |   |       |        |
| No                                                               | t   | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
| 1                                                                | 25  | 0     | Inf. Dilution | lgK:539.2 (4SF)      | 1 | ENS   | 6422   |
| 2                                                                | 25  | 0     | Inf. Dilution | lgK:541.3 (4SF)      | 1 | ENS   | 6494   |
| 3                                                                | 27  | 0     | Inf. Dilution | lgK:535.6 (4SF)      | 0 | ENS   | 6422   |
| 4                                                                | 27  | 0     | Inf. Dilution | lgK:537.8 (4SF)      | 0 | ENS   | 6494   |
| 5                                                                | 127 | 0     | Inf. Dilution | lgK:391.8 (4SF)      | 0 | ENS   | 6422   |
| 6                                                                | 127 | 0     | Inf. Dilution | lgK:393.6 (4SF)      | 0 | ENS   | 6494   |
| 7                                                                | 227 | 0     | Inf. Dilution | lgK:305.5 (4SF)      | 0 | ENS   | 6422   |
| 8                                                                | 227 | 0     | Inf. Dilution | lgK:307.1 (4SF)      | 0 | ENS   | 6494   |
| 9                                                                | 327 | 0     | Inf. Dilution | lgK:248.0 (4SF)      | 0 | ENS   | 6422   |
| 10                                                               | 327 | 0     | Inf. Dilution | lgK:249.5 (4SF)      | 0 | ENS   | 6494   |
| 11                                                               | 25  | 0     | Inf. Dilution | dG:-735.5 (4SF)      | 1 | EFE   | 6422   |
| 12                                                               | 25  | 0     | Inf. Dilution | dG:-738.5 (4SF)      | 1 | EFE   | 6494   |
| 13                                                               | 25  | 0     | Inf. Dilution | dG:-740.73 (5SF)     | 0 | CRV   | 10174  |
| 14                                                               | 25  | 0     | Inf. Dilution | dG:-3078.90 (6SF) kJ | 3 | CRV   | 25435  |
| 15                                                               | 25  | 0     | Inf. Dilution | dG:-3084. (4SF) kJ   | 3 | EFE   | 29327  |

|    |     |   |               |                       |   |     |       |
|----|-----|---|---------------|-----------------------|---|-----|-------|
| 16 | 50  | 0 | Inf. Dilution | dG:-742.18 (5SF)      | 0 | CRV | 10174 |
| 17 | 75  | 0 | Inf. Dilution | dG:-743.73 (5SF)      | 0 | CRV | 10174 |
| 18 | 100 | 0 | Inf. Dilution | dG:-745.38 (5SF)      | 0 | CRV | 10174 |
| 19 | 125 | 0 | Inf. Dilution | dG:-747.13 (5SF)      | 0 | CRV | 10174 |
| 20 | 150 | 0 | Inf. Dilution | dG:-748.96 (5SF)      | 0 | CRV | 10174 |
| 21 | 175 | 0 | Inf. Dilution | dG:-750.87 (5SF)      | 0 | CRV | 10174 |
| 22 | 200 | 0 | Inf. Dilution | dG:-752.86 (5SF)      | 0 | CRV | 10174 |
| 23 | 225 | 0 | Inf. Dilution | dG:-754.93 (5SF)      | 0 | CRV | 10174 |
| 24 | 250 | 0 | Inf. Dilution | dG:-757.06 (5SF)      | 0 | CRV | 10174 |
| 25 | 300 | 0 | Inf. Dilution | dG:-761.51 (5SF)      | 0 | CRV | 10174 |
| 26 | 25  | 0 | Inf. Dilution | dH:-789.4 (4SF)       | 3 | MTD | 6422  |
| 27 | 25  | 0 | Inf. Dilution | dH:-791.1 (4SF)       | 1 | MTD | 6494  |
| 28 | 25  | 0 | Inf. Dilution | dH:-3296.9 (5SF) kJ   | 3 | CRV | 25435 |
| 29 | 25  | 0 | Inf. Dilution | dH:-3308.00 (6SF) kJ  | 3 | CRV | 27383 |
| 30 | 25  | 0 | Inf. Dilution | dH:-3303.2 (2.9SD) kJ | 4 | ETR | 29327 |
| 31 | 127 | 0 | Inf. Dilution | dH:-790.0 (4SF)       | 0 | MTD | 6422  |
| 32 | 127 | 0 | Inf. Dilution | dH:-791.9 (4SF)       | 0 | MTD | 6494  |
| 33 | 227 | 0 | Inf. Dilution | dH:-789.4 (4SF)       | 0 | MTD | 6422  |
| 34 | 227 | 0 | Inf. Dilution | dH:-791.6 (4SF)       | 0 | MTD | 6494  |
| 35 | 327 | 0 | Inf. Dilution | dH:-788.5 (4SF)       | 0 | MTD | 6422  |
| 36 | 327 | 0 | Inf. Dilution | dH:-790.9 (4SF)       | 0 | MTD | 6494  |

| Reaction No. 67105                         |     |       |               |                     |   |       |        |
|--------------------------------------------|-----|-------|---------------|---------------------|---|-------|--------|
| 2<Al(s)> + Ni(s) + 2<O2(g)> = NiO.Al2O3(s) |     |       |               |                     |   |       |        |
| No                                         | t   | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                          | 25  | 0     | Inf. Dilution | lgK:314.8 (4SF)     | 4 | ENS   | 6422   |
| 2                                          | 27  | 0     | Inf. Dilution | lgK:312.8 (4SF)     | 0 | ENS   | 6422   |
| 3                                          | 127 | 0     | Inf. Dilution | lgK:229.4 (4SF)     | 2 | ENS   | 6422   |
| 4                                          | 227 | 0     | Inf. Dilution | lgK:179.4 (4SF)     | 1 | ENS   | 6422   |
| 5                                          | 327 | 0     | Inf. Dilution | lgK:146.1 (4SF)     | 0 | ENS   | 6422   |
| 6                                          | 25  | 0     | Inf. Dilution | dG:-429.5 (4SF)     | 4 | EFE   | 6422   |
| 7                                          | 25  | 0     | Inf. Dilution | dH:-457.9 (4SF)     | 4 | MTD   | 6422   |
| 8                                          | 25  | 0     | Inf. Dilution | dH:-1915.9 (5SF) kJ | 4 | CRV   | 9015   |
| 9                                          | 127 | 0     | Inf. Dilution | dH:-457.8 (4SF)     | 1 | MTD   | 6422   |
| 10                                         | 227 | 0     | Inf. Dilution | dH:-457.5 (4SF)     | 0 | MTD   | 6422   |
| 11                                         | 327 | 0     | Inf. Dilution | dH:-457.3 (4SF)     | 0 | MTD   | 6422   |

| Reaction No. 67365                                           |     |       |               |                |   |       |        |
|--------------------------------------------------------------|-----|-------|---------------|----------------|---|-------|--------|
| Al(s) + 2<Cl <sub>2</sub> (g)> + K(s) = Al+3_K+1_Cl-1(4)_(s) |     |       |               |                |   |       |        |
| No                                                           | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                            | 25  | 0     | Inf. Dilution | lgK:191.8(4SF) | 4 | ENS   | 6422   |
| 2                                                            | 27  | 0     | Inf. Dilution | lgK:190.5(4SF) | 0 | ENS   | 6422   |
| 3                                                            | 127 | 0     | Inf. Dilution | lgK:138.4(4SF) | 2 | ENS   | 6422   |
| 4                                                            | 227 | 0     | Inf. Dilution | lgK:107.2(4SF) | 1 | ENS   | 6422   |
| 5                                                            | 256 | 0     | Inf. Dilution | lgK:100.4(4SF) | 0 | ENS   | 6422   |
| 6                                                            | 25  | 0     | Inf. Dilution | dG:-261.6(4SF) | 4 | EFE   | 6422   |
| 7                                                            | 25  | 0     | Inf. Dilution | dH:-286.0(4SF) | 2 | MTD   | 6422   |
| 8                                                            | 127 | 0     | Inf. Dilution | dH:-285.7(4SF) | 1 | MTD   | 6422   |
| 9                                                            | 227 | 0     | Inf. Dilution | dH:-284.7(4SF) | 0 | MTD   | 6422   |
| 10                                                           | 256 | 0     | Inf. Dilution | dH:-284.4(4SF) | 0 | MTD   | 6422   |

| Reaction No. 67366                                                 |     |       |               |                |   |       |        |
|--------------------------------------------------------------------|-----|-------|---------------|----------------|---|-------|--------|
| Al(s) + 3<Cl <sub>2</sub> (g)> + 3<K(s)> = Al+3_K+1(3)_Cl-1(6)_(s) |     |       |               |                |   |       |        |
| No                                                                 | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                  | 25  | 0     | Inf. Dilution | lgK:339.6(4SF) | 4 | ENS   | 6422   |
| 2                                                                  | 27  | 0     | Inf. Dilution | lgK:337.3(4SF) | 0 | ENS   | 6422   |
| 3                                                                  | 127 | 0     | Inf. Dilution | lgK:246.2(4SF) | 2 | ENS   | 6422   |
| 4                                                                  | 227 | 0     | Inf. Dilution | lgK:191.5(4SF) | 1 | ENS   | 6422   |
| 5                                                                  | 327 | 0     | Inf. Dilution | lgK:155.1(4SF) | 0 | ENS   | 6422   |
| 6                                                                  | 25  | 0     | Inf. Dilution | dG:-463.3(4SF) | 4 | EFE   | 6422   |
| 7                                                                  | 25  | 0     | Inf. Dilution | dH:-500.0(4SF) | 2 | MTD   | 6422   |
| 8                                                                  | 127 | 0     | Inf. Dilution | dH:-500.9(4SF) | 1 | MTD   | 6422   |
| 9                                                                  | 227 | 0     | Inf. Dilution | dH:-500.0(4SF) | 0 | MTD   | 6422   |
| 10                                                                 | 327 | 0     | Inf. Dilution | dH:-498.9(4SF) | 0 | MTD   | 6422   |

| Reaction No. 67367                                               |     |       |               |                |   |       |        |
|------------------------------------------------------------------|-----|-------|---------------|----------------|---|-------|--------|
| Al(s) + 3<F <sub>2</sub> (g)> + 3<K(s)> = Al+3_K+1(3)_F-1(6)_(s) |     |       |               |                |   |       |        |
| No                                                               | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                | 25  | 0     | Inf. Dilution | lgK:554.2(4SF) | 4 | ENS   | 6422   |
| 2                                                                | 27  | 0     | Inf. Dilution | lgK:550.6(4SF) | 0 | ENS   | 6422   |
| 3                                                                | 127 | 0     | Inf. Dilution | lgK:405.7(4SF) | 2 | ENS   | 6422   |
| 4                                                                | 227 | 0     | Inf. Dilution | lgK:318.7(4SF) | 1 | ENS   | 6422   |
| 5                                                                | 327 | 0     | Inf. Dilution | lgK:260.8(4SF) | 0 | ENS   | 6422   |

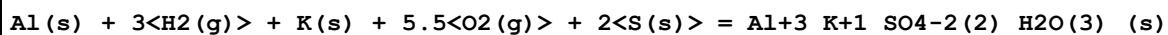
|    |     |   |               |                  |   |     |      |
|----|-----|---|---------------|------------------|---|-----|------|
| 6  | 25  | 0 | Inf. Dilution | dG: -756.1 (4SF) | 4 | EFE | 6422 |
| 7  | 25  | 0 | Inf. Dilution | dH: -795.0 (4SF) | 2 | MTD | 6422 |
| 8  | 127 | 0 | Inf. Dilution | dH: -796.3 (4SF) | 1 | MTD | 6422 |
| 9  | 227 | 0 | Inf. Dilution | dH: -795.6 (4SF) | 0 | MTD | 6422 |
| 10 | 327 | 0 | Inf. Dilution | dH: -794.7 (4SF) | 0 | MTD | 6422 |

## Reaction No. 67368



| No | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|-----|-------|---------------|------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK: 392.4 (4SF) | 4 | ENS   | 6422   |
| 2  | 25  | 0     | Inf. Dilution | lgK: 392.5 (4SF) | 4 | ENS   | 6494   |
| 3  | 27  | 0     | Inf. Dilution | lgK: 389.7 (4SF) | 0 | ENS   | 6422   |
| 4  | 27  | 0     | Inf. Dilution | lgK: 389.8 (4SF) | 0 | ENS   | 6494   |
| 5  | 127 | 0     | Inf. Dilution | lgK: 282.1 (4SF) | 3 | ENS   | 6422   |
| 6  | 127 | 0     | Inf. Dilution | lgK: 282.2 (4SF) | 3 | ENS   | 6494   |
| 7  | 227 | 0     | Inf. Dilution | lgK: 217.4 (4SF) | 2 | ENS   | 6422   |
| 8  | 227 | 0     | Inf. Dilution | lgK: 217.4 (4SF) | 2 | ENS   | 6494   |
| 9  | 327 | 0     | Inf. Dilution | lgK: 174.2 (4SF) | 0 | ENS   | 6422   |
| 10 | 327 | 0     | Inf. Dilution | lgK: 174.2 (4SF) | 0 | ENS   | 6494   |
| 11 | 25  | 0     | Inf. Dilution | dG: -535.3 (4SF) | 3 | EFE   | 6422   |
| 12 | 25  | 0     | Inf. Dilution | dG: -535.5 (4SF) | 4 | EFE   | 6494   |
| 13 | 25  | 0     | Inf. Dilution | dH: -590.4 (4SF) | 3 | MTD   | 6422   |
| 14 | 25  | 0     | Inf. Dilution | dH: -590.6 (4SF) | 3 | MTD   | 6494   |
| 15 | 127 | 0     | Inf. Dilution | dH: -592.2 (4SF) | 2 | MTD   | 6422   |
| 16 | 127 | 0     | Inf. Dilution | dH: -592.4 (4SF) | 2 | MTD   | 6494   |
| 17 | 227 | 0     | Inf. Dilution | dH: -592.6 (4SF) | 1 | MTD   | 6422   |
| 18 | 227 | 0     | Inf. Dilution | dH: -592.8 (4SF) | 1 | MTD   | 6494   |
| 19 | 327 | 0     | Inf. Dilution | dH: -592.5 (4SF) | 0 | MTD   | 6422   |
| 20 | 327 | 0     | Inf. Dilution | dH: -592.7 (4SF) | 0 | MTD   | 6494   |

## Reaction No. 67369



| No | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|-----|-------|---------------|------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK: 521.1 (4SF) | 3 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK: 517.5 (4SF) | 0 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK: 370.2 (4SF) | 0 | ENS   | 6422   |

|   |     |   |               |                |   |     |      |
|---|-----|---|---------------|----------------|---|-----|------|
| 4 | 272 | 0 | Inf. Dilution | lgK:281.6(4SF) | 0 | ENS | 6422 |
| 5 | 25  | 0 | Inf. Dilution | dG:-711.0(4SF) | 4 | EFE | 6422 |
| 6 | 25  | 0 | Inf. Dilution | dH:-808.1(4SF) | 2 | MTD | 6422 |
| 7 | 127 | 0 | Inf. Dilution | dH:-810.1(4SF) | 0 | MTD | 6422 |
| 8 | 272 | 0 | Inf. Dilution | dH:-810.9(4SF) | 0 | MTD | 6422 |

| Reaction No. 67370                                                                                                                                                         |    |       |               |                |   |       |        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $\text{Al(s)} + 12\text{H}_2\text{(g)} + \text{K(s)} + 10\text{O}_2\text{(g)} + 2\text{S(s)} = \text{Al}_3\text{K}_4\text{SO}_4\cdot 2\text{(H}_2\text{O)}_{12}\text{(s)}$ |    |       |               |                |   |       |        |
| No                                                                                                                                                                         | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                                                                                          | 25 | 0     | Inf. Dilution | lgK:900.6(4SF) | 4 | ENS   | 6422   |
| 2                                                                                                                                                                          | 27 | 0     | Inf. Dilution | lgK:894.1(4SF) | 0 | ENS   | 6422   |
| Note (2): Low wgt for internal consistency                                                                                                                                 |    |       |               |                |   |       |        |
| 3                                                                                                                                                                          | 91 | 0     | Inf. Dilution | lgK:708.5(4SF) | 4 | ENS   | 6422   |
| 4                                                                                                                                                                          | 25 | 0     | Inf. Dilution | dG:-1229.(4SF) | 4 | EFE   | 6422   |
| 5                                                                                                                                                                          | 25 | 0     | Inf. Dilution | dH:-1449.(4SF) | 4 | MTD   | 6422   |
| 6                                                                                                                                                                          | 91 | 0     | Inf. Dilution | dH:-1449.(4SF) | 4 | MTD   | 6422   |

| Reaction No. 67383                                                                                              |     |       |               |                     |   |       |        |
|-----------------------------------------------------------------------------------------------------------------|-----|-------|---------------|---------------------|---|-------|--------|
| $\text{Al(s)} + \text{K(s)} + 4\text{O}_2\text{(g)} + 3\text{Si(s)} = \text{KAlSi}_3\text{O}_8\text{(Micr.,s)}$ |     |       |               |                     |   |       |        |
| No                                                                                                              | t   | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                                                                               | 25  | 0     | Inf. Dilution | lgK:656.(3SF)       | 1 | ELC   |        |
| 2                                                                                                               | 25  | 0     | Inf. Dilution | lgK:655.7(4SF)      | 2 | ENS   | 6422   |
| 3                                                                                                               | 25  | 0     | Inf. Dilution | lgK:656.8(4SF)      | 2 | ENS   | 6494   |
| 4                                                                                                               | 27  | 0     | Inf. Dilution | lgK:651.4(4SF)      | 0 | ENS   | 6422   |
| 5                                                                                                               | 27  | 0     | Inf. Dilution | lgK:652.5(4SF)      | 0 | ENS   | 6494   |
| 6                                                                                                               | 127 | 0     | Inf. Dilution | lgK:478.6(4SF)      | 1 | ENS   | 6422   |
| 7                                                                                                               | 127 | 0     | Inf. Dilution | lgK:479.4(4SF)      | 1 | ENS   | 6494   |
| 8                                                                                                               | 227 | 0     | Inf. Dilution | lgK:374.9(4SF)      | 0 | ENS   | 6422   |
| 9                                                                                                               | 227 | 0     | Inf. Dilution | lgK:375.5(4SF)      | 0 | ENS   | 6494   |
| 10                                                                                                              | 327 | 0     | Inf. Dilution | lgK:305.8(4SF)      | 0 | ENS   | 6422   |
| 11                                                                                                              | 327 | 0     | Inf. Dilution | lgK:306.3(4SF)      | 0 | ENS   | 6494   |
| 12                                                                                                              | 25  | 0     | Inf. Dilution | dG:-894.5(4SF)      | 0 | EFE   | 6422   |
| 13                                                                                                              | 25  | 0     | Inf. Dilution | dG:-895.18(5SF)     | 0 | EFE   | 6482   |
| 14                                                                                                              | 25  | 0     | Inf. Dilution | dG:-896.1(4SF)      | 0 | EFE   | 6494   |
| 15                                                                                                              | 25  | 0     | Inf. Dilution | dG:-3749.50(6SF) kJ | 0 | EFE   | 8798   |
| 16                                                                                                              | 25  | 0     | Inf. Dilution | dG:-897.07(5SF)     | 0 | CRV   | 10174  |

|                                             |     |   |               |                      |   |     |       |
|---------------------------------------------|-----|---|---------------|----------------------|---|-----|-------|
| 17                                          | 50  | 0 | Inf. Dilution | dG:-898.40 (5SF)     | 0 | CRV | 10174 |
| 18                                          | 75  | 0 | Inf. Dilution | dG:-899.82 (5SF)     | 0 | CRV | 10174 |
| 19                                          | 100 | 0 | Inf. Dilution | dG:-901.35 (5SF)     | 0 | CRV | 10174 |
| 20                                          | 125 | 0 | Inf. Dilution | dG:-902.97 (5SF)     | 0 | CRV | 10174 |
| 21                                          | 150 | 0 | Inf. Dilution | dG:-904.67 (5SF)     | 0 | CRV | 10174 |
| 22                                          | 175 | 0 | Inf. Dilution | dG:-906.46 (5SF)     | 0 | CRV | 10174 |
| 23                                          | 200 | 0 | Inf. Dilution | dG:-908.33 (5SF)     | 0 | CRV | 10174 |
| 24                                          | 225 | 0 | Inf. Dilution | dG:-910.27 (5SF)     | 0 | CRV | 10174 |
| 25                                          | 250 | 0 | Inf. Dilution | dG:-912.28 (5SF)     | 0 | CRV | 10174 |
| 26                                          | 300 | 0 | Inf. Dilution | dG:-916.48 (5SF)     | 0 | CRV | 10174 |
| 27                                          | 25  | 0 | Inf. Dilution | dH:-948.4 (4SF)      | 0 | MTD | 6422  |
| 28                                          | 25  | 0 | Inf. Dilution | dH:-949.04 (5SF)     | 0 | EFE | 6482  |
| 29                                          | 25  | 0 | Inf. Dilution | dH:-950.0 (4SF)      | 0 | MTD | 6494  |
| 30                                          | 25  | 0 | Inf. Dilution | dH:-3974.90 (6SF) kJ | 0 | EFE | 8798  |
| 31                                          | 127 | 0 | Inf. Dilution | dH:-949.1 (4SF)      | 0 | MTD | 6422  |
| Note (31): Low Wgt for internal consistency |     |   |               |                      |   |     |       |
| 32                                          | 127 | 0 | Inf. Dilution | dH:-950.9 (4SF)      | 0 | MTD | 6494  |
| 33                                          | 227 | 0 | Inf. Dilution | dH:-948.7 (4SF)      | 0 | MTD | 6422  |
| 34                                          | 227 | 0 | Inf. Dilution | dH:-950.9 (4SF)      | 0 | MTD | 6494  |
| 35                                          | 327 | 0 | Inf. Dilution | dH:-947.9 (4SF)      | 0 | MTD | 6422  |
| 36                                          | 327 | 0 | Inf. Dilution | dH:-950.6 (4SF)      | 0 | MTD | 6494  |

| Reaction No. 67384                                     |     |       |               |                 |   |       |        |
|--------------------------------------------------------|-----|-------|---------------|-----------------|---|-------|--------|
| Al(s) + K(s) + 4<O2(g)> + 3<Si(s)> = KAlSi3O8(Adul.,s) |     |       |               |                 |   |       |        |
| No                                                     | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                      | 25  | 0     | Inf. Dilution | lgK:654.3 (4SF) | 4 | ENS   | 6422   |
| 2                                                      | 27  | 0     | Inf. Dilution | lgK:650.0 (4SF) | 4 | ENS   | 6422   |
| 3                                                      | 127 | 0     | Inf. Dilution | lgK:477.8 (4SF) | 4 | ENS   | 6422   |
| 4                                                      | 227 | 0     | Inf. Dilution | lgK:374.4 (4SF) | 4 | ENS   | 6422   |
| 5                                                      | 327 | 0     | Inf. Dilution | lgK:305.5 (4SF) | 4 | ENS   | 6422   |
| 6                                                      | 25  | 0     | Inf. Dilution | dG:-892.6 (4SF) | 4 | EFE   | 6422   |
| 7                                                      | 25  | 0     | Inf. Dilution | dH:-945.1 (4SF) | 4 | MTD   | 6422   |
| 8                                                      | 127 | 0     | Inf. Dilution | dH:-946.1 (4SF) | 4 | MTD   | 6422   |
| 9                                                      | 227 | 0     | Inf. Dilution | dH:-945.9 (4SF) | 4 | MTD   | 6422   |
| 10                                                     | 327 | 0     | Inf. Dilution | dH:-945.4 (4SF) | 4 | MTD   | 6422   |

| Reaction No. 67401                                                                                                          |     |       |               |                |   |       |        |
|-----------------------------------------------------------------------------------------------------------------------------|-----|-------|---------------|----------------|---|-------|--------|
| $3\text{Al(s)} + 7\text{F}_2\text{(g)} + 5\text{Na(s)} = \text{Al}_3\text{(3)}_{\text{Na+1(5)}}\text{F-1(14)}_{\text{(s)}}$ |     |       |               |                |   |       |        |
| No                                                                                                                          | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                                           | 25  | 0     | Inf. Dilution | lgK:1257.(4SF) | 4 | ENS   | 6494   |
| 2                                                                                                                           | 27  | 0     | Inf. Dilution | lgK:1249.(4SF) | 0 | ENS   | 6494   |
| Note (2): Low wgt for internal consistency                                                                                  |     |       |               |                |   |       |        |
| 3                                                                                                                           | 127 | 0     | Inf. Dilution | lgK:920.2(4SF) | 4 | ENS   | 6494   |
| 4                                                                                                                           | 227 | 0     | Inf. Dilution | lgK:722.9(4SF) | 4 | ENS   | 6494   |
| 5                                                                                                                           | 327 | 0     | Inf. Dilution | lgK:591.5(4SF) | 4 | ENS   | 6494   |
| 6                                                                                                                           | 25  | 0     | Inf. Dilution | dG:-1715.(4SF) | 4 | EFE   | 6494   |
| 7                                                                                                                           | 25  | 0     | Inf. Dilution | dH:-1804.(4SF) | 4 | MTD   | 6494   |
| 8                                                                                                                           | 127 | 0     | Inf. Dilution | dH:-1806.(4SF) | 4 | MTD   | 6494   |
| 9                                                                                                                           | 227 | 0     | Inf. Dilution | dH:-1805.(4SF) | 4 | MTD   | 6494   |
| 10                                                                                                                          | 327 | 0     | Inf. Dilution | dH:-1804.(4SF) | 4 | MTD   | 6494   |

| Reaction No. 67403                                                                                                                                                           |     |       |               |                    |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------|---------------|--------------------|---|-------|--------|
| $\text{C(s)} + \text{H}_2\text{(g)} + \text{Al(s)} + \text{Na(s)} + 2.5\text{O}_2\text{(g)} = \text{Al}_3\text{(3)}_{\text{Na+1}}\text{OH-1(2)}_{\text{CO3-2}}_{\text{(s)}}$ |     |       |               |                    |   |       |        |
| Note: Tutolo et al. [14TKS_29553, p. 224, footnote Table 4 indicate that the reported values for Cp coeffs.in Robie & Hemingway (1995) have a typo                           |     |       |               |                    |   |       |        |
| No                                                                                                                                                                           | t   | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                                                                                                                            | 25  | 0     | Inf. Dilution | lgK:312.9(4SF)     | 2 | ENS   | 6494   |
| 2                                                                                                                                                                            | 27  | 0     | Inf. Dilution | lgK:310.8(4SF)     | 0 | ENS   | 6494   |
| 3                                                                                                                                                                            | 127 | 0     | Inf. Dilution | lgK:225.0(4SF)     | 4 | ENS   | 6494   |
| 4                                                                                                                                                                            | 227 | 0     | Inf. Dilution | lgK:173.1(4SF)     | 0 | ENS   | 6494   |
| 5                                                                                                                                                                            | 25  | 0     | Inf. Dilution | dG:-426.9(4SF)     | 2 | EFE   | 6494   |
| 6                                                                                                                                                                            | 25  | 0     | Inf. Dilution | dG:-1782.(2.MD)kJ  | 6 | MSL   | 26787  |
| 7                                                                                                                                                                            | 25  | 0     | Inf. Dilution | dG:-1782.3(5SF)kJ  | 5 | CRV   | 29482  |
| 8                                                                                                                                                                            | 25  | 0     | Inf. Dilution | dG:-425.91(5SF)    | 5 | CRV   | 29553  |
| 9                                                                                                                                                                            | 25  | 0     | Inf. Dilution | dH:-469.4(4SF)     | 0 | MTD   | 6494   |
| 10                                                                                                                                                                           | 25  | 0     | Inf. Dilution | dH:-1960.(7.MD)kJ  | 3 | MSL   | 26787  |
| 11                                                                                                                                                                           | 25  | 0     | Inf. Dilution | dH:-1939.1(5SF)kJ  | 0 | CRV   | 29482  |
| 12                                                                                                                                                                           | 25  | 0     | Inf. Dilution | dH:-468.45(5SF)    | 5 | CRV   | 29553  |
| 13                                                                                                                                                                           | 27  | 0     | Inf. Dilution | dH:-469.5(4SF)     | 0 | MTD   | 6494   |
| 14                                                                                                                                                                           | 127 | 0     | Inf. Dilution | dH:-473.3(4SF)     | 0 | MTD   | 6494   |
| 15                                                                                                                                                                           | 227 | 0     | Inf. Dilution | dH:-476.7(4SF)     | 0 | MTD   | 6494   |
| 16                                                                                                                                                                           | 25  | 0     | Inf. Dilution | Cp:142.6(0.3MD)J/K | 4 | MSL   | 26787  |



| Reaction No. 67405                                                                                                                                                           |     |       |               |                |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------|---------------|----------------|---|-------|--------|
| $3\text{Al(s)} + 3\text{H}_2\text{(g)} + \text{K(s)} + 7\text{O}_2\text{(g)} + 2\text{S(s)} = \text{Al}_3\text{(3)}_{\text{K}+1}\text{OH-1(6)}_{\text{SO}_4-2(2)}\text{(s)}$ |     |       |               |                |   |       |        |
| No                                                                                                                                                                           | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                                                                                            | 25  | 0     | Inf. Dilution | lgK:815.1(4SF) | 1 | ERG   |        |
| Note (1): Was 815.2,815.5 prev iters                                                                                                                                         |     |       |               |                |   |       |        |
| 2                                                                                                                                                                            | 25  | 0     | Inf. Dilution | lgK:817.0(4SF) | 0 | ENS   | 6494   |
| 3                                                                                                                                                                            | 27  | 0     | Inf. Dilution | lgK:811.4(4SF) | 0 | ENS   | 6494   |
| 4                                                                                                                                                                            | 127 | 0     | Inf. Dilution | lgK:585.9(4SF) | 0 | ENS   | 6494   |
| 5                                                                                                                                                                            | 227 | 0     | Inf. Dilution | lgK:450.5(4SF) | 0 | ENS   | 6494   |
| 6                                                                                                                                                                            | 327 | 0     | Inf. Dilution | lgK:360.2(4SF) | 0 | ENS   | 6494   |
| 7                                                                                                                                                                            | 25  | 0     | Inf. Dilution | dG:-1115.(4SF) | 0 | EFE   | 6494   |
| 8                                                                                                                                                                            | 25  | 0     | Inf. Dilution | dH:-1237.(4SF) | 1 | MTD   | 6494   |
| 9                                                                                                                                                                            | 127 | 0     | Inf. Dilution | dH:-1240.(4SF) | 0 | MTD   | 6494   |
| 10                                                                                                                                                                           | 227 | 0     | Inf. Dilution | dH:-1240.(4SF) | 0 | MTD   | 6494   |
| 11                                                                                                                                                                           | 327 | 0     | Inf. Dilution | dH:-1239.(4SF) | 0 | MTD   | 6494   |

| Reaction No. 67407                                                                                                                                                       |     |       |               |                   |   |       |        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------|---------------|-------------------|---|-------|--------|
| $\text{Al(s)} + \text{F}_2\text{(g)} + \text{K(s)} + 3\text{Mg(s)} + 5\text{O}_2\text{(g)} + 3\text{Si(s)} = \text{KMg}_3\text{AlSi}_3\text{O}_{10}\text{F}_2\text{(s)}$ |     |       |               |                   |   |       |        |
| No                                                                                                                                                                       | t   | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                                                                        | 25  | 0     | Inf. Dilution | lgK:1056.(4SF)    | 4 | ENS   | 6494   |
| 2                                                                                                                                                                        | 27  | 0     | Inf. Dilution | lgK:1050.(4SF)    | 0 | ENS   | 6494   |
| 3                                                                                                                                                                        | 127 | 0     | Inf. Dilution | lgK:771.9(4SF)    | 4 | ENS   | 6494   |
| 4                                                                                                                                                                        | 227 | 0     | Inf. Dilution | lgK:605.4(4SF)    | 4 | ENS   | 6494   |
| 5                                                                                                                                                                        | 327 | 0     | Inf. Dilution | lgK:494.4(4SF)    | 4 | ENS   | 6494   |
| 6                                                                                                                                                                        | 25  | 0     | Inf. Dilution | dG:-1441.(4SF)    | 4 | EFE   | 6494   |
| 7                                                                                                                                                                        | 25  | 0     | GAMSPATH      | dG:-6031.(4SF) kJ | 3 | CRV   | 12638  |
| 8                                                                                                                                                                        | 25  | 0     | Inf. Dilution | dH:-1524.(4SF)    | 4 | MTD   | 6494   |
| 9                                                                                                                                                                        | 127 | 0     | Inf. Dilution | dH:-1525.(4SF)    | 4 | MTD   | 6494   |
| 10                                                                                                                                                                       | 227 | 0     | Inf. Dilution | dH:-1524.(4SF)    | 4 | MTD   | 6494   |
| 11                                                                                                                                                                       | 327 | 0     | Inf. Dilution | dH:-1523.(4SF)    | 4 | MTD   | 6494   |

| Reaction No. 67408                                                                                      |    |       |               |                |   |       |        |
|---------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $\text{Al(s)} + \text{Na(s)} + 2\text{O}_2\text{(g)} + \text{Si(s)} = \text{NaAlSiO}_4\text{(glass,s)}$ |    |       |               |                |   |       |        |
| No                                                                                                      | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                       | 25 | 0     | Inf. Dilution | lgK:346.4(4SF) | 4 | ENS   | 6494   |
| 2                                                                                                       | 27 | 0     | Inf. Dilution | lgK:344.2(4SF) | 0 | ENS   | 6494   |

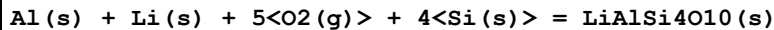
|    |     |   |               |                |   |     |      |
|----|-----|---|---------------|----------------|---|-----|------|
| 3  | 127 | 0 | Inf. Dilution | lgK:253.2(4SF) | 4 | ENS | 6494 |
| 4  | 227 | 0 | Inf. Dilution | lgK:198.6(4SF) | 3 | ENS | 6494 |
| 5  | 327 | 0 | Inf. Dilution | lgK:162.2(4SF) | 0 | ENS | 6494 |
| 6  | 25  | 0 | Inf. Dilution | dG:-472.6(4SF) | 4 | EFE | 6494 |
| 7  | 25  | 0 | Inf. Dilution | dH:-499.3(4SF) | 4 | MTD | 6494 |
| 8  | 127 | 0 | Inf. Dilution | dH:-500.0(4SF) | 4 | MTD | 6494 |
| 9  | 227 | 0 | Inf. Dilution | dH:-499.9(4SF) | 3 | MTD | 6494 |
| 10 | 327 | 0 | Inf. Dilution | dH:-499.6(4SF) | 0 | MTD | 6494 |

| Reaction No. 67409                                   |     |       |               |                  |   |       |        |
|------------------------------------------------------|-----|-------|---------------|------------------|---|-------|--------|
| Al(s) + Na(s) + 2<O2(g)> + Si(s) = NaAlSiO4(Carn.,s) |     |       |               |                  |   |       |        |
| No                                                   | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                    | 25  | 0     | Inf. Dilution | lgK:345.7(4SF)   | 4 | ENS   | 6494   |
| 2                                                    | 27  | 0     | Inf. Dilution | lgK:343.4(4SF)   | 0 | ENS   | 6494   |
| 3                                                    | 127 | 0     | Inf. Dilution | lgK:252.4(4SF)   | 4 | ENS   | 6494   |
| 4                                                    | 227 | 0     | Inf. Dilution | lgK:197.8(4SF)   | 3 | ENS   | 6494   |
| 5                                                    | 327 | 0     | Inf. Dilution | lgK:161.3(4SF)   | 0 | ENS   | 6494   |
| 6                                                    | 25  | 0     | Inf. Dilution | dG:-471.6(4SF)   | 4 | EFE   | 6494   |
| 7                                                    | 25  | 0     | Inf. Dilution | dG:-1958.(4SF)kJ | 2 | EFE   | 24802  |
| 8                                                    | 25  | 0     | Inf. Dilution | dH:-499.4(4SF)   | 4 | MTD   | 6494   |
| 9                                                    | 127 | 0     | Inf. Dilution | dH:-500.1(4SF)   | 4 | MTD   | 6494   |
| 10                                                   | 227 | 0     | Inf. Dilution | dH:-500.0(4SF)   | 3 | MTD   | 6494   |
| 11                                                   | 327 | 0     | Inf. Dilution | dH:-499.8(4SF)   | 0 | MTD   | 6494   |

| Reaction No. 67410                                       |     |       |               |                |   |       |        |
|----------------------------------------------------------|-----|-------|---------------|----------------|---|-------|--------|
| Al(s) + Na(s) + 4<O2(g)> + 3<Si(s)> = NaAlSi3O8(glass,s) |     |       |               |                |   |       |        |
| No                                                       | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                        | 25  | 0     | Inf. Dilution | lgK:642.1(4SF) | 4 | ENS   | 6494   |
| 2                                                        | 27  | 0     | Inf. Dilution | lgK:637.9(4SF) | 4 | ENS   | 6494   |
| 3                                                        | 127 | 0     | Inf. Dilution | lgK:469.2(4SF) | 4 | ENS   | 6494   |
| 4                                                        | 227 | 0     | Inf. Dilution | lgK:367.9(4SF) | 4 | ENS   | 6494   |
| 5                                                        | 327 | 0     | Inf. Dilution | lgK:300.4(4SF) | 4 | ENS   | 6494   |
| 6                                                        | 25  | 0     | Inf. Dilution | dG:-876.0(4SF) | 4 | EFE   | 6494   |
| 7                                                        | 25  | 0     | Inf. Dilution | dH:-926.3(4SF) | 4 | MTD   | 6494   |
| 8                                                        | 127 | 0     | Inf. Dilution | dH:-927.1(4SF) | 4 | MTD   | 6494   |
| 9                                                        | 227 | 0     | Inf. Dilution | dH:-926.8(4SF) | 4 | MTD   | 6494   |

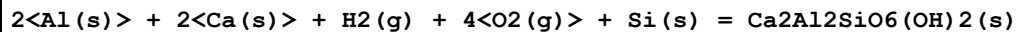
|    |     |   |               |                |   |     |      |
|----|-----|---|---------------|----------------|---|-----|------|
| 10 | 327 | 0 | Inf. Dilution | dH:-926.3(4SF) | 4 | MTD | 6494 |
|----|-----|---|---------------|----------------|---|-----|------|

## Reaction No. 67411



| No | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|-----|-------|---------------|----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:807.7(4SF) | 4 | ENS   | 6494   |
| 2  | 27  | 0     | Inf. Dilution | lgK:802.5(4SF) | 0 | ENS   | 6494   |
| 3  | 127 | 0     | Inf. Dilution | lgK:589.7(4SF) | 3 | ENS   | 6494   |
| 4  | 227 | 0     | Inf. Dilution | lgK:462.1(4SF) | 2 | ENS   | 6494   |
| 5  | 327 | 0     | Inf. Dilution | lgK:376.9(4SF) | 0 | ENS   | 6494   |
| 6  | 25  | 0     | Inf. Dilution | dG:-1102.(4SF) | 4 | EFE   | 6494   |
| 7  | 25  | 0     | Inf. Dilution | dH:-1168.(4SF) | 4 | MTD   | 6494   |
| 8  | 127 | 0     | Inf. Dilution | dH:-1168.(4SF) | 3 | MTD   | 6494   |
| 9  | 227 | 0     | Inf. Dilution | dH:-1169.(4SF) | 2 | MTD   | 6494   |
| 10 | 327 | 0     | Inf. Dilution | dH:-1168.(4SF) | 0 | MTD   | 6494   |

## Reaction No. 67412



| No | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|-----|-------|---------------|----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:713.6(4SF) | 4 | ENS   | 6494   |
| 2  | 27  | 0     | Inf. Dilution | lgK:708.9(4SF) | 4 | ENS   | 6494   |
| 3  | 127 | 0     | Inf. Dilution | lgK:519.9(4SF) | 4 | ENS   | 6494   |
| 4  | 227 | 0     | Inf. Dilution | lgK:406.5(4SF) | 4 | ENS   | 6494   |
| 5  | 327 | 0     | Inf. Dilution | lgK:330.9(4SF) | 4 | ENS   | 6494   |
| 6  | 25  | 0     | Inf. Dilution | dG:-973.5(4SF) | 4 | EFE   | 6494   |
| 7  | 25  | 0     | Inf. Dilution | dH:-1038.(4SF) | 4 | MTD   | 6494   |
| 8  | 127 | 0     | Inf. Dilution | dH:-1038.(4SF) | 4 | MTD   | 6494   |
| 9  | 227 | 0     | Inf. Dilution | dH:-1038.(4SF) | 4 | MTD   | 6494   |
| 10 | 327 | 0     | Inf. Dilution | dH:-1037.(4SF) | 4 | MTD   | 6494   |

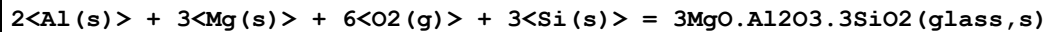
## Reaction No. 67413



| No | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|-----|-------|---------------|----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:691.7(4SF) | 4 | ENS   | 6494   |
| 2  | 27  | 0     | Inf. Dilution | lgK:687.2(4SF) | 4 | ENS   | 6494   |
| 3  | 127 | 0     | Inf. Dilution | lgK:506.0(4SF) | 4 | ENS   | 6494   |
| 4  | 227 | 0     | Inf. Dilution | lgK:397.3(4SF) | 4 | ENS   | 6494   |

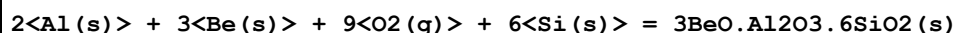
|    |     |   |               |                 |   |     |      |
|----|-----|---|---------------|-----------------|---|-----|------|
| 5  | 327 | 0 | Inf. Dilution | lgK:324.8 (4SF) | 4 | ENS | 6494 |
| 6  | 25  | 0 | Inf. Dilution | dG:-943.7 (4SF) | 4 | EFE | 6494 |
| 7  | 25  | 0 | Inf. Dilution | dH:-995.0 (4SF) | 4 | MTD | 6494 |
| 8  | 127 | 0 | Inf. Dilution | dH:-995.2 (4SF) | 4 | MTD | 6494 |
| 9  | 227 | 0 | Inf. Dilution | dH:-994.9 (4SF) | 4 | MTD | 6494 |
| 10 | 327 | 0 | Inf. Dilution | dH:-994.4 (4SF) | 4 | MTD | 6494 |

## Reaction No. 67417



| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:1022. (4SF) | 4 | ENS   | 6494   |
| 2  | 27  | 0     | Inf. Dilution | lgK:1016. (4SF) | 0 | ENS   | 6494   |
| 3  | 127 | 0     | Inf. Dilution | lgK:747.6 (4SF) | 4 | ENS   | 6494   |
| 4  | 227 | 0     | Inf. Dilution | lgK:586.8 (4SF) | 4 | ENS   | 6494   |
| 5  | 327 | 0     | Inf. Dilution | lgK:479.7 (4SF) | 4 | ENS   | 6494   |
| 6  | 25  | 0     | Inf. Dilution | dG:-1395. (4SF) | 4 | EFE   | 6494   |
| 7  | 25  | 0     | Inf. Dilution | dH:-1473. (4SF) | 4 | MTD   | 6494   |
| 8  | 127 | 0     | Inf. Dilution | dH:-1472. (4SF) | 4 | MTD   | 6494   |
| 9  | 227 | 0     | Inf. Dilution | dH:-1471. (4SF) | 4 | MTD   | 6494   |
| 10 | 327 | 0     | Inf. Dilution | dH:-1470. (4SF) | 4 | MTD   | 6494   |

## Reaction No. 67423



| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:1489. (4SF) | 4 | ENS   | 6494   |
| 2  | 27  | 0     | Inf. Dilution | lgK:1479. (4SF) | 0 | ENS   | 6494   |
| 3  | 127 | 0     | Inf. Dilution | lgK:1087. (4SF) | 4 | ENS   | 6494   |
| 4  | 227 | 0     | Inf. Dilution | lgK:852.1 (4SF) | 4 | ENS   | 6494   |
| 5  | 327 | 0     | Inf. Dilution | lgK:695.2 (4SF) | 4 | ENS   | 6494   |
| 6  | 25  | 0     | Inf. Dilution | dG:-2032. (4SF) | 4 | EFE   | 6494   |
| 7  | 25  | 0     | Inf. Dilution | dH:-2153. (4SF) | 4 | MTD   | 6494   |
| 8  | 27  | 0     | Inf. Dilution | dH:-2153. (4SF) | 0 | MTD   | 6494   |
| 9  | 127 | 0     | Inf. Dilution | dH:-2153. (4SF) | 4 | MTD   | 6494   |
| 10 | 227 | 0     | Inf. Dilution | dH:-2153. (4SF) | 4 | MTD   | 6494   |
| 11 | 327 | 0     | Inf. Dilution | dH:-2152. (4SF) | 4 | MTD   | 6494   |

## Reaction No. 67424

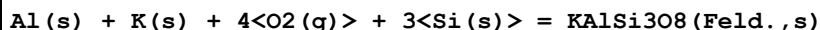
| Al(s) + Be(s) + 0.5<H2(g)> + 2.5<O2(g)> + Si(s) = BeAlSiO4(OH)(s) |     |       |               |                |   |       |        |
|-------------------------------------------------------------------|-----|-------|---------------|----------------|---|-------|--------|
| No                                                                | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                 | 25  | 0     | Inf. Dilution | lgK:415.2(4SF) | 4 | ENS   | 6494   |
| 2                                                                 | 27  | 0     | Inf. Dilution | lgK:412.5(4SF) | 4 | ENS   | 6494   |
| 3                                                                 | 127 | 0     | Inf. Dilution | lgK:302.2(4SF) | 4 | ENS   | 6494   |
| 4                                                                 | 227 | 0     | Inf. Dilution | lgK:236.0(4SF) | 4 | ENS   | 6494   |
| 5                                                                 | 327 | 0     | Inf. Dilution | lgK:191.9(4SF) | 4 | ENS   | 6494   |
| 6                                                                 | 25  | 0     | Inf. Dilution | dG:-566.5(4SF) | 4 | EFE   | 6494   |
| 7                                                                 | 25  | 0     | Inf. Dilution | dH:-605.4(4SF) | 4 | MTD   | 6494   |
| 8                                                                 | 127 | 0     | Inf. Dilution | dH:-605.6(4SF) | 4 | MTD   | 6494   |
| 9                                                                 | 227 | 0     | Inf. Dilution | dH:-605.5(4SF) | 4 | MTD   | 6494   |
| 10                                                                | 327 | 0     | Inf. Dilution | dH:-605.1(4SF) | 4 | MTD   | 6494   |

| Reaction No. 67425                                                                   |     |       |               |                |   |       |        |
|--------------------------------------------------------------------------------------|-----|-------|---------------|----------------|---|-------|--------|
| 6.75<Al(s)> + B(s) + 0.375<H2(g)> + 9<O2(g)> + 3<Si(s)> = Al6.75Si3BO17.25(OH).75(s) |     |       |               |                |   |       |        |
| No                                                                                   | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                    | 25  | 0     | Inf. Dilution | lgK:1501.(4SF) | 3 | ENS   | 6494   |
| 2                                                                                    | 27  | 0     | Inf. Dilution | lgK:1491.(4SF) | 0 | ENS   | 6494   |
| 3                                                                                    | 127 | 0     | Inf. Dilution | lgK:1095.(4SF) | 3 | ENS   | 6494   |
| 4                                                                                    | 227 | 0     | Inf. Dilution | lgK:856.7(4SF) | 3 | ENS   | 6494   |
| 5                                                                                    | 327 | 0     | Inf. Dilution | lgK:698.1(4SF) | 3 | ENS   | 6494   |
| 6                                                                                    | 25  | 0     | Inf. Dilution | dG:-2048.(4SF) | 3 | EFE   | 6494   |
| 7                                                                                    | 25  | 0     | Inf. Dilution | dH:-2177.(4SF) | 3 | MTD   | 6494   |
| 8                                                                                    | 127 | 0     | Inf. Dilution | dH:-2178.(4SF) | 3 | MTD   | 6494   |
| 9                                                                                    | 227 | 0     | Inf. Dilution | dH:-2178.(4SF) | 3 | MTD   | 6494   |
| 10                                                                                   | 327 | 0     | Inf. Dilution | dH:-2177.(4SF) | 3 | MTD   | 6494   |

| Reaction No. 67426                                 |     |       |               |                |   |       |        |
|----------------------------------------------------|-----|-------|---------------|----------------|---|-------|--------|
| 2<Al(s)> + F2(g) + 2<O2(g)> + Si(s) = Al2SiO4F2(s) |     |       |               |                |   |       |        |
| No                                                 | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                  | 25  | 0     | Inf. Dilution | lgK:509.9(4SF) | 4 | ENS   | 6494   |
| 2                                                  | 27  | 0     | Inf. Dilution | lgK:506.6(4SF) | 4 | ENS   | 6494   |
| 3                                                  | 127 | 0     | Inf. Dilution | lgK:372.3(4SF) | 4 | ENS   | 6494   |
| 4                                                  | 227 | 0     | Inf. Dilution | lgK:291.8(4SF) | 4 | ENS   | 6494   |
| 5                                                  | 327 | 0     | Inf. Dilution | lgK:238.1(4SF) | 4 | ENS   | 6494   |
| 6                                                  | 25  | 0     | Inf. Dilution | dG:-695.7(4SF) | 4 | EFE   | 6494   |

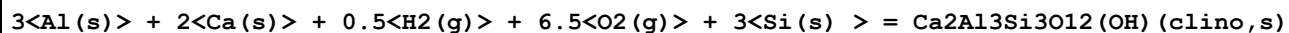
|    |     |   |               |                 |   |     |      |
|----|-----|---|---------------|-----------------|---|-----|------|
| 7  | 25  | 0 | Inf. Dilution | dH:-737.2 (4SF) | 4 | MTD | 6494 |
| 8  | 127 | 0 | Inf. Dilution | dH:-737.3 (4SF) | 4 | MTD | 6494 |
| 9  | 227 | 0 | Inf. Dilution | dH:-737.0 (4SF) | 4 | MTD | 6494 |
| 10 | 327 | 0 | Inf. Dilution | dH:-736.5 (4SF) | 4 | MTD | 6494 |

## Reaction No. 67430



| No | t   | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
|----|-----|-------|---------------|----------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:658. (3SF)       | 1 | ELC   |        |
| 2  | 25  | 0     | Inf. Dilution | dG:-895.18 (5SF)     | 0 | EFE   | 6482   |
| 3  | 25  | 0     | Inf. Dilution | dG:-897.07 (5SF)     | 0 | CRV   | 10174  |
| 4  | 25  | 0     | Inf. Dilution | dG:-896.44 (5SF)     | 5 | CRV   | 29553  |
| 5  | 25  | 0     | GAMSPATH      | dG:-3737. (4SF) kJ   | 0 | CRV   | 12638  |
| 6  | 50  | 0     | Inf. Dilution | dG:-898.39 (5SF)     | 0 | CRV   | 10174  |
| 7  | 75  | 0     | Inf. Dilution | dG:-899.81 (5SF)     | 0 | CRV   | 10174  |
| 8  | 100 | 0     | Inf. Dilution | dG:-901.33 (5SF)     | 0 | CRV   | 10174  |
| 9  | 125 | 0     | Inf. Dilution | dG:-902.93 (5SF)     | 0 | CRV   | 10174  |
| 10 | 150 | 0     | Inf. Dilution | dG:-904.63 (5SF)     | 0 | CRV   | 10174  |
| 11 | 175 | 0     | Inf. Dilution | dG:-906.42 (5SF)     | 0 | CRV   | 10174  |
| 12 | 200 | 0     | Inf. Dilution | dG:-908.28 (5SF)     | 0 | CRV   | 10174  |
| 13 | 225 | 0     | Inf. Dilution | dG:-910.23 (5SF)     | 0 | CRV   | 10174  |
| 14 | 250 | 0     | Inf. Dilution | dG:-912.25 (5SF)     | 0 | CRV   | 10174  |
| 15 | 300 | 0     | Inf. Dilution | dG:-916.48 (5SF)     | 0 | CRV   | 10174  |
| 16 | 25  | 0     | Inf. Dilution | dH:-949.04 (5SF)     | 0 | EFE   | 6482   |
| 17 | 25  | 0     | Inf. Dilution | dH:-950.30 (5SF)     | 5 | CRV   | 29553  |
| 18 | 25  | 0     | GAMSPATH      | dH:-3962.40 (6SF) kJ | 0 | CRV   | 12638  |

## Reaction No. 67432



| No | t  | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:1137. (4SF)      | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | dG:-1552.7 (5SF)     | 0 | EFE   | 6482   |
| 3  | 25 | 0     | GAMSPATH      | dG:-6484. (4SF) kJ   | 0 | CRV   | 12638  |
| 4  | 25 | 0     | Inf. Dilution | dH:-1647.9 (5SF)     | 2 | EFE   | 6482   |
| 5  | 25 | 0     | GAMSPATH      | dH:-6879.40 (6SF) kJ | 0 | CRV   | 12638  |

## Reaction No. 67433

| Al+3_OH-1(3)_(Micr.Gibb.,s) + 3<H+1> = Al+3 + 3<H2O> |    |       |               |               |   |       |        |
|------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| No                                                   | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                    | 25 | 0     | Inf. Dilution | lgK:9.35(3SF) | 4 | ENS   | 6496   |
| 2                                                    | 25 | 0     | Inf. Dilution | dH:-24.5(3SF) | 4 | ENS   | 6496   |

| Reaction No. 67434                                                       |    |       |               |                  |   |       |        |
|--------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Al2Si2O5(OH)4(Kaol.,s) + 6<H+1> = 2<Al+3> + 2<H+1(2)_SiH2O4-2> + H2O     |    |       |               |                  |   |       |        |
| No                                                                       | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                        | 25 | 0     | Inf. Dilution | lgK:7.0(2SF)     | 1 | ELC   |        |
| 2                                                                        | 25 | 0     | Inf. Dilution | lgK:7.428(4SF)   | 0 | CRV   | 440    |
| 3                                                                        | 25 | 0     | Inf. Dilution | lgK:5.706(4SF)   | 0 | CRV   | 5619   |
| 4                                                                        | 25 | 0     | Inf. Dilution | lgK:11.122(5SF)  | 0 | CRV   | 6421   |
| 5                                                                        | 25 | 0     | Inf. Dilution | lgK:7.435(4SF)   | 0 | ENS   | 6496   |
| 6                                                                        | 25 | 0     | Inf. Dilution | lgK:5.45(3SF)    | 0 | MGL   | 6498   |
| Note (6): Possibly for reaction with halved stoichiometry, possibly not! |    |       |               |                  |   |       |        |
| 7                                                                        | 25 | 0     | Inf. Dilution | lgK:7.41(3SF)    | 0 | CRV   | 8872   |
| 8                                                                        | 25 | 0     | Inf. Dilution | lgK:11.122(5SF)  | 0 | EOD   | 8872   |
| Note (8): Data for Naumov, Ryzhenko & KKhodakovsky, 1971                 |    |       |               |                  |   |       |        |
| 9                                                                        | 25 | 0     | Inf. Dilution | lgK:7.028(4SF)   | 0 | CRV   | 8873   |
| 10                                                                       | 25 | 0     | Inf. Dilution | lgK:7.604(4SF)   | 0 | CRV   | 8874   |
| 11                                                                       | 25 | 0     | Inf. Dilution | lgK:7.435(4SF)   | 3 | EOD   | 29487  |
| Note (11): Presumed reference to Kaolinite                               |    |       |               |                  |   |       |        |
| 12                                                                       | 25 | 0     | Inf. Dilution | lgK:6.5(2SF)     | 3 | MSL   | 31760  |
| Note (12): Slightly impure solid & average                               |    |       |               |                  |   |       |        |
| 13                                                                       | 25 | 0     | Inf. Dilution | lgK:7.446(4SF)   | 0 | CRV   | 32224  |
| 14                                                                       | 40 | 0     | Inf. Dilution | lgK:6.(1SF)      | 5 | MSL   | 31760  |
| Note (14): Slightly impure solid & average                               |    |       |               |                  |   |       |        |
| 15                                                                       | 90 | 0     | Inf. Dilution | lgK:2.80(3SF)    | 3 | EOD   | 29487  |
| Note (15): Presumed reference to Kaolinite                               |    |       |               |                  |   |       |        |
| 16                                                                       | 25 | 0     | Inf. Dilution | dH:-35.3(3SF)    | 2 | ENS   | 6496   |
| 17                                                                       | 25 | 0     | Inf. Dilution | dH:-147.7(4SF)kJ | 3 | EOD   | 29487  |
| Note (17): Presumed reference to Kaolinite                               |    |       |               |                  |   |       |        |

| Reaction No. 67454                                      |   |       |        |             |   |       |        |
|---------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| Al+3_OH-1(3)_(Nordstrandite,s) + 3<H+1> = Al+3 + 3<H2O> |   |       |        |             |   |       |        |
| No                                                      | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

|   |     |   |               |               |   |     |      |
|---|-----|---|---------------|---------------|---|-----|------|
| 1 | 25  | 0 | Inf. Dilution | lgK:8.13(3SF) | 4 | MGL | 6498 |
| 2 | 100 | 0 | Inf. Dilution | lgK:6.5(2SF)  | 1 | ELC |      |

| Reaction No. 67455                                                                                                                                                                                              |    |       |               |               |   |       |        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4(\text{Hall.}, \text{s}) + 6\langle\text{H}+1\rangle = 2\langle\text{Al}+3\rangle + 2\langle\text{H}+1(2)\rangle_{\text{SiH}_2\text{O}_4-2} + \text{H}_2\text{O}$ |    |       |               |               |   |       |        |
| No                                                                                                                                                                                                              | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                                                                                                                                               | 25 | 0     | Inf. Dilution | lgK:10.4(3SF) | 1 | ELC   |        |
| 2                                                                                                                                                                                                               | 25 | 0     | Inf. Dilution | lgK:8.72(3SF) | 0 | MGL   | 6498   |

| Reaction No. 67456                                                                                                                                                                                |    |       |               |                |   |       |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $\text{Al}_2\text{O}_3\text{SiOH}(\text{OH})_3(\text{s}) + 6\langle\text{H}+1\rangle = 2\langle\text{Al}+3\rangle + \text{H}+1(2)_{\text{SiH}_2\text{O}_4-2} + 3\langle\text{H}_2\text{O}\rangle$ |    |       |               |                |   |       |        |
| No                                                                                                                                                                                                | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                                                                                                                 | 25 | 0     | Inf. Dilution | lgK:14.1(3SF)  | 1 | ELC   |        |
| 2                                                                                                                                                                                                 | 25 | 0     | Inf. Dilution | lgK:12.00(4SF) | 0 | MGL   | 6498   |

| Reaction No. 67597                                                                     |    |       |        |                 |   |       |        |
|----------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $2\langle 2\text{OxButan-1}\rangle + \text{Al}+3 = \text{Al}+3_{2\text{OxButan-1}(2)}$ |    |       |        |                 |   |       |        |
| No                                                                                     | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                      | 25 | 0.1   | KCl    | lgK:3.7(0.03SD) | 5 | MGL   | 6523   |

| Reaction No. 67606                                                                              |    |       |        |                  |   |       |        |
|-------------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $2\langle\text{aKetoglutaric-2}\rangle + \text{Al}+3 = \text{Al}+3_{\text{aKetoglutaric-2}(2)}$ |    |       |        |                  |   |       |        |
| No                                                                                              | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                               | 25 | 0.1   | KCl    | lgK:10.8(0.05SD) | 5 | MGL   | 6523   |

| Reaction No. 67950                                                                |    |       |               |                |   |       |        |
|-----------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $2\langle\text{Acetic-1}\rangle + \text{Al}+3 = \text{Al}+3_{\text{Acetic-1}(2)}$ |    |       |               |                |   |       |        |
| No                                                                                | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                 | 0  | 0     | Inf. Dilution | lgK:4.26(3SF)  | 4 | CRV   | 8888   |
| 2                                                                                 | 25 | 0     | Inf. Dilution | lgK:7.35(3SF)  | 0 | ENS   | 6953   |
| 3                                                                                 | 25 | 0     | Inf. Dilution | lgK:4.60(3SF)  | 4 | CRV   | 8888   |
| 4                                                                                 | 25 | 0     | NaCl/m        | lgK:4.6(0.6MD) | 3 | MEF   | 4277   |
| 5                                                                                 | 25 | 0.1   | NaCl/m        | lgK:3.5(0.6MD) | 4 | MEF   | 4277   |
| 6                                                                                 | 25 | 0.2   | NaCl/m        | lgK:3.2(0.6MD) | 7 | MEF   | 4277   |
| 7                                                                                 | 25 | 0.5   | NaCl/m        | lgK:3.0(0.6MD) | 7 | MEF   | 4277   |
| 8                                                                                 | 25 | 1     | NaCl/m        | lgK:3.1(0.6MD) | 7 | MEF   | 4277   |
| 9                                                                                 | 50 | 0     | Inf. Dilution | lgK:5.17(3SF)  | 4 | CRV   | 8888   |

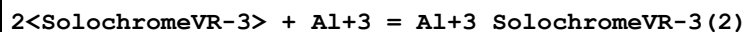


|              |     |      |               |                   |   |     |      |
|--------------|-----|------|---------------|-------------------|---|-----|------|
| 10           | 50  | 0    | NaCl/m        | lgK:5.1 (0.3MD)   | 4 | MEF | 4277 |
| 11           | 50  | 0.1  | NaCl/m        | lgK:3.9 (0.3MD)   | 7 | MEF | 4277 |
| 12           | 50  | 0.2  | NaCl/m        | lgK:3.7 (0.3MD)   | 7 | MEF | 4277 |
| 13           | 50  | 0.5  | NaCl/m        | lgK:3.5 (0.3MD)   | 7 | MEF | 4277 |
| 14           | 50  | 1    | NaCl/m        | lgK:3.6 (0.3MD)   | 7 | MEF | 4277 |
| 15           | 60  | 0.02 | NaCl/m        | lgK:4.10 (0.06SD) | 0 | MSL | 6039 |
| 16           | 75  | 0    | Inf. Dilution | lgK:5.84 (3SF)    | 4 | CRV | 8888 |
| 17           | 75  | 0    | NaCl/m        | lgK:5.8 (0.2MD)   | 4 | MEF | 4277 |
| 18           | 75  | 0.1  | NaCl/m        | lgK:4.4 (0.2MD)   | 7 | MEF | 4277 |
| 19           | 75  | 0.2  | NaCl/m        | lgK:4.2 (0.2MD)   | 7 | MEF | 4277 |
| 20           | 75  | 0.5  | NaCl/m        | lgK:4.0 (0.1MD)   | 7 | MEF | 4277 |
| 21           | 75  | 1    | NaCl/m        | lgK:4.2 (0.1MD)   | 7 | MEF | 4277 |
| 22           | 80  | 0    | Inf. Dilution | lgK:7.7 (2SF)     | 0 | ENS | 6953 |
| 23           | 100 | 0    | Inf. Dilution | lgK:6.57 (3SF)    | 4 | CRV | 8888 |
| 24           | 100 | 0    | NaCl/m        | lgK:6.4 (0.2MD)   | 4 | MEF | 4277 |
| 25           | 100 | 0.1  | NaCl/m        | lgK:5.1 (0.2MD)   | 7 | MEF | 4277 |
| 26           | 100 | 0.2  | NaCl/m        | lgK:4.8 (0.2MD)   | 7 | MEF | 4277 |
| 27           | 100 | 0.5  | NaCl/m        | lgK:4.6 (0.1MD)   | 7 | MEF | 4277 |
| 28           | 100 | 1    | NaCl/m        | lgK:4.78 (0.09MD) | 7 | MEF | 4277 |
| 29           | 120 | 0.01 | NaCl/m        | lgK:6.15 (0.07SD) | 5 | MSL | 6039 |
| 30           | 125 | 0    | Inf. Dilution | lgK:7.34 (3SF)    | 4 | CRV | 8888 |
| 31           | 125 | 0    | NaCl/m        | lgK:7.2 (0.2MD)   | 4 | MEF | 4277 |
| 32           | 125 | 0.1  | NaCl/m        | lgK:5.7 (0.2MD)   | 7 | MEF | 4277 |
| 33           | 125 | 0.2  | NaCl/m        | lgK:5.4 (0.2MD)   | 7 | MEF | 4277 |
| 34           | 125 | 0.5  | NaCl/m        | lgK:5.2 (0.1MD)   | 7 | MEF | 4277 |
| 35           | 125 | 1    | NaCl/m        | lgK:5.38 (0.07MD) | 7 | MEF | 4277 |
| 36           | 150 | 0    | Inf. Dilution | lgK:8.15 (3SF)    | 4 | CRV | 8888 |
| 37           | 150 | 0    | NaCl/m        | lgK:7.9 (0.3MD)   | 4 | MEF | 4277 |
| Note (37): n |     |      |               |                   |   |     |      |
| 38           | 150 | 0.1  | NaCl/m        | lgK:6.4 (0.3MD)   | 7 | MEF | 4277 |
| 39           | 150 | 0.2  | NaCl/m        | lgK:6.0 (0.2MD)   | 4 | MEF | 4277 |
| 40           | 150 | 0.5  | NaCl/m        | lgK:5.7 (0.1MD)   | 4 | MEF | 4277 |
| 41           | 150 | 1    | NaCl/m        | lgK:5.98 (0.07MD) | 7 | MEF | 4277 |
| 42           | 170 | 0.01 | NaCl/m        | lgK:7.43 (3SF)    | 0 | MSL | 6039 |
| 43           | 175 | 0    | Inf. Dilution | lgK:8.99 (3SF)    | 4 | CRV | 8888 |
| 44           | 200 | 0    | Inf. Dilution | lgK:9.85 (3SF)    | 4 | CRV | 8888 |

|    |     |         |               |                      |   |     |      |
|----|-----|---------|---------------|----------------------|---|-----|------|
| 45 | 200 | 0.1     | NaCl/m        | lgK:0.024 (0.18SD)   | 0 | MSL | 6039 |
| 46 | 225 | 0       | Inf. Dilution | lgK:10.7 (3SF)       | 4 | CRV | 8888 |
| 47 | 250 | 0       | Inf. Dilution | lgK:11.7 (3SF)       | 4 | CRV | 8888 |
| 48 | 275 | 0       | Inf. Dilution | lgK:12.7 (3SF)       | 4 | CRV | 8888 |
| 49 | 300 | 0       | Inf. Dilution | lgK:13.8 (3SF)       | 0 | CRV | 8888 |
| 50 | 325 | 0       | Inf. Dilution | lgK:14.9 (3SF)       | 0 | CRV | 8888 |
| 51 | 350 | 0       | Inf. Dilution | lgK:16.0 (3SF)       | 0 | CRV | 8888 |
| 52 | 25  | 0       | NaCl/m        | dH:30. (30.MD) kJ    | 3 | MTD | 4277 |
| 53 | 25  | 0.1:1   | NaCl/m        | dH:40. (30.MD) kJ    | 0 | MTD | 4277 |
| 54 | 25  | 0.2:1   | NaCl/m        | dH:40. (30.MD) kJ    | 0 | MTD | 4277 |
| 55 | 50  | 0:0.5   | NaCl/m        | dH:50. (20.MD) kJ    | 3 | MTD | 4277 |
| 56 | 50  | 1       | NaCl/m        | dH:60. (20.MD) kJ    | 0 | MTD | 4277 |
| 57 | 75  | 0:0.2   | NaCl/m        | dH:60. (10.MD) kJ    | 5 | MTD | 4277 |
| 58 | 75  | 0.5:1   | NaCl/m        | dH:70. (10.MD) kJ    | 0 | MTD | 4277 |
| 59 | 100 | 0       | NaCl/m        | dH:74. (7.MD) kJ     | 3 | MTD | 4277 |
| 60 | 100 | 0.1     | NaCl/m        | dH:76. (7.MD) kJ     | 0 | MTD | 4277 |
| 61 | 100 | 0.2     | NaCl/m        | dH:77. (7.MD) kJ     | 0 | MTD | 4277 |
| 62 | 100 | 0.5     | NaCl/m        | dH:81. (7.MD) kJ     | 0 | MTD | 4277 |
| 63 | 100 | 1       | NaCl/m        | dH:88. (6.MD) kJ     | 0 | MCL | 4277 |
| 64 | 125 | 0       | NaCl/m        | dH:90. (8.SD) kJ     | 3 | MTD | 4277 |
| 65 | 125 | 0.1     | NaCl/m        | dH:91. (9.MD) kJ     | 0 | MTD | 4277 |
| 66 | 125 | 0.2     | NaCl/m        | dH:93. (8.MD) kJ     | 0 | MTD | 4277 |
| 67 | 125 | 0.5     | NaCl/m        | dH:98. (8.MD) kJ     | 0 | MTD | 4277 |
| 68 | 125 | 1       | NaCl/m        | dH:105. (7.MD) kJ    | 0 | MTD | 4277 |
| 69 | 150 | 0.1:0.2 | NaCl/m        | dH:110. (20.MD) kJ   | 0 | MTD | 4277 |
| 70 | 150 | 0       | NaCl/m        | dH:100. (20.MD) kJ   | 3 | MTD | 4277 |
| 71 | 150 | 0.5:1   | NaCl/m        | dH:120. (20.MD) kJ   | 0 | MTD | 4277 |
| 72 | 25  | 0:1     | NaCl/m        | Cp:500. (300.MD) J/K | 7 | MTD | 4277 |
| 73 | 50  | 0:0.5   | NaCl/m        | Cp:500. (300.MD) J/K | 7 | MTD | 4277 |
| 74 | 50  | 1       | NaCl/m        | Cp:600. (300.MD) J/K | 7 | MTD | 4277 |
| 75 | 75  | 0:1     | NaCl/m        | Cp:600. (300.MD) J/K | 7 | MTD | 4277 |
| 76 | 100 | 0:0.5   | NaCl/m        | Cp:600. (300.MD) J/K | 7 | MTD | 4277 |
| 77 | 100 | 1       | NaCl/m        | Cp:700. (300.MD) J/K | 7 | MTD | 4277 |
| 78 | 125 | 0:0.1   | NaCl/m        | Cp:600. (400.MD) J/K | 7 | MTD | 4277 |
| 79 | 125 | 0.2:1   | NaCl/m        | Cp:700. (400.MD) J/K | 7 | MTD | 4277 |
| 80 | 150 | 0:0.5   | NaCl/m        | Cp:700. (400.MD) J/K | 7 | MTD | 4277 |

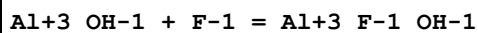
|    |     |   |        |                     |   |     |      |
|----|-----|---|--------|---------------------|---|-----|------|
| 81 | 150 | 1 | NaCl/m | Cp:800.(400.MD) J/K | 7 | MTD | 4277 |
|----|-----|---|--------|---------------------|---|-----|------|

## Reaction No. 67951



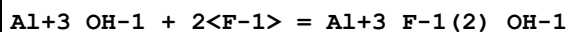
| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:35.4(3SF) | 4 | MPS   | 7000   |

## Reaction No. 67997



| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:6.13(3SF) | 4 | ENS   | 6953   |

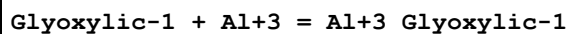
## Reaction No. 67998



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:10.36(4SF) | 1 | ENS   | 6953   |

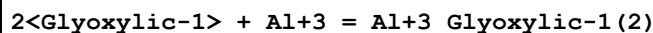
Note (1): p. 52, eqn (12)

## Reaction No. 68000



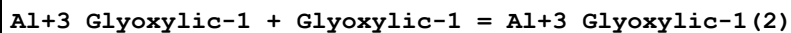
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:13.5(3SF) | 5 | MGL   | 3187   |

## Reaction No. 68001



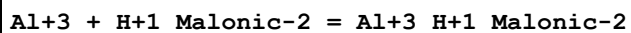
| No | t  | I-Str | Medium    | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|-----------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | Cl-1 salt | lgK:22.80(4SF) | 4 | ENS   | 6953   |

## Reaction No. 68002



| No | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
|----|----|-------|--------|--------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:9.3(2SF) | 5 | MGL   | 3187   |

## Reaction No. 68009



| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 1     | NaClO4 | lgK:3.58(3SF) | 5 | MGL   | 1243   |
| 2  | 35 | 1     | NaClO4 | lgK:4.04(3SF) | 5 | MGL   | 1243   |

| Reaction No. 68010                                                           |    |       |        |                |   |       |        |
|------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Al}+3 + 2\text{<H+1\_Malonic-2>} = \text{Al}+3\_H+1(2)\_Malonic-2(2)$ |    |       |        |                |   |       |        |
| No                                                                           | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                            | 25 | 1     | NaClO4 | lgK:5.99 (3SF) | 5 | MGL   | 1243   |
| 2                                                                            | 35 | 1     | NaClO4 | lgK:6.32 (3SF) | 5 | MGL   | 1243   |

| Reaction No. 68011                                                           |    |       |        |                |   |       |        |
|------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Al}+3 + 3\text{<H+1\_Malonic-2>} = \text{Al}+3\_H+1(3)\_Malonic-2(3)$ |    |       |        |                |   |       |        |
| No                                                                           | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                            | 25 | 1     | NaClO4 | lgK:8.58 (3SF) | 5 | MGL   | 1243   |
| 2                                                                            | 35 | 1     | NaClO4 | lgK:8.98 (3SF) | 5 | MGL   | 1243   |

| Reaction No. 68012                                                    |    |       |               |                |   |       |        |
|-----------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $\text{Al}+3 + \text{H}+1\_Succinic-2 = \text{Al}+3\_H+1\_Succinic-2$ |    |       |               |                |   |       |        |
| No                                                                    | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                     | 25 | 0     | Inf. Dilution | lgK:2.1 (2SF)  | 1 | ELC   |        |
| 2                                                                     | 35 | 1     | NaClO4        | lgK:3.84 (3SF) | 2 | MGL   | 1243   |
| 3                                                                     | 35 | 1     | NaClO4        | lgK:3.78 (3SF) | 3 | MGL   | 1243   |

| Reaction No. 68013                                                             |    |       |        |                |   |       |        |
|--------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Al}+3 + 2\text{<H+1\_Succinic-2>} = \text{Al}+3\_H+1(2)\_Succinic-2(2)$ |    |       |        |                |   |       |        |
| No                                                                             | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                              | 25 | 1     | NaClO4 | lgK:5.93 (3SF) | 5 | MGL   | 1243   |
| 2                                                                              | 35 | 1     | NaClO4 | lgK:6.40 (3SF) | 5 | MGL   | 1243   |

| Reaction No. 68014                                                             |    |       |        |                |   |       |        |
|--------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Al}+3 + 3\text{<H+1\_Succinic-2>} = \text{Al}+3\_H+1(3)\_Succinic-2(3)$ |    |       |        |                |   |       |        |
| No                                                                             | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                              | 25 | 1     | NaClO4 | lgK:8.98 (3SF) | 5 | MGL   | 1243   |
| 2                                                                              | 35 | 1     | NaClO4 | lgK:9.52 (3SF) | 5 | MGL   | 1243   |

| Reaction No. 68015                                                            |    |       |        |               |   |       |        |
|-------------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| $\text{Al}+3 + \text{H}+1\_Succinic-2 = \text{Al}+3\_Succinic-2 + \text{H}+1$ |    |       |        |               |   |       |        |
| No                                                                            | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                             | 25 | 0.5   | NaClO4 | lgK:4.2 (2SF) | 0 | ENS   | 6953   |

| Reaction No. 68016 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| Al+3 + H+1(2)_Malic-2 = Al+3_Malic-2 + 2<H+1> |       |       |               |                 |   |       |        |
|-----------------------------------------------|-------|-------|---------------|-----------------|---|-------|--------|
| No                                            | t     | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                             | 25    | 0     | Inf. Dilution | lgK:-2.52 (3SF) | 2 | EAE   |        |
| 2                                             | 27:30 | 1     | NaClO4        | lgK:-3.43 (3SF) | 3 | ENS   | 6953   |

| Reaction No. 68017                    |    |       |               |                |   |       |        |
|---------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| 2<Maleic-2> + Al+3 = Al+3_Maleic-2(2) |    |       |               |                |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 25 | 0     | Inf. Dilution | lgK:8.78 (3SF) | 4 | ENS   | 6953   |

| Reaction No. 68018                                                              |    |       |        |                 |   |       |        |
|---------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 3<H+1(2)_Cat-2> + 3<Al+3> + 3<OH-1> = 9<H+1> + Al+3(3)_H+1(-3)_OH-1(3)_Cat-2(3) |    |       |        |                 |   |       |        |
| No                                                                              | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                               | 25 | 0.6   | NaCl   | lgK:29.91 (4SF) | 5 | MGL   | 2357   |

| Reaction No. 68019                                                  |    |       |        |                  |   |       |        |
|---------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 2<H+1(2)_Cat-2> + Al+3 + OH-1 = 5<H+1> + Al+3_H+1(-1)_OH-1_Cat-2(2) |    |       |        |                  |   |       |        |
| No                                                                  | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                   | 25 | 0.6   | NaCl   | lgK:-23.45 (4SF) | 5 | MGL   | 2357   |

| Reaction No. 68020                  |    |       |        |                 |   |       |        |
|-------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 3NitrCat-2 + Al+3 = Al+3_3NitrCat-2 |    |       |        |                 |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | NaCl   | lgK:14.17 (4SF) | 5 | ENS   | 6953   |

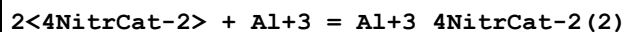
| Reaction No. 68021                        |    |       |        |                 |   |       |        |
|-------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2<3NitrCat-2> + Al+3 = Al+3_3NitrCat-2(2) |    |       |        |                 |   |       |        |
| No                                        | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                         | 25 | 0.1   | NaCl   | lgK:26.30 (4SF) | 5 | ENS   | 6953   |

| Reaction No. 68022                        |    |       |        |                 |   |       |        |
|-------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 3<3NitrCat-2> + Al+3 = Al+3_3NitrCat-2(3) |    |       |        |                 |   |       |        |
| No                                        | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                         | 25 | 0.1   | NaCl   | lgK:35.81 (4SF) | 5 | ENS   | 6953   |

| Reaction No. 68023                  |   |       |        |             |   |       |        |
|-------------------------------------|---|-------|--------|-------------|---|-------|--------|
| 4NitrCat-2 + Al+3 = Al+3_4NitrCat-2 |   |       |        |             |   |       |        |
| No                                  | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

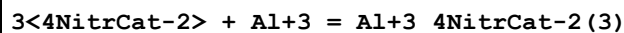
|   |    |     |      |                 |   |     |      |
|---|----|-----|------|-----------------|---|-----|------|
| 1 | 25 | 0.1 | NaCl | lgK:13.74 (4SF) | 5 | ENS | 6953 |
|---|----|-----|------|-----------------|---|-----|------|

## Reaction No. 68024



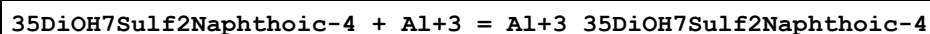
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaCl   | lgK:25.39 (4SF) | 5 | ENS   | 6953   |

## Reaction No. 68025



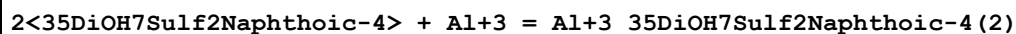
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaCl   | lgK:34.31 (4SF) | 5 | ENS   | 6953   |

## Reaction No. 68026



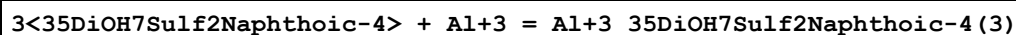
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.5   | Unknown | lgK:16.00 (4SF) | 4 | ENS   | 6953   |

## Reaction No. 68027



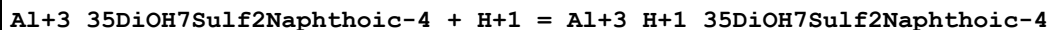
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.5   | Unknown | lgK:24.09 (4SF) | 4 | ENS   | 6953   |

## Reaction No. 68028



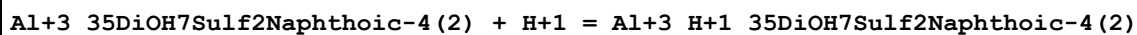
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.5   | Unknown | lgK:30.85 (4SF) | 4 | ENS   | 6953   |

## Reaction No. 68029



| No | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------|---------------|---|-------|--------|
| 1  | 25 | 0.5   | Unknown | lgK:4.5 (2SF) | 4 | ENS   | 6953   |

## Reaction No. 68030



| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 25 | 0.5   | Unknown | lgK:8.35 (3SF) | 4 | ENS   | 6953   |

## Reaction No. 68031

| Al+3 + OH-1 + 4<H2O> = Al+3_OH-1(5) + 4<H+1> |    |       |        |                   |   |       |        |
|----------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| No                                           | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                            | 25 | 0.6   | NaCl   | lgK:-23.5 (0.1SD) | 0 | ENS   | 966    |

| Reaction No. 68032                                |    |       |        |                 |   |       |        |
|---------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2<AlizarinRedS-3> + Al+3 = Al+3_AlizarinRedS-3(2) |    |       |        |                 |   |       |        |
| No                                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                 | 25 | 0.1   | NaClO4 | lgK:26.92 (4SF) | 4 | ENS   | 6953   |

| Reaction No. 68033              |    |       |               |                 |   |       |        |
|---------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| 2<NTA-3> + Al+3 = Al+3_NTA-3(2) |    |       |               |                 |   |       |        |
| No                              | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                               | 25 | 0     | Inf. Dilution | lgK:22.8 (3SF)  | 1 | ELC   |        |
| 2                               | 25 | 0.1   | NaClO4        | lgK:19.08 (4SF) | 0 | MGL   | 1411   |
| 3                               | 35 | 0.1   | NaClO4        | lgK:18.57 (4SF) | 0 | MGL   | 1411   |
| 4                               | 45 | 0.1   | NaClO4        | lgK:18.32 (4SF) | 0 | MGL   | 1411   |

| Reaction No. 68036                          |    |       |        |                |   |       |        |
|---------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 3<Picolinic-1> + Al+3 = Al+3_Picolinic-1(3) |    |       |        |                |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                           | 25 | 0.5   | NaClO4 | lgK:12.3 (3SF) | 5 | ENS   | 6953   |

| Reaction No. 68037              |    |       |               |                 |   |       |        |
|---------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| 3<Gly-1> + Al+3 = Al+3_Gly-1(3) |    |       |               |                 |   |       |        |
| No                              | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                               | 25 | 0     | Inf. Dilution | lgK:19.9 (3SF)  | 2 | EAE   |        |
| 2                               | 35 | 0.01  | NaClO4        | lgK:19.40 (4SF) | 4 | ENS   | 6953   |

| Reaction No. 68038          |    |       |           |                 |   |       |        |
|-----------------------------|----|-------|-----------|-----------------|---|-------|--------|
| EHPG-4 + Al+3 = Al+3_EHPG-4 |    |       |           |                 |   |       |        |
| No                          | t  | I-Str | Medium    | Value (Dev)     | W | Tech. | Ref. # |
| 1                           | 25 | 0.12  | Cl-1 salt | lgK:24.48 (4SF) | 4 | ENS   | 6953   |

| Reaction No. 68039                                |    |       |        |                |   |       |        |
|---------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 23PyridineDiCOO-2 + Al+3 = Al+3_23PyridineDiCOO-2 |    |       |        |                |   |       |        |
| No                                                | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                 | 25 | 0.5   | NaClO4 | lgK:4.50 (3SF) | 5 | ENS   | 6953   |

| Reaction No. 68040                                      |    |       |        |                |   |       |        |
|---------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 2<23PyridineDiCOO-2> + Al+3 = Al+3_23PyridineDiCOO-2(2) |    |       |        |                |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.5   | NaClO4 | lgK:7.00 (3SF) | 5 | ENS   | 6953   |

| Reaction No. 68041                                |    |       |        |                |   |       |        |
|---------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 24PyridineDiCOO-2 + Al+3 = Al+3_24PyridineDiCOO-2 |    |       |        |                |   |       |        |
| No                                                | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                 | 25 | 0.5   | NaClO4 | lgK:4.35 (3SF) | 5 | ENS   | 6953   |

| Reaction No. 68042                                      |    |       |        |                |   |       |        |
|---------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 2<24PyridineDiCOO-2> + Al+3 = Al+3_24PyridineDiCOO-2(2) |    |       |        |                |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.5   | NaClO4 | lgK:7.30 (3SF) | 5 | ENS   | 6953   |

| Reaction No. 68043                          |    |       |         |                |   |       |        |
|---------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| 2<PhenylSer-1> + Al+3 = Al+3_PhenylSer-1(2) |    |       |         |                |   |       |        |
| No                                          | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | Unknown | lgK:9.98 (3SF) | 5 | ENS   | 6953   |

| Reaction No. 68080                                    |    |       |        |                |   |       |        |
|-------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Al+3 + H+1_XylenolOrange-6 = Al+3_H+1_XylenolOrange-6 |    |       |        |                |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                     | 25 | 0.1   | NaClO4 | lgK:16.3 (3SF) | 5 | ENS   | 6953   |

| Reaction No. 68081                                    |    |       |               |                 |   |       |        |
|-------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| H+1 + MeThymolBlue-6 + Al+3 = Al+3_H+1_MeThymolBlue-6 |    |       |               |                 |   |       |        |
| No                                                    | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                     | 25 | 0     | Inf. Dilution | lgK:29.5 (3SF)  | 1 | ESC   |        |
| 2                                                     | 25 | 0.1   | NaClO4        | lgK:29.14 (4SF) | 5 | ENS   | 6953   |

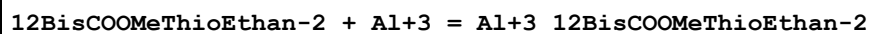
| Reaction No. 68082                    |    |       |        |                |   |       |        |
|---------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Quercetin-5 + Al+3 = Al+3_Quercetin-5 |    |       |        |                |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 25 | 0.5   | NaClO4 | lgK:18.3 (3SF) | 5 | ENS   | 6953   |

| Reaction No. 68083                                  |  |  |  |  |  |  |  |
|-----------------------------------------------------|--|--|--|--|--|--|--|
| Al+3_Pyridox5OPO3-3 + H+1 = Al+3_H+1_Pyridox5OPO3-3 |  |  |  |  |  |  |  |



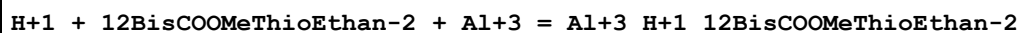
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 1     | KCl    | lgK:4.23 (3SF) | 5 | MGL   | 1585   |

## Reaction No. 68086



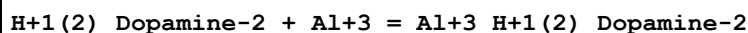
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.5   | KCl    | lgK:2.05 (3SF) | 5 | ENS   | 6953   |

## Reaction No. 68087



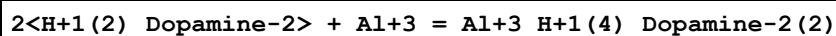
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.5   | KCl    | lgK:12.46 (4SF) | 5 | ENS   | 6953   |

## Reaction No. 68088



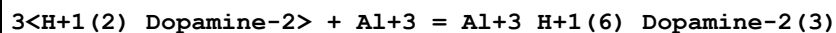
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:15.63 (4SF) | 5 | ENS   | 6953   |

## Reaction No. 68089



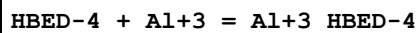
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:28.61 (4SF) | 5 | ENS   | 6953   |

## Reaction No. 68090



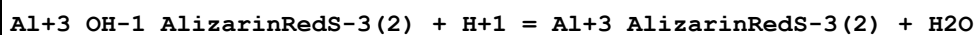
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:37.56 (4SF) | 5 | ENS   | 6953   |

## Reaction No. 68091



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.12  | KCl    | lgK:24.78 (4SF) | 4 | ENS   | 6953   |

## Reaction No. 68342



| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 20 | 0.1   | Unknown       | lgK:7.2 (2SF) | 4 | CRV   | 7271   |
| 2  | 25 | 0.1   | Inf. Dilution | lgK:7.5 (2SF) | 1 | EES   | 7271   |

| Reaction No. 68344                                                                                                                     |    |       |         |                              |   |       |        |
|----------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------|------------------------------|---|-------|--------|
| $\text{Al}+3\_ \text{AlizarinRedS-3} + \text{H}+1(2)\_ \text{AlizarinRedS-3} = \text{Al}+3\_ \text{AlizarinRedS-3}(2) + 2<\text{H}+1>$ |    |       |         |                              |   |       |        |
| No                                                                                                                                     | t  | I-Str | Medium  | Value (Dev)                  | W | Tech. | Ref. # |
| 1                                                                                                                                      | 25 | 0.1   | Unknown | $\lg K: -3.97(3\text{SF})$   | 4 | CRV   | 7271   |
| 2                                                                                                                                      | 25 | 0.1   | Unknown | $\text{dH}: -2.(1\text{SF})$ | 4 | CRV   | 7271   |

| Reaction No. 68346                                                                                          |    |       |         |                             |   |       |        |
|-------------------------------------------------------------------------------------------------------------|----|-------|---------|-----------------------------|---|-------|--------|
| $\text{Al}+3 + \text{H}+1(2)\_ \text{AlizarinRedS-3} = \text{Al}+3\_ \text{AlizarinRedS-3} + 2<\text{H}+1>$ |    |       |         |                             |   |       |        |
| No                                                                                                          | t  | I-Str | Medium  | Value (Dev)                 | W | Tech. | Ref. # |
| 1                                                                                                           | 25 | 0.1   | Unknown | $\lg K: -2.44(3\text{SF})$  | 4 | CRV   | 7271   |
| 2                                                                                                           | 25 | 0.1   | Unknown | $\text{dH}: 6.(1\text{SF})$ | 4 | CRV   | 7271   |

| Reaction No. 68366                                                                                |    |       |         |                               |   |       |        |
|---------------------------------------------------------------------------------------------------|----|-------|---------|-------------------------------|---|-------|--------|
| $\text{H}+1 + \text{Glyphosate-3} + \text{Al}+3 = \text{Al}+3\_ \text{H}+1\_ \text{Glyphosate-3}$ |    |       |         |                               |   |       |        |
| No                                                                                                | t  | I-Str | Medium  | Value (Dev)                   | W | Tech. | Ref. # |
| 1                                                                                                 | 25 | 0.1   | KNO3    | $\lg K: 16.18(0.02\text{SD})$ | 4 | MGL   | 6861   |
| 2                                                                                                 | 25 | 0.1   | Unknown | $\lg K: 16.2(0.1\text{SD})$   | 6 | CIU   | 7242   |

| Reaction No. 68368                                                                                      |    |       |        |                               |   |       |        |
|---------------------------------------------------------------------------------------------------------|----|-------|--------|-------------------------------|---|-------|--------|
| $2<\text{H}+1> + \text{Glyphosate-3} + \text{Al}+3 = \text{Al}+3\_ \text{H}+1(2)\_ \text{Glyphosate-3}$ |    |       |        |                               |   |       |        |
| No                                                                                                      | t  | I-Str | Medium | Value (Dev)                   | W | Tech. | Ref. # |
| 1                                                                                                       | 25 | 0.1   | KNO3   | $\lg K: 27.76(0.02\text{SD})$ | 4 | MGL   | 6861   |

| Reaction No. 68483                                      |    |       |                           |                              |   |       |        |
|---------------------------------------------------------|----|-------|---------------------------|------------------------------|---|-------|--------|
| $\text{Cl-1} + \text{Al}+3 = \text{Al}+3\_ \text{Cl-1}$ |    |       |                           |                              |   |       |        |
| No                                                      | t  | I-Str | Medium                    | Value (Dev)                  | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.2   | DMF<br>Et4NC1O4           | $\lg K: 1.73(0.03\text{MD})$ | 5 | MCL   | 7237   |
| 2                                                       | 25 | 0.2   | DMF<br>Et4NC1O4           | $\text{dH}: 1.2(2\text{SF})$ | 5 | MCL   | 7237   |
| 3                                                       | 25 | 0.2   | DiMeAcetamide<br>Bu4NC1O4 | $\lg K: 3.17(0.01\text{MD})$ | 5 | MCL   | 7237   |
| 4                                                       | 25 | 0.2   | DiMeAcetamide<br>Bu4NC1O4 | $\text{dH}: 3.5(2\text{SF})$ | 5 | MCL   | 7237   |

| Reaction No. 68484                                            |   |       |        |             |   |       |        |
|---------------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| $2<\text{Cl-1}> + \text{Al}+3 = \text{Al}+3\_ \text{Cl-1}(2)$ |   |       |        |             |   |       |        |
| No                                                            | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

|   |    |     |                           |                   |   |     |      |
|---|----|-----|---------------------------|-------------------|---|-----|------|
| 1 | 25 | 0.2 | DMF<br>Et4NC1O4           | lgK:2.2 (0.08MD)  | 5 | MCL | 7237 |
| 2 | 25 | 0.2 | DMF<br>Et4NC1O4           | dH:5.8 (2SF)      | 5 | MCL | 7237 |
| 3 | 25 | 0.2 | DiMeAcetamide<br>Bu4NC1O4 | lgK:5.56 (0.02MD) | 5 | MCL | 7237 |
| 4 | 25 | 0.2 | DiMeAcetamide<br>Bu4NC1O4 | dH:8.1 (2SF)      | 5 | MCL | 7237 |

| Reaction No. 68564                                                                                     |    |       |        |                 |   |       |        |
|--------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $6\text{Al}+3\text{OH}^{-1}(3) + 4\text{H}_2\text{O} = \text{Al}+3(6)\text{OH}^{-1}(22) + 4\text{H}+1$ |    |       |        |                 |   |       |        |
| No                                                                                                     | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                      | 25 | 8     | NaClO4 | lgK:-35.4 (3SF) | 5 | MGL   | 7188   |

| Reaction No. 69124                                                                                                |    |       |        |                  |   |       |        |
|-------------------------------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $\text{H}+1 + 35\text{DiOH7Sulf2Naphthoic-4} + \text{Al}+3 = \text{Al}+3\text{H}+1\text{35DiOH7Sulf2Naphthoic-4}$ |    |       |        |                  |   |       |        |
| No                                                                                                                | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                                 | 25 | 0.5   | NaClO4 | lgK:20.651 (5SF) | 6 | MGL   | 5987   |

| Reaction No. 69125                                                                                                    |    |       |        |                  |   |       |        |
|-----------------------------------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $\text{H}+1 + 2\text{35DiOH7Sulf2Naphthoic-4} + \text{Al}+3 = \text{Al}+3\text{H}+1\text{35DiOH7Sulf2Naphthoic-4}(2)$ |    |       |        |                  |   |       |        |
| No                                                                                                                    | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                                     | 25 | 0.5   | NaClO4 | lgK:32.443 (5SF) | 6 | MGL   | 5987   |

| Reaction No. 69132                                                    |    |       |                 |                   |   |       |        |
|-----------------------------------------------------------------------|----|-------|-----------------|-------------------|---|-------|--------|
| $\text{Al}+3\text{SCN-1} + \text{SCN-1} = \text{Al}+3\text{SCN-1}(2)$ |    |       |                 |                   |   |       |        |
| No                                                                    | t  | I-Str | Medium          | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                     | 25 | 0.2   | DMF<br>Et4NC1O4 | lgK:1.5 (0.3MD)   | 5 | MCL   | 6237   |
| 2                                                                     | 25 | 0.4   | DMF<br>Et4NC1O4 | lgK:1.2 (0.2MD)   | 5 | MCL   | 6237   |
| 3                                                                     | 25 | 0.2   | DMF<br>Et4NC1O4 | dH:13. (0.6MD) kJ | 5 | MCL   | 6237   |
| 4                                                                     | 25 | 0.4   | DMF<br>Et4NC1O4 | dH:13. (0.6MD) kJ | 5 | MCL   | 6237   |

| Reaction No. 69133                                                       |    |       |                 |                 |   |       |        |
|--------------------------------------------------------------------------|----|-------|-----------------|-----------------|---|-------|--------|
| $\text{Al}+3\text{SCN-1}(2) + \text{SCN-1} = \text{Al}+3\text{SCN-1}(3)$ |    |       |                 |                 |   |       |        |
| No                                                                       | t  | I-Str | Medium          | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                        | 25 | 0.2   | DMF<br>Et4NC1O4 | lgK:1.4 (0.4MD) | 5 | MCL   | 6237   |

|   |    |     |                 |                 |   |     |      |
|---|----|-----|-----------------|-----------------|---|-----|------|
| 2 | 25 | 0.4 | DMF<br>Et4NC1O4 | lgK:1.3(0.3MD)  | 5 | MCL | 6237 |
| 3 | 25 | 0.2 | DMF<br>Et4NC1O4 | dH:0.0(10.MD)kJ | 5 | MCL | 6237 |
| 4 | 25 | 0.4 | DMF<br>Et4NC1O4 | dH:-3.(0.9MD)kJ | 5 | MCL | 6237 |

| Reaction No. 69134                    |    |       |                 |                 |   |       |        |
|---------------------------------------|----|-------|-----------------|-----------------|---|-------|--------|
| Al+3_SCN-1(3) + SCN-1 = Al+3_SCN-1(4) |    |       |                 |                 |   |       |        |
| No                                    | t  | I-Str | Medium          | Value (Dev)     | W | Tech. | Ref. # |
| 1                                     | 25 | 0.2   | DMF<br>Et4NC1O4 | lgK:1.0(0.3MD)  | 5 | MCL   | 6237   |
| 2                                     | 25 | 0.4   | DMF<br>Et4NC1O4 | lgK:0.8(0.2MD)  | 5 | MCL   | 6237   |
| 3                                     | 25 | 0.2   | DMF<br>Et4NC1O4 | dH:12.(0.9MD)kJ | 5 | MCL   | 6237   |
| 4                                     | 25 | 0.4   | DMF<br>Et4NC1O4 | dH:17.(0.7MD)kJ | 5 | MCL   | 6237   |

| Reaction No. 69191        |    |       |               |                   |   |       |        |
|---------------------------|----|-------|---------------|-------------------|---|-------|--------|
| Al(s) - 3<e-1> = Al+3_(g) |    |       |               |                   |   |       |        |
| No                        | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                         | 25 | 0     | Inf. Dilution | dG:5000.(4SF)kJ   | 1 | ENS   |        |
| 2                         | 25 | 0     | Inf. Dilution | dH:5483.20(6SF)kJ | 3 | CRV   | 10802  |

| Reaction No. 69202                           |     |       |               |                |   |       |        |
|----------------------------------------------|-----|-------|---------------|----------------|---|-------|--------|
| AlOOH(Boehmite,s) + 2<H+1> = Al+3_OH-1 + H2O |     |       |               |                |   |       |        |
| No                                           | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                            | 25  | 0     | Inf. Dilution | lgK:3.3(2SF)   | 1 | ELC   |        |
| 2                                            | 150 |       |               | lgK:-0.55(2SF) | 3 | MSL   | 4281   |
| 3                                            | 200 |       |               | lgK:-1.01(3SF) | 3 | MSL   | 4281   |
| 4                                            | 250 |       |               | lgK:-1.28(3SF) | 3 | MSL   | 4281   |

| Reaction No. 69203                     |     |       |               |                |   |       |        |
|----------------------------------------|-----|-------|---------------|----------------|---|-------|--------|
| AlOOH(Boehmite,s) + H+1 = Al+3_OH-1(2) |     |       |               |                |   |       |        |
| No                                     | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                      | 25  | 0     | Inf. Dilution | lgK:-2.2(2SF)  | 1 | ELC   |        |
| 2                                      | 150 |       |               | lgK:-3.24(3SF) | 1 | MSL   | 4281   |
| 3                                      | 200 |       |               | lgK:-3.32(3SF) | 2 | MSL   | 4281   |
| 4                                      | 250 |       |               | lgK:-3.41(3SF) | 3 | MSL   | 4281   |

| Reaction No. 69204                     |     |       |               |                |   |       |        |
|----------------------------------------|-----|-------|---------------|----------------|---|-------|--------|
| AlOOH(Boehmite,s) + H2O = Al+3_OH-1(3) |     |       |               |                |   |       |        |
| No                                     | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                      | 25  | 0     | Inf. Dilution | lgK:-9.0(2SF)  | 1 | ELC   |        |
| 2                                      | 150 |       |               | lgK:-6.30(3SF) | 3 | MSL   | 4281   |
| 3                                      | 200 |       |               | lgK:-6.00(3SF) | 3 | MSL   | 4281   |
| 4                                      | 250 |       |               | lgK:-6.00(3SF) | 3 | MSL   | 4281   |

| Reaction No. 69248                    |    |       |        |                 |   |       |        |
|---------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| RibosePO3-2 + Al+3 = Al+3_RibosePO3-2 |    |       |        |                 |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                     | 25 | 0.2   | KCl    | lgK:5.6(0.03SD) | 5 | MGL   | 7048   |

| Reaction No. 69249                                  |    |       |        |                  |   |       |        |
|-----------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| RibosePO3-2 + Al+3 = H+1 + Al+3_H+1(-1)_RibosePO3-2 |    |       |        |                  |   |       |        |
| No                                                  | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.2   | KCl    | lgK:1.69(0.02SD) | 5 | MGL   | 7048   |

| Reaction No. 69250                                     |    |       |        |                  |   |       |        |
|--------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| RibosePO3-2 + Al+3 = 2<H+1> + Al+3_H+1(-2)_RibosePO3-2 |    |       |        |                  |   |       |        |
| No                                                     | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                      | 25 | 0.2   | KCl    | lgK:-4.8(0.05SD) | 5 | MGL   | 7048   |

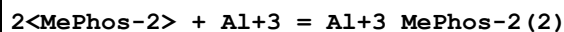
| Reaction No. 69251              |    |       |         |                  |   |       |        |
|---------------------------------|----|-------|---------|------------------|---|-------|--------|
| MePhos-2 + Al+3 = Al+3_MePhos-2 |    |       |         |                  |   |       |        |
| No                              | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
| 1                               | 25 | 0.2   | KCl     | lgK:6.5(0.03SD)  | 5 | MGL   | 7048   |
| 2                               | 25 | 0.2   | Unknown | lgK:6.48(0.03SD) | 6 | CIU   | 7242   |

| Reaction No. 69252                            |    |       |        |                 |   |       |        |
|-----------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| MePhos-2 + Al+3 = H+1 + Al+3_H+1(-1)_MePhos-2 |    |       |        |                 |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                             | 25 | 0.2   | KCl    | lgK:2.3(0.03SD) | 5 | MGL   | 7048   |

| Reaction No. 69253                               |   |       |        |             |   |       |        |
|--------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| MePhos-2 + Al+3 = 2<H+1> + Al+3_H+1(-2)_MePhos-2 |   |       |        |             |   |       |        |
| No                                               | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

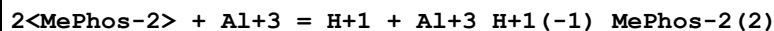
|   |    |     |     |                  |   |     |      |
|---|----|-----|-----|------------------|---|-----|------|
| 1 | 25 | 0.2 | KCl | lgK:-3.9(0.06SD) | 5 | MGL | 7048 |
|---|----|-----|-----|------------------|---|-----|------|

## Reaction No. 69254



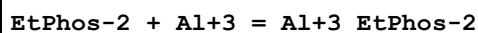
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl     | lgK:12.3(0.1SD) | 5 | MGL   | 7048   |
| 2  | 25 | 0.2   | Unknown | lgK:12.3(0.1SD) | 6 | CIU   | 7242   |

## Reaction No. 69255



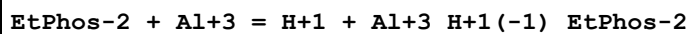
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:5.8(0.2SD) | 5 | MGL   | 7048   |

## Reaction No. 69256



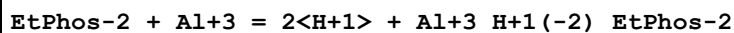
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:6.6(0.03SD) | 5 | MGL   | 7048   |

## Reaction No. 69257



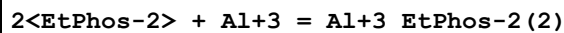
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:2.34(0.02SD) | 5 | MGL   | 7048   |

## Reaction No. 69258



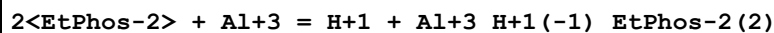
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:-3.9(0.06SD) | 5 | MGL   | 7048   |

## Reaction No. 69259



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:12.1(0.2SD) | 5 | MGL   | 7048   |

## Reaction No. 69260



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:6.1(0.2SD) | 5 | MGL   | 7048   |

| Reaction No. 69261                  |    |       |        |                 |   |       |        |
|-------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| PhenPhos-2 + Al+3 = Al+3_PhenPhos-2 |    |       |        |                 |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                   | 25 | 0.2   | KCl    | lgK:5.3(0.03SD) | 5 | MGL   | 7048   |

| Reaction No. 69262                                |    |       |        |                  |   |       |        |
|---------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| PhenPhos-2 + Al+3 = H+1 + Al+3_H+1(-1)_PhenPhos-2 |    |       |        |                  |   |       |        |
| No                                                | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                 | 25 | 0.2   | KCl    | lgK:1.51(0.02SD) | 5 | MGL   | 7048   |

| Reaction No. 69263                                   |    |       |        |                 |   |       |        |
|------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| PhenPhos-2 + Al+3 = 2<H+1> + Al+3_H+1(-2)_PhenPhos-2 |    |       |        |                 |   |       |        |
| No                                                   | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                    | 25 | 0.2   | KCl    | lgK:-4.5(0.1SD) | 5 | MGL   | 7048   |

| Reaction No. 69264                            |    |       |        |                  |   |       |        |
|-----------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 4NitrPhenOPO3-2 + Al+3 = Al+3_4NitrPhenOPO3-2 |    |       |        |                  |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                             | 25 | 0.2   | KCl    | lgK:4.80(0.02SD) | 5 | MGL   | 7048   |

| Reaction No. 69265                                          |    |       |        |                  |   |       |        |
|-------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 4NitrPhenOPO3-2 + Al+3 = H+1 + Al+3_H+1(-1)_4NitrPhenOPO3-2 |    |       |        |                  |   |       |        |
| No                                                          | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                           | 25 | 0.2   | KCl    | lgK:1.19(0.02SD) | 5 | MGL   | 7048   |

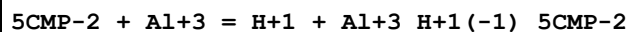
| Reaction No. 69266                                             |    |       |        |                |   |       |        |
|----------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 4NitrPhenOPO3-2 + Al+3 = 2<H+1> + Al+3_H+1(-2)_4NitrPhenOPO3-2 |    |       |        |                |   |       |        |
| No                                                             | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                              | 25 | 0.2   | KCl    | lgK:-5.(0.4SD) | 5 | MGL   | 7048   |

| Reaction No. 69268                    |    |       |        |                |   |       |        |
|---------------------------------------|----|-------|--------|----------------|---|-------|--------|
| H+1 + 5CMP-2 + Al+3 = Al+3_H+1_5CMP-2 |    |       |        |                |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 25 | 0.2   | KCl    | lgK:9.0(0.2SD) | 5 | MGL   | 7048   |

| Reaction No. 69269          |   |       |        |             |   |       |        |
|-----------------------------|---|-------|--------|-------------|---|-------|--------|
| 5CMP-2 + Al+3 = Al+3_5CMP-2 |   |       |        |             |   |       |        |
| No                          | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

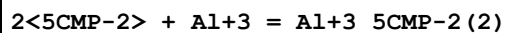
|   |    |     |     |                   |   |     |      |
|---|----|-----|-----|-------------------|---|-----|------|
| 1 | 25 | 0.2 | KCl | lgK:6.08 (0.02SD) | 5 | MGL | 7048 |
|---|----|-----|-----|-------------------|---|-----|------|

## Reaction No. 69270



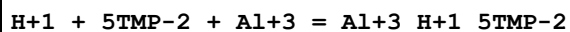
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:0.4 (0.06SD) | 5 | MGL   | 7048   |

## Reaction No. 69271



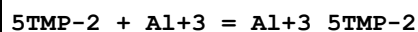
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:9.7 (0.2SD) | 5 | MGL   | 7048   |

## Reaction No. 69274



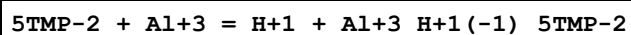
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:15.4 (0.2SD) | 5 | MGL   | 7048   |

## Reaction No. 69275



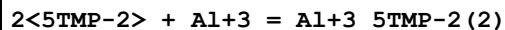
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:11.92 (0.01SD) | 5 | MGL   | 7048   |

## Reaction No. 69276



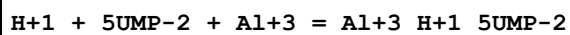
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:5.6 (0.2SD) | 5 | MGL   | 7048   |

## Reaction No. 69277



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:18.7 (0.06SD) | 5 | MGL   | 7048   |

## Reaction No. 69279



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:14.8 (0.03SD) | 5 | MGL   | 7048   |

## Reaction No. 69280



| 5UMP-2 + Al+3 = Al+3_5UMP-2 |    |       |        |                   |   |       |        |
|-----------------------------|----|-------|--------|-------------------|---|-------|--------|
| No                          | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                           | 25 | 0.2   | KCl    | lgK:11.18(0.02SD) | 5 | MGL   | 7048   |

| Reaction No. 69281                |    |       |        |                  |   |       |        |
|-----------------------------------|----|-------|--------|------------------|---|-------|--------|
| 2<5UMP-2> + Al+3 = Al+3_5UMP-2(2) |    |       |        |                  |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                 | 25 | 0.2   | KCl    | lgK:17.5(0.09SD) | 5 | MGL   | 7048   |

| Reaction No. 69282                    |    |       |        |                 |   |       |        |
|---------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| H+1 + 5GMP-2 + Al+3 = Al+3_H+1_5GMP-2 |    |       |        |                 |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                     | 25 | 0.2   | KCl    | lgK:14.9(0.2SD) | 5 | MGL   | 7048   |

| Reaction No. 69283          |    |       |        |                  |   |       |        |
|-----------------------------|----|-------|--------|------------------|---|-------|--------|
| 5GMP-2 + Al+3 = Al+3_5GMP-2 |    |       |        |                  |   |       |        |
| No                          | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                           | 25 | 0.2   | KCl    | lgK:11.7(0.03SD) | 5 | MGL   | 7048   |

| Reaction No. 69457                  |    |       |         |               |   |       |        |
|-------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Al+3_5ATP-4 + H+1 = Al+3_H+1_5ATP-4 |    |       |         |               |   |       |        |
| No                                  | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                   | 25 | 0.5   | Unknown | lgK:3.38(3SF) | 4 | MMR   | 7163   |

| Reaction No. 69458                         |    |       |         |                |   |       |        |
|--------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Al+3_5ATP-4 + H2O = Al+3_OH-1_5ATP-4 + H+1 |    |       |         |                |   |       |        |
| No                                         | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                          | 25 | 0.5   | Unknown | lgK:-5.46(3SF) | 4 | MMR   | 7163   |

| Reaction No. 69459                    |    |       |         |               |   |       |        |
|---------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Al+3_5ATP-4 + 5ATP-4 = Al+3_5ATP-4(2) |    |       |         |               |   |       |        |
| No                                    | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                     | 25 | 0.5   | Unknown | lgK:4.55(3SF) | 4 | MMR   | 7163   |

| Reaction No. 69460                               |    |       |         |                |   |       |        |
|--------------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Al+3_5ATP-4(2) + H2O = Al+3_OH-1_5ATP-4(2) + H+1 |    |       |         |                |   |       |        |
| No                                               | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                | 25 | 0.5   | Unknown | lgK:-7.63(3SF) | 4 | MMR   | 7163   |

| Reaction No. 69623                                                                                                |    |       |        |                              |   |       |        |
|-------------------------------------------------------------------------------------------------------------------|----|-------|--------|------------------------------|---|-------|--------|
| $\text{Al}+3 + \text{H}+1\_ \text{Catechin-4} = \text{Al}+3\_ \text{H}+1(-1)\_ \text{Catechin-4} + 2<\text{H}+1>$ |    |       |        |                              |   |       |        |
| No                                                                                                                | t  | I-Str | Medium | Value (Dev)                  | W | Tech. | Ref. # |
| 1                                                                                                                 | 25 | 0.1   | KCl    | $\lg K: -5.6(0.04\text{SD})$ | 5 | MPT   | 1214   |

| Reaction No. 69624                                                                                                                                      |    |       |        |                               |   |       |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|-------------------------------|---|-------|--------|
| $\text{Al}+3\_ \text{H}+1\_ \text{Catechin-4} + \text{H}+1(3)\_ \text{Catechin-4} = \text{Al}+3\_ \text{H}+1(2)\_ \text{Catechin-4}(2) + 2<\text{H}+1>$ |    |       |        |                               |   |       |        |
| No                                                                                                                                                      | t  | I-Str | Medium | Value (Dev)                   | W | Tech. | Ref. # |
| 1                                                                                                                                                       | 25 | 0.1   | KCl    | $\lg K: -8.79(0.02\text{SD})$ | 5 | MPT   | 1214   |

| Reaction No. 69625                                                                                                                                  |    |       |        |                              |   |       |        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|------------------------------|---|-------|--------|
| $\text{Al}+3\_ \text{H}+1(2)\_ \text{Catechin-4} + \text{H}_2\text{O} = \text{Al}+3\_ \text{H}+1(2)\_ \text{OH-1}\_ \text{Catechin-4} + \text{H}+1$ |    |       |        |                              |   |       |        |
| No                                                                                                                                                  | t  | I-Str | Medium | Value (Dev)                  | W | Tech. | Ref. # |
| 1                                                                                                                                                   | 25 | 0.1   | KCl    | $\lg K: -6.0(0.09\text{SD})$ | 5 | MPT   | 1214   |

| Reaction No. 69715                                                                                    |    |       |        |                               |   |       |        |
|-------------------------------------------------------------------------------------------------------|----|-------|--------|-------------------------------|---|-------|--------|
| $\text{H}+1 + \text{bAlaHydroxam-1} + \text{Al}+3 = \text{Al}+3\_ \text{H}+1\_ \text{bAlaHydroxam-1}$ |    |       |        |                               |   |       |        |
| No                                                                                                    | t  | I-Str | Medium | Value (Dev)                   | W | Tech. | Ref. # |
| 1                                                                                                     | 25 | 0.2   | KCl    | $\lg K: 16.72(0.02\text{SD})$ | 5 | MGL   | 4764   |

| Reaction No. 69716                                                                                                |    |       |        |                               |   |       |        |
|-------------------------------------------------------------------------------------------------------------------|----|-------|--------|-------------------------------|---|-------|--------|
| $2<\text{H}+1> + 2<\text{bAlaHydroxam-1}> + \text{Al}+3 = \text{Al}+3\_ \text{H}+1(2)\_ \text{bAlaHydroxam-1}(2)$ |    |       |        |                               |   |       |        |
| No                                                                                                                | t  | I-Str | Medium | Value (Dev)                   | W | Tech. | Ref. # |
| 1                                                                                                                 | 25 | 0.2   | KCl    | $\lg K: 32.07(0.02\text{SD})$ | 5 | MGL   | 4764   |

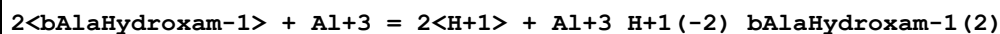
| Reaction No. 69717                                                                                          |    |       |        |                               |   |       |        |
|-------------------------------------------------------------------------------------------------------------|----|-------|--------|-------------------------------|---|-------|--------|
| $\text{H}+1 + 2<\text{bAlaHydroxam-1}> + \text{Al}+3 = \text{Al}+3\_ \text{H}+1\_ \text{bAlaHydroxam-1}(2)$ |    |       |        |                               |   |       |        |
| No                                                                                                          | t  | I-Str | Medium | Value (Dev)                   | W | Tech. | Ref. # |
| 1                                                                                                           | 25 | 0.2   | KCl    | $\lg K: 27.04(0.02\text{SD})$ | 5 | MGL   | 4764   |

| Reaction No. 69718                                                                |    |       |        |                               |   |       |        |
|-----------------------------------------------------------------------------------|----|-------|--------|-------------------------------|---|-------|--------|
| $2<\text{bAlaHydroxam-1}> + \text{Al}+3 = \text{Al}+3\_ \text{bAlaHydroxam-1}(2)$ |    |       |        |                               |   |       |        |
| No                                                                                | t  | I-Str | Medium | Value (Dev)                   | W | Tech. | Ref. # |
| 1                                                                                 | 25 | 0.2   | KCl    | $\lg K: 19.87(0.02\text{SD})$ | 5 | MGL   | 4764   |

| Reaction No. 69719                                                                                              |  |  |  |  |  |  |  |
|-----------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|
| $2<\text{bAlaHydroxam-1}> + \text{Al}+3 = \text{H}+1 + \text{Al}+3\_ \text{H}+1(-1)\_ \text{bAlaHydroxam-1}(2)$ |  |  |  |  |  |  |  |

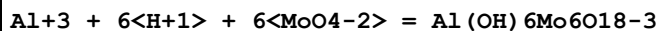
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:10.74 (0.02SD) | 5 | MGL   | 4764   |

## Reaction No. 69720



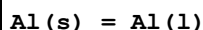
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:0.04 (0.3SD) | 5 | MGL   | 4764   |

## Reaction No. 70124



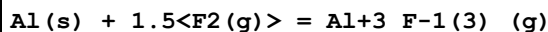
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:50.96 (0.06SD) | 5 | MMR   | 6953   |
| 2  | 25 | 0.6   | NaCl   | lgK:50.95 (0.04SD) | 5 | MPT   | 6953   |

## Reaction No. 70147



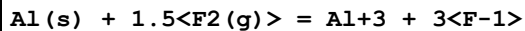
| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dG:7.20 (3SF) kJ  | 5 | EFE   | 7275   |
| 2  | 25 | 0     | Inf. Dilution | dH:10.56 (4SF) kJ | 5 | EFE   | 7275   |

## Reaction No. 70523



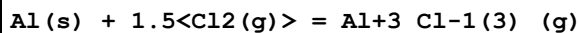
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dG:-284.0 (4SF) | 4 | CRV   | 7382   |
| 2  | 25 | 0     | Inf. Dilution | dH:-287.9 (4SF) | 4 | CRV   | 7382   |

## Reaction No. 70524



| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:233.2 (4SF) | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | dG:-316. (3SF)  | 0 | CRV   | 7382   |
| 3  | 25 | 0     | Inf. Dilution | dH:-366. (3SF)  | 2 | CRV   | 7382   |

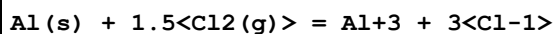
## Reaction No. 70525



| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dG:-500. (3SF) kJ | 1 | ENS   |        |

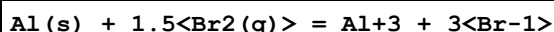
|   |    |   |               |                  |   |     |       |
|---|----|---|---------------|------------------|---|-----|-------|
| 2 | 25 | 0 | Inf. Dilution | dH:-139.4(4SF)   | 4 | CRV | 7382  |
| 3 | 25 | 0 | Inf. Dilution | dH:-583.2(4SF)kJ | 3 | CRV | 10802 |

## Reaction No. 70526



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:155.1(4SF) | 1 | ERG   |        |
| 2  | 25 | 0     | Inf. Dilution | dG:-210.(3SF)  | 1 | CRV   | 7382   |
| 3  | 25 | 0     | Inf. Dilution | dH:-247.(3SF)  | 1 | CRV   | 7382   |

## Reaction No. 70527



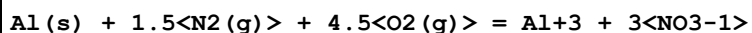
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:140.7(4SF) | 1 | ERG   |        |
| 2  | 25 | 0     | Inf. Dilution | dG:-191.(3SF)  | 0 | CRV   | 7382   |
| 3  | 25 | 0     | Inf. Dilution | dH:-214.(3SF)  | 1 | CRV   | 7382   |

## Reaction No. 70528



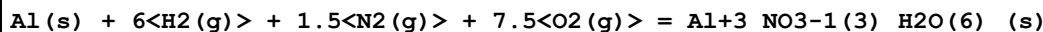
| No | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:809.9(4SF)     | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | dG:-1104.80(6SF)   | 0 | CRV   | 7382   |
| 3  | 25 | 0     | Inf. Dilution | dG:-4622.10(6SF)kJ | 0 | CRV   | 9015   |
| 4  | 25 | 0     | Inf. Dilution | dG:-1104.80(6SF)   | 0 | CRV   | 12011  |
| 5  | 25 | 0     | Inf. Dilution | dH:-1269.50(6SF)   | 1 | CRV   | 7382   |
| 6  | 25 | 0     | Inf. Dilution | dH:-5311.70(6SF)kJ | 1 | CRV   | 9015   |
| 7  | 25 | 0     | Inf. Dilution | dH:-1269.50(6SF)   | 2 | CRV   | 12011  |

## Reaction No. 70529



| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dG:-196.(3SF) | 4 | CRV   | 7382   |
| 2  | 25 | 0     | Inf. Dilution | dH:-276.(3SF) | 4 | CRV   | 7382   |

## Reaction No. 70530



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

|   |    |   |               |                  |   |     |       |
|---|----|---|---------------|------------------|---|-----|-------|
| 1 | 25 | 0 | Inf. Dilution | dG:-526.74 (5SF) | 4 | CRV | 7382  |
| 2 | 25 | 0 | Inf. Dilution | dG:-526.74 (5SF) | 4 | CRV | 12011 |
| 3 | 25 | 0 | Inf. Dilution | dH:-681.28 (5SF) | 4 | CRV | 7382  |
| 4 | 25 | 0 | Inf. Dilution | dH:-681.28 (5SF) | 4 | CRV | 12011 |

| Reaction No. 70531                                                                                                                                                                     |    |       |               |                 |   |       |        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{Al(s)} + 1.5\langle\text{Cl}_2(\text{g})\rangle + 1.5\langle\text{H}_2(\text{g})\rangle + 0.5\langle\text{N}_2(\text{g})\rangle = \text{Al+3\_NH}_3\text{Cl-1(3)}_{\text{(s)}}$ |    |       |               |                 |   |       |        |
| No                                                                                                                                                                                     | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                                                                                      | 25 | 0     | Inf. Dilution | dH:-212.6 (4SF) | 4 | CRV   | 7382   |

| Reaction No. 70532                                                                                                                                                                        |    |       |               |                 |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{Al(s)} + 1.5\langle\text{Cl}_2(\text{g})\rangle + 4.5\langle\text{H}_2(\text{g})\rangle + 1.5\langle\text{N}_2(\text{g})\rangle = \text{Al+3\_NH}_3(3)\text{Cl-1(3)}_{\text{(s)}}$ |    |       |               |                 |   |       |        |
| No                                                                                                                                                                                        | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                                                                                         | 25 | 0     | Inf. Dilution | dG:-260. (3SF)  | 1 | EES   |        |
| 2                                                                                                                                                                                         | 25 | 0     | Inf. Dilution | dH:-283.0 (4SF) | 4 | CRV   | 7382   |

| Reaction No. 70533                                                                                                                                                                    |    |       |               |                 |   |       |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{Al(s)} + 1.5\langle\text{Cl}_2(\text{g})\rangle + 9\langle\text{H}_2(\text{g})\rangle + 3\langle\text{N}_2(\text{g})\rangle = \text{Al+3\_NH}_3(6)\text{Cl-1(3)}_{\text{(s)}}$ |    |       |               |                 |   |       |        |
| No                                                                                                                                                                                    | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                                                                                     | 25 | 0     | Inf. Dilution | dG:-363.4 (4SF) | 4 | CRV   | 7382   |

| Reaction No. 70534                                                                                                                                                                     |    |       |               |                 |   |       |        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{Al(s)} + 1.5\langle\text{Br}_2(\text{l})\rangle + 1.5\langle\text{H}_2(\text{g})\rangle + 0.5\langle\text{N}_2(\text{g})\rangle = \text{Al+3\_NH}_3\text{Br-1(3)}_{\text{(s)}}$ |    |       |               |                 |   |       |        |
| No                                                                                                                                                                                     | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                                                                                      | 25 | 0     | Inf. Dilution | dH:-177.6 (4SF) | 4 | CRV   | 7382   |

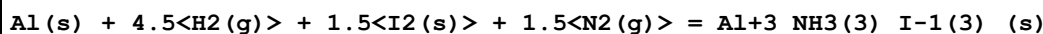
| Reaction No. 70535                                                                                                                                                                        |    |       |               |                 |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{Al(s)} + 1.5\langle\text{Br}_2(\text{l})\rangle + 4.5\langle\text{H}_2(\text{g})\rangle + 1.5\langle\text{N}_2(\text{g})\rangle = \text{Al+3\_NH}_3(3)\text{Br-1(3)}_{\text{(s)}}$ |    |       |               |                 |   |       |        |
| No                                                                                                                                                                                        | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                                                                                         | 25 | 0     | Inf. Dilution | dH:-252.7 (4SF) | 4 | CRV   | 7382   |

| Reaction No. 70536                                                                                                                                                                    |    |       |               |                 |   |       |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{Al(s)} + 1.5\langle\text{Br}_2(\text{l})\rangle + 9\langle\text{H}_2(\text{g})\rangle + 3\langle\text{N}_2(\text{g})\rangle = \text{Al+3\_NH}_3(6)\text{Br-1(3)}_{\text{(s)}}$ |    |       |               |                 |   |       |        |
| No                                                                                                                                                                                    | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                                                                                     | 25 | 0     | Inf. Dilution | dH:-343.0 (4SF) | 4 | CRV   | 7382   |

| Reaction No. 70537                                                                                                                                                                   |  |  |  |  |  |  |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|
| $\text{Al(s)} + 1.5\langle\text{H}_2(\text{g})\rangle + 1.5\langle\text{I}_2(\text{s})\rangle + 0.5\langle\text{N}_2(\text{g})\rangle = \text{Al+3\_NH}_3\text{I-1(3)}_{\text{(s)}}$ |  |  |  |  |  |  |  |

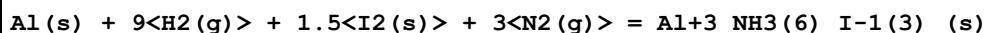
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dH:-119.0 (4SF) | 4 | CRV   | 7382   |

## Reaction No. 70538



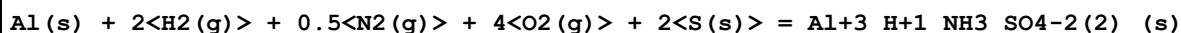
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dH:-205.9 (4SF) | 4 | CRV   | 7382   |

## Reaction No. 70539



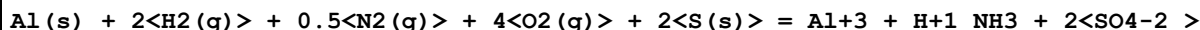
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dH:-312.9 (4SF) | 4 | CRV   | 7382   |

## Reaction No. 70540



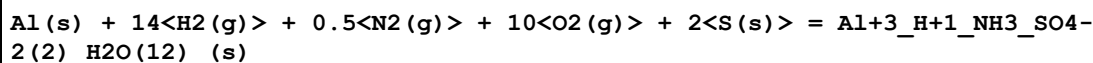
| No | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dG:-487.2 (4SF)     | 3 | CRV   | 7382   |
| 2  | 25 | 0     | Inf. Dilution | dG:-2038.2 (5SF) kJ | 4 | CRV   | 9015   |
| 3  | 25 | 0     | Inf. Dilution | dG:-487.2 (4SF)     | 4 | CRV   | 12011  |
| 4  | 25 | 0     | Inf. Dilution | dH:-562.2 (4SF)     | 3 | CRV   | 7382   |
| 5  | 25 | 0     | Inf. Dilution | dH:-2352.2 (5SF) kJ | 4 | CRV   | 9015   |
| 6  | 25 | 0     | Inf. Dilution | dH:-562.2 (4SF)     | 4 | CRV   | 12011  |

## Reaction No. 70541



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dG:-491. (3SF) | 4 | CRV   | 7382   |
| 2  | 25 | 0     | Inf. Dilution | dH:-593. (3SF) | 4 | CRV   | 7382   |

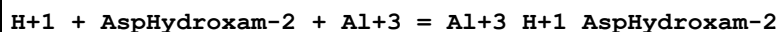
## Reaction No. 70542



| No | t  | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dG:-1180.20 (6SF)    | 4 | CRV   | 7382   |
| 2  | 25 | 0     | Inf. Dilution | dG:-4937.20 (6SF) kJ | 3 | CRV   | 9015   |
| 3  | 25 | 0     | Inf. Dilution | dG:-1180.20 (6SF)    | 4 | CRV   | 12011  |
| 4  | 25 | 0     | Inf. Dilution | dH:-1420.30 (6SF)    | 4 | CRV   | 7382   |
| 5  | 25 | 0     | Inf. Dilution | dH:-5942.40 (6SF) kJ | 3 | CRV   | 9015   |

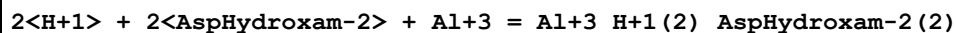
|   |    |   |               |                   |   |     |       |
|---|----|---|---------------|-------------------|---|-----|-------|
| 6 | 25 | 0 | Inf. Dilution | dH:-1420.30 (6SF) | 4 | CRV | 12011 |
|---|----|---|---------------|-------------------|---|-----|-------|

## Reaction No. 70597



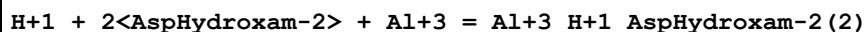
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:16.27 (0.05SD) | 5 | MGL   | 4764   |

## Reaction No. 70598



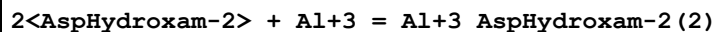
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:31.76 (0.07SD) | 5 | MGL   | 4764   |

## Reaction No. 70599



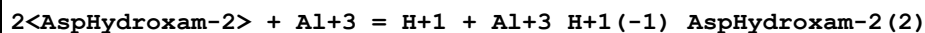
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:26.80 (0.04SD) | 5 | MGL   | 4764   |

## Reaction No. 70600



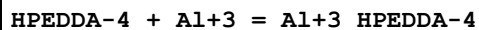
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:19.26 (0.03SD) | 5 | MGL   | 4764   |

## Reaction No. 70601



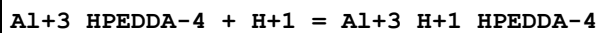
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:10.36 (0.04SD) | 5 | MGL   | 4764   |

## Reaction No. 70642



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:25.78 (0.5SD) | 4 | MGL   | 2462   |

## Reaction No. 70643



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:3.73 (0.5SD) | 4 | MGL   | 2462   |

## Reaction No. 70683

| 2<Al+3_H+1(-1)_Citric-3> = Al+3(2)_H+1(-2)_Citric-3(2) |    |       |        |                  |   |       |        |
|--------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| No                                                     | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                      | 25 | 0.5   | NaNO3  | lgK:-14.8(0.2SD) | 0 | MGL   | 2261   |

| Reaction No. 70692                                             |    |       |        |                 |   |       |        |
|----------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Al+3 + H+1(3)_Tartaric-2 + H2O = Al+3_OH-1_Tartaric-2 + 4<H+1> |    |       |        |                 |   |       |        |
| No                                                             | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                              | 25 | 0.5   | NaNO3  | lgK:-7.8(0.2SD) | 0 | MGL   | 2261   |

| Reaction No. 70693                                         |    |       |        |                  |   |       |        |
|------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 2<Al+3_H+1(-1)_Tartaric-2> = Al+3(2)_H+1(-2)_Tartaric-2(2) |    |       |        |                  |   |       |        |
| No                                                         | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                          | 25 | 0.5   | NaNO3  | lgK:-18.5(0.3SD) | 3 | MGL   | 2261   |

| Reaction No. 70701                                    |    |       |        |                  |   |       |        |
|-------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Al+3 + H+1(3)_Malic-2 = Al+3_H+1(-1)_Malic-2 + 4<H+1> |    |       |        |                  |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                     | 25 | 0.5   | NaNO3  | lgK:-10.5(0.3SD) | 3 | MGL   | 2261   |

| Reaction No. 70702                                   |    |       |        |                 |   |       |        |
|------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2<Al+3_H+1(-1)_Malic-2> = Al+3(2)_H+1(-2)_Malic-2(2) |    |       |        |                 |   |       |        |
| No                                                   | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                    | 25 | 0.5   | NaNO3  | lgK:22.2(0.4SD) | 0 | MGL   | 2261   |

| Reaction No. 70703                |    |           |        |                |   |       |        |
|-----------------------------------|----|-----------|--------|----------------|---|-------|--------|
| 2<Al+3_OH-1(4)> = Al+3(2)_OH-1(8) |    |           |        |                |   |       |        |
| No                                | t  | I-Str     | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                 | 25 | 4.4:4.7   | NaOH   | lgK:-1.43(3SF) | 2 | MRM   | 7693   |
| 2                                 | 25 | 5.9:6.3   | NaOH   | lgK:-1.32(3SF) | 2 | MRM   | 7693   |
| 3                                 | 25 | 6.9:7.1   | CsOH   | lgK:-1.38(3SF) | 2 | MRM   | 7693   |
| 4                                 | 25 | 7.3:7.8   | NaOH   | lgK:-1.24(3SF) | 2 | MRM   | 7693   |
| 5                                 | 25 | 7.7:7.9   | KOH    | lgK:-1.29(3SF) | 2 | MRM   | 7693   |
| 6                                 | 25 | 8.1:8.4   | NaOH   | lgK:-1.20(3SF) | 2 | MRM   | 7693   |
| 7                                 | 25 | 8.7:9.1   | NaOH   | lgK:-1.15(3SF) | 2 | MRM   | 7693   |
| 8                                 | 25 | 10.2:10.5 | NaOH   | lgK:-1.02(3SF) | 2 | MRM   | 7693   |

| Reaction No. 71192 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|



| Al+3_F-1(6) + 3<e-1> = Al(s) + 6<F-1> |    |       |               |                 |   |       |        |
|---------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| No                                    | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                     | 25 | 0     | Inf. Dilution | lgK:-106.3(4SF) | 1 | ELC   |        |
| 2                                     | 25 | 0     | Inf. Dilution | E:-2.069(4SF)   | 0 | CRV   | 6330   |
| 3                                     | 25 | 0     | Inf. Dilution | E:-2.067(4SF)   | 0 | CRV   | 22551  |

| Reaction No. 71194                                               |    |       |               |                  |   |       |        |
|------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Al+3_OH-1(4) + 3<e-1> = Al(s) + 4<OH-1>                          |    |       |               |                  |   |       |        |
| No                                                               | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                | 25 | 0     | Inf. Dilution | lgK:-119.3(4SF)  | 1 | ELC   |        |
| 2                                                                | 25 | 0     | Inf. Dilution | E:-2.328(4SF)    | 0 | CRV   | 6330   |
| 3                                                                | 25 | 0     | Inf. Dilution | E:-2.328(4SF)    | 0 | CRV   | 7761   |
| 4                                                                | 25 | 0     | Inf. Dilution | E:-2.31(3SF)     | 0 | CRV   | 22551  |
| Note (4): Typo in rxn from source; value also off a bit it seems |    |       |               |                  |   |       |        |
| 5                                                                | 25 | 0     | Inf. Dilution | dH:576.3(4SF) kJ | 3 | CRV   | 7761   |

| Reaction No. 71195                                                          |    |       |               |                  |   |       |        |
|-----------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Al+3_OH-1(3)_(am.,s) + 3<e-1> = Al(s) + 3<OH-1>                             |    |       |               |                  |   |       |        |
| No                                                                          | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                           | 25 | 0     | Inf. Dilution | lgK:-117.2(4SF)  | 1 | ELC   |        |
| 2                                                                           | 25 | 0     | Inf. Dilution | E:-2.31(3SF)     | 0 | CRV   | 6330   |
| Note (2): Low wgt for inter-reaction consistency (corresponds to lgK=-117.) |    |       |               |                  |   |       |        |
| 3                                                                           | 25 | 0     | Inf. Dilution | E:-2.31(3SF)     | 0 | CRV   | 7761   |
| 4                                                                           | 25 | 0     | Inf. Dilution | E:-2.3(2SF)      | 0 | CRV   | 22551  |
| 5                                                                           | 25 | 0     | Inf. Dilution | dH:584.9(4SF) kJ | 3 | CRV   | 7761   |

| Reaction No. 71320                     |    |       |               |                  |   |       |        |
|----------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Al+3_OH-1 + H+1 + 3<e-1> = Al(s) + H2O |    |       |               |                  |   |       |        |
| No                                     | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                      | 25 | 0     | Inf. Dilution | E:-1.579(4SF)    | 0 | CRV   | 7761   |
| 2                                      | 25 | 0     | Inf. Dilution | dH:481.2(4SF) kJ | 0 | CRV   | 7761   |

| Reaction No. 71321                                    |    |       |               |                  |   |       |        |
|-------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Al+3(2)_OH-1(2) + 2<H+1> + 6<e-1> = 2<Al(s)> + 2<H2O> |    |       |               |                  |   |       |        |
| No                                                    | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                     | 25 | 0     | Inf. Dilution | E:-1.60(3SF)     | 0 | CRV   | 7761   |
| 2                                                     | 25 | 0     | Inf. Dilution | dH:981.5(4SF) kJ | 0 | CRV   | 7761   |

| Reaction No. 71322                                                   |    |       |               |                   |   |       |        |
|----------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $\text{Al}+3 + 2\text{H}_2(\text{g}) + 4\text{e}^- = \text{AlH}_4^-$ |    |       |               |                   |   |       |        |
| No                                                                   | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                    | 25 | 0     | Inf. Dilution | E:-1.78 (3SF)     | 3 | CRV   | 7761   |
| 2                                                                    | 25 | 0     | Inf. Dilution | dH:662.8 (4SF) kJ | 3 | CRV   | 7761   |

| Reaction No. 71323                                                             |    |       |               |                   |   |       |        |
|--------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $\text{Al}+3 + 1.5\text{H}_2(\text{g}) + 3\text{e}^- = \text{AlH}_3(\text{s})$ |    |       |               |                   |   |       |        |
| No                                                                             | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                              | 25 | 0     | Inf. Dilution | lgK:-94.3 (3SF)   | 1 | ELC   |        |
| 2                                                                              | 25 | 0     | Inf. Dilution | E:-1.838 (4SF)    | 0 | CRV   | 7761   |
| 3                                                                              | 25 | 0     | Inf. Dilution | dH:520.1 (4SF) kJ | 1 | CRV   | 7761   |

| Reaction No. 71324                                                                                                       |    |       |               |                    |   |       |        |
|--------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| $\text{Al}_2\text{O}_3(\text{alpha},\text{s}) + 3\text{H}_2\text{O} + 6\text{e}^- = 2\text{Al}(\text{s}) + 6\text{OH}^-$ |    |       |               |                    |   |       |        |
| No                                                                                                                       | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                                                                        | 25 | 0     | Inf. Dilution | E:-2.332 (4SF)     | 0 | CRV   | 7761   |
| 2                                                                                                                        | 25 | 0     | Inf. Dilution | dH:1153.3 (5SF) kJ | 0 | CRV   | 7761   |

| Reaction No. 71325                                                                                         |    |       |               |                   |   |       |        |
|------------------------------------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $\text{Al}+3\text{OH}^- (3)_{\text{Bayerite},\text{s}} + 3\text{e}^- = \text{Al}(\text{s}) + 3\text{OH}^-$ |    |       |               |                   |   |       |        |
| No                                                                                                         | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                          | 25 | 0     | Inf. Dilution | E:-2.338 (4SF)    | 0 | CRV   | 7761   |
| 2                                                                                                          | 25 | 0     | Inf. Dilution | dH:596.7 (4SF) kJ | 0 | CRV   | 7761   |

| Reaction No. 71326                                                                                               |    |       |               |                   |   |       |        |
|------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $\text{AlOOH}(\text{Diaspore},\text{s}) + \text{H}_2\text{O} + 3\text{e}^- = \text{Al}(\text{s}) + 3\text{OH}^-$ |    |       |               |                   |   |       |        |
| No                                                                                                               | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                | 25 | 0     | Inf. Dilution | lgK:-120.0 (4SF)  | 1 | ELC   |        |
| 2                                                                                                                | 25 | 0     | Inf. Dilution | E:-2.371 (4SF)    | 0 | CRV   | 7761   |
| 3                                                                                                                | 25 | 0     | Inf. Dilution | dH:595.3 (4SF) kJ | 0 | CRV   | 7761   |

| Reaction No. 71687                                                                             |    |       |        |                    |   |       |        |
|------------------------------------------------------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| $\text{Al}+3 + \text{H}+1(-1)_{\text{Desfer-2}} = \text{Al}+3\text{H}+1(-1)_{\text{Desfer-2}}$ |    |       |        |                    |   |       |        |
| No                                                                                             | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                                              | 25 | 0.1   | KCl    | lgK:24.14 (0.01SD) | 3 | MGL   | 1382   |

|   |    |      |      |                     |   |     |      |
|---|----|------|------|---------------------|---|-----|------|
| 2 | 37 | 0.15 | NaCl | lgK:23.754(0.012SD) | 5 | MGL | 8370 |
|---|----|------|------|---------------------|---|-----|------|

| Reaction No. 71688                            |    |       |               |                     |   |       |        |
|-----------------------------------------------|----|-------|---------------|---------------------|---|-------|--------|
| Al+3 + H+1(-1)_Desfer-2 + H+1 = Al+3_Desfer-2 |    |       |               |                     |   |       |        |
| No                                            | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                             | 25 | 0     | Inf. Dilution | lgK:35.4(3SF)       | 1 | ELC   |        |
| 2                                             | 37 | 0.15  | NaCl          | lgK:33.708(0.009SD) | 5 | MGL   | 8370   |

| Reaction No. 71689                                   |    |       |               |                     |   |       |        |
|------------------------------------------------------|----|-------|---------------|---------------------|---|-------|--------|
| Al+3 + H+1(-1)_Desfer-2 + 2<H+1> = Al+3_H+1_Desfer-2 |    |       |               |                     |   |       |        |
| No                                                   | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                    | 25 | 0     | Inf. Dilution | lgK:36.3(3SF)       | 1 | ESC   |        |
| 2                                                    | 37 | 0.15  | NaCl          | lgK:35.946(0.015SD) | 5 | MGL   | 8370   |

| Reaction No. 71694                             |    |       |               |                     |   |       |        |
|------------------------------------------------|----|-------|---------------|---------------------|---|-------|--------|
| Al+3 + H+1_23DHB-3 + H+1 = Al+3_H+1(2)_23DHB-3 |    |       |               |                     |   |       |        |
| No                                             | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                              | 25 | 0     | Inf. Dilution | lgK:12.3(3SF)       | 1 | ESC   |        |
| 2                                              | 37 | 0.15  | NaCl          | lgK:12.066(0.069SD) | 6 | MGL   | 8370   |

| Reaction No. 71695                          |    |       |               |                     |   |       |        |
|---------------------------------------------|----|-------|---------------|---------------------|---|-------|--------|
| 2<Al+3> + H+1_23DHB-3 = Al+3(2)_H+1_23DHB-3 |    |       |               |                     |   |       |        |
| No                                          | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                           | 25 | 0     | Inf. Dilution | lgK:12.4(3SF)       | 1 | ESC   |        |
| 2                                           | 37 | 0.15  | NaCl          | lgK:12.156(0.054SD) | 6 | MGL   | 8370   |

| Reaction No. 71698                          |    |       |               |                     |   |       |        |
|---------------------------------------------|----|-------|---------------|---------------------|---|-------|--------|
| Al+3 + H+1(2)_Tiron-4 = Al+3_H+1(2)_Tiron-4 |    |       |               |                     |   |       |        |
| No                                          | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                           | 25 | 0     | Inf. Dilution | lgK:16.3(3SF)       | 1 | ESC   |        |
| 2                                           | 37 | 0.15  | NaCl          | lgK:15.955(0.004SD) | 6 | MGL   | 8370   |

| Reaction No. 71699                                |    |       |               |                     |   |       |        |
|---------------------------------------------------|----|-------|---------------|---------------------|---|-------|--------|
| Al+3 + 2<H+1(2)_Tiron-4> = Al+3_H+1(4)_Tiron-4(2) |    |       |               |                     |   |       |        |
| No                                                | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                 | 25 | 0     | Inf. Dilution | lgK:30.0(3SF)       | 1 | ESC   |        |
| 2                                                 | 37 | 0.15  | NaCl          | lgK:28.612(0.015SD) | 6 | MGL   | 8370   |

| Reaction No. 71700                                                                               |    |       |               |                     |   |       |        |
|--------------------------------------------------------------------------------------------------|----|-------|---------------|---------------------|---|-------|--------|
| $\text{Al}+3 + 3\text{<H}+1(2)\text{\_Tiron-4>} = \text{Al}+3\text{\_H}+1(6)\text{\_Tiron-4}(3)$ |    |       |               |                     |   |       |        |
| No                                                                                               | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                                                                | 25 | 0     | Inf. Dilution | lgK:38.2(3SF)       | 1 | ESC   |        |
| 2                                                                                                | 37 | 0.15  | NaCl          | lgK:37.376(0.048SD) | 6 | MGL   | 8370   |

| Reaction No. 71701                                                                                            |    |       |               |                     |   |       |        |
|---------------------------------------------------------------------------------------------------------------|----|-------|---------------|---------------------|---|-------|--------|
| $\text{Al}+3 + 3\text{<H}+1(2)\text{\_Tiron-4>} = \text{Al}+3\text{\_H}+1(5)\text{\_Tiron-4}(3) + \text{H}+1$ |    |       |               |                     |   |       |        |
| No                                                                                                            | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                                                                             | 25 | 0     | Inf. Dilution | lgK:26.3(3SF)       | 1 | ESC   |        |
| 2                                                                                                             | 37 | 0.15  | NaCl          | lgK:25.925(0.103SD) | 6 | MGL   | 8370   |

| Reaction No. 71703                                                                                      |    |       |               |                    |   |       |        |
|---------------------------------------------------------------------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| $\text{Al}+3 + \text{H}+1(3)\text{\_Tiron-4} = \text{Al}+3\text{\_H}+1(2)\text{\_Tiron-4} + \text{H}+1$ |    |       |               |                    |   |       |        |
| No                                                                                                      | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                                                       | 25 | 0     | Inf. Dilution | lgK:4.4(2SF)       | 1 | ESC   |        |
| 2                                                                                                       | 37 | 0.15  | NaCl          | lgK:4.168(0.005SD) | 6 | MGL   | 8370   |

| Reaction No. 71704                                                                                                      |    |       |               |                    |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| $\text{Al}+3 + 2\text{<H}+1(3)\text{\_Tiron-4>} = \text{Al}+3\text{\_H}+1(4)\text{\_Tiron-4}(2) + 2\text{<H}+1\text{>}$ |    |       |               |                    |   |       |        |
| No                                                                                                                      | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                                                                       | 25 | 0     | Inf. Dilution | lgK:6.7(2SF)       | 1 | ELC   |        |
| 2                                                                                                                       | 37 | 0.15  | NaCl          | lgK:5.036(0.020SD) | 6 | MGL   | 8370   |

| Reaction No. 71705                                                                                                      |    |       |               |                    |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| $\text{Al}+3 + 3\text{<H}+1(3)\text{\_Tiron-4>} = \text{Al}+3\text{\_H}+1(6)\text{\_Tiron-4}(3) + 3\text{<H}+1\text{>}$ |    |       |               |                    |   |       |        |
| No                                                                                                                      | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                                                                       | 25 | 0     | Inf. Dilution | lgK:3.3(2SF)       | 1 | ELC   |        |
| 2                                                                                                                       | 37 | 0.15  | NaCl          | lgK:2.019(0.069SD) | 6 | MGL   | 8370   |

| Reaction No. 71706                                                                                                      |    |       |               |                     |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|---------------------|---|-------|--------|
| $\text{Al}+3 + 3\text{<H}+1(3)\text{\_Tiron-4>} = \text{Al}+3\text{\_H}+1(5)\text{\_Tiron-4}(3) + 4\text{<H}+1\text{>}$ |    |       |               |                     |   |       |        |
| No                                                                                                                      | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                                                                                       | 25 | 0     | Inf. Dilution | lgK:-8.6(2SF)       | 1 | ESC   |        |
| 2                                                                                                                       | 37 | 0.15  | NaCl          | lgK:-8.809(0.107SD) | 6 | MGL   | 8370   |

| Reaction No. 71715 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| Al+3_OH-1 + H2O = Al+3_OH-1(2) + H+1 |     |       |        |                |   |       |        |
|--------------------------------------|-----|-------|--------|----------------|---|-------|--------|
| No                                   | t   | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                    | 0   | 0     | NaCl/m | lgK:-6.4 (2SF) | 4 | MSL   | 7025   |
| 2                                    | 0   | 0.1   | NaCl/m | lgK:-6.6 (2SF) | 5 | MSL   | 7025   |
| 3                                    | 0   | 0.5   | NaCl/m | lgK:-6.7 (2SF) | 5 | MSL   | 7025   |
| 4                                    | 0   | 1     | NaCl/m | lgK:-6.8 (2SF) | 5 | MSL   | 7025   |
| 5                                    | 0   | 5     | NaCl/m | lgK:-7.5 (2SF) | 5 | MSL   | 7025   |
| 6                                    | 25  | 0     | NaCl/m | lgK:-5.6 (2SF) | 4 | MSL   | 7025   |
| 7                                    | 25  | 0.1   | NaCl/m | lgK:-5.8 (2SF) | 5 | MSL   | 7025   |
| 8                                    | 25  | 0.5   | NaCl/m | lgK:-6.0 (2SF) | 5 | MSL   | 7025   |
| 9                                    | 25  | 1     | NaCl/m | lgK:-6.0 (2SF) | 5 | MSL   | 7025   |
| 10                                   | 25  | 5     | NaCl/m | lgK:-6.6 (2SF) | 5 | MSL   | 7025   |
| 11                                   | 50  | 0     | NaCl/m | lgK:-5.0 (2SF) | 4 | MSL   | 7025   |
| 12                                   | 50  | 0.1   | NaCl/m | lgK:-5.2 (2SF) | 5 | MSL   | 7025   |
| 13                                   | 50  | 0.5   | NaCl/m | lgK:-5.3 (2SF) | 5 | MSL   | 7025   |
| 14                                   | 50  | 1     | NaCl/m | lgK:-5.4 (2SF) | 5 | MSL   | 7025   |
| 15                                   | 50  | 5     | NaCl/m | lgK:-6.0 (2SF) | 5 | MSL   | 7025   |
| 16                                   | 75  | 0     | NaCl/m | lgK:-4.5 (2SF) | 4 | MSL   | 7025   |
| 17                                   | 75  | 0.1   | NaCl/m | lgK:-4.7 (2SF) | 5 | MSL   | 7025   |
| 18                                   | 75  | 0.5   | NaCl/m | lgK:-4.8 (2SF) | 5 | MSL   | 7025   |
| 19                                   | 75  | 1     | NaCl/m | lgK:-4.8 (2SF) | 5 | MSL   | 7025   |
| 20                                   | 75  | 5     | NaCl/m | lgK:-5.4 (2SF) | 5 | MSL   | 7025   |
| 21                                   | 100 | 0     | NaCl/m | lgK:-4.0 (2SF) | 4 | MSL   | 7025   |
| 22                                   | 100 | 0.1   | NaCl/m | lgK:-4.2 (2SF) | 5 | MSL   | 7025   |
| 23                                   | 100 | 0.5   | NaCl/m | lgK:-4.3 (2SF) | 5 | MSL   | 7025   |
| 24                                   | 100 | 1     | NaCl/m | lgK:-4.3 (2SF) | 5 | MSL   | 7025   |
| 25                                   | 100 | 5     | NaCl/m | lgK:-5.0 (2SF) | 5 | MSL   | 7025   |
| 26                                   | 125 | 0     | NaCl/m | lgK:-3.6 (2SF) | 4 | MSL   | 7025   |
| 27                                   | 150 | 0     | NaCl/m | lgK:-3.2 (2SF) | 4 | MSL   | 7025   |
| 28                                   | 175 | 0     | NaCl/m | lgK:-2.9 (2SF) | 4 | MSL   | 7025   |
| 29                                   | 200 | 0     | NaCl/m | lgK:-2.6 (2SF) | 4 | MSL   | 7025   |
| 30                                   | 225 | 0     | NaCl/m | lgK:-2.4 (2SF) | 4 | MSL   | 7025   |
| 31                                   | 250 | 0     | NaCl/m | lgK:-2.1 (2SF) | 4 | MSL   | 7025   |
| 32                                   | 275 | 0     | NaCl/m | lgK:-1.9 (2SF) | 4 | MSL   | 7025   |

Reaction No. 71716

| Al+3_OH-1(2) + H2O = Al+3_OH-1(3) + H+1 |     |       |        |               |   |       |        |
|-----------------------------------------|-----|-------|--------|---------------|---|-------|--------|
| No                                      | t   | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                       | 0   | 0     | NaCl/m | lgK:-8.0(2SF) | 4 | MSL   | 7025   |
| 2                                       | 0   | 0.1   | NaCl/m | lgK:-8.0(2SF) | 5 | MSL   | 7025   |
| 3                                       | 0   | 0.5   | NaCl/m | lgK:-8.0(2SF) | 5 | MSL   | 7025   |
| 4                                       | 0   | 1     | NaCl/m | lgK:-8.0(2SF) | 5 | MSL   | 7025   |
| 5                                       | 0   | 5     | NaCl/m | lgK:-8.1(2SF) | 5 | MSL   | 7025   |
| 6                                       | 25  | 0     | NaCl/m | lgK:-6.7(2SF) | 4 | MSL   | 7025   |
| 7                                       | 25  | 0.1   | NaCl/m | lgK:-6.8(2SF) | 5 | MSL   | 7025   |
| 8                                       | 25  | 0.5   | NaCl/m | lgK:-6.8(2SF) | 5 | MSL   | 7025   |
| 9                                       | 25  | 1     | NaCl/m | lgK:-6.8(2SF) | 5 | MSL   | 7025   |
| 10                                      | 25  | 5     | NaCl/m | lgK:-6.8(2SF) | 5 | MSL   | 7025   |
| 11                                      | 50  | 0     | NaCl/m | lgK:-5.8(2SF) | 4 | MSL   | 7025   |
| 12                                      | 50  | 0.1   | NaCl/m | lgK:-5.8(2SF) | 5 | MSL   | 7025   |
| 13                                      | 50  | 0.5   | NaCl/m | lgK:-5.8(2SF) | 5 | MSL   | 7025   |
| 14                                      | 50  | 1     | NaCl/m | lgK:-5.8(2SF) | 5 | MSL   | 7025   |
| 15                                      | 50  | 5     | NaCl/m | lgK:-5.8(2SF) | 5 | MSL   | 7025   |
| 16                                      | 75  | 0     | NaCl/m | lgK:-5.0(2SF) | 4 | MSL   | 7025   |
| 17                                      | 75  | 0.1   | NaCl/m | lgK:-5.0(2SF) | 5 | MSL   | 7025   |
| 18                                      | 75  | 0.5   | NaCl/m | lgK:-5.0(2SF) | 5 | MSL   | 7025   |
| 19                                      | 75  | 1     | NaCl/m | lgK:-5.0(2SF) | 5 | MSL   | 7025   |
| 20                                      | 75  | 5     | NaCl/m | lgK:-5.0(2SF) | 5 | MSL   | 7025   |
| 21                                      | 100 | 0     | NaCl/m | lgK:-4.3(2SF) | 4 | MSL   | 7025   |
| 22                                      | 100 | 0.1   | NaCl/m | lgK:-4.3(2SF) | 5 | MSL   | 7025   |
| 23                                      | 100 | 0.5   | NaCl/m | lgK:-4.3(2SF) | 5 | MSL   | 7025   |
| 24                                      | 100 | 1     | NaCl/m | lgK:-4.3(2SF) | 5 | MSL   | 7025   |
| 25                                      | 100 | 5     | NaCl/m | lgK:-4.4(2SF) | 5 | MSL   | 7025   |
| 26                                      | 125 | 0     | NaCl/m | lgK:-3.8(2SF) | 4 | MSL   | 7025   |
| 27                                      | 150 | 0     | NaCl/m | lgK:-3.3(2SF) | 4 | MSL   | 7025   |
| 28                                      | 175 | 0     | NaCl/m | lgK:-3.0(2SF) | 4 | MSL   | 7025   |
| 29                                      | 200 | 0     | NaCl/m | lgK:-2.7(2SF) | 4 | MSL   | 7025   |
| 30                                      | 225 | 0     | NaCl/m | lgK:-2.4(2SF) | 4 | MSL   | 7025   |
| 31                                      | 250 | 0     | NaCl/m | lgK:-2.2(2SF) | 4 | MSL   | 7025   |
| 32                                      | 275 | 0     | NaCl/m | lgK:-2.0(2SF) | 4 | MSL   | 7025   |
| 33                                      | 300 | 0     | NaCl/m | lgK:-1.9(2SF) | 4 | MSL   | 7025   |

| Reaction No. 71717                      |     |       |        |               |   |       |        |
|-----------------------------------------|-----|-------|--------|---------------|---|-------|--------|
| Al+3_OH-1(3) + H2O = Al+3_OH-1(4) + H+1 |     |       |        |               |   |       |        |
| No                                      | t   | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                       | 0   | 0     | NaCl/m | lgK:-5.7(2SF) | 4 | MSL   | 7025   |
| 2                                       | 0   | 0.1   | NaCl/m | lgK:-5.5(2SF) | 5 | MSL   | 7025   |
| 3                                       | 0   | 0.5   | NaCl/m | lgK:-5.4(2SF) | 5 | MSL   | 7025   |
| 4                                       | 0   | 1     | NaCl/m | lgK:-5.4(2SF) | 5 | MSL   | 7025   |
| 5                                       | 0   | 5     | NaCl/m | lgK:-5.8(2SF) | 5 | MSL   | 7025   |
| 6                                       | 25  | 0     | NaCl/m | lgK:-5.6(2SF) | 4 | MSL   | 7025   |
| 7                                       | 25  | 0.1   | NaCl/m | lgK:-5.4(2SF) | 5 | MSL   | 7025   |
| 8                                       | 25  | 0.5   | NaCl/m | lgK:-5.3(2SF) | 5 | MSL   | 7025   |
| 9                                       | 25  | 1     | NaCl/m | lgK:-5.3(2SF) | 5 | MSL   | 7025   |
| 10                                      | 25  | 5     | NaCl/m | lgK:-5.7(2SF) | 5 | MSL   | 7025   |
| 11                                      | 50  | 0     | NaCl/m | lgK:-5.5(2SF) | 4 | MSL   | 7025   |
| 12                                      | 50  | 0.1   | NaCl/m | lgK:-5.2(2SF) | 5 | MSL   | 7025   |
| 13                                      | 50  | 0.5   | NaCl/m | lgK:-5.1(2SF) | 5 | MSL   | 7025   |
| 14                                      | 50  | 1     | NaCl/m | lgK:-5.1(2SF) | 5 | MSL   | 7025   |
| 15                                      | 50  | 5     | NaCl/m | lgK:-5.5(2SF) | 5 | MSL   | 7025   |
| 16                                      | 75  | 0     | NaCl/m | lgK:-5.3(2SF) | 4 | MSL   | 7025   |
| 17                                      | 75  | 0.1   | NaCl/m | lgK:-5.1(2SF) | 5 | MSL   | 7025   |
| 18                                      | 75  | 0.5   | NaCl/m | lgK:-5.0(2SF) | 5 | MSL   | 7025   |
| 19                                      | 75  | 1     | NaCl/m | lgK:-5.0(2SF) | 5 | MSL   | 7025   |
| 20                                      | 75  | 5     | NaCl/m | lgK:-5.2(2SF) | 5 | MSL   | 7025   |
| 21                                      | 100 | 0     | NaCl/m | lgK:-5.2(2SF) | 4 | MSL   | 7025   |
| 22                                      | 100 | 0.1   | NaCl/m | lgK:-5.0(2SF) | 5 | MSL   | 7025   |
| 23                                      | 100 | 0.5   | NaCl/m | lgK:-4.8(2SF) | 5 | MSL   | 7025   |
| 24                                      | 100 | 1     | NaCl/m | lgK:-4.8(2SF) | 5 | MSL   | 7025   |
| 25                                      | 100 | 5     | NaCl/m | lgK:-5.0(2SF) | 5 | MSL   | 7025   |
| 26                                      | 125 | 0     | NaCl/m | lgK:-5.1(2SF) | 4 | MSL   | 7025   |
| 27                                      | 150 | 0     | NaCl/m | lgK:-5.0(2SF) | 4 | MSL   | 7025   |
| 28                                      | 175 | 0     | NaCl/m | lgK:-4.9(2SF) | 4 | MSL   | 7025   |
| 29                                      | 200 | 0     | NaCl/m | lgK:-4.8(2SF) | 4 | MSL   | 7025   |
| 30                                      | 225 | 0     | NaCl/m | lgK:-4.7(2SF) | 4 | MSL   | 7025   |
| 31                                      | 250 | 0     | NaCl/m | lgK:-4.7(2SF) | 4 | MSL   | 7025   |
| 32                                      | 275 | 0     | NaCl/m | lgK:-4.6(2SF) | 4 | MSL   | 7025   |
| 33                                      | 300 | 0     | NaCl/m | lgK:-4.5(2SF) | 3 | MSL   | 7025   |

| Reaction No. 71723                                                                                          |     |       |               |                              |   |       |        |
|-------------------------------------------------------------------------------------------------------------|-----|-------|---------------|------------------------------|---|-------|--------|
| $\text{AlOOH}(\text{Diaspore}, \text{s}) + \text{OH}^- + \text{H}_2\text{O} = \text{Al} + 3\text{OH}^- (4)$ |     |       |               |                              |   |       |        |
| No                                                                                                          | t   | I-Str | Medium        | Value (Dev)                  | W | Tech. | Ref. # |
| 1                                                                                                           | 25  | 0     | Inf. Dilution | $\lg K: -0.7 (1\text{SF})$   | 1 | ELC   |        |
| 2                                                                                                           | 135 | 0     | Inf. Dilution | $\lg K: -1.017 (4\text{SF})$ | 3 | MSL   | 6028   |
| 3                                                                                                           | 150 | 0     | Inf. Dilution | $\lg K: -0.858 (3\text{SF})$ | 3 | MSL   | 6028   |
| 4                                                                                                           | 200 | 0     | Inf. Dilution | $\lg K: -0.400 (3\text{SF})$ | 2 | MSL   | 6028   |
| 5                                                                                                           | 250 | 0     | Inf. Dilution | $\lg K: -0.030 (2\text{SF})$ | 1 | MSL   | 6028   |
| 6                                                                                                           | 300 | 0     | Inf. Dilution | $\lg K: 0.276 (3\text{SF})$  | 0 | MSL   | 6028   |

| Reaction No. 71757                                                                          |    |       |               |                                               |   |       |        |
|---------------------------------------------------------------------------------------------|----|-------|---------------|-----------------------------------------------|---|-------|--------|
| $\text{Al}(\text{s}) + 0.5\text{F}_2(\text{g}) - 2\text{e}^- = \text{Al} + 3\text{F}^- (1)$ |    |       |               |                                               |   |       |        |
| No                                                                                          | t  | I-Str | Medium        | Value (Dev)                                   | W | Tech. | Ref. # |
| 1                                                                                           | 25 | 0     | Inf. Dilution | $\lg K: 141.7 (4\text{SF})$                   | 1 | ELC   |        |
| 2                                                                                           | 25 | 0     | GAMSPATH      | $\text{dG}: -803.9 (4\text{SF}) \text{ kJ}$   | 0 | CRV   | 12638  |
| 3                                                                                           | 25 | 0     | GAMSPATH      | $\text{dH}: -860.340 (6\text{SF}) \text{ kJ}$ | 2 | CRV   | 12638  |

| Reaction No. 71758                                                                      |    |       |               |                                               |   |       |        |
|-----------------------------------------------------------------------------------------|----|-------|---------------|-----------------------------------------------|---|-------|--------|
| $\text{Al}(\text{s}) + \text{F}_2(\text{g}) - \text{e}^- = \text{Al} + 3\text{F}^- (2)$ |    |       |               |                                               |   |       |        |
| No                                                                                      | t  | I-Str | Medium        | Value (Dev)                                   | W | Tech. | Ref. # |
| 1                                                                                       | 25 | 0     | Inf. Dilution | $\lg K: 196.5 (4\text{SF})$                   | 1 | ELC   |        |
| 2                                                                                       | 25 | 0     | GAMSPATH      | $\text{dG}: -1115. (4\text{SF}) \text{ kJ}$   | 0 | CRV   | 12638  |
| 3                                                                                       | 25 | 0     | GAMSPATH      | $\text{dH}: -1189.10 (6\text{SF}) \text{ kJ}$ | 2 | CRV   | 12638  |

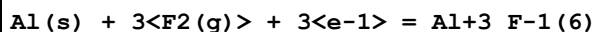
| Reaction No. 71759                                                                       |    |       |               |                                               |   |       |        |
|------------------------------------------------------------------------------------------|----|-------|---------------|-----------------------------------------------|---|-------|--------|
| $\text{Al}(\text{s}) + 2\text{F}_2(\text{g}) + \text{e}^- = \text{Al} + 3\text{F}^- (4)$ |    |       |               |                                               |   |       |        |
| No                                                                                       | t  | I-Str | Medium        | Value (Dev)                                   | W | Tech. | Ref. # |
| 1                                                                                        | 25 | 0     | Inf. Dilution | $\lg K: 301.4 (4\text{SF})$                   | 1 | ELC   |        |
| 2                                                                                        | 25 | 0     | GAMSPATH      | $\text{dG}: -1709. (4\text{SF}) \text{ kJ}$   | 0 | CRV   | 12638  |
| 3                                                                                        | 25 | 0     | GAMSPATH      | $\text{dH}: -1853.10 (6\text{SF}) \text{ kJ}$ | 2 | CRV   | 12638  |

| Reaction No. 71760                                                                          |    |       |               |                                             |   |       |        |
|---------------------------------------------------------------------------------------------|----|-------|---------------|---------------------------------------------|---|-------|--------|
| $\text{Al}(\text{s}) + 2.5\text{F}_2(\text{g}) + 2\text{e}^- = \text{Al} + 3\text{F}^- (5)$ |    |       |               |                                             |   |       |        |
| No                                                                                          | t  | I-Str | Medium        | Value (Dev)                                 | W | Tech. | Ref. # |
| 1                                                                                           | 25 | 0     | Inf. Dilution | $\lg K: 352.1 (4\text{SF})$                 | 1 | ELC   |        |
| 2                                                                                           | 25 | 0     | GAMSPATH      | $\text{dG}: -1996. (4\text{SF}) \text{ kJ}$ | 0 | CRV   | 12638  |



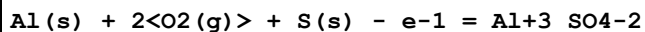
|   |    |   |          |                      |   |     |       |
|---|----|---|----------|----------------------|---|-----|-------|
| 3 | 25 | 0 | GAMSPATH | dH:-2187.10 (6SF) kJ | 2 | CRV | 12638 |
|---|----|---|----------|----------------------|---|-----|-------|

## Reaction No. 71761



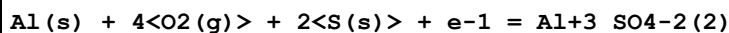
| No | t  | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:401.7 (4SF)      | 1 | ELC   |        |
| 2  | 25 | 0     | GAMSPATH      | dG:-2277. (4SF) kJ   | 0 | CRV   | 12638  |
| 3  | 25 | 0     | GAMSPATH      | dH:-2525.80 (6SF) kJ | 2 | CRV   | 12638  |

## Reaction No. 71762



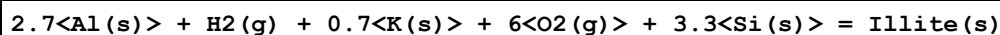
| No | t  | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:219.8 (4SF)      | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | dG:-1250.40 (6SF) kJ | 0 | EOD   | 23101  |
| 3  | 25 | 0     | GAMSPATH      | dG:-1247. (4SF) kJ   | 0 | CRV   | 12638  |
| 4  | 25 | 0     | Inf. Dilution | dH:-1422.6 (5SF) kJ  | 3 | EOD   | 23101  |
| 5  | 25 | 0     | GAMSPATH      | dH:-1431.10 (6SF) kJ | 3 | CRV   | 12638  |

## Reaction No. 71763



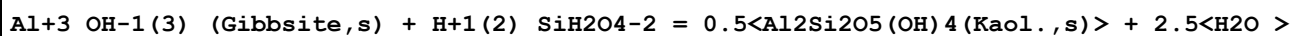
| No | t  | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:351.1 (4SF)      | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | dG:-2006.30 (6SF) kJ | 0 | EOD   | 23101  |
| 3  | 25 | 0     | GAMSPATH      | dG:-2002. (4SF) kJ   | 0 | CRV   | 12638  |
| 4  | 25 | 0     | Inf. Dilution | dH:-2338.40 (6SF) kJ | 0 | EOD   | 23101  |
| 5  | 25 | 0     | GAMSPATH      | dH:-2337.00 (6SF) kJ | 0 | CRV   | 12638  |

## Reaction No. 71820



| No | t  | I-Str | Medium   | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|----------|--------------------|---|-------|--------|
| 1  | 25 | 0     | GAMSPATH | dG:-5491. (4SF) kJ | 3 | CRV   | 12638  |
| 2  | 25 | 0     | GAMSPATH | dH:-5870. (4SF) kJ | 3 | CRV   | 12638  |

## Reaction No. 71872



| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:4.1 (2SF) | 1 | ELC   |        |

|                  |    |   |               |               |   |     |      |
|------------------|----|---|---------------|---------------|---|-----|------|
| 2                | 25 | 0 | Inf. Dilution | lgK:4.25(3SF) | 0 | CRV | 6405 |
| Note (2): p. 246 |    |   |               |               |   |     |      |

| Reaction No. 71873                                                                                                                                                                                                                                                                                        |    |       |               |                |   |       |        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4(\text{Kaol.},s) + 2\langle\text{K}+1\rangle + 6\langle\text{Mg}+2\rangle + 4\langle\text{H}+1(2)\text{SiH}_2\text{O}_4-2\rangle - \text{H}_2\text{O} = 2\langle\text{KMg}_3\text{AlSi}_3\text{O}_{10}(\text{OH})_2(s)\rangle + 14\langle\text{H}+1\rangle$ |    |       |               |                |   |       |        |
| No                                                                                                                                                                                                                                                                                                        | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                                                                                                                                                                                                                         | 25 | 0     | Inf. Dilution | lgK:-69.2(3SF) | 1 | ELC   |        |
| 2                                                                                                                                                                                                                                                                                                         | 25 | 0     | Inf. Dilution | lgK:70.14(4SF) | 0 | CRV   | 6405   |
| Note (2): p. 246                                                                                                                                                                                                                                                                                          |    |       |               |                |   |       |        |

| Reaction No. 71874                                                                                                                                                                                      |    |       |               |               |   |       |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $\text{KAl}_3\text{Si}_3\text{O}_{10}(\text{OH})_2(s) + \text{H}+1 + 1.5\langle\text{H}_2\text{O}\rangle = 1.5\langle\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4(\text{Kaol.},s)\rangle + \text{K}+1$ |    |       |               |               |   |       |        |
| No                                                                                                                                                                                                      | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                                                                                                                                       | 25 | 0     | Inf. Dilution | lgK:1.4(2SF)  | 1 | ELC   |        |
| 2                                                                                                                                                                                                       | 25 | 0     | Inf. Dilution | lgK:3.51(3SF) | 2 | CRV   | 6405   |

| Reaction No. 71875                                                                                                                                                                 |    |       |               |                |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $\text{CaO}.\text{Al}_2\text{O}_3.2\text{SiO}_2(s) + 2\langle\text{H}+1\rangle + \text{H}_2\text{O} = \text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4(\text{Kaol.},s) + \text{Ca}+2$ |    |       |               |                |   |       |        |
| No                                                                                                                                                                                 | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                                                                                                  | 25 | 0     | Inf. Dilution | lgK:19.3(3SF)  | 1 | ELC   |        |
| 2                                                                                                                                                                                  | 25 | 0     | Inf. Dilution | lgK:19.66(4SF) | 0 | CRV   | 6405   |

| Reaction No. 71876                                                                                                                                                                                                                                                           |    |       |               |               |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $2\langle\text{KAlSi}_3\text{O}_8(\text{Feld.},s)\rangle + 2\langle\text{H}+1\rangle + 9\langle\text{H}_2\text{O}\rangle = \text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4(\text{Kaol.},s) + 4\langle\text{H}+1(2)\text{SiH}_2\text{O}_4-2\rangle + 2\langle\text{K}+1\rangle$ |    |       |               |               |   |       |        |
| No                                                                                                                                                                                                                                                                           | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                                                                                                                                                                                                            | 25 | 0     | Inf. Dilution | lgK:-7.0(2SF) | 3 | CRV   | 6405   |

| Reaction No. 71877                                                                                                                                                                                                                                                                                                                 |    |       |               |                 |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{Na}_2\text{O}.( \text{Al}_2\text{O}_3 )_7.( \text{SiO}_2 )_{22}.6\text{H}_2\text{O}(s) + 2\langle\text{H}+1\rangle + 23\langle\text{H}_2\text{O}\rangle = 7\langle\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4(\text{Kaol.},s)\rangle + 8\langle\text{H}+1(2)\text{SiH}_2\text{O}_4-2\rangle + 2\langle\text{Na}+1\rangle$ |    |       |               |                 |   |       |        |
| No                                                                                                                                                                                                                                                                                                                                 | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                                                                                                                                                                                                                                  | 25 | 0     | Inf. Dilution | lgK:-18.22(4SF) | 5 | CRV   | 6405   |
| Note (1): p. 246 - comment re inconsistency with Stumm & Morgan data                                                                                                                                                                                                                                                               |    |       |               |                 |   |       |        |

| Reaction No. 71878                                                                                                                                                                                                                                                                                         |  |  |  |  |  |  |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|
| $\text{CaO}.( \text{Al}_2\text{O}_3 )_7.( \text{SiO}_2 )_{22}.6\text{H}_2\text{O}(s) + 2\langle\text{H}+1\rangle + 23\langle\text{H}_2\text{O}\rangle = 7\langle\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4(\text{Kaol.},s)\rangle + 8\langle\text{H}+1(2)\text{SiH}_2\text{O}_4-2\rangle + \text{Ca}+2$ |  |  |  |  |  |  |  |

| No                                                                   | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| 1                                                                    | 25 | 0     | Inf. Dilution | lgK:-13.3(3SF) | 1 | ELC   |        |
| 2                                                                    | 25 | 0     | Inf. Dilution | lgK:-15.4(3SF) | 0 | CRV   | 6405   |
| Note (2): p. 246 - comment re inconsistency with Stumm & Morgan data |    |       |               |                |   |       |        |

| Reaction No. 72237                                                                |    |       |               |                 |   |       |        |
|-----------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Al <sub>2</sub> O <sub>3</sub> (alpha,s) + 6<H+1> = 2<Al+3> + 3<H <sub>2</sub> O> |    |       |               |                 |   |       |        |
| No                                                                                | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                 | 25 | 0     | Inf. Dilution | lgK:18.7(3SF)   | 1 | ELC   |        |
| 2                                                                                 | 25 | 0     | Inf. Dilution | lgK:21.376(5SF) | 0 | CRV   | 440    |
| 3                                                                                 | 25 | 0     | Inf. Dilution | lgK:18.902(5SF) | 0 | CRV   | 5619   |
| 4                                                                                 | 25 | 0     | Inf. Dilution | lgK:17.492(5SF) | 0 | CRV   | 5627   |
| 5                                                                                 | 25 | 0     | Inf. Dilution | lgK:23.442(5SF) | 0 | CRV   | 6421   |
| 6                                                                                 | 25 | 0     | Inf. Dilution | lgK:19.266(5SF) | 0 | CRV   | 8872   |
| 7                                                                                 | 25 | 0     | Inf. Dilution | lgK:19.104(5SF) | 0 | EOD   | 8872   |
| Note (7): Data from Khodakovsky, 1983                                             |    |       |               |                 |   |       |        |
| 8                                                                                 | 25 | 0     | Inf. Dilution | lgK:23.442(5SF) | 0 | EOD   | 8872   |
| Note (8): Data from Naumov, Ryzhenko & Khodakovsky, 1971                          |    |       |               |                 |   |       |        |
| 9                                                                                 | 25 | 0     | Inf. Dilution | lgK:17.038(5SF) | 0 | CRV   | 8873   |
| 10                                                                                | 25 | 0     | Inf. Dilution | lgK:22.98(4SF)  | 0 | EOD   | 23102  |
| 11                                                                                | 25 | 0     | Inf. Dilution | lgK:21.415(5SF) | 0 | CRV   | 32224  |

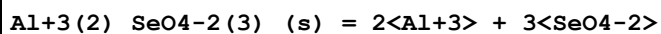
| Reaction No. 72448                                                   |    |       |               |                 |   |       |        |
|----------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Al+3 + 2<H+1_Malonic-2> + 2<OH-1> = Al+3_H+1(2)_OH-1(2)_Malonic-2(2) |    |       |               |                 |   |       |        |
| No                                                                   | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                    | 25 | 0     | Inf. Dilution | lgK:27.2(3SF)   | 2 | EAE   |        |
| 2                                                                    | 35 | 0     | Inf. Dilution | lgK:27.3(1.0SD) | 4 | MSL   | 7062   |
| 3                                                                    | 80 | 0     | Inf. Dilution | lgK:27.6(0.5SD) | 4 | MSL   | 7062   |

| Reaction No. 72848                    |    |       |               |                   |   |       |        |
|---------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| 2<Al+3> + Citric-3 = Al+3(2)_Citric-3 |    |       |               |                   |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                     | 25 | 0     | Inf. Dilution | lgK:12.3(3SF)     | 2 | EAE   |        |
| 2                                     | 31 | 0     | Unknown       | lgK:12.26(0.28SD) | 3 | EOD   | 6004   |

| Reaction No. 72943                                                    |  |  |  |  |  |  |  |
|-----------------------------------------------------------------------|--|--|--|--|--|--|--|
| Al+3(2)_SeO <sub>3</sub> -2(3)_(s) = 2<Al+3> + 3<SeO <sub>3</sub> -2> |  |  |  |  |  |  |  |

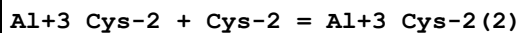
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-32.47(4SF) | 4 | ENS   | 12931  |

## Reaction No. 72967



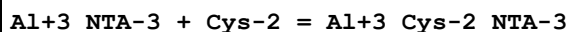
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-21.46(4SF) | 4 | ENS   | 12931  |

## Reaction No. 73216



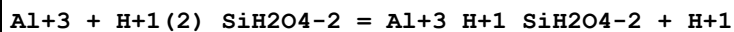
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:16.1(3SF)  | 1 | ESC   |        |
| 2  | 35 | 0.1   | Unknown       | lgK:15.78(4SF) | 0 | EOD   | 6223   |

## Reaction No. 73217



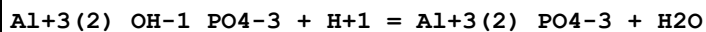
| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:6.4(2SF)  | 1 | ESC   |        |
| 2  | 35 | 0.1   | Unknown       | lgK:6.23(3SF) | 0 | EOD   | 6223   |

## Reaction No. 73370



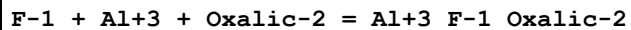
| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-2.50(0.05SD) | 4 | MGL   | 7023   |

## Reaction No. 73374



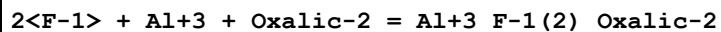
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:14.21(0.04SD) | 5 | MGL   | 7048   |

## Reaction No. 73482



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.6   | NaCl   | lgK:11.53(0.03MD) | 7 | MMR   | 14695  |

## Reaction No. 73483



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

|   |    |     |      |                   |   |     |       |
|---|----|-----|------|-------------------|---|-----|-------|
| 1 | 25 | 0.6 | NaCl | lgK:15.67(0.03MD) | 7 | MMR | 14695 |
|---|----|-----|------|-------------------|---|-----|-------|

| Reaction No. 73484                              |    |       |        |                |   |       |        |
|-------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| F-1 + Al+3 + 2<Oxalic-2> = Al+3_F-1_Oxalic-2(2) |    |       |        |                |   |       |        |
| No                                              | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                               | 25 | 0.6   | NaCl   | lgK:15.74(4SF) | 7 | MMR   | 14695  |

| Reaction No. 73485                                    |    |       |        |                   |   |       |        |
|-------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| 2<F-1> + Al+3 + 2<Oxalic-2> = Al+3_F-1(2)_Oxalic-2(2) |    |       |        |                   |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                     | 25 | 0.6   | NaCl   | lgK:19.10(0.04MD) | 7 | MMR   | 14695  |

| Reaction No. 73531                                           |    |       |               |                     |   |       |        |
|--------------------------------------------------------------|----|-------|---------------|---------------------|---|-------|--------|
| 2<Al+3> + 6<H2O> + 3<SeO3-2> = Al+3(2)_SeO3-2(3)_H2O(6)_ (s) |    |       |               |                     |   |       |        |
| No                                                           | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                            | 25 | 0     | Inf. Dilution | lgK:31.000(2.000SD) | 6 | CRV   | 14923  |

| Reaction No. 74318 |    |       |               |              |   |       |        |
|--------------------|----|-------|---------------|--------------|---|-------|--------|
| Al+1 + e-1 = Al(s) |    |       |               |              |   |       |        |
| No                 | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
| 1                  | 25 | 0     | Inf. Dilution | lgK:9.3(2SF) | 1 | EFE   |        |
| 2                  | 25 | 0     | Inf. Dilution | E:0.55(2SF)  | 3 | CRV   | 22551  |

| Reaction No. 74319          |    |       |               |              |   |       |        |
|-----------------------------|----|-------|---------------|--------------|---|-------|--------|
| Al+3 + 2<e-1> + H+1 = AlH+2 |    |       |               |              |   |       |        |
| No                          | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
| 1                           | 25 | 0     | Inf. Dilution | E:-0.62(2SF) | 4 | CRV   | 22551  |

| Reaction No. 74320                              |    |       |               |               |   |       |        |
|-------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Al+3_OH-1(4) + 6<e-1> + 3<H2O> = AlH3 + 7<OH-1> |    |       |               |               |   |       |        |
| No                                              | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                               | 25 | 0     | Inf. Dilution | E:-1.748(4SF) | 4 | CRV   | 22551  |

| Reaction No. 74663                                   |    |       |        |                    |   |       |        |
|------------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| Al+3 + H+1(2)_Ascorbic-2 = Al+3_H+1_Ascorbic-2 + H+1 |    |       |        |                    |   |       |        |
| No                                                   | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                    | 25 | 0.6   | NaCl   | lgK:-2.59(0.036SD) | 7 | MGL   | 22912  |

| Reaction No. 75028                                  |      |       |               |                   |   |       |        |
|-----------------------------------------------------|------|-------|---------------|-------------------|---|-------|--------|
| $\text{CrO4-2} + \text{Al+3} = \text{Al+3\_CrO4-2}$ |      |       |               |                   |   |       |        |
| No                                                  | t    | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                   | 0.01 | 0     | Inf. Dilution | lgK:6.7347 (5SF)  | 3 | EOD   | 22958  |
| 2                                                   | 25   | 0     | Inf. Dilution | lgK:7.2333 (5SF)  | 3 | EOD   | 22958  |
| 3                                                   | 60   | 0     | Inf. Dilution | lgK:7.9720 (5SF)  | 3 | EOD   | 22958  |
| 4                                                   | 100  | 0     | Inf. Dilution | lgK:8.8113 (5SF)  | 3 | EOD   | 22958  |
| 5                                                   | 150  | 0     | Inf. Dilution | lgK:9.8647 (5SF)  | 3 | EOD   | 22958  |
| 6                                                   | 200  | 0     | Inf. Dilution | lgK:10.9730 (6SF) | 3 | EOD   | 22958  |

| Reaction No. 75131                                                        |    |       |         |                    |   |       |        |
|---------------------------------------------------------------------------|----|-------|---------|--------------------|---|-------|--------|
| $\text{Al+3} + \text{H+1\_HexCOO-1} = \text{Al+3\_HexCOO-1} + \text{H+1}$ |    |       |         |                    |   |       |        |
| No                                                                        | t  | I-Str | Medium  | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                         | 25 | 0.6   | Na (Cl) | lgK:-3.48 (0.25MD) | 7 | MGL   | 17107  |

| Reaction No. 75132                                                                                          |    |       |         |                     |   |       |        |
|-------------------------------------------------------------------------------------------------------------|----|-------|---------|---------------------|---|-------|--------|
| $2\text{<Al+3>} + \text{H+1\_HexCOO-1} + 2\text{<H2O>} = \text{Al+3(2)\_HexCOO-1\_OH-1(2)} + 3\text{<H+1>}$ |    |       |         |                     |   |       |        |
| No                                                                                                          | t  | I-Str | Medium  | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                                                                           | 25 | 0.6   | Na (Cl) | lgK:-8.04 (0.011MD) | 7 | MGL   | 17107  |

| Reaction No. 75133                                                          |    |       |         |                    |   |       |        |
|-----------------------------------------------------------------------------|----|-------|---------|--------------------|---|-------|--------|
| $\text{Al+3} + \text{H+1\_Benzoic-1} = \text{Al+3\_Benzoic-1} + \text{H+1}$ |    |       |         |                    |   |       |        |
| No                                                                          | t  | I-Str | Medium  | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                           | 25 | 0.6   | Na (Cl) | lgK:-2.67 (0.07MD) | 7 | MGL   | 17107  |

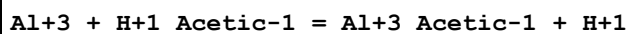
| Reaction No. 75134                                                                                            |    |       |         |                      |   |       |        |
|---------------------------------------------------------------------------------------------------------------|----|-------|---------|----------------------|---|-------|--------|
| $2\text{<Al+3>} + \text{H+1\_Benzoic-1} + 2\text{<H2O>} = \text{Al+3(2)\_Benzoic-1\_OH-1(2)} + 3\text{<H+1>}$ |    |       |         |                      |   |       |        |
| No                                                                                                            | t  | I-Str | Medium  | Value (Dev)          | W | Tech. | Ref. # |
| 1                                                                                                             | 25 | 0.6   | Na (Cl) | lgK:-7.446 (0.008MD) | 7 | MGL   | 17107  |

| Reaction No. 75135                                                                  |    |       |         |                    |   |       |        |
|-------------------------------------------------------------------------------------|----|-------|---------|--------------------|---|-------|--------|
| $\text{Al+3} + \text{H+1(2)\_3OHBenz-2} = \text{Al+3\_H+1\_3OHBenz-2} + \text{H+1}$ |    |       |         |                    |   |       |        |
| No                                                                                  | t  | I-Str | Medium  | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                                   | 25 | 0.6   | Na (Cl) | lgK:-2.59 (0.06MD) | 7 | MGL   | 17107  |

| Reaction No. 75136                                                                                         |  |  |  |  |  |  |  |
|------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|
| $2\text{<Al+3>} + \text{H+1(2)\_3OHBenz-2} + \text{H2O} = \text{Al+3(2)\_OH-1\_3OHBenz-2} + 3\text{<H+1>}$ |  |  |  |  |  |  |  |

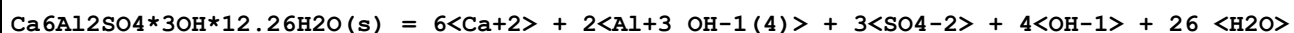
| No | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
|----|----|-------|--------|---------------------|---|-------|--------|
| 1  | 25 | 0.6   | Na(Cl) | lgK:-7.453(0.007MD) | 7 | MGL   | 17107  |

## Reaction No. 75137



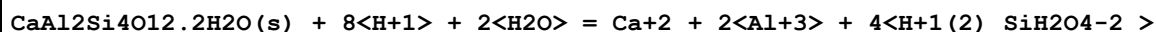
| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-2.1(2SF)     | 1 | ELC   |        |
| 2  | 25 | 0.6   | Na(Cl)        | lgK:-3.97(0.54MD) | 5 | MGL   | 17107  |

## Reaction No. 75219



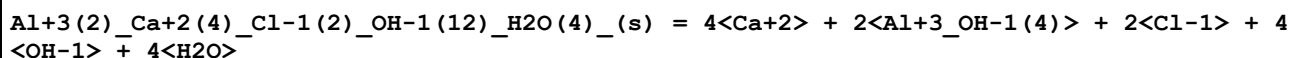
| No                                                                    | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|-----------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| 1                                                                     | 5  | 0     | Inf. Dilution | lgK:-47.3(3SF) | 4 | MSL   | 8775   |
| 2                                                                     | 25 | 0     | Inf. Dilution | lgK:-44.7(3SF) | 3 | MSL   | 8775   |
| 3                                                                     | 25 | 0     | Inf. Dilution | lgK:-44.9(3SF) | 3 | MSL   | 8775   |
| Note (3): Value in text appears to differ slightly from Eqn in Figure |    |       |               |                |   |       |        |
| 4                                                                     | 25 | 0     | Inf. Dilution | lgK:-44.9(3SF) | 3 | EOD   | 23102  |
| 5                                                                     | 25 | 0     | Inf. Dilution | lgK:-44.9(3SF) | 5 | CRV   | 31308  |
| 6                                                                     | 50 | 0     | Inf. Dilution | lgK:-41.9(3SF) | 4 | MSL   | 8775   |
| 7                                                                     | 75 | 0     | Inf. Dilution | lgK:-39.6(3SF) | 4 | MSL   | 8775   |

## Reaction No. 75220



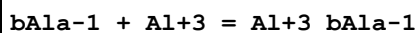
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:18.87(4SF) | 0 | EOD   | 23102  |
| 2  | 25 | 0     | Inf. Dilution | lgK:14.42(4SF) | 3 | CRV   | 27383  |
| 3  | 25 | 0     | Inf. Dilution | lgK:18.68(4SF) | 0 | CRV   | 32224  |

## Reaction No. 75225



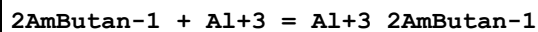
| No                                                    | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|-------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| 1                                                     | 25 | 0     | Inf. Dilution | lgK:-28.(2SF)   | 1 | MSL   | 23099  |
| Note (1): Value wuithin the range -28.8 < lgK < -27.6 |    |       |               |                 |   |       |        |
| 2                                                     | 25 | 0     | Inf. Dilution | lgK:-27.27(4SF) | 5 | CRV   | 31308  |

## Reaction No. 75273



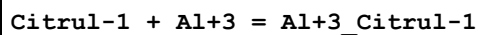
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:6.4 (2SF)   | 1 | EAE   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:6.4 (2SF)   | 1 | ESC   |        |
| 3  | 37 | 0.15  | Blood Plasma  | lgK:6.226 (4SF) | 5 | CRV   | 22388  |

## Reaction No. 75280



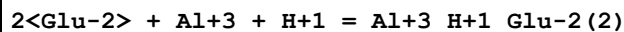
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:2.29 (3SF) | 2 | EAE   |        |
| 2  | 37 | 0.15  | Blood Plasma  | lgK:1.5 (2SF)  | 5 | CRV   | 22388  |

## Reaction No. 75282



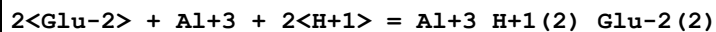
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:7.01 (3SF)  | 2 | EAE   |        |
| 2  | 37 | 0.15  | Blood Plasma  | lgK:6.226 (4SF) | 5 | CRV   | 22388  |

## Reaction No. 75285



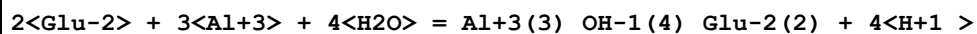
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:21.1 (3SF)  | 2 | EAE   |        |
| 2  | 37 | 0.15  | Blood Plasma  | lgK:18.85 (4SF) | 5 | CRV   | 22388  |

## Reaction No. 75286



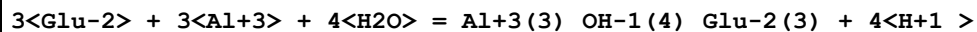
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:25.6 (3SF)   | 2 | EAE   |        |
| 2  | 37 | 0.15  | Blood Plasma  | lgK:23.425 (5SF) | 5 | CRV   | 22388  |

## Reaction No. 75287



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:7.47 (3SF) | 2 | EAE   |        |
| 2  | 37 | 0.15  | Blood Plasma  | lgK:3.81 (3SF) | 5 | CRV   | 22388  |

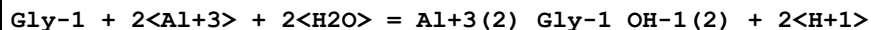
## Reaction No. 75288





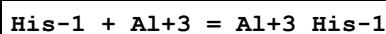
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:15.4 (3SF)  | 2 | EAE   |        |
| 2  | 37 | 0.15  | Blood Plasma  | lgK:11.22 (4SF) | 5 | CRV   | 22388  |

## Reaction No. 75289



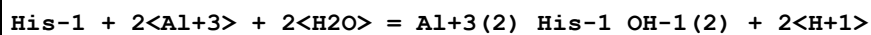
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:1.39 (3SF)  | 2 | EAE   |        |
| 2  | 37 | 0.15  | Blood Plasma  | lgK:0.361 (3SF) | 5 | CRV   | 22388  |

## Reaction No. 75290



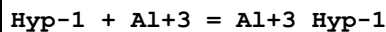
| No                                                         | t  | I-Str | Medium       | Value (Dev)     | W | Tech. | Ref. # |
|------------------------------------------------------------|----|-------|--------------|-----------------|---|-------|--------|
| 1                                                          | 25 | 0.1   | NaNO3        | lgK:10.37 (4SF) | 0 | MGL   | 22560  |
| Note (1): Khalil & Fazary, Monat. Chem., 2004, 135, 1455   |    |       |              |                 |   |       |        |
| 2                                                          | 25 | 0.1   | Unknown      | lgK:7.21 (3SF)  | 4 | MGL   | 22560  |
| 3                                                          | 25 | 0.5   | NaClO4       | lgK:8.45 (3SF)  | 3 | MGL   | 22560  |
| Note (3): Charlet et al, Bull. Soc. Chim. Fr., 1984, I-222 |    |       |              |                 |   |       |        |
| 4                                                          | 25 | 0.5   | Unknown      | lgK:8.11 (3SF)  | 3 | CRV   | 2556   |
| Note (4): Species given as M(OH)HL                         |    |       |              |                 |   |       |        |
| 5                                                          | 37 | 0.15  | Blood Plasma | lgK:7.08 (3SF)  | 5 | CRV   | 22388  |

## Reaction No. 75291



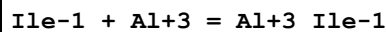
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:2.19 (3SF)  | 2 | EAE   |        |
| 2  | 37 | 0.15  | Blood Plasma  | lgK:1.163 (4SF) | 5 | CRV   | 20085  |

## Reaction No. 75292



| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:9.01 (3SF)  | 2 | EAE   |        |
| 2  | 37 | 0.15  | Blood Plasma  | lgK:8.226 (4SF) | 5 | CRV   | 22388  |

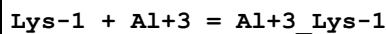
## Reaction No. 75293



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

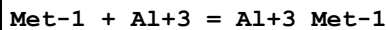
|   |    |      |               |                |   |     |       |
|---|----|------|---------------|----------------|---|-----|-------|
| 1 | 25 | 0    | Inf. Dilution | lgK:7.01(3SF)  | 2 | EAE |       |
| 2 | 37 | 0.15 | Blood Plasma  | lgK:6.226(4SF) | 5 | CRV | 22388 |

## Reaction No. 75294



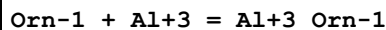
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:7.8(2SF)   | 1 | ERG   |        |
| 2  | 37 | 0.15  | Blood Plasma  | lgK:6.226(4SF) | 5 | CRV   | 22388  |

## Reaction No. 75296



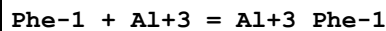
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:7.01(3SF)  | 2 | EAE   |        |
| 2  | 37 | 0.15  | Blood Plasma  | lgK:6.226(4SF) | 5 | CRV   | 22388  |

## Reaction No. 75297



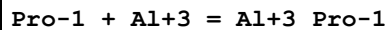
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:7.01(3SF)  | 2 | EAE   |        |
| 2  | 37 | 0.15  | Blood Plasma  | lgK:6.226(4SF) | 5 | CRV   | 22388  |

## Reaction No. 75298



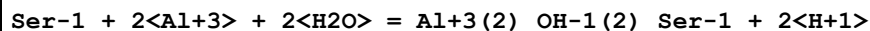
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:7.01(3SF)  | 2 | EAE   |        |
| 2  | 37 | 0.15  | Blood Plasma  | lgK:6.226(4SF) | 5 | CRV   | 22388  |

## Reaction No. 75299



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:7.01(3SF)  | 2 | EAE   |        |
| 2  | 37 | 0.15  | Blood Plasma  | lgK:6.226(4SF) | 5 | CRV   | 22388  |

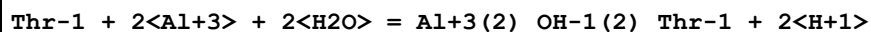
## Reaction No. 75300



| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:1.22(3SF) | 2 | EAE   |        |

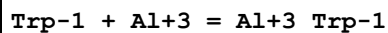
|   |    |      |              |                 |   |     |       |
|---|----|------|--------------|-----------------|---|-----|-------|
| 2 | 37 | 0.15 | Blood Plasma | lgK:0.198 (3SF) | 5 | CRV | 22388 |
|---|----|------|--------------|-----------------|---|-----|-------|

## Reaction No. 75301



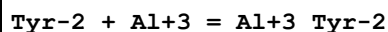
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:0.85 (3SF)   | 2 | EAE   |        |
| 2  | 37 | 0.15  | Blood Plasma  | lgK:-0.177 (3SF) | 5 | CRV   | 22388  |

## Reaction No. 75302



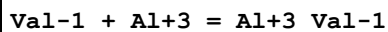
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:7.01 (3SF)  | 2 | EAE   |        |
| 2  | 37 | 0.15  | Blood Plasma  | lgK:6.226 (4SF) | 5 | CRV   | 22388  |

## Reaction No. 75303



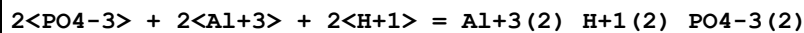
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:7.73 (3SF)  | 2 | EAE   |        |
| 2  | 37 | 0.15  | Blood Plasma  | lgK:6.226 (4SF) | 5 | CRV   | 22388  |

## Reaction No. 75304



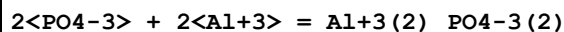
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:7.01 (3SF)  | 2 | EAE   |        |
| 2  | 37 | 0.15  | Blood Plasma  | lgK:6.226 (4SF) | 5 | CRV   | 22388  |

## Reaction No. 75305



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:43.3 (3SF) | 2 | EAE   |        |
| 2  | 37 | 0.15  | Blood Plasma  | lgK:39.2 (3SF) | 5 | CRV   | 22388  |

## Reaction No. 75306



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:37.8 (3SF) | 2 | EAE   |        |
| 2  | 37 | 0.15  | Blood Plasma  | lgK:33.4 (3SF) | 5 | CRV   | 22388  |

| Reaction No. 75307                                                                |    |       |               |               |   |       |        |
|-----------------------------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $\text{PO4-3} + \text{Al+3} + \text{H2O} = \text{Al+3\_OH-1\_PO4-3} + \text{H+1}$ |    |       |               |               |   |       |        |
| No                                                                                | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                 | 25 | 0     | Inf. Dilution | lgK:8.37(3SF) | 2 | EFE   | 21910  |
| Note (1): Based on Al+3_PO4-3 and probably wrong                                  |    |       |               |               |   |       |        |
| 2                                                                                 | 37 | 0.15  | Blood Plasma  | lgK:6.3(2SF)  | 2 | CRV   | 22388  |
| Note (2): Derived from Harris [92Har_21910]                                       |    |       |               |               |   |       |        |

| Reaction No. 75310                                                                                  |    |       |               |                |   |       |        |
|-----------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $3\text{<Lactic-1>} + 2\text{<Al+3>} + \text{H2O} = \text{Al+3(2)\_Lactic-1(3)\_OH-1} + \text{H+1}$ |    |       |               |                |   |       |        |
| No                                                                                                  | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                   | 25 | 0     | Inf. Dilution | lgK:8.53(3SF)  | 2 | EAE   |        |
| 2                                                                                                   | 37 | 0.15  | Blood Plasma  | lgK:6.555(4SF) | 5 | CRV   | 22388  |

| Reaction No. 75311                                                                                           |    |       |               |                |   |       |        |
|--------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $3\text{<Lactic-1>} + 2\text{<Al+3>} + 2\text{<H2O>} = \text{Al+3(2)\_Lactic-1(3)\_OH-1(2)} + 2\text{<H+1>}$ |    |       |               |                |   |       |        |
| No                                                                                                           | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                            | 25 | 0     | Inf. Dilution | lgK:5.54(3SF)  | 2 | EAE   |        |
| 2                                                                                                            | 37 | 0.15  | Blood Plasma  | lgK:3.324(4SF) | 5 | CRV   | 22388  |

| Reaction No. 75312                                                                                   |    |       |               |                |   |       |        |
|------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $\text{Malic-2} + 2\text{<Al+3>} + 2\text{<H2O>} = \text{Al+3(2)\_OH-1(2)\_Malic-2} + 2\text{<H+1>}$ |    |       |               |                |   |       |        |
| No                                                                                                   | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                    | 25 | 0     | Inf. Dilution | lgK:2.55(3SF)  | 2 | EAE   |        |
| 2                                                                                                    | 37 | 0.15  | Blood Plasma  | lgK:0.564(3SF) | 5 | CRV   | 22388  |

| Reaction No. 75313                                                                                         |    |       |               |                 |   |       |        |
|------------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $2\text{<Malic-2>} + 2\text{<Al+3>} + 4\text{<H2O>} = \text{Al+3(2)\_OH-1(4)\_Malic-2(2)} + 4\text{<H+1>}$ |    |       |               |                 |   |       |        |
| No                                                                                                         | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                          | 25 | 0     | Inf. Dilution | lgK:-2.24(3SF)  | 2 | EAE   |        |
| 2                                                                                                          | 37 | 0.15  | Blood Plasma  | lgK:-4.462(4SF) | 5 | CRV   | 22388  |

| Reaction No. 75314                                                                                |    |       |               |                 |   |       |        |
|---------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $3\text{<Malic-2>} + 2\text{<Al+3>} + \text{H2O} = \text{Al+3(2)\_OH-1\_Malic-2(3)} + \text{H+1}$ |    |       |               |                 |   |       |        |
| No                                                                                                | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                 | 25 | 0     | Inf. Dilution | lgK:16.2(3SF)   | 2 | EAE   |        |
| 2                                                                                                 | 37 | 0.15  | Blood Plasma  | lgK:12.789(5SF) | 5 | CRV   | 22388  |

| Reaction No. 75315                                                                                                         |    |       |               |                 |   |       |        |
|----------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $4\text{<Malic-2>} + 3\text{<Al+3>} + 4\text{<H2O>} = \text{Al+3(3)}_{\text{OH-1(4)}}_{\text{Malic-2(4)}} + 4\text{<H+1>}$ |    |       |               |                 |   |       |        |
| No                                                                                                                         | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                          | 25 | 0     | Inf. Dilution | lgK:13.8(3SF)   | 2 | EAE   |        |
| 2                                                                                                                          | 37 | 0.15  | Blood Plasma  | lgK:10.132(5SF) | 5 | CRV   | 22388  |

| Reaction No. 75316                                                                                                         |    |       |               |                 |   |       |        |
|----------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $4\text{<Malic-2>} + 4\text{<Al+3>} + 5\text{<H2O>} = \text{Al+3(4)}_{\text{OH-1(5)}}_{\text{Malic-2(4)}} + 5\text{<H+1>}$ |    |       |               |                 |   |       |        |
| No                                                                                                                         | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                          | 25 | 0     | Inf. Dilution | lgK:16.1(3SF)   | 2 | EAE   |        |
| 2                                                                                                                          | 37 | 0.15  | Blood Plasma  | lgK:10.537(5SF) | 5 | CRV   | 22388  |

| Reaction No. 75319                                                |    |       |               |               |   |       |        |
|-------------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $\text{Pyruvic-1} + \text{Al+3} = \text{Al+3}_{\text{Pyruvic-1}}$ |    |       |               |               |   |       |        |
| No                                                                | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                 | 25 | 0     | Inf. Dilution | lgK:2.29(3SF) | 2 | EAE   |        |
| 2                                                                 | 37 | 0.15  | Blood Plasma  | lgK:1.5(2SF)  | 5 | CRV   | 22388  |

| Reaction No. 75320                                                          |    |       |               |                 |   |       |        |
|-----------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{Salicylic-2} + 2\text{<Al+3>} = \text{Al+3(2)}_{\text{Salicylic-2}}$ |    |       |               |                 |   |       |        |
| No                                                                          | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                           | 25 | 0     | Inf. Dilution | lgK:15.4(3SF)   | 2 | EAE   |        |
| 2                                                                           | 37 | 0.15  | Blood Plasma  | lgK:14.665(5SF) | 5 | CRV   | 22388  |

| Reaction No. 75321                                                                              |    |       |               |                 |   |       |        |
|-------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{Salicylic-2} + \text{Al+3} + \text{H+1} = \text{Al+3}_{\text{H+1}}_{\text{Salicylic-2}}$ |    |       |               |                 |   |       |        |
| No                                                                                              | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                               | 25 | 0     | Inf. Dilution | lgK:15.9(3SF)   | 2 | EAE   |        |
| 2                                                                                               | 37 | 0.15  | Blood Plasma  | lgK:14.641(5SF) | 5 | CRV   | 22388  |

| Reaction No. 75322                                                                                                         |    |       |               |                 |   |       |        |
|----------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{Succinic-2} + 2\text{<Al+3>} + 2\text{<H2O>} = \text{Al+3(2)}_{\text{OH-1(2)}}_{\text{Succinic-2}} + 2\text{<H+1>}$ |    |       |               |                 |   |       |        |
| No                                                                                                                         | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                          | 25 | 0     | Inf. Dilution | lgK:0.882(3SF)  | 2 | EAE   |        |
| 2                                                                                                                          | 37 | 0.15  | Blood Plasma  | lgK:-1.103(4SF) | 5 | CRV   | 22388  |

| Reaction No. 75323 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| 2<Succinic-2> + 3<Al+3> + H2O = Al+3(3)_OH-1_Succinic-2(2) + H+1 |    |       |               |                |   |       |        |
|------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| No                                                               | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                | 25 | 0     | Inf. Dilution | lgK:11.3(3SF)  | 2 | EAE   |        |
| 2                                                                | 37 | 0.15  | Blood Plasma  | lgK:9.114(4SF) | 5 | CRV   | 22388  |

| Reaction No. 75324                                                        |    |       |               |                |   |       |        |
|---------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| 2<Succinic-2> + 3<Al+3> + 3<H2O> = Al+3(3)_OH-1(3)_Succinic-2(2) + 3<H+1> |    |       |               |                |   |       |        |
| No                                                                        | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                         | 25 | 0     | Inf. Dilution | lgK:4.8(3SF)   | 2 | EAE   |        |
| 2                                                                         | 37 | 0.15  | Blood Plasma  | lgK:1.385(4SF) | 5 | CRV   | 22388  |

| Reaction No. 75325                                                |    |       |               |                 |   |       |        |
|-------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| 3<Succinic-2> + 4<Al+3> + 6<OH-1> = Al+3(4)_OH-1(6)_Succinic-2(3) |    |       |               |                 |   |       |        |
| No                                                                | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                 | 25 | 0     | Inf. Dilution | lgK:82.4(3SF)   | 2 | EAE   |        |
| 2                                                                 | 37 | 0.15  | Blood Plasma  | lgK:75.838(5SF) | 5 | CRV   | 22388  |

| Reaction No. 75326                                                  |    |       |               |                  |   |       |        |
|---------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 4<Succinic-2> + 8<Al+3> + 12<OH-1> = Al+3(8)_OH-1(12)_Succinic-2(4) |    |       |               |                  |   |       |        |
| No                                                                  | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                   | 25 | 0     | Inf. Dilution | lgK:158.0(3SF)   | 2 | EAE   |        |
| 2                                                                   | 37 | 0.15  | Blood Plasma  | lgK:148.410(6SF) | 5 | CRV   | 22388  |

| Reaction No. 75531                        |    |       |               |                 |   |       |        |
|-------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Al+3_OH-1_SO4-2_(s) = Al+3 + OH-1 + SO4-2 |    |       |               |                 |   |       |        |
| No                                        | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                         | 25 | 0     | Inf. Dilution | lgK:-17.23(4SF) | 5 | MSL   | 23912  |

| Reaction No. 75559                                                       |    |       |               |                      |   |       |        |
|--------------------------------------------------------------------------|----|-------|---------------|----------------------|---|-------|--------|
| 6<Al(s)> + Cl2(g) + 8<Na(s)> + 12<O2(g)> + 6<Si(s)> = Na8Al6Si6O24Cl2(s) |    |       |               |                      |   |       |        |
| No                                                                       | t  | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
| 1                                                                        | 25 | 0     | Inf. Dilution | dG:-1.2697E+4(5SF)kJ | 2 | EFE   | 24802  |

| Reaction No. 75632                                 |   |       |        |             |   |       |        |
|----------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| Al2O3(alpha,s) + 5<H2O> = 2<Al+3_OH-1(4)> + 2<H+1> |   |       |        |             |   |       |        |
| Note: Some log Ks in source should be ln Ks?       |   |       |        |             |   |       |        |
| No                                                 | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

|                     |     |   |               |                   |   |     |       |
|---------------------|-----|---|---------------|-------------------|---|-----|-------|
| 1                   | 0   | 0 | Inf. Dilution | lgK:-63.369 (5SF) | 0 | CRV | 25435 |
| 2                   | 25  | 0 | Inf. Dilution | lgK:-26.5 (3SF)   | 1 | ELC |       |
| 3                   | 25  | 0 | Inf. Dilution | lgK:-26.45 (4SF)  | 0 | CRV | 25435 |
| Note (3): Corrected |     |   |               |                   |   |     |       |
| 4                   | 25  | 0 | Inf. Dilution | lgK:-60.908 (5SF) | 0 | CRV | 25435 |
| 5                   | 60  | 0 | Inf. Dilution | lgK:-58.656 (5SF) | 0 | CRV | 25435 |
| 6                   | 100 | 0 | Inf. Dilution | lgK:-56.998 (5SF) | 0 | CRV | 25435 |
| 7                   | 150 | 0 | Inf. Dilution | lgK:-55.809 (5SF) | 0 | CRV | 25435 |
| 8                   | 200 | 0 | Inf. Dilution | lgK:-55.425 (5SF) | 0 | CRV | 25435 |
| 9                   | 250 | 0 | Inf. Dilution | lgK:-55.928 (5SF) | 0 | CRV | 25435 |
| 10                  | 300 | 0 | Inf. Dilution | lgK:-57.757 (5SF) | 0 | CRV | 25435 |

| Reaction No. 75633                                                       |    |       |               |                     |   |       |        |
|--------------------------------------------------------------------------|----|-------|---------------|---------------------|---|-------|--------|
| 1.7<Al(s)> + 0.7<Ca(s)> + 0.3<Na(s)> + 4<O2(g)> + Si(s) = Plagioclase(s) |    |       |               |                     |   |       |        |
| No                                                                       | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                                        | 25 | 0     | Inf. Dilution | dG:-3906.5 (5SF) kJ | 5 | CRV   | 25435  |
| 2                                                                        | 25 | 0     | Inf. Dilution | dH:-4108.2 (5SF) kJ | 5 | CRV   | 25435  |

| Reaction No. 75636                                             |    |       |               |                      |   |       |        |
|----------------------------------------------------------------|----|-------|---------------|----------------------|---|-------|--------|
| 2<Al(s)> + 2<H2(g)> + 3.72<O2(g)> + 1.22<Si(s)> = Allophane(s) |    |       |               |                      |   |       |        |
| No                                                             | t  | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
| 1                                                              | 25 | 0     | Inf. Dilution | dG:-3227.90 (6SF) kJ | 3 | CRV   | 25435  |
| 2                                                              | 25 | 0     | Inf. Dilution | dH:-3512.20 (6SF) kJ | 3 | CRV   | 25435  |

| Reaction No. 75825                  |    |       |        |                 |   |       |        |
|-------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Al+3_Asp-2 + Asp-2 = Al+3_Asp-2 (2) |    |       |        |                 |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | NaClO4 | lgK:14.40 (4SF) | 0 | MGL   | 5763   |

| Reaction No. 75911                           |    |       |        |                |   |       |        |
|----------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Al+3_Lactic-1 + Lactic-1 = Al+3_Lactic-1 (2) |    |       |        |                |   |       |        |
| No                                           | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                            | 20 | 0.2   | NaNO3  | lgK:2.18 (3SF) | 3 | ENS   | 1242   |
| 2                                            | 25 | 0.6   | NaCl   | lgK:2.06 (3SF) | 3 | MGL   | 916    |
| 3                                            | 25 | 1     | NaClO4 | lgK:0.54 (2SF) | 0 | MGL   | 4204   |

| Reaction No. 75983 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| Al+3_Malic-2 + Malic-2 = Al+3_Malic-2(2)                     |    |       |               |               |   |       |        |
|--------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| No                                                           | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                            | 20 | 0.2   | KNO3          | lgK:3.98(3SF) | 5 | MEF   | 11516  |
| Note (1): Pavlinova A et al., Ukr. Khim. Zhur., 1969, 35, 37 |    |       |               |               |   |       |        |
| 2                                                            | 25 | 0     | Inf. Dilution | lgK:4.42(3SF) | 2 | EAE   |        |
| 3                                                            | 37 | 0.15  | NaCl          | lgK:3.80(3SF) | 5 | MGL   | 10029  |

| Reaction No. 76190                 |    |       |        |               |   |       |        |
|------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Al+3_NTA-3 + NTA-3 = Al+3_NTA-3(2) |    |       |        |               |   |       |        |
| No                                 | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                  | 25 | 0.1   | NaClO4 | lgK:8.55(3SF) | 5 | MGL   | 1411   |

| Reaction No. 76225                       |    |       |        |               |   |       |        |
|------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Al+3_23DHB-3 + 23DHB-3 = Al+3_23DHB-3(2) |    |       |        |               |   |       |        |
| No                                       | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                        | 25 | 0.2   | KCl    | lgK:7.94(3SF) | 5 | MGL   | 2470   |

| Reaction No. 76236                          |    |       |         |               |   |       |        |
|---------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Al+3_Formic-1 + Formic-1 = Al+3_Formic-1(2) |    |       |         |               |   |       |        |
| No                                          | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 25 | 1     | Unknown | lgK:1.20(3SF) | 5 | MSC   | 2103   |

| Reaction No. 76253                          |    |       |         |               |   |       |        |
|---------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Al+3_Acetic-1 + Acetic-1 = Al+3_Acetic-1(2) |    |       |         |               |   |       |        |
| No                                          | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | NaCl    | lgK:1.48(3SF) | 5 | MEF   | 4277   |
| 2                                           | 60 | 0.02  | Unknown | lgK:1.77(3SF) | 5 | MSL   | 6039   |

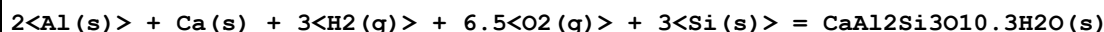
| Reaction No. 76276                                   |    |       |        |               |   |       |        |
|------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Al+3_Propanoic-1 + Propanoic-1 = Al+3_Propanoic-1(2) |    |       |        |               |   |       |        |
| No                                                   | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                    | 25 | 1     | NaClO4 | lgK:1.62(3SF) | 5 | MGL   | 4204   |

| Reaction No. 76501                                                                |    |       |               |                |   |       |        |
|-----------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| 2<Al(s)> + 2<H2(g)> + 2<Na(s)> + 6<O2(g)> + 3<Si(s)> = Na2Al2Si3O10.2H2O(Natr.,s) |    |       |               |                |   |       |        |
| No                                                                                | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                 | 25 | 0     | Inf. Dilution | lgK:931.4(4SF) | 3 | CRV   | 6494   |



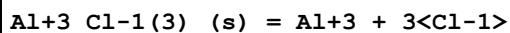
|   |    |   |               |                     |   |     |       |
|---|----|---|---------------|---------------------|---|-----|-------|
| 2 | 25 | 0 | Inf. Dilution | dG:-5316.5(5SF)kJ   | 3 | CRV | 6494  |
| 3 | 25 | 0 | Inf. Dilution | dG:-5316.6(5.0SD)kJ | 5 | MCL | 29317 |
| 4 | 25 | 0 | Inf. Dilution | dH:-5718.6(5SF)kJ   | 3 | CRV | 6494  |
| 5 | 25 | 0 | Inf. Dilution | dH:-5718.6(5.0SD)kJ | 5 | CRV | 29317 |

## Reaction No. 76503



| No | t  | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:980.7(4SF)       | 3 | CRV   | 6494   |
| 2  | 25 | 0     | Inf. Dilution | dG:-5597.6(5SF)kJ    | 3 | CRV   | 6494   |
| 3  | 25 | 0     | Inf. Dilution | dG:-5597.9(5.0SD)kJ  | 5 | MCL   | 29317  |
| 4  | 25 | 0     | Inf. Dilution | dH:-6049.0(5SF)kJ    | 3 | CRV   | 6494   |
| 5  | 25 | 0     | Inf. Dilution | dH:-6049.00(5.5SD)kJ | 4 | CRV   | 27383  |
| 6  | 25 | 0     | Inf. Dilution | dH:-6049.0(5.0SD)kJ  | 5 | CRV   | 29317  |

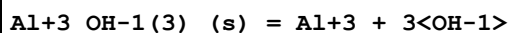
## Reaction No. 76546



| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:44.1(3SF) | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:4.31(3SF) | 0 | CRV   |        |

Note (2): <http://www.saltlakemetals.com/SolubilityProducts.htm> (15/7/14)

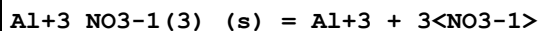
## Reaction No. 76547



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-34.3(3SF) | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:-32.3(3SF) | 0 | CRV   |        |

Note (2): <http://www.saltlakemetals.com/SolubilityProducts.htm> (15/7/14)

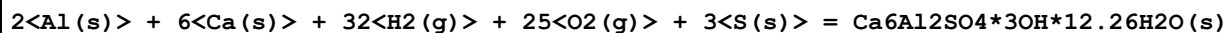
## Reaction No. 76548



| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:3.33(3SF) | 1 | CRV   |        |

Note (1): <http://www.saltlakemetals.com/SolubilityProducts.htm> (15/7/14)

## Reaction No. 76684



| No | t  | I-Str | Medium        | Value (Dev)               | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:2665. (4SF)           | 0 | EDH   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:2667. (4SF)           | 1 | ELC   |        |
| 3  | 25 | 0     | Inf. Dilution | dG:-1.5211E+4 (18.5MD) kJ | 0 | MSL   | 8775   |
| 4  | 25 | 0     | Inf. Dilution | dG:-1.5205E+4 (5SF) kJ    | 0 | CRV   | 31308  |
| 5  | 25 | 0     | Inf. Dilution | dH:-1.7550E+4 (12.MD) kJ  | 0 | CRV   | 8775   |
| 6  | 25 | 0     | Inf. Dilution | dH:-1.7535E+4 (5SF) kJ    | 3 | CRV   | 31308  |
| 7  | 25 | 0     | Inf. Dilution | Cp:590. (143.MD) J/K      | 2 | CRV   | 8775   |

| Reaction No. 76840                                   |    |       |         |                 |   |       |        |
|------------------------------------------------------|----|-------|---------|-----------------|---|-------|--------|
| Al+3 + 2<H+1_Dopamine-2> = Al+3_H+1(2)_Dopamine-2(2) |    |       |         |                 |   |       |        |
| No                                                   | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                    | 25 | 0.1   | Unknown | lgK:23.99 (4SF) | 3 | CRV   | 552    |

| Reaction No. 76841                                   |    |       |         |                 |   |       |        |
|------------------------------------------------------|----|-------|---------|-----------------|---|-------|--------|
| Al+3 + 3<H+1_Dopamine-2> = Al+3_H+1(3)_Dopamine-2(3) |    |       |         |                 |   |       |        |
| No                                                   | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                    | 25 | 0.1   | Unknown | lgK:30.63 (4SF) | 3 | CRV   | 552    |

| Reaction No. 76960                                    |    |       |           |                |   |       |        |
|-------------------------------------------------------|----|-------|-----------|----------------|---|-------|--------|
| SemiXylenolOrange-4 + Al+3 = Al+3_SemiXylenolOrange-4 |    |       |           |                |   |       |        |
| No                                                    | t  | I-Str | Medium    | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                     | 25 | 0.1   | Na+1 salt | lgK:16.7 (3SF) | 3 | CRV   | 7271   |

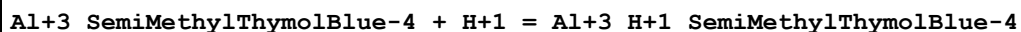
| Reaction No. 76961                                            |    |       |         |               |   |       |        |
|---------------------------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Al+3_SemiXylenolOrange-4 + H+1 = Al+3_H+1_SemiXylenolOrange-4 |    |       |         |               |   |       |        |
| No                                                            | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                             | 25 | 0.1   | Unknown | lgK:3.3 (2SF) | 3 | CRV   | 7271   |

| Reaction No. 76962                                                   |    |       |         |               |   |       |        |
|----------------------------------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Al+3_OH-1_SemiXylenolOrange-4 + H+1 = Al+3_SemiXylenolOrange-4 + H2O |    |       |         |               |   |       |        |
| No                                                                   | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                    | 25 | 0.1   | Unknown | lgK:7.2 (2SF) | 3 | CRV   | 7271   |

| Reaction No. 76988                                          |   |       |        |             |   |       |        |
|-------------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| SemiMethylThymolBlue-4 + Al+3 = Al+3_SemiMethylThymolBlue-4 |   |       |        |             |   |       |        |
| No                                                          | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

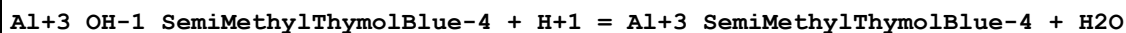
|   |    |     |           |               |   |     |      |
|---|----|-----|-----------|---------------|---|-----|------|
| 1 | 25 | 0.1 | Na+1 salt | lgK:17.9(3SF) | 3 | CRV | 7271 |
|---|----|-----|-----------|---------------|---|-----|------|

## Reaction No. 76989



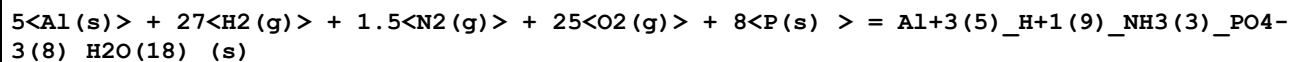
| No | t  | I-Str | Medium  | Value (Dev)  | W | Tech. | Ref. # |
|----|----|-------|---------|--------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:3.5(2SF) | 3 | CRV   | 7271   |

## Reaction No. 76990



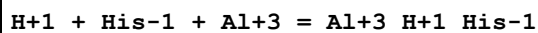
| No | t  | I-Str | Medium  | Value (Dev)  | W | Tech. | Ref. # |
|----|----|-------|---------|--------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:7.6(2SF) | 3 | CRV   | 7271   |

## Reaction No. 76997



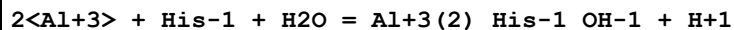
| No | t  | I-Str | Medium        | Value (Dev)           | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:2830.(4SF)        | 1 | ERG   |        |
| 2  | 25 | 0     | Inf. Dilution | dG:-1.61650E+4(6SF)kJ | 1 | CRV   | 9015   |
| 3  | 25 | 0     | Inf. Dilution | dG:-3863.9(5SF)       | 4 | CRV   | 12011  |
| 4  | 25 | 0     | Inf. Dilution | dH:-1.85480E+4(6SF)kJ | 1 | CRV   | 9015   |
| 5  | 25 | 0     | Inf. Dilution | dH:-4433.0(5SF)       | 4 | CRV   | 12011  |

## Reaction No. 77284



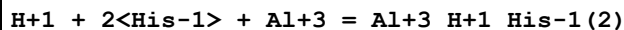
| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:12.15(4SF) | 4 | MGL   | 22560  |

## Reaction No. 77285



| No | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:6.15(3SF) | 4 | MGL   | 22560  |

## Reaction No. 77286



| No | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:20.1(3SF) | 4 | MGL   | 22560  |

## Reaction No. 77287

| Al+3_His-1 + His-1 = Al+3_His-1(2)                       |    |       |               |               |   |       |        |
|----------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| No                                                       | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                        | 25 | 0     | Inf. Dilution | lgK:6.9(2SF)  | 1 | ELC   |        |
| 2                                                        | 25 | 0.1   | NaNO3         | lgK:5.33(3SF) | 0 | MGL   | 22560  |
| Note (2): Khalil & Fazary, Monat. Chem., 2004, 135, 1455 |    |       |               |               |   |       |        |

| Reaction No. 77288                                       |    |       |               |                |   |       |        |
|----------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| 2<His-1> + Al+3 = Al+3_His-1(2)                          |    |       |               |                |   |       |        |
| No                                                       | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                        | 25 | 0     | Inf. Dilution | lgK:16.(2SF)   | 1 | ELC   |        |
| 2                                                        | 25 | 0.1   | NaNO3         | lgK:15.70(4SF) | 0 | MGL   | 22560  |
| Note (2): Khalil & Fazary, Monat. Chem., 2004, 135, 1455 |    |       |               |                |   |       |        |

| Reaction No. 77289                                                                   |    |       |               |               |   |       |        |
|--------------------------------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Al+3_His-1_OH-1 + H+1 = Al+3_His-1 + H2O                                             |    |       |               |               |   |       |        |
| No                                                                                   | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                    | 25 | 0     | Inf. Dilution | lgK:6.4(2SF)  | 1 | ERG   |        |
| 2                                                                                    | 25 | 0.5   | Unknown       | lgK:4.36(3SF) | 0 | CRV   | 7271   |
| Note (2): Species given as M(OH)HL. Charlet et al, Bull. Soc. Chim. Fr., 1984, I-222 |    |       |               |               |   |       |        |

| Reaction No. 77440                                |    |       |               |               |   |       |        |
|---------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| 2<Al+3> + Gly-1 + 2<OH-1> = Al+3(2)_Gly-1_OH-1(2) |    |       |               |               |   |       |        |
| No                                                | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                 | 25 | 0     | Inf. Dilution | lgK:29.4(3SF) | 1 | ELC   |        |

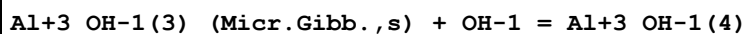
| Reaction No. 77448                                |    |       |               |               |   |       |        |
|---------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Al+3 + 2<Citric-3> + OH-1 = Al+3_OH-1_Citric-3(2) |    |       |               |               |   |       |        |
| No                                                | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                 | 25 | 0     | Inf. Dilution | lgK:22.3(3SF) | 1 | ELC   |        |

| Reaction No. 77449                              |    |       |               |               |   |       |        |
|-------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Al+3 + 3<OH-1> = Al+3_OH-1(3)_(Nordstrandite,s) |    |       |               |               |   |       |        |
| No                                              | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                               | 25 | 0     | Inf. Dilution | lgK:33.9(3SF) | 1 | ELC   |        |

| Reaction No. 77450                      |  |  |  |  |  |  |  |
|-----------------------------------------|--|--|--|--|--|--|--|
| Al+3 + 3<OH-1> + F-1 = Al+3_F-1_OH-1(3) |  |  |  |  |  |  |  |

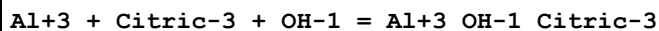
| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:31.0(3SF) | 1 | ELC   |        |

## Reaction No. 77451



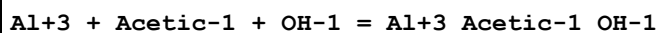
| No | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:0.6(1SF) | 1 | ELC   |        |

## Reaction No. 77452



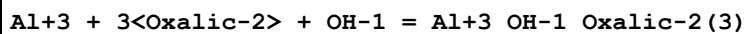
| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:20.6(3SF) | 1 | ELC   |        |

## Reaction No. 77453



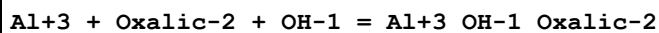
| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:13.8(3SF) | 1 | ELC   |        |

## Reaction No. 77454



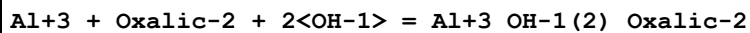
| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:24.0(3SF) | 1 | ELC   |        |

## Reaction No. 77455



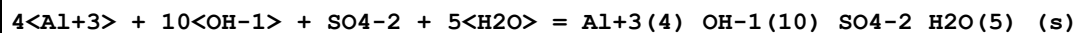
| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:16.5(3SF) | 1 | ELC   |        |

## Reaction No. 77456



| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:24.9(3SF) | 1 | ELC   |        |

## Reaction No. 77457



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:117.7(4SF) | 1 | ELC   |        |

| Reaction No. 77458                                                                                                                                               |    |       |               |               |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $\text{Al}+3 + \text{SO}_4-2 + \text{OH}-1 + 5\langle\text{H}_2\text{O}\rangle = \text{Al}+3\_ \text{OH}-1\_ \text{SO}_4-2\_ \text{H}_2\text{O}(5)\_ (\text{s})$ |    |       |               |               |   |       |        |
| No                                                                                                                                                               | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                                                                                                | 25 | 0     | Inf. Dilution | lgK:22.0(3SF) | 1 | ELC   |        |

| Reaction No. 77459                                                                                                                                                                                                 |    |       |               |                |   |       |        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $4\langle\text{Al}+3\rangle + \text{SO}_4-2 + 10\langle\text{OH}-1\rangle + 5\langle\text{H}_2\text{O}\rangle = \text{Al}+3(4)\_ \text{OH}-1(10)\_ \text{SO}_4-2\_ \text{H}_2\text{O}(5)\_ (\text{am.}, \text{s})$ |    |       |               |                |   |       |        |
| No                                                                                                                                                                                                                 | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                                                                                                                                  | 25 | 0     | Inf. Dilution | lgK:116.0(4SF) | 1 | ELC   |        |

| Reaction No. 77460                                                                          |    |       |               |               |   |       |        |
|---------------------------------------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $2\langle\text{Al}+3\rangle + 4\langle\text{OH}-1\rangle = \text{Al}+3(2)\_ \text{OH}-1(4)$ |    |       |               |               |   |       |        |
| No                                                                                          | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                           | 25 | 0     | Inf. Dilution | lgK:40.4(3SF) | 1 | ELC   |        |

| Reaction No. 77461                                                                                                                                 |    |       |               |               |   |       |        |
|----------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $\text{Al}+3 + \text{Na}+1 + 2\langle\text{OH}-1\rangle + \text{CO}_3-2 = \text{Al}+3\_ \text{Na}+1\_ \text{OH}-1(2)\_ \text{CO}_3-2\_ (\text{s})$ |    |       |               |               |   |       |        |
| No                                                                                                                                                 | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                                                                                  | 25 | 0     | Inf. Dilution | lgK:33.3(3SF) | 1 | ELC   |        |

| Reaction No. 77462                                                                            |    |       |               |                |   |       |        |
|-----------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $7\langle\text{Al}+3\rangle + 17\langle\text{OH}-1\rangle = \text{Al}+3(7)\_ \text{OH}-1(17)$ |    |       |               |                |   |       |        |
| No                                                                                            | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                             | 25 | 0     | Inf. Dilution | lgK:181.3(4SF) | 1 | ELC   |        |

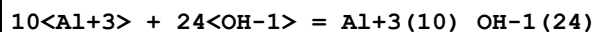
| Reaction No. 77463                                                                            |    |       |               |                |   |       |        |
|-----------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $8\langle\text{Al}+3\rangle + 20\langle\text{OH}-1\rangle = \text{Al}+3(8)\_ \text{OH}-1(20)$ |    |       |               |                |   |       |        |
| No                                                                                            | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                             | 25 | 0     | Inf. Dilution | lgK:213.9(4SF) | 1 | ELC   |        |

| Reaction No. 77464                                                                                                                                                                    |    |       |               |                |   |       |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $9\langle\text{Al}(\text{s})\rangle + 11.5\langle\text{H}_2(\text{g})\rangle + 11.5\langle\text{O}_2(\text{g})\rangle - 4\langle\text{e}-1\rangle = \text{Al}+3(9)\_ \text{OH}-1(23)$ |    |       |               |                |   |       |        |
| No                                                                                                                                                                                    | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                                                                                                     | 25 | 0     | Inf. Dilution | lgK:1653.(4SF) | 1 | ELC   |        |

| Reaction No. 77465                                                                            |   |       |        |             |   |       |        |
|-----------------------------------------------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| $9\langle\text{Al}+3\rangle + 23\langle\text{OH}-1\rangle = \text{Al}+3(9)\_ \text{OH}-1(23)$ |   |       |        |             |   |       |        |
| No                                                                                            | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

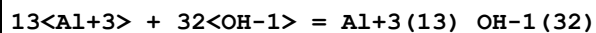
|   |    |   |               |                |   |     |  |
|---|----|---|---------------|----------------|---|-----|--|
| 1 | 25 | 0 | Inf. Dilution | lgK:246.2(4SF) | 1 | ELC |  |
|---|----|---|---------------|----------------|---|-----|--|

## Reaction No. 77466



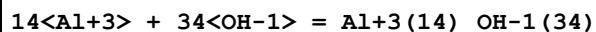
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:257.9(4SF) | 1 | ELC   |        |

## Reaction No. 77467



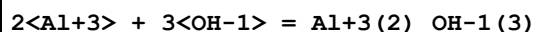
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:348.6(4SF) | 1 | ELC   |        |

## Reaction No. 77468



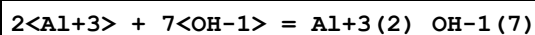
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:365.8(4SF) | 1 | ELC   |        |

## Reaction No. 77469



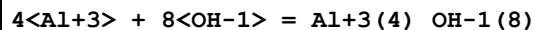
| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:32.5(3SF) | 1 | ELC   |        |

## Reaction No. 77470



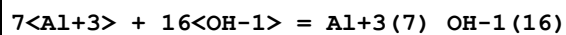
| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:65.4(3SF) | 1 | ELC   |        |

## Reaction No. 77471



| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:88.1(3SF) | 1 | ELC   |        |

## Reaction No. 77472



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:171.8(4SF) | 1 | ELC   |        |

## Reaction No. 77473

| 2<Al+3> + OH-1 = Al+3(2)_OH-1 |    |       |               |               |   |       |        |
|-------------------------------|----|-------|---------------|---------------|---|-------|--------|
| No                            | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                             | 25 | 0     | Inf. Dilution | lgK:10.2(3SF) | 1 | ELC   |        |

| Reaction No. 77474                                |    |       |               |               |   |       |        |
|---------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| 2<Al+3> + PO4-3 + 3<OH-1> = Al+3(2)_OH-1(3)_PO4-3 |    |       |               |               |   |       |        |
| No                                                | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                 | 25 | 0     | Inf. Dilution | lgK:49.5(3SF) | 1 | ELC   |        |

| Reaction No. 77475                    |    |       |               |               |   |       |        |
|---------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Al+3 + PO4-3 + OH-1 = Al+3_OH-1_PO4-3 |    |       |               |               |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                     | 25 | 0     | Inf. Dilution | lgK:22.3(3SF) | 1 | ELC   |        |

| Reaction No. 77595                            |    |       |               |                |   |       |        |
|-----------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| PO4-3 + 2<H2O> + Al+3 = Al+3_PO4-3_H2O(2)_(s) |    |       |               |                |   |       |        |
| No                                            | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                             | 25 | 0     | Inf. Dilution | lgK:22.02(4SF) | 1 | ELC   |        |

| Reaction No. 77620                        |    |       |               |                |   |       |        |
|-------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| PO4-3 + H+1 + 2<Al+3> = Al+3(2)_H+1_PO4-3 |    |       |               |                |   |       |        |
| No                                        | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                         | 25 | 0     | Inf. Dilution | lgK:19.67(4SF) | 1 | ELC   |        |

| Reaction No. 77924                                                                         |    |       |               |                |   |       |        |
|--------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| EDTA-4 + S-2 + Al+3 = Al+3_S-2_EDTA-4                                                      |    |       |               |                |   |       |        |
| Note: The species S-2 does not exist in aqueous solution See reaction: H+1 + S-2 = H+1_S-2 |    |       |               |                |   |       |        |
| No                                                                                         | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                          | 25 | 0     | Inf. Dilution | lgK:28.81(4SF) | 0 | ELC   |        |

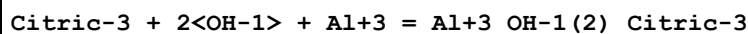
| Reaction No. 78053                    |    |       |               |                |   |       |        |
|---------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| EDTA-4 + F-1 + Al+3 = Al+3_F-1_EDTA-4 |    |       |               |                |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 25 | 0     | Inf. Dilution | lgK:23.61(4SF) | 1 | ELC   |        |

| Reaction No. 78213                                  |  |  |  |  |  |  |  |
|-----------------------------------------------------|--|--|--|--|--|--|--|
| 2<Citric-3> - H+1 + Al+3 = Al+3_H+1(-1)_Citric-3(2) |  |  |  |  |  |  |  |



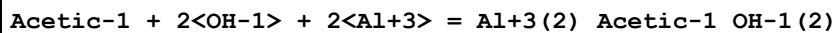
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:8.204 (4SF) | 1 | ELC   |        |

## Reaction No. 78308



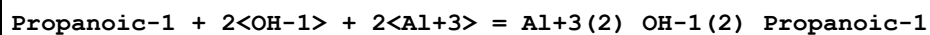
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:27.31 (4SF) | 1 | ELC   |        |

## Reaction No. 78389



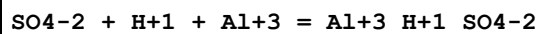
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:25.68 (4SF) | 1 | ELC   |        |

## Reaction No. 78409



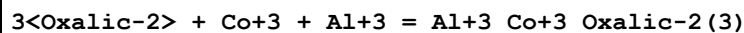
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:25.73 (4SF) | 1 | ELC   |        |

## Reaction No. 78427



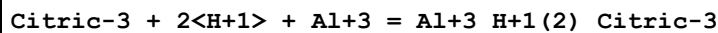
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:2.403 (4SF) | 1 | ELC   |        |

## Reaction No. 78468



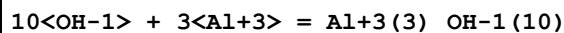
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-7.986 (4SF) | 1 | ELC   |        |

## Reaction No. 78551



| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:14.02 (4SF) | 1 | ELC   |        |

## Reaction No. 78582



| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:95.54 (4SF) | 1 | ELC   |        |

| Reaction No. 78617                                                |    |       |               |                |   |       |        |
|-------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Oxalic-2 + 2<OH-1> + Na+1 + Al+3 = Al+3_Na+1_OH-1(2)_Oxalic-2_(s) |    |       |               |                |   |       |        |
| No                                                                | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                 | 25 | 0     | Inf. Dilution | lgK:58.35(4SF) | 1 | ELC   |        |

| Reaction No. 78618                                          |    |       |               |                |   |       |        |
|-------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Oxalic-2 + 7<OH-1> + 3<Al+3> = Al+3(3)_OH-1(7)_Oxalic-2_(s) |    |       |               |                |   |       |        |
| No                                                          | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                           | 25 | 0     | Inf. Dilution | lgK:113.3(4SF) | 1 | ELC   |        |

| Reaction No. 78643                          |    |       |               |                |   |       |        |
|---------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| OH-1 + 2<F-1> + Al+3 = Al+3_F-1(2)_OH-1_(s) |    |       |               |                |   |       |        |
| No                                          | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                           | 25 | 0     | Inf. Dilution | lgK:22.72(4SF) | 1 | ELC   |        |

| Reaction No. 78650                                |    |       |               |               |   |       |        |
|---------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| PO4-3 + 2<OH-1> + 2<Al+3> = Al+3(2)_OH-1(2)_PO4-3 |    |       |               |               |   |       |        |
| No                                                | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                 | 25 | 0     | Inf. Dilution | lgK:38.1(3SF) | 1 | ELC   |        |

| Reaction No. 78891                                |    |       |               |                |   |       |        |
|---------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| 2<OH-1> + CO3-2 + 2<Al+3> = Al+3(2)_OH-1(2)_CO3-2 |    |       |               |                |   |       |        |
| No                                                | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                 | 25 | 0     | Inf. Dilution | lgK:32.32(4SF) | 1 | ELC   |        |

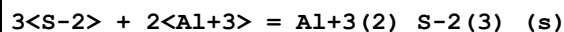
| Reaction No. 78892                                          |    |       |               |                |   |       |        |
|-------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| 4<OH-1> + H+1 + CO3-2 + 3<Al+3> = Al+3(3)_H+1_OH-1(4)_CO3-2 |    |       |               |                |   |       |        |
| No                                                          | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                           | 25 | 0     | Inf. Dilution | lgK:57.69(4SF) | 1 | ELC   |        |

| Reaction No. 78930                |    |       |               |                 |   |       |        |
|-----------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| 3<Br-1> + Al+3 = Al+3_Br-1(3)_(s) |    |       |               |                 |   |       |        |
| No                                | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 25 | 0     | Inf. Dilution | lgK:-55.24(4SF) | 1 | ELC   |        |

| Reaction No. 78931                                |   |       |        |             |   |       |        |
|---------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| 6<H2O> + 3<Cl-1> + Al+3 = Al+3_Cl-1(3)_H2O(6)_(s) |   |       |        |             |   |       |        |
| No                                                | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

|   |    |   |               |                 |   |     |  |
|---|----|---|---------------|-----------------|---|-----|--|
| 1 | 25 | 0 | Inf. Dilution | lgK:-8.197(4SF) | 1 | ELC |  |
|---|----|---|---------------|-----------------|---|-----|--|

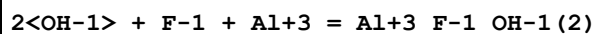
## Reaction No. 78932



Note: The species S-2 does not exist in aqueous solution See reaction: H+1 + S-2 = H+1\_S-2

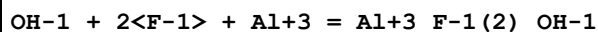
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-1.707(4SF) | 0 | ELC   |        |

## Reaction No. 78973



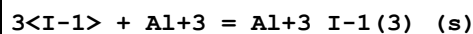
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-190.8(4SF) | 1 | ELC   |        |

## Reaction No. 78974



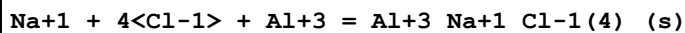
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:20.11(4SF) | 1 | ELC   |        |

## Reaction No. 78976



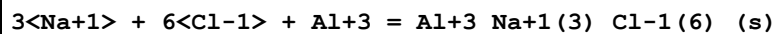
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-60.85(4SF) | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:-60.35(4SF) | 4 | CRV   | 32224  |

## Reaction No. 79001



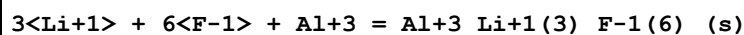
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-41.48(4SF) | 1 | ELC   |        |

## Reaction No. 79004



| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-41.29(4SF) | 1 | ELC   |        |

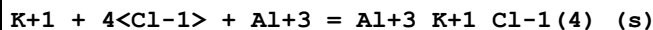
## Reaction No. 79030



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

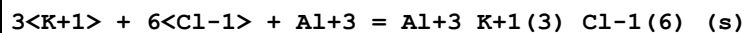
|   |    |   |               |               |   |     |  |
|---|----|---|---------------|---------------|---|-----|--|
| 1 | 25 | 0 | Inf. Dilution | lgK:30.0(3SF) | 1 | ELC |  |
|---|----|---|---------------|---------------|---|-----|--|

## Reaction No. 79042



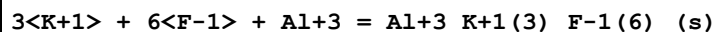
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-35.78(4SF) | 1 | ELC   |        |

## Reaction No. 79043



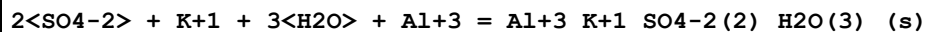
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-32.8(3SF) | 1 | ELC   |        |

## Reaction No. 79044



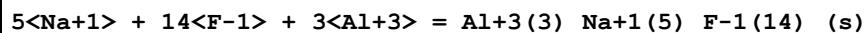
| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:24.2(3SF) | 1 | ELC   |        |

## Reaction No. 79045



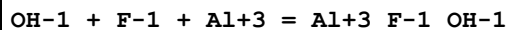
| No | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:0.3(1SF) | 1 | ELC   |        |

## Reaction No. 79051



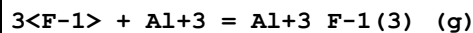
| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:81.8(3SF) | 1 | ELC   |        |

## Reaction No. 79072



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:15.67(4SF) | 1 | ELC   |        |

## Reaction No. 79127



| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-25.91(4SF) | 1 | ELC   |        |

## Reaction No. 79128

| 3<Cl-1> + Al+3 = Al+3_Cl-1(3)_(g) |    |       |               |                 |   |       |        |
|-----------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| No                                | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 25 | 0     | Inf. Dilution | lgK:-67.51(4SF) | 1 | ELC   |        |

| Reaction No. 79129                                  |    |       |               |                 |   |       |        |
|-----------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| 3<NO3-1> + 6<H2O> + Al+3 = Al+3_NO3-1(3)_H2O(6)_(s) |    |       |               |                 |   |       |        |
| No                                                  | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                   | 25 | 0     | Inf. Dilution | lgK:-7.581(4SF) | 1 | ELC   |        |

| Reaction No. 79130                                      |    |       |               |              |   |       |        |
|---------------------------------------------------------|----|-------|---------------|--------------|---|-------|--------|
| 2<SO4-2> + NH3 + H+1 + Al+3 = Al+3_H+1_NH3_SO4-2(2)_(s) |    |       |               |              |   |       |        |
| No                                                      | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                       | 25 | 0     | Inf. Dilution | lgK:5.8(2SF) | 1 | ELC   |        |

| Reaction No. 79253 |    |       |               |                 |   |       |        |
|--------------------|----|-------|---------------|-----------------|---|-------|--------|
| Al+3 = Al+3_(g)    |    |       |               |                 |   |       |        |
| No                 | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                  | 25 | 0     | Inf. Dilution | lgK:-962.1(4SF) | 1 | ELC   |        |

| Reaction No. 79305                          |    |       |               |               |   |       |        |
|---------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| PO4-3 + OH-1 + 2<Al+3> = Al+3(2)_OH-1_PO4-3 |    |       |               |               |   |       |        |
| No                                          | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 25 | 0     | Inf. Dilution | lgK:17.1(3SF) | 1 | ELC   |        |

| Reaction No. 79314                                            |    |       |               |                |   |       |        |
|---------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| 3<Oxalic-2> + 3<OH-1> + 3<Al+3> = Al+3(3)_OH-1(3)_Oxalic-2(3) |    |       |               |                |   |       |        |
| No                                                            | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                             | 25 | 0     | Inf. Dilution | lgK:57.27(4SF) | 1 | ELC   |        |

| Reaction No. 79315                                            |    |       |               |                |   |       |        |
|---------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| 4<Oxalic-2> + 2<OH-1> + 2<Al+3> = Al+3(2)_OH-1(2)_Oxalic-2(4) |    |       |               |                |   |       |        |
| No                                                            | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                             | 25 | 0     | Inf. Dilution | lgK:44.63(4SF) | 1 | ELC   |        |

| Reaction No. 79382                                |    |       |               |                |   |       |        |
|---------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| 3<OH-1> + CO3-2 + 3<Al+3> = Al+3(3)_OH-1(3)_CO3-2 |    |       |               |                |   |       |        |
| No                                                | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                 | 25 | 0     | Inf. Dilution | lgK:43.63(4SF) | 1 | ELC   |        |

| Reaction No. 79429                                                                                                    |    |       |               |                |   |       |        |
|-----------------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $\text{SiH}_2\text{O}_4\text{-2} + \text{H}+1 + \text{Al}+3 = \text{Al}+3\text{-H}+1\text{-SiH}_2\text{O}_4\text{-2}$ |    |       |               |                |   |       |        |
| No                                                                                                                    | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                                     | 25 | 0     | Inf. Dilution | lgK:20.65(4SF) | 1 | ELC   |        |

| Reaction No. 79482                                                                            |    |       |               |              |   |       |        |
|-----------------------------------------------------------------------------------------------|----|-------|---------------|--------------|---|-------|--------|
| $\text{H}+1\text{-CO}_3\text{-2} + \text{Al}+3 = \text{Al}+3\text{-H}+1\text{-CO}_3\text{-2}$ |    |       |               |              |   |       |        |
| No                                                                                            | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                                                             | 25 | 0     | Inf. Dilution | lgK:5.3(2SF) | 1 | ELC   |        |
| Note (1): x VEWCMN                                                                            |    |       |               |              |   |       |        |

| Reaction No. 79505                                                                                                                                           |    |       |               |                   |   |       |        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $\text{Al(s)} + 1.5\text{<H}_2\text{(g)>} + 2\text{<O}_2\text{(g)>} + \text{Si(s)} - 2\text{<e-1>} = \text{Al}+3\text{-H}+1\text{-SiH}_2\text{O}_4\text{-2}$ |    |       |               |                   |   |       |        |
| No                                                                                                                                                           | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                                                            | 25 | 0     | Inf. Dilution | dG:-1782.5(5SF)kJ | 3 | EFE   | 28693  |

| Reaction No. 79576                                                                             |    |       |               |                |   |       |        |
|------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $\text{Al}+3 + \text{H}+1(2)\text{-EDTA-4} = \text{Al}+3\text{-EDTA-4} + 2\text{<H}+1\text{>}$ |    |       |               |                |   |       |        |
| No                                                                                             | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                              | 25 | 0     | Inf. Dilution | lgK:1.49(3SF)  | 1 | ELC   |        |
| 2                                                                                              | 25 | 0     | Inf. Dilution | dH:96.6(3SF)kJ | 1 | ELC   |        |

| Reaction No. 79600                                                                                               |    |       |               |                     |   |       |        |
|------------------------------------------------------------------------------------------------------------------|----|-------|---------------|---------------------|---|-------|--------|
| $\text{Al}+3 + \text{Tricarballic-3} + \text{H}_2\text{O} = \text{Al}+3\text{-OH-1-Tricarballic-3} + \text{H}+1$ |    |       |               |                     |   |       |        |
| No                                                                                                               | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                                                                                | 25 | 0     | Inf. Dilution | lgK:1.97(0.07SD)    | 2 | EDH   | 29190  |
| 2                                                                                                                | 25 | 0.15  | NaCl          | lgK:0.02(1SF)       | 3 | MGL   | 29190  |
| 3                                                                                                                | 25 | 0.5   | NaCl          | lgK:-0.418(0.007SD) | 3 | MGL   | 29190  |
| 4                                                                                                                | 25 | 1     | NaCl          | lgK:-0.680(0.007SD) | 3 | MGL   | 29190  |

| Reaction No. 79601                                                                                  |    |       |        |                   |   |       |        |
|-----------------------------------------------------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| $\text{H}+1 + \text{Bu1234TetrCOO-4} + \text{Al}+3 = \text{Al}+3\text{-H}+1\text{-Bu1234TetrCOO-4}$ |    |       |        |                   |   |       |        |
| No                                                                                                  | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                   | 25 | 0.15  | NaCl   | lgK:10.17(0.01SD) | 0 | MGL   | 29190  |
| 2                                                                                                   | 25 | 0.5   | NaCl   | lgK:8.71(0.05SD)  | 2 | MGL   | 29190  |
| 3                                                                                                   | 25 | 1     | NaCl   | lgK:8.03(0.05SD)  | 2 | MGL   | 29190  |

| Reaction No. 79602                            |    |       |        |                    |   |       |        |
|-----------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| Bu1234TetrCOO-4 + Al+3 = Al+3_Bu1234TetrCOO-4 |    |       |        |                    |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                             | 25 | 0.15  | NaCl   | lgK:5.87(3SF)      | 3 | MGL   | 29190  |
| 2                                             | 25 | 0.5   | NaCl   | lgK:5.26(0.01SD)   | 3 | MGL   | 29190  |
| 3                                             | 25 | 1     | NaCl   | lgK:4.959(0.004SD) | 3 | MGL   | 29190  |

| Reaction No. 79609                                      |    |       |               |                   |   |       |        |
|---------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| 2<H+1> + BenzHexCOO-6 + Al+3 = Al+3_H+1(2)_BenzHexCOO-6 |    |       |               |                   |   |       |        |
| No                                                      | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.15  | Inf. Dilution | lgK:14.69(0.01SD) | 3 | ENS   | 29190  |

| Reaction No. 79610                                |    |       |               |                 |   |       |        |
|---------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| H+1 + BenzHexCOO-6 + Al+3 = Al+3_H+1_BenzHexCOO-6 |    |       |               |                 |   |       |        |
| No                                                | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                 | 25 | 0.15  | Inf. Dilution | lgK:10.788(5SF) | 3 | ENS   | 29190  |

| Reaction No. 79611                      |    |       |        |               |   |       |        |
|-----------------------------------------|----|-------|--------|---------------|---|-------|--------|
| BenzHexCOO-6 + Al+3 = Al+3_BenzHexCOO-6 |    |       |        |               |   |       |        |
| No                                      | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                       | 25 | 0.15  | NaCl   | lgK:5.89(3SF) | 3 | ENS   | 29190  |

| Reaction No. 79612                           |    |       |               |                   |   |       |        |
|----------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| Al+3 + EDDS-4 + H2O = Al+3_OH-1_EDDS-4 + H+1 |    |       |               |                   |   |       |        |
| No                                           | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                            | 25 | 0     | Inf. Dilution | lgK:14.6(0.2SD)   | 2 | EDH   | 29190  |
| 2                                            | 25 | 0     | Inf. Dilution | lgK:14.6(0.02SD)  | 2 | ENS   | 29190  |
| 3                                            | 25 | 0.15  | NaCl          | lgK:12.24(0.04SD) | 2 | ENS   | 29190  |
| 4                                            | 25 | 0.5   | NaCl          | lgK:11.15(0.02SD) | 3 | ENS   | 29190  |
| 5                                            | 25 | 1     | NaCl          | lgK:10.87(0.02SD) | 3 | ENS   | 29190  |

| Reaction No. 79614                                                    |    |       |               |                  |   |       |        |
|-----------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Al+3_Ca+2(2)_OH-1(7)_H2O_(s) = 2<Ca+2> + Al+3_OH-1(4) + 3<OH-1> + H2O |    |       |               |                  |   |       |        |
| No                                                                    | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                     | 25 | 0     | Inf. Dilution | lgK:-11.4(0.5SD) | 3 | MSL   | 28806  |
| 2                                                                     | 50 | 0     | Inf. Dilution | lgK:-11.6(0.5SD) | 3 | MSL   | 28806  |
| 3                                                                     | 75 | 0     | Inf. Dilution | lgK:-12.1(3SF)   | 3 | MSL   | 28806  |

| Reaction No. 79662                                                                                                                                                                          |    |       |               |               |   |       |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $\text{Al}^{+3} + \text{Na}^{+1} + 2\langle\text{OH}^{-1}\rangle + \text{H}^{+1}\text{CO}_3^{-2} = \text{Al}^{+3}\text{Na}^{+1}\text{OH}^{-1}(2)\text{CO}_3^{-2}(\text{s}) + \text{H}^{+1}$ |    |       |               |               |   |       |        |
| No                                                                                                                                                                                          | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                                                                                                                           | 25 | 0     | Inf. Dilution | lgK:22.5(3SF) | 1 | ELC   |        |

| Reaction No. 79667                                                                                |    |       |               |               |   |       |        |
|---------------------------------------------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $\text{H}^{+1}\text{CO}_3^{-2} + \text{Al}^{+3} = \text{Al}^{+3}\text{CO}_3^{-2} + \text{H}^{+1}$ |    |       |               |               |   |       |        |
| No                                                                                                | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                                 | 25 | 0     | Inf. Dilution | lgK:-3.1(2SF) | 1 | ELC   |        |

| Reaction No. 79668                                                                                                                                                                                                                                                                 |    |       |               |                     |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|---------------------|---|-------|--------|
| $1.99\langle\text{Al}(\text{s})\rangle + 0.657\langle\text{Ca}(\text{s})\rangle + 2.147\langle\text{H}_2(\text{g})\rangle + 0.676\langle\text{Na}(\text{s})\rangle + 6.3235\langle\text{O}_2(\text{g})\rangle + 3.01\langle\text{Si}(\text{s})\rangle = \text{Mesolite}(\text{s})$ |    |       |               |                     |   |       |        |
| No                                                                                                                                                                                                                                                                                 | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                                                                                                                                                                                                                                                  | 25 | 0     | Inf. Dilution | dG:-5513.2(5.4SD)kJ | 5 | MCL   | 29317  |
| 2                                                                                                                                                                                                                                                                                  | 25 | 0     | Inf. Dilution | dH:-5947.1(5.4SD)kJ | 5 | CRV   | 29317  |

| Reaction No. 79669                                                                                                                                                                                                                                          |    |       |               |                |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $\text{CaAl}_2\text{Si}_4\text{O}_{12}\cdot 6\text{H}_2\text{O}(\text{s}) + 8\langle\text{H}^{+1}\rangle = \text{Ca}^{+2} + 2\langle\text{Al}^{+3}\rangle + 4\langle\text{H}^{+1}(2)\text{SiH}_2\text{O}_4^{-2}\rangle + 2\langle\text{H}_2\text{O}\rangle$ |    |       |               |                |   |       |        |
| No                                                                                                                                                                                                                                                          | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                                                                                                                                                                           | 25 | 0     | Inf. Dilution | lgK:11.51(4SF) | 4 | CRV   | 27383  |

| Reaction No. 79670                                                                                                                                                                                                                                                                   |    |       |               |                |   |       |        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $\text{Ca}_2\text{Al}_4\text{Si}_4\text{O}_{16}\cdot 9\text{H}_2\text{O}(\text{s}) + 16\langle\text{H}^{+1}\rangle = 2\langle\text{Ca}^{+2}\rangle + 4\langle\text{Al}^{+3}\rangle + 4\langle\text{H}^{+1}(2)\text{SiH}_2\text{O}_4^{-2}\rangle + 9\langle\text{H}_2\text{O}\rangle$ |    |       |               |                |   |       |        |
| No                                                                                                                                                                                                                                                                                   | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                                                                                                                                                                                                    | 25 | 0     | Inf. Dilution | lgK:38.96(4SF) | 4 | CRV   | 27383  |

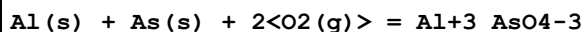
| Reaction No. 79750                                                                                                                                                                        |    |       |               |                    |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| $\text{Al}(\text{s}) + \text{As}(\text{s}) + \text{H}_2(\text{g}) + 2\langle\text{O}_2(\text{g})\rangle - 2\langle\text{e}^{-1}\rangle = \text{Al}^{+3}\text{H}^{+1}(2)\text{AsO}_4^{-3}$ |    |       |               |                    |   |       |        |
| No                                                                                                                                                                                        | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                                                                                                                                         | 25 | 0     | Inf. Dilution | dG:-1255.10(6SF)kJ | 5 | CRV   | 29482  |
| 2                                                                                                                                                                                         | 25 | 0     | Inf. Dilution | dH:-1467.40(6SF)kJ | 5 | CRV   | 29482  |

| Reaction No. 79762                                                                                                                                                                       |    |       |               |                    |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| $\text{Al}(\text{s}) + \text{As}(\text{s}) + 0.5\langle\text{H}_2(\text{g})\rangle + 2\langle\text{O}_2(\text{g})\rangle - \text{e}^{-1} = \text{Al}^{+3}\text{H}^{+1}\text{AsO}_4^{-3}$ |    |       |               |                    |   |       |        |
| No                                                                                                                                                                                       | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                                                                                                                                        | 25 | 0     | Inf. Dilution | dG:-1238.00(6SF)kJ | 5 | CRV   | 29482  |



|   |    |   |               |                      |   |     |       |
|---|----|---|---------------|----------------------|---|-----|-------|
| 2 | 25 | 0 | Inf. Dilution | dH:-1436.70 (6SF) kJ | 5 | CRV | 29482 |
|---|----|---|---------------|----------------------|---|-----|-------|

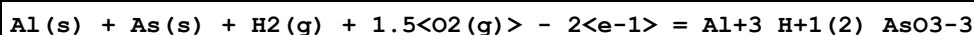
## Reaction No. 79775



Note: Italian supremo disagrees [DR]

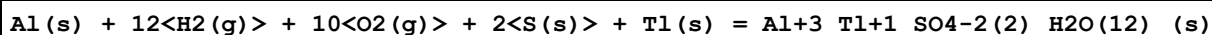
| No | t  | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:213.6 (4SF)      | 1 | EES   |        |
| 2  | 25 | 0     | Inf. Dilution | dG:-1194.80 (6SF) kJ | 0 | CRV   | 29482  |
| 3  | 25 | 0     | Inf. Dilution | dH:-1380.80 (6SF) kJ | 2 | CRV   | 29482  |

## Reaction No. 79784



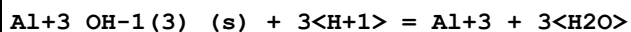
| No | t  | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dG:-1115.50 (6SF) kJ | 5 | CRV   | 29482  |
| 2  | 25 | 0     | Inf. Dilution | dH:-1295.40 (6SF) kJ | 5 | CRV   | 29482  |

## Reaction No. 79799



| No | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dG:-4948.2 (5SF) kJ | 4 | EOD   | 29486  |

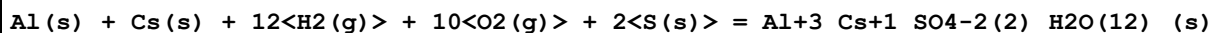
## Reaction No. 79805



Note: Probably refers to Gibbsite but doesn't say so

| No | t  | I-Str | Medium        | Value (Dev)          | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:7.756 (4SF)      | 3 | EOD   | 29487  |
| 2  | 25 | 0     | Inf. Dilution | lgK:8.28 (0.2SD)     | 4 | EDH   | 30072  |
| 3  | 90 | 0     | Inf. Dilution | lgK:4.53 (3SF)       | 3 | EOD   | 29487  |
| 4  | 25 | 0     | Inf. Dilution | dH:-102.780 (6SF) kJ | 3 | EOD   | 29487  |

## Reaction No. 79814



| No | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dG:-5168.1 (5SF) kJ | 2 | EOD   | 29489  |
| 2  | 25 | 0     | Inf. Dilution | dH:-6094.8 (5SF) kJ | 2 | EOD   | 29489  |

## Reaction No. 79858

| Al+3 + Na+1 + 4<H2O> = Al+3_Na+1_OH-1(4) + 4<H+1> |     |       |                        |                  |   |       |        |
|---------------------------------------------------|-----|-------|------------------------|------------------|---|-------|--------|
| No                                                | t   | I-Str | Medium                 | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                 | 200 | 0     | P=300<br>Inf. Dilution | lgK:-10.748(5SF) | 4 | EOD   | 29559  |
| 2                                                 | 25  | 0     | Inf. Dilution          | lgK:-22.90(4SF)  | 4 | EOD   | 29559  |
| 3                                                 | 90  | 0     | Inf. Dilution          | lgK:-16.941(5SF) | 3 | EOD   | 29559  |
| 4                                                 | 150 | 0     | Inf. Dilution          | lgK:-13.097(5SF) | 3 | EOD   | 29559  |
| 5                                                 | 210 | 0     | Inf. Dilution          | lgK:-10.153(5SF) | 2 | EOD   | 29559  |

| Reaction No. 80047                                  |    |       |               |                   |   |       |        |
|-----------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| H+1 + Transexamic-2 + Al+3 = Al+3_H+1_Transexamic-2 |    |       |               |                   |   |       |        |
| No                                                  | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                   | 25 | 0     | Inf. Dilution | lgK:13.57(4SF)    | 2 | EDH   | 29906  |
| 2                                                   | 25 | 0.15  | NaCl          | lgK:13.31(0.01SD) | 4 | MGL   | 29906  |
| 3                                                   | 25 | 0.15  | NaCl          | lgK:13.31(0.06SD) | 5 | MMR   | 29906  |
| 4                                                   | 25 | 0.5   | NaCl          | lgK:13.33(0.02SD) | 4 | MGL   | 29906  |
| 5                                                   | 25 | 1     | NaCl          | lgK:13.26(0.02SD) | 4 | MGL   | 29906  |
| 6                                                   | 25 | 0.15  | NaCl          | dG:-76.0(3SF)kJ   | 5 | MCL   | 29906  |
| 7                                                   | 25 | 0.15  | NaCl          | dH:-25.(3.SD)kJ   | 5 | MCL   | 29906  |

| Reaction No. 80048                                         |    |       |               |                  |   |       |        |
|------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Al+3 + Transexamic-2 + H2O = Al+3_OH-1_Transexamic-2 + H+1 |    |       |               |                  |   |       |        |
| No                                                         | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                          | 25 | 0     | Inf. Dilution | lgK:5.6(0.1SD)   | 2 | EDH   | 29906  |
| 2                                                          | 25 | 0.15  | NaCl          | lgK:4.61(0.01SD) | 4 | MGL   | 29906  |
| 3                                                          | 25 | 0.15  | NaCl          | lgK:5.39(0.05SD) | 0 | MMR   | 29906  |
| 4                                                          | 25 | 0.5   | NaCl          | lgK:4.66(0.02SD) | 4 | MGL   | 29906  |
| 5                                                          | 25 | 1     | NaCl          | lgK:4.52(0.01SD) | 4 | MGL   | 29906  |

| Reaction No. 80049                  |    |       |        |                     |   |       |        |
|-------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| H+1 + Gly-1 + Al+3 = Al+3_H+1_Gly-1 |    |       |        |                     |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                   | 25 | 0.15  | NaCl   | lgK:11.698(0.007SD) | 4 | MGL   | 29906  |
| 2                                   | 25 | 0.5   | NaCl   | lgK:11.55(0.06SD)   | 4 | MGL   | 29906  |
| 3                                   | 25 | 1     | NaCl   | lgK:11.694(0.004SD) | 4 | MGL   | 29906  |

| Reaction No. 80050 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| Al+3 + Cys-2 + H2O = Al+3_OH-1_Cys-2 + H+1 |    |       |        |                  |   |       |        |
|--------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| No                                         | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                          | 25 | 0.15  | NaCl   | lgK:6.96(0.02SD) | 4 | MGL   | 29906  |
| 2                                          | 25 | 0.15  | NaCl   | lgK:7.0(0.5SD)   | 3 | MMR   | 29906  |
| 3                                          | 25 | 0.15  | NaCl   | lgK:6.93(3SF)    | 4 | MSP   | 29906  |
| 4                                          | 25 | 0.5   | NaCl   | lgK:6.39(0.04SD) | 4 | MGL   | 29906  |
| 5                                          | 25 | 1     | NaCl   | lgK:6.70(0.05SD) | 4 | MGL   | 29906  |

| Reaction No. 80051              |    |       |        |                   |   |       |        |
|---------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Cys-2 + 2<Al+3> = Al+3(2)_Cys-2 |    |       |        |                   |   |       |        |
| No                              | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                               | 25 | 0.15  | NaCl   | lgK:15.17(0.01SD) | 3 | MGL   | 29906  |
| 2                               | 25 | 0.15  | NaCl   | lgK:15.0(0.4SD)   | 2 | MMR   | 29906  |
| 3                               | 25 | 0.15  | NaCl   | lgK:15.17(4SF)    | 3 | MSP   | 29906  |
| 4                               | 25 | 0.5   | NaCl   | lgK:14.89(0.01SD) | 3 | MGL   | 29906  |
| 5                               | 25 | 1     | NaCl   | lgK:14.24(0.07SD) | 2 | MGL   | 29906  |

| Reaction No. 80052                                              |    |       |        |                  |   |       |        |
|-----------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 2<Al+3> + 2<Gly-1> + 2<H2O> = Al+3(2)_Gly-1(2)_OH-1(2) + 2<H+1> |    |       |        |                  |   |       |        |
| No                                                              | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                               | 25 | 0.15  | NaCl   | lgK:8.62(0.02SD) | 2 | MGL   | 29906  |
| 2                                                               | 25 | 0.5   | NaCl   | lgK:7.84(3SF)    | 3 | MGL   | 29906  |
| 3                                                               | 25 | 1     | NaCl   | lgK:8.29(3SF)    | 3 | MGL   | 29906  |

| Reaction No. 80089                                 |    |       |        |                |   |       |        |
|----------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Al+3 + 2<5ADP-3> + H2O = Al+3_OH-1_5ADP-3(2) + H+1 |    |       |        |                |   |       |        |
| No                                                 | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                  | 25 | 0.15  | NaCl   | lgK:5.2(0.1SD) | 2 | MGL   | 30067  |

| Reaction No. 80090                                                    |    |       |               |                   |   |       |        |
|-----------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| 3<Al+3> + 3<Citric-3> + 4<H2O> = Al+3(3)_OH-1(4)_Citric-3(3) + 4<H+1> |    |       |               |                   |   |       |        |
| No                                                                    | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                     | 10 | 0.249 | NaCl          | lgK:7.75(0.20SD)  | 2 | MGL   | 30072  |
| 2                                                                     | 25 | 0     | Inf. Dilution | lgK:18.21(0.12SD) | 2 | EDH   | 30072  |
| 3                                                                     | 25 | 0.249 | NaCl          | lgK:11.68(0.12SD) | 2 | MGL   | 30072  |
| 4                                                                     | 25 | 0.956 | NaCl          | lgK:10.24(0.03SD) | 2 | MGL   | 30072  |
| 5                                                                     | 45 | 0.249 | NaCl          | lgK:16.08(4SF)    | 0 | MGL   | 30072  |

|   |    |      |         |                    |   |     |       |
|---|----|------|---------|--------------------|---|-----|-------|
| 6 | 25 | 0.25 | Unknown | dG:-66.7(0.68SD)kJ | 3 | EFE | 30072 |
| 7 | 25 | 0.25 | Unknown | dH:412.(12.SD)kJ   | 3 | EFE | 30072 |

| Reaction No. 80091                                   |    |       |               |                     |   |       |        |
|------------------------------------------------------|----|-------|---------------|---------------------|---|-------|--------|
| Al+3 + Gluconic-1 + H2O = Al+3_Gluconic-1_OH-1 + H+1 |    |       |               |                     |   |       |        |
| No                                                   | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                    | 25 | 0     | Inf. Dilution | lgK:0.15(0.10SD)    | 3 | EDH   | 30072  |
| 2                                                    | 25 | 0.107 | NaCl          | lgK:-0.718(0.008SD) | 3 | MGL   | 30072  |
| 3                                                    | 25 | 0.254 | NaCl          | lgK:-0.98(0.01SD)   | 3 | MGL   | 30072  |
| 4                                                    | 25 | 0.486 | NaCl          | lgK:-1.00(0.01SD)   | 3 | MGL   | 30072  |
| 5                                                    | 25 | 0.73  | NaCl          | lgK:-1.15(0.01SD)   | 3 | MGL   | 30072  |
| 6                                                    | 25 | 0.963 | NaCl          | lgK:-1.26(0.01SD)   | 3 | MGL   | 30072  |

| Reaction No. 80092                                            |    |       |               |                   |   |       |        |
|---------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| Al+3 + Gluconic-1 + 2<H2O> = Al+3_Gluconic-1_OH-1(2) + 2<H+1> |    |       |               |                   |   |       |        |
| No                                                            | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                             | 25 | 0     | Inf. Dilution | lgK:-4.91(0.16SD) | 3 | EDH   | 30072  |
| 2                                                             | 25 | 0.107 | NaCl          | lgK:-5.85(0.03SD) | 3 | MGL   | 30072  |
| 3                                                             | 25 | 0.254 | NaCl          | lgK:-5.92(0.05SD) | 3 | MGL   | 30072  |
| 4                                                             | 25 | 0.486 | NaCl          | lgK:-6.18(0.05SD) | 3 | MGL   | 30072  |
| 5                                                             | 25 | 0.73  | NaCl          | lgK:-6.11(0.04SD) | 3 | MGL   | 30072  |
| 6                                                             | 25 | 0.963 | NaCl          | lgK:-6.33(0.03SD) | 3 | MGL   | 30072  |

| Reaction No. 80093                                            |    |       |               |                    |   |       |        |
|---------------------------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| Al+3 + Gluconic-1 + 3<H2O> = Al+3_Gluconic-1_OH-1(3) + 3<H+1> |    |       |               |                    |   |       |        |
| No                                                            | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                             | 25 | 0     | Inf. Dilution | lgK:-11.23(0.15SD) | 3 | EDH   | 30072  |
| 2                                                             | 25 | 0.107 | NaCl          | lgK:-11.90(0.03SD) | 3 | MGL   | 30072  |
| 3                                                             | 25 | 0.254 | NaCl          | lgK:-12.02(0.03SD) | 3 | MGL   | 30072  |
| 4                                                             | 25 | 0.486 | NaCl          | lgK:-11.91(0.02SD) | 2 | MGL   | 30072  |
| 5                                                             | 25 | 0.73  | NaCl          | lgK:-11.98(4SF)    | 3 | MGL   | 30072  |
| 6                                                             | 25 | 0.963 | NaCl          | lgK:-12.21(0.03SD) | 3 | MGL   | 30072  |

| Reaction No. 80094                               |    |       |        |                     |   |       |        |
|--------------------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| Al+3 + Lactic-1 + H2O = Al+3_Lactic-1_OH-1 + H+1 |    |       |        |                     |   |       |        |
| No                                               | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                | 15 | 0.105 | NaCl   | lgK:-1.365(0.007SD) | 3 | MGL   | 30072  |

|   |    |       |               |                     |   |     |       |
|---|----|-------|---------------|---------------------|---|-----|-------|
| 2 | 15 | 0.961 | NaCl          | lgK:-1.87(0.01SD)   | 3 | MGL | 30072 |
| 3 | 25 | 0.105 | NaCl          | lgK:-1.095(0.004SD) | 3 | MGL | 30072 |
| 4 | 25 | 0.488 | NaCl          | lgK:-1.40(0.01SD)   | 3 | MGL | 30072 |
| 5 | 25 | 0.962 | NaCl          | lgK:-1.52(0.01SD)   | 3 | MGL | 30072 |
| 6 | 45 | 0.107 | NaCl          | lgK:-0.77(0.01SD)   | 3 | MGL | 30072 |
| 7 | 45 | 0.941 | NaCl          | lgK:-1.12(0.02SD)   | 3 | MGL | 30072 |
| 8 | 25 | 0     | Inf. Dilution | dG:1.43(0.34SD)kJ   | 3 | EFE | 30072 |
| 9 | 25 | 0     | Inf. Dilution | dH:31.(3.SD)kJ      | 3 | EFE | 30072 |

| Reaction No. 80095                                     |    |       |               |                    |   |       |        |
|--------------------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| Al+3 + 2<Lactic-1> + H2O = Al+3_Lactic-1(2)_OH-1 + H+1 |    |       |               |                    |   |       |        |
| No                                                     | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                      | 15 | 0.105 | NaCl          | lgK:-0.19(0.05SD)  | 3 | MGL   | 30072  |
| 2                                                      | 15 | 0.961 | NaCl          | lgK:-0.61(0.10SD)  | 3 | MGL   | 30072  |
| 3                                                      | 25 | 0.105 | NaCl          | lgK:0.55(0.04SD)   | 3 | MGL   | 30072  |
| 4                                                      | 25 | 0.488 | NaCl          | lgK:0.31(0.05SD)   | 3 | MGL   | 30072  |
| 5                                                      | 25 | 0.962 | NaCl          | lgK:0.05(0.09SD)   | 3 | MGL   | 30072  |
| 6                                                      | 45 | 0.107 | NaCl          | lgK:1.59(0.04SD)   | 3 | MGL   | 30072  |
| 7                                                      | 45 | 0.941 | NaCl          | lgK:1.26(0.03SD)   | 3 | MGL   | 30072  |
| 8                                                      | 25 | 0     | Inf. Dilution | dG:-9.07(0.45SD)kJ | 3 | EFE   | 30072  |
| 9                                                      | 25 | 0     | Inf. Dilution | dH:95.(4.SD)kJ     | 3 | EFE   | 30072  |

| Reaction No. 80096                                              |    |       |               |                    |   |       |        |
|-----------------------------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| Al+3 + 2<Lactic-1> + 2<H2O> = Al+3_Lactic-1(2)_OH-1(2) + 2<H+1> |    |       |               |                    |   |       |        |
| No                                                              | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                               | 15 | 0.105 | NaCl          | lgK:-4.14(0.03SD)  | 3 | MGL   | 30072  |
| 2                                                               | 15 | 0.961 | NaCl          | lgK:-4.65(0.02SD)  | 3 | MGL   | 30072  |
| 3                                                               | 25 | 0.105 | NaCl          | lgK:-3.75(0.03SD)  | 3 | MGL   | 30072  |
| 4                                                               | 25 | 0.488 | NaCl          | lgK:-3.92(0.04SD)  | 3 | MGL   | 30072  |
| 5                                                               | 25 | 0.962 | NaCl          | lgK:-4.11(0.04SD)  | 3 | MGL   | 30072  |
| 6                                                               | 45 | 0.107 | NaCl          | lgK:-2.75(0.03SD)  | 3 | MGL   | 30072  |
| 7                                                               | 45 | 0.941 | NaCl          | lgK:-3.03(0.03SD)  | 3 | MGL   | 30072  |
| 8                                                               | 25 | 0     | Inf. Dilution | dG:16.03(0.34SD)kJ | 3 | EFE   | 30072  |
| 9                                                               | 25 | 0     | Inf. Dilution | dH:82.(2.SD)kJ     | 3 | EFE   | 30072  |

| Reaction No. 80097                        |  |  |  |  |  |  |  |
|-------------------------------------------|--|--|--|--|--|--|--|
| Lactic-1 + Al+3 + F-1 = Al+3_F-1_Lactic-1 |  |  |  |  |  |  |  |

| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaCl   | lgK:8.96(0.02SD) | 3 | MGL   | 30072  |
| 2  | 25 | 0.5   | NaCl   | lgK:7.98(0.05SD) | 2 | MGL   | 30072  |
| 3  | 25 | 1     | NaCl   | lgK:8.34(0.01SD) | 3 | MGL   | 30072  |

| Reaction No. 80247          |    |       |               |                   |   |       |        |
|-----------------------------|----|-------|---------------|-------------------|---|-------|--------|
| AsO4-3 + Al+3 = Al+3_AsO4-3 |    |       |               |                   |   |       |        |
| No                          | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                           | 25 | 0     | Inf. Dilution | lgK:14.6(0.1SD)   | 3 | EDH   | 29939  |
| 2                           | 25 | 0.109 | NaCl          | lgK:12.68(0.03SD) | 2 | MGL   | 29939  |
| 3                           | 25 | 0.25  | NaCl          | lgK:12.00(0.06SD) | 4 | MGL   | 29939  |
| 4                           | 25 | 0.502 | NaCl          | lgK:11.42(4SF)    | 3 | MGL   | 29939  |
| 5                           | 25 | 0.697 | NaCl          | lgK:11.39(0.06SD) | 4 | MGL   | 29939  |
| 6                           | 25 | 0.1   | NaCl          | dH:7.(2.SD)kJ     | 4 | MTD   | 29939  |

| Reaction No. 80248                           |    |       |               |                  |   |       |        |
|----------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Al+3 + AsO4-3 + H2O = Al+3_OH-1_AsO4-3 + H+1 |    |       |               |                  |   |       |        |
| No                                           | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                            | 25 | 0     | Inf. Dilution | lgK:9.9(0.1SD)   | 3 | EDH   | 29939  |
| 2                                            | 25 | 0.109 | NaCl          | lgK:8.29(0.05SD) | 4 | MGL   | 29939  |
| 3                                            | 25 | 0.25  | NaCl          | lgK:7.87(0.08SD) | 4 | MGL   | 29939  |
| 4                                            | 25 | 0.502 | NaCl          | lgK:7.44(0.10SD) | 4 | MGL   | 29939  |
| 5                                            | 25 | 0.697 | NaCl          | lgK:7.66(0.08SD) | 4 | MGL   | 29939  |

| Reaction No. 80293          |    |       |        |                   |   |       |        |
|-----------------------------|----|-------|--------|-------------------|---|-------|--------|
| P2O7-4 + Al+3 = Al+3_P2O7-4 |    |       |        |                   |   |       |        |
| No                          | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                           | 25 | 0.15  | NaCl/m | lgK:13.53(0.03SD) | 2 | MGL   | 29916  |
| 2                           | 25 | 0.5   | NaCl/m | lgK:12.15(0.04SD) | 3 | MGL   | 29916  |
| 3                           | 25 | 1     | NaCl/m | lgK:12.01(4SF)    | 3 | MGL   | 29916  |

| Reaction No. 80294                          |    |       |        |                   |   |       |        |
|---------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| H+1 + 2<P2O7-4> + Al+3 = Al+3_H+1_P2O7-4(2) |    |       |        |                   |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                           | 25 | 0.15  | NaCl/m | lgK:25.17(0.03SD) | 0 | MGL   | 29916  |
| 2                                           | 25 | 0.5   | NaCl/m | lgK:23.33(0.04SD) | 2 | MGL   | 29916  |
| 3                                           | 25 | 1     | NaCl/m | lgK:22.94(0.02SD) | 2 | MGL   | 29916  |

| Reaction No. 80295                    |    |       |        |                   |   |       |        |
|---------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| H+1 + P207-4 + Al+3 = Al+3_H+1_P207-4 |    |       |        |                   |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                     | 25 | 0.15  | NaCl/m | lgK:16.81(0.04SD) | 2 | MGL   | 29916  |
| 2                                     | 25 | 0.5   | NaCl/m | lgK:15.17(0.04SD) | 2 | MGL   | 29916  |
| 3                                     | 25 | 1     | NaCl/m | lgK:15.05(0.03SD) | 2 | MGL   | 29916  |

| Reaction No. 80296                |    |       |        |                   |   |       |        |
|-----------------------------------|----|-------|--------|-------------------|---|-------|--------|
| 2<P207-4> + Al+3 = Al+3_P207-4(2) |    |       |        |                   |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                 | 25 | 0.15  | NaCl/m | lgK:19.29(0.02SD) | 1 | MGL   | 29916  |
| 2                                 | 25 | 0.5   | NaCl/m | lgK:17.98(0.02SD) | 2 | MGL   | 29916  |
| 3                                 | 25 | 1     | NaCl/m | lgK:17.67(0.02SD) | 2 | MGL   | 29916  |

| Reaction No. 80297                           |    |       |        |                  |   |       |        |
|----------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Al+3 + P207-4 + H2O = Al+3_OH-1_P207-4 + H+1 |    |       |        |                  |   |       |        |
| No                                           | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                            | 25 | 0.15  | NaCl/m | lgK:6.69(0.06SD) | 2 | MGL   | 29916  |
| 2                                            | 25 | 0.5   | NaCl/m | lgK:6.46(0.03SD) | 2 | MGL   | 29916  |
| 3                                            | 25 | 1     | NaCl/m | lgK:6.06(0.02SD) | 2 | MGL   | 29916  |

| Reaction No. 80298                            |    |       |        |                   |   |       |        |
|-----------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| 2<H+1> + P3010-5 + Al+3 = Al+3_H+1(2)_P3010-5 |    |       |        |                   |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                             | 25 | 0.15  | NaCl/m | lgK:19.84(0.05SD) | 2 | MGL   | 29916  |
| 2                                             | 25 | 0.5   | NaCl/m | lgK:18.49(0.07SD) | 2 | MGL   | 29916  |
| 3                                             | 25 | 1     | NaCl/m | lgK:17.37(0.05SD) | 1 | MGL   | 29916  |

| Reaction No. 80299                      |    |       |        |                   |   |       |        |
|-----------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| H+1 + P3010-5 + Al+3 = Al+3_H+1_P3010-5 |    |       |        |                   |   |       |        |
| No                                      | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                       | 25 | 0.15  | NaCl/m | lgK:17.79(0.04SD) | 2 | MGL   | 29916  |
| 2                                       | 25 | 0.5   | NaCl/m | lgK:16.45(0.06SD) | 2 | MGL   | 29916  |
| 3                                       | 25 | 1     | NaCl/m | lgK:15.41(0.04SD) | 1 | MGL   | 29916  |

| Reaction No. 80300 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| P3O10-5 + Al+3 = Al+3_P3O10-5 |    |       |        |                    |   |       |        |
|-------------------------------|----|-------|--------|--------------------|---|-------|--------|
| No                            | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                             | 25 | 0.15  | NaCl/m | lgK:14.11 (0.03SD) | 2 | MGL   | 29916  |
| 2                             | 25 | 0.5   | NaCl/m | lgK:12.75 (0.04SD) | 2 | MGL   | 29916  |
| 3                             | 25 | 1     | NaCl/m | lgK:11.92 (0.03SD) | 1 | MGL   | 29916  |

| Reaction No. 80301                   |    |       |        |                    |   |       |        |
|--------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| 2<P3O10-5> + Al+3 = Al+3_P3O10-5 (2) |    |       |        |                    |   |       |        |
| No                                   | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                    | 25 | 0.15  | NaCl/m | lgK:17.95 (0.02SD) | 2 | MGL   | 29916  |
| 2                                    | 25 | 0.5   | NaCl/m | lgK:17.05 (0.03SD) | 2 | MGL   | 29916  |
| 3                                    | 25 | 1     | NaCl/m | lgK:16.74 (0.03SD) | 2 | MGL   | 29916  |

| Reaction No. 80302                             |    |       |        |                   |   |       |        |
|------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Al+3 + P3O10-5 + H2O = Al+3_OH-1_P3O10-5 + H+1 |    |       |        |                   |   |       |        |
| No                                             | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                              | 25 | 0.15  | NaCl/m | lgK:6.64 (0.03SD) | 2 | MGL   | 29916  |
| 2                                              | 25 | 0.5   | NaCl/m | lgK:6.30 (0.02SD) | 2 | MGL   | 29916  |
| 3                                              | 25 | 1     | NaCl/m | lgK:5.84 (0.02SD) | 2 | MGL   | 29916  |

| Reaction No. 80303                      |    |       |        |                    |   |       |        |
|-----------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| H+1 + P6O18-6 + Al+3 = Al+3_H+1_P6O18-6 |    |       |        |                    |   |       |        |
| No                                      | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                       | 25 | 0.15  | NaCl/m | lgK:11.10 (0.04SD) | 1 | MGL   | 29916  |
| 2                                       | 25 | 0.5   | NaCl/m | lgK:11.41 (0.04SD) | 1 | MGL   | 29916  |
| 3                                       | 25 | 1     | NaCl/m | lgK:12.68 (0.05SD) | 0 | MGL   | 29916  |

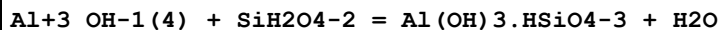
| Reaction No. 80304            |    |       |        |                   |   |       |        |
|-------------------------------|----|-------|--------|-------------------|---|-------|--------|
| P6O18-6 + Al+3 = Al+3_P6O18-6 |    |       |        |                   |   |       |        |
| No                            | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                             | 25 | 0.15  | NaCl/m | lgK:9.04 (0.04SD) | 1 | MGL   | 29916  |
| 2                             | 25 | 0.5   | NaCl/m | lgK:8.81 (0.03SD) | 2 | MGL   | 29916  |
| 3                             | 25 | 1     | NaCl/m | lgK:9.18 (0.04SD) | 2 | MGL   | 29916  |

| Reaction No. 80305                             |   |       |        |             |   |       |        |
|------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| Al+3 + P6O18-6 + H2O = Al+3_OH-1_P6O18-6 + H+1 |   |       |        |             |   |       |        |
| No                                             | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |



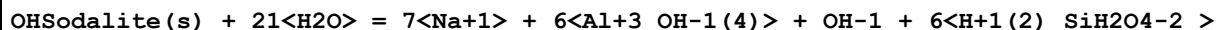
|   |    |      |        |                   |   |     |       |
|---|----|------|--------|-------------------|---|-----|-------|
| 1 | 25 | 0.15 | NaCl/m | lgK:3.87 (0.03SD) | 2 | MGL | 29916 |
| 2 | 25 | 0.5  | NaCl/m | lgK:3.14 (0.03SD) | 2 | MGL | 29916 |
| 3 | 25 | 1    | NaCl/m | lgK:2.64 (0.04SD) | 2 | MGL | 29916 |

## Reaction No. 80307



| No | t   | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|-----|-------|---------------|--------------------|---|-------|--------|
| 1  | 20  | 0     | Inf. Dilution | lgK:-0.42 (0.10SD) | 4 | EOD   | 30201  |
| 2  | 25  | 0     | Inf. Dilution | lgK:-0.42 (0.15SD) | 4 | EOD   | 30201  |
| 3  | 50  | 0     | Inf. Dilution | lgK:-0.38 (0.25SD) | 4 | EOD   | 30201  |
| 4  | 75  | 0     | Inf. Dilution | lgK:-0.35 (0.25SD) | 4 | EOD   | 30201  |
| 5  | 100 | 0     | Inf. Dilution | lgK:-0.33 (0.25SD) | 4 | EOD   | 30201  |

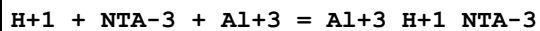
## Reaction No. 80308



Note: Stoichiometric analog of actual reported solid

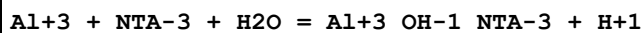
| No | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|-----|-------|---------------|------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:-60.34 (4SF) | 4 | EOD   | 30201  |
| 2  | 30  | 0     | Inf. Dilution | lgK:-61.18 (4SF) | 4 | EOD   | 30201  |
| 3  | 45  | 0     | Inf. Dilution | lgK:-64.20 (4SF) | 4 | EOD   | 30201  |
| 4  | 60  | 0     | Inf. Dilution | lgK:-66.69 (4SF) | 4 | EOD   | 30201  |
| 5  | 75  | 0     | Inf. Dilution | lgK:-68.89 (4SF) | 3 | EOD   | 30201  |
| 6  | 90  | 0     | Inf. Dilution | lgK:-71.26 (4SF) | 4 | EOD   | 30201  |
| 7  | 100 | 0     | Inf. Dilution | lgK:-72.68 (4SF) | 4 | EOD   | 30201  |

## Reaction No. 80358



| No | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:16.8 (3SF)     | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:21.5 (0.2SD)   | 0 | EDH   | 30680  |
| 3  | 25 | 0.15  | NaCl          | lgK:13.51 (0.08SD) | 0 | MGL   | 30680  |

## Reaction No. 80359



| No | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:10.91 (0.08SD) | 1 | EDH   | 30680  |
| 2  | 25 | 0.15  | NaCl          | lgK:6.83 (0.02SD)  | 3 | MGL   | 30680  |

| Reaction No. 80360                                                                                                        |    |       |               |                                 |   |       |        |
|---------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|---------------------------------|---|-------|--------|
| $\text{Al}+3 + \text{NTA}-3 + 2\text{<H}_2\text{O>} = \text{Al}+3_{\text{OH}-1(2)}_{\text{NTA}-3} + 2\text{<H}+1\text{>}$ |    |       |               |                                 |   |       |        |
| No                                                                                                                        | t  | I-Str | Medium        | Value (Dev)                     | W | Tech. | Ref. # |
| 1                                                                                                                         | 25 | 0     | Inf. Dilution | $\lg K: 0.4(0.1\text{SD})$      | 2 | EDH   | 30680  |
| 2                                                                                                                         | 25 | 0.15  | NaCl          | $\lg K: -0.95(0.02\text{SD})$   | 3 | MGL   | 30680  |
| 3                                                                                                                         | 25 | 0.15  | NaCl          | $dG: 5.4(2\text{SF})\text{kJ}$  | 3 | EDH   | 30680  |
| 4                                                                                                                         | 25 | 0.15  | NaCl          | $dH: 55.(2.\text{SD})\text{kJ}$ | 3 | EDH   | 30680  |

| Reaction No. 80361                                                                                                                                  |    |       |        |                                  |   |       |        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|----------------------------------|---|-------|--------|
| $2\text{<Al}+3\text{>} + 2\text{<NTA}-3\text{>} + 2\text{<H}_2\text{O>} = \text{Al}+3(2)_{\text{OH}-1(2)}_{\text{NTA}-3(2)} + 2\text{<H}+1\text{>}$ |    |       |        |                                  |   |       |        |
| No                                                                                                                                                  | t  | I-Str | Medium | Value (Dev)                      | W | Tech. | Ref. # |
| 1                                                                                                                                                   | 25 | 0.15  | NaCl   | $\lg K: 16.82(0.05\text{SD})$    | 3 | MGL   | 30680  |
| 2                                                                                                                                                   | 25 | 0.15  | NaCl   | $dG: -95.9(3\text{SF})\text{kJ}$ | 3 | EDH   | 30680  |
| 3                                                                                                                                                   | 25 | 0.15  | NaCl   | $dH: 85.(4.\text{SD})\text{kJ}$  | 3 | EDH   | 30680  |

| Reaction No. 80410                                                                                                                                                                                                      |    |       |               |                                    |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|------------------------------------|---|-------|--------|
| $\text{Al}(\text{s}) + 2\text{<H}_2(\text{g})\text{>} + 0.5\text{<N}_2(\text{g})\text{>} + 4\text{<O}_2(\text{g})\text{>} + 3\text{<Si}(\text{s})\text{>} = \text{NH}_4\text{AlSi}_3\text{O}_8(\text{disord},\text{s})$ |    |       |               |                                    |   |       |        |
| No                                                                                                                                                                                                                      | t  | I-Str | Medium        | Value (Dev)                        | W | Tech. | Ref. # |
| 1                                                                                                                                                                                                                       | 25 | 0     | Inf. Dilution | $dG: -3525.1(5\text{SF})\text{kJ}$ | 3 | ETR   | 30953  |

| Reaction No. 80661                                                                                                                                 |    |       |               |                                    |   |       |        |
|----------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|------------------------------------|---|-------|--------|
| $\text{Al}(\text{s}) + 1.5\text{<H}_2(\text{g})\text{>} + 1.5\text{<O}_2(\text{g})\text{>} = \text{Al}+3_{\text{OH}-1(3)}_{\text{(Micr.Gibb.,s)}}$ |    |       |               |                                    |   |       |        |
| No                                                                                                                                                 | t  | I-Str | Medium        | Value (Dev)                        | W | Tech. | Ref. # |
| 1                                                                                                                                                  | 25 | 0     | Inf. Dilution | $dG: -1148.4(5\text{SF})\text{kJ}$ | 5 | CRV   | 31308  |
| 2                                                                                                                                                  | 25 | 0     | Inf. Dilution | $dH: -1265.3(5\text{SF})\text{kJ}$ | 5 | CRV   | 31308  |

| Reaction No. 80663                                                                                                                                                                                                                                                                     |    |       |               |                             |   |       |        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------------------|---|-------|--------|
| $\text{Al}+3(2)_{\text{Ca}+2(4)}_{\text{Cl}-1}_{\text{OH}-1(12)}_{\text{SO}_4-2(0.5)}_{\text{H}_2\text{O}(6)}_{\text{(s)}} = 4\text{<Ca}+2\text{>} + 2\text{<Al}+3_{\text{OH}-1(4)}\text{>} + \text{Cl}-1 + 0.5\text{<SO}_4-2\text{>} + 4\text{<OH}-1\text{>} + 6\text{<H}_2\text{O>}$ |    |       |               |                             |   |       |        |
| No                                                                                                                                                                                                                                                                                     | t  | I-Str | Medium        | Value (Dev)                 | W | Tech. | Ref. # |
| 1                                                                                                                                                                                                                                                                                      | 25 | 0     | Inf. Dilution | $\lg K: -28.53(4\text{SF})$ | 5 | CRV   | 31308  |

| Reaction No. 80666                                                                                                                                                                                                                       |    |       |               |                             |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------------------|---|-------|--------|
| $\text{Na}_2\text{Al}_2\text{Si}_3\text{O}_{10} \cdot 2\text{H}_2\text{O}(\text{Natr.,s}) + 8\text{<H}_2\text{O>} = 2\text{<Na}+1\text{>} + 2\text{<Al}+3_{\text{OH}-1(4)}\text{>} + 3\text{<H}+1(2)_{\text{SiH}_2\text{O}_4-2}\text{>}$ |    |       |               |                             |   |       |        |
| No                                                                                                                                                                                                                                       | t  | I-Str | Medium        | Value (Dev)                 | W | Tech. | Ref. # |
| 1                                                                                                                                                                                                                                        | 25 | 0     | Inf. Dilution | $\lg K: -25.19(4\text{SF})$ | 1 | ELC   |        |
| 2                                                                                                                                                                                                                                        | 25 | 0     | Inf. Dilution | $\lg K: -7.20(3\text{SF})$  | 0 | CRV   | 31308  |

Note (2): For species SiO<sub>2</sub>(aq) !

Reaction No. 80880



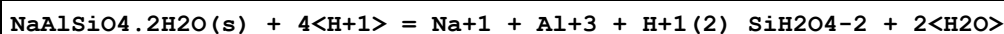
| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:11.5(3SF) | 5 | MSL   | 31760  |

Note (1): Slightly impure solid

|   |    |   |               |              |   |     |       |
|---|----|---|---------------|--------------|---|-----|-------|
| 2 | 40 | 0 | Inf. Dilution | lgK:9.6(2SF) | 5 | MSL | 31760 |
|---|----|---|---------------|--------------|---|-----|-------|

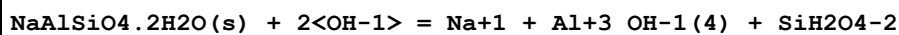
Note (2): Slightly impure solid

Reaction No. 80886



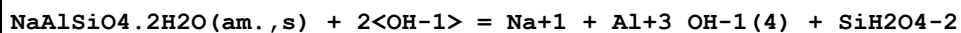
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:12.1(3SF)   | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:11.1(0.2SD) | 0 | MSL   | 31772  |

Reaction No. 80887



| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-5.74(0.2SD) | 5 | MSL   | 31772  |

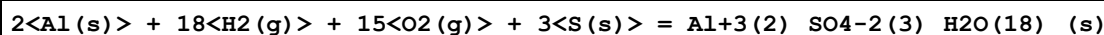
Reaction No. 80888



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-3.15(3SF) | 5 | MSL   | 31772  |

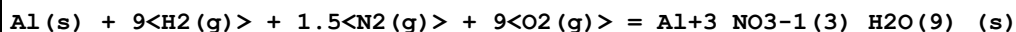
Note (1): Table 4

Reaction No. 81186



| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dH:-2122.1(5SF) | 4 | CRV   | 12011  |

Reaction No. 81187



| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dH:-897.96(5SF) | 4 | CRV   | 12011  |

Reaction No. 81188

| 6<C(s)> + 4.5<H2(g)> + Al(s) + 3<O2(g)> = Al+3_Acetic-1(3)_(s) |    |       |               |                |   |       |        |
|----------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| No                                                             | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                              | 25 | 0     | Inf. Dilution | dH:-452.3(4SF) | 4 | CRV   | 12011  |

| Reaction No. 81307                                                 |    |       |               |                 |   |       |        |
|--------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| AlO2-1 + 4<H+1> = Al+3 + 2<H2O>                                    |    |       |               |                 |   |       |        |
| Note: The species AlO2-1 is an artefact; it should be Al+3_OH-1(4) |    |       |               |                 |   |       |        |
| No                                                                 | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                  | 25 | 0     | Inf. Dilution | lgK:24.601(5SF) | 0 | CRV   | 32224  |

| Reaction No. 81308                         |    |       |               |                 |   |       |        |
|--------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Al2O3(gamma,s) + 6<H+1> = 2<Al+3> + 3<H2O> |    |       |               |                 |   |       |        |
| No                                         | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                          | 25 | 0     | Inf. Dilution | lgK:22.390(5SF) | 4 | CRV   | 32224  |

| Reaction No. 81349                                                  |    |       |               |                   |   |       |        |
|---------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| CaAl2Si4O12.4H2O(s) + 8<H+1> = Ca+2 + 2<Al+3> + 4<H+1(2)_SiH2O4-2 > |    |       |               |                   |   |       |        |
| No                                                                  | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                   | 25 | 0     | Inf. Dilution | lgK:11.66(4SF)    | 6 | CRV   | 27383  |
| 2                                                                   | 25 | 0     | Inf. Dilution | lgK:14.25(4SF)    | 0 | CRV   | 32224  |
| 3                                                                   | 25 | 0     | CHEMVAL1      | lgK:11.85(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                   | 25 | 0     | CHEMVAL2      | lgK:11.85(0.01SD) | 0 | ENS   | 5156   |
| 5                                                                   | 25 | 0     | CHEMVAL3      | lgK:11.85(0.01SD) | 0 | ENS   | 5156   |
| 6                                                                   | 25 | 0     | CHEMVAL4      | lgK:11.80(0.01SD) | 1 | ENS   | 5156   |
| Note (6): query mineral name "laumontite"                           |    |       |               |                   |   |       |        |

| Reaction No. 81350                                                           |    |       |               |                |   |       |        |
|------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| CaAl4Si2O10(OH)2(s) + 14<H+1> = Ca+2 + 4<Al+3> + 2<H+1(2)_SiH2O4-2> + 4<H2O> |    |       |               |                |   |       |        |
| No                                                                           | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                            | 25 | 0     | Inf. Dilution | lgK:42.33(4SF) | 4 | CRV   | 32224  |

| Reaction No. 81351                                                                 |    |       |               |                |   |       |        |
|------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Ca2Al3Si3O12(OH)(clino,s) + 13<H+1> = 2<Ca+2> + 3<Al+3> + 3<H+1(2)_SiH2O4-2> + H2O |    |       |               |                |   |       |        |
| No                                                                                 | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                  | 25 | 0     | Inf. Dilution | lgK:44.16(4SF) | 4 | CRV   | 32224  |

| Reaction No. 81352 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| Ca <sub>2</sub> FeAl <sub>2</sub> Si <sub>3</sub> O <sub>12</sub> (OH) (s) + 13<H+1> + e-1 = 2<Ca+2> + 2<Al+3> + Fe+2 + 3<H+1(2)_SiH <sub>2</sub> O <sub>4</sub> -2 > + H <sub>2</sub> O |    |       |               |                |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| No                                                                                                                                                                                       | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                                                                                                        | 25 | 0     | Inf. Dilution | lgK:44.29(4SF) | 4 | CRV   | 32224  |

| Reaction No. 81452                                                                                                                      |    |       |               |               |   |       |        |
|-----------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| KAlSi <sub>3</sub> O <sub>8</sub> (Feld.,s) + 4<H+1> + 4<H <sub>2</sub> O> = K+1 + Al+3 + 3<H+1(2)_SiH <sub>2</sub> O <sub>4</sub> -2 > |    |       |               |               |   |       |        |
| No                                                                                                                                      | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                                                                       | 25 | 0     | Inf. Dilution | lgK:0.05(1SF) | 4 | CRV   | 32224  |

| Reaction No. 81453                                                                                                                     |    |       |               |               |   |       |        |
|----------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| KAlSi <sub>3</sub> O <sub>8</sub> (Mur.,s) + 4<H <sub>2</sub> O> + 4<H+1> = K+1 + Al+3 + 3<H+1(2)_SiH <sub>2</sub> O <sub>4</sub> -2 > |    |       |               |               |   |       |        |
| No                                                                                                                                     | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                                                                      | 25 | 0     | Inf. Dilution | lgK:1.25(3SF) | 4 | CRV   | 32224  |

| Reaction No. 81454                                                                                                                                        |    |       |               |                |   |       |        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| KFe <sub>3</sub> AlSi <sub>3</sub> O <sub>10</sub> (OH) <sub>2</sub> (s) + 10<H+1> = K+1 + 3<Fe+2> + Al+3 + 3<H+1(2)_SiH <sub>2</sub> O <sub>4</sub> -2 > |    |       |               |                |   |       |        |
| No                                                                                                                                                        | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                                                                         | 25 | 0     | Inf. Dilution | lgK:23.13(4SF) | 4 | CRV   | 32224  |

| Reaction No. 81455                                                                                                                                        |    |       |               |                |   |       |        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| KMg <sub>3</sub> AlSi <sub>3</sub> O <sub>10</sub> (OH) <sub>2</sub> (s) + 10<H+1> = K+1 + 3<Mg+2> + Al+3 + 3<H+1(2)_SiH <sub>2</sub> O <sub>4</sub> -2 > |    |       |               |                |   |       |        |
| No                                                                                                                                                        | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                                                                         | 25 | 0     | Inf. Dilution | lgK:38.17(4SF) | 4 | CRV   | 32224  |

| Reaction No. 81473                                                                                                                                                            |    |       |               |                |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| 2MgO.2Al <sub>2</sub> O <sub>3</sub> .5SiO <sub>2</sub> .H <sub>2</sub> O(s) + 16<H+1> + H <sub>2</sub> O = 2<Mg+2> + 4<Al+3> + 5<H+1(2)_SiH <sub>2</sub> O <sub>4</sub> -2 > |    |       |               |                |   |       |        |
| No                                                                                                                                                                            | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                                                                                             | 25 | 0     | Inf. Dilution | lgK:51.39(4SF) | 4 | CRV   | 32224  |

| Reaction No. 81488                                                                                       |    |       |               |               |   |       |        |
|----------------------------------------------------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Al+3_H+1_NH <sub>3</sub> _SO <sub>4</sub> -2(2)_(s) = H+1_NH <sub>3</sub> + Al+3 + 2<SO <sub>4</sub> -2> |    |       |               |               |   |       |        |
| No                                                                                                       | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                                        | 25 | 0     | Inf. Dilution | lgK:3.47(3SF) | 4 | CRV   | 32224  |

| Reaction No. 81489                                                                                                                                   |   |       |        |             |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| Al+3_H+1_NH <sub>3</sub> _SO <sub>4</sub> -2(2)_H <sub>2</sub> O(12)_(s) = H+1_NH <sub>3</sub> + Al+3 + 2<SO <sub>4</sub> -2> + 12<H <sub>2</sub> O> |   |       |        |             |   |       |        |
| No                                                                                                                                                   | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

|   |    |   |               |                 |   |     |       |
|---|----|---|---------------|-----------------|---|-----|-------|
| 1 | 25 | 0 | Inf. Dilution | lgK:-5.97 (3SF) | 4 | CRV | 32224 |
|---|----|---|---------------|-----------------|---|-----|-------|

## Reaction No. 81491

$$\text{Al}+3(5)\text{H}+1(9)\text{NH}_3(3)\text{PO}_4-3(8)\text{H}_2\text{O}(18)\text{(s)} = 6\text{H}+1 + 3\text{H}+1\text{NH}_3 + 5\text{Al}+3 + 8\text{PO}_4-3 + 18\text{H}_2\text{O}$$

| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-175.8 (4SF)  | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:-185.74 (5SF) | 0 | CRV   | 32224  |

## Reaction No. 81506

$$\text{NaCa}_2\text{Mg}_4\text{Al}_3\text{Si}_6\text{O}_{22}(\text{OH})_2(\text{s}) + 22\text{H}+1 = \text{Na}+1 + 2\text{Ca}+2 + 4\text{Mg}+2 + 3\text{Al}+3 + 6\text{H}+1(2)\text{SiH}_2\text{O}_4-2$$

| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:103.45 (5SF) | 4 | CRV   | 32224  |

## Reaction No. 81529

$$\text{Al}+3(3)\text{Pb}+2\text{OH}-1(6)\text{SO}_4-2\text{PO}_4-3(\text{s}) + 6\text{H}+1 = \text{Pb}+2 + 3\text{Al}+3 + \text{PO}_4-3 + \text{SO}_4-2 + 6\text{H}_2\text{O}$$

| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-2.5 (2SF) | 4 | CRV   | 32224  |

## Reaction No. 81539

$$\text{Al}+3(3)\text{Pb}+2\text{OH}-1(5)\text{PO}_4-3(2)\text{H}_2\text{O}(\text{s}) + 5\text{H}+1 = \text{Pb}+2 + 3\text{Al}+3 + 2\text{PO}_4-3 + 6\text{H}_2\text{O}$$

| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-32.79 (4SF) | 4 | CRV   | 32224  |

## Data Assessment

| Weight | Assessment                                    |
|--------|-----------------------------------------------|
| 9      | highest accuracy                              |
| 8      | multi-determined; excellent dependability     |
| 7      | trustworthy; outstanding single determination |
| 6      | better than average reliability               |
| 5      | average single determination                  |
| 4      | some doubt                                    |
| 3      | poorly known                                  |
| 2      | guess to achieve consistency (or just as bad) |

|   |                               |
|---|-------------------------------|
| 1 | no information but consistent |
| 0 | problematic/inconsistent      |

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## Techniques

| Abbrev | Method                                          |
|--------|-------------------------------------------------|
| CIU    | Critical IUPAC publication (crit.)              |
| CMN    | Several experimental values (crit.)             |
| CRV    | Critical review                                 |
| EAE    | Automated extrapolation                         |
| EBU    | Brown's Unified Theory                          |
| EDH    | Debye-Hueckel corrections                       |
| EDT    | Debye-Hueckel & Temperature corrections         |
| EES    | Personal estimate                               |
| EFE    | Free Energy relations                           |
| EIE    | Interpolation/Extrapolation (not Debye-Hueckel) |
| EIU    | Critical IUPAC review (estim.)                  |
| ELC    | Linear Combination of Reactions                 |
| EMU    | Method unknown                                  |
| ENS    | Not specified                                   |
| EOD    | Calculated from data of others                  |
| EOR    | Other Reaction Constants                        |
| ERG    | Relative Gibbs energy                           |
| ESC    | Standard conditions anchor                      |
| ESL    | Similar Ligand                                  |
| ETR    | Thermodynamic relationships                     |
| ETS    | Ternary (statistical)                           |
| MAM    | Amperometry                                     |
| MCL    | Calorimetry                                     |
| MCN    | Conductivity                                    |
| MCO    | Colorimetry (often for measuring pH)            |
| MDS    | Distribution between two phases                 |
| MEF    | Electromotoric force, not specified             |
| MEP    | Electrophoresis                                 |

|     |                                     |
|-----|-------------------------------------|
| MFP | Freezing point                      |
| MGC | Gas chromatography                  |
| MGL | Glass electrode                     |
| MHG | Emf with amalgam electrode          |
| MHY | Emf with hydrogen electrode         |
| MIS | Ion selective electrode             |
| MIX | Ion exchange                        |
| MKN | Rate of reaction                    |
| MLC | Ligand competition                  |
| MMR | Nuclear magnetic resonance          |
| MMX | Mixed techniques                    |
| MOR | Optical Rotary Dispersion           |
| MPH | pH method, not specified            |
| MPL | Polarography                        |
| MPS | Potentiometry and Spectrophotometry |
| MPT | Potentiometry                       |
| MQH | Quinhydrone electrode               |
| MRM | Raman spectra                       |
| MRX | Emf with redox electrode            |
| MSC | Miscellaneous                       |
| MSF | Stopped Flow                        |
| MSL | Solubility                          |
| MSP | Spectroscopy                        |
| MTD | Temperature difference              |
| MTN | Tyndallometry, nephelometry         |
| MVM | Voltametric Methods                 |

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